Class–Theoretic Foundations of Mathematics & Applications (Very Early Draft)

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Introduction

This book develops class-theoretic foundations of mathematics alongside a prototype proof assistant. Its central aim is to make familiar mathematical reasoning explicit enough to support both readable proofs and mechanized checking. A second aim is exploratory: descriptors, class-valued families, and types offer several ways to express constructions that ordinary notation leaves implicit.

The chapters record more than one foundational approach. Each section should be read with its own stated language, assumptions, and interpretation of terms. In particular, a change from total evaluation to partial descriptors, or from classes to conglomerates, changes the context in which formulas are understood. The guide that follows the contents identifies these approaches and suggests a route through them. The implementation's chosen semantics is one thread of the book; the other threads retain their own mathematical purpose.

This is a developing manuscript. Some sections contain detailed arguments; others formulate proposals or record proof sketches. Local notes identify these differences so that an unfinished argument can be studied without being mistaken for an established result.

One motivating proposal concerns the relation between ZFC and a system consisting of the logical axioms, class extensionality, unbounded class construction, and the assertion that at least one set exists:

> If ZFC is contradictory, then this class-theoretic system is contradictory.

The direction of this proposed implication matters: it would transfer consistency from the class-theoretic system to ZFC. Its justification requires precise versions of the listed axioms and a derivation or interpretation; the formulation here introduces the question rather than supplying that argument. Theorem 17.1 gives a two-object model of the bare relational comprehension principles just listed. That model has no infinite set, so a direct derivation of the usual set axioms needs additional principles. It does not address a richer language with extra constructors or reflection rules. Identifying those extra principles is part of making the motivating proposal precise.

Another central idea is *predicates-as-types, classes-as-expressions*. It connects the language used to state mathematical properties with the representations used to check proofs. The later type-theoretic sections explore this connection, including earlier variants with different interpretations of terms and types.

The accompanying work has three strands:

1. the LaTeX source of this book;
2. a Standard ML prototype for representing the theory, its metatheory, and proof checking;
3. experiments with interpretable dependent types for Poly/ML, intended to explore a more expressive metatheory.

The files in this repository document the available work. Code excerpts in the book are identified as prototypes where their implementation is incomplete.

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Guide to the Five Foundational Approaches

The book explores several ways to express classes, choice, partiality, and types. This guide locates those approaches and distinguishes their local assumptions. The term *active* describes the proof-compiler development; it is not a ranking of the mathematical value of the other sections. Inclusion in one book does not assert a common semantics or a proved translation between approaches.

How to read a section. First identify what its variables and terms denote, which quantifiers it uses, and how it treats an expression with no unique value. Then read its claims under those conventions. A total default value, an undefined term, and an uncertain truth value are different responses to different questions. When a section introduces a new convention, that convention governs its argument; definitions from another approach apply only when explicitly carried over. Labels such as `[OUTDATED]` record the history of a draft, not a reason to dismiss the idea it investigates.

Notation whose scope matters.

- X set is the assertion that X is a set. Every class is a subclass of V, but only sets are members of V.
- $!x, \varphi$ means that at most one x satisfies φ; $\exists!x, \varphi$ additionally asserts existence. A semi-singleton may therefore be empty.
- In the class-axiom chapter, $\mathrm{Dom}(F)$ records represented arguments, including those with proper-class values; $\mathrm{dom}(F)$ selects arguments whose guarded evaluations are sets. In ordinary graph contexts, the usual domain is the set or class of first coordinates. Check the local definition when moving between these settings.
- A relational image collects second coordinates of records. An image formed from evaluated values is a different operation for families encoding proper-class values. The image chapter states which one it uses.
- Pair notation and descriptor notation are interfaces, not a guarantee that two sections use identical codes or off-domain values. The corresponding local definitions determine their interpretation.

1 Active Mainline: Approaches 1 and 2

The active development uses classical two-valued class theory, with Morse–Kelley (MK) and NBG fragments distinguished by their comprehension principles. In MK, comprehension permits quantification over classes; the NBG restriction permits only set quantifiers within the defining formula, while allowing class parameters. One total evaluation convention, recorded in the local Standard ML prototype, is unconditional union evaluation:

$$F(x) \stackrel{\text{def.}}{:=} \bigcup \{\, s \mid (x, s) \in F \,\}.$$

The book explores the recursive Dunaev tagged construction for pairs of arbitrary classes, with pairing and product defined simultaneously:

$$(a, b) \coloneqq (\{0\} \times a) \cup (\{1\} \times b),$$

This is not an outer Kuratowski pair of proper classes. The newer Obelisk entry `DefDunaev OrderedPair` uses a different marker-tagged encoding. The ordered-pair section distinguishes these constructions and their empty pairs; a projection proof for one does not certify the recursive definition of the other.

Two total evaluation conventions. The guarded convention in the class-axiom section and in the newer Obelisk export `DefValueAtArg` uses

$$F_V(x) = \begin{cases} \bigcup \{\, s \mid (x, s) \in F \,\}, & \text{if } \exists s,\ (x, s) \in F, \\ V, & \text{otherwise.} \end{cases}$$

The subscript here distinguishes conventions for the reader. Unconditional union evaluation returns $\varnothing$ on an absent fibre; guarded evaluation returns V. Both are total and agree when a fibre is represented. Neither is the partial description used in PFOL. The newer export also displays an unguarded family-evaluation theorem that conflicts with its guarded definition on the empty family. Until the source is reconciled, that theorem must be read with an explicit domain hypothesis, not as evidence that the two defaults coincide.
The principal sources are `presentation.ML`, `060_axioms.tex`, and `070_ordered_pair.tex`.
The accompanying type discipline is *predicates/formulas-as-types, classes-as-expressions.* Class expressions are syntactic objects; their representations in SML are not object-theoretic classes. The chapter `130_classes_as_types.tex` preserves older proposals under local `[OUTDATED]` labels alongside its explicit set models. Its scoped predicate/type interface treats $\mathrm{Emb}(A)$ as a supplied schematic predicate, not as an object containing all values. The collection operation κ retains only set instances: collecting and then testing membership recovers a predicate exactly when all its instances are sets. Option storage has explicit sethood guards, and a countable-syntax diagonal argument limits what finite closed expressions can denote. These results do not supply a uniform internal truth predicate or a soundness theorem for every proposed constructor. The local discussion explains why identifying every type with $\top$ conflicts with the intended uniqueness of the term–type relation. This objection concerns that interpretation, rather than every possible use of classes or predicates as types.

2 Preserved Alternative 3: Uncertain Descriptors

`020_descriptors.tex` and `180_many-valued.tex` preserve a three-valued/uncertain calculus. Its three conventions are distinct:

$$\tau \coloneqq \tau^{\mathcal{U}}, \qquad \epsilon \coloneqq \tau^{\{\varnothing\}}, \qquad \varepsilon \coloneqq \tau^{\varnothing}.$$

The last descriptor is partial: no default value is supplied. The uncertain value U belongs to this alternative, not to the classical active kernel.

3 Preserved Alternative 4: Farmer's PFOL

`210_partial_functions_semantics.tex` presents Farmer's Partial First-Order Logic. Its iota descriptor satisfies

$$(\iota x, \varphi)\downarrow \longleftrightarrow (\exists! x, \varphi),$$

and partial values are compared with Kleene equality $\simeq$. These notions must not be silently identified with the active total union evaluation.

4 Preserved Alternative 5: Conglomerate-Valued Semantics

`030_alternative_semantics.tex` retains the outdated subset semantics with assignments of the form

$$A : \mathrm{Var} \to \mathcal{P}(K).$$

It is an independent historical semantics, not an interpretation of the active formulas-as-types discipline.

5 Reading and Maintenance Rules

For the class-theoretic development, begin with the logical framework, axioms, ordered pairs, and sethood, then continue to class operations, ordinals, and recursion. For descriptors and partiality, read the uncertain descriptor notes, the many-valued experiments, and the PFOL section as separate approaches. For the connection with implementation, read the classes-as-types discussion together with the language and proof-tree notes.

The active logical metatheory. The assembled logic chapter proves transport of quantification along syntax coding and identifies the extra surjectivity needed for semantic quantification. It gives a conservative total witness extension on nonempty set domains, a deduction proof retaining context freshness, and the diagonal construction and Löb theorem under explicit hypotheses. The internal-truth obstruction distinguishes a same-language truth scheme from external or stratified evaluation. Its opaque-constant result also separates normalization from Boolean canonicity and from the decision to generate executable code. Separate classical-fragment and candidate-selection lemmas are preserved in a disabled historical block of the source; they are not part of the assembled chapter.

A dependency route through the constructions. For the recursive constructions, first read the elementary ordinal comparison results in the ordinal chapter, then ordinal and membership recursion in the induction chapter. Return to the ordinal chapter for set-like relation recursion, collapse, and Hartogs' theorem; the recursive pairing proof uses the membership-recursion result. The book retains its historical chapter order, so some of these prerequisites occur after their first application. In particular, the set-valued transfinite-recursion theorem at the start of the induction chapter is the prerequisite for the ordinal chapter's later founded-recursion and ordinal-addition arguments; it uses elementary ordinal ordering, not those later

constructions. The W-type height arguments then apply founded recursion to individual address sets before establishing foundedness of the relation between trees. The union argument has a different role: its ordinal facts are supplied as local assumptions, and its proof does not invoke later recursion results that themselves assume union. The finite and hereditarily finite models in the class overview help distinguish these assumptions from bare comprehension.

From concrete types to their universal properties. The classes-as-types chapter constructs Boolean, option, list, W-, and M-type models in explicit set-sized settings. The following chapter extends the finite-address models to a possibly proper class of set labels with set arities. Its initiality and finality quantify over class structure maps in the stated NBG setting; individual trees remain sets. The language chapter separately studies a specified simple syntax: type checking, substitution, preservation, normalization, and confluence. Its Boolean operational extension adds a two-point type and conditionals with both branches typed; the new substitution, progress, normalization, and confluence cases are proved separately, yielding decidable typed conversion. The category chapter explains recursion and corecursion through initial algebras and final coalgebras, and derives fusion, induction, and coinduction principles. A semantic model and a syntactic decision procedure are distinct results, even when they use the same type names.

The analytic route. Metric compactness and finite-dimensional normed spaces supply the initial topological preliminaries. The topology chapter also proves proper ultrafilter extension, the ultrafilter criterion for compactness, and set-indexed Tychonoff in an explicit classical setting with Choice. Measure theory constructs the integral and proves its convergence theorems and function-space completeness. Functional analysis then develops duality and Banach-space principles, compact operators, uniform integrability, and the sequential finite-measure Dunford–Pettis criterion. A further argument proves the full topological equivalence for real L^1 on a finite positive measure space: uniform integrability, compactness of the weak closure, and the weak subsequence criterion are equivalent. Its prerequisites include Hilbert-ball weak compactness via products and Riesz representation, preservation of uniform integrability under weak closure, and weak subsequences in weakly compact subsets of normed spaces; uniformly controlled truncations provide the ultrafilter proof. The probability applications compare weak, setwise, in-measure, and total-variation convergence and prove that probability densities form a weakly closed set, giving compactness results for their uniformly integrable families. The examples distinguish sequences from nets and finite measure from infinite counting measure; those changes of setting matter to the conclusions. Each chapter records the choice assumptions actually used.
The Schrödinger chapters give a further analytic route: the free heat equation and its uniqueness class, an abstract criterion for separately continuous kernels, bounded-potential semigroups, and quantitative regularity of dynamically defined kernels. They explicitly import Fourier–Plancherel and the self-adjoint spectral theorem, then prove the perturbation, kernel-continuity and parameter estimates. The same chapters form the standalone thesis in `thesis_schrodinger.tex`. They also use ordinary multivariable calculus and Lebesgue product integration, including Tonelli and Fubini, as analytic background; the preceding measure chapter does not construct product measures. First spatial derivatives and smooth positive-time dependence are proved without differentiating the potential; a discontinuous-potential example explains why classical second spatial derivatives require more care. The dynamical discussion distinguishes weighted regularity of a canonical kernel from strong or norm continuity of the complete operator family. Its final spectral application proves a Euclidean specialization of the Beckus–Takase localization method from the Gaussian first moment and the localized operator estimate. It compares closed heat-spectrum unions over invariant subsystems, rather than spectra at arbitrary individual phases; the general group theorem is a separate comparison, not a proof premise.

The assembled book also includes exploratory sections from approaches 3–5; these do not inherit the active implementation's conventions merely by appearing after a mainline chapter. Examples include `180_many-valued.tex`, `210_partial_functions_semantics.tex`, and `030_alternative_semantics.tex`. The chapter `020_descriptors.tex` supplies the descriptor comparison and its finite selection model. The historical iota and family constructions introduce further local variants. A heading marked as an outline or a proof sketch indicates the extent of the current development. The Alexandrov section now proves the correspondence between preorders and topologies closed under arbitrary intersections, including its continuity and separation criteria. The unspecified Strang placeholder is retained as source but excluded from the assembled book. The files `025_unified_descriptors.tex` and `035_unified_type_semantics.tex` are retained only as rejected, non-normative drafts marked `[OUTDATED]`. `050_unified_interface.tex` provides navigation only: changing a `\topic` title does not and cannot switch foundations.

Chapter

Logical framework

6 Example of the proof assistant's output

This chapter develops several related projects: a notation for manipulating syntax in a proof assistant, a classical predicate calculus, and a framework for comparing effective proof with an ideal, non-effective foundational theory. Each project has its own assumptions. In particular, a computation on syntax, a derivation in a theory, and truth in an intended interpretation are different notions; the notation introduced below is intended to make their interaction explicit. The historical semantic experiments retained in the source use additional conventions of their own.
The screenshot below illustrates tests of a kernel that stores proof objects. Storing the entire derivation is an implementation choice; it is distinct from checking that each inference is licensed by the kernel.
The motivating question is how much metatheory can be represented *syntactically* inside the theory. Encoding a metatheoretic statement is only the first step: a rule that reflects that statement back into the theory needs its own justification. A related approach represents a theory *semantically*, through an inner model. These approaches serve different purposes. The later sections explore a framework for discussing incompleteness while keeping these levels visible.

Informal definition 6.1 (syntactic and semantic types)**.**
A guest type A in a [conglomerate–level] type theory of this book is called *syntactic* if its intended meaning – as a "collection" of "elements" of some size – is "bijectively representable" in the host language. That is, there exists a host type T such that:

1. There is a bijection between the "intended meanings" of the terms of the guest type A and the terms of the host type T.

2. In particular, the interpretation of A as a "collection" is "effectively" given by its image in T: the representation must specify how a term of type T is read as an element of A. This is an informal representability requirement; a claim of computability would additionally require an explicit encoding and decoding procedure.

```
main_DEMO ();
Some elementary theorems:
th5 : F,‹[‹(B0∈x0)›]›⊢‹(∃(x0:Class),(B0∈x0))›
th1 : F,‹[‹(B0∈x0)›]›⊢‹(∀(B0:Class),((B0 set)⟷(∃(x0:Class),(B0∈x0))))›
th2 : F,‹[‹(B0∈x0)›]›⊢‹((B0 set)⟷(∃(x0:Class),(B0∈x0)))›
An inference rule:
(((((((((
th5 : F,‹[‹(B0∈x0)›]›⊢‹(∃(x0:Class),(B0∈x0))›
---------------------------------------  (def_to_intro th2)
res : F,‹[‹(B0∈x0)›]›⊢‹(B0 set)›
)))))))))
An application of a deduction rule:
res2 : F ⊢‹((B0∈x0)→(B0 set))›
Using: shen_test ();
A simple theorem is being proven using tactics:
F,‹[‹(∃(y0:Class),(∀(x0:Class),(x0∈y0)))›]›⊢‹(∃(y0:Class),(a0∈y0))›
F,‹[‹(∀(x0:Class),(x0∈b0))›]›⊢‹(∃(y0:Class),(a0∈y0))›
F,‹[‹(a0∈b0)›]›⊢‹(∃(y0:Class),(a0∈y0))›
F,‹[‹(a0∈b0)›]›⊢‹(a0∈b0)›
Displaying the proof of the theorem:
Proof.
1       |F,‹[‹(a0∈b0)›]›⊢‹(a0∈b0)›
2       |F,‹[‹(a0∈b0)›]›⊢‹(∃(y0:Class),(a0∈y0))›
3       |F,‹[‹(∀(x0:Class),(x0∈b0))›]›⊢‹(∃(y0:Class),(a0∈y0))›
4       |F,‹[‹(∃(y0:Class),(∀(x0:Class),(x0∈y0)))›]›⊢‹(∃(y0:Class),(a0∈y0))›
5       |F,‹[‹(∃(y0:Class),(∀(x0:Class),(x0∈y0)))›]›⊢‹(∀(x0:Class),(∃(y0:Class),(x0∈y0))))›
6       |F ⊢‹((∃(y0:Class),(∀(x0:Class),(x0∈y0)))→(∀(x0:Class),(∃(y0:Class),(x0∈y0))))›
Q.E.D.
Incompleteness-related statements:
Let Q be Robinson's arithmetic.
Let T be a recursively axiomatizable theory of classes. (The base theory of the proof assistant.)
Let F be an extension(not recursively axiomatizable) of T (in the extension
 of the language that has all Henkin constants), which is complete,
 and admits the reflection axiom.
The introducing of the types allows to hide both Gödel's encoding and decoding,
 and to keep all variables on the same metalevel.
Löb's theorem:
1) partially unfolded version, for demonstration purposes
(∀(φ^code0:Class),((φ^code0 Emb(PropTerm))→(Q ⊢‹((T ⊢‹((T ⊢EVal(DecPropTerm(φ^code0)))→⟦EVal(DecPro
pTerm(φ^code0))⟧)›)→(T ⊢EVal(DecPropTerm(φ^code0))))›)))
2) in the most laconic higher-order notation
(∀(φ0:PropTerm),(Q ⊢‹((T ⊢‹((T ⊢φ0)→⟦φ0⟧)›)→(T ⊢φ0))›))
Gödel's second incompleteness theorem:
(Q ⊢‹(¬(T ⊢‹⊥›)→¬(T ⊢‹¬(T ⊢‹⊥›)›))›)
Host functional symbols: (DecProp : ClassTerm -> PropTerm), (EncProp : PropTerm -> ClassTerm)
Example of computation of antiquotations:
‹(⟦a0⟧∈⟦‹b0›⟧)›  ⇝  ‹(⟦a0⟧∈b0)›
val it = (): unit
>
```

7 Type theory of the proof assistant

The book also documents a proof assistant. Its architecture is inspired by the LCF approach: a small kernel controls the construction of certified theorems, while higher-level code organises proofs. The type notation below describes syntactic objects manipulated by that implementation. It must be distinguished from the classes and types discussed inside an object theory.

Note 7.1 (Checking, normalization, and semantic values)**.**
The companion compiler documentation specifies verification of derivations, without executable code generation. This is compatible with the present book's use of classical axioms and declarative meanings. It does not identify three different mathematical questions: whether a proposed finite derivation can be checked, whether a specified reduction relation terminates, and whether a normal form computes a semantic witness. In particular, an opaque axiom can have no computation rule without causing an infinite reduction. The following elementary consequence of the language chapter makes the distinction precise.

Proposition 7.1 (Opaque constants and normalization)**.**
Take the simple Boolean calculus of Theorem 47.9, and add constants with fixed simple types and no new reduction rules. Every well-typed term is still strongly normalizing for its full reduction relation. However, adding an opaque constant $b : \mathsf{Bool}$ invalidates the claim that every closed Boolean normal form is true or false.

Proof.
A finite term t contains only finitely many new constants. Replace each such constant $c_i : \alpha_i$ by a distinct fresh free variable $z_i : \alpha_i$, first renaming bound variables to avoid these names. Induction on typing gives a term of the original calculus in the context extended by the z_i. This replacement commutes with capture-avoiding substitution: constants are never bound, and the fresh variables cannot be captured. Consequently every beta, projection, or Boolean contraction is carried to the same contraction, and the assertion extends to contractions inside contexts. No reduction introduces a new constant. An infinite reduction from t would therefore give an infinite reduction of its translated original term, contradicting the cited normalization theorem for open well-typed terms. But b itself is a closed normal form of Boolean type and is neither of the two Boolean constructors. □

This is a syntactic result for the stated calculus. It does not assert semantic soundness of arbitrary added axioms, nor normalization of an unspecified dependent calculus. Likewise the absence of code generation does not establish a complexity bound for checking: that requires a particular checker, a representation of proofs and types, conversion rules, and a cost analysis. The author's choice to verify certificates without executing their denotations is retained throughout.

```
(*** TO PDF PART BEGIN ***)
type nat = Nat.t;

(**
A variable is free in two cases: when it has no binding level, and
when its binding level exceeds the amount of binders above it.
So technically, it is an unnamed unbound variable.
(Processing it must be stopped in most cases via the error message.)
**)

type Name = string;
type MetaLevel = nat;
type BindLevel = (nat option);

datatype Variable =
  namelvl of string * nat * (nat option) (* actual variable *)
(*
| VariableVar of Variable (* variable of a variable *)
*)
;

type LVariable = Variable; (* specifically for the case when the variable is of type TClass *)

(*
fun LVar (x:Variable) : LVariable = x;
```

```
datatype LVariable =
  LVar of Variable
;
*)

(** SML-type of bindings: **)
(* 1) for cases when the type is known automatically to be a "Class": *)
datatype 'at LBound =
    LBind of LVariable * 'at
;

(* 2) for cases when the type is not known: *)

type 'CongType GTyVariable = Variable * 'CongType;

datatype ('CongType,'at) GTyBound =
    TyBind of ('CongType GTyVariable) * 'at
;


(*

datatype ('CongType,'at) TyBound = (* todo: actually use it *)
    CTyBind of 'CongType * ('at LBound)
;

OR

datatype ('CongType,'at) GenericTypedBound = (* todo: actually use it *)
    CTyBind of ('CongType GenericTypedVariable) * 'at
;
type 'at TypedBound = (CongType,'at) GenericTypedBound;

*)

(**

The Standard ML, and PolyML don't support general polymorphic recursion.
Due this limitation, even though it is possible to define the following type,"
datatype 'a Term =
    EVar of Variable
  | EQuote of 'a
  | EAntiquote of 'a Term Term
;
", but the implementation of nested functions on it is signinficantly complicated
by neccessity to have the maximal number of steps for ANY each such function,
what in its turn will lead to change of all infrastructure of the proof assistant.

Using METAPROGAMMING allows to implement similar functions up to arbitrary finite
number of level.
(currently even metaprogramming is not needed due to 3 levels, and the small amount of code)

The best use is to instantiate 'a with some primitive type.
**)

(* Extend further if necessary: *)
(*
datatype 'a TermTermTerm =
    EEEVar of Variable
  | EEEQuote of 'a TermTerm
;
*)

(*because Value is not yet defined here
type 'a Term = (Value, 'a) GTerm;
type 'a TermTerm = (Value, 'a) GTermTerm;
*)
datatype ('Value,'a) GTermTerm =
    EEVar of Variable
  | EEQuote of ('Value,'a) GTerm
```

```
(*
| EEAntiquote of 'a TermTermTerm
*)
  | EEVal of 'Value
and ('Value,'a) GTerm =
    EVar of Variable (* variables for terms, like x t phi *)
  | EQuote of 'a (* "quote" *) (* constants*)
  | EAntiquote of ('Value,'a) GTermTerm (* "antiquote"/"cite" *)
  | EVal of 'Value
(*| EDefault (No need for this.)*)
(** experimental:
  | EEval of (('a -> 'b) Term) * ('a Term)
  ... OR
  | EExpr of ('a Term) Expr
  ... OR nothing
  Or something else.
**)
;
(*
signature
sig

end
struct (:> )
  val (ty:CongType);(*opaque, cannot be changed*)
end

datatype ('ty,'a) LBound =
    LBind of ('ty Variable) * 'a
;
*)


signature SMLTYPES =
sig
eqtype Variable
eqtype 'a LBound
(*
eqtype 'b Term (* maybe get rid of this! (by moving defs of Variable, LBound, Term upward) *)
*)
eqtype Stmt
eqtype PropTerm
eqtype Ctx
eqtype TheoryTerm
eqtype Value
eqtype ClassTerm
eqtype CongType
eqtype Prf
eqtype Restr
end

functor Types2Tree (A:SMLTYPES) =
struct
open A;

structure TypeNames =
struct
type TyVariable = CongType GTyVariable;
type 'b TyBound = (CongType,'b) GTyBound;

type BOUND = TyVariable list;

type 'a Term = (Value, 'a) GTerm;
type 'a TermTerm = (Value, 'a) GTermTerm;


type CtxTerm = Ctx Term;
type PrfTerm = Prf Term;
type StmtTerm = Stmt Term;
type RestrTerm = Restr Term;
```

```
type PropTermTerm = PropTerm Term;
(* etc *)
end;
open TypeNames;

datatype 'a Tree =
       PCheat_tree of Stmt
  | PWeakening_tree of PropTerm * ('a Tree)
  | PWeak_tree of (Ctx Term) * ('a Tree)
  | PInCtx_tree of (Ctx Term) * ('a Tree)
  | PA0_tree of Variable
  | PS1_tree of TheoryTerm * (Ctx Term) * PropTerm * Value * Variable
  | PCredo_tree of ('a Tree)
  | POmnis_tree of ('a Tree)
  | PHenkin_tree of ('a Tree)

  | PMP_tree of ('a Tree) * ('a Tree)
  | PIdent_tree of TheoryTerm * PropTerm
  | PCut_tree of ('a Tree) * ('a Tree)

  | PAnd1L_tree of PropTerm * ('a Tree)
  | PAnd2L_tree of PropTerm * ('a Tree)
  | PAndR_tree of ('a Tree) * ('a Tree)

  | POr1R_tree of PropTerm * ('a Tree)
  | POr2R_tree of PropTerm * ('a Tree)
  | POrL_tree of ('a Tree) * ('a Tree)

  | PImplL_tree of ('a Tree) * ('a Tree)
  | PImplR_tree of ('a Tree)

  | PNegL_tree of PropTerm * ('a Tree)
  | PNegR_tree of ('a Tree)

  | PForaL_tree of PropTerm * Value * Variable * ('a Tree)
  | PExisR_tree of PropTerm * Value * Variable * ('a Tree)
  | PForaR_tree of PropTerm * Variable * Variable * CongType * ('a Tree)
  | PExisL_tree of PropTerm * Variable * Variable * CongType * ('a Tree)

  | PComprAx_tree of PropTerm LBound
  | PEpsIndAx_tree of ClassTerm
  | PPowSetAx_tree of ClassTerm
  | PEqDef_tree of LVariable * LVariable
  | PIffDef_tree of PropTerm * PropTerm
  | PSubDef_tree of LVariable * LVariable * LVariable
  | PSetDef_tree of LVariable * LVariable
  | PPowDef_tree of LVariable * ClassTerm
;
end;




(** an obsolete sml-type of metalevel shifts: **)
datatype 'a Shift =
    Quote of 'a (** Object-level shift / descent **)
  | Antiquote of 'a (** Meta-level shift / ascent **)
;







(**
Principle:
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+Metatheory of the class theory should not say anything about itself.+
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
```

```
(otherwise the complexity of the system grows infinitely during the implentation)
**)

(* DEPRECATED CODE: (the more permissive "shift-to-term" approach is used instead).
(it is never fully defined, so the naive induction on it leads to contradtictions)
datatype Class =
    ClAntiquote of ClassTerm
and
*)
(** SML-type of class terms: **)
datatype (*ClassExpre*) ClassTerm = (* needed for representing classes *)
  (** An arbitrary class constant(for demo): **)
    Cc
  (**============TEMPORARY:==============**)(*it can be deleted later*)
  (** Class constructors, which are useful for formulating axioms, just built-in: **)
  | CEmptySet
  (** The class of all sets: **)
  | CSets
  (** Powerclass: **)
  | CPower of (ClassTerm)
  (** Image of a functional class: **) (* It is of the first-order. *)
  | CImage of (ClassTerm * ClassTerm) (* It can be defined. *)
  (* etc *)
  (**============NECESSARY:==============**)
  (** Class comprehension: **)
  | CKappa of PropTerm LBound
  (* in this case, it is automatically assumed that type of the binding variable is OClass *)
(*CLam
  (** Lambda-abstraction: **)
  | CLam of ClassTerm LBound (* avoid using it, maybe delete it*)
*)
  (** Indefinite description: **)
  | CEps of PropTerm LBound
  (** Definite description: **)
  | CIota of PropTerm LBound
  (**===============META:=================**)
(*META-*)
  (** Quotation and antiquotation for classes: **)
(* OLDSHIFT
  | CShift of ClassTerm Shift
*)
(**)
  (** Class variable: **)
  | CVar of LVariable
  (*new: remark CVar NOT EQUAL TO CAntiquote o EVar *)
  | CAntiquote of (Value,ClassTerm) GTerm
(**)
(*
  (** New alternative quotation: (two constructors) **)
  | CUse of ClassTerm (* import and interprete the term in the current environment *)
  | CRef of ClassTerm Obj (* antiquotation: open locally outer context *)
*)
(*
  (** no need in CObj -- broken! **)
  | CObj of ClassTerm (* a model of a code *)
*)
  (********** Encoding the syntactic entities: **********)
  (**
  Basically, if (t:A), then (Enc(t) \in Emb(A)).
  Is "Emb" a good name? It is a class of certain type's terms.
  It is an embedding of some of SML-types.
  It may be called an "almost inner model", because
  it interpretes terms with classes, so
  it can't send "Type" anywhere inside the conglomerate of all classes.
  (Moreover, the requirement "Emb(A) = Enc(A)" may not hold too.)
  **)
  (**
  A constructor "CEnc" simulates the encoding function "Enc".
  The output of such function Enc is a class,
  which is a class-theoretic model of syntactic entity's code.
```

```
Note, that a value can have free variables, and then the encoding
will also depend on these variables.
(because for "shortness" and "flatness" of the formulas, all variables
are considered as metavaribles by default, and one needs to use Quotation,
to actually produce a purely symbolic formula/variable/term/etc.)

const(x) = Antiquote(x) =

"the standard numeral of the code of phi"
In diagonal lemma:
Numbering is a function that takes a QUOTED formula and produces a natural number out of it.
Numeral is a

"standard" refers to the idea that phi was implicitly quoted;
 and the assumption that it doesn't have zero level antiquotations.
So the term that is produces is closed.

7 9 = Quote o Enc?

Numeration is a function, that produces a number.
**)
(*Variants:
1) (using the membership relation)

(in case of term-type pair)
"low-level"
CEnc of SetTerms
CEmb of SetType

2) (also using the membership relation)

(in case of type-typeclass pair)
"middle-level"
CEmb of SetType
( ) of ClassTerm


2a) Any interesting examples?...
Functions(in sense of "class of functions")
Universes
SetTypes

2b) But it all can be done with using predicates on classes! What for?
IDK

2c)
SetType are valuated as sets anyway.
So in MK, there is no need for this "middle-level" type theory!
SetType are sets, and a class is availible for _any_ formula.

2d) Why do I need it?
In MK, the only thing that is special about "middle-level type theory"
is that it's types are guarantied to be classes,
but also it restricts its terms to necessarily be sets.

In NBG it is weaker that (3), because it has less types and because it has less terms.
In MK it is weaker that (3), because it has less types("\phi(x)" vs. "\phi(x)/\(x set)") and because it has less
      terms.


3) (using the unary predicates on conglomerate of all classes) "high-level"
  () of ClassTerm
app of CongType ?

Not possible to include...
*)
| CEnc of SetTerms (* the output here is a term of a model of a syntactic entity *)
(** Embedding of types: **)
(**
 For every (fixed) interpretation of variables, all encodings of terms
```

```
   are memebers of the embedding class of the corresponding type.
  **)
(* CEmb is DEPRECATED; "| CSetEmb of SetType" is possible, but it seems unnecessary.
  | CEmb of (*TODO: SetType-not enough?Enough... ClassTerm is also enough! DEPRECATED *)
  (*CURRENTLY WRONG:*)CongType(*SetType*) (* the output here is a small set of syntactic things/expressions/terms
      *)
*)
  (********** Embedding of the semantic entities: **********)
  (**
     "CVal"
     Simulacrum of the function which converts any Value to ClassTerm.
     It helps to abstract from particular implementation of a value.
     Since it is implemented in SML, then there is such class,
     because SML has a model. No need to redo its construction.
     Example:
       CVal(VNatTerm(max(x,y))) = union({CVal(VNatTerm(x)),CVal(VNatTerm(y))})
  **)
  | CVal of Value (* semantic; it is both interpretation and "type conversion". *)
(*| CSetType of SetType ( CSetType === (CVal o SESetType) ) *)
(*
  | CInterpr of Value ( * semantic * )
( **
 A constructor CInterpr simulates the interpretation function from Value to Class.
Let's call this function "Val2Class".
It is not implemented but it should exist. (And it does exist.)
     Everything is a class in the class theory, so CVal simulates building of a class.
** )
  | CVal of Value ( * semantic * )
( ** TYPE CONVERSION ("ClassTerm" works very much like "void*" from C++)
[Technical]
 A constructor "CVal" simulates a partially defined function
 of type conversion from Value to Class.
     If CVal is given a class as an argument, then it simply returns the same class.
If the argument(a value) of Val2Class is a class of my theory, then it returns that class.
Otherwise, it is an unspecified default Class, e.g. the empty class.
It is implemented as a constructor, and not as a function,
because Value has a constructor of form (VDec t), which makes
it impossible to know the value, e.g. in case if the class t is just a class--variable.
** )
*)
(*-META*)
(*
and PropTermListTerm =
    PELEVar of Variable
  | PELEConst of (PropTerm list)
  | PELEAntiquote of PropTermListTerm Term
*)
and VoidTerm =
    VoidVar of Variable
(*| VoidQuote of Void.t (It both is useless and complicates the theory with SML-related things, so it is
     certainly forbidden.) *)
  | VoidAntiquote of (Value,VoidTerm) GTerm
(*| VoidDefault (This is forbidden since there are no constants of type Void: that creates
necessity of variable use, e.g. sth like "default void variable", so it is also forbidden) *)
(*
In general, there is no need in "default xxx term", because
the type which semantics contains undefinable objects, usually have constants.
Also a quote of a constant term is also a constant term,
so there is no need for it to be in 'a Term definition too.
*)
(*
and
  ClassTerm =
    RRR of ClassExpre sion
*)
and (** Type of contexts: **)
  Ctx = (*CtxTermFirstHalf*)
    XCtx of (((Value,PropTerm) GTerm) list)
(* Type Ctx already consists of terms, so no need to make it a term artificially:
  (Ctx Term) works as intended. *)
```

```
(* NO NEED FOR
and (** Type of contexts' terms: **)
  CtxTerm =
    XECtx of (PropTerm list)
  | XEVar of Variable
  | XEAntiquote of CtxTerm Term
( *| CVal2Ctx of Value
PropGround
* )
*)
and
  (** Type of statements: **)
  Stmt = (* it is actually a Quote part of StmtTerm ... "StmtHalfterm" *)
    AStmt of TheoryTerm * ((Value,Ctx) GTerm) * ((Value,PropTerm) GTerm)
  (* Type Stmt already consists of terms, so no need to make it a term artificially:
  (Stmt Term) works as intended. *)
(*
Originally it is: "AStmt of TheoryTerm* (Ctx Term) * PropTerm".
But due to metalevel type-to-term shift, so it needs to be changed.
*)
and
 (** Type of theories: ("au" is a postfix for a theory names) **)
  TheoryTerm = (* todo: rename to TheoryTerm? *)
  (* Predefined constants: *)
    Tau (** consistent, recursively axiomatizable foundational theory,
           it is being partially implemented, it is a metatheory of Fau **)
  | Fau (** completion of a foundational theory, not recursively axiomatizable **)
  | Qau (** Robinson's arithmetic theory **)
  | Wau (** arbitrary first order theory **)
  (* Meta: *)
  | HVar of Variable (* variable for an arbitrary theory *)
  | HAntiquote of (Value,TheoryTerm) GTerm
and
  Prop = (* NEW!!! (underdefined?) *)
    PropVar of Variable
  | PropAntiquote of PropTerm (* either "Trueprop" or "totype" meaning. No. Just "totype". *)
and
  Class = (* NEW!!! (underdefined?) *)
    ClassVar of Variable
  | ClassAntiquote of ClassTerm
and
  (** Definition: a formula is a term of a proposition. **)
  (** Type of formulas: **)
 PropTerm = (*it's main usage is to store some of Props *)
  (**============NECESSARY:==============**)
    FBot
  | FTop
  | FNeg of PropTerm
  | FImpl of PropTerm * PropTerm
  | FIff of PropTerm * PropTerm
  | FAnd of PropTerm * PropTerm
  | FOr of PropTerm * PropTerm
  (* Simple variant of quantifiers don't need a type.
     For them "Class" is assumed to be the type of the bound variable.
  | FSimpleFora of (PropTerm LBound)
  | FSimpleExis of (PropTerm LBound)
  *)
  | FFora of CongType * (PropTerm LBound)
  | FExis of CongType * (PropTerm LBound)
(** For quantifiers, "a type of variable" must be stated explicitly for variable binding. **)
(*| FTruth of (ClassTerm * ClassTerm * ClassTerm) *)
(**L(* a boring first-order version of $\vdash$ *)L**)
  | FIn of ClassTerm * ClassTerm
  (** **)
  | FIs of ClassTerm * Predicate (*PropTerm LBound*)
  (*NEW; FIs means that a predicate on classes is true on a class. *)
  (**============TEMPORARY:==============**)
  (*| FSubst of*)
  | FEq of ClassTerm * ClassTerm
```

```
  | FSet of ClassTerm
  | FFunc of ClassTerm
  | FSub of ClassTerm * ClassTerm
  | FCF of string (* closed formulas *)
  (**==============META:================**)
  | FTruth of Stmt (** Other names: "Der", "True", etc **)
(**L\ ||
Set/Class--theoretic part of the definition:\\
Some necessary propositional symbols of class theory, which are built--in for the time being.
L**)
  (** Closure operator (it just bounds all variables as universal quantifier)**)
  | FClosure of PropTerm (* no need?, since "$\Wau \vdash \phi$" works, so todo: remove*)
(*META-*)
  | FVal of Value (** this is here because VDec is one of constructors of Value **)
(**L\ ||
  If one gets rid of Val, then it will be necessary to add DecPropTerm here, DecClass in Class, etc.
\ But also I will be able to get rid of FVal2PropTerm.
\ Why Value may be useful? Since Enc and Dec are declared,
\ but not computed anyway, I will have to try to call a procedure which attempts a "normalisation"
\ whenever it is needed and hope that it will do the job. Since the task is futile, it can stay,
\ because it creates a setting for developing a better type theory. (with functions, etc)
\ A Greek letter $\phi$ with a dot is just a notation for Antiquote (FVar $\phi$).
\ Antiquote is more useful than just a variables metalevel, because it allows to include
\ structures, bigger than just a variable.\\

 It it useful for example, in the context of the Diagonal Lemma:\\
 Upper corner brackets: "(AntiquoteProp o EncProp)" , so $\ulcorner \varphi \urcorner$ is by definition $\
     overline{\alpha(\varphi)}$.
\ Lower corner brackets "(DecProp o QuoteProp)" are usually necessary in the proof assistant on the right side of
     the "|-" symbol($\vdash$), but they are
\ not usually written in the standard mathematical notation
(see \url{ https://en.wikipedia.org/wiki/Diagonal_lemma }).
L**)
  (** Variables AND Quotation and antiquotation for formulas: **)
  (*
  | FShift of PropTerm Shift (** this is for expressive meta **)
  *)
  (*
  FVar is a constructor of proposition
  which argument is a variable for PROPOSITIONS e.g.: p1, p2, p3, ...
  Do not confuse it with variables for FORMULAS : phi1, phi2, phi3, ...,
  which are implemented as EVar, a part of the sml-type 'Term'.
  *)
  | FVar of Variable
  | FAntiquote of (Value,PropTerm) GTerm
(*-META*)
(** Theory **)
(***** ================================BORDER================================ *****)
(** Metatheory **)
and
  (** Type of substitutions: (it is not part of the language) **)
  Restr =
    CSubst of Variable * Value
and
(*
Types on classes are modelled as unary predicates.
So all of the dependent types, etc, may be done in CongType, because it is also about classes.
The idea of small types is in saving the information that a type itself is also a set.

It may be possible to track this. (But also, why it is important for my tasks?..)

*)
  (** Type of small types (of the kind named "Set"): **)
  SetType = (* IT IS RATHER AN SML-TYPE OF BASIC TYPES OF CONSTANTS! *)
  (* SYVar of Variable ( ** Variables for small types. ** ) *)
  (** Q: Must they be (lazy-)typed? A: Not here. **)
  (** Q: "Cumulative hierarchy" OR "Universes don't share variables"? A: The second. **)

  (*** CONSTANTS: ***)
  (**** Theory-related constants of types: ****)
```

```
  (* storable: *)
    SYCtx (* not only a list of formulas, but also variables for it! *)
  | SYPrf
  | SYRestr
  | SYStmt
  (* etc... *)
  (* non-storable: *)
  | SYTheory (*?*)
  (*
  | SYProp
  *)
  (* This is a type of variables, it will be obsolete(for the best) after implementation of De Bruijn indices.
  | SYVari of (*Small?*)CongType (* automatically labeled variables for each type. No implicit subtyping! *) (* [
      it will soon be obsolete!] *)
  *)
  (**** Some basic programming-related constants of types: ****)
  | SYVoid (** It is implemented via structures. **)
  | SYUnit
  | SYBool
  | SYString

  (*** NON-CONSTANTS ***)
  | SYPi of SetType LBound (* todo: move to CongType *)
  (** It is here instead of "Expr LBound", since range must be a small type, not a function. **)
  (**** Universes: (Assuming Tarski's axiom.) ****)
  | SYUn of nat (** A simple variant of implementation. ToDo: index type for universes. **)
  (** SYCongTypeTerms is a set-level type of terms of any particular conglomerate-level type. **)
  | SYCongTypeTerms of (Value,CongType) GTerm (** a set of ALL terms of...; (SYCongTypeTerms OClass)=("SYCongType
      OClass") **)
  (*| SYTermsOf of ... no need, it is via conglmerates: (SYCongTypeTerms o OSmallType) *)
  (*| SYTypeTerm*)
  (*
    A quote of a proposition is a formula.
    A quote of a formula is "PropTerm Term".
    TODO: There is a need for notation agreement in order to distinguish SML-types and MK-types.
    TODO: be more careful with use of quotations in comments, they mean something...
  *)
and Sort = (* it is not used. just for understanding *)
    sets
  | classes
  | conglomerates
(** SML-type of conglomerate-level types(i.e. types of classes): **)
and CongType = (* NAMES ARE AS IN THEORY *)
    (* no variables here! *)
    (* Terms of ? *)
  (** user-defined conglomerates: **)
    CongDef of PropTerm LBound
  (** pre-defined conglomerates: **)
  | OProp (* (lam C, (C subset of { {} } ) ) , but may be better to keep underspecified *)
  (* size of OProp is unspecified *)
  | OSet (* (lam C, (C set)), aka ((lam C, exists Q, C in Q) *)
  (* OSet has class size *)
  | OClass (* corresponds to predicate (lam C, Top) *)
  (* OClass has conglomerate size *)
  | OFamily (* OFamily has a big definition, see much later *)
  (* OFamily has conglomerate size *)
  (* OSubtype (TyBind (x,X) ) *)
  | OSubtype of (CongType,PropTerm) GTyBound
(**** Rocq-related things (* TODO: MOVE IT*)
  | OSetTypei
  (* OTypei has set size for each i *)
  | OSetType (* in Rocq sense of "TypeInfinity" *)
  (* OType has class size *)
  | OSetTypeclass (* in Rocq sense*)
  (* OTypeclass has conglomerate size *)
****)

  (* the following is unnecessary:
  (** Subconglomerates of OSet:
  (no need! just inherit it from SetTerms )
```

```
  **)
  | OType of Nat.t (* denotes the universes *)
  (** Subconglomerates of OClass:
  (no need! just inherit it from SetTerms )
  **)
  | OType (* (denotes a complex subconglomerate OSet) *)
  *)
  | OSmallType of SetType (* a kind of a constructor OSmallType is (SetType -> CongType) *)
(*
and Type = (*it's not a type, it's a CongType! *)
    OProp
  | OClass (* a model of this type is a conglomerate of all classes *)
  | OSmallType of SetType (* a kind of a constructor OSmallType is (SetType -> Type) *)
*)
  (* not sure if it is "small" or not *)
(* NEW:
It is not possible to store a class, only a class term.
It is not possible to stora a proposition, only a formula. ((formula) = (proposion term))

Values and their types:

(* values with a quotation-induced shift of type: *)
to_string (typeof (VProp (f:PropTerm) )) = "Prop"
to_string (typeof (VClass (c:ClassTerm))) = "Class"

(* values without a quotation-induced shift of type: *)
to_string (typeof (VPropTerm (f:PropTerm) )) = "OProp"
to_string (typeof (VClassTerm (c:ClassTerm))) = "OClassTerm"

Idea: "Syntactic objects coincide (or almost coincide in case of bindings) with their terms."
(if De Bruijn's index is used, then it seems to coincide, but the display of formmulas is more old-fashioned,
     with named variables. But it still needs to be of different type i suppose, otherewise confusion on the lower
      level(on a quote-level).)

*)
and SetTerms = (*
(not the whole)
NO TYPE SHIFT HERE, because elements are already terms. Is it a SML-level subtype? *)
  (**** Tiny syntactic objects: ****)
    (*SETypedVariable of Variable*)
(*
PREFIX "E" MEANS IT IS A TERM
...
*)
    SECtx of (Value, Ctx) GTerm (* PropTerm list & variables of type "Ctx"*)
  | SETheory of TheoryTerm (*todo: add variables and antiquotation *)
  | SEPrf of (Value, Prf) GTerm(* Prf = ThmTerm (maybe)*)
  | SERestr of (Value,Restr) GTerm
  | SEStmt of (Value, Stmt) GTerm
(*? | SEProp of PropTerm *)
  (*
     Idea: ALL: Stmt and 'CtxTerm' and TheoryTherm must have variables!
     Answer: NO, WRONG! Only something that is not "storable".
        So, only PropTerm, ClassTerm, TheoryTerm. (And, MAYBE, CongType, or maybe not.)
        Also RealTerm, ComplexTerm, etc. (anything that is uncountable is stored as terms)
        (also something that is countable, but that is hard or impossible to define.)
  *)
  (* HOPEFULLY OBSOLETE
  | SEVari of Variable
  *)
  (** (inheritance of some standard SML types as small objects...) **)
  | SEUnit
  | SEBool of (Value, bool) GTerm
  | SEString of (Value,string) GTerm

  | SESetType of SetType(* no modifictors *)
(*
  | SECongType of CongType
*)
  (* Is SEDec useful? *)
```

```
  | SEDec of ClassTerm * SetType (* What is a default value if cannot decode emptyset or better V and move it to
      Value?*)
and Predicate = (* this SMLtype is given only for demonstration purposes of "type unfolding" *)
    PDef of PropTerm LBound
  | PEmb of CongType
(* PEmb is optional; It allows avoiding implementing all of the semantics immediately. *)
and FamilyTerm =
    FEId (*just a constant, identity class-functions, as a family. Not in use. *)
and Value = (* THERE IS A TYPE SHIFT HERE! *) (* the most general sml-type of terms*)
  (**** Arbitrary, potentially large objects: ****)
    VClass of ClassTerm
  | VSet of SetTerms
  | VProp of PropTerm
  | VFamily of FamilyTerm
(* "Value" = "AnyTerm", except from itself and other either technical or paradox-risky things: conglomerates,
     simulations, encoder-decoders, quotations-antiquotations *)
(*
There are two typings:
1) Typing of a SetTerms with a SetTerms, such that their the valueation of judgment is set-theoretically true.
     ("(ta:tA)" |-> "for any (v:Variable->Class), F |- < v(ta) in v(tA) >" )

*)
  | VDec of ClassTerm * CongType
  (** VDec is here because implementing a decoder for an arbitrary class is too hard **)
(**
partial [almost-]impossible-to-implement function as a constructor
it considers a class as code of a value, and decode it as an actual value (non-specified one).
It is very non-constructive: we just assume that there is such a function.

What is guaranteed is that Dec(Enc(va)) === va.
I simulate that this function is implimented.
The expressions "(Dec t)" for arbitrary class "t" should be understood
 and treated as if it is a value, which is obtained by one of the
 previously written constructors.
It is useful to convert it to an explicit value, when it is possible,
 and to leave it ambiguous, when it is not possible.
**)
and Expr = (** to be developed... Make it abstract ('E, 'Y) Expr. **)
  (**** Lambda-calculus: ****)
  (** General **)
    var of Variable (** variables **) (*it used to be Variable *)
  | con of Value (** constants **)
  | abs of CongType * (Expr LBound) (** abstraction **) (*it used to be SetType*)
  | app of Expr * Expr (** application **)
(*
and 'a Tree =
       PCheat_cmd of Stmt
  | PWeakening_cmd of PropTerm * ('a Tree)
  | PWeak_cmd of (Ctx Term) * ('a Tree)
  | PInCtx_cmd of (Ctx Term) * ('a Tree)
  | PA0_cmd of Variable
  | PS1_cmd of TheoryTerm * (Ctx Term) * PropTerm * Value * Variable
  | PCredo_cmd of ('a Tree)
  | POmnis_cmd of ('a Tree)
  | PHenkin_cmd of ('a Tree)

  | PMP_cmd of ('a Tree) * ('a Tree)
  | PIdent_cmd of TheoryTerm * PropTerm
  | PCut_cmd of ('a Tree) * ('a Tree)

  | PAnd1L_cmd of PropTerm * ('a Tree)
  | PAnd2L_cmd of PropTerm * ('a Tree)
  | PAndR_cmd of ('a Tree) * ('a Tree)

  | POr1R_cmd of PropTerm * ('a Tree)
  | POr2R_cmd of PropTerm * ('a Tree)
  | POrL_cmd of ('a Tree) * ('a Tree)

  | PImplL_cmd of ('a Tree) * ('a Tree)
  | PImplR_cmd of ('a Tree)
```

```
  | PNegL_cmd of PropTerm * ('a Tree)
  | PNegR_cmd of ('a Tree)

  | PForaL_cmd of PropTerm * Value * Variable * ('a Tree)
  | PExisR_cmd of PropTerm * Value * Variable * ('a Tree)
  | PForaR_cmd of PropTerm * Variable * Variable * CongType * ('a Tree)
  | PExisL_cmd of PropTerm * Variable * Variable * CongType * ('a Tree)

  | PComprAx_cmd of PropTerm LBound
  | PEpsIndAx_cmd of ClassTerm
  | PPowSetAx_cmd of ClassTerm
  | PEqDef_cmd of LVariable * LVariable
  | PIffDef_cmd of PropTerm * PropTerm
  | PSubDef_cmd of LVariable * LVariable * LVariable
  | PSetDef_cmd of LVariable * LVariable
  | PPowDef_cmd of LVariable * ClassTerm
*)
(*
and
  Cmd = (* one-level tree of commands, without underlying proof objects *)
          RawStmt of Stmt
  (*=====================================================================*)
  | PCheat_cmd of Stmt
  | PWeakening_cmd of PropTerm * Cmd
  | PWeak_cmd of (Ctx Term) * Cmd
  | PInCtx_cmd of (Ctx Term) * Cmd
  | PA0_cmd of Variable
  | PS1_cmd of TheoryTerm * (Ctx Term) * PropTerm * Value * Variable
  | PCredo_cmd of Cmd
  | POmnis_cmd of Cmd
  | PHenkin_cmd of Cmd

  | PMP_cmd of Cmd * Cmd
  | PIdent_cmd of TheoryTerm * PropTerm
  | PCut_cmd of Cmd * Cmd

  | PAnd1L_cmd of PropTerm * Cmd
  | PAnd2L_cmd of PropTerm * Cmd
  | PAndR_cmd of Cmd * Cmd

  | POr1R_cmd of PropTerm * Cmd
  | POr2R_cmd of PropTerm * Cmd
  | POrL_cmd of Cmd * Cmd

  | PImplL_cmd of Cmd * Cmd
  | PImplR_cmd of Cmd

  | PNegL_cmd of PropTerm * Cmd
  | PNegR_cmd of Cmd

  | PForaL_cmd of PropTerm * Value * Variable * Cmd
  | PExisR_cmd of PropTerm * Value * Variable * Cmd
  | PForaR_cmd of PropTerm * Variable * Variable * CongType * Cmd
  | PExisL_cmd of PropTerm * Variable * Variable * CongType * Cmd

  | PComprAx_cmd of PropTerm LBound
  | PEpsIndAx_cmd of ClassTerm
  | PPowSetAx_cmd of ClassTerm
  | PEqDef_cmd of LVariable * LVariable
  | PIffDef_cmd of PropTerm * PropTerm
  | PSubDef_cmd of LVariable * LVariable * LVariable
  | PSetDef_cmd of LVariable * LVariable
  | PPowDef_cmd of LVariable * ClassTerm
*)
and
  Prf = (* Prf = ThmTerm *)
       PCheat_prf of Stmt
  | PWeakening_prf of ((Value,PropTerm) GTerm) * Prf
  | PWeak_prf of ((Value,Ctx) GTerm) * Prf
```

```
  | PInCtx_prf of ((Value,Ctx) GTerm) * Prf
  | PA0_prf of (* (Ctx Term) * *) Variable (** The variable here is intended for classs **)
  (*
  | CA1_prf of (Ctx Term) * PropTerm * PropTerm
  | CA2_prf of (Ctx Term) * PropTerm * PropTerm * PropTerm
  Todo: just write all axioms from the PDF here.
  *)
  | PS1_prf of TheoryTerm * ((Value,Ctx) GTerm) * ((Value,PropTerm) GTerm) * Value * Variable
 (* during the checking it must confirm, that a value has a type(i.e. CongType),
 which is the same as a variable type(i.e. CongType) *)
  | PCredo_prf of Prf
  | POmnis_prf of Prf
  | PHenkin_prf of Prf (* (Variable) * PropTerm *)
(**
     Fau, Gamma |- exists x, phi
     ---------------------------
     Fau, Gamma |- phi[Witness("exists x, phi")/x]
**)
(**L
\ ||
In general:
\[
 \text{Witness}(\Wau, \Gamma, f) := (\varepsilon (t: \ClassTerm), \exists p\in \Prf, (p:(\Wau, \Gamma \vdash f)))
     )
\]
\noindent In particular:
\[
\text{Witness}(\Wau, \Gamma, \exists x, \phi) := \text{HenkinConst}(\exists x, \phi)
\]
But it already encodes proofs...
L**)

  | PMP_prf of Prf * Prf
  | PIdent_prf of TheoryTerm * ((Value,PropTerm) GTerm)
  | PCut_prf of Prf * Prf

  | PAnd1L_prf of ((Value,PropTerm) GTerm) * Prf
  | PAnd2L_prf of ((Value,PropTerm) GTerm) * Prf
  | PAndR_prf of Prf * Prf

  | POr1R_prf of ((Value,PropTerm) GTerm) * Prf
  | POr2R_prf of ((Value,PropTerm) GTerm) * Prf
  | POrL_prf of Prf * Prf

  | PImplL_prf of Prf * Prf
  | PImplR_prf of Prf

  | PNegL_prf of ((Value,PropTerm) GTerm) * Prf
  | PNegR_prf of Prf

  | PForaL_prf of ((Value,PropTerm) GTerm) * Value * Variable * Prf
  | PExisR_prf of ((Value,PropTerm) GTerm) * Value * Variable * Prf
  | PForaR_prf of ((Value,PropTerm) GTerm) * Variable * Variable * CongType * Prf
  | PExisL_prf of ((Value,PropTerm) GTerm) * Variable * Variable * CongType * Prf
  (*
  | PSaved_prf of string
  | PSaved_prf of Prf ref
  *)
  (*
  | PWSelect_prf of Prf * PropTerm
  | PDouble_prf of Prf
  | PReduct_prf of Prf
  *)
  (** Built-in Axioms: **)
  | PComprAx_prf of PropTerm LBound (* (W:ClassTerm).phi *)
  (**L(* Epsilon induction: $((\power K \subseteq K) \implies (\V \subseteq K))$ *)L**)
  | PEpsIndAx_prf of ClassTerm(*K*)
  (* it is ok to use Class here, since no side conditions required. *)
  | PPowSetAx_prf of ClassTerm (* (A set) -> (Pow(A) set) *)
  (** Built-in Definitions: **)
```

```
  (* forall A: forall B: ((A = B) <-> ((A $\subseteq$ B) $\land$ (A $\supseteq$ B)) ) *)
  | PEqDef_prf of (LVariable) * (LVariable)
  | PIffDef_prf of PropTerm * PropTerm
  (* forall A: forall B: ((A $\subseteq$ B) <-> forall x, x in A -> x in B) *)
  | PSubDef_prf of (LVariable) * (LVariable) * (LVariable)
  (* forall B: ( (B set) <-> exists x, x in B) *)
  | PSetDef_prf of (LVariable) * (LVariable)
  (* Pow K = CKappa v (Sub v K) *) (* if v is not free in K *)
  | PPowDef_prf of LVariable(*(v:ClassTerm)*) * ClassTerm(*K*)
;
(*
open TypeNames;
*)
(*
type Term = Term
*)
(*type TheoryTerm = Theory Term*)


structure SMLTypes : SMLTYPES =
struct
  type Variable = Variable
  type 'a LBound = 'a LBound
(*
  type 'a Term = 'a Term
*)
  type Stmt = Stmt
  type PropTerm = PropTerm
  type Ctx = Ctx
  type TheoryTerm = TheoryTerm
  type Value = Value
  type ClassTerm = ClassTerm
  type CongType = CongType
  type Prf = Prf
  type Restr = Restr
end;

structure Tree = Types2Tree(SMLTypes);

open Tree;


datatype Cmd = (* one-level tree of commands, without underlying proof objects *)
    RawStmt of Stmt
  | Application of Cmd Tree.Tree
;

(*
val OClass = TLarge LClass;
*)

(*val TyVariable = OSmallType SYTypedVariable;*)
(*
val TProp = OSmallType SYProp ;
*)

val TProp = OProp ; (*?*)
val TClass = OClass;

val TCtx = OSmallType SYCtx ;
val TTheory = OSmallType SYTheory ;
val TPrf = OSmallType SYPrf ;
val TRestr = OSmallType SYRestr ;
val TStmt = OSmallType SYStmt ;

(*val VVariable = VSet o SETypedVariable;*)
(*
val VProp = VSet o SEProp ;
*)
val VCtx = VSet o SECtx ;
```

```
val VTheory = VSet o SETheory ;
val VPrf = VSet o SEPrf ;
val VRestr = VSet o SERestr ;
val VStmt = VSet o SEStmt ;

(*???
val VSetType = VSet o SESetType ;
*)
(*
    SEVariable of Variable
  | SEProp of PropTerm
  | SECtx of CtxTerm
  | SETheory of TheoryTerm
  | SEPrf of PrfTerm
  | SERestr of RestrTerm
  | SEStmt of StmtTerm
  (** (inheritance of some standard SML types as small objects...) **)
  | SEUnit
  | SEBool of bool
  | SEString of string
  (**** All types up to medium are also values. ****)
  (*| VTinyType of TinyType *)
  | SESetType of SetType
*)



(*META-*)
(*
datatype 'a Term =
  EVar of Variable (* variables for terms, like x t phi *)
| EQuote of 'a
| EAntiquote of 'a Term Term
;
| CAntiquote of ClassTerm Term

*)

val QuoteClass : ClassTerm -> ClassTerm Term = (fn t => (EQuote t));
val AntiquoteClass : ClassTerm Term -> ClassTerm = (fn t => (CAntiquote t));

val QuotePropTerm : PropTerm-> PropTerm Term = (fn f => ( EQuote f));
val AntiquotePropTerm : PropTerm Term -> PropTerm = (fn f => (FAntiquote f));

(* BEFORE
val QuoteClass : ClassTerm -> ClassTerm = (fn t => (CShift (Quote t)));
val AntiquoteClass : ClassTerm -> ClassTerm = (fn t => (CShift (Antiquote t)));

val QuotePropTerm : PropTerm -> PropTerm = (fn f => (FShift ( Quote f)));
val AntiquotePropTerm : PropTerm -> PropTerm = (fn f => (FShift (Antiquote f)));
*)
(*-META*)

(*** Structure Thm ***)
signature ThmWithoutPrf =
sig
  type Thm
  val stmt : Thm -> Stmt

  (** Inference rules' signatures: **)
  val cheat : TheoryTerm -> (Ctx Term) -> PropTerm Term -> Thm
  val Weakening : PropTerm Term -> Thm -> Thm
  val Weak : (Ctx Term) -> Thm -> Thm
  val InCtx : (Ctx Term) -> Thm -> Thm

  val Omnis : Thm -> Thm
  val Credo : Thm -> Thm

  val Ident : TheoryTerm -> PropTerm Term -> Thm
```

```
  val Cut : Thm -> Thm -> Thm
  val ImplL : Thm -> Thm -> Thm
  val ImplR : Thm -> Thm
  val Or1R : PropTerm Term -> Thm -> Thm
  val Or2R : PropTerm Term -> Thm -> Thm
  val OrL : Thm -> Thm -> Thm
  val And1L : PropTerm Term -> Thm -> Thm
  val And2L : PropTerm Term -> Thm -> Thm
  val AndR : Thm -> Thm -> Thm
  val NegL : PropTerm Term -> Thm -> Thm
  val NegR : Thm -> Thm
  val ForaL : PropTerm Term -> Value -> Variable -> Thm -> Thm
  val ExisR : PropTerm Term -> Value -> Variable -> Thm -> Thm
  val ForaR : PropTerm Term -> Variable -> Variable -> CongType -> Thm -> Thm
  val ExisL : PropTerm Term -> Variable -> Variable -> CongType -> Thm -> Thm
  val SubDef : LVariable -> LVariable -> LVariable -> Thm
  val SetDef : LVariable -> LVariable -> Thm

  val A0 : Variable -> Thm
  val MP : Thm -> Thm -> Thm
  val IffDef : PropTerm -> PropTerm -> Thm
  val EqDef : LVariable -> LVariable -> Thm
  val ComprAx : (PropTerm LBound) -> Thm
  val EpsIndAx : ClassTerm -> Thm
  val PowSetAx : ClassTerm -> Thm
  val PowDef : LVariable -> ClassTerm -> Thm

  val show_proof : Thm -> unit; (* it shows IF it can *)
  val prf_show : Thm -> unit; (* it shows IF it can, different presenttation *)
end;

signature ThmWithPrf =
sig
  include ThmWithoutPrf
  val prf : Thm -> Prf
  val derive : Prf -> Stmt
  val check : Prf -> Thm
  val slow_check : Prf -> Thm
end;

(** Tactics **)
(*
The definition of Validation and Tactic are commented here,
because the definition of them requires type Thm, which requires
definition of all inference rules, and also depend on
whether kernel should or should not store proof objects, so all of that
is defined later in code.
*)
type Goal = Stmt;
(**
type Validation = Thm list -> Thm;
type Tactic = Goal -> Goal list * Validation;
**)

(** ==== Variables ===== **)
fun zero_name (v:string) : Variable = (namelvl (v, Nat.zero, SOME Nat.zero));

(*META-*)

(** Encoding **)
(*?
val EncProp: PropTerm -> ClassTerm = CEnc o SEProp;
*)
(*val EncClass : ClassTerm -> ClassTerm = CEnc o SEClass; DEPRECATED*)

val EncCtx : Ctx Term -> ClassTerm = CEnc o SECtx ;
(* ... *)

(** Decoding **)
val DecPropTerm: ClassTerm -> PropTerm = (fn x => FVal (VDec (x, TProp)));
```

```
(*val DecClass : ClassTerm -> ClassTerm = (fn x => CVal (VDec (x,OClass)));*)
(*
fun DecCtx : ClassTerm -> CtxTerm = (fn x => CVal2Ctx (VDec (x,TCtx )));
...*)

(** Bold notation **)
(*?
val BoldPropTerm: PropTerm -> ClassTerm = EncPropTerm;
*)
(*val BoldClass : ClassTerm -> ClassTerm = EncClass;*)
val BoldCtx : Ctx Term -> ClassTerm = EncCtx ;
(*...*)

(** Dot notation **)
(***
val Dot : ClassTerm -> ClassTerm = AntiquoteClass;


(** Lower corner **)
val LCornerPropTerm= DecPropTermo QuoteClass;
(*val LCornerClass = DecClass o QuoteClass;*)
(*
val LCornerCtx = DecCtx o QuoteClass;
...*)

(** Upper corner **)
val UCornerPropTerm= AntiquoteClass o EncPropTerm;
(*val UCornerClass = AntiquoteClass o EncClass;*)
val UCornerCtx = AntiquoteClass o EncCtx ;
(*...*)
***)
(*-META*)

structure GlobalNames =
struct
(** Default names **)

(** (1) Untyped variables: **)

(** Greek lower-case variables for formulas **)
val vphi:Variable = zero_name sphi;
val vpsi:Variable = zero_name spsi;
val vchi:Variable = zero_name schi;
(* ... *)

(** Latin lower-case variables for sets **)
val va:LVariable = zero_name "a";
val vb:LVariable = zero_name "b";
val vc:LVariable = zero_name "c";
(* ... *)

(** Latin higher-case variables for classes **)
val vA:LVariable = zero_name "A";
val vB:LVariable = zero_name "B";
val vC:LVariable = zero_name "C";
(* ... *)

(** (2) Default classes: **)

(** Sets **)
val cta : ClassTerm = CVar va;
val ctb : ClassTerm = CVar vb;
val ctc : ClassTerm = CVar vc;

(* term of a set: *)
val ctta : ClassTerm Term = EVar (zero_name "a");
(* ... *)

(** Classes **)
val ctA : ClassTerm = CVar vA;
```

```
val ctB : ClassTerm = CVar vB;
val ctC : ClassTerm = CVar vC;
(* ... *)

(** Formulas **)
val fphi : PropTerm = (FVar vphi);(*???p1,p2,p3?*)
val fpsi : PropTerm = (FVar vpsi);
val fchi : PropTerm = (FVar vchi);
(* ... *)

(** ==== Bold symbols ==== **)
(*
(* Due to "Bold = Encode o Quote", every type is Class here. *)

(** Sets **)
val ab : ClassTerm = BoldClass ta;
val bb : ClassTerm = BoldClass tb;
val cb : ClassTerm = BoldClass tc;
(* ... *)

(** Classes **)
val Ab : ClassTerm = BoldClass tA;
val Bb : ClassTerm = BoldClass tB;
val Cb : ClassTerm = BoldClass tC;
(* ... *)
*)
(** Formulas **)
(*?
val phib : ClassTerm = BoldProp fphi;
val psib : ClassTerm = BoldProp fpsi;
val chib : ClassTerm = BoldProp fchi;
*)
(* ... *)

end; (*GlobalNames*)

(*** TO PDF PART END ***)
```

7.1 Primitive definitions and notation

Definition 7.1 (variable)**.**
A *variable* is an ordered pair consisting of a string, called its *name*, and a natural number, called its *level*. The two components record the identifier and its metalevel in the syntactic representation.

The companion code gives the corresponding datatype constructors. The mathematical notation below suppresses some of those constructors, so the translation conventions are stated explicitly. The following examples connect the typed notation of the implementation with ordinary mathematical notation.
The comparison begins with representability in arithmetic, then records the proposed encoding in the notation of this book:

Definition (representability in the traditional notation)**.**
Let $\mathbb{N}$ be the set of natural numbers. A theory T in the language of arithmetic containing Q "represents" the k–ary recursive function $f : \mathbb{N}^k \to \mathbb{N}$ if there is a formula $\varphi_f(x_1, \ldots, x_k, y)$ in the language of T s.t. for all $m_1, \ldots, m_k \in \mathbb{N}$, if $f(m_1, \ldots, m_k) = n$ then

$$T \vdash \forall y(\varphi_f(\overline{m_1}, \ldots, \overline{m_k}, y) \longleftrightarrow y = \overline{n}).$$

(https://en.wikipedia.org/wiki/Diagonal_lemma#Representation_Theorem)

Example 1 (representability in the proposed higher-order notation)**.**
The displays below explore how to encode the preceding definition. Their intended common meaning is the preceding representability condition. A fully typed translation still requires the domains of quotation, antiquotation, and substitution to be fixed; the displays are not separate representation theorems. Let $\mathbb{N}$ be the set of natural numbers. Atheory T in the language of arithmetic containing Q "represents" the k–ary recursive function $f : \mathbb{N}^k \to \mathbb{N}$ if there is a formula $\varphi_f(x_1, \ldots, x_k, y)$ in the language of T s.t. for all $m_1, \ldots, m_k \in \mathbb{N}$, if $f(m_1, \ldots, m_k) = n$ then

$$T \vdash \mathrm{QuoteFm}(\forall y(\mathrm{AntiquoteFm}(\varphi_f[m_1/x_1, \ldots, m_k/x_k, \mathrm{Quote}(y)/y]) \leftrightarrow (y = \mathrm{Antiquote}(n)))).$$

A proposed alternative display, whose typing remains to be checked, is

$$T \vdash (\forall \mathrm{Quote}(y), ((\varphi_f[m_1/x_1, \ldots, m_k/x_k, \mathrm{QuoteClass}(y)/y]) \longleftrightarrow (\mathrm{QuoteClass}(y) = n))).$$

Another proposed display is

$$T \vdash (\forall y, ((\varphi_f[m_1/x_1, \ldots, m_k/x_k, \mathrm{QuoteClass}(y)/y]) \longleftrightarrow (\mathrm{QuoteClass}(y) = n))).$$

For comparison, the shorter assertion notation is:

$$T \vdash \llcorner \boldsymbol{\varphi} \lrcorner$$

The corresponding expanded display is

$$T \vdash \llcorner \boldsymbol{\forall y(\varphi_f(\dot{m}_1, \ldots, \dot{m}_k, y) \leftrightarrow y = \dot{n})} \lrcorner,$$

where

$$\ulcorner \cdot \urcorner \overset{\mathrm{def.}}{:\equiv} (\mathrm{AntiquoteClass} \circ \mathrm{EncFm}),$$
$$\ulcorner \cdot \urcorner : \mathrm{Fm} \to \mathrm{Class}$$
$$\llcorner \cdot \lrcorner \overset{\mathrm{def.}}{:\equiv} (\mathrm{DecFm} \circ \mathrm{QuoteClass}).$$
$$\llcorner \cdot \lrcorner : \mathrm{ClassTerm} \to \mathrm{Fm}$$

For any formula φ,

$$\llcorner \ulcorner \varphi \urcorner \lrcorner \equiv \varphi.$$

Example 7.1 (extension rule)**.**
The following proposed *extension rule* records a further design question. The question marks mark domains that have not yet been specified:

$$\forall(\varphi : \mathrm{Fm}),\ (\mathcal{F} \vdash \forall(n :?),\ \llcorner \boldsymbol{\varphi}(\dot{\boldsymbol{n}})\lrcorner) \implies (\mathcal{F} \vdash \forall(n :?),\ \varphi(n)),$$

$$\forall(\varphi : \mathrm{Fm}),\ (\forall(n : \mathrm{ClassTerm}),\ \mathcal{F} \vdash \llcorner \boldsymbol{\varphi}(\dot{\boldsymbol{n}})\lrcorner) \implies (\mathcal{F} \vdash \forall(n : \mathrm{Class}),\ \varphi(n)),$$

Here the operation denoted by $\varphi(n)$, the domains of the two quantifiers, and the hypotheses licensing passage from all instances to a universal statement remain to be defined. This example is a proposed rule, not an admissibility result.

7.2 Meaning of dots and bold symbols

The intended translation represents higher-order syntax by first-order codes, using encoding/decoding together with quotation/antiquotation. The identities below specify the intended behaviour on well-formed inputs; they do not by themselves establish a translation theorem for the entire framework.
This subsection uses Fm as the running example. Analogous conventions are intended for the other syntactic types: Ctx, Class, Theory, etc.

Definition 7.2 (functions Bold and Dot)**.**
Let $\circ$ be an infix operator of composition of functional symbols here. Then

$$\mathrm{BoldFm} \overset{\mathrm{def.}}{:\equiv} \mathrm{EncFm}$$

$$\mathrm{BoldFm} : \mathrm{Fm} \to \mathrm{ClassTerm}$$

and

$$\mathrm{Dot} \overset{\mathrm{def.}}{:\equiv} (\mathrm{AntiquoteClass}).$$

$$\mathrm{Dot} : \mathrm{ClassTerm} \to \mathrm{ClassTerm}$$

Notation 7.1 (bold and/or dotted variables for formulas)**.**
For a formula variable φ, its encoding is denoted as a bold $\boldsymbol{\varphi}$:

$$\boldsymbol{\varphi} \overset{\mathrm{notation}}{:\equiv} \mathrm{BoldFm}(\varphi).$$

For any class variable t, its antiquotation is denoted with a dot:

$$\dot{t} \overset{\mathrm{notation}}{:\equiv} \mathrm{Dot}(t).$$

Note 7.2 (non–commutativity of notation)**.**
Boldface and a dot denote operations applied in a specified order:

$$\dot{\boldsymbol{\varphi}} \overset{\mathrm{notation}}{:\equiv} \mathrm{Dot}(\mathrm{BoldFm}(\varphi)).$$

After the corresponding decoding step, the round trip is intended to recover the original formula:

$$\llcorner \dot{\boldsymbol{\varphi}} \lrcorner \equiv ((\mathrm{DecFm} \circ \mathrm{QuoteClass}) \circ (\mathrm{AntiquoteClass}) \circ (\mathrm{EncFm}))(\varphi) \equiv \varphi$$

So the following computational equality is an identity:

$$\boxed{\llcorner \dot{\boldsymbol{\varphi}} \lrcorner \equiv \varphi}$$

Notation for variables

Notation 7.2 (bold variables for terms)**.**
For a lower-case Latin letter, such as "x", boldface denotes the encoded variable:

```
val xb : Class = (EncClass (CV (namelvl ("x", Nat.zero))))
val yb : Class = (EncClass (CV (namelvl ("y", Nat.zero))))
...
```

Equivalently, suppressing datatype constructors in the mathematical notation,

$$\boldsymbol{x} \stackrel{\text{def.}}{:\equiv} \mathrm{EncClass}(x_{(0)}),$$

$$\boldsymbol{y} \stackrel{\text{def.}}{:\equiv} \mathrm{EncClass}(y_{(0)}),$$

$$\ldots$$

Notation 7.3 (bold variables for formulas)**.**
For any lower–case Greek letter('φ','ψ', 'χ', ...), define a bold lower-case Greek letter to denote

```
val phib : Class = (EncFm (FmVar vphi))
val psib : Class = (EncFm (FmVar vpsi))
...
```

Equivalently, suppressing datatype constructors in the mathematical notation,

$$\boldsymbol{\varphi} \stackrel{\text{def.}}{:\equiv} \mathrm{EncFm}(\varphi_{(0)}),$$

$$\boldsymbol{\psi} \stackrel{\text{def.}}{:\equiv} \mathrm{EncFm}(\psi_{(0)}),$$

$$\ldots$$

Notation for functional and predicate symbols

Bold function and predicate notation denotes constructors acting on codes. Encoding is therefore described inductively, one syntactic constructor at a time:

Definition 7.3 (computation of the encoding of statements)**.**
For a theory term $\mathcal{W}$, a context term Γ, and a formula term φ,

$$\mathrm{EncStmt}(\mathcal{W},\, \Gamma \vdash \varphi) \equiv \mathbf{(}\mathrm{EncTheory}(\mathcal{W})\mathbf{,}\, \mathrm{EncCtx}(\Gamma)\, \mathbf{\vdash}\, \mathrm{EncFm}(\varphi)\mathbf{)}$$

The right-hand side applies a first-order function symbol to the codes of the three arguments. Its value represents the code of the whole statement. Other syntactic constructors are encoded in the same way.

7.3 Definitions of inference–related constants and their notation

Definition 7.4 (sets of codes)**.**
Fix a finite-tree coding for the syntactic datatypes under discussion. For a syntactic datatype label A, write $\mathrm{Code}(A)$ for its carrier of valid set codes, with a specified bijection between its syntax objects and those codes. For the finite syntax over countably encoded identifiers considered here, these carriers may be taken as subsets of $\mathbb{N}$. The following constants denote code carriers:

$$\mathrm{FmCode} \overset{\text{def.}}{:\equiv} \mathrm{CFm}, \text{ where } \mathrm{CFm} \overset{\text{def.}}{:\equiv} \mathrm{Code}(\mathrm{TFm})$$

$$\mathrm{ClassCode} \overset{\text{def.}}{:\equiv} \mathrm{CClass}, \text{ where } \mathrm{CClass} \overset{\text{def.}}{:\equiv} \mathrm{Code}(\mathrm{TClass})$$

...

Here CClass codes finite *class expressions*; it is not a carrier containing every semantic class. The old notation $\mathrm{Emb}(A)$ for these carriers is replaced by $\mathrm{Code}(A)$ to distinguish it from the predicate interpretation of types in Section 31.2. In that interface, a carrier C corresponds to the membership predicate $c \mapsto c \in C$; collecting that predicate returns C. In particular, $\mathrm{Code}(\mathrm{TClass})$ is not the collected extension of the semantic type Class, which would be the whole class V of sets. An expression's code and the class it denotes are different objects.

Note 7.3 (Quotation, decoding, and valuation have different signatures)**.**
Three operations must be distinguished even when a source uses "quotation" or "antiquotation" for more than one of them. Syntactic encoding sends an expression to its code; decoding recovers an expression from a valid code; semantic valuation interprets that expression in a fixed model and assignment. In the type chapter's notation, μ_A is a type-indexed translation into class expressions, so evaluation of a valid code c has the composite form

$$\mathrm{Eval}_{A,\rho}(c) = \mathrm{Val}_\rho\big(\mu_A(\mathrm{Dec}_A(c))\big).$$

The newer symbol pages instead use μ_A for valuation under an assignment called A. That is a different local signature; its version of $\mu_A(\mathrm{Dec}(c))$ already includes semantic evaluation. The two notations agree only after their roles have been translated, not by deleting the outer valuation in the displayed formula.
The round trip from syntax to its code and back is a coding identity. A round trip from an arbitrary semantic value to a finite expression denoting it would additionally require surjectivity of interpretation. That stronger property fails in the function-space setting of Proposition 31.1. Distinct syntax can also have the same value: in ordinary arithmetic, $0 + 0$ and 0 have different raw syntax codes but equal denotations. Thus decoding is not inversion of semantic equality. For formula codes, truth in the chosen model is again a separate semantic operation, subject to the language boundary in Proposition 13.1.

7.4 Typed quantification and its notation

Typed quantification over syntactic objects is represented below by quantification over their codes. A code variable must be fresh for the body, and every displayed substitution is capture-avoiding. Quantification over semantic classes is a separate issue: the set- and class-quantifier schemes later in this subsection are proposed axioms, not consequences of the coding convention alone.

Proposition 7.2 (Quantification transported along a coding)**.**
Let S be a set of syntax objects, let C be its code carrier, and let $\mathrm{enc} : S \to C$ and $\mathrm{dec} : C \to S$ be mutually inverse maps. For any metatheoretic predicate P on S,

$$(\forall s \in S,\ P(s)) \longleftrightarrow (\forall c \in C,\ P(\mathrm{dec}(c))),$$
$$(\exists s \in S,\ P(s)) \longleftrightarrow (\exists c \in C,\ P(\mathrm{dec}(c))).$$

Unique existence is preserved as well. No interpretation of P as an object-language truth predicate is needed.

Proof.
For the universal forward implication, apply P to $\mathrm{dec}(c) \in S$. For the reverse implication, use $c = \mathrm{enc}(s)$ and $\mathrm{dec}(\mathrm{enc}(s)) = s$. For existence, encode a witness in one direction and decode a witness in the other. If two codes satisfy the decoded predicate and P has a unique witness, their decoded objects are equal; applying enc makes the codes equal. Conversely, equal codes of any two witnesses give equal witnesses after decoding. Together with existence, these arguments prove preservation of unique existence. □
The displayed formula substitution below implements this transport only when the decoding and substitution operations have the specified types and meanings. Decoding is required on valid codes; a guarded formula in a total language additionally needs a total extension of the decoder outside that carrier. One may use any fixed syntax object as default when the datatype is nonempty. Alternatively the display can be read as a bounded metatheoretic quantifier, which never applies the decoder to an invalid code. These conventions do not give a total semantic evaluator for arbitrary expressions or a code for every semantic class.

Proposition 7.3 (When codes can test semantic quantification)**.**
Let C, A be sets and let $v : C \to A$ be an interpretation of codes. The equivalence

$$(\forall a \in A,\ P(a)) \longleftrightarrow (\forall c \in C,\ P(v(c)))$$

holds for every predicate P on A exactly when v is surjective. The same criterion holds for the corresponding existential equivalence.

Proof.
Surjectivity supplies a code for each a, proving the reverse universal implication and the forward existential implication; the other two implications hold for every map. If $a_0 \in A$ has no code, take $P(a)$ to be $a \neq a_0$ for the universal assertion. It holds at every interpreted code but not throughout A. For the existential assertion, take $P(a)$ to be $a = a_0$: it has a semantic witness and no coded witness. These examples also cover an empty code carrier with nonempty A. □
Thus a syntax-coding bijection alone does not justify replacing a quantifier over semantic values by one over expressions. A further surjectivity hypothesis is needed for the interpretation being used. Even under surjectivity, unique *codes* need not correspond to unique semantic values: two different codes can denote the same value. For example, the constant map from $\{0, 1\}$ to $\{a\}$ gives a unique value satisfying $x = a$ but two codes satisfying its pullback. The countable-syntax obstruction in Proposition 31.1 gives another reason to keep these quantifiers distinct in a fixed function-space interpretation. None of these observations precludes a separately specified model in which the required surjectivity holds.

Definition 7.5 (quantifiers with a bound variable of type Fm)**.**
Let φ be a variable for formulas.

Let ψ be a formula.
Choose a fresh code variable φ_{code}. Assume also that the substitution $\psi[(\text{DecFm}(\varphi_{\text{code}}))/\varphi]$ is computed successfully, i.e. without raising an exception or error. Then

$$(\forall(\varphi : \text{TFm}),\ \psi) \overset{\text{def.}}{\longleftrightarrow}$$

$$\overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi_{\text{code}} : \text{TClass}),\ (\varphi_{\text{code}} \in \text{Code}(\text{TFm})) \longrightarrow \psi[\text{DecFm}(\varphi_{\text{code}})/\varphi])$$

Notation 7.4 (naming prefix of labels)**.**
Standard ML distinguishes type names from values. Here the datatype named Type has value constructors with the prefix "T", such as "TFm", "TClass", and "TCtx". The prefix helps distinguish a syntactic type label from the datatype of expressions that carry that label.
In mathematical displays we suppress the prefix when the context determines its meaning. Thus, instead of the explicit label

$$\text{“}(\forall(\varphi : \text{TFm}),\ \ldots)\text{”},$$

starting from the next chapter, it will be written

$$\text{“}(\forall(\varphi : \text{Fm}),\ \ldots)\text{”}.$$

Other changes between syntactic and semantic notation will be stated explicitly.

Axiom 1 (set terms quantification unfolds via itself)**.**
The proposed set-quantifier scheme retains a set-typed code quantifier:

$$(\forall(t : \text{TSet}),\ \psi) \overset{\text{axm.}}{\longleftrightarrow}$$

$$\overset{\text{axm.}}{\longleftrightarrow} (\forall(t_{\text{code}} : \text{TSet}),\ (t_{\text{code}} \in \text{Code}(\text{TSet})) \longrightarrow \psi[\text{DecSet}(t_{\text{code}})/t])$$

Here t_{code} is fresh for ψ. The scheme is stated as an axiom because the same quantifier appears on both sides; it is not a recursive definition. Its semantic justification is an additional obligation of this proposed model.

Notation 7.5 (overloading class bindings)**.**
Again, as previously for formulas, a formula

$$\text{“}(\forall(K : \text{TSet}),\ \ldots)\text{”}$$

can be written as

$$\text{“}(\forall(K : \text{Set}),\ \ldots)\text{”}.$$

Axiom 2 (class terms quantification unfolds via itself)**.**
The analogous proposed class-quantifier scheme retains a class-typed code quantifier. The code guard below concerns syntax, not the predicate that accepts all semantic class values.

$$(\forall(t : \text{TClass}),\ \psi) \overset{\text{axm.}}{\longleftrightarrow}$$

$$\overset{\text{axm.}}{\longleftrightarrow} (\forall(t_{\text{code}} : \text{TClass}),\ (t_{\text{code}} \in \text{Code}(\text{TClass})) \longrightarrow \psi[\text{DecClass}(t_{\text{code}})/t])$$

Here t_{code} is fresh for ψ. The scheme is stated as an axiom because the same quantifier appears on both sides; it is not a recursive definition. Its semantic justification is an additional obligation of this proposed model.

Notation 7.6 (overloading class bindings)**.**
Again, as previously for formulas, a formula

$$\text{“}(\forall(K : \mathrm{TClass}),\ \ldots)\text{”}$$

can be written as

$$\text{“}(\forall(K : \mathrm{Class}),\ \ldots)\text{”}.$$

Notation 7.7 (the default type)**.**
In class-theoretic formulas, an omitted type annotation means “Class”:

$$(\forall K,\ \psi) \overset{\text{notation}}{\longleftrightarrow} (\forall(K : \mathrm{Class}),\ \psi)$$

Thus an unqualified “term” in that context means a class term. In discussions of implementation or syntax, the relevant datatype is stated separately.

7.5 Axioms for the predicate calculus

This subsection specifies the classical calculus used locally. A theory $\mathcal{W}$ includes the logical axiom schemes and is closed under the stated inference rules. A displayed scheme stands for its well-formed instances; where axioms are required to be closed, take their universal closures.
Both $\mathcal{T}$ and $\mathcal{F}$ are assumed to satisfy these requirements. Their additional principles are introduced separately below.

Propositional part

Metaaxiom 1 (classical propositional axiom schemes)**.**
For all formulas A, B, C, for all contexts, the following formulas are derivable in $\mathcal{W}$:

$$(A \longrightarrow (B \longrightarrow A)) \tag{axm:A1}$$

$$((A \longrightarrow (B \longrightarrow C)) \longrightarrow ((A \longrightarrow B) \longrightarrow (A \longrightarrow C))) \tag{axm:A2}$$

$$((A \wedge B) \longrightarrow A) \tag{axm:A3}$$

$$((A \wedge B) \longrightarrow B) \tag{axm:A4}$$

$$(A \longrightarrow (B \longrightarrow (A \wedge B))) \tag{axm:A5}$$

$$(A \longrightarrow (A \vee B)) \tag{axm:A6}$$

$$(B \longrightarrow (A \vee B)) \tag{axm:A7}$$

$$((A \longrightarrow C) \longrightarrow ((B \longrightarrow C) \longrightarrow ((A \vee B) \longrightarrow C))) \tag{axm:A8}$$

$$(\neg A \longrightarrow (A \longrightarrow \bot)) \tag{axm:A9}$$

$$((A \longrightarrow B) \longrightarrow ((A \longrightarrow \neg B) \longrightarrow \neg A)) \tag{axm:A10}$$

$$(\bot \longrightarrow B) \tag{axm:exfalso}$$

$$(A \vee \neg A) \tag{axm:LEM}$$

Metaaxiom 2 (Modus Ponens)**.**
The theory $\mathcal{W}$ must be closed w.r.t. the *modus ponens* inference rule:

$$\frac{A \longrightarrow B \qquad A}{B} \tag{rule:MP}$$

Propositional part expressed via metastatements.

Notation 7.8 ("antiquotation" of a term)**.**
The bullet above a variable $\dot{A}$ (or $A_{(1)}$) indicates that the variable belongs to the metalevel; $\ddot{A}$ (or $A_{(2)}$) belongs to the metametalevel, and so on. An antiquoted term t of any type can be written as "Antiquote(t)", without the surrounding quotation marks.

Note 7.4 (requirement for the implementation)**.**
For a formula φ to be well–formed, all antiquotations in it must be quoted.
(A free antiquotation would instead describe a template that still needs an argument.)
The proof assistant needs a term that is ground at the metalevel, with no free antiquotations, to check syntactic side conditions such as "is free in" and "is free for substitution".
This agreement makes the side conditions decidable.
The check computes a Boolean answer at the formula's highest metalevel. If unbound metavariables prevent that check, it raises an exception.

The axiom schemes are expressed on the metalevel in this way:

$$\forall(A, B : \mathrm{Fm}),\ (\mathcal{T} \vdash (A \longrightarrow (B \longrightarrow A)))$$

or, equivalently:

$$\forall(A, B : \mathrm{Fm}),\ (\mathcal{T} \vdash \llcorner \dot{\boldsymbol{A}} \longrightarrow (\dot{\boldsymbol{B}} \longrightarrow \dot{\boldsymbol{A}}) \lrcorner)$$

$$\ldots \text{ (other axioms are transformed similarly) } \ldots$$

$$\forall(A : \mathrm{Fm}),\ (\mathcal{T} \vdash \llcorner \dot{\boldsymbol{A}} \vee \neg \dot{\boldsymbol{A}} \lrcorner)$$

Everything that is derivable in $\mathcal{T}$ is also derivable in $\mathcal{F}$:

$$\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{T} \vdash \varphi) \longrightarrow (\mathcal{F} \vdash \varphi))$$

Modus ponens is required for each derivability relation separately: inclusion of the theorems of $\mathcal{T}$ in those of $\mathcal{F}$ does not itself specify how derivations with arbitrary contexts behave.

$$\forall(A, B : \mathrm{Fm}),\ \forall(\Gamma : \mathrm{Ctx}),\ (((\mathcal{T}, \Gamma \vdash (A \longrightarrow B)) \wedge (\mathcal{T}, \Gamma \vdash A)) \longrightarrow (\mathcal{T}, \Gamma \vdash B))$$

$$\forall(A, B : \mathrm{Fm}),\ \forall(\Gamma : \mathrm{Ctx}),\ (((\mathcal{F}, \Gamma \vdash (A \longrightarrow B)) \wedge (\mathcal{F}, \Gamma \vdash A)) \longrightarrow (\mathcal{F}, \Gamma \vdash B))$$

(The symbol $\equiv$ is an unnecessary informal notation for $=$, for a single purpose of giving a hint, that equality is not only propositional, but also computational. For the purpose of this book, their difference can be safely ignored by the reader.)

Quantifier rules

The variable and the substituting term in each rule have the same type. Contexts are denoted by Γ. Freshness conditions apply to all undischarged assumptions, not merely to the displayed conclusion.

Specification axiom scheme:

For any variable x, for any type X, for any term t of that type, for any formula φ such that term t is free for [the substitution of it instead of the free occurrences of the variable] x in φ, the following formulas are derivable in $\mathcal{W}$:

$$(\forall(x : X), \varphi) \longrightarrow \varphi[t/x] \qquad \text{(axm:S1)}$$

$$\varphi[t/x] \longrightarrow (\exists(x : X), \varphi) \qquad \text{(axm:S2)}$$

$$\forall(\mathcal{W} : \mathrm{Theory}),\ \text{“}\forall(v : \mathrm{Var}),\text{”}\ \forall(t : X),\ \forall(\varphi : \mathrm{Fm}),$$
$$((t \text{ free for } v \text{ in } \varphi) \longrightarrow (\mathcal{W} \vdash ((\forall(\mathrm{var}(v) : X), \varphi) \longrightarrow \varphi[t/v]))) \qquad \text{(axm:MetaS1)}$$

$$\forall(X : \mathrm{T}(\mathrm{TTerm})),\ \mathcal{T} \vdash \forall(\mathcal{W} : \mathrm{Theory}),\ \forall(v : \mathrm{Var}(\mathrm{TClass})),\ \forall(t : [\![X]\!]),\ \forall(\varphi : \mathrm{Fm}),$$
$$((t \text{ free for } [\![v]\!] \text{ in } \varphi) \longrightarrow (\mathcal{W} \vdash (\varphi[t/v] \longrightarrow (\exists([\![v]\!] : [\![X]\!]), \varphi))))) \qquad \text{(axm:MetaMetaS2)}$$

$$\forall(X : \mathrm{Types}(\mathrm{Type})),\ \mathcal{T} \vdash \forall(\mathcal{W} : \mathrm{Theory}),\ \forall(v : \mathrm{Var}(\mathrm{Class})),\ \forall(t : \dot{X}),\ \forall(\varphi : \mathrm{Fm}),$$
$$((t \text{ free for } \dot{v} \text{ in } \varphi) \longrightarrow (\mathcal{W} \vdash \langle(\varphi[t/v] \longrightarrow (\exists(\dot{v} : \dot{X}), \varphi))\rangle)) \qquad \text{(axm:MetaMetaS2Bold)}$$

The signatures of the functions implementing the predicates

$$\text{“}(\cdot \text{ free for } \cdot \text{ in } \cdot)\text{”}$$

and

$$\text{“}(\cdot \text{ occurs free in } \cdot)\text{”}$$

are given in the listing below:

```
FFI : Value * Variable * Value -> bool
isFree : Variable * Value -> bool
```

(For simplicity, "for any type X" is metatheoretic quantification here. It could be included in the object language if dependent types were implemented.)
Epsilon calculus axiom scheme:
For any type X, for any variable x and the formula φ such that the term $(\epsilon(x : X), \varphi)$ is free for [the substitution of it instead of the free occurrences of the variable] x in φ, the following formula is derivable in $\mathcal{F}$:

$$(\exists(x : X), \varphi) \longrightarrow (\varphi[(\epsilon(x : X), \varphi)/x]) \tag{axm:H}$$

"$\forall(X : \text{SmallType}),$" $\forall(x : \text{Var}),\ \forall(\varphi : \text{Fm}),\ \mathcal{F} \vdash ((\exists(x : X), \varphi) \longrightarrow (\varphi[(\epsilon(x : X), \varphi)/x]))$ (axm:MetaH)

The Henkin property defined later supplies closed witnesses for provable existential sentences. It does not by itself specify a uniform descriptor for formulas with parameters. Moreover, in total semantics a descriptor of type X must denote even when its defining predicate has no witness; therefore an empty interpretation of X is not permitted by this scheme. The following result gives a precise local justification for nonempty set-sized sorts. The proposed class-sized and higher-order instances require their own interpretation and are not consequences of this result.

Theorem 7.1 (A conservative total witness extension)**.**
Work in classical set metatheory with Choice. Let T be a theory in an ordinary one-sorted first-order language, with total function symbols and nonempty set-sized domains. Extend its finite term and formula syntax by the binding constructor $\epsilon x\, \varphi$, with capture-avoiding substitution, and add all instances of the witness scheme

$$(\exists x\, \varphi) \longrightarrow \varphi[\epsilon x\, \varphi/x].$$

Retain the original nonlogical axioms of T, without extending their schemata to new formulas, and the classical logical axioms with their usual context and freshness conditions. Every model of T expands to a model of this extended calculus. Consequently, if an old-language sentence is derivable in the extension, it is a semantic consequence of T. If the original calculus is complete for these structures, the extension is also conservative for old-language derivability.

Proof.
Fix a model M of T with nonempty domain D. Choose $d_0 \in D$ and a choice function c on the nonempty subsets of D, and put

$$\widehat{c}(S) = \begin{cases} c(S), & S \neq \varnothing, \\ d_0, & S = \varnothing. \end{cases}$$

Interpret the new terms and formulas together by induction on their finite syntax trees. Keep every old interpretation clause, and define

$$[\![\epsilon x\, \varphi]\!]_s = \widehat{c}\big(\{a \in D : M, s[x := a] \models \varphi\}\big).$$

The formula inside this clause is a proper subexpression of the term, so mutual recursion is well founded even with nested descriptors. Its truth set is a subset of D in the metatheory. The same fixed $\widehat{c}$ is used for every assignment and every descriptor.

We check the new binder case needed for the logical substitution rules. First, coincidence of assignments extends by mutual induction: assignments agreeing on the free variables of $\epsilon x\,\varphi$ give, after updating x to each a, assignments agreeing on the free variables of φ. The induction hypothesis makes the two truth sets equal, so the selected values agree. The other clauses are the ordinary first-order clauses.
For substitution of a term t for a variable z, consider $e = \epsilon x\,\varphi$. If $x = z$, substitution does not enter this binder and coincidence gives $[\![e]\!]_s = [\![e]\!]_{s[z:=[\![t]\!]_s]}$. Otherwise rename x fresh for t and distinct from z. For each $a \in D$, the induction hypothesis for φ gives

$$M, s[x := a] \models \varphi[t/z] \longleftrightarrow M, s[x := a][z := [\![t]\!]_{s[x:=a]}] \models \varphi.$$

Coincidence and freshness give $[\![t]\!]_{s[x:=a]} = [\![t]\!]_s$; the two updates commute. Thus the set selected when evaluating $e[t/z]$ at s equals the set selected when evaluating e at $s[z := [\![t]\!]_s]$. Applying $\widehat{c}$ proves the term substitution clause. The formula clauses follow by the same mutual induction, with bound variables renamed fresh for t. Renaming a bound identifier itself preserves the truth set, since both versions range over the same $a \in D$; hence these choices of fresh names do not change the interpretation.
If $M, s \models \exists x\,\varphi$, the set in the descriptor clause is nonempty. Its selected element satisfies φ by the definition of c, and the substitution clause just proved identifies this with truth of $\varphi[\epsilon x\,\varphi/x]$ at s. If the existential is false, the witness implication is true by classical logic. All instances of the new scheme are therefore valid. Totality, substitution, and coincidence give the ordinary soundness checks for the logical axioms and inference rules, including the freshness condition for generalisation. The old axioms remain true because their interpretations have not changed. This proves the expansion assertion.
An old-language countermodel to a proposed semantic consequence of T therefore expands to a countermodel in which every extended derivation is sound. No old-language sentence derivable in the extension can have such a countermodel. Completeness of the original calculus, when assumed, converts this semantic consequence into an original derivation. □
The construction fixes a choice operation as semantic data; it does not claim that the expansion is unique or that the choice is definable in M. For finitely or set-many sorts the same argument works with nonempty set carriers and specified choice operations for those carriers. It cannot be applied to an empty carrier: even $\epsilon x\,(x \neq x)$ would have to denote an element of that carrier. For descriptors intended to range over classes, the separate ranked class-parameter construction in the relational-image chapter supplies a different, explicitly stronger setting.
Abstraction rules:
For a variable x of type X and formulas ψ, φ, assume that x is free in neither ψ nor the context Γ. The first rule moves a universal quantifier into the consequent; the second moves an existential quantifier into the antecedent. Notice the different order of ψ and φ in their premises.

$$\frac{\psi \longrightarrow \varphi}{\psi \longrightarrow (\forall(x : X), \varphi)} \qquad (\text{where } x \notin \mathrm{FV}(\psi :: \Gamma)) \qquad \text{(axm:B1)}$$

$$\frac{\varphi \longrightarrow \psi}{(\exists(x : X), \varphi) \longrightarrow \psi} \qquad (\text{where } x \notin \mathrm{FV}(\psi :: \Gamma)) \qquad \text{(axm:B2)}$$

(The sign $\notin$ inside the side conditions is always used on the metalevel in the sense "not in the list", because they are metatheoretical statements about terms, formulas, and other syntactic entities, not the statements within the object theory.)

Simplification of rules B1 and B2

In the classical setting, the intended presentation uses generalisation and the dual definition of the existential quantifier. The type-size conventions below belong to this particular proposed typed calculus.

Note 7.5 (sizes of types terminology)**.**
This local proposal distinguishes three sizes of types:

1. "small": types interpreted as sets.
2. "medium": types interpreted as classes.
3. "large": all mentioned types, extended by types "Class" and "Type", which is interpreted as a conglomerate of all classes.

Note 7.6 (judgements)**.**
The following typing conventions are assumed in this proposal:

- All small terms have small types.
- All small types are also small terms, but not vice versa.
- Thus each small type has a small type of its own.
  The type of a small type is required to be a universe. Consequently, the type of a universe is also a universe. All these universes are small types.
- All medium types have the large type "Type".
- Large types have no types. There are only two of them: "Type" and "Class".

Note 7.7 (on quantifications over all types)**.**
To avoid assuming a type of all types, the proposal separates a scheme over medium types from explicitly stated instances for the two large types, "Class" and "Type". This is the intended reading of "for any type" in the generalisation rule below.

Definition 7.6 (generalisation rule(s))**.**
For $\mathcal{W}$ to be a theory, the following rule must hold for any type X:
" For any variable x, for any formula φ such that the variable x is not free in the context, then the derivability of φ implies the derivability of $(\forall(x : X), \varphi)$ in $\mathcal{W}$. "

$$\frac{\varphi}{(\forall(x : X), \varphi)} \qquad (\text{where } x \notin \mathrm{FV}(\Gamma)) \qquad (\text{Gen})$$

Note 7.8 (Theorem rules and rules under assumptions)**.**
The companion catalog's short generalisation display (2.06) lists no freshness condition, whereas its context-indexed version (4.02) explicitly requires that the variable not occur free in Γ. These presentations agree when the short rule acts on theorems with no undischarged assumptions and the background axioms are closed. They do not justify an unrestricted rule from an arbitrary assumption φ to $\forall x\, \varphi$. For example, on $\{0, 1\}$ with $P = \{0\}$, the context $P(x)$ is true at $x = 0$, but $\forall x\, P(x)$ is false. The abstraction rules B1 and B2 likewise require the context freshness stated above, in addition to freshness for the unchanged formula. The proof of Proposition 12.3 retains these restrictions explicitly.

Definition 7.7 (generalisation rule, written on the meta–level)**.**

$$\forall(X :: \mathrm{Type}),\ \forall(v : \mathrm{Var}),\ \forall(\varphi : \mathrm{Fm}),\ \forall(\Gamma : \mathrm{Ctx}),$$
$$(v \notin \mathrm{FV}(\Gamma)) \longrightarrow ((\mathcal{F}, \Gamma \vdash \varphi) \longrightarrow (\mathcal{F}, \Gamma \vdash \langle(\forall(\dot{v} : X), [\![\varphi]\!])\rangle)) \qquad \text{(MetaGen)}$$

Specialising the metatheoretic variable v supplies an object-language identifier, including at the binder position. This is different from substituting a term for a free occurrence in the resulting object formula. The following distinction explains the relevant freshness checks.

Remark 7.1 (Metatheoretic variables (v_x, v_y, v_C, v_s) vs. metatheoretic constants (c_x, c_y, c_C) and capture avoidance)**.**
In metalevel rules such as (MetaGen), symbols v, v_x, v_y, v_C, v_s : Var are *metatheoretic variables* universally quantified over the datatype Var of object-language variable identifiers. Fixed identifiers (such as the literal identifier for "x") are instead *metatheoretic constants*, denoted c_x, c_y, c_C : Var.
Inside a quoted formula $\langle(\forall(\dot{v} : X), [\![\varphi]\!])\rangle$, the antiquotation $\dot{v}$ splices the variable identifier denoted by v into the binder position. Decidable Boolean checks on the syntax trees enforce the following freshness and capture-avoidance conditions:

1. In (MetaGen), the side condition $(v \notin \mathrm{FV}(\Gamma))$ (checked by $\mathrm{NFV}(v, \gamma) = \mathrm{True}$ for every $\gamma \in \Gamma$) guarantees that the generalized variable identifier v does not occur free in any assumption of Γ.

2. In universal instantiation and existential elimination, the decidable predicate

$$\mathrm{FFI}(t, v_x, \varphi) = \mathrm{True}$$

("term expression t is free for metavariable v_x in φ") recursively verifies that no free variable identifier occurring in t : Class.Expr becomes bound by an inner quantifier when t is substituted for free occurrences of v_x in φ : Fm.

Definition 7.8 (classical existence)**.**

$$(\exists(x : X), \varphi) \overset{\text{def.}}{\longleftrightarrow} (\neg\forall(x : X),\ (\neg\varphi))$$

Definition 7.9 (a global choice function)**.**
If $x \notin \mathrm{FV}(S)$ then

$$c(S) \coloneqq (\epsilon(x : \mathrm{Class}),\ (x \in S))$$

This is a class-parameter choice operation under the proposed class-valued witness scheme. When S is nonempty, that scheme gives $c(S) \in S$, so the selected value is a set. When S is empty, the witness scheme prescribes no value; a total interpretation still needs a default. An ordinary global-choice *class graph* has nonempty sets as its arguments, not arbitrary proper classes. Collecting this operation's restriction into such a graph additionally requires the corresponding comprehension instance in its expanded language. The set-domain conservativity theorem does not establish that additional class-theoretic assertion. A concrete class-parameter construction with its own hypotheses is given in the relational-image chapter.
The following schemes describe *provable equivalence* of syntactic objects. The term "effective" would require a separate decision procedure; the definitions themselves only refer to derivability in $\mathcal{F}$. Every substitution below is subject to the same typing and capture-avoidance conditions as the quantifier rules. The equality-congruence statements apply to the ordinary total fragment with equality substitutivity and extensional term constructors. They do not assert congruence

through a constructor that inspects raw syntax, such as quotation or an injective coding of expressions. Such a constructor may distinguish terms with equal denotations.

$$(t_1 \sim t_2) \overset{\text{def.}}{\longleftrightarrow} (\mathcal{F} \vdash (t_1 = t_2))$$

$$(\varphi_1 \sim \varphi_2) \overset{\text{def.}}{\longleftrightarrow} (\mathcal{F} \vdash (\varphi_1 \longleftrightarrow \varphi_2))$$

$$(y \sim x) \longrightarrow (\varphi[y/x] \sim \varphi)$$

$$\forall (y : \mathrm{Var}),\ ((y \notin \mathrm{FV}(\varphi)) \longrightarrow ((y, \varphi[y/x]) \equiv_\alpha (x, \varphi)))$$

Here (x, φ) denotes a binding expression and $\equiv_\alpha$ means capture-avoiding renaming of its bound identifier, not equality of the two raw ordered syntax pairs. The variable y has the same sort as x; inner binders are renamed before substitution when necessary. The renaming changes exactly the occurrences bound by the displayed binder. If syntax is stored modulo alpha-equivalence, these expressions have the same stored representative under a canonical coding. An injective coding of raw named trees need not assign them the same code.

$$\forall (x : \mathrm{Var}),\ (x \notin (\mathrm{FV}(t_1) \cup \mathrm{FV}(t_2))) \wedge (t_1 \sim t_2) \longrightarrow (q[t_1/x] \sim q[t_2/x])$$
$$\forall (x : \mathrm{Var}),\ (x \notin (\mathrm{FV}(t_1) \cup \mathrm{FV}(t_2))) \wedge (t_1 \sim t_2) \longrightarrow (\varphi[t_1/x] \sim \varphi[t_2/x])$$

$$(\varphi_1 \sim \varphi_2) \longrightarrow \big((\mathcal{F}, \Gamma \vdash \varphi_1) \longleftrightarrow (\mathcal{F}, \Gamma \vdash \varphi_2)\big)$$

This last implication is a metatheorem, with both turnstiles read externally. Indeed, a theorem $\varphi_1 \longleftrightarrow \varphi_2$ supplies both implications; weaken either into Γ and apply modus ponens to a derivation of its antecedent. This transforms derivations in both directions. It does not by itself prove inside $\mathcal{F}$ a formula about the equivalence of two quoted sequents. That internal version requires specified representations and a proof that the two transformations are represented correctly.
The preceding equality-congruence displays similarly use the equality axioms: from a theorem $t_1 = t_2$, substitution of equals proves the equation between $q[t_1/x]$ and $q[t_2/x]$, and the biconditional between $\varphi[t_1/x]$ and $\varphi[t_2/x]$. The first display follows directly from equality substitution applied to $y = x$; it does not instantiate the later display's extra condition that its placeholder be absent from both replacement terms. These conclusions concern the stated extensional fragment and do not erase the distinctions just made for binders or quotations. The interaction between provable equivalence and encoding is a separate design question. The draft records the possibility that

$$(\vec{\exists})((t_1 \sim t_2) \not\longleftrightarrow (\mathrm{encode}(t_1) \sim_{\mathcal{T}} \mathrm{encode}(t_2))),$$

A proposed encoding that identifies equivalent terms would need a precise construction. Searching for a least equivalent code need not be an effective procedure. The use of de Bruijn notation addresses variable binding, but does not by itself decide provable equivalence.
For $\mathcal{F}$, the intended additional compatibility principle is

$$(t_1 \sim t_2) \longrightarrow (\mathrm{encode}(t_1) \sim_{\mathcal{F}} \mathrm{encode}(t_2)).$$

Its justification depends on the chosen encoding and equivalence relation; the corresponding principle for $\mathcal{T}$ is a separate question.

7.6 Admissible rules

An admissibility claim says that every derivation of the premises can be transformed into a derivation of the conclusion. The context and the substitution side conditions are therefore part of the claim. Inference rules are metaaxioms.
Admissible rules are metatheorems.

Metatheorem 7.1 (simultaneous substitution)**.**
Work in the ordinary total first-order fragment. Assume its nonlogical axiom schemes and inference rules are stable under substitution. Terms denote values of their sorts and universal instantiation is available for them. Let $x_1, \ldots, x_n$ be distinct variables and let $t_1, \ldots, t_n$ be terms of their respective types, free for simultaneous substitution in φ. With no undischarged assumptions, the following rule

$$\frac{\varphi}{\varphi[t_1, \ldots, t_n / x_1, \ldots, x_n]}\ \text{SimSubst}$$

is admissible. With a nonempty context, the same substitution must also be applied to that context.

Proof. For an empty context, generalise over the free variables to be replaced, using fresh bound identifiers if necessary. Instantiate the resulting universal quantifiers with the specified terms. Freshness and the free-for conditions ensure that the result is the simultaneous substitution rather than a sequence that changes an earlier replacement. For a derivation from assumptions, first rename generalisation variables fresh for the substitution terms, then substitute throughout the derivation. Axiom stability and the substitution-compatible rule schemes preserve every step and transform the assumptions to the substituted context. □
This does not assert unrestricted substitution for a partial or uncertain language. For example, a denoting variable satisfies x defined, but replacing it by an undefined term makes that assertion false. A term with several candidates similarly need not preserve a variable's uniqueness. Such languages require their own guarded rules and semantic substitution lemmas, as discussed in the evaluation sections.

Metatheorem 7.2 (substitution property / Leibniz's law)**.**
Assume equality has its substitutivity axioms, and that both substitutions below are well-typed and capture-avoiding. Informally:
"If $a = b$, then, for any well–formed formula φ, $\varphi(a) \implies \varphi(b)$."
Formally:

$$\forall(a, b : \text{ClassTerm}), (\mathcal{F} \vdash \langle([\![a]\!] = [\![b]\!])\rangle) \longrightarrow (\forall(x : \text{ClassVar}),\ \mathcal{F} \vdash (\varphi[a/x] \longrightarrow \varphi[b/x]))$$

Proof. Apply the assumed equality-substitution scheme to a, b, and φ. It gives $(a = b) \longrightarrow (\varphi[a/x] \longrightarrow \varphi[b/x])$. Modus ponens with the given derivation of $a = b$ yields the required implication. □

8 Incompleteness, consistency, serial proofs

This section adopts an ideal theory $\mathcal{F}$ that is complete, consistent, and not recursively enumerable. It is considered alongside an effective theory $\mathcal{T}$. The purpose is to distinguish what an ideal foundation may assert from what a program can enumerate or certify.

The additional reflection principles below concern truth in an intended interpretation. Completeness and consistency alone do not specify that interpretation. Statements about truth are read in a metatheory that can interpret the object language; displaying them using the book's compact quotation notation does not remove this semantic assumption.

1. The usual effective incompleteness argument does not rule out such a non-enumerable complete theory. Applications of Löb's theorem also require specified provability predicates and derivability conditions; these must be checked separately for the proposed framework.

2. The intended semantics additionally motivates reflection principles such as

$$((\mathcal{F} \vdash \varphi) \longrightarrow [\![\varphi]\!]), \qquad \text{(Reflection)}$$

$$\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{F} \vdash \varphi) \longrightarrow [\![\varphi]\!]), \qquad \text{(MetaReflection)}$$

   In the terminology used later, this direction is *credibility*; omniscience is the converse direction. The proposed obstruction to making reflection effective is recorded schematically as

$$\mathcal{T} \vdash \langle(\neg(\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{T} \vdash \varphi) \longrightarrow [\![\varphi]\!])))\rangle \qquad \text{(Proposed effective-reflection obstruction)}$$

   To read this as a theorem of $\mathcal{T}$, one must first specify an internal provability predicate and the meaning of formula evaluation in the display. The ordinary second incompleteness theorem concerns the resulting arithmetised consistency statement; the displayed unrestricted truth notation is not, by itself, such an arithmetisation.

3. The ideal theory is intended to reduce repeated shifts of metalevel in mathematical exposition, while the quotation conventions continue to record where those shifts occur.

4. An implementation can certify a specified fragment of the ideal theory. It cannot enumerate all of its theorems. The correspondence between the implemented rules and the intended semantics must therefore be established for that fragment.

9 Informal definitions

We now distinguish the effective calculus from its proposed ideal extension. These conventions apply to the following discussion of $\mathcal{T}$ and $\mathcal{F}$; they do not replace the local foundations of the other chapters.

Definition (informal definition of language $\mathcal{L}$)**.**
Let $\mathcal{L}$ be the object language of the chosen foundational development, including its function and predicate symbols. Its effective syntax must be fixed before the claims about enumerability below are applied.

Definition (informal definition of theory $\mathcal{T}$)**.**
Let $\mathcal{T}$ be the deductive closure of the chosen effective foundational axioms under their effective inference rules. The ideal reflection principles for $\mathcal{F}$ are additional metatheoretic assumptions. Under the consistency and arithmetic-strength hypotheses used below, $\mathcal{T}$ is incomplete.

9.1 Theory parameters and explicit evaluation

The companion Obelisk presentation distinguishes four roles that are useful here. $\mathcal{W}$ is a schematic variable ranging over theories; $\mathcal{T}$ is the chosen effective base theory; $\mathcal{F}$ is the proposed ideal extension; and $\mathcal{E}$ denotes a metatheory in which a particular result about $\mathcal{T}$ is formalised. Choosing $\mathcal{E}$ requires the hypotheses of that result. These roles do not by themselves assign a fixed universe level to each theory, or identify the schematic $\mathcal{W}$ with a fourth distinguished theory.
Write $\mathrm{eval}_M(\varphi)$ for the truth of a sentence φ in the chosen interpretation M. For an open formula, evaluation also needs an assignment, written $\mathrm{eval}_{M,s}(\varphi)$. This makes explicit the semantic operation that some of the earlier displays leave implicit. The companion entries 3.72–3.75 then have the intended readings

$$(\mathcal{T} \vdash \varphi) \longrightarrow \mathrm{eval}_M(\varphi), \qquad (\mathcal{F} \vdash \varphi) \longrightarrow \mathrm{eval}_M(\varphi),$$

$$\mathrm{eval}_M(\varphi) \longrightarrow (\mathcal{F} \vdash \varphi), \qquad (\mathcal{T}, \Gamma \vdash \varphi) \longrightarrow (\mathcal{F}, \Gamma \vdash \varphi).$$

The first three are semantic principles in the surrounding setting; the last preserves a context when passing to an extension. In particular, none of these displays asserts that $\mathcal{T}$ proves its own soundness. An internal version must name its ambient theory and the language interpreted by eval_M.
The third display is called *omniscience* below. The distinct notion of *negation-completeness* decides each sentence or its negation. In a fixed classical interpretation, soundness together with negation-completeness implies omniscience: if φ is true, soundness rules out a proof of $\neg\varphi$, so completeness supplies a proof of φ. Conversely, omniscience and semantic bivalence supply a proof of one of the two sentences. These are conditional connections between the notions, not interchangeable definitions without their hypotheses.

10 Formal statements

In the definitions of completeness and disjunctiveness below, formulas are *sentences*, that is, formulas with no free variables. The displayed quantifiers over Fm are restricted accordingly. This avoids confusing a decision of a sentence with a uniform decision of an open formula for every possible assignment. The same sentence convention is used in the later truth and provability comparison.

Definition 10.1 (consistent theory)**.**
Let $\mathcal{W}$ be a theory. It is called *consistent*, if "false" is not derivable in $\mathcal{W}$.

$$(\mathcal{W} \text{ consistent}) \overset{\text{def.}}{\longleftrightarrow} (\neg(\mathcal{W} \vdash \bot))$$

Axiom 3 ($\mathcal{T}$ is consistent)**.**
We assume, in the surrounding metatheory, that $\mathcal{T}$ does not derive a contradiction. This assumption is not asserted to be a theorem of $\mathcal{T}$.

$$\neg(\mathcal{T} \vdash \bot)$$

Definition 10.2 (complete theory)**.**
Let $\mathcal{W}$ be a theory. It is called *complete*, if for every formula, either its negation is derivable, or the formula itself is derivable.

$$(\mathcal{W} \text{ complete}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{W} \vdash \varphi) \vee (\mathcal{W} \vdash \neg\varphi)))$$

Definition 10.3 (Henkin theory)**.**
Let $\mathcal{W}$ be a theory in language $\mathcal{L}$. It has the *Henkin property* for classes if for every existential statement there is a witnessing term in $\mathcal{L}$, with a well-formed, capture-avoiding substitution. In the display, t ranges over closed class terms, and x is the only possible free variable of φ.

$$(\mathcal{W} \text{ Henkin}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi : \text{Fm}),\ ((\mathcal{W} \vdash \exists(x : \text{Class}),\ \varphi) \longrightarrow (\exists(t : \text{Class}),\ (\mathcal{W} \vdash \varphi[t/x]))))$$

Definition 10.4 (disjunctive theory)**.**
Let $\mathcal{W}$ be a theory in language $\mathcal{L}$. It is *disjunctive* if, whenever a disjunction of sentences is derivable, at least one of its disjuncts is derivable.

$$(\mathcal{W} \text{ disjunctive}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\chi_1 : \text{Fm}),\ \forall(\chi_2 : \text{Fm}),\ ((\mathcal{W} \vdash (\chi_1 \vee \chi_2)) \longrightarrow ((\mathcal{W} \vdash \chi_1) \vee (\mathcal{W} \vdash \chi_2))))$$

The disjunction property concerns which sentences the theory derives. Any further conclusion about standard elements requires a specified model and an additional argument; it is not part of this definition.

Lemma 10.1 (disjunctive property implies completeness)**.**
For any classical theory $\mathcal{W}$ satisfying excluded middle,

$$(\mathcal{W} \text{ disjunctive}) \longrightarrow (\mathcal{W} \text{ complete}) \qquad \text{(lem:DisjImplCompl)}$$

Proof. Fix $(\mathcal{W} : \text{Theory})$ and $(\varphi : \text{Fm})$.

$$(\mathcal{W} \text{ disjunctive}) \implies (\forall(\chi_1 : \text{Fm}),\ \forall(\chi_2 : \text{Fm}),\ ((\mathcal{W} \vdash (\chi_1 \vee \chi_2)) \longrightarrow ((\mathcal{W} \vdash \chi_1) \vee (\mathcal{W} \vdash \chi_2))))$$

Let $\chi_1 :\equiv \varphi$ and $\chi_2 :\equiv \neg\varphi$. Then

$$(\mathcal{W} \vdash (\varphi \vee \neg\varphi)) \qquad \text{(by axm:LEM)}$$

Therefore, it follows by Modus Ponens from the assumption, that

$$((\mathcal{W} \vdash \varphi) \vee (\mathcal{W} \vdash \neg\varphi)))$$

□

Definition (foundational theory $\mathcal{F}$)**.**
The intended foundational theory $\mathcal{F}$ is a complete, consistent extension of $\mathcal{T}$, together with the stronger semantic requirement of reflexivity defined below. The following choice notation records that requirement; it presupposes that a theory satisfying it is available in the chosen metatheory.

$$\mathcal{F} \overset{\text{def.}}{:=} (\epsilon(F : \text{Theory}),\ (F \text{ reflexive}))$$

For a consistent theory in the usual classical setting, Lindenbaum's lemma supplies a complete consistent extension. The displayed reflexivity condition is stronger: it also connects derivability with truth in a chosen interpretation. Its existence is an additional assumption of this framework, not a consequence of completion alone. One semantic reading is to take the sentences true in a fixed intended structure, provided that structure satisfies $\mathcal{T}$ and the relevant language is fixed.

11 Gödel–Löb polymodal logic

The following is a proposed hierarchy of theories and reflection rules, inspired by polymodal provability logic. The notation $\Box_n$ refers to provability at level n. The stage construction and its serial-property operator must be specified before this sketch yields a formal calculus or a soundness theorem. In particular, the next displayed rule is a proposal for relating levels, rather than the usual necessitation rule.

Inference rule 1 (proposed transfer between levels)**.**

$$\frac{\Theta_0 \vdash \Box_n \psi}{\Theta_{\max(n,1)} \vdash \psi} \qquad \text{(Effective Necessitation)}$$

The abbreviation "GLP" stands for "Gödel–Löb polymodal logic".

Notation 11.1 (provability)**.**
In this section, the unindexed box abbreviates provability in the fixed theory $\mathcal{T}$:

$$\Box\varphi \overset{\text{def.}}{\longleftrightarrow} (\mathcal{T} \vdash \varphi)$$

Here it is defined for *the* theory $\mathcal{T}$, which is a constant.

Definition 11.1 (proposed standardisation of a theory)**.**
For a theory Θ, the intended recursion has the following form. The successor-stage operator is schematic until "axiom generated by a serial property" has been defined, and the last line denotes the union of all ordinal stages in a metatheory capable of forming it:

$$\Theta_0 \overset{\text{def.}}{\coloneqq} \Theta$$

$$\Theta_{n+1} \overset{\text{def.}}{\coloneqq} \Theta_n \cup \left\{ A \;\middle|\; \begin{array}{c} A \text{ is an axiom generated by a serial property} \\ \text{of all terms in the language of } \Theta_n \end{array} \right\}$$

$$\Theta_\lambda \overset{\text{def.}}{\coloneqq} \bigcup_{\alpha<\lambda} \Theta_\alpha$$

$$\Theta_{\mathrm{On}} \overset{\text{def.}}{\coloneqq} \bigcup_{\alpha\in\mathrm{On}} \Theta_\alpha$$

(following the ideas of Japaridze, 1986)

12 Incompleteness of the theory $\mathcal{T}$

For the incompleteness theorem in this section, fix a consistent, effectively axiomatised theory at each stage to which the theorem is applied, with enough arithmetic to represent proofs and a standard provability predicate satisfying the required derivability conditions. A transfinite union from the preceding sketch need not retain these effective properties. Thus the theorem is applied stage by stage under its stated hypotheses.

Note 12.1 (where the incompleteness statement is proved)**.**
The companion notation makes the ambient metatheory explicit. For example, with $\mathrm{Con}(\mathcal{T})$ abbreviating $\neg\,\mathrm{Pr}_{\mathcal{T}}(\ulcorner\bot\urcorner)$, the formalised statement has the shape

$$\mathcal{E} \vdash \mathrm{Con}(\mathcal{T}) \longrightarrow \neg\,\mathrm{Pr}_{\mathcal{T}}(\ulcorner\mathrm{Con}(\mathcal{T})\urcorner).$$

Here $\mathrm{Pr}_{\mathcal{T}}$ is the specified arithmetised provability predicate, and corner brackets denote syntactic codes. The metatheory proves an implication; it need not prove its consistency premise. If $\mathcal{E} = \mathcal{T}$ is allowed by the formalisation, this still does not assert $\mathcal{T} \vdash \mathrm{Con}(\mathcal{T})$. The accompanying proof representation and derivability conditions must be checked for that choice.

Definition 12.1 (Arithmetised provability in this subsection)**.**
Fix a theory T and write

$$\Box\varphi := \mathrm{Pr}_T(\ulcorner\varphi\urcorner), \qquad \mathrm{Con}(T) := \neg\Box\bot.$$

Here $\Box\varphi$ is a sentence of arithmetic, whereas $T \vdash \varphi$ is the external assertion that a derivation exists. The notation $\Box_n$ uses $T = \mathcal{T}_n$. In the intended arithmetic interpretation, the chosen predicate represents finite T-proofs; its internal reasoning properties must also be verified, not inferred from the name "provability".

Proposition 12.1 (Direction of a relative-consistency argument)**.**
Let S, T be theories. Suppose a metatheoretically specified transformation sends every S-derivation of a sentence φ to a T-derivation of its translation φ^*, and suppose $T \vdash \bot^* \longrightarrow \bot$. Then inconsistency of S implies inconsistency of T; equivalently,

$$\mathrm{Con}(T) \longrightarrow \mathrm{Con}(S)$$

in that metatheory. To claim the displayed implication as a theorem inside a particular metatheory, the transformation and its correctness must themselves be formalized there.

Proof.
An S-proof of $\bot$ translates into a T-proof of $\bot^*$. Append the assumed implication and modus ponens to obtain a T-proof of $\bot$. Contraposition gives the consistency implication. □
This is a criterion for a relative-consistency proof, not a construction of an interpretation. In particular, to establish the motivating implication from inconsistency of ZFC to inconsistency of a proposed class theory, one needs a translation from ZFC into that theory with these properties. A translation in the reverse direction proves the reverse consistency implication.

Lemma 12.1 (Diagonal construction under a represented substitution operation)**.**
Work in a first-order arithmetical language with numerals $\overline{n}$, classical logic, and equality with substitution. Fix an effective coding of finite formulas by natural numbers, with decidable valid codes and an effective decoding procedure. Write $\#\theta$ for the external natural-number code of a formula θ, distinguishing it from the term $\overline{\#\theta}$ denoting that number inside arithmetic. Call

a code n eligible when it represents a formula $\theta(x)$ whose only possible free variable is x. For eligible n, let $d(n)$ be the code of the sentence $\theta(\overline{n})$. On all other inputs, including codes of formulas with other free variables, put $d(n) = 0$. This is an effective operation on finite syntax. Assume T has a formula $D(x, y)$, with no other free variables, such that for every eligible standard code n,

$$T \vdash \forall y \left(D(\overline{n}, y) \longleftrightarrow y = \overline{d(n)} \right).$$

This numeralwise exactness is a hypothesis on the chosen coding and its representation in T; it is not inferred just from having a numeral notation. Then for every formula $\psi(y)$ with no other free variables there is a sentence G such that

$$T \vdash G \longleftrightarrow \psi(\overline{\#G}).$$

Proof.
Rename bound variables as necessary, and choose x fresh for ψ. Use distinct names x, y, renaming the two free argument positions of D consistently if needed. Form the formula

$$\theta(x) := \exists y \left(D(x, y) \wedge \psi(y) \right).$$

Its only possible free variable is x. Put $k = \#\theta$ and let G be the sentence $\theta(\overline{k})$. By the definition of the external syntactic operation, $d(k) = \#G$. Numerals are closed, so this substitution introduces no free variables and captures none. The assumed representation property at the particular code k gives

$$T \vdash \forall y \left(D(\overline{k}, y) \longleftrightarrow y = \overline{\#G} \right).$$

Consequently, by first-order logic and substitution of equals,

$$\begin{aligned} T \vdash G &\longleftrightarrow \exists y \left(D(\overline{k}, y) \wedge \psi(y) \right) \\ &\longleftrightarrow \exists y \left(y = \overline{\#G} \wedge \psi(y) \right) \\ &\longleftrightarrow \psi(\overline{\#G}). \end{aligned}$$

For the final equivalence, a witness on the left yields the instance on the right by equality substitution; conversely the numeral $\overline{\#G}$ supplies a witness, using equality reflexivity. The first equivalence simply unfolds the sentence constructed above. Thus the construction proves the required equivalence inside T. It introduces no truth predicate: D describes a syntactic code operation, and ψ is the given arithmetic formula. □
For the provability notation used here, an occurrence $\mathrm{Pr}_T(\ulcorner \chi \urcorner)$ inside an arithmetic formula means $\mathrm{Pr}_T(\overline{\#\chi})$. For a sentence φ, apply the lemma to $\psi(y) = (\mathrm{Pr}_T(y) \longrightarrow \varphi)$, choosing y fresh for φ. It gives a sentence L with

$$T \vdash L \longleftrightarrow (\Box L \longrightarrow \varphi),$$

which is the fixed point used in the next proof. This application still requires the specified representation property for D in the language of T; quotation of class expressions alone does not provide it.

Theorem 12.1 (Löb's theorem with explicit hypotheses)**.**
Suppose T contains classical propositional logic, is closed under modus ponens, and its specified box satisfies, for sentences α, β:

1. if $T \vdash \alpha$, then $T \vdash \Box\alpha$;

2. $T \vdash \Box(\alpha \longrightarrow \beta) \longrightarrow (\Box\alpha \longrightarrow \Box\beta)$;

3. $T \vdash \Box\alpha \longrightarrow \Box\Box\alpha$.

Assume also the fixed-point lemma in the required language: for each sentence φ there is a sentence L such that $T \vdash L \longleftrightarrow (\Box L \longrightarrow \varphi)$. Then

$$T \vdash (\Box\varphi \longrightarrow \varphi) \quad \Longrightarrow \quad T \vdash \varphi.$$

Proof.
Choose the displayed fixed point. From $T \vdash L \longrightarrow (\Box L \longrightarrow \varphi)$, conditions 1 and 2 give

$$T \vdash \Box L \longrightarrow \Box(\Box L \longrightarrow \varphi).$$

Another application of condition 2 gives

$$T \vdash \Box L \longrightarrow (\Box\Box L \longrightarrow \Box\varphi).$$

Condition 3 and propositional reasoning therefore give $T \vdash \Box L \longrightarrow \Box\varphi$. Under the hypothesis $T \vdash \Box\varphi \longrightarrow \varphi$, it follows that $T \vdash \Box L \longrightarrow \varphi$. The reverse direction of the fixed-point equivalence now yields $T \vdash L$. Condition 1 yields $T \vdash \Box L$, and modus ponens yields $T \vdash \varphi$. All uses of condition 1 apply to theorems of T, not to formulas under an undischarged extra assumption. □
The proof isolates the syntactic ingredients; consistency and semantic soundness are not needed for Löb's implication itself. The preceding lemma constructs the fixed point when the diagonal code operation has the stated representation. Establishing that coding hypothesis and the three derivability conditions for a proposed theory such as $\mathcal{T}$ remains necessary; neither follows merely from quotation notation. The arithmetical setting and the second incompleteness theorem are discussed in Beklemishev, 2010, Section 2.

Corollary 12.1 (Gödel's second incompleteness theorem)**.**
If T is consistent and satisfies the preceding hypotheses, then

$$T \nvdash \mathrm{Con}(T).$$

In particular this applies to $T = \mathcal{T}_n$ whenever that stage has the stated proof predicate and fixed-point properties.

Proof.
A proof of $\mathrm{Con}(T)$ is a proof of $\Box\bot \longrightarrow \bot$. Löb's theorem with $\varphi = \bot$ would then give $T \vdash \bot$, contrary to consistency. □

Corollary 12.2 (Internal reflection is not an unrestricted axiom)**.**
Under the same hypotheses and consistency, T cannot prove $\Box\varphi \longrightarrow \varphi$ for any sentence φ unprovable in T. Conversely, if $T \vdash \varphi$, propositional logic proves that implication. Thus provability of this particular reflection instance is equivalent to provability of φ itself.

Proof.
The first implication is Löb's theorem. For the converse apply the propositional theorem $\varphi \longrightarrow (\Box\varphi \longrightarrow \varphi)$. □

Proposition 12.2 (An external extraction rule has different hypotheses)**.**
Suppose the intended standard interpretation makes $\mathrm{Pr}_T(\ulcorner\varphi\urcorner)$ true exactly when a finite T-proof of φ exists. If every theorem of T of this provability form is true in that interpretation, then the theorem-level rule

$$\frac{T \vdash \Box\varphi}{T \vdash \varphi}$$

is admissible. For a usual Σ_1 proof predicate, Σ_1-soundness is a sufficient condition for this hypothesis.

Proof.
The premise and assumed soundness give truth of the specified provability sentence. Its intended interpretation supplies an actual finite proof of φ, which is exactly the conclusion. This metatheoretic argument does not construct a T-proof of the internal implication $\Box\varphi \longrightarrow \varphi$ and does not permit applying the rule beneath arbitrary additional assumptions. Consistency alone is not the soundness hypothesis used in this argument. □

Conjecture 12.1 (generalisation of Artemov's serial consistency theorem from PA to $\mathcal{T}$)**.**

$$\Box_{n+1}\neg\Box_n\bot$$

(Artemov, 2024) This is a proposed statement about the next theory in the hierarchy. A selector proof for a sequence of instances must be distinguished from a proof of the single arithmetised consistency sentence. Establishing the connection with this successor-stage claim is part of the conjecture.

Note 12.2 (structural rules convention)**.**
The rules in this development are intended to be structural: admissible substitutions give further instances of a rule. "Admissible" includes the typing, freshness, and capture-avoidance conditions of that rule.

Inference rule 2 (deduction)**.**
For the ordinary predicate-calculus rules, the deduction theorem applies when no generalisation in the derivation uses a variable free in the assumption being discharged. Any added reflection or serial rule requires a separate check that this transformation remains valid.

$$\frac{\mathcal{W}, \Gamma, \psi \vdash \varphi}{\mathcal{W}, \Gamma \vdash (\psi \longrightarrow \varphi)} \qquad \text{(rule:Deduction)}$$

Proposition 12.3 (Deduction for the specified predicate-calculus rules)**.**
Consider derivations using assumptions, logical and nonlogical axioms, modus ponens, generalisation, and the abstraction rules B1 and B2 above. Suppose every generalisation or abstraction step has its stated freshness conditions, including freshness for the assumption ψ to be discharged. Then a derivation of φ from $\Gamma \cup \{\psi\}$ can be transformed into a derivation of $\psi \longrightarrow \varphi$ from Γ. Adding witness axioms of the preceding total extension does not change this conclusion; adding arbitrary new inference rules is not covered.

Proof.
The propositional rewrites used below are consequences of A1–A5 and modus ponens (implication composition and permutation, and conjunction introduction and elimination); they do not use the deduction theorem being proved. Induct on the given derivation, transforming a line χ into

$\psi \longrightarrow \chi$. If χ is an axiom or an assumption in Γ, retain it and apply A1, namely $\chi \longrightarrow (\psi \longrightarrow \chi)$. If the line is ψ itself, derive $\psi \longrightarrow \psi$ from A1 and A2: A1 gives $\psi \longrightarrow ((\psi \longrightarrow \psi) \longrightarrow \psi)$ and $\psi \longrightarrow (\psi \longrightarrow \psi)$; A2 combines these to the result. For a modus-ponens step from χ and $\chi \longrightarrow \eta$, the induction hypotheses and A2 give $\psi \longrightarrow \eta$.
For generalisation from χ to $\forall x\, \chi$, the induction hypothesis gives $\psi \longrightarrow \chi$. Rule B1 gives $\psi \longrightarrow \forall x\, \chi$ because x is free in neither ψ nor Γ.
For a B1 step from $\chi \longrightarrow \eta$ to $\chi \longrightarrow \forall x\, \eta$, the transformed premise is $\psi \longrightarrow (\chi \longrightarrow \eta)$. Propositional logic, using conjunction introduction and elimination, rewrites this as $(\psi \wedge \chi) \longrightarrow \eta$. The original side conditions make x fresh for χ, ψ, Γ, so B1 gives $(\psi \wedge \chi) \longrightarrow \forall x\, \eta$. Propositional logic rewrites this as $\psi \longrightarrow (\chi \longrightarrow \forall x\, \eta)$.
For a B2 step from $\chi \longrightarrow \eta$ to $(\exists x\, \chi) \longrightarrow \eta$, the transformed premise $\psi \longrightarrow (\chi \longrightarrow \eta)$ propositionally implies $\chi \longrightarrow (\psi \longrightarrow \eta)$. The B2 side conditions make x fresh for η, ψ, Γ, so B2 yields $(\exists x\, \chi) \longrightarrow (\psi \longrightarrow \eta)$. Reordering the two antecedents gives $\psi \longrightarrow ((\exists x\, \chi) \longrightarrow \eta)$. These are all allowed inference steps. Witness axioms use the same axiom case as any other axiom; an additional inference rule would require another induction case, which is why no assertion about reflection or serial rules follows from this proof. □
For a finite context Γ, and subject to the same restrictions,

$$\frac{\mathcal{W}, \Gamma \vdash \varphi}{\mathcal{W} \vdash \bigwedge \Gamma \longrightarrow \varphi} \qquad \text{(rule:Full Deduction)}$$

Inference rule 3 (effective necessitation)**.**
The intended rule represents a given certified derivation inside $\mathcal{T}$. It requires a representation of the theory $\mathcal{W}$ and of the particular proof, together with a proof-checking relation available in $\mathcal{T}$. More precisely, for each actual certified derivation p of ψ in the represented fragment, require the numeralwise verification $\mathcal{T} \vdash \mathrm{Prf}_{\mathcal{W}}(\overline{\#p}, \overline{\#\psi})$. Read the represented assertion $(\mathcal{W} \vdash \psi)$ in the conclusion as $\exists q\ \mathrm{Prf}_{\mathcal{W}}(q, \overline{\#\psi})$, or require $\mathcal{T}$ to prove its equivalence to that formula. Existential introduction with the certificate numeral then justifies the rule; merely having a formula named "proof checking" does not suffice:

$$\frac{\mathcal{W} \vdash \psi}{\mathcal{T} \vdash (\mathcal{W} \vdash \psi)} \qquad \text{(rule:Effective Necessitation)}$$

The author also proposes taking $\mathcal{W} = \mathcal{F}$. For that instance, the representation requirement applies to the certified fragment being used; non-enumerability prevents treating all $\mathcal{F}$-derivations as the output of one effective proof enumeration. The extension to arbitrary ideal derivations remains an additional requirement of the framework.

Bad inference rule (not an effective rule for the fixed theory)**.**
The following unrestricted passage from truth to effective provability is excluded from the effective calculus. Unlike the preceding rule, its premise does not provide a certified derivation that the implementation can encode.

$$\frac{\psi}{\mathcal{T} \vdash \psi} \qquad \text{(badrule:"Effective Omniscience")}$$

Under the arithmetic and consistency hypotheses above, the fixed effective theory $\mathcal{T}$ cannot capture all truth in the intended standard interpretation. This is not a claim that adjoining true sentences must produce a contradiction: starting with a sound theory, adjoining all such sentences preserves soundness and consistency, but in this setting the resulting theory is no longer effectively axiomatised. That enlarged theory must be distinguished from the fixed $\mathcal{T}$ in the displayed rule.

Note 12.3 (philosophical)**.**
A computer stores a representation of a proof and checks it against specified rules. The connection between that certificate and truth depends on the interpretation and soundness of those rules. Keeping the certificate and its semantic interpretation distinct is the point of the effective/ideal comparison.

Bad axiom 1 ("cotransitivity")**.**
It is not possible to have an "axiom" scheme like

$$\Box\Box_n\psi \longrightarrow \Box_n\psi, \qquad \text{(badaxm:"cotransitivity")}$$

as an unrestricted scheme in a consistent effective theory whose theorems of provability form are true in the intended interpretation. The exact obstruction is worth distinguishing from consistency alone. For the self-instance, Löb's theorem applied to the sentence $\Box\psi$ turns a proof of $\Box\Box\psi \longrightarrow \Box\psi$ into a proof of $\Box\psi$. Taking $\psi = \bot$ contradicts provability-soundness for a consistent theory. Bare consistency does not exclude a proof of $\Box\bot$: that is a sentence asserting inconsistency, not itself an actual proof of $\bot$. Applications to different indices require the interpretation of each box and the relation between the stages to be fixed.

Proposition 12.4 (Consistency does not imply provability-soundness)**.**
Let S be a consistent, effectively axiomatised classical arithmetic theory satisfying the hypotheses of second incompleteness, and put $T = S + \neg\,\mathrm{Con}(S)$. Use standard proof predicates such that inclusion of S-proofs among T-proofs is formalised in S:

$$S \vdash \mathrm{Pr}_S(\ulcorner\bot\urcorner) \longrightarrow \mathrm{Pr}_T(\ulcorner\bot\urcorner).$$

Assume the usual derivability conditions for the predicate of T. Then T is consistent but $T \vdash \mathrm{Pr}_T(\ulcorner\bot\urcorner)$. Indeed, writing $\Box_T$ for this predicate, for every sentence ψ,

$$T \vdash \Box_T\Box_T\psi \longrightarrow \Box_T\psi.$$

Thus consistency alone cannot rule out all these cotransitivity instances.

Proof.
If T were inconsistent, the ordinary deduction theorem would give $S \vdash \neg\,\mathrm{Con}(S) \longrightarrow \bot$. Classical logic would then give $S \vdash \mathrm{Con}(S)$, contrary to second incompleteness and consistency of S. Hence T is consistent. Its added axiom, with classical double-negation elimination, gives $T \vdash \mathrm{Pr}_S(\ulcorner\bot\urcorner)$. The specified proof-inclusion implication gives $T \vdash \Box_T\bot$. For any ψ, the propositional theorem $\bot \longrightarrow \psi$ and conditions 1 and 2 yield $T \vdash \Box_T\bot \longrightarrow \Box_T\psi$. Therefore $T \vdash \Box_T\psi$, and propositional logic gives the displayed cotransitivity implication. Its conclusion is already a theorem of T; no reflection or extraction of an actual proof of ψ was used. □
Externally, consistency of T makes its asserted provability of contradiction false in the intended interpretation. The example concerns ordinary deductive consistency, not credibility. The formalised inclusion hypothesis is justified in usual arithmetical presentations by viewing a finite S-derivation as a T-derivation; it must be checked for a different choice of proof coding.

13 Completeness of the theory $\mathcal{F}$

We return to the ideal semantic setting. In this section, "validity" means truth in the fixed intended interpretation, not truth in every structure. When a formula φ appears on the truth side of an implication, read it as its metatheoretic assertion of truth. Quantification over Fm is restricted to sentences, as above. This convention explains the compact displays; a fully internal truth predicate would require its own language, typing, and consistency analysis.
The iterated displays below record the author's proposed treatment of metalevels. Their internal readings are additional principles to be justified for that treatment; they do not follow merely by suppressing quotation marks in an external semantic equivalence.

Proposition 13.1 (The obstruction to unrestricted internal truth)**.**
Let T be a classical theory in a language containing a unary predicate Tr and the sentence codes needed for the diagonal construction proved above. Assume that construction applies also to formulas containing Tr. If, for every sentence θ of this same language,

$$T \vdash \mathrm{Tr}(\overline{\#\theta}) \longleftrightarrow \theta,$$

then T is inconsistent. No effective enumeration of the axioms of T is required for this conclusion.

Proof.
Apply the diagonal construction to $\psi(y) = \neg\,\mathrm{Tr}(y)$. It gives a sentence L with

$$T \vdash L \longleftrightarrow \neg\,\mathrm{Tr}(\overline{\#L}).$$

The truth biconditional for L gives $T \vdash \mathrm{Tr}(\overline{\#L}) \longleftrightarrow L$. Substitution of these propositional equivalents yields $T \vdash L \longleftrightarrow \neg L$. Its forward implication gives $T \vdash \neg L$ by propositional reasoning: an assumption L would yield both L and $\neg L$. Its reverse implication then gives $T \vdash L$, hence $T \vdash \bot$. □
The proposition concerns an internal predicate that applies to its own full language and the specified diagonal coding. It does not apply to truth asserted externally for a fixed object language, or to a higher-level truth predicate restricted to a lower-level language. In those settings the self-referential sentence used in the proof is outside the domain of the relevant truth scheme. Non-enumerability of $\mathcal{F}$ alone does not resolve this particular obstruction; the interpretation and metalevel of each evaluation and quotation must supply the boundary.

Definition 13.1 (credible theory)**.**
Let $\mathcal{W}$ be a theory. It is called *credible*, if the derivability of any formula in it implies the validity of the formula.

$$(\mathcal{W}\text{ credible}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{W} \vdash \varphi) \longrightarrow \varphi))$$

Definition 13.2 (omniscient theory)**.**
Let $\mathcal{W}$ be a theory. It is *omniscient* if every sentence true in the intended interpretation is derivable in $\mathcal{W}$.

$$(\mathcal{W}\text{ omniscient}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi : \mathrm{Fm}),\ (\varphi \longrightarrow (\mathcal{W} \vdash \varphi)))$$

Definition 13.3 (reflexive theory)**.**
Let $\mathcal{W}$ be a theory. It is called *reflexive*, if the derivability of any formula in it is equivalent to the validity of the formula.

$$(\mathcal{W}\text{ reflexive}) \overset{\text{def.}}{\longleftrightarrow} (\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{W} \vdash \varphi) \longleftrightarrow \varphi))$$

The definitions immediately give:

$$(\mathcal{W} \text{ reflexive}) \longleftrightarrow ((\mathcal{W} \text{ credible}) \wedge (\mathcal{W} \text{ omniscient})).$$

Axiom 4 ($\mathcal{F}$ is reflexive)**.**

$$(\mathcal{F} \text{ reflexive})$$

The following statement holds because of the properties of $\mathcal{F}$:

$$((\mathcal{F} \vdash \varphi) \longleftrightarrow \varphi) \qquad \text{(axm:reflexivity)}$$

$$\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{F} \vdash \varphi) \longleftrightarrow \varphi) \qquad \text{(axm:metareflexivityorig)}$$

$$\forall(\varphi : \mathrm{Fm}),\ (\mathcal{F} \vdash ((\mathcal{F} \vdash \varphi) \longleftrightarrow \varphi)) \qquad \text{(axm:metareflexivity1)}$$

$$\mathcal{F} \vdash (\forall(\varphi : \mathrm{Fm}),\ ((\mathcal{F} \vdash \varphi) \longleftrightarrow \varphi)) \qquad \text{(axm:metareflexivity2)}$$

Note 13.1 (a nonstandard writing convention)**.**
The author's predicate $\vdash$ differs from the traditional notation: it is higher-order, taking a theory (a set of formulas), a context (a list of formulas), and a formula as arguments.

$$\mathrm{Der} : \mathrm{Theory} \times \mathrm{Ctx} \times \mathrm{Fm} \to \mathrm{Fm}$$

The function isFree treats quotation specially: when checking whether a variable occurs free, it increases the variable's metalevel as it enters a quotation. This convention avoids writing the encoding and decoding of syntactic objects explicitly and simplifies the implementation.

Under the stated reflexivity convention, provability in $\mathcal{F}$ and truth coincide for the sentences in question. The notation $(\mathcal{F} \vdash \dots)$ still records a syntactic assertion about derivability and makes the metalevel explicit. This distinction remains useful even when the intended semantics validates their equivalence.

Inference rule 4 (creed)**.**
The context must be empty: only "$\mathcal{F}$" appears on the left-hand side of "$\vdash$". The rule applies only to the constant "$\mathcal{F}$".

$$\frac{\mathcal{F} \vdash \psi}{\psi} \qquad \text{(rule:Credo)}$$

$$\forall(\psi : \mathrm{Fm}),\ ((\mathcal{F} \vdash \psi) \longrightarrow \psi) \qquad \text{(axm:Credo)}$$

$$\forall(\psi : \mathrm{Fm}),\ (\mathcal{F} \vdash (\mathcal{F} \vdash \psi)) \longrightarrow (\mathcal{F} \vdash \psi) \qquad \text{(rule:MetaCredo)}$$

Theorem 13.1 (credibility implies consistency)**.**

$$(\mathcal{W} \text{ credible}) \longrightarrow (\mathcal{W} \text{ consistent}).$$

Proof. Assume that $\mathcal{W}$ is credible. If $\mathcal{W} \vdash \bot$, credibility applied to the sentence $\bot$ would make $\bot$ true in the intended interpretation. Since $\bot$ is false there, $\mathcal{W} \nvdash \bot$. This is precisely the definition of consistency. The argument uses credibility of $\mathcal{W}$ itself; it does not apply a rule reserved for $\mathcal{F}$ to an arbitrary theory. □

Remark 13.1 (co-necessitation and the two contexts)**.**
The companion rule 4.03 retains a context inside the quoted sequent:

$$\frac{\mathcal{F}, [] \vdash \langle \mathcal{F}, \Gamma \vdash \varphi \rangle}{\mathcal{F}, \Gamma \vdash \varphi}.$$

Here $\langle \mathcal{F}, \Gamma \vdash \varphi \rangle$ denotes the formula representing that particular sequent. The *outer* derivation has empty context; Γ is the *inner* context being represented. Consequently this rule does not remove the empty-context restriction on creed and does not assert φ independently of Γ.
Its intended justification is to instantiate soundness of $\mathcal{F}$ at the quoted sequent, then use the specified evaluation law

$$\mathrm{eval}_M(\langle \mathcal{F}, \Gamma \vdash \varphi \rangle) \longleftrightarrow (\mathcal{F}, \Gamma \vdash \varphi).$$

This argument is conditional on the quotation being in the domain of the chosen valuation, with all parameters interpreted. In a stratified language, soundness of a lower-level language cannot automatically be applied to a new formula about its own truth or provability.
For an arbitrary effective theory, admissibility of an external rule that extracts a derivation must be distinguished from an implication proved inside that theory. Under the usual derivability conditions, an unrestricted internal self-reflection scheme has the familiar Löb obstruction; this observation does not forbid the syntactic representation of a theory's own proofs. Soundness assumptions sufficient for an external extraction argument must be stated for its particular proof predicate.

Inference rule 5 (omniscience)**.**

$$\frac{\psi}{\mathcal{F} \vdash \psi} \qquad \text{(rule:Omniscience)}$$

$$\forall(\psi : \mathrm{Fm}),\ (\psi \longrightarrow (\mathcal{F} \vdash \psi)) \qquad \text{(axm:Omniscience)}$$

$$\forall(\psi : \mathrm{Fm}),\ ((\mathcal{F} \vdash \psi) \longrightarrow (\mathcal{F} \vdash (\mathcal{F} \vdash \psi))) \qquad \text{(axm:MetaOmniscience)}$$

Lemma 13.1 (credibility and omniscience imply the disjunction property)**.**

$$((\mathcal{W} \text{ credible}) \wedge (\mathcal{W} \text{ omniscient})) \longrightarrow (\mathcal{W} \text{ disjunctive})$$

Proof. Fix $(\mathcal{W} : \mathrm{Theory})$. By definition,

$$(\mathcal{W} \text{ disjunctive}) \longleftrightarrow (\forall(\chi_1 : \mathrm{Fm}),\ \forall(\chi_2 : \mathrm{Fm}),\ ((\mathcal{W} \vdash (\chi_1 \vee \chi_2)) \longrightarrow ((\mathcal{W} \vdash \chi_1) \vee (\mathcal{W} \vdash \chi_2))))$$

So fix $(\chi_1 : \mathrm{Fm})$, $(\chi_2 : \mathrm{Fm})$, and assume $(\mathcal{W} \vdash (\chi_1 \vee \chi_2))$.
Using the omniscience property and specification, it is shown for these formulas that:

$$(\chi_1 \longrightarrow (\mathcal{W} \vdash \chi_1)),$$

$$(\chi_2 \longrightarrow (\mathcal{W} \vdash \chi_2)).$$

Therefore, via several propositional axioms, the following holds:

$$(\chi_1 \vee \chi_2) \longrightarrow ((\mathcal{W} \vdash \chi_1) \vee (\mathcal{W} \vdash \chi_2)).$$

By virtue of credibility rule,

$$(\mathcal{W} \vdash (\chi_1 \vee \chi_2)) \longrightarrow (\chi_1 \vee \chi_2).$$

Finally, by transitivity of implication,

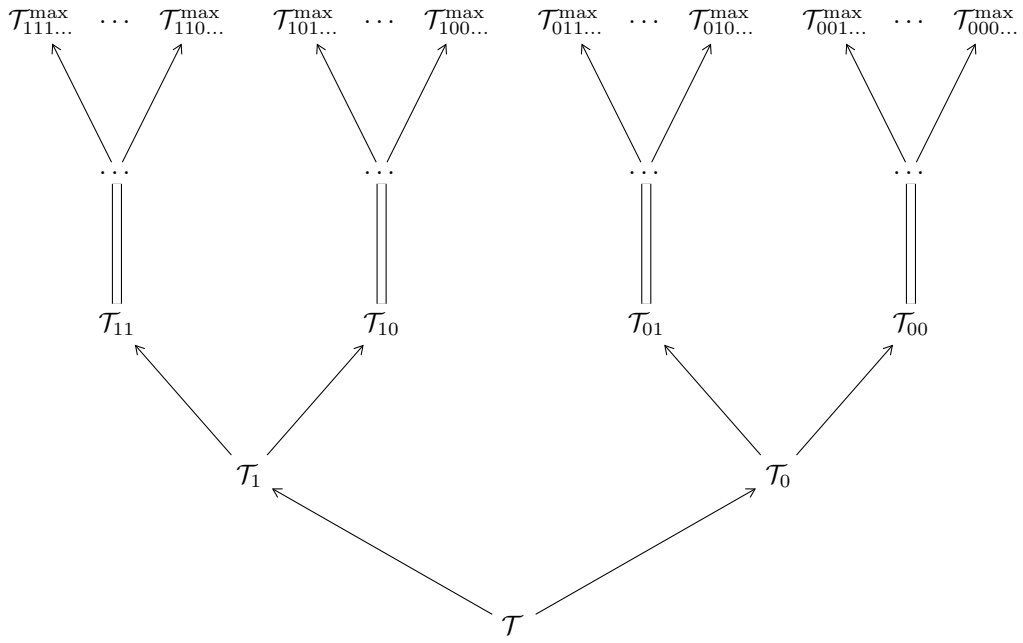
$$(\mathcal{W} \vdash (\chi_1 \vee \chi_2)) \longrightarrow ((\mathcal{W} \vdash \chi_1) \vee (\mathcal{W} \vdash \chi_2)).$$

□

Theorem 13.2 (reflexivity implies consistency and completeness)**.**

$$(\mathcal{W} \text{ reflexive}) \longrightarrow ((\mathcal{W} \text{ consistent}) \wedge (\mathcal{W} \text{ complete}))$$

Proof. Reflexivity gives credibility and omniscience. Credibility implies consistency, and credibility together with omniscience gives the disjunction property. The disjunction property, combined with classical excluded middle, gives completeness. □


$\mathcal{T}^{\max}_{111\ldots}$ $\cdots$ $\mathcal{T}^{\max}_{110\ldots}$
$\mathcal{T}^{\max}_{101\ldots}$ $\cdots$ $\mathcal{T}^{\max}_{100\ldots}$
$\mathcal{T}^{\max}_{011\ldots}$ $\cdots$ $\mathcal{T}^{\max}_{010\ldots}$
$\mathcal{T}^{\max}_{001\ldots}$ $\cdots$ $\mathcal{T}^{\max}_{000\ldots}$
$\mathcal{T}_{11}$
$\mathcal{T}_{10}$
$\mathcal{T}_{01}$
$\mathcal{T}_{00}$
$\mathcal{T}_1$
$\mathcal{T}_0$
$\mathcal{T}$

Chapter

Classes

14 Setting

In the author's motivating picture, sets are "terminal" objects: they are the objects on which class membership is grounded. Classes are properties of sets. The formal theory expresses this picture through membership, comprehension, and the distinction between sets and proper classes.
Class theory lets us describe a collection before deciding whether it is a set. A class is determined by its set members; a class is itself a set when it can occur as a member of another class. Thus a definition by class comprehension need not first supply an ambient set. The separate task of proving sethood can be postponed until a construction requires it.
The distinction between Morse–Kelley (MK) and von Neumann–Bernays–Gödel (NBG) comprehension concerns the formulas admitted in that construction. In MK, those formulas may quantify over classes. In NBG comprehension, quantified variables range over sets, although class parameters are allowed. In the one-sorted notation used here, set quantifiers are expressed by restriction to V. A restriction to another class also forces its bound variable to be a set, since only sets can be members of classes.

How to read the axioms. This section develops class comprehension, elementary class operations, and several additional principles governing sethood. The displayed axioms specify the local assumptions; they are not all consequences of comprehension alone. In particular, heredity, powerset, regularity, and limitation of size are stated separately so that their uses remain visible.
The discussion of functions and families also records a local evaluation convention with default value V. That convention should be read with its own definitions, rather than silently substituted into sections using empty-class defaults or partial evaluation.
Types provide a second way to describe which objects a construction accepts. A predicate on classes can serve as a type even when its instances cannot be collected into a class. The section on types develops this distinction between a class of sets and a predicate that may also hold of proper classes.

15 Axioms and definitions of class theory

Constructor 15.1 (membership relation)**.**
If A and B are terms, then $(A \in B)$ is a formula.

Constructor 15.2 (subclass relation)**.**
If A and B are terms, then $(A \subseteq B)$ is a formula.

Definition 15.1 (subclass relation)**.**
Let A and B be terms, and x a variable free in neither A nor B, then

$$(A \subseteq B) \overset{\text{def.}}{\longleftrightarrow} (\forall x \in A,\ x \in B) \tag{def:subclass}$$

An equality experiment. The next display records an attempt to define equality by membership profiles in a language without primitive equality. Its scope must be kept separate from the extensional identity used below: two proper classes have no containing classes, so the displayed membership-profile condition alone does not distinguish them. A language using this experiment needs an additional account of identity before it can support ordinary class equality.

Constructor 15.3 (equality)**.**
If A and B are terms, then $(A = B)$ is a formula.

Definition 15.2 (equality)**.**
This proposed definition concerns first-order logic without primitive equality.
If Y is free in neither A nor B, then:

$$(A = B) \overset{\text{def.}}{\longleftrightarrow} (\forall Y,\ A \in Y \longleftrightarrow B \in Y) \tag{def:equality}$$

15.1 Extensionality

Axiom 5 (extensionality)**.**

$$(A = B) \longleftrightarrow ((A \subseteq B) \wedge (B \subseteq A)) \tag{axm:extensionality}$$

Axiom (extensionality)**.**
Equivalently, classes are equal exactly when they have the same members. The variable X must be free in neither term A nor term B.

$$(A = B) \longleftrightarrow (\forall X,\ (X \in A \longleftrightarrow X \in B)) \tag{axm:extensionality2}$$

Constructor 15.4 (setness predicate)**.**
If A is a term, then $(A \text{ set})$ is a formula.

Definition 15.3 (setness predicate)**.**
If Y is a variable, which is not free in term X, then:

$$(X \text{ set}) \overset{\text{def.}}{\longleftrightarrow} (\exists Y,\ X \in Y) \tag{def:isSet}$$

15.2 Class comprehension

Notation 15.1 (bound term)**.**

$$t_x \overset{\text{notation}}{:\equiv} (x, t)$$

Notation 15.2 (bound formula)**.**
$(\varphi : \text{Fm})$,$(x : \text{Variable})$

$$\varphi_x \overset{\text{notation}}{:\equiv} (x, \varphi)$$

Notation 15.3 (restricted binding)**.**
A pair of a bound formula φ_x and a term t is called a restricted (by a formula φ) binding of a term t.

$$(\varphi_x, t)$$

A pair of a bound formula φ_x and a formula ψ is called a restricted (by a formula φ) binding of a formula ψ.

$$(\varphi_x, \psi)$$

$$(\forall \varphi_{y::\vec{x}},\ \psi) \stackrel{\text{def.}}{:\equiv} (\forall y, (\forall \varphi_{\vec{x}},\ \psi))$$

Definition 15.4 (restricted universal quantifier)**.**

$$(\forall \varphi_x,\ \psi) \stackrel{\text{def.}}{\longleftrightarrow} (\forall x,\ \varphi \longrightarrow \psi)$$

Definition 15.5 (restricted existential quantifier)**.**

$$(\exists \varphi_x,\ \psi) \stackrel{\text{def.}}{\longleftrightarrow} (\exists x,\ \varphi \wedge \psi)$$

Definition 15.6 (informal)**.**
A class is an arbitrary collection of sets.

Note 15.1 (definability and comprehension)**.**
A comprehension term describes a class using a formula admitted by the chosen schema. What counts as definable depends on the language, permitted parameters, and surrounding model. A descriptor may specify an object without providing a comprehension formula in that same language.

Note 15.2 (constructibility)**.**
The constructible universe L is a particular hierarchy of sets. The axiom $\mathrm{V} = L$ should not be read as saying that every set is definable without parameters. Constructibility and definability are distinct notions; this section makes no assumption that $\mathrm{V} = L$.

Constructor 15.5 (class comprehension)**.**
If φ is a formula, $\vec{x}$ is a list of variables(they are used for classes), t is a class term, then $\{\, t \mid^{\vec{x}} \varphi \,\}$ is a class term.

Notation 15.4 (clearer notation)**.**

$$\{\, t \mid^{\vec{x}} \varphi \,\} \stackrel{\text{notation}}{:\equiv} \{\, t_{\vec{x}} \mid \varphi_{\vec{x}} \,\}$$

If the list of variables is empty, generalized comprehension gives the following singleton expression. As always, class comprehension retains only set members; thus $\{r\} = \varnothing$ when r is a proper class:

$$\{r\} = \{\, p \mid (p = r) \wedge \top \,\} = \{\, r \mid^{[]} \top \,\},$$

For a set r, it also gives a set-level conditional:

$$\{\, r \mid^{[]} \varphi \,\} = \{\, p \mid (p = r) \wedge \varphi \,\} = (\text{if } \varphi \text{ then } \{r\} \text{ else } \varnothing \text{ fi})$$

$$\bigcup \{\, r \mid^{[]} \varphi \,\} = \bigcup (\text{if } \varphi \text{ then } \{r\} \text{ else } \varnothing \text{ fi}) = (\text{if } \varphi \text{ then } r \text{ else } \varnothing \text{ fi})$$

Note 15.3 (MK and NBG comprehension)**.**
In MK, φ may contain unrestricted class quantifiers. In the NBG fragment, its internal quantifiers must range over sets; class parameters may still occur. These restrictions concern the comprehension schema, not the availability of class quantifiers elsewhere in the language.

Notation 15.5 (list of variables)**.**
Assume $v_1, \dots, v_n$ are variables, assume $T_1, \dots, T_n$ are arbitrary types. Then

$$(\forall(\vec{v} : \vec{T}),\ \varphi) \overset{\text{def.}}{\longleftrightarrow} (\forall(v_1 : T_1),\ \dots\ , \forall(v_n : T_n),\ \varphi)$$

$$(\exists(\vec{v} : \vec{T}),\ \varphi) \overset{\text{def.}}{\longleftrightarrow} (\exists(v_1 : T_1),\ \dots\ , \exists(v_n : T_n),\ \varphi)$$

Definition 15.7 (universal closure)**.**
There is an order on a set of variables Var.
List the free variables of φ as $v_1, \dots, v_n$ in that order.
Then define the closure via the formula

$$\text{closure}(\varphi) \overset{\text{def.}}{\longleftrightarrow} (\forall v_1, \dots, \forall v_n, \varphi)$$

$$((\vec{\forall})(\varphi) \overset{\text{def.}}{\longleftrightarrow} (\forall v_1, \dots, \forall v_n, \varphi))$$

Also the closure can be easier to use as a part of the definition of a formula.
Every displayed axiom schema is understood with its universal closure.
This convention only specifies the binding of free variables. The choice of an axiom collection $\mathcal{F}$ remains substantive: different collections can prove different statements. Whether a particular collection is recursively axiomatizable depends on its definition; no nonaxiomatizability claim follows from the closure notation itself.

Notation 15.6 (the case if a term is a variable)**.**
If a term is just a variable, then there is no need to specify it twice:

$$\{\, x \mid \varphi \,\} \overset{\text{notation}}{:\equiv} \{\, x \mid^{x} \varphi \,\}. \qquad \text{(notation:cccxx)}$$

Axiom 6 (class comprehension)**.**
For any formula φ admitted by the chosen comprehension schema and any variable w,

$$(\vec{\forall})(\forall(w : \text{Class}),\ ((w \in \{\, w \mid \varphi \,\}) \longleftrightarrow ((w \text{ set}) \wedge \varphi))) \qquad \text{(axm:varComprehension)}$$

For a small type whose interpretation is a set, the typing condition already ensures sethood. In that case the explicit sethood guard can be omitted:

$$(\vec{\forall})(\forall(W \text{ isSmallType})\forall(w : W),\ ((w \in \{\, w : W \mid \varphi \,\}) \longleftrightarrow \varphi)) \qquad \text{(axm:varSmallTypeComprehension)}$$

Notation 15.7 (omitting types)**.**
Due to lack of space in class–builder notation, it may be written just $\vec{x}$ instead of $(\vec{x} : \vec{X})$, if it is easy to restore the types from the context.

Axiom 7 (class term comprehension)**.**
Let s, t be class terms and φ an admitted formula. Rename the distinct bound variables $\vec{x}$, if necessary, so that $\text{FV}(t) \cap \{\vec{x}\} = \varnothing$. Then

$$(\vec{\forall})((t \in \{\, s \mid^{\vec{x}} \varphi \,\}) \longleftrightarrow ((t \text{ set}) \wedge (\exists(\vec{x} : \vec{X}),\ (t = s) \wedge \varphi))) \qquad \text{(axm:termComprehension)}$$

It may be called "class–builder notation". The terms s and t may share free parameters; disjointness of their free-variable sets is neither needed nor a substitute for freshness of the actual binders. Every substitution in this schema avoids capture. Schema admissibility applies to the complete defining formula after expanding generalized comprehension and defined notation. In particular, the witnesses in $\exists(\vec{x} : \vec{X})$ must be set-quantified in an NBG instance, unless the formula is separately shown equivalent to an admitted one. A sethood guard on the collected value does not remove quantification over arbitrary classes.

Notation 15.8 (class comprehension with a term's evaluation agreement)**.**
If a term is an evaluation of a function f (technically, it is an arbitrary term f), and if the argument is a variable, then it is agreed to be restored from the argument in the following way:

$$\{\, f(x) \mid \varphi \,\} \overset{\text{notation}}{:\equiv} \{\, f(x) \mid^{x} \varphi \,\}.$$

Note 15.4 (restoring one version of comprehension from the other)**.**
If necessary, the class constructor can be restored from its truncated version via

$$\{\, t \mid^{x} \varphi \,\} \coloneqq \{\, w \mid \exists(x : X),\ (w = t) \wedge \varphi \,\}.$$

$$\{\, t \mid^{\vec{x}} \varphi \,\} \coloneqq \left\{\, w \;\middle|\; \exists(\vec{x} : \vec{X}),\ (w = t) \wedge \varphi \,\right\}.$$

Here w is fresh for the bound-variable list, the terms, the formula, and the type expressions. Renaming bound variables before expansion keeps the same parameters free on both sides.

$$\{\, x \mid^{x} \varphi \,\} = \{\, w \mid \exists x,\ (w = x) \wedge \varphi \,\} = \{\, w \mid \varphi[w/x] \,\} = \{\, x \mid \varphi \,\}.$$

Here w is fresh, substitution is capture-avoiding, and the final equality renames the bound variable. Keeping these conditions explicit prevents an inner quantifier from accidentally capturing the comprehension variable.
Implementation principle. A proved equality with a preferred direction can motivate a simplification rule in a proof assistant. Promoting it to definitional equality is a further design decision: the resulting conversion procedure must still have the properties required by the kernel.

Theorem 15.1 (example of changing the variables)**.**
Assume that X and Y are fresh for φ except at the indicated substitution positions, and that both substitutions avoid capture. Then:

$$(Y \in \{\, X \mid \varphi[X/W] \,\}) \longleftrightarrow ((Y \text{ set}) \wedge (\varphi[Y/W])).$$

Proof.
($\longrightarrow$)
Assume $Y \in \{\, X \mid \varphi[X/W] \,\}$.

$$(Y \in \{\, X \mid \varphi[X/W] \,\}) \implies ((Y \text{ set}) \quad \& \quad \varphi[X/W][Y/X])$$

The substitution composition law gives $\varphi[X/W][Y/X] = \varphi[Y/W]$. Substituting this into the comprehension axiom proves the displayed equivalence, including its reverse implication. □

15.3 Empty class

Constructor 15.6 (empty class)**.**
The empty set symbol $\varnothing$ is a constant closed term.

$$(\varnothing : \mathrm{Class}) \tag{constr:emptyclass}$$

Definition 15.8 (empty class)**.**

$$\varnothing \coloneqq^{\mathrm{def.}} \{\, x \mid \bot \,\} \tag{def:emptyclass}$$

Definition 15.9 (empty set predicate)**.**

$$((A \text{ empty}) \overset{\mathrm{def.}}{\longleftrightarrow} (A = \varnothing)) \tag{def:empty}$$

Definition 15.10 (nonempty set predicate)**.**

$$(A \text{ nonempty}) \overset{\mathrm{def.}}{\longleftrightarrow} (\neg(A \text{ empty})) \tag{def:nonempty}$$

Axiom 8 (the empty class is a set)**.**

$$(\varnothing \text{ set}) \tag{axm:emptyset}$$

15.4 Universal class

Metadefinition 15.1 (constant)**.**
A functional symbol of valency zero is called a *constant*.

(A constant is always a closed term, due to vacuous argument.)

Constructor 15.7 (class constant V)**.**
Let the symbol V be a constant of the language of the class theory being discussed.

Definition 15.11 (universal class)**.**

$$\mathrm{V} \coloneqq^{\mathrm{def.}} \{\, x \mid \top \,\} \tag{def:universal class}$$

In the local convention used in the next subsection, V serves as a distinguished default for an argument outside a family's domain. The resulting expression still denotes a class; the default records a domain failure rather than a third truth value.

Definition 15.12 (default nonset)**.**

$$\mathrm{nonset} \coloneqq^{\mathrm{def.}} \mathrm{V} \tag{def:nonset}$$

Definition 15.13 ("is not set" predicate)**.**

$$(t \text{ not set}) \overset{\mathrm{def.}}{\longleftrightarrow} (t \notin \mathrm{V}) \tag{def:not set}$$

This convention keeps evaluation within a two-valued class language. It is one possible treatment of domain failures, alongside the separate partial and many-valued experiments. In this subsection, "not set" is a precise predicate; it should not be used as a synonym for every ordinary use of "undefined".

Example 15.1 (intersection of the empty set)**.**

$$\left(\left(\bigcap \emptyset\right) \text{ not set}\right)$$

Example 15.2 (value of a function)**.**

$$(f \text{ function}) \ \& \ (x \notin \text{dom}(f)) \implies (f(x) \text{ not set})$$

15.5 Functions and families

The local encoding distinguishes a set value, stored as one fiber entry, from a proper-class value, represented by its singleton subclasses. $\mathrm{Dom}(F)$ records represented arguments, while $\mathrm{dom}(F)$ records arguments with set values. The proposed extensionality statements require canonical fibers and a relation consisting entirely of encoded pairs; arbitrary extra members must be excluded. These representation conditions are part of the family construction, not consequences of the default value alone. For this subsection, fix an injective ordered-pair encoding on sets with set-valued pairs and definable projections; ordinary Kuratowski pairs suffice. Every stored record has two set coordinates. The off-domain value nonset is one fixed proper class, for example V. We use the set-image principle and union when proving that a small family has only set values.

Definition 15.14 (big domain "Dom" including proper values)**.**
For any set-record relation $F \subseteq \mathrm{V} \times \mathrm{V}$, define its represented domain by

$$\mathrm{Dom}(F) \overset{\text{def.}}{\coloneqq} \{\, x \mid \exists (b \in \mathrm{V}),\ (x, b) \in F \,\} \tag{def:bigdomain}$$

The preliminary domain, image, and evaluation operations apply to arbitrary such relations. The family predicate below selects their canonical representations, so its definition does not presuppose that evaluation is already restricted to families. For a canonical family, the second coordinate will either be its set value or a singleton member-record of its proper-class value, as proved below.

Notation 15.9 ("in the big domain of" relation)**.**
Represented definedness is the following binary predicate on a relation and its argument. It is definable in the ordinary two-valued class language and should be distinguished from a predicate on the evaluated value alone:

$$(F(x) \text{ defined}) \overset{\text{notation}}{\iff} (x \in \mathrm{Dom}(F))$$

This is a predicate symbol of two arguments, in a sense that "$F(x)$" is not a term in this notation, but rather a way of writing the arguments. So in the code it is written as something like "inDomOf(F,x)"), but it is displayed as "$(F(x)$ defined)".
In this formula, "$(\cdot(\cdot)$ defined)" is a predicate symbol of the valence two, and the internal parentheses are an essential part of its name. The coloring of the internal parentheses may be omitted as an abuse of notation.

Definition 15.15 (image)**.**
Let $F \subseteq \mathrm{V} \times \mathrm{V}$ be any relation and A any class. This is its relational image: it collects the stored second coordinates at arguments from A. For a set-valued family these coordinates are the values themselves. At a proper-class value they are singleton records, not the evaluated value:

$$F[A] \overset{\text{def.}}{\coloneqq} \{\, b \mid \exists (a \in A),\ (a, b) \in F \,\}. \tag{def:image}$$

The newer Obelisk catalog's `DefRelationalImage` (3.26) instead uses the evaluated-image expression $\{\, b \mid \exists a \in A,\ F(a) = b \,\}$. These agree for class functions with the proper-class off-domain default, but not for general families. If the value at x is a proper class P, the evaluated image of $\{x\}$ is empty, while its relational image contains the singleton records $\{q\}$ for $q \in P$. The union in the following evaluation formula therefore uses the relational-image convention explicitly defined here.

Definition 15.16 (evaluation of a family)**.**
Let $F \subseteq \mathrm{V} \times \mathrm{V}$ be any relation and x a class. Evaluation is defined by:

$$F(x) \stackrel{\text{def.}}{\coloneqq} \Big(\text{if } (x \in \mathrm{Dom}(F)) \text{ then } \bigcup(F[\{x\}]) \text{ else nonset fi}\Big)$$

Definition 15.17 (small domain "dom")**.**
For any set-record relation f, put

$$\mathrm{dom}(f) \stackrel{\text{def.}}{\coloneqq} \{\, x \mid f(x) \text{ set} \,\}$$

(note: if f is a function or a class–function, then $\mathrm{dom}(f) = \mathrm{Dom}(f)$)

Definition 15.18 (graph of the (class–)function)**.**

$$\Gamma(f) \stackrel{\text{def.}}{\coloneqq} \{\, (x, f(x)) \mid x \in \mathrm{dom}(f) \,\}$$

Definition 15.19 (class–function predicate)**.**

$$(f \text{ class–function}) \stackrel{\text{def.}}{\longleftrightarrow} (f = \Gamma(f))$$

Definition 15.20 (range)**.**

$$\mathrm{ran}(f) \stackrel{\text{def.}}{\coloneqq} \{\, f(x) \mid x \in \mathrm{dom}(f) \,\}$$

Definition 15.21 (signature of a function)**.**

$$(f : X \to Y) \stackrel{\text{def.}}{\longleftrightarrow} ((\mathrm{dom}(f) = X) \wedge (\mathrm{ran}(f) \subseteq Y) \wedge (f \text{ class–function}))$$

Thus arrows between arbitrary classes denote set-valued class functions. When X is a set, the set-image principle applied to $x \mapsto (x, f(x))$ makes the graph a set, so this notation also gives an ordinary small function. Requiring a set graph for every arrow would incorrectly exclude the class functions $\mathrm{V} \to \mathrm{V}$ used in recursion.

Definition 15.22 (family)**.**
A class f is called a *family* if $f \subseteq \mathrm{V} \times \mathrm{V}$ and it satisfies the following two conditions:

$$\forall(x \in \mathrm{Dom}(f)),\ ((f(x) \text{ set}) \longrightarrow (\exists!(y \in \mathrm{V}),\ ((x, y) \in f))) \qquad \text{(concentrated set values)}$$

$$\forall(x \in \mathrm{Dom}(f)),\ ((f(x) \text{ proper}) \longrightarrow (\forall(b \in \mathrm{V}),\ (((x, b) \in f) \longrightarrow (b \text{ singleton})))) \qquad \text{(flat proper values)}$$

Lemma 15.1 (Canonical fibers and graph reconstruction)**.**
Let f be a family and $x \in \mathrm{Dom}(f)$. If $v = f(x)$ is a set, then

$$f[\{x\}] = \{v\}.$$

If v is proper, then

$$f[\{x\}] = \{\, \{q\} \mid q \in v \,\}.$$

Consequently, the family is determined by its domain and evaluated values. It is set-valued exactly when $f = \Gamma(f)$.

Proof. In the set-valued case there is exactly one second coordinate b. Evaluation gives $v = \bigcup\{b\} = b$, proving the first formula. In the proper-valued case every second coordinate is a singleton. If that coordinate is $\{q\}$, then q belongs to their union v. Conversely, $q \in v$ supplies a second coordinate containing q; as that coordinate is a singleton, it must be $\{q\}$. This proves the second formula. Injectivity of the pair coding and the absence of nonrecord members now reconstruct f from these fibers.
If all values are sets, the first formula says precisely that the records are $(x, f(x))$ for $x \in \mathrm{Dom}(f) = \mathrm{dom}(f)$, hence $f = \Gamma(f)$. Conversely, the graph $\Gamma(f)$ stores only the set values at the arguments in $\mathrm{dom}(f)$; equality with f therefore leaves no represented proper-class value. This also verifies agreement between the two local definitions of class function. □

Definition 15.23 (set–valued)**.**
A class f is called *set–valued*, if all of its values are sets.

$$(f \text{ set–valued}) \overset{\text{def.}}{\longleftrightarrow} (\forall (x \in \mathrm{Dom}(f)),\ (f(x) \text{ set}))$$

Definition 15.24 (class–function)**.**
A family is called a *class–function* if it is set–valued.

$$(F \text{ class–function}) \overset{\text{def.}}{\longleftrightarrow} ((F \text{ family}) \wedge (F \text{ set–valued}))$$

Definition 15.25 (function)**.**
A family is called a *function* if it is a set.

$$(F \text{ function}) \overset{\text{def.}}{\longleftrightarrow} ((F \text{ family}) \wedge (F \text{ set}))$$

Note 15.5 (category–theory inspired naming)**.**
A function can be called "a small family".
A class–function can be called "a locally small family".
An arbitrary family can be called "a (large) family".

Lemma 15.2 (proper value not from dom)**.**
For every set-record relation f and every class argument x,

$$(f(x) \text{ proper}) \longrightarrow (x \notin \mathrm{dom}(f))$$

Proof. Assume the opposite: $(f(x)$ proper$)$ and $(x \in \mathrm{dom}(f))$.

$$(x \in \mathrm{dom}(f)) \implies (x \in \{\, x \mid f(x) \text{ set} \,\})$$

$$(x \in \{\, x \mid f(x) \text{ set} \,\}) \implies (f(x) \text{ set})$$

The contradiction is obtained, so $(f(x)$ proper$) \longrightarrow (x \notin \mathrm{dom}(f))$. □

Theorem 15.2 (small implies locally small)**.**
Every function is a class–function(but not vice versa).

$$\forall f,\ ((f \text{ function}) \longrightarrow (f \text{ class–function}))$$

Proof. Project the second coordinates of the set of records f. The projection is a class function, so its range D is a set by the set-image principle. For $x \in \mathrm{Dom}(f)$, the fiber $f[\{x\}] \subseteq D$ is a set by heredity. Its union is a set by the union axiom, and that union is $f(x)$. Thus every represented value is a set, so f is a class function. The conclusion that represented values are sets holds for every set relation under union evaluation; concluding that the original records form a class function also uses the canonical-family hypothesis. Neither conclusion here requires limitation of size.
The converse fails: the identity graph on V is a class function. If it were a set, projecting its first coordinates would make V a set, a contradiction. □

Theorem 15.3 (family extensionality)**.**

$$\forall(F_1 \text{ family})\forall(F_2 \text{ family}),\ (\mathrm{Dom}(F_1) = \mathrm{Dom}(F_2)) \longrightarrow ((\forall x,\ (F_1(x) = F_2(x))) \longrightarrow (F_1 = F_2))$$

Proof. Equal domains specify exactly the represented arguments. At each such argument, equal values have identical canonical fibers by the preceding lemma, whether those values are sets or proper classes. Every member of either family is a record, so the two families are equal by extensionality. □

Example 15.3 (Why the domain is retained for proper values)**.**
Take the default to be V, fix a set x, and let $F = \{\, (x, \{q\}) \mid q \in \mathrm{V} \,\}$. This is a family with domain $\{x\}$ and proper value $F(x) = \mathrm{V}$. At every other argument its default is also V. Thus F and the empty family have identical evaluations everywhere, but different domains and different records. Family extensionality needs the domain hypothesis; class-function extensionality below does not. In particular, no unary predicate D on class values can satisfy $D(H(y)) \longleftrightarrow y \in \mathrm{Dom}(H)$ uniformly for all families H and arguments y: at the fixed x, the two families give the same value V, but only F represents that argument. This obstruction concerns definedness from a value alone. It does not affect the binary predicate already defined by domain membership. For set-valued class functions, sethood of the evaluated value does recover domain membership, because the off-domain default is proper.

Theorem 15.4 (class–function extensionality)**.**

$$\forall(F_1 \text{ class–function}),\ \forall(F_2 \text{ class–function}),\ ((\forall x,\ (F_1(x) = F_2(x))) \longrightarrow (F_1 = F_2))$$

Proof. For a class function, the domain is exactly the class of arguments where evaluation is a set; the fixed proper-class default excludes every other argument. Equality of all evaluations therefore implies equality of the domains, and family extensionality applies. □

Theorem 15.5 (function extensionality)**.**

$$\forall(f_1 \text{ function}),\ \forall(f_2 \text{ function}),\ ((\forall x,\ (f_1(x) = f_2(x))) \longrightarrow (f_1 = f_2))$$

Proof. Small families are class functions by the preceding theorem, so the class-function extensionality result applies. □

Theorem 15.6 (function extensionality for a set–theoretic foundations)**.**

$$\forall(f_1 \text{ function}),\ \forall(f_2 \text{ function}),\ \forall X,$$
$$(X = \mathrm{dom}(f_1) = \mathrm{dom}(f_2)) \longrightarrow \big((\forall(x \in X),\ (f_1(x) = f_2(x))) \longrightarrow (f_1 = f_2)\big)$$

Proof. Both graphs consist precisely of the pairs $(x, f_i(x))$ with $x \in X$. Agreement on X gives the same pairs in each graph. Unlike the preceding formulation, this argument needs no convention for off-domain evaluation. □

Axiom 9 (ordered pair of proper classes is a proper class)**.**
The following additional pairing assumption is recorded as an option for this local encoding. Its use must be stated when a result depends on it:

$$(((A \text{ proper}) \wedge (B \text{ proper})) \longrightarrow ((A, B) \text{ proper}))$$

15.6 Unions

In the unary union and intersection formulas below, auxiliary variables x, a, y are chosen distinct and fresh for the argument class term A. For a restricted binder with fixed indexing term s, its bound variable y is not free in s. These conventions preserve the free parameters of the argument when the binders are expanded.

Definition 15.26 (elementary union)**.**
The elementary union of a class A collects the members of its members. Below we recover the same expression from the more general binder notation.

$$\bigcup A \overset{\text{def.}}{\coloneqq} \{\, x \mid \exists(a \in A),\ x \in a \,\} \qquad \text{(def:elementary union)}$$

Definition 15.27 (union of a definable conglomerate of classes)**.**

$$\bigcup_{\psi}^{y} t \overset{\text{def.}}{\coloneqq} \{\, x \mid \exists y, \psi \wedge (x \in t) \,\} \qquad \text{(def:conglomerate union)}$$

Side conditions:
1) formula ψ must not depend on variable x,
2) term t must not depend on variable x,
3) variable y is different from variable x.
(Note: this definition includes a second–order class construction, which is available in MK, but not available in NBG.)

Definition 15.28 (union of terms built out of the elements of a class)**.**
Let y be a variable, let s and t be terms.
If a formula has form "$(y \in s)$", and if y is also its bound variable, then it is allowed to skip writing a variable on top.

$$\bigcup_{y \in s} t \overset{\text{def.}}{\coloneqq} \bigcup_{y \in s}^{y} t$$

Definition 15.29 (union of the elements of a class)**.**

$$\bigcup A \overset{\text{def.}}{\coloneqq} \bigcup_{y \in A} y$$

Theorem 15.7 (Elementary membership description of union)**.**

$$\bigcup A = \{\, x \mid \exists a \in A,\ x \in a \,\} \tag{thm:union}$$

Proof.

$$\bigcup A = \bigcup_{y \in A} y = \bigcup_{y \in A}^{y} y = \{\, x \mid \exists y, (y \in A) \wedge (x \in y) \,\} = \{\, x \mid \exists a \in A,\ x \in a \,\}$$

□

15.7 Intersections

Definition 15.30 (elementary intersection)**.**
The elementary intersection of A collects the sets belonging to every member of A. In particular, the empty intersection is V, since its membership condition is vacuous.

$$\bigcap A \stackrel{\text{def.}}{:=} \{\, x \mid \forall (a \in A),\ x \in a \,\} \tag{def:intersection}$$

Definition 15.31 (intersection of a definable conglomerate of classes)**.**
For any formula ψ, any class term t, any variables x and y, in case all of them satisfy the side conditions, the following holds

$$\bigcap_{\psi}^{y} t \stackrel{\text{def.}}{:=} \{\, x \mid \forall y,\ (\psi \longrightarrow (x \in t)) \,\} \tag{def:conglomerate intersection}$$

Side conditions:
1) formula ψ must not depend on variable x,
2) term t must not depend on variable x,
3) variable y is different from variable x.
(Note: this definition includes a second–order class construction, which is available in MK, but not available in NBG.)

Notation 15.10 (intersection of terms built out of the elements of a class)**.**
Let y be a variable, let s and t be terms.
If formula has form "$(y \in s)$", and if y is also its bound variable, then it is allowed to skip writing a variable on top.

$$\bigcap_{y \in s} t \stackrel{\text{def.}}{:=} \bigcap_{y \in s}^{y} t$$

Notation 15.11 (intersection of the elements of a class)**.**

$$\bigcap A \stackrel{\text{def.}}{:=} \bigcap_{y \in A} y$$

Theorem 15.8 ("elementary def of intersection")**.**

$$\bigcap A = \{\, x \mid \forall (a \in A),\ x \in a \,\}$$

Proof.

$$\bigcap A = \bigcap_{y \in A} y = \bigcap_{y \in A}^{y} y = \{\, x \mid \forall y,\ ((y \in A) \longrightarrow (x \in y)) \,\} = \{\, x \mid \forall a \in A . x \in a \,\}$$

□

15.8 Powerclass

Definition 15.32 (powerclass)**.**
A powerclass collects all set subclasses. For a class term X, choose a fresh bound variable x not free in X, and define

$$\mathcal{P}X \stackrel{\text{def.}}{\coloneqq} \{\, x \mid x \subseteq X \,\} \tag{def:powerclass}$$

15.9 Heredity

Axiom 10 (heredity)**.**
A subclass of a set is also a set.

$$\forall (A : \text{Class}),\ \forall (B : \text{Class}),\ (((A \text{ set}) \wedge (B \subseteq A)) \longrightarrow (B \text{ set})) \tag{axm:heredity}$$

15.10 Powerset

Axiom 11 (powerset)**.**
The powerclass of a set is also a set.

$$\forall (A : \text{Class}),\ ((A \text{ set}) \longrightarrow (\mathcal{P}A \text{ set})) \tag{axm:powerset}$$

Note 15.6 (the powerset axiom is weak here)**.**
The powerclass contains only set subclasses by comprehension. Heredity ensures that every subclass of a set occurs in this powerclass, so the two axioms together express the usual powerset requirement.

15.11 Regularity in form of epsilon–induction

Axiom 12 ($\in$–induction)**.**

$$\forall (\Psi : \text{Class}),\ ((\mathcal{P}\Psi \subseteq \Psi) \longrightarrow (\text{V} \subseteq \Psi)) \tag{axm:epsilon induction}$$

15.12 Limitation of size

Axiom 13 (limitation of size (of a set))**.**

$$(\neg (X \text{ set})) \longleftrightarrow (\exists (T : \text{Class}),\ ((T \text{ class–function}) \wedge (\text{V} \subseteq T[X]))) \tag{axm:limitation of size}$$

Note 15.7 (properties)**.**
One direction of the equivalence extends what it means for class to be a set, i.e. it "extends the inductive definition of a set", so more classes are called sets as a result. Let such properties be called *positive*.

$$(\neg (X \text{ set})) \longrightarrow (\exists (T : \text{Class}),\ ((T \text{ class–function}) \wedge (\text{V} \subseteq T[X])))$$

or

$$\Big(\neg\exists(T : \text{Class}),\ ((T \text{ class–function}) \wedge (\mathrm{V} \subseteq T[X]))\Big) \longrightarrow (X \text{ set})$$

The opposite direction forbids certain situations. It makes proofs shorter. It reduces the amount of models of the theory. Let such properties be called *negative*.

$$\Big(\exists(T : \text{Class}),\ ((T \text{ class–function}) \wedge (\mathrm{V} \subseteq T[X]))\Big) \longrightarrow \neg(X \text{ set})$$

This terminology distinguishes two roles an axiom can play in the proposed development. It is a heuristic for organizing assumptions, not a consistency theorem: compatibility of a collection of axioms still requires a separate argument.

15.13 Set images under the surjective-image size axiom

The following consequences use precisely the limitation-of-size axiom just stated: a proper class admits a class-function image containing V. They are not inferred from assigning the same default ordinal cardinality to proper classes. Use the earlier graph-image convention, class comprehension, and injective set-pair coding with set-valued pairs. In particular, a class function may be defined on only some members of an input class; its graph image retains only represented outputs.

Theorem 15.9 (Class-function images of sets are sets)**.**
Under the preceding assumptions, if A is a set and f is a class function, then $f[A]$ is a set. It is not necessary that $A \subseteq \mathrm{dom}(f)$.

Proof. Suppose $B = f[A]$ is proper. The surjective-image size axiom supplies a class function T with $\mathrm{V} \subseteq T[B]$. By class comprehension form the composite graph

$$H = \{(x, z) : \exists y \in \mathrm{V},\ (x, y) \in f \wedge (y, z) \in T\}.$$

All coordinates are sets, and uniqueness of the outputs in f and T makes H single-valued. Thus H is a class function with its actual represented domain. This construction collects a class of set records; it does not use set-image or assert that the graph is a set.
A set z belongs to $H[A]$ exactly when there are $x \in A$ and $y \in \mathrm{V}$ with $(x, y) \in f$ and $(y, z) \in T$. This is exactly $z \in T[f[A]] = T[B]$. Therefore $\mathrm{V} \subseteq H[A]$, and the same size axiom implies that A is proper, a contradiction. Classical logic gives that $f[A]$ is a set. No off-domain evaluation, choice of preimages, or union axiom was used. □

Corollary 15.1 (A set domain gives a set graph)**.**
Under the same assumptions, a class function f whose domain D is a set has a set graph and a set range.

Proof. The range is $f[D]$, so the theorem gives its sethood. For each $a \in D$, the unique represented value $f(a)$ is a set, and so is the pair $(a, f(a))$. Class comprehension gives the class function

$$q : D \to \mathrm{V}, \qquad q(a) = (a, f(a)).$$

Its image on D is exactly the graph f. Apply the same theorem to q to conclude that f is a set. This also covers $D = \varnothing$: both graph and range are empty. Uniqueness of the values, rather than an axiom of choice, supplies the assignment used in the graph construction. □

These results justify the set-image and graph-collection steps used in the local small type models when the present strong size axiom is part of their background. In settings with a different size principle, replacement or the corresponding set-image assumption must still be stated separately.

16 Ordered pairs of sets and classes

16.1 Introduction

An ordered-pair construction must recover both components from the result. For classes, there is an additional question: can either component be a proper class? We first compare set-based constructions, then develop the author's recursive tagged construction. The same notation (a, b) is reused locally; subscripts and subsection boundaries identify the construction in use. The three displayed axioms are the intended interface, whose verification is a separate task for each proposed model.

16.2 Axioms of ordered pair

$$((a \in \mathrm{Term}) \wedge (b \in \mathrm{Term})) \longrightarrow ((a, b) \in \mathrm{Term}) \qquad \text{(axm:DunaevTaggedOPairConstructor)}$$

$$((a \text{ set}) \wedge (b \text{ set})) \longrightarrow ((a, b) \text{ set}) \qquad \text{(axm:DunaevTaggedOPairSize)}$$

$$((a_1, b_1) = (a_2, b_2)) \longleftrightarrow ((a_1 = a_2) \wedge (b_1 = b_2)) \qquad \text{(axm:OPairUniqueness)}$$

16.3 Kuratowski's ordered pair of sets

For sets a and b, the usual Kuratowski construction is

Definition 16.1 (Kuratowski pair)**.**

$$(a, b)_K \coloneqq \{\{a, a\}, \{a, b\}\}.$$

The inner collections distinguish the first component from the second. The following comprehensions recover the singleton classes $\{a\}$ and $\{b\}$; taking their unions recovers the components themselves.

Definition 16.2 (Kuratowski projections)**.**

$$\mathrm{fst}_K(p) \coloneqq \bigcup \{\, x \mid \forall y \in p,\ x \in y \,\}, \qquad \mathrm{snd}_K(p) \coloneqq \bigcup \{\, x \mid \exists! y \in p,\ x \in y \,\}.$$

These projection equations are intended for $p = (a, b)_K$ with set components. The equal-component case is included: then $p = \{\{a\}\}$.

Remark 16.1 (recall the definition)**.**

$$(\exists! y \in Q,\ \psi) \overset{\text{def}}{\iff} (\exists y \in Q,\ \psi) \& (!y \in Q,\ \psi)$$

Theorem 16.1 (Kuratowski projections recover the components)**.**
For sets a, b, $\mathrm{fst}_K((a, b)_K) = a$ and $\mathrm{snd}_K((a, b)_K) = b$.

Proof. If $a \neq b$, the elements common to $\{a\}$ and $\{a, b\}$ form $\{a\}$, and the elements belonging to exactly one of these two sets form $\{b\}$. If $a = b$, both comprehensions give $\{a\}$. Taking unions proves the two identities in either case. □

16.4 Definition of Metamath's ordered pair of sets

The following convention deliberately restricts pairing to set components:

Definition (Metamath's ordered pair)**.**

$$\langle A, B\rangle = \{\, x \mid (A \text{ set}) \wedge (B \text{ set}) \wedge (x \in \{\{A\}, \{A, B\}\})\,\} \qquad \text{(def:MM_ordered pair)}$$

When a component is proper, the defining guard fails and the result is empty. Thus this convention serves set relations, but does not provide an injective pairing of arbitrary classes.

16.5 Definition of Morse's ordered pair

The next construction extends pairing to class arguments by first coding their elements. The products in this display use the preceding set-pair construction; the resulting class is then taken as a new pair:

$$s(x) = \{\varnothing\} \cup \{\{t\} \mid t \in x\}$$
$$(x, y) = (\{0\} \times s(x)) \cup (\{1\} \times s(y))$$

and also defines 3–tuple:

$$(x, y, z) = (\{0\} \times s(x)) \cup (\{1\} \times s(y)) \cup (\{2\} \times s(z))$$

Similar tuples which are presented in the section "Classes as Types" are called lists in this text. This approach distinguishes the auxiliary set pairs from the resulting class pairs, and correspondingly distinguishes their products and projections.
The following proposal instead seeks a uniform recursive definition for sets and classes.

16.6 Recursive Dunaev tagged pairs

Remark 16.2 (comparison with the newer marker construction)**.**
The Obelisk catalog's entry `DefDunaevOrderedPair` (3.76) uses a separate marker-based encoding. Writing it as D to avoid ambiguity,

$$S_3(a) = \{\{\{a\}\}\}, \qquad M(A, n) = \{\{0, n\}\} \cup \{\, \{S_3(a), n\} \mid a \in A\,\}, D(A, B) = M(A, 1) \cup M(B, 2).$$

In contrast, this subsection develops the recursive pair P given below. Already $P(0, 0) = \varnothing$, whereas $D(0, 0) = \{\{0, 1\}, \{0, 2\}\}$. The newer catalog's projection entries 3.77–3.78 concern D, so they should not be read as proofs of the recursive construction. Both approaches are retained in their own setting.

Local assumptions. We work with well-founded sets and use extensionality, separation, replacement, and well-founded set recursion. These are available in the usual ZF set part of NBG or MK. Class comprehension then extends the construction to arbitrary classes. In particular, the argument below is not an existence proof from extensionality and class comprehension alone. No choice principle is needed. The required membership-recursion theorem is proved in 25.4; regularity identifies its domain W with V in the present setting.
The simultaneous equations we want to justify are

$$(a, b) \coloneqq (\{0\} \times a) \cup (\{1\} \times b), \qquad X \times Y \coloneqq \{\, (x, y) \mid x \in X \wedge y \in Y\,\}.$$

The apparent mutual recursion disappears once the two tag maps are isolated. We use $0 = \varnothing$ and $1 = \{0\}$. Auxiliary graphs used to invoke the recursion theorem may be coded by the already available Kuratowski pair; this does not assume the new pair construction.

Definition 16.3 (The two recursive tag maps)**.**
On sets, define R by well-founded recursion and then define L by

$$R(x) = \{0\} \cup \{\, R(u) \mid u \in x \,\}, \qquad L(x) = \{\, R(u) \mid u \in x \,\}.$$

Here each image is a set by replacement. The recursion step takes the already constructed values on the members of x, forms their image, and adjoins the empty set.

Lemma 16.1 (Existence and separation of the tags)**.**
There is a unique class function $R : \mathrm{V} \to \mathrm{V}$ satisfying its displayed recursion. For every set x, $0 \in R(x)$ and $0 \notin L(x)$. Consequently, $L(x) \neq R(y)$ for all sets x, y.

Proof. Apply well-founded set recursion to the operation sending a set-sized partial-value graph g to $\{0\} \cup \operatorname{ran}(g)$. Its restriction to the members of any set is a set graph by replacement, so the operation has a set value at every stage. The recursion theorem gives existence; membership induction gives uniqueness, since agreement on every member of x forces agreement at x.
The definition puts 0 in every $R(u)$, so $R(u)$ is never empty. Every member of $L(x)$ is some $R(u)$, hence is different from 0. Thus $L(x)$ omits the marker that every $R(y)$ contains. □

Lemma 16.2 (Injectivity of each tag)**.**
For sets x, y,

$$R(x) = R(y) \longleftrightarrow x = y, \qquad L(x) = L(y) \longleftrightarrow x = y.$$

Proof. For R, use membership induction on x with the induction assertion

$$\forall (y \text{ set}),\ R(x) = R(y) \longrightarrow x = y.$$

Assume this assertion holds for every $u \in x$, and suppose $R(x) = R(y)$. If $u \in x$, then $R(u) \in R(x) = R(y)$. Since $R(u) \neq 0$, it is not the adjoined marker; therefore $R(u) = R(v)$ for some $v \in y$. The induction hypothesis for u, which quantifies over *every* set v, gives $u = v$. This proves $x \subseteq y$.
Conversely, if $v \in y$, then the nonempty set $R(v)$ belongs to $R(x)$, so $R(v) = R(u)$ for some $u \in x$. The same induction hypothesis gives $u = v$, proving $y \subseteq x$. Extensionality yields $x = y$ and completes the induction. No induction hypothesis about the possibly unrelated set y was needed.
If $L(x) = L(y)$, adjoining $\{0\}$ to both sides gives $R(x) = R(y)$, hence $x = y$. The reverse implications follow by substitution of equals. □

Definition 16.4 (The recursive pair of classes)**.**
For arbitrary classes A, B, put

$$P(A, B) = \{\, L(a) \mid a \in A \,\} \cup \{\, R(b) \mid b \in B \,\}.$$

All members of A and B are sets, so the tags in this comprehension have already been defined. No graph whose members have proper-class components is being assumed. Write (A, B) for $P(A, B)$ within this construction, and use this pair in the associated Cartesian product.

Theorem 16.2 (The simultaneous equations have a unique solution)**.**
The construction satisfies

$$P(0, x) = L(x), \qquad P(1, x) = R(x) \quad (x \text{ set}),$$

and, for arbitrary classes A, B,

$$P(A,B) = \{\, P(0,a) \mid a \in A \,\} \cup \{\, P(1,b) \mid b \in B \,\} = (\{0\} \times A) \cup (\{1\} \times B).$$

It is the unique class-valued operation satisfying these equations, with set values on pairs of sets.

Proof. Since $L(0) = 0$, the definition gives $P(0,x) = \{\, R(u) \mid u \in x \,\} = L(x)$ and $P(1,x) = \{L(0)\} \cup \{\, R(u) \mid u \in x \,\} = R(x)$. Substituting these identities proves the equations for P and the product.
For uniqueness, let Q be another such operation. Its equations force $Q(0,0) = 0$ and then

$$Q(1,x) = \{0\} \cup \{\, Q(1,u) \mid u \in x \,\}.$$

Uniqueness of set recursion gives $Q(1,x) = R(x)$ for every set x. The equation at $(0,x)$ now gives $Q(0,x) = L(x)$. Finally the equation for arbitrary $Q(A,B)$ expresses it using these set tags, so $Q(A,B) = P(A,B)$. □

Remark 16.3.
Do not confuse any of previous definitions with a family $F : 2 \to \mathcal{U}$.

$$\tilde{K} \coloneqq \{\varnothing\} \cup \{\, \{k\} \mid k \in K \,\}$$

$$[a,b] \coloneqq (\{0\} \times \tilde{a}) \cup (\{1\} \times \tilde{b})$$

Generally speaking,

$$(a,b) :\neq [a,b]$$

The extra empty marker in $\tilde{K}$ distinguishes an empty value from an absent fiber. The relation between this family encoding and the recursive pair construction requires its own verification.

16.7 Implementation sketch in Coq

The following code illustrates the recursive construction for readers familiar with Coq. It is an excerpt, rather than a standalone proof certificate; its auxiliary definitions and proof obligations belong to the surrounding development.
The datatype `Ens` represents sets in the ambient type theory. A construction on this datatype does not by itself establish the proper-class extension; that extension uses class comprehension in the mathematical development.

```
(* The type of sets in a universe. *)
Inductive Ens : Type :=
    sup : forall A : Type, (A -> Ens) -> Ens.
```

```
(* The empty set *)
Definition Vide : Ens :=
  sup False (fun x : False => match x return Ens with
                              end).
Definition Pair (A B:Ens) : Ens
 := sup bool (fun b : bool => if b then A else B).
(* The singleton set *)
Definition Sing (E : Ens) := Pair E E.
```

```
Definition Zero := Vide.
Definition One := Sing Vide.
```

```
Definition img (f:Ens -> Ens) (c:Ens) : Ens :=
match c with
| sup C h => sup _ (fun d => f (h d))
end.
```

```
Definition bu A B := Union (Pair A B).
Infix "u" := bu (at level 60, right associativity).
```

The following straightforward implementation of the ordered pair is not accepted by the proof assistant, because the built–in termination checker does not accept the recursion as written.

```
Fail
Program Fixpoint OPair_Zero (c : Ens) : Ens
                  := bu (img OPair_Zero Zero) (img OPair_One c)
           with OPair_One (c : Ens) : Ens
                  := bu (img OPair_Zero One ) (img OPair_One c)
.
```

The first values help expose the recursion:

```
Definition img_OPair_Zero_on_Zero := Vide.
(* (img OPair_Zero Zero) === {(0,k)|k in 0} = emptyset *)
Definition OPair_Zero_Zero := Vide.
(* OPair Zero Zero === (0,0) = {(0,j)|j in 0}u{(1,k)|k in 0} = emptyset *)
Definition img_OPair_Zero_on_One := Sing Vide.
(*(img OPair_Zero One) === {(0,k)|k in 1} = {(0,0)} = {emptyset}*)
Definition OPair_Zero_One := Sing (Sing Vide).
(* OPair Zero One === (0,1) = {} u {(1,0)} = {{emptyset}} *)
```

The mathematical reduction above uses only one recursive function. A structural version of its right tag can be written as follows:

```
Fixpoint OPair_One (c : Ens) {struct c} : Ens :=
  match c with
  | sup C h =>
      sup (unit + C)%type
        (fun i => match i with
                  | inl _ => Vide
                  | inr u => OPair_One (h u)
                  end)
  end.

Definition OPair_Zero (c : Ens) : Ens := img OPair_One c.
```

The left summand supplies the empty marker; the right summand recurses on members of the input. This makes the decrease visible in the constructor instead of hiding it inside a call to `img`. The excerpt illustrates the construction; a mechanized certificate must additionally verify the universe declarations and compatibility with the representation's extensional equality. No compilation result for this excerpt is asserted here.
The pair operation then uses the two images:

```
Definition OPair (a b: Ens) : Ens := (img OPair_Zero a) u (img OPair_One b).
```

16.8 Projections and the ordered-pair property

The preceding construction explains both the recursion and the tags. The projections now recover arbitrary class components by comprehension.

Definition 16.5 (Recursive tagged projections)**.**
For any class C, define

$$\mathrm{pr}_1(C) = \{\, a \mid L(a) \in C \,\}, \qquad \mathrm{pr}_2(C) = \{\, b \mid R(b) \in C \,\}.$$

These are the original proposed formulas

$$\mathrm{pr}_1(C) = \{\, a \mid (0,a) \in C \,\}, \qquad \mathrm{pr}_2(C) = \{\, b \mid (1,b) \in C \,\},$$

since comprehension ranges over set witnesses.

Theorem 16.3 (Projection identities)**.**
For all classes A, B,

$$\mathrm{pr}_1(P(A,B)) = A, \qquad \mathrm{pr}_2(P(A,B)) = B.$$

Proof. For a set a, membership $L(a) \in P(A,B)$ says either $L(a) = L(u)$ for some $u \in A$, or $L(a) = R(v)$ for some $v \in B$. The second alternative is excluded by the marker distinction. The first is equivalent to $a \in A$ by injectivity of L. Thus the first projection has exactly the members of A.
Likewise, $R(b)$ can belong only to the right tagged image, and injectivity of R makes $R(b) \in P(A,B)$ equivalent to $b \in B$. Class extensionality gives both identities. □

Theorem 16.4 (The ordered-pair property)**.**
For arbitrary classes A_1, B_1, A_2, B_2,

$$P(A_1,B_1) = P(A_2,B_2) \longleftrightarrow (A_1 = A_2) \wedge (B_1 = B_2).$$

Proof. Apply the two projections to the asserted pair equality. The projection identities give equality of the first and of the second components. The converse follows directly from the definition of P. □

Theorem 16.5 (Sethood of a recursive pair)**.**
For arbitrary classes A, B,

$$(P(A,B) \text{ set}) \longleftrightarrow (A \text{ set}) \wedge (B \text{ set}).$$

Proof. If A, B are sets, replacement makes their two tagged images sets, and pairing and union make their union a set.
Conversely, suppose $P(A,B)$ is a set. Separation gives the subset

$$\{\, c \in P(A,B) \mid \exists (a \text{ set}),\ c = L(a) \,\}.$$

Each of its members has a unique inverse image under L by tag injectivity. Replacement applied to this inverse relation therefore makes A a set. The same argument with R makes B a set. These inverse images are uniquely determined, so choice is not needed. □

Why the height argument is unnecessary here. The recursive definition uses well-founded set recursion, and its injectivity uses membership induction. Both are explicit assumptions of this local construction. There is no need to assign a new numerical height to the pair codes. In particular, the bounded depth measure discussed in the height section need not be substituted for ordinal rank: its failure to decrease strictly at limit values has no bearing on the tag-injectivity proof above. In a setting without regularity, the same argument applies to its well-founded part when the required recursion and replacement principles are available; it makes no assertion about arbitrary non-well-founded sets.

16.9 Finite examples and Cartesian products

The recursive tags have particularly small initial values:

$$L(0) = 0, \quad R(0) = 1, \quad L(1) = \{1\}, \quad R(1) = 2,$$

where $0 = \varnothing$, $1 = \{0\}$, and $2 = \{0, 1\}$ are von Neumann ordinals. Thus the pair table is

P	0	1	2
0	0	$\{1\}$	$\{1, 2\}$
1	1	2	3
2	$\{0, \{1\}\}$	$\{0, 1, \{1\}\}$	$\{0, 1, \{1\}, 2\}$

The row is the first component. For example,

$$P(2, 2) = \{L(0), L(1), R(0), R(1)\} = \{0, \{1\}, 1, 2\} = \mathcal{P}2.$$

The number of members of a pair of finite sets is the sum of the numbers of members of its components, because the two tag maps are injective and have disjoint images. This does not make the pair operation ordinal addition: $P(1, 2) = 3$, whereas $P(2, 1) = \{0, 1, \{1\}\}$ is not an ordinal.

The Cartesian product instead collects a pair for each choice of a member of each factor. Its corresponding table is

$\times$	0	1	2
0	0	0	0
1	0	1	$\{0, \{1\}\}$
2	0	2	$\{0, 1, \{1\}, 2\}$

Thus $P(2, 2) = 2 \times 2 = \mathcal{P}2$ happens for this small example; pairing and Cartesian product remain different operations.

On the truth-value representatives Prop $= \{0, 1\}$, the product table restricts to Boolean conjunction: a product is 1 exactly when both factors are 1. In particular,

$$1 \times 1 = \{P(0, 0)\} = \{0\} = 1.$$

Intersection gives the same conjunction on these two representatives.

17 Class Theory Overview

This section summarizes the Morse–Kelley (MK) and von Neumann–Bernays–Gödel (NBG) principles used in the classical class-theoretic chapters. It connects the axiomatic kernel of Section 15 with family evaluation, sethood, and the union and intersection operations.

17.1 Classes, Sets, and Proper Classes

Every mathematical object in the first-order discourse is a *class*. The universal class of all sets is defined by

$$V \coloneqq^{\text{def.}} \{\, x \mid \top \,\}.$$

A class x is a *set* if and only if it belongs to some class ($x \in V \longleftrightarrow \exists Y,\ x \in Y$). Every set is a class; a *proper class* is a class X that is not a set ($X \notin V$). Consequently, only sets can appear on the left-hand side of membership ($z \in X \longrightarrow z \in V$).

17.2 Morse–Kelley Class Comprehension and Separation

For an admitted formula $\varphi(x)$, the class comprehension term $\{\, x \mid \varphi(x) \,\}$ satisfies the comprehension equivalence:

$$\forall z, \quad z \in \{\, x \mid \varphi(x) \,\} \longleftrightarrow (z \in V) \wedge \varphi(z).$$

The default sort of variables is Class. Full MK permits class quantification inside φ; NBG comprehension restricts its internal quantifiers to sets while allowing class parameters. In one-sorted syntax, a set quantifier can be written as a class quantifier restricted to V.

Notice the mandatory sethood guard $z \in V$: substituting the Russell class $R \coloneqq^{\text{def.}} \{\, x \mid x \notin x \,\}$ yields $R \in R \longleftrightarrow (R \in V) \wedge (R \notin R)$, which proves $\neg(R \in V)$ (i.e., R is a proper class) rather than a contradiction.

Given any set $a \in V$ and any class P (or formula φ), the *Separation Principle* (Subsets / Intersection with a Set) states that the intersection

$$a \cap P = \{\, x \in a \mid x \in P \,\} = \{\, x \mid (x \in a) \wedge (x \in P) \,\}$$

is again a set: $(a \in V) \longrightarrow (a \cap P \in V)$. In the axiom presentation used here, this is the heredity principle. In presentations based on limitation of size it can instead be derived from the relevant size axioms. A derivation of limitation of size must keep track of which of these principles it assumes, to avoid circularity.

17.3 What the minimal comprehension principles do not imply

Theorem 17.1 (A finite model of the minimal relational fragment)**.**
Consider classical one-sorted first-order logic in the language of equality and membership, with sethood defined by $\mathrm{Set}(x) \longleftrightarrow \exists Y\,(x \in Y)$. Take the following nonlogical assumptions only: class extensionality, the schema of set-guarded class comprehension, and existence of at least one set. This fragment has a model with two class objects and only one set. Consequently it does not imply the usual pairing or infinity axioms.

Proof.
Let the domain consist of two distinct objects e, v, and interpret membership by the single ordered pair (e, v). Thus e has no members, v has the sole member e, and the objects satisfying

Set are exactly e. Extensionality holds because the two objects have different members; the existence-of-a-set assertion holds because $e \in v$.
Fix any comprehension formula $\varphi(x)$ and any assignment of its parameters. The required extension $\{x : \mathrm{Set}(x) \wedge \varphi(x)\}$ is either empty or $\{e\}$, according as $\varphi(e)$ is false or true in this structure. It is represented respectively by e or v. This verifies every instance of the schema, including formulas with unrestricted quantifiers over the two class objects. The comprehension output variable is, as usual, fresh for φ; no satisfaction predicate is added to the language.
The only set is the empty object e. A set containing e, as required by pairing applied to e, e, would have to be v, which is not a set. Thus pairing fails. No set contains even the empty set, so no inductive set exists and the usual infinity axiom fails. Soundness of first-order logic gives the nonderivability conclusions. □
This is a diagnostic for exactly the listed relational assumptions, not a model of every system proposed in this book. Stronger principles about constructors, typed interpretation, reflection, or additional set existence can exclude this model. Any proposed interpretation of ZFC in a richer system must identify those additional principles and prove the required interpretation properties. Merely naming unrestricted class comprehension and one set does not supply the missing set-existence axioms.

Theorem 17.2 (Finite closure principles do not supply infinity)**.**
Work externally in ZF set theory, with its axiom of infinity. There is a set-sized model of the one-sorted class language satisfying full set-guarded class comprehension, extensionality, empty set, pairing, union, powerset, separation, and replacement for class functions, but not the axiom of infinity. In this model every internal set is externally finite. The result concerns these explicitly listed principles, not an unspecified stronger reflection or universe principle.

Proof.
Let $H = V_\omega$ be the set of hereditarily finite sets in the external theory, and take the class-object domain to be $\mathcal{P}(H)$. Interpret membership as ordinary membership restricted to that domain. Since H is transitive, every member of H is a subset of H and therefore an object of the structure. An object x belongs to another object exactly when $x \in H$: one implication follows because every object is a subset of H; for the other, the object $\{x\}$ witnesses membership. Thus internal sethood is precisely membership in H.
All members of an object are themselves in the domain, so external extensionality gives internal class extensionality. For any formula $\varphi(x)$ and parameters, including formulas quantifying over all class objects, its required comprehension extension is

$$\{x \in H : \varphi(x) \text{ holds in the structure}\}.$$

This is a subset of H, hence an object. The verification is external and applies instance by instance; it introduces no internal truth predicate.
The empty set belongs to H. Pairs of members of H, unions of members of H, and powersets of members of H again belong to H: all these constructions yield finite sets of hereditarily finite objects. Every subclass of an internal set a is an external subset of a finite hereditarily finite set, hence is again in H. Finally, a class function with internal set domain a has only finitely many inputs. Its image is a finite set of members of H, and therefore is in H. This proves class replacement, regardless of how the class function was defined. Ordinary finite graphs are also in H.
If an internal set were inductive, it would contain the external empty set and, by external induction, every finite von Neumann ordinal. Those ordinals are distinct, contradicting external finiteness of the internal set. Thus infinity fails. □

Corollary 17.1 (Even class limitation of size does not force infinity)**.**
The preceding hereditarily finite model also satisfies the following class-size principle: an internal class is proper if and only if it is in internal bijection with the universal class. Thus adding this principle to the listed closure axioms still does not imply infinity.

Proof.
Externally, H has a fixed bijection with ω. One explicit coding is the recursive finite-set code $c(x) = \sum_{y \in x} 2^{c(y)}$. Induction on finite ranks and uniqueness of binary expansion prove injectivity. Conversely, strong induction on n constructs a set with code n from the sets coding its binary exponents, all of which are smaller than n when $n > 0$. Hence c is bijective. This coding and the increasing enumeration below use no external choice axiom. The natural numbers here are external indices; they are not being asserted to form an internal set.
An object $A \subseteq H$ is an internal set exactly when it is externally finite: every finite subset of H is hereditarily finite, and every member of H is finite. If A is proper, its codes form an infinite subset of ω. Listing them in increasing order gives an external bijection $H \to A$. Its graph consists of ordinary ordered pairs of hereditarily finite sets, which are themselves hereditarily finite; hence that graph is a subset of H and is an internal class object. It witnesses the required internal bijection. Conversely, an internal bijection from H to an internal set would externally biject an infinite set with a finite one, which is impossible.
This also verifies the book's surjective-image formulation of limitation of size: for proper A, the inverse of the coded bijection $H \to A$ is an internal class function T with $H = T[A]$. If A is an internal set, every set-valued class function has finite image on A, so no such image can contain the infinite universal class H. □

Corollary 17.2 (Regularity and global choice in the same model)**.**
The hereditarily finite model also satisfies class foundation and has a class choice function on all nonempty internal sets. These principles therefore do not repair its failure of infinity.

Proof.
For a nonempty internal class $A \subseteq H$, the external finite ranks of its members form a nonempty subset of ω and have a least value. Choose one member a of that rank. If $b \in a \cap A$, then b has strictly smaller rank, a contradiction. Thus $a \cap A$ is empty. This verifies class foundation, and in particular set regularity. It makes no assertion that the class of all finite ordinals is an internal set.
For a nonempty $a \in H$, select its member with least Ackermann code $c(x)$. The code is injective, so the selected member is unique. The graph of this operation consists of ordered pairs of members of H, hence is a subset of H and an internal class. Its domain is exactly the nonempty internal sets and its value at a belongs to a. This is the claimed global choice function. All choices used to define it were uniquely determined by the fixed coding. □
The two models isolate different missing assumptions. The two-object model shows that comprehension and one set do not even give pairing. The hereditarily finite model shows that adding the listed set-closure principles still does not give infinity. These are relative model constructions in an external theory; neither is a claim that the book's stronger proposed systems have the same models. They provide concrete tests for any claimed derivation of set-existence principles.

17.4 Binders: Quantifiers, Class Comprehension, and Descriptors

A *binder* is a syntactic construction that binds a variable. The following list is a guide to the notation encountered in different sections, not a claim that every listed binder has the same

semantics in all of them. The default class sort applies to the classical setting; partial and uncertain semantics supply their own interpretation rules.

1. **Formula-Valued Logical Quantifiers**: universal $(\forall x,\ \varphi(x))$, existential $(\exists x,\ \varphi(x))$, uniqueness $(\exists! x,\ \varphi(x)$ and $!x,\ \varphi(x))$, and bounded quantifiers $(\forall(x \in Y),\ \varphi(x)$ and $\exists(x \in Y),\ \varphi(x))$.
2. **Class Comprehension Binder (κ-binder)**: $\{\, x \mid \varphi(x) \,\}$ (and generalized comprehension $\{\, t(\vec{x}) \mid^{\vec{x}} \varphi(\vec{x}) \,\}$), binding a class variable $x : \mathsf{Class}$ to form a class term.
3. **Definite Description (ι-binder)**: $\iota x,\ \varphi(x)$, denoting the unique class satisfying $\varphi(x)$ when $\exists! x,\ \varphi(x)$ holds (with its behavior in other cases specified by the local descriptor convention).
4. **Choice and Partial Descriptors** (τ, ϵ, ε; Section 51): the descriptor experiment distinguishes an arbitrary-class default for τ, an empty-class default for ϵ, and no default for partial ε. These are local conventions. In particular, an undefined expression is not automatically identified with a term denoting the empty class.

18 Sethood and elementary class reasoning

This section derives basic membership and inclusion facts from comprehension, extensionality, and the definition of sethood. The notation X set means that X is a member of some class. In particular, V contains all sets, while every class is a subclass of V; these are statements about membership and inclusion respectively.

Theorem 18.1 (example of class builder use)**.**

$$(\{\, y \mid \psi \,\} \text{ nonempty}) \longleftrightarrow (\exists (y \text{ set}),\ \psi)$$

Proof. A class is nonempty exactly when it has a member. By comprehension, $z \in \{\, y \mid \psi \,\}$ holds exactly when z is a set and $\psi[z/y]$ holds. Thus a member of the comprehension supplies the required witness, and any such set witness is a member of the comprehension. □

Theorem 18.2 (V is maximal)**.**
Every class is a subclass of the class of all sets.

$$(C \subseteq \mathrm{V})$$

Proof.
Claim: $(\forall x \in C,\ x \in \mathrm{V})$.
Fix x. Assume $(x \in C)$. To show $(x \in \mathrm{V})$.

$$(x \in C) \implies (x \text{ set}) \qquad \text{(by thm:in class then set)}$$

$$(x \text{ set}) \ \&\ \top \implies (x \in \mathrm{V}) \qquad \text{(by def:universal class)}$$

The claim is proven.

$$(\forall x \in C,\ x \in \mathrm{V}) \implies (C \subseteq \mathrm{V}) \qquad \text{(by def:subclass)}$$

□

Theorem 18.3 (sets are in V)**.**

$$(X \text{ set}) \longleftrightarrow (X \in \mathrm{V})$$

Proof.
($\Leftarrow$) Assume $(X \in \mathrm{V})$.

$$(X \in \mathrm{V}) \implies (\exists W,\ X \in W).$$
$$(\exists W,\ X \in W) \implies (X \text{ set}).$$

($\Rightarrow$) Assume $(X \text{ set})$.

$$(X \text{ set}) \ \ \& \ \ \top[X/W] \implies (X \in \{\, x \mid \top[x/W] \,\})$$
$$(X \in \{\, x \mid \top[x/W] \,\} = \{\, x \mid \top \,\} = \mathrm{V}) \implies (X \in \mathrm{V})$$

□

Theorem 18.4 (antisymmetry of inclusion)**.**

$$(A \subseteq B) \wedge (B \subseteq A) \longrightarrow (A = B)$$

Proof.
Claim: $(\forall x,\ x \in A \longleftrightarrow x \in B)$.
Fix x.
($\Rightarrow$)
Assume $(x \in A)$. To show: $(x \in B)$

$$(x \in A) \ \ \& \ \ (A \subseteq B) \implies (x \in B) \qquad \text{(by thm:element of subclass)}$$

($\Leftarrow$)
Assume $(x \in B)$. To show: $(x \in A)$

$$(x \in B) \ \ \& \ \ (B \subseteq A) \implies (x \in A) \qquad \text{(by thm thm:element of subclass)}$$

The claim has been proven.

$$(\forall x,\ x \in A \longleftrightarrow x \in B) \implies (A = B) \qquad \text{(by axm:extensionality)}$$

□

Theorem 18.5 (empty set is minimal)**.**
For every class A,

$$(\varnothing \subseteq A) \qquad \text{(thm:empty class is subclass)}$$

Proof.
Claim:

$$(\forall x \in \varnothing,\ x \in A)$$

Assume $(x \in \varnothing)$.

$$(x \in \varnothing) \implies \bot$$
$$\bot \implies (x \in A)$$

The claim is proven.

$$(\forall x \in \varnothing, x \in A) \implies (\varnothing \subseteq A) \qquad \text{(by def:emptyclass)}$$

□

Theorem 18.6.

$$\neg(\exists w, w \in A) \longrightarrow (A = \varnothing)$$

Proof.
Assume $\neg\exists w,\ w \in A$. To show: $(A = \varnothing)$.
$(\supseteq)$

$$(A \supseteq \varnothing) \qquad \text{(by thm:empty class is subclass)}$$

$(\subseteq)$
Claim: $(\forall x \in A,\ x \in \varnothing)$
Fix x. Assume $(x \in A)$. To show: $\bot$

$$(x \in A) \implies (\exists x,\ x \in A)$$

$$(\exists x,\ x \in A)\ \&\ (\neg\exists x,\ x \in A) \implies \bot$$

$$\bot \implies (x \in \varnothing)$$

The claim is proven.

$$(\forall x \in A,\ x \in \varnothing) \implies (A \subseteq \varnothing) \qquad \text{(by def of subclass def:subclass)}$$

$$(A \subseteq \varnothing)\ \&\ (A \supseteq \varnothing) \implies (A = \varnothing) \qquad \text{(by axm axm:extensionality)}$$

□

Theorem 18.7 (emptiness criterion)**.**

$$\neg\exists w,\ w \in A \longleftrightarrow (A \text{ empty})$$

Proof.
By the preceding theorem, having no member implies equality with $\varnothing$. Conversely, equality with $\varnothing$ implies having no member, since the empty class has none. This is exactly the stated criterion for A empty. □

Theorem 18.8 (nonemptiness criterion)**.**

$$\exists w,\ w \in A \longleftrightarrow (A \text{ nonempty})$$

Proof.
Negate the emptiness criterion and use classical double-negation elimination. Thus $A \neq \varnothing$ holds exactly when $\exists w\,(w \in A)$. □

Theorem 18.9 (membership in a subclass)**.**
Any element of a subclass is in the class.

$$(x \in A) \wedge (A \subseteq B) \longrightarrow (x \in B)$$

Proof.

$$(A \subseteq B) \implies (\forall x \in A,\ x \in B) \qquad \text{(by def of subclass def:subclass)}$$

$$(x \in A)\ \&\ (\forall x \in A,\ x \in B) \implies (x \in B) \qquad \text{(by restricted specification elimination)}$$

□

Theorem 18.10 (is in class then is set)**.**
For any terms A and B,

$$(B \in A) \longrightarrow (B \text{ set})$$

Proof.
Assume $(B \in A)$. To show: $(B$ set$)$. Choose a variable Y not free in either term A or B.

$$(B \in A) \implies (\exists Y,\ B \in Y). \qquad \text{(by right–exists–intro)}$$

$$(\exists Y,\ B \in Y) \implies (B \text{ set}). \qquad \text{(introduced by set's definition)}$$

□

Theorem 18.11 (a class with exactly one member)**.**

$$(w \in A) \wedge (!a,\ a \in A) \longrightarrow (\{w\} = A)$$

Proof.
Assume $(w \in A)$ and $(!a,\ a \in A)$. Membership implies that w and every other member of A are sets, so singleton introduction is legitimate in the following argument.
$(\subseteq)$ Assume $(q \in \{w\})$.

$$(q \in \{w\}) \implies (q = w).$$

$$(w \in A)\ \&\ (q = w) \implies (q \in A).$$

$(\supseteq)$ Assume $(a \in A)$.

$$(!a,\ a \in A)\ \&\ (a \in A)\ \&\ (w \in A) \implies (a = w).$$

$$(a = w) \implies (\{a\} = \{w\})$$

$$(a \in \{a\})\ \&\ (\{a\} = \{w\}) \implies (a \in \{w\})$$

□

19 Elementary theorems about union and intersection

These results use class comprehension with sets as members. Thus $\bigcup A$ consists of the sets belonging to some member of A, whereas $\bigcap A$ consists of the sets belonging to every member of A. In particular, $\bigcup \varnothing = \varnothing$ and $\bigcap \varnothing = \mathrm{V}$. The nonemptiness hypotheses below account for this difference.

Note 19.1 (Comparison with the newer Obelisk catalog)**.**
The catalog definitions `DefUnaryUnion` (3.19) and `DefUnaryIntersection` (3.20) use the same existential and universal membership conditions as this section. Its library result `NBG-093` states the binary inclusion $A \cap B \subseteq A \cup B$. That binary result requires no nonemptiness hypothesis, whereas the unary inclusion $\bigcap A \subseteq \bigcup A$ below does: at $A = \varnothing$ the two sides are V and $\varnothing$.
There is a separate difference between catalog variants concerning singletons. The primary comprehension definition `DefSingletonSet` (3.48) agrees with this section. The exported library entries `NBG-060` and `NBG-071` omit the sethood condition when introducing membership in a singleton. Under the present set-member semantics, only the restricted assertion w set $\longrightarrow w \in \{w\}$ is used. For a proper class w, comprehension instead gives $\{w\} = \varnothing$. Thus the singleton-intersection theorem below explicitly assumes that w is a set.

Theorem 19.1 (member then subset of union)**.**

$$\Big(B \in A\Big) \longrightarrow \Big(B \subseteq \bigcup A\Big)$$

Proof.
Assume $(b \in B)$.
To show: $(b \in \bigcup A)$.

$$(b \in B) \implies (b \text{ set}) \qquad \text{(by thm:in class then set)}$$
$$(B \in A) \;\&\; (b \in B) \implies (\exists a \in A,\ b \in a)$$
$$(b \text{ set}) \;\&\; (\exists a \in A,\ b \in a) \implies \Big(b \in \bigcup A\Big)$$

□

Theorem 19.2 (intersection is contained in each member)**.**

$$\Big(B \in A\Big) \longrightarrow \Big(\bigcap A \subseteq B\Big)$$

Proof.
Assume $(B \in A)$, $(x \in \bigcap A)$. To show: $(x \in B)$.

$$\Big(x \in \bigcap A\Big) \implies \Big(x \text{ set}\Big) \;\&\; \Big(\forall a \in A : x \in a\Big)$$
$$(\forall a \in A,\ x \in a) \;\&\; (B \in A) \implies (x \in B)$$

□

Theorem 19.3 (intersection of a singleton set)**.**
For a set w,

$$\Big(\bigcap \{w\} = w\Big)$$

Proof.
$(\subseteq)$
$(x \in \bigcap\{w\}) \implies (\forall m \in \{w\} : x \in m)$.
$(\forall m \in \{w\} : x \in m)$ & $(w \in \{w\}) \implies (x \in w)$.
$(\supseteq)$
Fix x. Assume $(x \in w)$. To show: $(x \in \bigcap\{w\})$.
$(x \in w) \implies (x \text{ set})$.
Claim: $(\forall m \in \{w\} : x \in m)$.
Fix m. Assume $m \in \{w\}$.
$(m \in \{w\}) \implies (m = w)$.
$(x \in w)$ & $(m = w) \implies (x \in m)$
The claim is proven.
$(x \text{ set})$ & $(\forall m \in \{w\} : x \in m) \implies (x \in \bigcap\{w\})$. □

Theorem 19.4 (intersection of singleton)**.**
Intersection of a singleton is in it.

$$\Big(A \text{ singleton}\Big) \longrightarrow \Big(\Big(\bigcap A\Big) \in A\Big)$$

Proof.

$$(A \text{ singleton}) \implies (\exists! w, w \in A)$$
$$(w \in A) \ \& \ (!a : a \in A) \implies \{w\} = A \qquad \text{(by extensionality and uniqueness)}$$
$$(\{w\} = A) \implies \Big(\bigcap\{w\} = \bigcap A\Big).$$
$$\Big(\bigcap\{w\} = w\Big) \ \& \ \Big(\bigcap\{w\} = \bigcap A\Big) \implies \Big(\bigcap A = w\Big)$$
$$\Big(\bigcap A = w\Big) \ \& \ (w \in A) \implies \Big(\bigcap A \in A\Big)$$

□

Theorem 19.5 (intersection of nonempty is subset of union)**.**

$$\Big(A \text{ nonempty}\Big) \longrightarrow \Big(\Big(\bigcap A\Big) \subseteq \Big(\bigcup A\Big)\Big)$$

Proof.

$$(A \text{ nonempty}) \implies (\exists B, B \in A)$$
$$(B \in A) \implies \Big(\bigcap A\Big) \subseteq B$$
$$(B \in A) \implies \Big(B \subseteq \bigcup A\Big)$$
$$\Big(\bigcap A\Big) \subseteq B \ \& \ \Big(B \subseteq \bigcup A\Big) \implies \Big(\bigcap A \subseteq \bigcup A\Big)$$

□

Theorem 19.6 (subclasses of a set belong to its powerclass)**.**

$$(A \text{ set}) \wedge (B \subseteq A) \longrightarrow (B \in \mathcal{P}A)$$

Proof.

$$(A \text{ set}) \;\&\; (B \subseteq A) \implies (B \text{ set}) \qquad \text{(by axm:heredity)}$$
$$(B \text{ set}) \;\&\; (B \subseteq A) \implies (B \in \mathcal{P}A) \qquad \text{(by def:powerclass)}$$

□

Theorem 19.7 (Union and powerclass comparison)**.**
For arbitrary classes A, B,

$$\bigcup A \subseteq B \longleftrightarrow A \subseteq \mathcal{P}B.$$

This concerns the powerclass of B; no assertion that it is a set is needed.

Proof.
If $\bigcup A \subseteq B$ and $a \in A$, then every element of a belongs to $\bigcup A$, hence to B. Membership already makes a a set, so $a \in \mathcal{P}B$. Conversely, if $A \subseteq \mathcal{P}B$ and $x \in \bigcup A$, choose $a \in A$ with $x \in a$. Since $a \in \mathcal{P}B$, we have $a \subseteq B$, and hence $x \in B$. The empty case is included: both inclusions hold when $A = \varnothing$. □
The transitivity equivalence below is the special case $B = A$. The general form explains the different roles of these operations: union collects members of members, while powerclass collects all set subclasses of the proposed bound.

Theorem 19.8 (Two descriptions of transitivity)**.**

$$\Big(A \subseteq \mathcal{P}A\Big) \longleftrightarrow \Big(\bigcup A \subseteq A\Big) \qquad \text{(thm:UACA=ACPA)}$$

Proof.
($\Leftarrow$)
Assume $(\bigcup A \subseteq A)$. To show:$(A \subseteq \mathcal{P}A)$.
Assume $(x \in A)$. To show: $(x \in \mathcal{P}A)$.
Claim: $(x \subseteq A)$.
Assume $(y \in x)$. To show: $(y \in A)$.

$$\Big(y \in x\Big) \;\&\; \Big(x \in A\Big) \implies \Big(y \in \bigcup A\Big)$$
$$\Big(y \in \bigcup A\Big) \;\&\; \Big(\bigcup A \subseteq A\Big) \implies \Big(y \in A\Big)$$

The claim is proven.

$$(x \in A) \implies (x \text{ set}) \qquad \text{(by thm:in class then set)}$$
$$(x \text{ set}) \;\&\; (x \subseteq A) \implies (x \in \mathcal{P}A)$$

($\Rightarrow$)
Assume ($A \subseteq \mathcal{P}A$). To show:($\bigcup A \subseteq A$).
Assume ($x \in \bigcup A$). To show:($x \in A$).

$$\left(x \in \bigcup A\right) \implies \left(\exists y \in A : x \in y\right)$$

$$(y \in A) \ \ \& \ \ (A \subseteq \mathcal{P}A) \implies (y \in \mathcal{P}A)$$

$$(y \in \mathcal{P}A) \implies (y \subseteq A)$$

$$(x \in y) \ \ \& \ \ (y \subseteq A) \implies (x \in A)$$

□

Theorem 19.9 (transitive through union)**.**

$$\left(A \text{ transitive}\right) \longleftrightarrow \left(\bigcup A \subseteq A\right)$$

Proof.
By thm:UACA=ACPA. □

Theorem 19.10 (intersection of a nonempty transitive class)**.**

$$\left(A \text{ nonempty}\right) \wedge \left(A \text{ transitive}\right) \longrightarrow \left(\left(\bigcap A\right) \subseteq A\right)$$

Proof.
By thm:intersection of nonempty is subset of union and thm:transitive through union. □

20 Relational image and glare

In this section, relations have set arguments: $F \subseteq \mathrm{V} \times \mathrm{V}$. The image uses an existential condition; the author's term *glare* uses the corresponding universal condition. We write F^{-1} and $F^{\top}$ interchangeably for the converse relation. The remarks about families examine a different encoding and should be read as comparisons, rather than as definitions of relational membership. Throughout the relational calculations, the domain and range are graph projections:

$$\mathrm{dom}(F) = \{\, x \mid \exists y \in \mathrm{V},\ (x, y) \in F \,\}, \qquad \mathrm{ran}(F) = \{\, y \mid \exists x \in \mathrm{V},\ (x, y) \in F \,\}.$$

These are not defined by testing the sethood of a totalized evaluation. In particular, the domain of ΣF below is its relational domain, even when that relation is not a functional or canonical family.

Definition 20.1 ((relational) membership relation)**.**

$$\mathrm{E} \coloneqq \{\, (x, y) \mid (y \in \mathrm{V}) \wedge (x \in y) \,\}$$

The guard on y is needed: $x \in y$ makes x a set, but y may be proper. Thus E is membership restricted to two set coordinates, agreeing with the ordinal chapter's relation E. It does not replace primitive membership of a set in an arbitrary class. Ordinary set-pair coding is applied only to the guarded set coordinates, without relying on any off-domain pairing convention.

Remark 20.1 (Membership and family evaluation)**.**
Membership is generally one-to-many, so it is not itself the graph of a single-valued function. Encoding all possible outputs by a family and then taking their union loses the individual membership relation: for a fixed set x, every set q belongs to some set Y containing x (take $Y = \{x, q\}$). Thus the union of those possible outputs is V.

Definition 20.2 ((relational) inverse membership relation)**.**

$$\ni \coloneqq \{\, (y, x) \mid (y \in \mathrm{V}) \wedge (x \in y) \,\} = \mathrm{E}^{-1}$$

Remark 20.2 (Inverse membership and family evaluation)**.**
Likewise, the inverse membership relation lists the individual members of a set y. Combining those outputs by union produces $\bigcup y$. This union-valued operation is a useful family evaluation, but it does not retain the graph of the original one-to-many relation.

Definition 20.3 (relational image)**.**

$$F[A] \coloneqq \{\, b \mid \exists a \in A,\ (a, b) \in F \,\}$$

Note 20.1 (Graph image and evaluation image)**.**
The newer Obelisk entry `DefRelationalImage` (3.26) defines an *evaluation image*,

$$\{\, b \mid \exists a \in A,\ f(a) = b \,\},$$

using its own evaluation operation. The definition immediately above is the *graph image*, which retains each output separately. They agree for an ordinary single-valued graph on inputs in its domain, but can differ for a relation with several outputs.
For example, put $0 = \varnothing$, $1 = \{0\}$, and $F = \{(0, 0), (0, 1)\}$. The graph image of $\{0\}$ is $\{0, 1\}$. Union evaluation combines the two outputs into $F(0) = \bigcup\{0, 1\} = 1$, so its evaluation image is $\{1\}$. The image–union and glare–intersection theorems here use the graph convention throughout; the catalog's evaluation image is a separately scoped construction.

Definition 20.4 (relational glare)**.**

$$F\langle A\rangle \coloneqq \{\, b \mid \forall a \in A,\ (a,b) \in F \,\}$$

Theorem 20.1 (Union as a relational image)**.**

$$\exists[A] = \bigcup A$$

$$(\text{i.e. } \mathrm{E}^{\top}[A] = \bigcup A)$$

Proof.
($\subseteq$)
Assume $w \in \mathrm{E}^{\top}[A]$.
Therefore $\exists a \in A,\ (a,w) \in \mathrm{E}^{\top}$. Choose such a witness.
By the definition of converse, $(w,a) \in \mathrm{E}$.
The membership relation then gives $w \in a$.
Since $w \in a$ and $a \in A$, $w \in \bigcup A$.
($\supseteq$)
Assume $w \in \bigcup A$.
Therefore $\exists a \in A,\ w \in a$. Choose such a witness.
By the definition of the membership relation, $(w,a) \in \mathrm{E}$. Taking the converse gives $(a,w) \in \mathrm{E}^{\top}$.
From $(a,w) \in \mathrm{E}^{\top}$ and $(a \in A)$, we have $(w \in \mathrm{E}^{\top}[A])$.
□

Theorem 20.2 (Intersection as a relational glare)**.**

$$\exists\langle A\rangle = \bigcap A$$

$$(\text{i.e. } \mathrm{E}^{\top}\langle A\rangle = \bigcap A)$$

Proof.
Fix a set v. This restriction supplies the sethood condition in each comprehension, including when A is empty. Then

$$\begin{aligned} v \in \mathrm{E}^{\top}\langle A\rangle &\iff \forall a \in A,\ (a,v) \in \mathrm{E}^{\top} \\ &\iff \forall a \in A,\ (v,a) \in \mathrm{E} \\ &\iff \forall a \in A,\ v \in a \\ &\iff v \in \bigcap A \end{aligned}$$

□

20.1 The order structure behind glare

The universal quantifier in glare reverses inclusion in the input. This is a different operation from relational image, which preserves it. The distinction yields a useful closure construction without a choice axiom.

Proposition 20.1 (Image and glare under unions)**.**
For classes A, A', B and a relation $F \subseteq \mathrm{V} \times \mathrm{V}$,

$$A \subseteq A' \longrightarrow F[A] \subseteq F[A'], \qquad A \subseteq A' \longrightarrow F\langle A' \rangle \subseteq F\langle A \rangle.$$

Moreover,

$$F[A \cup B] = F[A] \cup F[B], \qquad F\langle A \cup B \rangle = F\langle A \rangle \cap F\langle B \rangle.$$

The empty-input identities are $F[\varnothing] = \varnothing$ and $F\langle \varnothing \rangle = \mathrm{V}$.

Proof.
An existential witness in A remains a witness in A'. A universal condition on A' implies its restriction to A. Existential quantification over a binary union is a disjunction of the two existential conditions, while universal quantification over that union is their conjunction. The empty cases follow from absence of witnesses and vacuity, respectively. All resulting memberships concern sets. □

Proposition 20.2 (The polarity induced by a relation)**.**
Write $p(A) = F\langle A \rangle$ and $q(B) = F^{\top}\langle B \rangle$. Then

$$B \subseteq p(A) \longleftrightarrow A \subseteq q(B).$$

The operator $c(A) = q(p(A))$ is extensive, monotone, and idempotent:

$$A \subseteq c(A), \qquad A \subseteq A' \longrightarrow c(A) \subseteq c(A'), \qquad c(c(A)) = c(A).$$

Proof.
Both sides of the equivalence say that $(a, b) \in F$ for every $a \in A$ and $b \in B$. Applying it to $B = p(A)$ gives extensivity. Since both p and q reverse inclusion, their composite preserves it. The symmetric argument gives $B \subseteq p(q(B))$ for every B. Taking $B = p(A)$ yields $p(A) \subseteq p(q(p(A)))$. Applying p to $A \subseteq q(p(A))$ gives the reverse inclusion. Hence $p(q(p(A))) = p(A)$, and applying q proves idempotence. □

Example 20.1 (An order relation)**.**
Let P be a partially ordered set and $F = \{(x, y) \in P \times P : x \leq y\}$. To keep bounds inside P, use $p_P(A) = P \cap F\langle A \rangle$ and $q_P(B) = P \cap F^{\top}\langle B \rangle$ for $A, B \subseteq P$. The same proof gives a closure operator $q_P p_P$ on subsets of P: it sends a subset to the lower bounds of its upper bounds. If A has a least upper bound s in P, this closure is exactly $\{x \in P : x \leq s\}$. Indeed, s is one of the upper bounds, so a lower bound of all upper bounds lies below s; conversely, s lies below every upper bound, so every $x \leq s$ lies below them all. The restrictions to P matter for empty inputs: unrestricted glare of an empty class is V, whereas the upper bounds of the empty subset in P are precisely P.

Proposition 20.3 (The right adjoint of relational image)**.**
For a class B define

$$F^{\forall}(B) = \{\, x \mid \forall y \in \mathrm{V},\ (x, y) \in F \longrightarrow y \in B \,\}.$$

Then, for all classes A, B,

$$F[A] \subseteq B \longleftrightarrow A \subseteq F^{\forall}(B).$$

Proof.
The left condition says that every output paired with any $a \in A$ belongs to B. This is exactly the statement that each such a satisfies the defining universal condition on the right. Both implications use the same pair witnesses, so no functionality or choice assumption is needed. □
Unlike glare, this operation is monotone in B: enlarging the permitted output class makes its universal condition easier to satisfy. It also contains every input outside $\mathrm{dom}(F)$ by vacuity. Thus glare, direct image, and this universal inverse operation are three distinct constructions. For example, for the identity relation on a set P and $B \subseteq P$, one has $F^{\forall}(B) = (\mathrm{V} \setminus P) \cup B$, whereas its restriction to P is just B.

20.2 Further identities for the relational encoding

For the following identities, fix the expansion convention explicitly. For a relation $F \subseteq \mathrm{V} \times \mathrm{V}$, define

$$\Sigma F = \{\, (x, z) \mid \exists y \in \mathrm{V},\ (x, y) \in F \wedge z \in y \,\}.$$

Thus $(\Sigma F)[\{x\}]$ is the union of all graph outputs at x. This definition uses only set arguments and outputs; it does not presume that an absent value has any particular total-evaluation default.

Theorem 20.3 (Domain decomposition)**.**
If F is functional, then

$$\mathrm{dom}(\Sigma F) \cup F^{\top}[\{\varnothing\}] = \mathrm{dom}(F), \qquad \mathrm{dom}(\Sigma F) \cap F^{\top}[\{\varnothing\}] = \varnothing.$$

Proof.
An input belongs to $\mathrm{dom}(\Sigma F)$ precisely when it has a nonempty output in F: an element of that output supplies the witness z in the definition. It belongs to $F^{\top}[\{\varnothing\}]$ precisely when $(x, \varnothing) \in F$. Every output is either empty or nonempty, so the union is $\mathrm{dom}(F)$, even without functionality. If an input belonged to both pieces, functionality would identify its nonempty output with the empty set, a contradiction. This proves disjointness. □

Theorem 20.4 (Recovery from nonempty values)**.**
If $F \subseteq \mathrm{V} \times \mathrm{V}$ is functional and $\varnothing \notin \mathrm{ran}(F)$, then

$$F = \{\, (x, (\Sigma F)[\{x\}]) \mid x \in \mathrm{dom}(\Sigma F) \,\}.$$

Proof.
For $(x, y) \in F$, functionality gives

$$z \in (\Sigma F)[\{x\}] \longleftrightarrow z \in y$$

for every set z: any other output used in the expansion must equal y. Thus the fibre is the set y. The nonempty-value hypothesis and domain decomposition give $\mathrm{dom}(\Sigma F) = \mathrm{dom}(F)$. Replacing each output by its equal fibre therefore gives exactly the displayed graph. □

Remark 20.3 (What expansion forgets)**.**
For an arbitrary functional F, the same recovery formula is valid if its indexing domain is replaced by $\mathrm{dom}(F)$. The domain is then extra data: expansion alone forgets empty values. In particular, the empty relation and $\{(\varnothing, \varnothing)\}$ both have empty expansion, although their domains differ. Functionality also matters: the relation $\{(0, 0), (0, 1)\}$, with $1 = \{0\}$, has the same expansion as $\{(0, 1)\}$ and cannot be recovered from that expansion.

Theorem 20.5 (Image of a composite relation)**.**

$$(F \circ G)[X] = F[G[X]]$$

Proof.
For a set z, membership in the left side means that there are $x \in X$ and a set y with $(x, y) \in G$ and $(y, z) \in F$. This is exactly the condition that $y \in G[X]$ for some y with $(y, z) \in F$, or equivalently $z \in F[G[X]]$. Class extensionality gives the equality. □

21 A choice function and the epsilon operator

This is a local proposal for a choice-based descriptor. Let c be a choice function on nonempty sets, and put $S_\varphi = \{\, y \mid \varphi \,\}$. The formula below defines a chosen witness whenever S_φ is a nonempty set. A nonempty proper class lies outside the stated domain of c; merely having a witness does not establish that S_φ is a set. For empty inputs, a separate partial-value or default-value convention is needed.

Definition 21.1 (Choice-based epsilon on nonempty sets)**.**
For $S_\varphi \in \mathrm{V} \setminus \{\varnothing\}$, define

$$(\varepsilon y,\ \varphi) = c\,(\{\, y \mid \varphi \,\}) \qquad \text{(def:epsilon operator for sets)}$$

Here the constant c denotes a global choice class function (see def:globalchoice):

$$\mathrm{c} : \mathrm{V} \setminus \{\varnothing\} \to \mathrm{V}$$

$$\mathrm{c}(X) \in X$$

In the class framework with the ordinal prerequisites used in the Lévy chapter, limitation of size supplies such a function by the construction below. The union and closure consequences have their own arguments and prerequisites in the corresponding chapters; they are not additional consequences of this definition alone.

Remark 21.1 (Why the original class extension needs more data)**.**
The intended equivalence

$$((\epsilon y, \varphi) \text{ set}) \longleftrightarrow (\exists (y \text{ set}), \varphi)$$

cannot follow merely by applying the preceding c to the class $\{y : \varphi\}$. A nonempty proper class is not an argument in the stated domain of that class function. Moreover, an empty-set default would itself be a set and would make the left side true even when there is no witness. The next construction proves the intended statement in a precise stronger setting, using a proper-class default. It is a class-parameter operation, not an ordinary class graph with proper classes as members.

Theorem 21.1 (An ordinal-ranked descriptor for arbitrary classes)**.**
Work in the classical class setting with comprehension, V a proper class, and a fixed injective class function $j : \mathrm{V} \to \mathrm{On}$. Assume the usual ordinal fact that every nonempty class of ordinals has a least member. For every class C, define $E_j(C)$ as follows:

$$E_j(C) = \begin{cases} \text{the unique } c \in C \text{ with least } j(c), & C \neq \varnothing, \\ \mathrm{V}, & C = \varnothing. \end{cases}$$

This operation is well defined and extensional in C, and

$$E_j(C) \text{ set} \longleftrightarrow C \neq \varnothing.$$

When $C \neq \varnothing$, it also satisfies $E_j(C) \in C$. Its restriction to nonempty sets is represented by an ordinary global choice class function.

Proof.
Suppose C is nonempty. Its ordinal image $j[C]$ is a nonempty class, so it has a least member α. Some $c \in C$ has $j(c) = \alpha$, and injectivity of j makes this c unique. It is a set because it is a member of C. If C is empty, the assigned value is V, which is not a set by hypothesis. This proves both the equivalence and the witness property. Equal classes have the same ordinal images and hence the same selected member, or are both empty, so the operation is extensional. For the restriction, class comprehension forms the graph

$$G = \{(A, c) : A \in \mathrm{V},\ A \neq \varnothing,\ c \in A,\ \forall z \in A,\ j(c) \leq j(z)\}.$$

Both coordinates here are sets, so ordinary set-pair coding suffices. The preceding existence and uniqueness show that G is a class function on all nonempty sets and $G(A) \in A$. □
The least-member fact used here does not assume that $j[C]$ is a set: given one $\alpha_0 \in j[C]$, separate $j[C] \cap (\alpha_0 + 1)$ inside the set ordinal $\alpha_0 + 1$ and take its least element. It is also least in the whole class. Under the prerequisites used in the Lévy chapter, the book's surjective-image limitation-of-size axiom supplies the injection j by choosing the least ordinal preimage of each set. Thus the present hypothesis is a specific established route, not an assertion that every arbitrary choice function already acts on proper classes.

Corollary 21.1 (The proposed witness equivalence, now with a setting)**.**
For a formula φ and its bound variable y, put

$$(\epsilon_j y, \varphi) := E_j(\{y : \varphi\}).$$

Under the preceding hypotheses,

$$((\epsilon_j y, \varphi) \text{ set}) \longleftrightarrow (\exists (y \text{ set}), \varphi).$$

Whenever these conditions hold, the selected value satisfies φ under capture-avoiding substitution for y.

Proof.
Guarded class comprehension says that $\{y : \varphi\}$ is nonempty exactly when there is a set witness for φ. Apply the theorem. In that case its membership conclusion and comprehension give φ of the selected member. The substitution assertion uses the ordinary total class-language semantics and its freshness conditions. □

Corollary 21.2 (Preserving an already chosen set selector)**.**
If c is the global choice function originally fixed on nonempty sets, then the class-parameter operation

$$E_{c,j}(C) = \begin{cases} c(C), & C \text{ is a nonempty set}, \\ E_j(C), & \text{otherwise} \end{cases}$$

agrees with c throughout its original domain and has the same sethood and witness properties as E_j.

Proof.
For a nonempty set, the choice property gives a member of C, which is a set. For a nonempty proper class or the empty class, the preceding theorem gives exactly the required conclusion. These cases exhaust the class arguments in the stated classical setting. □
The ranked operator depends on the fixed j; a different ranking may select a different witness. Its restriction G need not equal a previously chosen c, whereas the patched operation $E_{c,j}$ does preserve that choice. Neither variation changes the proved witness property. The empty-case value is a genuine proper class, so this construction is not an interpretation of a descriptor whose values are required always to be sets. It also makes no claim that one class graph can collect pairs with arbitrary proper-class inputs. These distinctions retain the proposed idea without identifying it with the finite candidate semantics or with PFOL undefinedness.

22 Ordinals and induction principles

This section organizes the ordinal constructions around an arbitrary relation, then specializes to membership. The aim is to make the assumptions behind each induction principle visible. The notation distinguishes a transitive relation from a class closed under taking predecessors, and an ordinal class from an ordinal set. The elementary comparison results come first; the later recursion, collapse, and cardinal results state their additional set-existence and choice assumptions explicitly. They are not consequences of ordinal notation alone.

22.1 Ordinal classes in general setting

Let A be a class, possibly proper, and let $R \subseteq \mathrm{V} \times \mathrm{V}$ be a relation on sets. Write xRy for $(x, y) \in R$. Relational images use the orientation $R[S] = \{\, y \mid \exists x \in S,\ xRy \,\}$, so $R^{\top}[\{y\}]$ is the class of predecessors of y. The experimental family-based notation immediately below is an alternative representation; subsequent relation proofs use the ordinary pair-membership interpretation unless explicitly stated.

Definition 22.1 (related)**.**

$$ARB \overset{\text{def.}}{\longleftrightarrow} (A, B) \in R \tag{def:related}$$

Definition (related (alternative family representation))**.**

$$ARB \overset{\text{def.}}{\longleftrightarrow} \exists (i \text{ set}),\ (R((i, 0)) = A) \wedge (R((i, 1)) = B) \tag{def:relatedexper}$$

Definition 22.2 (epsilon relation)**.**

$$\mathrm{E} \overset{\text{def.}}{\coloneqq} \{\, (x, y) \mid (y \text{ set}) \wedge (x \in y) \,\} \tag{def:epsilonRelation}$$

Definition 22.3 (irreflexive on)**.**

$$(R \text{ Irr } A) \overset{\text{def.}}{\longleftrightarrow} (\forall x \in A,\ \neg(xRx)) \tag{def:irreflexiveOn}$$

Definition 22.4 (relation is transitive on)**.**

$$(R \text{ Trans } A) \overset{\text{def.}}{\longleftrightarrow} (\forall x \in A \forall y \in A \forall z \in A,\ xRy \wedge yRz \longrightarrow xRz) \tag{def:transitiveOn}$$

Definition 22.5 (relation is strictly partially ordered on)**.**

$$(R \text{ Po } A) \overset{\text{def.}}{\longleftrightarrow} ((R \text{ Irr } A) \wedge (R \text{ Trans } A)) \tag{def:partiallyOrderedOn}$$

Definition 22.6 (relation is linear on)**.**

$$(R \text{ Lin } A) \overset{\text{def.}}{\longleftrightarrow} (\forall x \in A \forall y \in A : (xRy) \vee (x = y) \vee (yRx)) \tag{def:linearOn}$$

Definition 22.7 (relation is strictly linearly ordered on)**.**

$$(R \text{ Or } A) \overset{\text{def.}}{\longleftrightarrow} ((R \text{ Po } A) \wedge (R \text{ Lin } A)) \tag{def:sOn}$$

Definition 22.8 (relation is founded on)**.**
The condition says that every nonempty subclass of A has an element with no predecessor in that subclass.

$$(R \text{ Fr } A) \overset{\text{def.}}{\longleftrightarrow} (\forall S \subseteq A,\ S \subseteq R[S] \longrightarrow S = \varnothing) \tag{def:foundedOn}$$

Definition 22.9 (relation is well–ordered on)**.**

$$(R \text{ We } A) \overset{\text{def.}}{\longleftrightarrow} ((R \text{ Or } A) \wedge (R \text{ Fr } A)) \tag{def:wellOrderedOn}$$

Definition 22.10 (R_1R_2 subclass)**.**

$$A \overset{R_1R_2}{\subseteq} B \overset{\text{def.}}{\longleftrightarrow} (\forall x,\ xR_1A \longrightarrow xR_2B) \tag{def:RRsubclass}$$

Definition 22.11 (R subclass)**.**

$$(A \overset{R}{\subseteq} B) \overset{\text{def.}}{\longleftrightarrow} (\forall x,\ xRA \longrightarrow x \in B) \tag{def:Rsubclass}$$

The target B may be a proper class. Therefore the final membership test is primitive membership, not $x\mathrm{E}B$: the relation E only contains records whose second coordinate is a set. When B is a set, this definition agrees with $A \overset{R\mathrm{E}}{\subseteq} B$; for a proper B those expressions need not agree. For example, a nonempty predecessor class is contained in V, although none of its elements is E-related to the proper class V.

Lemma 22.1 (R subclass)**.**

$$k \overset{R}{\subseteq} X \longleftrightarrow R^{\top}[\{k\}] \subseteq X \tag{lem:Rsubset}$$

Proof. Unfolding the definition, both sides say that every R-predecessor of k belongs to X. □

Definition 22.12 (powerclass w.r.t. relation R)**.**

$$\mathcal{P}_R X \overset{\text{def.}}{:=} \left\{ k \;\middle|\; k \overset{R}{\subseteq} X \right\} \tag{def:RpowerClass}$$

Lemma 22.2 (Membership predecessors and the powerclass)**.**
For every class X, including proper classes,

$$\mathcal{P}_{\mathrm{E}} X = \mathcal{P} X.$$

Proof. Both comprehensions contain only sets. For a set k, the defining condition for $k \in \mathcal{P}_{\mathrm{E}}X$ is that every x with $x\mathrm{E}k$ belongs to X. Since k is a set, $x\mathrm{E}k$ is equivalent to $x \in k$. The condition is therefore exactly $k \subseteq X$, which defines $k \in \mathcal{P}X$. Class extensionality proves the equality. □

Definition 22.13 (class is closed under R-predecessors)**.**
This is the meaning of "R-transitive class" here. It is different from transitivity of R as a binary relation.

$$(R \text{ Tr } K) \overset{\text{def.}}{\longleftrightarrow} (K \subseteq \mathcal{P}_R K) \tag{def:RtransitiveClass}$$

Theorem 22.1 (properties are preserved on subclasses(except transitivity w.r.t. relation))**.**

$$((R \text{ Irr } A) \wedge (B \subseteq A)) \longrightarrow (R \text{ Irr } B) \tag{thm:IrrInheritable}$$
$$((R \text{ Trans } A) \wedge (B \subseteq A)) \longrightarrow (R \text{ Trans } B) \tag{thm:TransInheritable}$$
$$((R \text{ Po } A) \wedge (B \subseteq A)) \longrightarrow (R \text{ Po } B) \tag{thm:PoInheritable}$$
$$((R \text{ Fr } A) \wedge (B \subseteq A)) \longrightarrow (R \text{ Fr } B) \tag{thm:FrInheritable}$$
$$((R \text{ We } A) \wedge (B \subseteq A)) \longrightarrow (R \text{ We } B) \tag{thm:WeInheritable}$$

Proof. The universally quantified conditions remain true after restricting the variables from A to B. For foundedness, every subclass of B is also a subclass of A. Combining these facts gives the claims for strict partial orders and well-orders. Closure under predecessors is excluded: a subclass can omit a predecessor that was present in A. □

Lemma 22.3 (predecessors of a minimal counterexample)**.**

$$(R^{\top}[\{m\}] \subseteq K) \;\;\&\;\; (m \notin R[K \setminus A]) \implies (R^{\top}[\{m\}] \subseteq A) \qquad \text{(lem:Rindlem)}$$

Proof.

$$(\forall q,\ qRm \longrightarrow q \in K) \;\;\&\;\; (\forall q \in K,\ qRm \longrightarrow q \in A) \implies (\forall q,\ qRm \longrightarrow q \in A)$$

□

Theorem 22.2 (induction on a R–transitive and R–well–founded class)**.**

$$(R \text{ Tr } K) \wedge (R \text{ Fr } K) \longrightarrow (((K \cap (\mathcal{P}_R A)) \subseteq A) \longrightarrow (K \subseteq A)) \tag{thm:Rind}$$

Proof. Assume K is closed under R-predecessors, R is founded on K, and $K \cap \mathcal{P}_R A \subseteq A$. If $K \nsubseteq A$, the nonempty class $S = K \setminus A$ has an R-minimal element m: $m \notin R[S]$. Since $m \in K$, all predecessors of m lie in K. None lies in S, so all lie in A. Thus $m \in \mathcal{P}_R A$, and the induction assumption implies $m \in A$, contradicting $m \in S$. □

Definition 22.14 (restriction of the predecessor coordinate)**.**
This local operation restricts the first coordinate only. It is chosen to make K closed under predecessors; it is not the usual two-coordinate restriction $R \cap (K \times K)$.

$$(x(R \text{ on } K)y) \overset{\text{def.}}{\longleftrightarrow} (x \in K) \wedge (xRy) \tag{def:restrrel}$$

Lemma 22.4 (the restricted relation makes K predecessor-closed)**.**
If R is a relation, then

$$((R \text{ on } K) \text{ Tr } K) \tag{lem:restrreltran}$$

Proof.
To Show: $K \subseteq \mathcal{P}_{(R \text{ on } K)} K$.
Fix $k \in K$. To show: $k \in \left\{ x \;\middle|\; (R \text{ on } K)^\top[\{x\}] \subseteq K \right\}$.
To show: $(R \text{ on } K)^\top[\{k\}] \subseteq K$.
Fix $w \in (R \text{ on } K)^\top[\{k\}]$. To show: $(w \in K)$.

$$\begin{aligned} (w \in (R \text{ on } K)^\top[\{k\}]) &\implies (k(R \text{ on } K)^\top w) \\ &\implies (w(R \text{ on } K)k) \\ &\implies (w \in K) \wedge (wRk) \end{aligned}$$

□

Corollary 22.1 (alternative induction principle on R_1–well–founded class)**.**
This formulation of induction corresponds to Exercise 9.22 of "Introduction to Axiomatic Set Theory" by Takeuti and Zaring.

$$(R_1 \text{ Fr } K) \longrightarrow (((K \cap (\mathcal{P}_{(R_1 \text{ on } K)} A)) \subseteq A) \longrightarrow (K \subseteq A)) \tag{thm:RindAlt}$$

Proof. Set $R = R_1$ on K. The preceding lemma makes K R-predecessor-closed. Since $R \subseteq R_1$, any nonempty subclass of K with an R_1-minimal element also has an R-minimal element. Thus R_1-foundedness implies R-foundedness. Apply thm:Rind to R and expand its definition. □

22.2 Specialization to membership

Now take $R = \mathrm{E}$. The predicate $\mathrm{Ord}(X)$ applies to classes, while On collects only the set instances of that predicate. The limit convention deliberately includes the empty ordinal. The operations sup and inf below first discard nonordinal members of their argument; in particular, $\sup(\varnothing) = \varnothing$ and $\inf(\varnothing) = \mathrm{On}$.

Definition 22.15 (class is transitive (w.r.t. epsilon relation))**.**

$$\mathrm{Tr}(X) \overset{\text{def.}}{\longleftrightarrow} (X \subseteq \mathcal{P}X) \tag{def:transitiveClass}$$

Definition 22.16 (class is ordinal (w.r.t. epsilon relation))**.**

$$\mathrm{Ord}(X) \overset{\text{def.}}{\longleftrightarrow} ((\mathrm{Tr}(X)) \wedge (\mathrm{E\ We}\ X)) \tag{def:ordinalClass}$$

Definition 22.17 (limit class)**.**
This convention includes the empty class; the usual limit convention also requires nonemptiness.

$$\mathrm{LimClass}(X) \overset{\text{def.}}{\longleftrightarrow} \left(X \subseteq \bigcup X\right) \tag{def:limitClass}$$

Definition 22.18 (limit ordinal)**.**
This convention includes zero; the usual definition of a limit ordinal also requires nonemptiness.

$$\mathrm{LimOrd}(X) \overset{\text{def.}}{\longleftrightarrow} ((\mathrm{Ord}(X)) \wedge (\mathrm{LimClass}(X))) \tag{def:limitOrdinal}$$

Definition 22.19 (ordinal numbers)**.**
These are also called "ordinal sets".

$$\mathrm{On} \overset{\text{def.}}{\coloneqq} \{\, x \mid \mathrm{Ord}(x) \,\} \tag{def:ordinalNumbers}$$

Comprehension for ordinal sets. The expanded predicate $\mathrm{Ord}(x)$ contains a class quantifier through foundedness, so its literal use in comprehension is permitted by MK. Under heredity, the displayed definition also has an elementary form suitable for NBG comprehension. Indeed, for a set x, every class $S \subseteq x$ is a set. Foundedness on x is therefore equivalent to saying that every nonempty *set* $S \subseteq x$ has a membership-minimal element. The remaining transitivity and ordering conditions quantify only over set elements. Replacing the foundedness clause by this equivalent set-quantified condition constructs On by elementary comprehension. This reduction concerns the set instances collected in On; it does not restrict the predicate Ord itself to sets or assume global regularity.

Definition 22.20 (supremum)**.**

$$\sup(A) \overset{\text{def.}}{\coloneqq} \bigcup(\mathrm{On} \cap A) \tag{def:supremum}$$

Definition 22.21 (infimum)**.**

$$\inf(A) \overset{\text{def.}}{\coloneqq} \mathrm{On} \cap \left(\bigcap(\mathrm{On} \cap A)\right) \tag{def:infimum}$$

Theorem 22.3 (infimum of a subset of ordinals is the unique minimum)**.**

$$((A \cap \mathrm{On}) \neq \varnothing) \longrightarrow (\inf(A) \in A) \tag{thm:infordmin}$$

Proof. Use the well-ordering of On under membership, recorded below as thm:ordon. The nonempty class $C = A \cap \mathrm{On}$ has a least member m. For every $q \in C$, ordinal comparison gives $m \subseteq q$. Hence $m \subseteq \bigcap C$. Since $m \in C$, also $\bigcap C \subseteq m$. Therefore $\inf(A) = \mathrm{On} \cap \bigcap C = \mathrm{On} \cap m = m \in A$. This argument makes the ordinal well-ordering prerequisite explicit. □

22.3 Properties of ordinals

The following lemmas establish the ordinal comparison and closure facts used elsewhere. Their proofs use the general well-order definition, so global regularity is not silently introduced.

Lemma 22.5 (epsilon relation criterion)**.**

$$((x \in y) \wedge (y \text{ set})) \longleftrightarrow ((x, y) \in \mathrm{E}) \tag{lem:ordlem0}$$

Proof. If $x \in y$ and y is a set, then x is a set as well; the set pair (x, y) belongs to the comprehension defining E. Conversely, membership in that relation supplies set components u, v with $u \in v$ and $(x, y) = (u, v)$. Injectivity of the chosen set-pair coding gives $x = u$ and $y = v$, hence $x \in y$ and y is a set. □

Lemma 22.6 (an element is a proper subset)**.**

$$((\mathrm{E\ Irr}\ A) \wedge (\mathrm{Tr}(A))) \longrightarrow ((B \in A) \longrightarrow (B \subsetneq A)) \tag{lem:ordlem1}$$

Proof. Let $B \in A$. Transitivity gives $B \subseteq A$. If $B = A$, then $A \in A$, so A is itself an element of the carrier A and irreflexivity on that carrier gives $A \notin A$. Thus $B \neq A$. □

Lemma 22.7 (a proper subclass is an element)**.**

$$((\mathrm{Tr}(A)) \wedge (\mathrm{E\ Fr}\ A) \wedge (\mathrm{E\ Lin}\ A) \wedge (\mathrm{Tr}(B))) \longrightarrow ((B \subsetneq A) \longrightarrow (B \in A)) \tag{lem:ordlem2}$$

Proof. Let $B \subsetneq A$. The nonempty class $A \setminus B$ has an E-minimal member m. Since $m \in A$ and A is transitive, $m \subseteq A$. Minimality excludes every member of m from $A \setminus B$, hence $m \subseteq B$. For the reverse inclusion, take $b \in B$. Both b and m belong to A, so linearity gives $b \in m$, $b = m$, or $m \in b$. The second alternative puts m in B; the third does so by transitivity of B. Both contradict $m \notin B$. Therefore $b \in m$, so $B \subseteq m$. Extensionality yields $B = m \in A$. □

Theorem 22.4 (proper subclass iff element)**.**

$$((\mathrm{Ord}(A)) \wedge (\mathrm{Tr}(B))) \longrightarrow ((B \in A) \longleftrightarrow (B \subsetneq A)) \tag{thm:subiffel}$$

Proof. An ordinal class is transitive, and membership on it is irreflexive, linear, and founded. Apply the preceding two lemmas. □

Theorem 22.5 (transitive subclass of ordinal numbers iff it is an ordinal class)**.**

$$(\mathrm{Ord}(A)) \longleftrightarrow ((A \subseteq \mathrm{On}) \wedge (\mathrm{Tr}(A))) \tag{thm:dford5}$$

https://us.metamath.org/mpeuni/dford5.html

Proof. If $\mathrm{Ord}(A)$, then every member of A is a set ordinal by thm:tz7.6; therefore $A \subseteq \mathrm{On}$, and transitivity is part of the definition. Conversely, if $A \subseteq \mathrm{On}$ and A is transitive, its membership order is inherited from the well-order of On, proved in thm:ordon. Hence $\mathrm{Ord}(A)$. The proof of that latter theorem uses the preceding proper-subclass criterion, not this characterization. □

Theorem 22.6 (element of ordinal is ordinal)**.**

$$\mathrm{Ord}(A) \wedge (a \in A) \longrightarrow \mathrm{Ord}(a) \tag{thm:tz7.6}$$

Proof. Let $a \in A$ and $\mathrm{Ord}(A)$. Transitivity of A gives $a \subseteq A$, so membership inherits its strict well-order from A. To check transitivity of a, take $z \in y \in a$. All three objects z, y, a lie in A, by transitivity of A. Transitivity of membership on A then gives $z \in a$. Thus a is transitive and well-ordered by membership. □

Theorem 22.7 (class of ordinal numbers is ordinal class)**.**

$$(\mathrm{Ord}(\mathrm{On})) \tag{thm:ordon}$$

Proof. First compare any two ordinal classes A, B. Their intersection $C = A \cap B$ is transitive. If C were a proper subclass of both, thm:subiffel would give $C \in A$ and $C \in B$, hence $C \in C$. Since $C \in A$, this contradicts irreflexivity on A. Thus $A \subseteq B$ or $B \subseteq A$; if the inclusion is proper, the smaller class is a member of the larger, again by thm:subiffel.
If $a \in \mathrm{On}$ and $b \in a$, then b is a set and is ordinal by thm:tz7.6, so $b \in \mathrm{On}$. Hence On is transitive. The comparison just proved gives linearity of membership on On; transitivity follows because its members are transitive. A self-member $a \in a$ would contradict irreflexivity on the ordinal carrier a.
Finally, let $S \subseteq \mathrm{On}$ be nonempty and choose $a \in S$. If $S \cap a$ is empty, then a is minimal in S. Otherwise choose its least member m using the well-order on a. Any $b \in S \cap m$ would also lie in a, by transitivity of a, contradicting minimality of m in $S \cap a$. Thus m is minimal in S. Membership on On is founded, completing the proof of Ord(On). No global regularity axiom was needed: foundedness was included in the definition of each ordinal class. □

Lemma 22.8 (union of subclass of ordinals is transitive class)**.**

$$(A \subseteq \mathrm{On}) \longrightarrow \left(\mathrm{Tr}\left(\bigcup A\right)\right) \tag{lem:trunonss}$$

Proof.
Fix $x \in \bigcup A$, and $y \in x$. To show: $\exists z \in A,\ y \in z$.

$$\left(x \in \bigcup A\right) \implies (\exists a \in A,\ x \in a)$$

$$(a \in A)\ \ \&\ \ (A \subseteq \mathrm{On}) \implies \mathrm{Ord}(a)$$

$$\mathrm{Ord}(a) \implies \mathrm{Tr}(a)$$

$$(x \in a)\ \ \&\ \ (\mathrm{Tr}(a)) \implies (x \subseteq a)$$

$$(y \in x)\ \ \&\ \ (x \subseteq a) \implies (y \in a)$$

$$z \coloneqq a$$

$$(a \in A)\ \ \&\ \ (y \in a) \implies (\exists z \in A : y \in z)$$

□

Theorem 22.8 (criterion in the context of the regularity axiom)**.**
Under global regularity, ordinality has the following shorter characterization.

$$(\text{E Fr } V) \longrightarrow (\text{Ord}(X) \longleftrightarrow \text{Tr}(X) \wedge (\forall Y \in X : \text{Tr}(Y))) \qquad \text{(thm:ordViaReg)}$$

Proof. The forward direction follows from transitivity and the fact that each member of an ordinal class is ordinal, hence transitive.
Conversely, suppose X and every member of X are transitive. Regularity makes membership founded on X. If $a \in X \cap \mathcal{P}\text{On}$, then a is transitive and $a \subseteq \text{On}$; thm:dford5 gives $\text{Ord}(a)$. Since a is a set, $a \in \text{On}$. Thus $X \cap \mathcal{P}\text{On} \subseteq \text{On}$. Apply the induction theorem thm:Rind to the transitive, founded class X, with target class On. It yields $X \subseteq \text{On}$, and thm:dford5 now gives $\text{Ord}(X)$. □

23 Cardinals

Here cardinality is extended from sets to classes by assigning On when no ordinal set receives an injection from the argument. This is a local convention for a class-valued functional symbol; On is not being declared a set cardinal. The intended set-cardinal behavior requires the stated well-ordering or size assumptions. Comparison of all classes additionally uses the relevant global choice or limitation-of-size principle.

23.1 Axioms

Axiom 1 (set cardinality is an ordinal set)**.**

$$|\cdot| : \text{V} \to \text{On}$$

Since $|\cdot|$ is a functional symbol, the intended assertion is:

$$(x \mapsto |x|) : \text{V} \to \text{On}$$

The functional symbol accepts any class as its argument, while $(x \mapsto |x|)$ is defined on sets only. Under the stated set-size assumption, this class function has graph

$$\{\, \langle x, |x| \rangle \mid x \text{ any set} \,\}.$$

Equivalently, we can state the axiom pointwise:

Axiom (alternative set-size formulation)**.**

$$(x \text{ set}) \longrightarrow (|x| \in \text{On})$$

We can also require only sethood of the value:

Axiom (alternative set-size formulation)**.**

$$(x \text{ set}) \longrightarrow (|x| \text{ set})$$

The following theorem explains why this last formulation suffices.

Theorem (cardinality is an ordinal)**.**

$$(x \text{ set}) \longrightarrow \text{Ord}(|x|)$$

Proof. Put $C = \{\, a \in \mathrm{On} \mid x \preccurlyeq a \,\}$. If C is nonempty, the infimum theorem identifies $|x| = \inf C$ with a member of C, hence an ordinal set. If C is empty, the definition gives $|x| = \mathrm{On}$, which is an ordinal class by thm:ordon. Thus ordinality holds in either case; the additional set-size axiom excludes the second case for set arguments. □

Axiom 2 (heredity)**.**
Sets are hereditary:

$$(A \subseteq B) \wedge (B \text{ set}) \longrightarrow (A \text{ set})$$

23.2 Implementation

The following definitions implement the proposed cardinality symbol.

Definition 23.1 (cardinality of a class)**.**

$$|K| \stackrel{\text{def.}}{\coloneqq} \mathrm{On} \cap \bigcap \{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$$

or even

$$|K| \stackrel{\text{def.}}{\coloneqq} \inf(\{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\})$$

in the sense of definition def:infimum.

Definition 23.2 (relation "not stronger in cardinality")**.**
Let X and Y be terms, then

$$(X \preccurlyeq Y) \longleftrightarrow (\exists f,\ (f : X \to Y) \wedge (f \text{ injective})) \qquad \text{(def:notstrongercard)}$$

Proposition 23.1 (Elementary comprehension for ordinal-set bounds)**.**
Assume heredity, the set-image principle, and ordinary set-pair formation. For every class K and set a, the assertion $K \preccurlyeq a$ is equivalent to the existence of a *set* graph that is an injection from K into a. Consequently the primary candidate class $\{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$ exists by elementary NBG comprehension under these assumptions. No choice principle is needed.

Proof. Let $f : K \to a$ be a class injection. Its range $D \subseteq a$ is a set by heredity. The inverse graph is a class function on D, with image K, so set-image makes K a set. Ordinary set-pair formation and elementary comprehension give the class function $x \mapsto (x, f(x))$ on K. Its image is exactly the graph f; set-image therefore makes f a set. The converse is immediate, since a set graph is also a class graph.
The set-graph assertion uses only set quantifiers, with K as a class parameter: its domain condition is $\forall x \in \mathrm{V},\ (x \in K \longleftrightarrow x \in \mathrm{dom}(f))$, and functionality, injectivity, and containment of its range in a also quantify only over sets. Combine this reduction with the preceding elementary definition of On. The resulting candidate class and its intersection are thus available by elementary comprehension. Without these reductions, the literal formula containing an unrestricted quantifier over class injections is directly licensed by MK, rather than being itself an instance of the NBG comprehension schema. □ Under the stated set-size assumption, this definition gives the cardinality of a set argument and the default value On for a proper-class argument.

Definition 23.3 (X can be injected in Y)**.**

$$(X \preccurlyeq Y) \longleftrightarrow (\exists f,\ (f : X \to Y) \wedge (f \text{ injective}))$$

Theorem 23.1 (comparability of classes under the size assumptions)**.**
Assume the set well-ordering theorem and the class-size principle that every proper class is in bijection with V. Then for arbitrary classes A, B,

$$((A \preccurlyeq B) \vee (B \preccurlyeq A))$$

Proof. If both classes are sets, well-order them and compare their ordinal cardinalities. If both are proper, their bijections with V give a bijection with each other. If exactly one is proper, the other is a subclass of V; compose its inclusion into V with the bijection from V to the proper class. In each case there is an injection in at least one direction. □

A predicate-indexed variant. The next proposal intersects ordinal classes, rather than only ordinal sets. Its agreement with the preceding definition depends on the class-size assumptions; it is not a purely syntactic reformulation. For the displayed comprehension, use MK: A ranges over classes, and the expanded injection predicate also quantifies over class relations. A weaker comprehension scheme needs a separate existence argument. Proposition 24.3 below characterizes the resulting class by cases in terms of the primary cardinality and V; whenever those classes exist, this also provides a pointwise existence justification without treating the displayed impredicative formula as an instance of NBG comprehension.

Definition 23.4 (cardinality of a class)**.**

$$|K| \overset{\text{def.}}{\coloneqq} \{\, x \mid \forall A,\ \mathrm{Ord}(A) \wedge (K \preccurlyeq A) \longrightarrow x \in A \,\}$$

$$|K| \overset{\text{def.}}{\coloneqq} \bigcap_{\mathrm{Ord}(A) \wedge (K \preccurlyeq A)}^{A} A$$

Definition 23.5 (cardinality of X is strictly smaller than cardinality of Y)**.**

$$(X \prec Y) \longleftrightarrow (\neg(Y \preccurlyeq X))$$

Here V denotes a class of all sets, and On denotes the class of all ordinal numbers.

Remark 23.1 (The default value and the chosen convention)**.**
For the primary ordinal-set definition, every proper class P has $|P| = \mathrm{On}$: an injection into an ordinal set would, by heredity and set-image applied to its inverse, make P a set. The empty-candidate intersection is V, and the explicit intersection with On gives the stated default. For the predicate-indexed variant, the conclusion additionally requires $P \preccurlyeq \mathrm{On}$. Under full limitation of size this follows from $P \subseteq \mathrm{V} \preccurlyeq \mathrm{On}$. Without that additional injection the variant instead returns V, as proved below. Thus the two displayed proposals are not interchangeable merely because they share a symbol. The assertion $(x \mapsto |x|) : \mathrm{V} \to \mathrm{On}$ separately uses the set well-ordering assumption; the default for proper classes does not prove it.

We next verify the basic properties of $|\cdot|$ under the primary convention.

23.3 Soundness

Lemma 23.1 (the universal class does not inject into an ordinal set)**.**
The universal class cannot inject into an ordinal set under the size assumptions used here.

$$\{\, a \in \mathrm{On} \mid \mathrm{V} \preccurlyeq a \,\} = \varnothing$$

Proof. Suppose $g : \mathrm{V} \to a$ is injective for an ordinal set a. The range $D = g[\mathrm{V}] \subseteq a$ is a set by heredity. The inverse graph gives a class function $g^{\top} : D \to \mathrm{V}$, and its image on its actual domain is $g^{\top}[D] = \mathrm{V}$. The set-image principle would make V a set, contradicting its properness. Thus no such a exists. Using the actual domain D makes the argument independent of off-domain default values. □

Theorem 23.2 (cardinality of the universal class)**.**

$$|\mathrm{V}| = \mathrm{On}$$

Proof.

$$|K| \overset{\text{def.}}{\coloneqq} \mathrm{On} \cap \bigcap \{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$$

$$|\mathrm{V}| = \mathrm{On} \cap \bigcap \{\, a \in \mathrm{On} \mid \mathrm{V} \preccurlyeq a \,\}$$

$$= \mathrm{On} \cap \bigcap \varnothing$$

$$= \mathrm{On} \cap \mathrm{V}$$

$$= \mathrm{On}$$

Therefore, by transitivity of equality, $|\mathrm{V}| = \mathrm{On}$.

□

23.4 Properties

Theorem 23.3 ($|\cdot|$ is a projection)**.**

$$(|\cdot| \text{ projection}) \tag{thm:cardproj}$$

Proof. Use the ordinal-set definition $|K| = \inf C$, where $C = \{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$. If C is empty, then $|K| = \mathrm{On}$. Under the size assumptions, the proper class On cannot inject into an ordinal set, so $|\mathrm{On}| = \mathrm{On}$ as well.
If C is nonempty, its least member $\kappa = |K|$ receives an injection from K. Let $\lambda = |\kappa|$. The identity injection of κ shows that $\lambda \subseteq \kappa$. Composing injections $K \preccurlyeq \kappa \preccurlyeq \lambda$ shows $\lambda \in C$, so minimality of κ gives $\kappa \subseteq \lambda$. Extensionality yields $\lambda = \kappa$, as required. □

Lemma 23.2 (property of intersection)**.**

$$\forall (a \in X),\ \left(\left(\bigcap X\right) \subseteq a\right)$$

Proof. Fix $(a \in X)$. Fix $(b \in (\bigcap X))$. To show: $(b \in a)$.

$$\left(b \in \left(\bigcap X\right)\right)\ \ \&\ \ \left(\left(\bigcap X\right) = \{\, w \mid \forall q \in X,\ w \in q \,\}\right) \implies (\forall q \in X,\ b \in q)$$

$$(a \in X)\ \ \&\ \ (\forall q \in X,\ b \in q) \implies (b \in a)$$

□

Theorem 23.4 (Schroeder–Bernstein for classes)**.**
Assume MK comprehension and let $f : A \to B$ and $g : B \to A$ be injective class functions. Then there is a class bijection $h : A \to B$.

Proof. Put $A_0 = A \setminus g[B]$. Let C be the least subclass of A containing A_0 and closed under $g \circ f$. It exists by the intersection construction for a monotone operation, and its fixed-point identity is

$$C = A_0 \cup g[f[C]].$$

For completeness, intersect all subclasses Q of A with $A_0 \subseteq Q$ and $g[f[Q]] \subseteq Q$. The intersection has these properties; applying minimality to $A_0 \cup g[f[C]]$ proves the identity. Define $h(a) = f(a)$ for $a \in C$ and $h(a) = g^{-1}(a)$ for $a \in A \setminus C$. The latter value exists uniquely because $A_0 \subseteq C$ and g is injective.
The two branches are injective. Their images are disjoint: if $f(c) = g^{-1}(a)$ with $c \in C$ and $a \notin C$, then $a = g(f(c)) \in C$, a contradiction. Thus h is injective. For surjectivity, take $b \in B$. If $b \in f[C]$, it is already in $h[C]$. Otherwise $g(b) \notin C$: if it belonged to C, it could not belong to A_0, so the displayed identity would give $g(b) = g(f(c))$ for some $c \in C$, contradicting injectivity of g and $b \notin f[C]$. Hence $h(g(b)) = b$ on the second branch. This proves surjectivity. All graphs use set coordinates; no proper class is collected as a member of another class. □

Corollary 23.1 (Bijection between the universal and ordinal classes)**.**
Under the limitation-of-size assumption supplying an injection $\mathrm{V} \to \mathrm{On}$, there is a class bijection $\mathrm{V} \to \mathrm{On}$.

Proof. The inclusion $\mathrm{On} \subseteq \mathrm{V}$ supplies an injection in the opposite direction. Apply the preceding theorem. The needed injection from V is derived from limitation of size in the union-theorem chapter; it does not follow merely from writing $|\mathrm{V}| = \mathrm{On}$ under the ordinal-valued cardinality convention. □

24 Various induction and recursion principles

24.1 Induction principles

Note 24.1 (successor–inductive (Peano–inductive) classes and ω as the $\subseteq$–minimal successor–inductive class)**.**
Because several distinct induction principles appear in class theory (countable Peano successor–induction, transfinite $\in$–induction on ordinals, and $\mathcal{P}$–induction on well–founded sets), the predicate $\mathrm{Ind}(X)$ (Entry 3.55 in the Obelisk catalog) uses "inductive" in its narrowest possible sense, namely **successor–inductive** (or **Peano–inductive** / 0–successor–closed):

$$\mathrm{Ind}(X) \overset{\text{def.}}{\longleftrightarrow} (\varnothing \in X) \wedge (\forall(a : \mathrm{Class}),\ (a \in X) \longrightarrow ((a \cup \{a\}) \in X)).$$

By its definition $\forall(x : \mathrm{Class}),\ (x \in \omega) \longleftrightarrow (\forall(Q : \mathrm{Class}),\ \mathrm{Ind}(Q) \longrightarrow (x \in Q))$ (Entry 3.56) together with $\mathrm{Ind}(\omega)$ (Entry 3.59), ω is not only the smallest successor–inductive ordinal, but also the **$\subseteq$–minimal (least) element among all successor–inductive classes in the subclass ($\subseteq$) order**:

$$\mathrm{Ind}(\omega) \wedge \forall(Q : \mathrm{Class}),\ \mathrm{Ind}(Q) \longrightarrow (\omega \subseteq Q),$$

which directly expresses simple countable mathematical (Peano) induction up to ω.

Proposition 24.1 (The least successor-inductive class and Infinity)**.**
Assume MK comprehension, empty set, pairing, union, and heredity. The class

$$\omega = \{x : \forall Q,\ \mathrm{Ind}(Q) \longrightarrow x \in Q\}$$

is successor-inductive and is contained in every successor-inductive class. It is a transitive class of ordinal sets. If additionally Infinity supplies a successor-inductive set, then ω is a set ordinal and is a nonzero limit ordinal. Conversely, sethood of this ω supplies the set required by Infinity.

Proof. Every successor-inductive class contains zero, so $0 \in \omega$. If $a \in \omega$, every such class contains a and therefore its successor $a \cup \{a\}$. Pairing and union make that successor a set, so it belongs to ω by comprehension. The definition also gives the asserted minimality.
Zero is an ordinal, and the successor of an ordinal set is an ordinal set: it is a transitive set of ordinal sets, so the ordinal characterization applies. Thus On is successor-inductive and minimality gives $\omega \subseteq \mathrm{On}$. For transitivity put

$$T = \{a \in \omega : a \subseteq \omega\}.$$

Zero belongs to T. If $a \in T$, then its successor belongs to ω and is a subclass of ω, because both $a \subseteq \omega$ and $a \in \omega$ hold. Thus T is successor-inductive. Minimality gives $\omega \subseteq T$, proving transitivity of ω.
If I is a successor-inductive set, minimality gives $\omega \subseteq I$, and heredity makes ω a set. Its transitivity and containment in On make it an ordinal set. It is nonempty because it contains zero. For each $a \in \omega$, its successor belongs to ω and contains a, so $a \in \bigcup\omega$. Transitivity gives the reverse inclusion, hence $\omega = \bigcup\omega$ is a nonzero limit ordinal. Finally, if ω is a set, its already proved successor-inductiveness is itself a witness to Infinity. □

Proposition 24.2 (From set regularity to class foundedness)**.**
Assume Infinity, pairing, union, heredity, set-image, class comprehension, and the set-valued ordinal recursion theorem in the induction and recursion chapter. Then ordinary set regularity, namely that every nonempty set has a membership-minimal element, is equivalent to foundedness of membership on V: every nonempty class has a membership-minimal element.

Proof. Only the forward implication needs proof. Let S be a nonempty class and choose $a \in S$. We first build a transitive set containing a as an element, without assuming that a already belongs to W. By recursion on the set ω, form

$$B_0 = \{a\}, \qquad B_{n+1} = \bigcup B_n, \qquad T = \bigcup\{B_n : n \in \omega\}.$$

Pairing and union make each B_n a set. Set-image on ω and union make T a set, with $a \in T$. If $z \in T$, then $z \in B_n$ for some n, and every member of z belongs to $B_{n+1} \subseteq T$. Thus T is transitive. This construction uses only ordinal recursion and the stated set-formation axioms, not membership induction or the regularity conclusion being established.
The class $D = S \cap T$ is a nonempty set by heredity, since $a \in D$. Set regularity supplies $m \in D$ with $m \cap D = \varnothing$. If $z \in m \cap S$, then $m \in T$ and transitivity give $z \in T$, so $z \in m \cap D$, a contradiction. Hence $m \cap S = \varnothing$, as required. Conversely, apply class foundedness to each nonempty set. □
This bridge permits a proof stated with set regularity to use a minimal element of a proper class only after the additional set-closure prerequisites have been supplied. The ordinal comparison proofs earlier in this chapter instead use foundedness built into their ordinal definition; they do not need this global regularity principle.

Corollary 24.1 (induction on a transitive and well–founded by membership relation class)**.**

$$(\mathrm{Tr}(K)) \wedge (\mathrm{E\ Fr}\ K) \longrightarrow (((K \cap (\mathcal{P}A)) \subseteq A) \longrightarrow (K \subseteq A))$$

Proof. Apply thm:Rind to membership. Transitivity of K says exactly that every member of any $x \in K$ again lies in K. For a set x, all membership predecessors of x lie in A exactly when $x \subseteq A$, which is the condition $x \in \mathcal{P}A$. Thus the relational induction hypothesis becomes precisely the displayed hypothesis. □

Corollary 24.2 (Epsilon induction and foundedness)**.**
Membership is founded on V if and only if, for every class A,

$$((\mathcal{P}A) \subseteq A) \longrightarrow (\mathrm{V} \subseteq A).$$

Here $\mathcal{P}A$ consists of the set subclasses of A, even when A is proper. The converse concerns the induction principle for all classes A, not one particular instance of the implication.

Proof. The class V is transitive: members of sets are sets. Also $\mathcal{P}A \subseteq \mathrm{V}$ by the definition of powerclass. Foundedness on V therefore makes the preceding corollary applicable with $K = \mathrm{V}$, giving the forward implication.
Conversely, assume the displayed induction principle for every class. Let $S \subseteq \mathrm{V}$ have no membership-minimal element, so every $x \in S$ has some $y \in S \cap x$. Put $A = \mathrm{V} \setminus S$. We show $\mathcal{P}A \subseteq A$. If $x \in \mathcal{P}A$, then x is a set and $x \subseteq A$. Were $x \in S$, the absence of minimal elements would supply $y \in S \cap x$, contradicting $x \subseteq A$. Thus $x \notin S$, and, since x is a set, $x \in A$. Induction gives $\mathrm{V} \subseteq A$, forcing $S = \varnothing$. This is exactly foundedness as defined above: every subclass with no minimal element is empty. The argument uses class complements and set elements; it never treats a proper class as a member of itself or of V. □

Corollary 24.3 (transfinite induction)**.**

$$((\mathrm{On} \cap (\mathcal{P}A)) \subseteq A) \longrightarrow (\mathrm{On} \subseteq A)$$

Proof. By thm:ordon, On is transitive and membership is founded on it. Apply thm:Rind with $R = \mathrm{E}$ and $K = \mathrm{On}$. □

Corollary 24.4 (induction up to ordinal class β)**.**
Assume $\mathrm{Ord}(\beta)$. Then

$$((\beta \cap (\mathcal{P}A)) \subseteq A) \longrightarrow (\beta \subseteq A)$$

Proof. By the definition of an ordinal class, β is transitive and membership is founded on it. Apply the first corollary of this subsection with $K = \beta$. The proof does not require β to be a set: every element considered in the induction is a set, even for a proper ordinal class such as On. □

24.2 Recursion on a set-like founded relation

The following generalization uses the class comprehension, Infinity, pairing, union, heredity, set-image, and ordinal-recursion principles used in the induction and recursion section. Global regularity is not needed: foundedness is imposed on the particular relation being used. Auxiliary graphs use ordinary Kuratowski pairs.

Definition 24.1 (Set-like relation)**.**
For $R \subseteq A \times A$ and $a \in A$, put

$$P_a = \{\, b \in A \mid bRa \,\}.$$

The relation is set-like on A when each P_a is a set. A subclass D of A is predecessor-closed when $a \in D$ implies $P_a \subseteq D$.

Theorem 24.1 (Set-valued recursion on a founded class)**.**
Let R be founded and set-like on A, and let $H : A \times \mathrm{V} \to \mathrm{V}$ be a class function. There is a unique class function $F : A \to \mathrm{V}$ such that

$$F(a) = H(a, F \restriction P_a) \qquad (a \in A).$$

Proof. First form the least predecessor-closed set T_a containing a. Starting with $B_0 = \{a\}$, put

$$B_{n+1} = \bigcup\{P_b : b \in B_n\}, \qquad T_a = \bigcup\{B_n : n < \omega\}.$$

Recursion on ω constructs this sequence independently of recursion on R. Set-likeness and set-image make the family of predecessor sets at each step a set; union and set-image then make T_a a set. It is predecessor-closed because the predecessors of a member of B_n belong to B_{n+1}. Induction on n proves minimality.
An approximation is a set function p on a predecessor-closed set $D \subseteq A$ satisfying $p(t) = H(t, p \restriction P_t)$ for $t \in D$. Any two approximations agree on their common domain. Otherwise an R-minimal disagreement has all its predecessors in both domains, with equal values there, and hence equal values after applying H, a contradiction.
Founded induction now constructs an approximation on T_a. Suppose the unique approximations p_b on T_b exist for all $b \in P_a$. Set-image on the set P_a collects these graphs, and their union q is a set function by compatibility. Its domain $D = \bigcup_{b \in P_a} T_b$ is predecessor-closed and contains P_a; compatibility also preserves all its recursion equations. If $a \in D$, retain q. Otherwise adjoin $(a, H(a, q \restriction P_a))$. The domain remains predecessor-closed, and old equations are unaffected because no member of D has a as a predecessor. Restrict to T_a, which is contained in this domain by minimality. This gives the desired approximation. The induction used here follows from a minimal counterexample, or equivalently from thm:RindAlt.
Collecting all approximation pairs by class comprehension produces F. Compatibility makes it a function, and the solutions on T_a give its domain and equation. Each restriction to P_a is part of such a set approximation. Finally, a minimal disagreement between two class solutions is impossible by the same argument, proving uniqueness. □

Theorem 24.2 (Ordinal ranks for a set-like founded relation)**.**
Under the preceding assumptions, a founded set-like relation R on A has a unique class function $r : A \to \mathrm{On}$ satisfying

$$r(a) = \sup\{r(b) + 1 : bRa\}.$$

In particular, bRa implies $r(b) < r(a)$. Conversely, for any relation on a class A, existence of a class function $q : A \to \mathrm{On}$ with $bRa \Rightarrow q(b) < q(a)$ implies foundedness. The converse does not require set-likeness.

Proof. For a set function graph g whose range consists of ordinals, let the recursion step be the supremum of the successors of its values; use zero on other inputs. Replacement makes that successor collection a set, and its union is an ordinal, including zero for the empty collection.

The preceding recursion theorem therefore gives a set-valued function. Founded induction shows that all its values are ordinals and that it satisfies the displayed equation: predecessor values are ordinals, so the intended branch of the recursion step applies. Uniqueness follows from the same recursion theorem. Each $r(b)+1$ is at most $r(a)$, giving strict increase along every R-edge.
Conversely, let S be a nonempty subclass of A. The nonempty class of ordinal sets $q[S]$ has a least element by the least-member property of On. Take $a \in S$ with that value. If $b \in S$ and bRa, then $q(b) < q(a)$, contradicting minimality. Thus a has no predecessor in S, proving foundedness. □
The rank need not be injective: two distinct minimal points both have rank zero. It measures the depth of a relation, whereas the subsequent collapse recovers its entire predecessor structure and therefore needs extensionality. For membership on W, the equation specializes to the ordinal rank introduced in the recursion chapter, not its separate left-addition depth convention.

24.3 Mostowski collapse

Theorem 24.3 (Collapse of a set-like extensional founded relation)**.**
Let $R \subseteq A \times A$ be founded and set-like. Assume it is extensional on A: $P_a = P_b$ implies $a = b$ for $a, b \in A$. There is a unique transitive class M and a unique isomorphism

$$\pi : (A, R) \longrightarrow (M, \in).$$

If A is a set, then M is a set.

Proof. Apply 24.1 with $H(a, g) = \operatorname{ran}(g)$ on set function graphs, extending it by the empty set on other inputs. Set-image makes these values sets. The result satisfies

$$\pi(a) = \{\pi(b) : bRa\}.$$

We prove injectivity by founded induction on a, with the assertion quantified over *every* $c \in A$: if $\pi(a) = \pi(c)$, then $a = c$. For bRa, equality of the displayed sets gives dRc with $\pi(b) = \pi(d)$. The induction hypothesis at b gives $b = d$, so bRc. Conversely, for dRc equality gives some bRa with the same image, and the same induction hypothesis gives $d = b$. Hence $P_a = P_c$, and extensionality gives $a = c$. No induction hypothesis at c has been assumed.
Let $M = \pi[A]$. The recursion equation makes M transitive, and injectivity gives both directions of $bRa \longleftrightarrow \pi(b) \in \pi(a)$. Thus π is the required isomorphism. If another isomorphism ρ has transitive range, every member of $\rho(a)$ belongs to that range and so is $\rho(b)$ for some $b \in A$. The isomorphism property then gives $\rho(a) = \{\rho(b) : bRa\}$. Recursion uniqueness yields $\rho = \pi$, and their ranges are equal. When A is a set, set-image makes M a set. □

Example 24.1 (Why the class collapse needs set-likeness)**.**
Order a copy of On and adjoin one new greatest point. More explicitly, use the carrier $(\{0\} \times \mathrm{On}) \cup \{(1, 0)\}$, order its first part by the ordinal coordinate, and put every member of that part below $(1, 0)$. This is a founded extensional class order, but the greatest point has a proper class of predecessors. In a transitive collapse the point $(0, \alpha)$ would have image α, by ordinal induction. The greatest point would therefore have image On, which is proper and cannot be the value of a class function into V. Thus dropping set-likeness would make the stated collapse theorem false.

Corollary 24.5 (Ordinal type of a set well-order)**.**
Every strictly well-ordered set (X, R) is isomorphic to a unique ordinal set. Its isomorphism to that ordinal is unique.

Proof. The predecessor sets are subclasses of the set X. They distinguish points: if xRy, then $x \in P_y$ but $x \notin P_x$. Thus the collapse theorem applies. Its image is a transitive set whose membership relation is a well-order, hence an ordinal set by definition. Any ordinal isomorphism is a collapse to a transitive range, so uniqueness also follows from that theorem. □

Lemma 24.1 (Increasing maps do not shorten ordinals)**.**
If $e : \alpha \to \beta$ is strictly increasing between ordinal sets, then $\xi \leq e(\xi)$ for every $\xi < \alpha$, and $\alpha \leq \beta$. Consequently the order type of a subset of β is at most β.

Proof. By ordinal induction, for each $\eta < \xi$ we have $\eta \leq e(\eta) < e(\xi)$. Thus every member of ξ belongs to $e(\xi)$, proving $\xi \subseteq e(\xi)$. If $\beta < \alpha$, applying this at $\xi = \beta$ contradicts $e(\beta) < \beta$. The increasing enumeration of a subset exists by its ordinal order type, so the same conclusion applies to it. □

Theorem 24.4 (The least ordinal bound is an ordinal representative)**.**
Let X be a set that injects into some ordinal set. For the primary ordinal-set definition, $\kappa = |X|$ is equinumerous with X and is an initial ordinal: no smaller ordinal is equinumerous with κ. Moreover $|\kappa| = \kappa$.

Proof. The nonempty class of candidate ordinal bounds has least member κ, so there is an injection $f : X \to \kappa$. Give its image S the inherited well-order. Its ordinal type β is at most κ by the lemma. The isomorphism $S \to \beta$ composed with f is a bijection $X \to \beta$. Hence β is a candidate bound and minimality gives $\kappa \leq \beta$. Thus $\beta = \kappa$ and the required bijection exists.
If an ordinal $\gamma < \kappa$ received an injection from κ, composition with this bijection would make γ a candidate for X, a contradiction. In particular no smaller ordinal is equinumerous with κ. The identity makes κ its own candidate bound, and the preceding exclusion of smaller bounds gives $|\kappa| = \kappa$. □

Corollary 24.6 (Comparison of well-orderable sets)**.**
For well-orderable sets X, Y,

$$X \preccurlyeq Y \longleftrightarrow |X| \leq |Y|.$$

They are equinumerous exactly when $|X| = |Y|$.

Proof. Transport an injection along the bijections to their ordinal representatives. If $|Y| < |X|$, such an injection contradicts the minimality of $|X|$. Conversely, ordinal inclusion gives an injection whenever $|X| \leq |Y|$. Equal representatives give a bijection by composition; a bijection gives injections in both directions and hence equality. □

Remark 24.1 (The comparison convention needs its hypotheses)**.**
The earlier notation $X \prec Y$ was defined by $\neg(Y \preccurlyeq X)$. Under comparability this is equivalent to $X \preccurlyeq Y$ together with the failure of the reverse injection. Without comparability, two incomparable classes would satisfy both $X \prec Y$ and $Y \prec X$. Thus the notation is not an unconditional strict-order assertion in a setting without the stated size principles. For arbitrary classes one may instead explicitly use $X \preccurlyeq Y \wedge \neg(Y \preccurlyeq X)$ when that asymmetric meaning is intended.

Proposition 24.3 (When the two cardinality proposals agree)**.**
Write $c(K)$ for the primary ordinal-set definition and $d(K)$ for the predicate-indexed intersection over ordinal classes. If $K \preccurlyeq \mathrm{On}$, then $c(K) = d(K)$. If no injection $K \to \mathrm{On}$ exists, then

$$c(K) = \mathrm{On}, \qquad d(K) = \mathrm{V}.$$

In particular, an assumption giving $\mathrm{V} \preccurlyeq \mathrm{On}$ makes the two proposals agree for all classes.

Proof. Every ordinal class is a subclass of On, since its members are ordinal sets. If such a class C is a proper subclass of On, choose $\alpha \in \mathrm{On} \setminus C$. Ordinal comparison gives $C \subseteq \alpha$ (the alternative $\alpha \subsetneq C$ would give $\alpha \in C$). Heredity makes C a set. Thus the ordinal classes are precisely the ordinal sets and On.
If there is an ordinal-set bound for K, its least member belongs to both collections of bounds and is contained in all their members. Both intersections therefore give that least member. If the only ordinal-class bound is On, then $d(K) = \mathrm{On}$ and the empty-candidate default also gives $c(K) = \mathrm{On}$. Finally, if K does not inject into On, it cannot inject into any ordinal class; the predicate-indexed intersection is then the empty intersection V, whereas the primary formula explicitly intersects its empty-intersection value with On. Every class embeds in V by inclusion, proving the last assertion. □

24.4 Hartogs' bound and the set well-ordering theorem

This subsection additionally assumes Powerset. The bound is constructed without choice; a choice function enters only in the subsequent enumeration.

Theorem 24.5 (An ordinal too large to inject into a given set)**.**
For every set X there is a least ordinal $\mathfrak{h}(X)$ which does not inject into X.

Proof. All pairs (B, R) with $B \subseteq X$ and $R \subseteq X \times X$ form a set, by powerset and the ordinary set-product construction. Separate those with $R \subseteq B \times B$ for which R is a strict well-order on B. Their unique ordinal types form a set H by 24.5 and set-image. An ordinal belongs to H exactly when it injects into X: an order isomorphism supplies such an injection, and any injection transports the ordinal order onto its image in X.
If $\alpha \in H$ and $\beta \in \alpha$, restrict the injection $\alpha \to X$ to β to get $\beta \in H$. Thus H is a transitive set of ordinals, hence itself an ordinal. It cannot inject into X, since that would give $H \in H$, contradicting ordinal irreflexivity. Every smaller ordinal is a member of H and does inject into X. Set $\mathfrak{h}(X) = H$. □

Theorem 24.6 (Well-ordering from a choice function)**.**
If a set X has a choice function c on its nonempty subsets, then X can be well-ordered. Conversely, a well-order of X supplies such a choice function.

Proof. Use transfinite recursion to select

$$f(\alpha) = \begin{cases} c(X \setminus \operatorname{ran}(f \restriction \alpha)), & X \setminus \operatorname{ran}(f \restriction \alpha) \neq \varnothing, \\ \varnothing, & \text{otherwise.} \end{cases}$$

The step is a set-valued rule on earlier set graphs, so the recursion theorem applies. If the remainder were nonempty at every $\alpha < \mathfrak{h}(X)$, these choices would be distinct members of X, giving an injection $\mathfrak{h}(X) \to X$, a contradiction. Let $\gamma < \mathfrak{h}(X)$ be the least stage with empty remainder. Before that stage the choices are distinct, and exhaustion says that they cover X. Hence $f \restriction \gamma$ is a bijection $\gamma \to X$. Transport the ordinal well-order along this bijection. The case $X = \varnothing$ has $\gamma = 0$.
Conversely, send each nonempty subset of a well-ordered X to its least element. Uniqueness and set-image give a choice function on the set $\mathcal{P}X \setminus \{\varnothing\}$. □

Corollary 24.7 (The strength of the set-size assumption)**.**
Under these set-formation assumptions, the following statements are equivalent: every set can be well-ordered; every set injects into an ordinal set; and every set X has $|X| \in \mathrm{On}$ for the ordinal-set cardinality convention of this chapter. They are also equivalent to the axiom of choice for set-indexed families of nonempty sets.

Proof. A set well-order gives an ordinal isomorphism; an injection into an ordinal pulls back a well-order. In the cardinality definition, the candidate class is nonempty exactly when such an injection exists. Its infimum is then its least member, an ordinal set; if it is empty, the value is the proper class On. Properness follows because Ord(On) together with sethood would give $\mathrm{On} \in \mathrm{On}$.
Choice applied to all nonempty subsets of X gives the choice function used above. Conversely, for a set-indexed family of nonempty sets, well-order their union and choose the least element of each member. The union is a set, and set-image collects the choices into a function. This proves the equivalence with set choice, without introducing a global choice function on proper classes. □

24.5 Ordinal addition and the height convention

We use the set-valued transfinite-recursion theorem from the induction and recursion chapter, together with pairing, union, and set-image. That theorem does not use ordinal addition. The following construction therefore also supplies the arithmetic facts used later in that chapter's height discussion, despite their present placement in the book.
The stages used in this recursion need no prior arithmetic. The successor $\alpha \cup \{\alpha\}$ is a transitive set of ordinal sets, hence an ordinal set. The union of a set of ordinals is likewise an ordinal set: it is transitive and contained in On, and union supplies sethood. Every nonzero ordinal δ is either a successor or a nonzero limit. Indeed, if $\delta \neq \bigcup \delta$, choose $\eta \in \delta \setminus \bigcup \delta$. Ordinal comparison forces every member of δ to be at most η, since $\eta \in \xi \in \delta$ would contradict the choice of η. Thus $\delta = \eta \cup \{\eta\}$, with unique predecessor η. Otherwise $\delta = \bigcup \delta$ is the limit case.

Theorem 24.7 (Ordinal addition by recursion)**.**
For each ordinal set α, there is a unique class function $\beta \mapsto \alpha + \beta$ from On to On satisfying

$$\begin{aligned} \alpha + 0 &= \alpha, \\ \alpha + (\beta + 1) &= (\alpha + \beta) \cup \{\alpha + \beta\}, \\ \alpha + \lambda &= \bigcup_{\beta < \lambda} (\alpha + \beta) \qquad (\lambda \text{ a nonzero limit ordinal}). \end{aligned}$$

Here $\beta + 1$ in the recursion index denotes the already defined successor $\beta \cup \{\beta\}$. After addition has been constructed, the equation at the index 1 identifies these two uses of $+1$.

Proof. Specify a total step on set graphs as follows. For a function graph p with ordinal domain δ and ordinal values, return α if $\delta = 0$, the successor of $p(\eta)$ if $\delta = \eta + 1$, and $\bigcup \mathrm{ran}(p)$ if δ is a nonzero limit. Return zero on every other set input. The successor and union outputs are sets by pairing, union, and set-image; they are ordinals by the successor and union properties of ordinals. Hence this is a set-valued class-function step, and transfinite recursion supplies its unique solution on On.
Ordinal induction shows that every value of the solution is an ordinal. At each stage its preceding restriction therefore satisfies the intended branch of the step, proving all three equations.

Conversely, any ordinal-valued solution of these equations satisfies the same total recursion, so uniqueness follows from the recursion theorem. The nonzero restriction on the limit clause is necessary: this book's limit convention includes zero, whose value is specified separately. □

Proposition 24.4 (Elementary addition identities)**.**
For every ordinal set β and every finite ordinal n,

$$0 + \beta = \beta, \qquad 1 + n = n + 1.$$

Assume Infinity for the assertions involving ω, so that Proposition 24.1 makes it a nonzero limit ordinal set. Then $1 + \omega = \omega$, whereas $\omega + 1$ is strictly greater than ω.

Proof. Induct on β for the first identity. The zero and successor clauses preserve the equality. At a nonzero limit λ, the induction hypothesis gives $0 + \lambda = \bigcup_{\beta<\lambda} \beta = \lambda$. Ordinary induction on n gives $1 + n = n + 1$: the base is $1 + 0 = 1$, and each successor step takes the successor of both sides. Continuity at the nonzero limit ω then gives

$$1 + \omega = \bigcup_{n<\omega} (n + 1) = \omega.$$

The last equality follows because every finite ordinal belongs to a later finite successor and every member of a finite successor is finite. Finally $\omega + 1 = \omega \cup \{\omega\}$ properly contains ω, since $\omega \notin \omega$. □

Thus left addition by 1 preserves the bound ω used for depth in the recursion chapter, whereas the successor operation strictly increases every ordinal rank. Multiplication and exponentiation require separate recursions and are not implicit in this construction.

24.6 Ordinals' definition if the axiom of regularity is present

This final comparison assumes regularity locally. The shorter characterization is justified by thm:ordViaReg; the earlier definition by transitivity and membership well-ordering remains available without global regularity.

The newer Obelisk export `DefOrdinalClass` (3.16) uses precisely this shorter condition, in a catalog that also assumes `AxmRegularity` (3.14). Its `ThmTransitiveClassInduction` (3.71) consequently need not repeat foundedness in each statement. The extra hypothesis in this chapter's general induction theorem records a different local setting, rather than a contradiction between the results. The site's prose dossier `docs/axioms/ordinals.md` additionally restricts "ordinal" to sets, whereas the generated predicate and this chapter also admit ordinal classes. The set instances are collected in On; the predicate itself must not be identified with membership in On.

Definition 24.2.
predicates on classes "is transitive" and "is ordinal"

$$\mathrm{tr}(X) \longleftrightarrow (\forall Z \in X,\ Z \subseteq X)$$

$$\mathrm{ord}(X) \longleftrightarrow \mathrm{tr}(X) \wedge (\forall Y \in X,\ \mathrm{tr}(Y))$$

the other possibilities are

$$\mathrm{ord}(X) \longleftrightarrow \mathrm{tr}(X) \wedge (\mathrm{E\ Wo}\ X)$$

$$\mathrm{ord}(X) \longleftrightarrow \mathrm{tr}(X) \wedge (\mathrm{E\ Wo}\ X) \wedge (\mathrm{E\ Fr}\ X)$$

etc ...

25 Well-founded sets and a reformulation of regularity

The first construction identifies the least class closed under forming set subclasses. It uses MK comprehension, since the defining predicate quantifies over classes. Regularity is introduced afterward as the claim that every set belongs to this class. The transfinite recursion theorem below uses the ordinary set-image, pairing, and union principles in addition to the ordinal results. These assumptions are stated here because the separate union-theorem chapter asks which of them can be derived from a smaller basis. All auxiliary graphs in this section use ordinary Kuratowski pairs of sets. Their formation does not depend on the recursive pairing construction that will use membership recursion.

Theorem 25.1 (Transfinite recursion with set-valued stages)**.**
Assume class comprehension, pairing of sets, the set-image principle, heredity (separation), the union axiom, and the ordinal ordering results. For a class function $G : \mathrm{V} \to \mathrm{V}$, there is a unique class function $F : \mathrm{On} \to \mathrm{V}$ such that

$$F(\alpha) = G(F \restriction \alpha) \qquad (\alpha \in \mathrm{On}).$$

The domain is On, the class of ordinal sets, rather than the predicate Ord. Each restriction $F \restriction \alpha$ is a set graph, so it is an admissible argument of G.

Proof. Call a set function p an approximation if its domain is an ordinal β and $p(\gamma) = G(p \restriction \gamma)$ for each $\gamma < \beta$. Every restriction of an approximation to an initial ordinal segment is an approximation. Any two approximations agree on their common domain: otherwise their least index of disagreement has identical earlier restrictions, and applying G gives identical values there, a contradiction.
By ordinal induction, there is an approximation on every ordinal domain. The empty graph handles zero. Given the approximation p on β, extend it by the pair $(\beta, G(p))$ to obtain one on $\beta + 1$. Pairing and union make this extended graph a set. At a nonzero limit λ, the unique approximations p_β for $\beta < \lambda$ form a set by the set-image principle. Their union is a set function by compatibility, has domain λ, and satisfies the same recurrence: for $\gamma < \lambda$, any stage beyond γ supplies its value and all earlier values. This completes the induction.
Define F by collecting the pairs appearing in some approximation. This is class comprehension over set graphs, not a collection of proper classes. Compatibility makes F a function, and the approximations on successor domains show that its domain is all of On. For each ordinal α, its restriction is the unique set approximation on α, so the recurrence holds. Finally, two class solutions would have a least ordinal of disagreement; the same earlier-restriction argument excludes that disagreement. Thus the solution is unique. □

Syntax and interpretation. A variable is a syntactic symbol, while its assigned value belongs to the semantic domain. Similarly, a functional symbol belongs to the language; a family is a mathematical object that can interpret suitable instances of function notation. Keeping these levels separate is useful when formalizing recursion.

Definition 25.1 (class of well–founded sets W)**.**
The class of well–founded sets W is defined as the intersection of all classes Q satisfying $\mathcal{P}Q \subseteq Q$. Here a subset means a subclass that is a set.

$$\mathrm{W} \stackrel{\text{def.}}{:=} \{\, x \mid \forall Q,\ ((\mathcal{P}Q \subseteq Q) \longrightarrow (x \in Q)) \,\}$$

Notation 25.1 (intersection over a class predicate)**.**
An intersection indexed by a predicate on classes can be written as follows. Its index range is not itself assumed to be a class:

$$\mathrm{W} \stackrel{\text{def.}}{=} \bigcap_{\mathcal{P}Q \subseteq Q}^{Q} Q.$$

$$\bigcap_{\psi(Q)} t(Q) \stackrel{\text{def.}}{=} \{\, x \mid \forall Q, \psi(Q) \longrightarrow x \in t(Q) \,\}$$

For the following image notation, assume every $t(y)$, for $y \in Y$, is a set. If values may be proper classes, use the predicate-indexed binder instead, since ordinary class comprehension retains only set values.

$$\bigcap_{y \in Y} t(y) = \bigcap \{\, t(y) \mid y \in Y \,\} = \{\, m \mid \forall y \in Y, m \in t(y) \,\} = \{\, m \mid \forall y, y \in Y \longrightarrow m \in t(y) \,\}$$

$$= \{\, x \mid \forall Q,\ Q \in Y \longrightarrow x \in t(Q) \,\}$$

$$\bigcap_{y \in Y} t(y) = \bigcap \{\, t(y) \mid y \in Y \,\} = \bigcap (y \mapsto t(y))[Y]$$

The membership-recursion theorem will be proved below, after establishing transitivity and induction on W. Its existence proof also needs the set-formation principles used to collect compatible local solutions.

Lemma 25.1 (minimality of W)**.**
The class W is contained in every class closed under taking set subclasses.

$$\forall Q,\ (\mathcal{P}Q \subseteq Q) \longrightarrow (\mathrm{W} \subseteq Q)$$

Proof.
Fix Q. Assume $(\mathcal{P}Q \subseteq Q)$. To show: $\mathrm{W} \subseteq Q$.
Fix m.
Assume $m \in \mathrm{W}$.
To show: $m \in Q$.

$$(m \in \mathrm{W}) \implies \forall D,\ ((\mathcal{P}D \subseteq D) \longrightarrow (m \in D))$$

Set $D := Q$.
The assumption $(\mathcal{P}Q \subseteq Q)$ and the preceding statement imply $(m \in Q)$ by specification and modus ponens. □

Note 25.1 (regularity as exhaustion of the universe)**.**
Using this minimality property, the axiom of regularity may be formulated as $(\mathrm{V} \subseteq \mathrm{W})$.

Theorem 25.2 (W contains its subsets)**.**
Every set subclass of W is a member of it.

$$(\mathcal{P}\mathrm{W} \subseteq \mathrm{W})$$

Proof.

Fix $(x \in \mathcal{P}\mathrm{W})$. To show: $x \in \mathrm{W}$.

$$(x \in \mathcal{P}\mathrm{W}) \implies (x \text{ set}) \wedge (x \subseteq \mathrm{W})$$

To show: $(x \text{ set}) \wedge (\forall Q,\ (\mathcal{P}Q \subseteq Q) \longrightarrow (x \in Q))$.
Fix Q.
Assume $(\mathcal{P}Q \subseteq Q)$.
To show $(x \in Q)$.
The preceding lemma and the assumption give $(\mathrm{W} \subseteq Q)$.

$$(x \subseteq \mathrm{W}) \ \ \& \ \ (\mathrm{W} \subseteq Q) \implies (x \subseteq Q)$$

$$(x \text{ set}) \ \ \& \ \ (x \subseteq Q) \implies (x \in \mathcal{P}Q)$$

$$(x \in \mathcal{P}Q) \ \ \& \ \ (\mathcal{P}Q \subseteq Q) \implies (x \in Q)$$

□

Theorem 25.3 (Transitivity and induction on W)**.**
Without assuming that every set is well-founded,

$$\mathrm{W} = \mathcal{P}\mathrm{W}.$$

In particular, every member of a member of W belongs to W. For every class A, the following local induction principle holds:

$$\mathrm{W} \cap \mathcal{P}A \subseteq A \quad \implies \quad \mathrm{W} \subseteq A.$$

Proof. We already know $\mathcal{P}\mathrm{W} \subseteq \mathrm{W}$. Monotonicity of the powerclass operation gives $\mathcal{P}(\mathcal{P}\mathrm{W}) \subseteq \mathcal{P}\mathrm{W}$. Thus $\mathcal{P}\mathrm{W}$ is one of the closed classes in the defining intersection, and minimality gives $\mathrm{W} \subseteq \mathcal{P}\mathrm{W}$. This proves the equality and transitivity.
For induction, suppose $\mathrm{W} \cap \mathcal{P}A \subseteq A$ and put $Q = A \cup (\mathrm{V} \setminus \mathrm{W})$. If a set x is a subclass of Q, either $x \notin \mathrm{W}$, in which case $x \in Q$, or $x \in \mathrm{W}$. In the latter case all members of x lie in W, so $x \subseteq Q$ implies $x \subseteq A$. The induction hypothesis gives $x \in A \subseteq Q$. Hence $\mathcal{P}Q \subseteq Q$. Minimality yields $\mathrm{W} \subseteq Q$, which is exactly $\mathrm{W} \subseteq A$. □

Corollary 25.1 (Foundedness on W)**.**
Every nonempty subclass S of W has a member m with $m \cap S = \varnothing$.

Proof. Otherwise every member of S has a member in S. Put $A = \mathrm{W} \setminus S$. If $x \in \mathrm{W}$ and $x \subseteq A$, then x cannot belong to S, so $x \in A$. Induction gives $\mathrm{W} \subseteq A$, contrary to nonemptiness of S. □

Lemma 25.2 (Set transitive closures inside W)**.**
Assume additionally Infinity, pairing, union, heredity, and the set-image principle. For each $x \in \mathrm{W}$ there is a least transitive set T_x containing x as an element, and $T_x \subseteq \mathrm{W}$.

Proof. Form $B_0 = \{x\}$ and $B_{n+1} = \bigcup B_n$ for $n < \omega$, and put $T_x = \bigcup\{B_n : n < \omega\}$. The sequence exists by the already proved transfinite-recursion theorem restricted to ω; its rule uses only the preceding term. Thus this step does not assume membership recursion. Pairing and union make each term a set, and Infinity, set-image, and union make T_x a set.
If $z \in T_x$, then $z \in B_n$ for some n, so every member of z belongs to $B_{n+1} \subseteq T_x$. Thus T_x is transitive. A transitive class containing x contains every B_n as a subclass, by induction on n, and therefore contains T_x. Finally, transitivity of W and $x \in \mathrm{W}$ give $B_n \subseteq \mathrm{W}$ for every n, hence $T_x \subseteq \mathrm{W}$. □

Theorem 25.4 (Membership recursion on W)**.**
Assume MK comprehension and the set-formation and ordinal principles used above. For every class function $G : \mathrm{V} \to \mathrm{V}$ there is a unique class function $F : \mathrm{W} \to \mathrm{V}$ satisfying

$$F(x) = G(F \restriction x) \qquad (x \in \mathrm{W}).$$

Every restriction in this equation is a set graph. The assertion does not require the global regularity axiom $\mathrm{W} = \mathrm{V}$.

Proof. An approximation is a set function p with a transitive domain $D \subseteq \mathrm{W}$ such that $p(t) = G(p \restriction t)$ for every $t \in D$. The restrictions are sets by heredity. Two approximations agree on their common domain: if their disagreement class were nonempty, foundedness on W would give a member t with no earlier disagreement. Both transitive domains contain every member of t, so the restrictions at t agree. Applying G contradicts disagreement at t.
We prove by induction on W that an approximation exists on T_x. Suppose this is known for every $y \in x$. Compatibility makes the approximation p_y on T_y unique. The set-image principle therefore collects $\{p_y : y \in x\}$ into a set; no choice of local solutions is needed. Its union q is a set function with transitive domain $D = \bigcup_{y \in x} T_y$. For each $t \in D$, one of the p_y contains t and every member of t in its domain, so compatibility gives $q(t) = G(q \restriction t)$. Thus q is an approximation, and $x \subseteq D$.
If $x \in D$, retain q. Otherwise adjoin the single pair $(x, G(q \restriction x))$. The enlarged domain $D \cup \{x\}$ is transitive, and the old equations remain valid: no member of D has x as a member, by transitivity of D. The new equation at x holds by construction. In either case the resulting transitive domain contains x, hence contains T_x by minimality. Restricting the graph to T_x gives the required approximation. This completes the induction.
Let F collect the pairs belonging to some approximation. Comprehension quantifies over set graphs, and compatibility makes this class a function with domain W. The approximation on T_x agrees with F on x and at x, so $F \restriction x$ is a set and the recurrence holds. If F_1, F_2 both satisfy the recurrence, induction on W makes them agree: agreement on the members of x gives equal restricted graphs and then equal values at x. This proves uniqueness. □

Axiom 14 (reformulation of regularity/epsilon induction axiom)**.**

$$(\mathrm{V} \subseteq \mathrm{W})$$

26 Height of a set

Two related measures occur in this investigation. The first records finite membership depth, with ω representing unbounded finite depth. The second records ordinal rank. They agree on finite ordinals, but their recursions differ at and above limit stages; they are kept separate below. Assume Infinity, regularity, well-founded recursion, and sethood of the ordinal images used in the recursions.

26.1 Depth with left ordinal addition

Definition 26.1 (height)**.**
The author's height convention is

$$h(\varnothing) \coloneqq 0, \qquad h(x) \coloneqq 1 + \sup\{\, h(y) \mid y \in x \,\} \quad (x \neq \varnothing).$$

The addition is ordinal addition with 1 on the left. In particular, $1 + \omega = \omega$, so once the depths of members are unbounded among the finite ordinals, this measure records ω. Section 24.5 constructs addition and proves this identity using the set-valued transfinite recursion proved above; that recursion does not depend on the present height construction.

Example 2 (finite values)**.**

$$h(\varnothing) = 0, \qquad h(\{\varnothing\}) = 1, \qquad h(\{\varnothing, \{\varnothing\}\}) = 2.$$

More generally, $h(n) = n$ for each finite ordinal n.

Lemma 26.1 (height is at most omega)**.**

$$h : \mathrm{V} \to \omega \cup \{\omega\}.$$

Proof. Use membership induction. The empty set has height 0. Suppose the heights of all members of a nonempty set x lie in $\omega \cup \{\omega\}$. Their supremum is either finite or ω. Adding 1 on the left gives a finite ordinal in the first case and ω in the second. Thus $h(x)$ also lies in the stated range. □

This measure need not decrease strictly along membership: for example, $h(\omega) = h(\{\omega\}) = \omega$. It therefore cannot automatically replace ordinal rank in an argument requiring a strictly decreasing measure.

26.2 The least strict ordinal bound: rank

The alternative construction from the draft is best given a separate symbol, r, so that its recurrence is not confused with that of h:

$$r(x) \coloneqq \inf \{\, \alpha \in \mathrm{On} \mid r[x] \subseteq \alpha \,\} = \sup \{\, r(y) + 1 \mid y \in x \,\}.$$

The condition $r[x] \subseteq \alpha$ says that every member-height is strictly below α. This is the usual rank recurrence, with $r(\varnothing) = 0$. For example, $r(\omega) = \omega$ but $r(\{\omega\}) = \omega + 1$. The displayed infimum requires the set of member-ranks to have an ordinal bound. Under the assumptions stated at the start of this section, recursion and the boundedness of sets of ordinals provide that bound.

27 "Paradoxes" of mathematics

The arguments below separate inconsistent unrestricted comprehension from valid class constructions. Each result states the logical or set-theoretic principle it uses. In particular, the first two arguments are intuitionistic; the regularity argument uses membership induction, the diagonal arguments use separation, and the result about nonempty intersections uses classical logic.

27.1 There is no formula equivalent to its negation.

In this subsection, F is an arbitrary formula and negation is the special case $C \longrightarrow \bot$. The arguments use implication introduction and modus ponens, without the law of excluded middle.

Lemma 27.1 (formula implies itself)**.**

$$(C \longrightarrow C)$$

Proof.

1. $$(C \longrightarrow (C \longrightarrow C)) \qquad \text{(by ax.1)}$$
2. $$(C \longrightarrow ((C \longrightarrow C) \longrightarrow C)) \longrightarrow (((C \longrightarrow (C \longrightarrow C)) \longrightarrow (C \longrightarrow C))) \qquad \text{(by ax.2)}$$
3. $$(C \longrightarrow ((C \longrightarrow C) \longrightarrow C)) \qquad \text{(by ax.1)}$$
4. $$((C \longrightarrow (C \longrightarrow C)) \longrightarrow (C \longrightarrow C)) \qquad \text{(by MP from steps 2 and 3)}$$
5. $$(C \longrightarrow C) \qquad \text{(by MP from steps 1 and 4)}$$

□

Note 27.1 (Implication and classical principles)**.**
Via MP and the deduction theorem:

$$C \longrightarrow ((C \longrightarrow F) \longrightarrow F)$$

For comparison, Peirce's law is a classical principle and is not used in the arguments here:

$$(((P \longrightarrow Q) \longrightarrow P) \longrightarrow P)$$

Lemma 27.2 (Contraction of an implication)**.**

$$(C \longrightarrow (C \longrightarrow F)) \longrightarrow (C \longrightarrow F)$$

Proof.
Assume $(C \longrightarrow (C \longrightarrow F))$, and assume C.
By modus ponens from the assumptions, $(C \longrightarrow F)$.
Then F is derived from C and $(C \longrightarrow F)$ via modus ponens. □

Theorem 27.1 (Curry's paradox)**.**

$$(C \longleftrightarrow (C \longrightarrow F)) \longrightarrow F$$

Proof.
Assume $(C \longleftrightarrow (C \longrightarrow F))$.
Therefore $(C \longrightarrow (C \longrightarrow F))$ and $((C \longrightarrow F) \longrightarrow C)$.
The formula $(C \longrightarrow F)$ holds by lem:CtoF from $(C \longrightarrow (C \longrightarrow F))$.
Modus ponens applied to $((C \longrightarrow F) \longrightarrow C)$ and $(C \longrightarrow F)$ first gives C. A second application, to C and $(C \longrightarrow F)$, gives F. □

27.2 There is no class containing all irreflexive classes

Theorem 27.2 (Russell's paradox)**.**
Both in classical and in intuitionistic logic, the following holds:

$$\neg(\exists W \forall Q,\ Q \in W \longleftrightarrow Q \notin Q)$$

Proof.
Assume that such a class W exists and substitute $Q = W$. Put $P \equiv (W \in W)$. The assumed equivalence gives $P \longrightarrow \neg P$ and $\neg P \longrightarrow P$. Assuming P therefore gives both P and $\neg P$, so $\neg P$ follows by implication introduction. The second implication now gives P, hence a contradiction. No case distinction between P and $\neg P$ is needed. □

Theorem 27.3 (The universal class is proper without regularity)**.**
Assume class comprehension and separation for subclasses of a set. Then V is not a set. No foundation or choice axiom is needed.

Proof.
Form $R = \{x \in \mathrm{V} : x \notin x\}$. If V were a set, separation would make R a set, hence $R \in \mathrm{V}$. Its defining condition would then give $R \in R \longleftrightarrow R \notin R$, contradicting the preceding intuitionistic argument. Thus V is proper. □
This conclusion is compatible with non-well-founded sets: it rules out a universal set under separation, without ruling out individual membership cycles. The next irreflexivity theorem uses the stronger hypothesis of membership induction and should not be imported into settings lacking it.

27.3 Regularity implies that all classes are irreflexive

Theorem 27.4 (a class is not in itself)**.**
With membership induction, the membership relation is irreflexive in both classical and intuitionistic logic:

$$\forall Y,\ Y \notin Y \qquad \text{(thm:not in itself)}$$

Proof.
Let $C \equiv \{\, W \mid W \notin W \,\}$.
Goal: to show that $(\mathcal{P}C \subseteq C)$.

Fix class B. Assume $B \in \mathcal{P}C$.

$$(B \in \mathcal{P}C) \Longrightarrow (B \text{ set}) \wedge (B \subseteq C).$$

Claim: $\forall b \in B,\ b \notin b$.
$(b \in B)\ \ \&\ \ (B \subseteq C) \Longrightarrow (b \in C)$.
$(b \in C) \Longrightarrow (b \notin b)$.
Claim: $B \notin B$.
Assume opposite $B \in B$ and derive a contradiction.

$$(B \in B) \Longrightarrow (B \notin B) \qquad \text{(by previous claim)}$$

$(B \in B)\ \ \&\ \ (B \notin B) \Longrightarrow \bot$.
Contradiction obtained, so the claim $B \notin B$ is proven.
$(B \text{ set})\ \ \&\ \ (B \notin B) \Longrightarrow (B \in C)$.
We are done proving the goal $\mathcal{P}C \subseteq C$.

$$\mathcal{P}C \subseteq C \Longrightarrow V \subseteq C \qquad \text{(by epsilon induction)}$$

It remains to cover arbitrary classes, including proper classes. Fix Y and assume $Y \in Y$. Membership implies that Y is a set, so $Y \in V$. Since $V \subseteq C$, we obtain $Y \in C$ and hence $Y \notin Y$, a contradiction. Thus $Y \notin Y$, without deciding whether Y is a set. □

Theorem 27.5 (intersection set then nonempty)**.**
In classical logic, if the intersection of a class A is a set, then A is nonempty. Here the empty intersection is the universal class V.

$$\left(\left(\bigcap A\right)\ \text{set}\right) \longrightarrow \left(\exists w,\ w \in A\right) \qquad \text{(thm:intersection set then nonempty)}$$

Proof.
If A has no member, extensionality gives $A = \varnothing$, so $\bigcap A = \mathrm{V}$. Theorem 27.3 says that V is not a set, contradicting the hypothesis. Thus A is not empty; classical logic converts this into an existential witness $w \in A$. The argument uses separation to establish properness of V, but does not use membership induction. □

27.4 The powerclass of a set A is not a subclass of A

Theorem 27.6 (powerset is not a subset)**.**
In classical and even in intuitionistic logic, the following holds:

$$\forall (Y \text{ set}),\ \neg(\mathcal{P}Y \subseteq Y) \qquad \text{(thm:powerset not a subset)}$$

Proof. Use separation to form the set $D = \{x \in Y : x \notin x\}$. If $\mathcal{P}Y \subseteq Y$, then $D \subseteq Y$ and sethood give $D \in \mathcal{P}Y \subseteq Y$. Consequently $D \in D \longleftrightarrow D \notin D$, a contradiction as in Russell's argument. This proof requires neither excluded middle nor membership induction; separation supplies the diagonal set. □

Theorem 27.7 (Cantor's diagonal theorem)**.**
For any set A, no function f with domain A can have every subset of A among its values. In particular, no function $A \to \mathcal{P}A$ is surjective. This uses separation but not choice or foundation.

Proof.
Define $D = \{x \in A : x \notin f(x)\}$ by separation. If $f(a) = D$ for some $a \in A$, then

$$a \in D \longleftrightarrow a \notin f(a) \longleftrightarrow a \notin D,$$

which is impossible intuitionistically. Thus the subset D is missing from the range. The proof does not need the powerclass to be a set; the powerset axiom is needed only when treating that whole collection as a set-valued codomain. □

Remark 27.1 (Why the domain in Cantor's theorem is a set)**.**
The book's powerclass $\mathcal{P}C$ collects the *set* subclasses of C. Consequently

$$\mathcal{P}\mathrm{V} = \mathrm{V} :$$

every set is a subclass of V, and every member of a powerclass is a set. The identity class function on V is therefore a surjection onto $\mathcal{P}$V. This does not contradict Cantor's theorem, whose domain is a set. For that identity map the diagonal class is $R = \{x \in \mathrm{V} : x \notin x\}$. It is a subclass of V but is proper: if R were a set, then $R \in \mathrm{V}$ and comprehension would give $R \in R \longleftrightarrow R \notin R$. Thus R is not a member of $\mathcal{P}$V and cannot serve as a missing value within that codomain. The collection of *all* subclasses of V, including proper ones, is a different, conglomerate-level object.

27.5 The conglomerate of all classes is not a class

Theorem 27.8 (conglomerate of classes not class)**.**
A conglomerate of all classes is not a class. (In the sense, that it "does not exist as class".)

$$\neg(\exists W \forall Q,\ Q \in W) \qquad \text{(thm:conglomerate of classes not class)}$$

$$(\forall W \exists Q,\ Q \notin W)$$

Proof.
For any class W, take $Q = \mathrm{V}$. Theorem 27.3 shows that Q is a proper class; since membership in any class implies sethood, $Q \notin W$. Thus the same witness works for every W. This strengthens the earlier regularity-based argument: separation suffices, even in the presence of non-well-founded sets. □

27.6 The class of all ordinals is not a set

Lemma 27.3 (On is not in On)**.**

$$\mathrm{On} \notin \mathrm{On} \qquad \text{(lem:OnnotinOn)}$$

Proof.
Assume On ∈ On.
Therefore ∈ well–orders On as a well–founded strict linear order.
In particular strict order is irreflexive.
Under the assumption, On is itself an element of this strictly ordered class, so irreflexivity gives On ∉ On.
Contradiction with assumption.
Therefore On ∉ On is proven. □

Theorem 27.9 (the class of ordinal sets is not a set)**.**
The class On of all ordinals is a proper class, i.e. it is not a set.

$$\mathrm{On} \notin \mathrm{V} \tag{thm:OnNotSet}$$

Proof.
Assume opposite: $\mathrm{On} \in \mathrm{V}$. To show: $\bot$.

$$(\mathrm{On} \in \mathrm{V}) \implies (\mathrm{On}\ \mathrm{set}).$$

By theorem thm:ordon, Ord(On). The rest is done as in the previous lemma:

$$\mathrm{Ord}(\mathrm{On}) \ \ \& \ \ (\mathrm{On}\ \mathrm{set}) \implies (\mathrm{On} \in \mathrm{On});$$

$$\mathrm{Ord}(\mathrm{On}) \implies (E\ \mathrm{Irr}\ \mathrm{On});$$
$$(E\ \mathrm{Irr}\ \mathrm{On}) \implies (\mathrm{On} \notin \mathrm{On});$$
$$(\mathrm{On} \in \mathrm{On}) \ \ \& \ \ (\mathrm{On} \notin \mathrm{On}) \implies \bot.$$

□

28 Azriel Levy's Union Theorem

This section expands the argument attributed to Azriel Levy in "On Von Neumann's Axiom System for Set Theory" Levy, 1968. The source is available at https://www.jstor.org/stable/2315201. The aim is to derive sethood of unions from other principles, so the union axiom must not be used implicitly in the intermediate steps.

Local assumptions and proof status. The argument uses classical logic, limitation of size, ordinal ordering, heredity, powerset, pairing of sets, and the principle that the image of a set under a class function is a set. Auxiliary graphs in this proof use ordinary Kuratowski pairs of sets, whose formation requires only pairing; they do not rely on the recursive pairing construction, whose existence proof elsewhere assumes the union axiom. Evaluation of an ordinary function means its unique graph value on its domain, with the empty class supplied outside that domain. This local totalization must not be replaced by the V-default convention from another section. The diagonal step below follows the author's handwritten reconstruction in `azriel_levy/2.jpg` and `azriel_levy/3.jpg`. Its essential point is to restrict the diagonal function to the supposed unbounded set before giving it an index in the enumeration of all sets. The proof remains relative to the stated ordinal and class-size prerequisites; it does not claim that those prerequisites have been derived here from a smaller list of axioms.

Theorem 28.1 (The ordinal-enumeration form of the union argument)**.**
Suppose there is a class surjection $F : \mathrm{On} \to \mathrm{V}$. Assume heredity, set-image, ordinary set pairing, and the following ordinal facts: On is proper, is transitively strictly well-ordered by membership, every nonempty subclass of On has a least member, and every ordinal set has a set successor. Then the union of every set is a set.

The boundedness and covering proofs below prove this statement directly. Full limitation of size is used only to obtain the displayed surjection; powerset is used below to ensure successors are sets without using union. In particular, neither transfinite recursion nor the later membership recursion theorem is invoked. Those later proofs assume union and hence cannot be used to establish the ordinal prerequisites of this theorem without a separate dependency argument. The local theorem remains valid with the ordinal facts explicitly supplied, irrespective of how they are established in another presentation.

28.1 Definitions

Theorem 28.2 (The class of sets is injectable into the class of ordinal numbers)**.**

$$\exists T : \mathrm{V} \to \mathrm{On} \text{ injective}$$

Proof.
Use the properness of On from the local ordinal prerequisites. The actual limitation-of-size axiom, axm:limitation of size, gives a set-valued class function H whose graph image $H[\mathrm{On}]$ contains every set. For each set y, the class

$$C_y = \{\alpha \in \mathrm{On} \cap \mathrm{dom}(H) : H(\alpha) = y\}$$

is therefore nonempty. By the least-member property of the ordinals, it has a unique least element, denoted $T(y)$. Class comprehension forms the graph of this assignment, so $T : \mathrm{V} \to \mathrm{On}$

is a class function. If $T(y) = T(z) = \alpha$, functionality of H gives $y = H(\alpha) = z$. Thus T is injective.
This derives the needed injection from the book's surjective-image axiom, rather than silently treating the two formulations as definitions of one another. Least ordinal witnesses require no separate global choice function. The subsequent inverse-extension lemma supplies the surjection $\mathrm{On} \to \mathrm{V}$ used in the diagonal argument. □

Lemma 28.1 (an injection admits a surjective inverse extension)**.**

$$(\exists T : B \to C \text{ injective}), \longrightarrow ((\exists F : C \to B \text{ surjective}) \vee (B = \varnothing))$$

Proof.
If $B = \varnothing$, the second alternative holds. Otherwise fix $b \in B$ and an injection $T : B \to C$. Define

$$F \coloneqq T^{\top} \cup ((C \setminus T[B]) \times \{b\}).$$

The inverse graph $T^{\top}$ assigns to each member of $T[B]$ its unique preimage; the remaining members of C are sent to b. Thus $F : C \to B$ is a function and $F(T(u)) = u$ for every $u \in B$, proving surjectivity. □

Lemma 28.2 (The class of ordinal numbers is mapped surjectively onto the class of sets)**.**

$$(\exists F,\ \mathrm{On} \to \mathrm{V} \text{ surjective})$$

Proof.
It follows from the previous lemma, with $C \coloneqq \mathrm{On}$ and $B \coloneqq \mathrm{V}$, that

$$(\exists T : \mathrm{V} \to \mathrm{On} \text{ injective}) \longrightarrow ((\exists F : \mathrm{On} \to \mathrm{V} \text{ surjective}) \vee (\mathrm{V} = \varnothing))$$

From the axiom of existence of a set, it follows that $(\mathrm{V} \neq \varnothing)$.
From this fact and the preceding injection theorem, it follows that

$$(\exists F : \mathrm{On} \to \mathrm{V} \text{ surjective})$$

□

Definition 28.1 (well–ordering of universal class)**.**

$$\mathrm{F} \stackrel{\text{def.}}{\coloneqq} (\epsilon F,\ (F : \mathrm{On} \to \mathrm{V} \text{ surjective})) \tag{def:FwoV}$$

It follows from Lemma lem:exOnToVSurj and Axiom axm:H that

$$\mathrm{F} : \mathrm{On} \to \mathrm{V} \text{ surjective.}$$

Definition 28.2 (the "minimal" section of F)**.**
For each set y, the section G chooses the least ordinal mapped to y by F. In particular, $\mathrm{F}(\mathrm{G}(y)) = y$.

$$\mathrm{G}(y) \stackrel{\text{def.}}{\coloneqq} \inf \{\, x \in \mathrm{On} \mid \mathrm{F}(x) = y \,\} \tag{def:Gsect}$$

Definition 28.3 (a global choice functional symbol)**.**
Here we define a functional symbol c as an alias for some bound term:

$$c(S) :\equiv \mathrm{F}(\inf \mathrm{G}[S]) \tag{def:globalchoice}$$

Lemma 28.3 (global choice chooses)**.**

$$(S \neq \varnothing) \longrightarrow (c(S) \in S) \tag{lem:j}$$

Proof.

$$(S \subseteq \mathrm{V}) \;\;\&\;\; (\mathrm{G} : \mathrm{V} \to \mathrm{On}) \implies (\mathrm{G}[S] \subseteq \mathrm{On})$$
$$(S \neq \varnothing) \;\;\&\;\; (S \subseteq \mathrm{V}) \;\;\&\;\; (\mathrm{G} : \mathrm{V} \to \mathrm{On}) \implies (\mathrm{G}[S] \neq \varnothing)$$

Since $(\mathrm{G}[S] \subseteq \mathrm{On})$ and $(\mathrm{G}[S] \neq \varnothing)$, we have $((\mathrm{G}[S] \bigcap \mathrm{On}) \neq \varnothing)$.
Therefore, by Theorem thm:infordmin, $(\inf(\mathrm{G}[S]) \in \mathrm{G}[S])$. Choose $s \in S$ with $\mathrm{G}(s) = \inf(\mathrm{G}[S])$.
Then $c(S) = \mathrm{F}(\mathrm{G}(s)) = s \in S$. □

Definition 28.4 (tilde)**.**

$$\tilde{f} \stackrel{\text{def.}}{\coloneqq} \begin{cases} f, & \text{if } f \text{ is a function,} \\ \varnothing, & \text{otherwise.} \end{cases} \tag{def:tilde}$$

Definition 28.5 (evaluation class–function)**.**

For each set f, first replace a nonfunction by $\varnothing$ using $\tilde{f}$. Then evaluate by graph lookup, supplying $\varnothing$ if x is outside the domain:

$$\mathrm{ev}_x \stackrel{\text{def.}}{\coloneqq} \left\{ (f, \tilde{f}(x)) \;\middle|\; f \in \mathrm{V} \right\} \tag{def:evclass}$$

For a function graph, an output is a set because it occurs as a component of a set pair; the default is also a set. Thus $\mathrm{ev}_x : \mathrm{V} \to \mathrm{V}$ is a class function without using the union axiom to evaluate an arbitrary multivalued relation.

Definition 28.6 (R function)**.**

$$\mathrm{R} \stackrel{\text{def.}}{\coloneqq} (\tilde{\cdot}) \circ \mathrm{F}) \tag{def:Rfunc}$$

Definition 28.7 (Q(x))**.**

$$Q(x) \stackrel{\text{def.}}{\coloneqq} \mathrm{ev}_x[R[x]] \tag{def:Qx}$$

Definition 28.8 (h function)**.**

$$h(x) \stackrel{\text{def.}}{\coloneqq} \inf(\mathrm{On} \setminus (Q(x))) \tag{def:hfunc}$$

Definition 28.9 (an enumeration index for a set function)**.**
For each set function u, put

$$\beta_u \stackrel{\text{def.}}{\coloneqq} \mathrm{G}(u). \tag{def:beta}$$

Then $\beta_u \in \mathrm{On}$ and $\mathrm{F}(\beta_u) = u$. This definition requires u to be a set; it will be applied to a restricted diagonal function.

Definition 28.10 (bounded subclasses of On)**.**
In the context of $(b \subseteq \mathrm{On})$,

$$(b \text{ bounded}) \stackrel{\text{def.}}{\longleftrightarrow} (\exists \beta \in \mathrm{On} : b \subseteq \beta)$$
$$(b \text{ unbounded}) \stackrel{\text{def.}}{\longleftrightarrow} (\neg(b \text{ bounded})) \tag{def:bddscOn}$$

28.2 Sets of ordinals are bounded

We first record why ordinal successors are available without the union axiom. If $\alpha \in \mathrm{On}$, transitivity gives $\alpha \subseteq \mathcal{P}\alpha$, and $\alpha \in \mathcal{P}\alpha$. Hence $\alpha \cup \{\alpha\} \subseteq \mathcal{P}\alpha$; powerset and heredity make this successor a set. In particular, an unbounded subclass of On has an element strictly above each specified ordinal.

Lemma 28.4 (sets of ordinals are bounded)**.**

$$(b \text{ set}) \wedge (b \subseteq \mathrm{On}) \longrightarrow (b \text{ bounded}) \tag{lem:ssOnBdd}$$

Proof.
Step 1: for an ordinal set x, the image $R[x]$ is a set by the set-image principle. The function $\mathrm{ev}_x : \mathrm{V} \to \mathrm{V}$ then makes $Q(x) = \mathrm{ev}_x[R[x]]$ a set. Since On is proper, it cannot be contained in $Q(x)$, by heredity. Therefore $h(x)$ exists as the least omitted ordinal and satisfies the following implications:

$$\begin{aligned}
(x \text{ set}) &\implies (Q(x) \text{ set}) \\
(Q(x) \text{ set}) &\implies ((\mathrm{On} \setminus Q(x)) \neq \varnothing) \\
&\implies \inf(\mathrm{On} \setminus Q(x)) \in (\mathrm{On} \setminus Q(x)) \\
&\implies h(x) \in (\mathrm{On} \setminus Q(x)) \\
&\implies (h(x) \in \mathrm{On}) \wedge (h(x) \notin Q(x))
\end{aligned}$$

Step 2:

$$\begin{aligned}
(y \text{ set}) \;\&\; (F : \mathrm{On} \to \mathrm{V} \text{ surjective}) &\implies F^{\top}[\{y\}] \neq \varnothing \\
&\implies \inf(F^{\top}[\{y\}]) \in F^{\top}[\{y\}] \\
&\implies G(y) \in F^{\top}[\{y\}] \\
&\implies (G(y) \in \mathrm{On}) \wedge (F(G(y)) = y)
\end{aligned}$$

Step 3: suppose, for a contradiction, that b is an unbounded set of ordinals. Form the restricted diagonal graph

$$u \coloneqq \{\, (x, h(x)) \mid x \in b \,\}.$$

Step 1 shows that each $h(x)$ is an ordinal set. Pairing makes $(x, h(x))$ a set, and the image principle applied to b makes u a set function with domain b. We may therefore put $\beta = \mathrm{G}(u)$. By Step 2, $\mathrm{F}(\beta) = u$, and since u is a function, $R(\beta) = \widetilde{\mathrm{F}(\beta)} = u$.
Choose $x \in b$ with $\beta \in x$, using unboundedness and the ordinal successor fact above. Then $u \in R[x]$, so

$$h(x) = u(x) = \mathrm{ev}_x(u) \in \mathrm{ev}_x[R[x]] = Q(x).$$

But Step 1 gives $h(x) \notin Q(x)$. This contradiction proves that b is bounded. Restriction to b is indispensable: the global rule h need not itself be a set and cannot be assigned an index by G merely by writing $\mathrm{G}(h)$. □

28.3 Union of a set is a set

By the preceding boundedness lemma, for each set b choose an ordinal $H(b)$ such that $\mathrm{G}[b] \subseteq H(b)$; the least such bound gives a class-function choice. If a is a set, the image principle makes $H[a]$ a set of ordinals. Choose $\alpha \in \mathrm{On}$ with $H[a] \subseteq \alpha$. The next lemma is stated in this context.

Lemma 28.5 (one ordinal image covers the union)**.**

$$F[\alpha] \supseteq \bigcup a \tag{lem:techlem}$$

Proof.
Fix $n \in \bigcup a$. To show: $n \in F[\alpha]$.

$$\left(n \in \bigcup a\right) \implies \exists b \in a,\ n \in b.$$

By Lemma lem:ssOnBdd, $H(b)$ is a strict upper bound of $G[b]$.

$$(b \text{ set}) \implies (G[b] \in \mathcal{P}(\mathrm{On}))$$

$$\implies (G[b] \text{ is bounded})$$

There is a bound; therefore, there is a strict bound.

$$(a \text{ set}) \ \&\ (H \text{ class–function}) \longrightarrow (H[a] \text{ set})$$

$$(H[a] \text{ set}) \ \&\ (H[a] \subseteq \mathrm{On}) \implies (\exists \alpha \in \mathrm{On} : H[a] \subseteq \alpha)$$

$$(b \in a) \implies (H(b) \in H[a])$$

$$H[a] \subseteq \alpha \implies H(b) \in \alpha$$

$$H(b) \in \alpha \implies H(b) \subsetneq \alpha$$

$$G[b] \subseteq H(b) \ \&\ H(b) \subsetneq \alpha \implies G[b] \subseteq \alpha$$

$$\implies (F[G[b]] \subseteq F[\alpha])$$

$$F[G[b]] = b$$

$$b \subseteq F[G[b]] \ \&\ (F[G[b]] \subseteq F[\alpha]) \implies (b \subseteq F[\alpha])$$

$$(n \in b) \ \&\ (b \subseteq F[\alpha]) \implies (n \in F[\alpha])$$

□

Theorem 28.3 (union of a set is a set)**.**

$$\left(A \text{ set}\right) \longrightarrow \left(\bigcup A \text{ set}\right) \tag{thm:unionisset}$$

Proof.
Let a be an arbitrary set (the argument denoted by A in the statement).

$$H(b) \stackrel{\text{def.}}{\coloneqq} \inf \{\, e \mid e \text{ strict upper bound of } G[b] \,\}$$

$$(a \text{ set}) \ \&\ (H \text{ class–function}) \longrightarrow (H[a] \text{ set})$$

$$(H[a] \text{ set}) \ \& \ (H[a] \subseteq \text{On}) \implies (H[a] \in \mathcal{P}(\text{On}))$$
$$(H[a] \in \mathcal{P}(\text{On})) \implies (\exists \alpha \in \text{On} : H[a] \subseteq \alpha)$$
$$(\alpha \text{ set}) \ \& \ (F \text{ class–function}) \implies (F[\alpha] \text{ set})$$

By Lemma lem:techlem, $F[\alpha] \supseteq \bigcup a$.

$$(F[\alpha] \text{ set}) \ \& \ \left(\left(\bigcup a\right) \subseteq (F[\alpha])\right) \implies \left(\left(\bigcup a\right) \text{ set}\right)$$

□

29 Theorems Related to Limitation of Size

The goal is to derive a cardinal form of limitation of size from a smaller collection of assumptions. Here V is the class of all sets, and $|K|$ is the ordinal-valued cardinality convention developed in the cardinal section, with $|V| = \mathrm{On}$. The displayed equivalences express the target; they are not additional premises of the derivation below.
For the argument that follows, assume heredity, preservation of sethood under class-function images, the ordinal-valued cardinality definition, and $(K \in V) \Rightarrow (|K| \in \mathrm{On})$. In particular, the last condition is a substantive set-size assumption, not a consequence of notation alone.

Axiom 3 (Limitation of Size)**.**

$$(K \in V) \longleftrightarrow (|K| < |V|)$$

With these cardinality and comparison conventions, the target can also be written:

$$(K \in V) \longleftrightarrow (|K| \subsetneq |V|)$$

$$(K \in V) \longleftrightarrow (|K| \neq |V|)$$

because $|V| = \mathrm{On}$.

29.1 Proof of the Limitation of Size via Other Axioms

We want to prove the following statement:

$$(K \in V) \longleftrightarrow (|K| \neq |V|)$$

We separate the two implications.

Lemma 29.1 (First Part of the Limitation of Size Statement)**.**

$$(K \in V) \longrightarrow (|K| \neq |V|)$$

Proof. Assume $(K \in V)$.
By the stated set-size assumption, $(|K| \in \mathrm{On})$.
By previous lemmas we have $|V| = \mathrm{On}$ and $\mathrm{On} \notin \mathrm{On}$.
From these three statements we deduce that $|K| \neq |V|$.
□

Lemma 29.2 (Second Part of the Limitation of Size Statement)**.**

$$(|K| \neq |V|) \longrightarrow (K \in V)$$

Proof. Assume $(|K| \neq |V|)$.
We know that $(|V| = \mathrm{On})$, so $(|K| \neq \mathrm{On})$.
Recalling the definition: $|K| \coloneqq \mathrm{On} \cap \bigcap \{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$.
If the class $C = \{\, a \in \mathrm{On} \mid K \preccurlyeq a \,\}$ were empty, then $|K| = \mathrm{On} \cap \bigcap \varnothing = \mathrm{On}$, contrary to the assumption. Thus choose $a \in C$ and an injection $f : K \to a$. The image $f[K] \subseteq a$ is a set by heredity. The inverse graph of f is a class function on $f[K]$ with image K. Preservation of sethood under class-function images therefore gives $K \in V$. □

29.2 What this cardinal criterion does and does not establish

The proved equivalence uses the primary cardinal convention. Under the ordinary set-formation assumptions in the cardinal chapter, its set-size premise is equivalent to the axiom of choice for sets. It does not by itself construct a class injection $\mathrm{V} \to K$ for every proper class K. Assigning the common default On to proper classes is not a proof that they are equinumerous.

Proposition 29.1 (The extra class-size assertion)**.**
Assume V is proper, heredity, set-image, and class Schroeder–Bernstein. The following two class-size statements are equivalent:

1. every proper class K receives an injection from V;
2. every proper class K is in bijection with V.

Together with heredity and set-image, either gives the injection form

$$K \text{ is a set} \longleftrightarrow \neg(\mathrm{V} \preccurlyeq K).$$

Proof. The inclusion $K \subseteq \mathrm{V}$ is always injective; combine it with the injection in the first statement using class Schroeder–Bernstein. The reverse implication follows by forgetting surjectivity. If K is a set and $f : \mathrm{V} \to K$ is injective, heredity makes its image a set and set-image applied to its inverse makes V a set, a contradiction. If K is proper, either class-size statement supplies the forbidden injection. Classical case distinction proves the displayed equivalence. □
This is the stronger assertion used in the union argument of the preceding section. That argument obtains an ordinal enumeration of all sets from $\mathrm{V} \preccurlyeq \mathrm{On}$; the ordinal-valued default convention alone cannot supply such an enumeration. Conversely, the present two elementary cardinality lemmas do not use the union theorem as an intermediate step: their stated premises are heredity, set-image, the ordinal comparison facts, and the set-size hypothesis.

30 Properties Preserved by Intersections of Universes

The relevant preservation principle concerns common operations in one ambient theory. It does not say that arbitrary axioms, or all properties called positive, survive intersection. We first isolate the closure argument, then apply it to candidate universes.

Theorem 30.1 (Least class closed under a monotone operation)**.**
Work in MK, so that the following quantification over classes is available. Let S be a class and let Φ be a definable operation taking classes to classes, monotone in the sense that

$$A \subseteq B \quad \Longrightarrow \quad \Phi(A) \subseteq \Phi(B).$$

Define

$$C = \{\, x \mid \forall Q,\ (S \subseteq Q \wedge \Phi(Q) \subseteq Q) \longrightarrow x \in Q \,\}.$$

Then C is the least class containing S and closed under Φ. Moreover,

$$C = S \cup \Phi(C).$$

The operation Φ here is specified by a formula or functional symbol; no class of proper classes is required.

Proof. The class V is an admissible Q, since every class is a subclass of V. Every admissible Q contains S, so $S \subseteq C$. For such a Q, the definition gives $C \subseteq Q$, and monotonicity gives $\Phi(C) \subseteq \Phi(Q) \subseteq Q$. Hence $\Phi(C) \subseteq C$. The same definition proves minimality among the admissible classes.
Put $D = S \cup \Phi(C)$. We have $D \subseteq C$, so monotonicity yields $\Phi(D) \subseteq \Phi(C) \subseteq D$. Since $S \subseteq D$, minimality gives $C \subseteq D$. Thus $C = D$. □
For $S = \varnothing$ and $\Phi(Q) = \mathcal{P}Q$, this recovers the construction of W and the identity $\mathrm{W} = \mathcal{P}\mathrm{W}$. The identity describes the least fixed point selected by the closure condition; it does not assert that every fixed point is least.

Theorem 30.2 (Intersection of candidate universes)**.**
Let I be a nonempty class of set indices, and let a predicate $U(i, x)$ specify a class U_i for each $i \in I$. Put

$$U = \{\, x \mid \forall i \in I,\ U(i, x) \,\}.$$

Each of the following properties of all the U_i is inherited by U:

1. transitivity;
2. containing a specified set c;
3. closure under pairs: $a, b \in U_i \Rightarrow \{a, b\} \in U_i$;
4. closure under union: $a \in U_i \Rightarrow \bigcup a \in U_i$;
5. closure under the ambient powerset: $a \in U_i \Rightarrow \mathcal{P}a \in U_i$;
6. closure under indexed unions: if $A \in U_i$ and $f : A \to \mathrm{V}$ is any class function with $f[A] \subseteq U_i$, then $\bigcup f[A] \in U_i$.

If at least one U_i is a set, then U is a set by heredity. All operations in these conditions have the same ambient interpretation for every index.

Proof. If $a \in U$ and $b \in a$, transitivity of every U_i gives $b \in U_i$ for every i, hence $b \in U$. A common specified member likewise belongs to the intersection. If $a, b \in U$, both are in each U_i, so their pair belongs to each U_i and therefore to U. Exactly the same argument works for the uniquely specified sets $\bigcup a$ and $\mathcal{P}a$.
For indexed unions, let $A \in U$ and $f[A] \subseteq U$. For each $i \in I$, the same set A belongs to U_i, and the same function f has all its values in U_i. Its union belongs to each U_i by the assumed closure, hence to U. Finally, $U \subseteq U_i$ for every index, so any set among these classes bounds U and heredity applies. □

Corollary 30.1 (Intersections with the usual universe closure properties)**.**
A nonempty intersection of transitive set universes closed under pairs, ambient powersets, and the indexed unions above has those same properties. If each contains ω, so does the intersection.

Proof. Apply the theorem to each defining closure property, and use any one of the intersected sets to establish sethood of the intersection. □
These statements distinguish external closure from satisfaction of an internally interpreted axiom. For example, an internal powerset $\{\, b \in U_i \mid b \subseteq a \,\}$ can depend on i; the powerset clause above instead requires the full ambient $\mathcal{P}a$. An existential requirement also needs separate treatment: two classes can each have a member while their intersection is empty. Thus a common specified witness, such as ω, is stronger than unrelated witnesses to an existence statement. No consistency or model-existence claim follows merely from the preservation results.

31 Types as predicates on classes

This section investigates interpretations of dependent sums, dependent products, and inductive and coinductive types using classes. Its central idea is that a type specifies a unary predicate on possible term values. The instances may include proper classes, so their collection need not itself be a class in the object theory.

Local models and earlier proposals. The subsections compare several representations; they do not assert that all displayed models are interchangeable. In particular, a class of set values, a predicate on arbitrary classes, and a syntactic type expression belong to different levels. A soundness claim must identify which of these is being interpreted and which family encoding it uses.

Note 31.1 (OUTDATED: the unrestricted class-as-type proposal)**.**
An earlier proposal assigned truth to every type declaration:

$$\text{semantics}(\text{“}(A : \text{Type})\text{”}) \stackrel{\text{def.}}{\coloneqq} \top.$$

This does not enforce the intended uniqueness of the term–type relation. It is retained as motivation for interpreting types by explicit predicates. In that earlier family-based proposal, a dependent declaration was written

$$\text{semantics}(\text{“}(x : A \vdash B(x) : \text{Type})\text{”}) \stackrel{\text{def.}}{\coloneqq} (B \text{ family}) \wedge (\text{Dom}(B) \subseteq A),$$

and the semisingleton encoding was

$$(\Lambda x,\ t) = \{\, (x, s) \mid (s \subseteq t) \wedge (s \text{ ss}) \,\}.$$

The domain condition and evaluation rules for a dependent type must be specified in the particular model; this sketch does not supply them all.

31.1 Explicit typed term interpretation (μ_A / I)

Throughout this chapter, fix a background model of the chosen class theory (NBG or MK) and a valuation of its free variables. Neither the model nor the valuation needs to be specified further here. With that context fixed, this chapter uses μ_A (written $I(a)$ or $\mu_A(a)$ below) for an *explicit typed term interpretation*, indexed by the type A, at the level of expressions:

$$\mu_A : A.\mathsf{Expression} \to \mathsf{Class.Expression},$$

which translates a typed expression $a : A$ into a class expression $\mu_A(a) \equiv I(a)$. Evaluating the translated expression in the fixed model is a further semantic operation; the expression itself is not a class object. The term *valuation homomorphism* can be used for the resulting semantic interpretation when preservation of the relevant constructors has been proved. That preservation is an obligation of the model, not a consequence of the map's displayed type. The companion catalog's comment on `DefMetaGeneralisation` (4.02) likewise describes a typed expression translation. Its symbol page for μ_A instead uses A for an assignment and describes semantic valuation. These are different local signatures; the notation here is the book's convention, not a claim that both exports or an uninspected kernel implementation already realize this same map.

Note 31.2 (advanced semantic setting: terms as classes, conglomerates (= predicates on classes) as types)**.**
Unlike the standard set-theoretic setting where terms are interpreted as sets and classes are interpreted as types, we work in a one-level-higher setting:

1. **Terms are interpreted as classes** (which may be either sets or proper classes such as V and On).

2. **Types are interpreted as conglomerates**. Here this means **predicates on the conglomerate of all classes** (Class $\to$ Bool). In particular, the type Class is the constant-true predicate on classes $\lambda(C : \text{Class}).\ \top$ (informally, a constant conglomerate-function to Bool, representing the conglomerate of all classes).

3. **Class is the default type**: it is implicitly assumed whenever a quantifier or comprehension omits an explicit type ($\forall x,\ \varphi(x) :\equiv \forall(x : \text{Class}),\ \varphi(x)$) and in **bound (bounded) quantifiers**:
$$\forall(x \in Y),\ \varphi(x) \overset{\text{def.}}{\longleftrightarrow} \forall(x : \text{Class}),\ (x \in Y) \longrightarrow \varphi(x),$$
$$\exists(x \in Y),\ \varphi(x) \overset{\text{def.}}{\longleftrightarrow} \exists(x : \text{Class}),\ (x \in Y) \wedge \varphi(x).$$

31.2 Predicate interpretation and collected extensions

We now give a uniform interface for the local predicate models. The letters P, Q stand schematically for formulas with a designated class argument C and specified parameters; they are not members of an object-theoretic class of all predicates. Work with MK comprehension, or restrict the formulas to the comprehension instances admitted in the chosen NBG fragment. The same restriction applies when a later comprehension or induction argument uses a schematic predicate. A fixed class parameter is permitted in an elementary NBG formula; a quantified variable ranging over arbitrary classes is different. Fix the background model and an assignment ρ of the free parameters.
For a type expression A, a specified interpretation supplies a predicate $P_A(C)$ on class values. Write $\mathrm{Emb}(A)$ for this predicate, not for a class that contains every value satisfying it. If $a : A$ has a target-language translation $\mu_A(a)$, put

$$v_\rho(a) = \mathrm{Val}_\rho(\mu_A(a)).$$

Thus $\mu_A(a)$ is syntax and $v_\rho(a)$ is its value. Soundness of a typing rule means that, for every assignment satisfying its context, its premises imply $P_A(v_\rho(a))$ for the conclusion $a : A$. This defines the obligation; it does not prove arbitrary proposed rules sound merely by assigning them this notation. When $I(a)$ occurs in a target-language equation below, its semantic reading is the value $v_\rho(a)$ of that class expression.
For a permitted predicate P, define its *collected extension* by guarded comprehension:

$$\kappa(P) = \{C : C \text{ is a set and } P(C)\}.$$

For a class X, define its membership predicate

$$\mathrm{Emb}_{\mathrm{cl}}(X)(C) \longleftrightarrow C \in X.$$

A type presented by the class carrier X uses this last predicate. These are schematic operations on formulas and class parameters, not an ordinary class-function graph whose arguments include all predicates.

Theorem 31.1 (What collection preserves)**.**
For every class C and every admitted predicate P,

$$C \in \kappa(P) \longleftrightarrow (C \text{ is a set and } P(C)).$$

Moreover $\kappa(\mathrm{Emb}_{\mathrm{cl}}(X)) = X$ for every class X. Consequently collecting a predicate and then taking the membership predicate recovers exactly its restriction to set values. It recovers P on all classes if and only if P has only set instances.

Proof.
The first equivalence is guarded comprehension. Every member of X is already a set, so the same equivalence with $P(C)$ given by $C \in X$ says that $\kappa(\mathrm{Emb}_{\mathrm{cl}}(X))$ and X have precisely the same members. Extensionality gives their equality. Finally,

$$\mathrm{Emb}_{\mathrm{cl}}(\kappa(P))(C) \longleftrightarrow (C \text{ is a set and } P(C)).$$

This agrees with $P(C)$ for all classes exactly when $P(C)$ implies sethood for every C. □

Example 31.1 (Collection forgets proper-class instances)**.**
Assume V is proper. The predicates $P(C) \longleftrightarrow \bot$ and $Q(C) \longleftrightarrow C = \mathrm{V}$ have the same collected extension, namely $\varnothing$, although $Q(\mathrm{V})$ holds and $P(\mathrm{V})$ does not. Likewise the intended type Class, interpreted by the constantly true predicate, has collected extension V. Its original predicate accepts proper classes as well, whereas membership in V accepts only sets. Thus the type Class cannot be replaced by its collected extension without changing its instances.

The interface therefore retains the full predicate P_A whenever proper-class values are allowed. Membership in $\kappa(\mathrm{Emb}(A))$ can replace a typing predicate only for a set-valued instance or a model whose instances are all sets. The option model below supplies such a model and proves its constructor clauses; it does not discard the other, genuinely class-valued interpretations in this chapter.

31.3 Finite storage and semantic functions

A stored expression is a finite tree with a constructor tag, finitely many immediate subexpressions, and finite encodings of identifiers and type annotations. For example, an abstraction stores its bound-variable name, its annotation, and its body. It need not store the graph of the function denoted by that body. The finite syntax also makes structural recursion available for renaming, substitution, and the specified type checker. Bound names must still be treated with the freshness conditions proved in the language chapter.
The language chapter verifies these obligations for its specified simple fragment: recovery of types terminates and matches the typing rules; substitution and reduction preserve types; denotational substitution and reduction preserve meanings; normalization and confluence justify its conversion procedure. Those are distinct results. A finite representation alone proves none of them, and their proofs do not automatically extend to every proposed dependent or class-valued constructor here.

Proposition 31.1 (Finite syntax does not name every semantic function)**.**
Suppose an expression language has an explicitly countable alphabet, and every expression is finite. Fix its interpretation. Among its closed expressions, consider those whose denotations are total functions $\mathbb{N} \to \{0, 1\}$. Not every such function can be denoted.

Proof.
Coding finite strings over the given alphabet by natural numbers gives a sequence of all possible expression codes. At code n, let f_n be the denoted binary-valued function if that code is a suitable closed expression; otherwise let f_n be the constant-zero function. This is a metatheoretic definition in the fixed interpretation, not a claim that semantic validity is decidable. Define

$$g(n) = 1 - f_n(n).$$

For every n, the function g differs from f_n at n. It therefore differs from the denotation of every suitable closed expression, proving the assertion. This argument takes place in the ordinary set metatheory of the fixed interpretation; it does not define an internal truth or universal evaluation predicate in the object language. □

The fixed interpretation and closed-expression restrictions matter. Allowing a free parameter to range over all functions can supply all those functions as values of one variable expression, but the varying parameter is additional semantic data. Similarly, a separate primitive name for every semantic function would violate the countable-alphabet hypothesis. Finite syntax is a faithful storage format for expressions, not an enumeration of all possible class functions or all set functions. Countability does not assert that each denoted function is computable: a fixed primitive constant may have a noncomputable interpretation.

31.4 Implementing the boolean type

Axioms of the boolean type

The class-valued conditional below has two computation laws. We then give a set model of the Boolean type with its dependent elimination rule. These are related constructions: a Boolean value is an element of a two-element set, whereas a proposition is a formula.

Model of boolean type

Formation of boolean type:
Introduction of terms of boolean type:
Elimination of terms of boolean type:

Definition 31.1 (conditional operator if–then–else / $\mathrm{if}_{\mathrm{dep}}$)**.**

$$(\text{if } \varphi \text{ then } t_1 \text{ else } t_2) \stackrel{\text{def.}}{\coloneqq} \mathrm{if}_{\mathrm{dep}}(\varphi, t_1, t_2) \stackrel{\text{def.}}{\coloneqq} (\{\, z \mid ((z \in t_1) \wedge \varphi) \vee ((z \in t_2) \wedge \neg\varphi) \,\})$$

Side conditions:
(z is not free in t_1),
(z is not free in t_2),
(z is not free in φ).

Computation of terms of boolean type:

Theorem 31.2 ($\mathrm{if}_{\mathrm{dep}}$–true–elim)**.**

$$(\varphi \longrightarrow (\mathrm{if}_{\mathrm{dep}}(\varphi, t_1, t_2) = t_1))$$

Theorem 31.3 ($\mathrm{if}_{\mathrm{dep}}$–false–elim)**.**

$$((\neg\varphi) \longrightarrow (\mathrm{if}_{\mathrm{dep}}(\varphi, t_1, t_2) = t_2))$$

Soundness of the boolean type

Under φ, the comprehension defining $\mathrm{if}_{\mathrm{dep}}$ has exactly the members of t_1; under $\neg\varphi$, it has exactly the members of t_2. Extensionality therefore proves the displayed computation laws. Each conditional implication holds intuitionistically; selecting a branch for an arbitrary formula requires deciding that formula.

Definition 31.2 (The two-element set model)**.**
Put $0 = \varnothing$, $1 = \{0\}$ and $\mathrm{Bool} = \{0, 1\}$. Interpret false by 0 and true by 1. Pairing and the empty-set axiom make these sets. They are distinct because $0 \in 1$ but 0 has no members. Thus every $b \in \mathrm{Bool}$ is exactly one of these constructors.

Theorem 31.4 (Dependent Boolean elimination)**.**
Let E be a set-valued family on Bool and suppose $e_0 \in E(0)$, $e_1 \in E(1)$. There is a unique function r with domain Bool such that $r(b) \in E(b)$ and

$$r(0) = e_0, \qquad r(1) = e_1.$$

For every dependent function q on Bool, elimination with branches $q(0)$ and $q(1)$ returns q itself.

Proof.
Use the graph $\{(0, e_0), (1, e_1)\}$ with ordinary injective set pairs. Distinctness of $0, 1$ makes it functional with precisely the desired domain, and its values have the required types by hypothesis. Any other function satisfying the equations agrees at both possible arguments and hence equals it by extensionality. Taking $e_i = q(i)$ proves the last assertion, the extensional Boolean eta law. □

Corollary 31.1 (Boolean induction)**.**
If a predicate P holds at 0 and 1, it holds at every element of Bool.

Proof.
Membership in $\{0, 1\}$ gives $b = 0$ or $b = 1$. Substitute in the corresponding premise. This uses the definition of the pair set, not a law deciding every proposition in the surrounding logic. □
For a decidable formula φ, the Boolean characteristic value is 1 when φ and 0 when $\neg\varphi$. Applying elimination with branches t_2, t_1 agrees with the earlier conditional when these branch values are sets. That agreement follows by the two cases. Arbitrary class-valued branches still have the earlier comprehension-based conditional, but they cannot both be stored as values of an ordinary set graph unless they are sets. This is the same size distinction encountered in family encodings.

31.5 Implementing option types

The semisingleton model concerns set-valued instances: the empty class represents no value, and a singleton represents one set value. It does not encode a proper-class value by placing it inside a singleton. For a predicate P, define the set-valued option predicate by

$$\begin{aligned}\mathrm{Opt}(P)(b) \longleftrightarrow\ & b \text{ is a set } \wedge \big(b = \varnothing \vee \\ & \exists(a \text{ a set}),\ P(a) \wedge b = \{a\}\big).\end{aligned}$$

For a type expression A interpreted by P_A, the local option interpretation is $P_{\mathrm{Option}(A)} = \mathrm{Opt}(P_A)$. This clause is uniform in the supplied predicate, but its Some branch admits only set instances. In the following displayed introduction rules, A is represented by a class carrier, so every instance is already a set. The class-valued types elsewhere in the chapter keep their full predicate interpretations.

Axioms of option types

Axiom 15 (None–ax)**.**

$$\mathrm{None} : \mathrm{Option}(A)$$
$$I(\mathrm{None}) \in \kappa(\mathrm{Emb}(\mathrm{Option}(A)))$$

Axiom 16 (Some–ax)**.**

$$\frac{(a : A)}{(\mathrm{Some}(a) : \mathrm{Option}(A))}$$
$$(I(a) \in \kappa(\mathrm{Emb}(A))) \longrightarrow (I(\mathrm{Some}(a)) \in \kappa(\mathrm{Emb}(\mathrm{Option}(A))))$$

The semantic constructor interpretations below verify these rules for the stated class-carrier model. They do not assert an unguarded Some rule for a proper-class instance of an arbitrary predicate type.

Model of option types

Definition 31.3 (None–def)**.**

$$I(\mathrm{None}) \stackrel{\mathrm{def.}}{:=} \varnothing$$

Definition 31.4 (Some–def)**.**

$$I(\mathrm{Some}(a)) \stackrel{\mathrm{def.}}{:=} \{I(a)\}$$

Definition 31.5 (Predicate and collected option interpretations)**.**
For a type expression A with supplied predicate interpretation,

$$\mathrm{Emb}(\mathrm{Option}(A)) = \mathrm{Opt}(\mathrm{Emb}(A)).$$

This is an equality of predicate interpretations. For a class carrier X, write $O(X) = \{b : b \text{ sss } X\}$ for its semisingleton extension.

Proposition 31.2 (Option collection and its constructor laws)**.**
In the classical class setting, with empty set and pairing,

$$\kappa(\mathrm{Opt}(P)) = O(\kappa(P)).$$

The empty set satisfies $\mathrm{Opt}(P)$, and if a is a set satisfying P, then $\{a\}$ satisfies $\mathrm{Opt}(P)$. In particular, for a class carrier X,

$$\kappa(\mathrm{Emb}(\mathrm{Option}(X))) = O(X),$$

where $\mathrm{Emb}(X) = \mathrm{Emb}_{\mathrm{cl}}(X)$ is the local carrier interpretation.

Proof.
The constructors are sets by the empty-set and pairing axioms, and each satisfies its corresponding disjunct in the definition of $\mathrm{Opt}(P)$. A set satisfying that definition is empty or a singleton of a set a satisfying P. Such an a belongs to $\kappa(P)$, so either case is a semisingleton subclass of $\kappa(P)$. Conversely, let $b \in O(\kappa(P))$. Classical logic distinguishes whether b is empty. If it is not, choose $a \in b$. The semisingleton condition gives $b = \{a\}$, and $a \in \kappa(P)$ gives sethood and $P(a)$. Thus b satisfies the option predicate in either case. Comprehension and extensionality prove the equality. The carrier case follows from $\kappa(\mathrm{Emb}_{\mathrm{cl}}(X)) = X$. □

Soundness of option types

The following calculations develop the same concrete model. The preceding interface identifies its predicate, its collected extension, and its two constructor expressions. The rule for Some is justified by membership in the class carrier, which guarantees a set argument. For a general predicate admitting proper classes, the formula $P_A(v_\rho(a))$ alone does not justify a singleton encoding: a proper class has no singleton containing it in the class theory. Such an option constructor would need a different representation, not merely the omission of this guard.

Theorem 31.5 (None in the semisingleton model)**.**
For every class A,

$$\varnothing \in O(A).$$

Proof.
The empty class is a set by the empty-set axiom. It is a subclass of A and has at most one member, since it has none. Thus $\varnothing$ sss A, and guarded class comprehension gives $\varnothing \in O(A)$. With $I(\mathrm{None}) = \varnothing$, this proves the intended None introduction law in this model. □

Theorem 31.6 (Some in the semisingleton model)**.**
For every class A and every $a \in A$,

$$\{a\} \in O(A).$$

Proof.
Membership $a \in A$ implies that a is a set. Pairing therefore makes $\{a\}$ a set. Its only member is a, so it is a semisingleton subclass of A. Comprehension gives $\{a\} \in O(A)$, proving the intended Some introduction law under $I(\mathrm{Some}(a)) = \{I(a)\}$. Moreover $a \in \{a\}$ makes this code nonempty, so it differs from None. Equality $\{a\} = \{a'\}$ for set values implies $a = a'$ by singleton membership. Thus the constructors are disjoint and Some is injective on the set values represented by A. □

Theorem 31.7 (The small option carrier and its cases)**.**
In the ordinary classical set model, if A is a set, then $O(A)$ is a set. Every $b \in O(A)$ is either $\varnothing$ or $\{a\}$ for a unique $a \in A$, and these alternatives are disjoint.

Proof.
Every member of $O(A)$ is a set subclass of A, so

$$O(A) = \{b \in \mathcal{P}(A) : b \text{ has at most one member}\}$$

is a set by powerset and separation. Given $b \in O(A)$, classical logic distinguishes whether b has a member. If it has none, extensionality gives $b = \varnothing$. Otherwise take $a \in b$. Then $a \in A$, and the at-most-one condition makes every member of b equal to a, so $b = \{a\}$. The earlier constructor properties give uniqueness of a and disjointness from the empty case. □
Thus this semisingleton carrier is a model of the tagged alternative "no value or one value" in the stated classical setting. The case split is part of the argument: for arbitrary semisingleton subsets in an intuitionistic setting, decidability of inhabitation cannot be silently assumed. When $A = \varnothing$, the carrier is $O(A) = \{\varnothing\}$. If $\varnothing \in A$, Some of that value is $\{\varnothing\}$, distinct from None, which is $\varnothing$.

Theorem 31.8 (Dependent option elimination and its computation laws)**.**
Let A be a set and $(E(b))_{b \in O(A)}$ a specified family of sets. Suppose an element $e_0 \in E(\varnothing)$ and a function e_1 on A are supplied, with $e_1(a) \in E(\{a\})$ for each $a \in A$. There is a unique function r on $O(A)$ such that

$$r(b) \in E(b), \qquad r(\varnothing) = e_0, \qquad r(\{a\}) = e_1(a) \quad (a \in A).$$

Consequently, reconstructing any section from its values on None and Some returns that section; this is the semantic eta law for elimination.

Proof.
Use ordinary injective set-pair coding to form the graph

$$r = \{(\varnothing, e_0)\} \cup \{(\{a\}, e_1(a)) : a \in A\}.$$

Pairing and replacement make this a set. The two cases have distinct inputs, and injectivity of singleton formation identifies a whenever two inputs in the second case agree. Hence the graph is functional. The case theorem says that its domain is exactly $O(A)$, and the assumed types of $e_0, e_1(a)$ give $r(b) \in E(b)$. Its graph directly proves the two computation equations. Any other section satisfying them agrees with r at the empty input and at every singleton input; those exhaust the domain, so extensionality proves uniqueness. For eta, take a given section q and supply $e_0 = q(\varnothing)$ and $e_1(a) = q(\{a\})$. The section q satisfies the same equations, so the reconstructed section equals q. □

The data e_0 and e_1 are supplied, rather than chosen from merely nonempty fibres; this elimination theorem uses no choice axiom. It proves equality of functions in the set model, not an additional judgmental conversion rule for the proposed type checker.

Corollary 31.2 (Option induction)**.**
Let A be a set and let $Q(b)$ be a predicate on $O(A)$. If $Q(\varnothing)$ holds and $Q(\{a\})$ holds for every $a \in A$, then $Q(b)$ holds for every $b \in O(A)$.

Proof.
The case theorem makes each b either the empty set or one of the singletons covered by the hypotheses. Substitute that equality into its corresponding hypothesis. □

31.6 Implementing ordered pairs

The ordered-pair section states the intended constructor, sethood condition, and injectivity property, and compares several candidate models. The type-theoretic presentation here requires the selected model to satisfy that interface.

Axioms of ordered pairs

Use the constructor, sethood, and injectivity interface stated in the ordered-pair section.

Model of ordered pairs

Choose one of the pair constructions described there, with its stated domain.

Soundness of ordered pairs

This requires the constructor and projection equations for that chosen model.

31.7 Implementing lists as a medium type

Each list of classes is modelled by a family whose domain is a finite initial segment of the ordinals. We introduce the type ClassList.

Axioms

Axiom 17 (nil)**.**

$$[] : \text{ClassList}$$

Axiom 18 (cons)**.**
The symbol "::" is infix notation for the constructor "cons". Enclosing it in parentheses allows it to be used in prefix position.

$$(::) : \ \text{Class} \to \text{ClassList} \to \text{ClassList}$$

Axiom 19 (concat)**.**

$$(+\!\!+) : \ \text{ClassList} \to \text{ClassList} \to \text{ClassList}$$

Model

For a medium type, some terms may be interpreted as proper classes. The model therefore uses a predicate on classes; it need not be a set or class containing those interpretations as members.

Definition 31.6 (inClassList)**.**
The historical representation proposal for class-list terms is:

$$(F \text{ in ClassList}) \overset{\text{def.}}{\longleftrightarrow} ((F \text{ family}) \wedge (\text{Dom}(F) \in \omega))$$

Interpretation of list expressions:

$$I : \text{ClassList.}\mathsf{Expression} \to \text{Class.}\mathsf{Expression}.$$

Definition 31.7 (nil)**.**

$$I([]) \overset{\text{def.}}{:=} \varnothing$$

A canonical model with arbitrary class entries

The preceding predicate needs a fixed meaning of "family." In the class-axiom chapter, a set value has a single concentrated record; the semisingleton encoding instead retains an empty record and all singleton member-records. Thus the latter's cons operation need not preserve the former family predicate. We now specify a local model using the canonicalization C_1 from the historical family chapter, with the same ordinary injective set-pair coding throughout this model. This makes the representation choice explicit while retaining arbitrary classes as possible entries. For a class relation $F \subseteq \mathrm{V} \times \mathrm{V}$, write

$$U_F(i) = \bigcup \{s : (i, s) \in F\}, \qquad N(F) = C_1(F).$$

By the canonicalization theorem, N preserves the represented domain and every union value, is idempotent, and satisfies

$$N(F) = \{(i, \varnothing) : i \in \mathrm{Dom}(F)\} \cup \{(i, \{q\}) : i \in \mathrm{Dom}(F),\ q \in U_F(i)\}.$$

Define the canonical class-list predicate by

$$\mathrm{CL}(F) \longleftrightarrow F \subseteq \mathrm{V} \times \mathrm{V} \ \wedge\ N(F) = F \ \wedge\ \mathrm{Dom}(F) \in \omega.$$

For such an F, its length is the unique natural ordinal $n = \mathrm{Dom}(F)$. Every relation with finite ordinal domain can be converted to this model by N, preserving its entries and length. In particular this applies to the historical family proposal, even though the resulting records can differ. Equal lengths and equal values at every represented index determine equal canonical lists, by the complete-normal-form theorem for C_1.

Definition 31.8 (Class-parameter cons and concatenation)**.**
For any class H and canonical class lists F, G, put

$$\mathrm{Cons}(H, F) = \{(0, \varnothing)\} \cup \{(0, \{q\}) : q \in H\} \cup \{(i+1, s) : (i, s) \in F\},$$
$$\mathrm{App}(F, G) = F \cup \{(n+i, s) : (i, s) \in G\}, \qquad n = \mathrm{Dom}(F).$$

Nil is $\varnothing$, agreeing with $I([])$ above. The numbers and their sums in these formulas are natural ordinals.

These are operations defined by formulas with class parameters. Every stored coordinate is a set: $q \in H$ makes q a set, hence $\{q\}$ and its record are sets, even when H is proper. Class comprehension forms the displayed results; it does not collect a pair whose coordinate is the proper class H or a proper class list. No ordinary class function graph on all such arguments is asserted.

Theorem 31.9 (Class-list domains and values)**.**
Nil is canonical with domain zero. If F, G are canonical with lengths n, m, respectively, then $\mathrm{Cons}(H, F)$ and $\mathrm{App}(F, G)$ are canonical, and

$$\begin{aligned} \mathrm{Dom}(\mathrm{Cons}(H, F)) = n+1, \qquad & U_{\mathrm{Cons}(H,F)}(0) = H, \\ & U_{\mathrm{Cons}(H,F)}(i+1) = U_F(i) \quad (i < n), \\ \mathrm{Dom}(\mathrm{App}(F, G)) = n+m, \qquad & U_{\mathrm{App}(F,G)}(i) = U_F(i) \quad (i < n), \\ & U_{\mathrm{App}(F,G)}(n+j) = U_G(j) \quad (j < m). \end{aligned}$$

Proof.
The empty relation has empty domain and is fixed by N. For cons, index zero has exactly the empty record and all singleton records from H. Their union is H: each $q \in H$ occurs in $\{q\}$, and no other element is contributed. For each $i < n$, the fibre at $i+1$ is exactly the fibre of F at i. Successor is injective on natural ordinals and never zero, so these fibres do not overlap. They are all nonempty because each contains the empty record. The domain is therefore $\{0\} \cup \{i+1 : i < n\} = n+1$. Every fibre has precisely the form prescribed by N, proving canonicality as well as the asserted values.
For concatenation, every $k < n+m$ lies uniquely either below n or has the form $n+j$ with $j < m$. Thus the first part of the union gives exactly the fibres of F below n, and the shifted part gives exactly the fibres of G at the remaining indices. Every such fibre is nonempty and canonical, and there are no others. This proves the domain, canonicality, and both value equations, including the cases $n = 0$ or $m = 0$. □

Corollary 31.3 (Class-list concatenation identities and associativity)**.**
For canonical class lists F, G, K,

$$\mathrm{App}(\varnothing, F) = F = \mathrm{App}(F, \varnothing), \qquad \mathrm{App}(\mathrm{App}(F, G), K) = \mathrm{App}(F, \mathrm{App}(G, K)).$$

Proof.
Concatenating with nil leaves the domain and every represented value unchanged by the theorem, so canonical extensionality gives both identities. For associativity, write the three lengths as n, m, p. Both outputs have length $(n + m) + p = n + (m + p)$. Splitting the indices into the consecutive intervals of lengths n, m, p, the theorem gives respectively the values of F, G, K on each side. On the final interval the two shifts agree because $(n + m) + j = n + (m + j)$. Hence all values and domains agree, and canonical extensionality proves equality. □

Proposition 31.3 (When a canonical class list is a set)**.**
With pairing, separation, replacement, and union, a canonical class list F of length n is a set if and only if every entry $U_F(i)$, $i < n$, is a set.

Proof.
If F is a set, separation gives its records at any fixed index. Replacement by the second-coordinate projection gives a set of stored values, and union makes $U_F(i)$ a set. Conversely, suppose all entries are sets. At index $i < n$, the canonical fibre $\{\varnothing\} \cup \{\{q\} : q \in U_F(i)\}$ is a set by replacement and pairing. Its records with first coordinate i form a set as well. Replacement collects these uniquely specified record sets over the set domain n, and their union is exactly F. No choice of representatives is involved. □
For example, $\mathrm{Cons}(\mathrm{V}, \varnothing)$ is a one-entry list whose value is a proper class, represented entirely by set records. The proposition shows that this list itself is proper. For the empty class as head, the one-entry list is instead $\{(0, \varnothing)\}$; it is distinct from nil despite its empty value. On noncanonical input relations, the corresponding operations first apply N, so an identity law returns $N(F)$ rather than necessarily the original records F. We have proved class-parameter construction and equality laws, not an initial-algebra property for a collection of all proper-class-valued lists. The set-valued model below uses these same canonical fibres and admits a separate set-level recursion theorem.
The newer Obelisk export's entry 3.51, `DefClassLists`, records another construction using explicit markers and one-based positions. The present model uses zero-based positions and ordinary set-pair records. These are different representations of the intended entries; neither literal equality of their codes nor interchangeability of their projection formulas follows. The ordered-pair chapter explains the separate marker convention. Transport between the encodings must preserve length and decoded entries explicitly.

Transport to the catalog's finite marker lists

The newer Obelisk catalog's `DefClassLists` (3.51), still marked `TBI`, proposes a different representation. Its displayed payload $\{\{\{a\}\}, k\}$ is an *unordered* pair: one member is $\{\{a\}\}$ and the other is the tag k. To keep the brace depth explicit, put

$$e(a) = \{\{a\}\}, \qquad \mathrm{mark}_2(H, k) = \{\{0, k\}\} \cup \{\{e(a), k\} : a \in H\}.$$

Here a ranges over set members of H, while k is a positive finite ordinal. The subscript records the two singleton operations in e. This is the convention of entry 3.51; the separate tagged-pair

entry `DefDunaevOrderedPair` (3.76) uses three singleton operations in its payload. The two formulas are not asserted to be literally equal. We prove the finite-list interpretation of 3.51, rather than inferring its correctness from the generated entry's label.

Lemma 31.1 (Separation of markers and payloads)**.**
For sets a, b and positive natural ordinals k, l, the following hold:

$$\{0, k\} = \{0, l\} \longleftrightarrow k = l,$$
$$\{e(a), k\} = \{e(b), l\} \longleftrightarrow (a = b \text{ and } k = l),$$
$$\{0, k\} \neq \{e(a), l\}.$$

Proof.
The set $e(a)$ is a singleton and cannot be an ordinal. Indeed, the only singleton ordinal is $1 = \{0\}$: a nonempty ordinal contains zero, so its sole member must be zero. Equality $e(a) = 1$ would force $\{a\} = 0$, contrary to $a \in \{a\}$. Thus a payload $\{e(a), k\}$ has exactly one natural-ordinal member, namely k, and exactly one nonordinal member, namely $e(a)$. Equality of two payloads identifies these members separately. Two applications of singleton injectivity then give $a = b$. A marker $\{0, k\}$ has only ordinal members, so cannot equal a payload; its unique nonzero member is k, proving marker injectivity. The reverse implications follow directly from equality substitution. □
For a canonical class list F of length n, define its marker encoding by a single class comprehension, written as the following union:

$$\mathcal{M}(F) = \{\{0, i+1\} : i < n\} \cup \{\{e(a), i+1\} : i < n,\ a \in U_F(i)\}.$$

All members of this class are sets. The expression uses the class parameter F and its definable values, rather than forming a set or class whose members are those possibly proper-class values. For a class L and positive natural k, define the decoded entry

$$D_k(L) = \{a : \{e(a), k\} \in L\}.$$

As usual comprehension retains set values of a; its auxiliary binders are fresh for the class parameters.

Theorem 31.10 (Faithful transport between the two list representations)**.**
For canonical class lists F, G, the marker encoding retains both length and entries. More explicitly, if $\mathrm{Dom}(F) = n$, then

$$\{k \in \omega \setminus \{0\} : \{0, k\} \in \mathcal{M}(F)\} = \{i + 1 : i < n\}, \qquad D_{i+1}(\mathcal{M}(F)) = U_F(i) \quad (i < n).$$

Consequently $\mathcal{M}(F) = \mathcal{M}(G)$ if and only if $F = G$. For arbitrary classes H, K it also satisfies

$$\mathcal{M}(\varnothing) = \varnothing,$$
$$\mathcal{M}(\mathrm{Cons}(H, \varnothing)) = \mathrm{mark}_2(H, 1),$$
$$\mathcal{M}(\mathrm{Cons}(H, \mathrm{Cons}(K, \varnothing))) = \mathrm{mark}_2(H, 1) \cup \mathrm{mark}_2(K, 2).$$

Proof.
By separation of markers and payloads, the marker $\{0, k\}$ occurs exactly when $k = i + 1$ for some $i < n$. The same lemma says that a payload $\{e(a), i + 1\}$ occurs exactly when $i < n$ and $a \in U_F(i)$. These are the first two assertions by extensionality. In particular an empty entry

still contributes its marker and cannot shorten the list. The finite tag set is empty exactly when $n = 0$, and otherwise has maximum n; hence it determines n uniquely.
If two encodings are equal, their marker sets give equal lengths and the decoding formulas give equal entries at every represented index. The C_1 normal-form theorem then gives $F = G$. Conversely equal canonical lists give equal marker comprehensions. Finally nil has no indices; the one-entry cons list has domain 1 and value H at zero; the two-entry cons list has domain 2 and values H, K at zero and one. Substituting these domain and value equations into $\mathcal{M}$ gives the three displayed identities. □
The last two identities give precisely the catalog's singleton-list and two-entry-list formulas, and D_1, D_2 give its stated projections on their represented indices. They concern a translation of representations, not equality of a semisingleton-record list with its marker encoding. The proof allows proper-class entries but only finite natural tags; it does not verify any unstated kernel conversion rules or an extension to arbitrary class-valued tags.

31.8 Implementing lists of values of a small type

Fix a set A interpreting the small type. We use the author's semisingleton family encoding, rather than the concentrated-fibre convention used in some other sections. Work with Infinity, powerset, replacement, union, and ordinary injective pairs of sets. Every value $h \in A$ is a set; no proper-class entry is being stored in this model. The displayed type interface below will follow from the construction.

The encoded list carrier

For a set h, write

$$\mathcal{S}(h) = \{s \subseteq h : s \text{ has at most one member}\}.$$

Thus s sss h means $s \in \mathcal{S}(h)$. This is a set by powerset and separation. It contains $\varnothing$ and every singleton $\{q\}$ for $q \in h$, and

$$\bigcup \mathcal{S}(h) = h.$$

Indeed each member of a semisingleton subclass lies in h, and each $q \in h$ occurs in its singleton subclass. In particular $\mathcal{S}(\varnothing) = \{\varnothing\}$, not the empty family.
For an ordinary function graph $f : n \to A$, where $n \in \omega$, define

$$L(n, f) = \{(i, s) : i \in n,\ s \in \mathcal{S}(f(i))\}.$$

Put

$$\mathcal{L}_n(A) = \{L(n, f) : f : n \to A\}, \qquad \mathrm{List}(A) = \bigcup_{n \in \omega} \mathcal{L}_n(A).$$

The graph f is an auxiliary code. The list value is $L(n, f)$, which generally has several records at each index and is not an ordinary single-valued graph. Here list evaluation means union of the fibre, and its represented domain means the indices having a record.

Lemma 31.2 (Sethood and recovery of the sequence code)**.**
The collection $\mathrm{List}(A)$ is a set. For every $f : n \to A$,

$$\mathrm{Dom}(L(n, f)) = n, \qquad \bigcup \{s : (i, s) \in L(n, f)\} = f(i) \quad (i \in n).$$

Consequently $L(n, f) = L(m, g)$ if and only if $n = m$ and $f = g$.

Proof.
For each $i \in n$, the set of records $\{(i, s) : s \in \mathcal{S}(f(i))\}$ is a set by replacement. Replacement over n and union make $L(n, f)$ a set. The ordinary function space A^n is a set, as a separated subclass of $\mathcal{P}(n \times A)$. Replacement therefore makes $\mathcal{L}_n(A)$ a set. Replacement over ω and union give $\mathrm{List}(A)$.
Every $i \in n$ has the record $(i, \varnothing)$, and no other index occurs, proving the domain equation. Injectivity of pair coding makes the fibre at i precisely $\mathcal{S}(f(i))$, whose union is $f(i)$. Equality of two encoded lists therefore gives equal domains, and then equal values at every index. Function extensionality gives $f = g$. The reverse implication follows from the defining comprehension. □
The unique n in this lemma is the length $\ell(t)$ of the list $t = L(n, f)$. An empty entry cannot disappear: its fibre contains the empty record. This domain information distinguishes the one-entry list with entry $\varnothing$ from the empty list.

Constructors and their computation rules

Definition 31.9 (Nil and the author's cons operation)**.**
Put $[] = \varnothing$ and retain the definitions

$$\mathrm{inc_fst}(r) = \{(i \cup \{i\}, s) : (i, s) \in r\}, \qquad h :: t = \{(\varnothing, s) : s \text{ sss } h\} \cup \mathrm{inc_fst}(t).$$

On a list's natural-number indices, $i \cup \{i\} = i + 1$.

Theorem 31.11 (List constructors and unique decomposition)**.**
One has $[] \in \mathrm{List}(A)$, and $h :: t \in \mathrm{List}(A)$ for every $h \in A$, $t \in \mathrm{List}(A)$. Their lengths satisfy

$$\ell([]) = 0, \qquad \ell(h :: t) = \ell(t) + 1.$$

Every list is either $[]$ or has the form $h :: t$ for unique $h \in A$ and $t \in \mathrm{List}(A)$, and these cases are disjoint. Thus the constructors realize the intended interface

$$[] : \mathrm{List}(A), \qquad (::) : A \to \mathrm{List}(A) \to \mathrm{List}(A).$$

Proof.
The unique graph $f : 0 \to A$ is empty and $L(0, f) = \varnothing$, so nil is a list. For $t = L(n, f)$, define the ordinary graph $g : n + 1 \to A$ by

$$g(0) = h, \qquad g(i + 1) = f(i) \quad (i \in n).$$

Every nonzero natural number below $n + 1$ has a unique predecessor below n. The record at index zero comes from the first term of cons, and the records at index $i + 1$ are precisely those in the fibre of t at i. Hence the defining equations give $h :: t = L(n + 1, g)$. This proves closure and the length formulas. In particular every cons has $(0, \varnothing)$ as a record, so it cannot equal nil.
Conversely, if $L(n, f)$ is not nil, its domain is not zero. Write $n = m + 1$, put $h = f(0)$, and let $f_-(i) = f(i + 1)$ for $i \in m$. The same fibre calculation gives $L(n, f) = h :: L(m, f_-)$. Recovery of the sequence code proves uniqueness: the head is the value at zero, and all tail entries are the subsequent values in their order. This also proves that $h :: t = h' :: t'$ implies $h = h'$ and $t = t'$. □
For example, if $\varnothing \in A$, then $\varnothing :: [] = \{(0, \varnothing)\}$ has length one, although its only evaluated entry is empty. No out-of-domain default is used to recover its length. If $A = \varnothing$, there are no positive-length sequence graphs into A, so $\mathrm{List}(A) = \{[]\}$.

Induction and recursion on length

Theorem 31.12 (List induction and recursion)**.**
A predicate that holds at nil and is preserved by cons holds at every member of $\mathrm{List}(A)$. Given a set X, an element $z \in X$, and a function $c : A \times X \to X$, there is a unique function $r : \mathrm{List}(A) \to X$ satisfying

$$r([]) = z, \qquad r(h :: t) = c(h, r(t)).$$

Proof.
For induction, induct on the natural-number length. The zero-length list is nil, and every list of length $n+1$ uniquely decomposes as $h :: t$ with $\ell(t) = n$. The corresponding induction premise and cons step prove the assertion at that length.
For recursion, define functions $r_n : \mathcal{L}_n(A) \to X$. On $\mathcal{L}_0(A) = \{[]\}$ set $r_0([]) = z$. Given r_n, the unique decomposition lets us define

$$r_{n+1}(h :: t) = c(h, r_n(t)) \qquad (t \in \mathcal{L}_n(A)).$$

Replacement collects this graph, and natural-number recursion produces the sequence (r_n). Distinct length classes are disjoint by code recovery, so $r = \bigcup_{n\in\omega} r_n$ is a function with domain $\mathrm{List}(A)$ and values in X. Replacement and union make its graph a set. The stage equations give the required computation rules. Any other solution agrees on length zero, and agreement on length n forces agreement on length $n+1$ by unique decomposition. Induction therefore gives equality everywhere, proving uniqueness. □
This is the usual initial-algebra property for the tagged sum functor $X \mapsto 1 + (A \times X)$ on sets: the nil and cons branches specify an algebra, and the two recursion equations specify its unique morphism from the encoded list algebra. The disjoint alternatives are supplied by the tags in that functor, while the represented list domain already distinguishes nil from cons in the concrete encoding. This result does not assert an initial-algebra property for the class-parameter model in the preceding subsection.

Concatenation and maps of small lists

For each $u \in \mathrm{List}(A)$, apply list recursion with target $\mathrm{List}(A)$, nil value u, and step $(h, v) \mapsto h :: v$. Write its value at t as $t +\!\!+ u$. Thus

$$[] +\!\!+ u = u, \qquad (h :: t) +\!\!+ u = h :: (t +\!\!+ u).$$

Uniqueness specifies the operation for every pair (t, u), and replacement on $\mathrm{List}(A) \times \mathrm{List}(A)$ makes it a set function. This realizes the earlier concatenation interface for small lists in this encoding.

Proposition 31.4 (Concatenation laws)**.**
For $t, u, v \in \mathrm{List}(A)$,

$$t +\!\!+ [] = t, \qquad (t +\!\!+ u) +\!\!+ v = t +\!\!+ (u +\!\!+ v),$$
$$\ell(t +\!\!+ u) = \ell(t) + \ell(u),$$
$$t +\!\!+ u = t +\!\!+ v \implies u = v.$$

Together with the defining nil equation, nil is a two-sided identity.

Proof.

For the right identity, the function $t \mapsto t +\!\!+ []$ and the identity on $\mathrm{List}(A)$ both take nil to nil and satisfy the same cons equation. Recursion uniqueness makes them equal.
Fix u, v. As functions of t, both sides of the associativity equation take nil to $u +\!\!+ v$. Applying either function to $h :: t$ gives h consed onto its value at t, by the two defining equations. Recursion uniqueness again proves equality.
For length, induct on t. At nil the equation is $\ell(u) = 0 + \ell(u)$. At $h :: t$, concatenation and the length formula give $\ell(t +\!\!+ u) + 1$; the induction hypothesis makes this $\ell(t) + \ell(u) + 1 = (\ell(t) + 1) + \ell(u)$, as required by natural-number arithmetic. Finally, induct on the common left list for cancellation. At nil the hypothesis is already $u = v$. At $h :: t$, constructor injectivity cancels h from $h :: (t +\!\!+ u) = h :: (t +\!\!+ v)$, and the induction hypothesis gives $u = v$. □
For a set function $f : A \to B$, list recursion with target $\mathrm{List}(B)$, nil value $[]$, and step $(h, v) \mapsto f(h) :: v$ defines $\mathrm{map}_f : \mathrm{List}(A) \to \mathrm{List}(B)$, with

$$\mathrm{map}_f([]) = [], \qquad \mathrm{map}_f(h :: t) = f(h) :: \mathrm{map}_f(t).$$

These equations use the same semisingleton list encoding on both sides; the operation replaces list entries, not individual records in a fibre.

Proposition 31.5 (Mapping identities, composition, and concatenation)**.**
For set functions $f : A \to B$, $g : B \to C$,

$$\mathrm{map}_{1_A} = 1_{\mathrm{List}(A)}, \qquad \mathrm{map}_{g \circ f} = \mathrm{map}_g \circ \mathrm{map}_f,$$
$$\mathrm{map}_f(t +\!\!+ u) = \mathrm{map}_f(t) +\!\!+ \mathrm{map}_f(u).$$

Proof.
For the identity equation, both functions send nil to nil and send $h :: t$ to h consed onto their value at t. For composition, both send nil to nil and send $h :: t$ to $g(f(h))$ consed onto their value at t. Each comparison is therefore an application of recursion uniqueness with the indicated target set and step operation.
For the last equation fix u. As functions of t, both sides take nil to $\mathrm{map}_f(u)$. The cons and mapping equations make both send $h :: t$ to $f(h)$ consed onto their value at t. Recursion uniqueness with that nil value proves the claimed compatibility. □
All constructions in this subsection have set-valued entries and set carriers. The medium-type concatenation laws above have their separate class-parameter proof; the present recursion and mapping arguments do not assert a recursion principle for arbitrary class-valued entries.

31.9 Implementing functions

This subsection records class operations and compares two canonical family encodings. A relation is a class of encoded set pairs. A family may use several entries in a fiber to represent a proper-class value, so relational composition and family evaluation must not be interchanged without checking the representation. The historical conversion laws below explicitly return to semisingleton fibers and total union evaluation.

Family basics

Definition 31.10 (restriction)**.**

$$F \restriction A \stackrel{\text{def.}}{\coloneqq} \{\, (a, s) \in F \mid a \in A \,\}$$

Definition 31.11 (image)**.**

$$\operatorname{image}_R(A) \stackrel{\text{def.}}{\coloneqq} \operatorname{ran}(R \restriction A)$$

Definition 31.12 (composition)**.**

$$F \circ G \stackrel{\text{def.}}{\coloneqq} \{\, (x, z) \mid x \in \operatorname{Dom}(G) \wedge (G(x), z) \in F \,\}$$

Definition 31.13 (membership class)**.**

$$\mathrm{E} \stackrel{\text{def.}}{\coloneqq} \{\, (x, y) \mid x \in y \,\}$$

Definition 31.14 (identity class)**.**

$$\mathrm{I} \stackrel{\text{def.}}{\coloneqq} \{\, (x, x) \mid^{x} \top \,\}$$

Notation 31.1 (containment relation)**.**

$$\beth \stackrel{\text{def.}}{\coloneqq} \mathrm{E}^{\top}$$

Definition 31.15 (transposition)**.**

$$F^{\intercal} \stackrel{\text{def.}}{\coloneqq} \{\, (F(x), x) \mid x \in \operatorname{dom}(F) \,\}$$

Family

A family combines its domain with the chosen fibre representation. The preceding family-evaluation section describes the two encodings compared here.

Lambda–abstraction

Failed definition (set–valued abstraction)**.**
This graph abstraction represents set values. It applies when t is set-valued on A; proper-class values require another encoding:

$$(x \in A \mapsto t) \stackrel{\text{bad def}}{\coloneqq} \{\, (x, t) \mid^{x} (x \in A) \,\} \qquad \text{(def:setvaluedabstraction)}$$

Here x is bound on both sides of the equality.

In each of the following abstractions, x is not free in the fixed domain A, and the auxiliary variable b is distinct from x and fresh for A, t. Rename auxiliary variables before substitution whenever needed.

Definition 31.16 (OUTDATED class–valued abstraction)**.**

$$(x \in A \mapsto t) \coloneqq^{\text{def.}} \{\, (x,b) \mid^{x,b} (x \in A) \wedge (b \text{ sss } t) \,\} \tag{def:classvaluedabstraction-old}$$

In the singleton-based variant, a set value is stored directly, while a proper-class value is represented by its singleton subclasses:

Definition 31.17 (class–valued abstraction)**.**

$$(x \in A \mapsto t) \coloneqq^{\text{def.}} \{\, (x,b) \mid^{x,b} (x \in A) \wedge (((t \text{ set}) \wedge (b = t)) \vee ((t \text{ proper}) \wedge (b \text{ singleton}) \wedge (b \subset t))) \,\} \tag{def:classvaluedabstraction}$$

Alpha–equivalence

Renaming the bound variable in an abstraction preserves its meaning, provided the new variable is fresh and the renaming avoids capture. This follows from the corresponding renaming property for class comprehension.

Beta–conversion

Definition 31.18 (conditional operator)**.**
For a formula φ and class terms t_1 and t_2, define the conditional operator by

$$(\text{if } \varphi \text{ then } t_1 \text{ else } t_2 \text{ fi}) \coloneqq^{\text{def.}} \{\, w \mid ((\varphi) \wedge (w \in t_1)) \vee ((\neg\varphi) \wedge (w \in t_2)) \,\}$$

Side conditions:
1) w is not free in φ,
2) w is not free in t_1,
3) w is not free in t_2.

Theorem 31.13 (OUTDATED β–conversion)**.**
This historical theorem uses the earlier union-evaluation convention:

$$F(x) \coloneqq^{\text{def.}} \bigcup \{\, c \mid ((x,c) \in F) \,\}$$

All auxiliary fibre variables below are chosen fresh for A, t, a, F; the inner abstraction uses a different variable c from the outer fibre variable b. Let A be a class term with x not free in A, let a be a set-valued argument term, and let substitution of a for x avoid capture. The body value $t[a/x]$ may be a proper class. For the semisingleton abstraction,

$$(x \in A \mapsto t)(a) = (\text{if } (a \in A) \text{ then } t[a/x] \text{ else } \varnothing \text{ fi}) \tag{thm:fam–beta}$$

Proof.

$$\begin{aligned} &(x \in A \mapsto t)(a) \\ &= \bigcup \{\, b \mid ((a,b) \in (x \in A \mapsto t)) \,\} \end{aligned}$$

$$= \bigcup \{\, b \mid ((a,b) \in \{\, (x,c) \mathbin{\overset{x,c}{\restriction}} (x \in A) \wedge (c \text{ sss } t) \,\}) \,\}$$

$$= \bigcup \{\, b \mid (a \in A) \wedge (b \text{ sss } t[a/x]) \,\}$$

$$= \bigcup (\text{if } (a \in A) \text{ then } \{\, b \mid (b \text{ sss } t[a/x]) \,\} \text{ else } \varnothing \text{ fi})$$

$$= (\text{if } (a \in A) \text{ then } \bigcup \{\, b \mid (b \text{ sss } t[a/x]) \,\} \text{ else } \bigcup \varnothing \text{ fi})$$

$$= (\text{if } (a \in A) \text{ then } t[a/x] \text{ else } \varnothing \text{ fi})$$

□

The set-argument hypothesis is the scope in which the chosen set-pair encoding justifies identifying graph coordinates in this calculation. It does not restrict the represented body values to sets. An extension of the equation to proper-class arguments would require a specified pair encoding and evaluation law for those arguments. The empty result outside A belongs to this local unguarded union convention.

Eta–conversion

Theorem 31.14 (OUTDATED η–conversion)**.**
Assume that F uses canonical semisingleton fibers on its represented domain A, and in this local notation write $\mathrm{dom}(F) = A$. Choose the abstraction variable x fresh for the class expressions F and A; choose the fibre variables b, c distinct from x and fresh for those expressions. Thus F remains a fixed family when the binder is introduced. The abstraction and evaluation use the same local semisingleton and unguarded union conventions as in the preceding theorem. Then

$$(x \in \mathrm{dom}(F) \mapsto F(x)) = F. \tag{thm:fam–eta}$$

Proof.

$$(x \in \mathrm{dom}(F) \mapsto F(x))$$

$$= \{\, (x,b) \mathbin{\overset{x,b}{\restriction}} (x \in A) \wedge (b \text{ sss } F(x)) \,\}$$

$$= \left\{\, (x,b) \mathbin{\overset{x,b}{\restriction}} \Big(x \in A\Big) \wedge \Big(b \text{ sss } \bigcup \{\, c \mid ((x,c) \in F) \,\}\Big) \right\}$$

$$= \{\, (x,b) \mathbin{\overset{x,b}{\restriction}} (x \in A) \wedge (b \in \{\, c \mid ((x,c) \in F) \,\}) \,\}$$

$$= \{\, (x,b) \mathbin{\overset{x,b}{\restriction}} ((x,b) \in F) \,\}$$

$$= F$$

□

Relations and their induced operators

Fix set carriers X, Y and a relation $R \subseteq X \times Y$. Write $R[A] = \{y \in Y : \exists x \in A,\ (x,y) \in R\}$ and $R^{\top}[B] = \{x \in X : \exists y \in B,\ (x,y) \in R\}$. The direct-image operator $R^{\rightarrow} : \mathcal{P}X \to \mathcal{P}Y$ and inverse-image operator $R^{\leftarrow} : \mathcal{P}Y \to \mathcal{P}X$ are the functions sending a subset to these respective images. They are functions even when R is not functional: each subset has a uniquely specified image, though an individual point may have many outputs. The identity relation $I_X = \{(x,x) : x \in X\}$ is a subset of $X \times X$, since both coordinates of each record belong to X. Restriction $R|_A$ retains exactly the records whose first coordinate belongs to A.

Theorem 31.15 (Images preserve unions)**.**
For $\mathcal{M} \subseteq \mathcal{P}X$ and $\mathcal{N} \subseteq \mathcal{P}Y$,

$$R[\bigcup \mathcal{M}] = \bigcup R^{\rightarrow}[\mathcal{M}], \qquad R^{\top}[\bigcup \mathcal{N}] = \bigcup R^{\leftarrow}[\mathcal{N}].$$

Neither functionality nor injectivity of R is required.

Proof.
Membership in the first left side means that some $A \in \mathcal{M}$ contains an x with $(x, y) \in R$. Equivalently, $y \in R[A]$ for some $A \in \mathcal{M}$, which is membership in the right side. Apply the same argument to $R^{\top}$ for the second equality. The empty indexing families give empty images on both sides. □
The fixed set carriers ensure that all image values used as members of the right-hand families are sets. For arbitrary class relations an image can be proper, and a class graph cannot store that value as a member. An extension to that setting must use the family's class-value encoding or retain the quantified membership formulation instead.

Theorem 31.16 (Functional images of countable sets)**.**
If M admits an injection $j : M \to \omega$ and g is a set-valued class function, then $g[M]$ is countable, where only arguments in the domain of g contribute to the graph image.

Proof.
First $j[M] \subseteq \omega$ is a set by separation, and the inverse class function j^{-1} on $j[M]$ has image M. Replacement therefore makes M a set and then makes $g[M]$ a set. For each $y \in g[M]$, let $k(y)$ be the least $j(x)$ among $x \in M \cap \operatorname{dom}(g)$ with $g(x) = y$. Such a natural number exists by the definition of image and well-ordering of ω. If $k(y) = k(z)$, injectivity of j identifies the two witnessing inputs, and functionality of g gives $y = z$. Thus $k : g[M] \to \omega$ is injective. The empty image is included. Least witnesses avoid any appeal to a choice axiom. □
The union–powerclass comparison $A \subseteq \mathcal{P}B \longleftrightarrow \bigcup A \subseteq B$ was proved in the union and intersection section. It is a statement about arbitrary classes; it is distinct from the set-carrier restrictions needed to form the image operators here.

31.10 Implementing Σ–types

Axioms of Σ–types

The formation rule for a dependent sum is:

$$\frac{(A : \mathrm{Type}) \qquad (x : A \vdash B(x) : \mathrm{Type})}{\vdash \Sigma_{(x:A)} B(x) : \mathrm{Type}}\ \Sigma\text{–form}$$

Model of Σ–types

Throughout this subsection, B is a family. The dependent sum consists of pairs containing an index in its domain and a member of the corresponding value. The explicit domain guard prevents an out-of-domain default from contributing extra pairs.

Definition 31.19 (dependent sum of a family)**.**

$$\Sigma B \stackrel{\text{def.}}{\coloneqq} \{\, (x, y) \mid (x \in \mathrm{Dom}(B)) \wedge (y \in B(x)) \,\}$$

Definition 31.20 (ordered pair projections)**.**

$$\mathrm{pr}_1 \stackrel{\text{def.}}{\coloneqq} \{\, ((a, b), a)\ {}^{\P,b}\, a \text{ any}, b \text{ any} \,\}$$
$$\mathrm{pr}_2 \stackrel{\text{def.}}{\coloneqq} \{\, ((a, b), b)\ {}^{\P,b}\, a \text{ any}, b \text{ any} \,\}$$

Soundness of Σ–types

Theorem 31.17 (Soundness of ΣB Formation, Introduction, Projections, and β/η Rules)**.** Work with guarded class comprehension and injective set-pair coding. Let B be a specified family of classes over the class $A = \mathrm{Dom}(B)$, and let

$$\Sigma B \stackrel{\text{def.}}{\coloneqq} \{\, p \mid \exists (x \in \mathrm{Dom}(B)),\ \exists (y \in B(x)),\ (p = (x, y)) \,\}.$$

This defines a class carrier. If A and every $B(x)$ are sets, replacement and union additionally make this carrier a set. Its pairing and projections satisfy:

1. **Introduction (Pairing):** If $(x \in \mathrm{Dom}(B))$ and $(y \in B(x))$, then $((x, y) \in \Sigma B)$.
2. **Elimination (Projections):** If $p \in \Sigma B$, then $\mathrm{pr}_1(p) \in A$ and $\mathrm{pr}_2(p) \in B(\mathrm{pr}_1(p))$.
3. **β- and η-Computation:** For $x \in A$ and $y \in B(x)$, one has $\mathrm{pr}_1((x, y)) = x$ and $\mathrm{pr}_2((x, y)) = y$. For every $p \in \Sigma B$, one has $(\mathrm{pr}_1(p), \mathrm{pr}_2(p)) = p$.

Proof.
The displayed predicate is an admitted class-comprehension instance. If $x \in A$ and $y \in B(x)$, both x and y are sets, so their pair code is a set. These two witnesses therefore put that pair in the guarded comprehension, proving introduction. Conversely, every $p \in \Sigma B$ has such a decomposition. Injectivity of the chosen pairing makes its two coordinates unique. The projection graphs restricted to these pair codes thus return precisely x and y, which have the asserted memberships. Returning a pair's coordinates proves the two beta equations, and reconstructing the unique decomposition of p proves eta.

For the additional set assertion, replacement over the set $B(x)$ forms the set $S_x = \{(x, y) : y \in B(x)\}$ for each $x \in A$. The family $x \mapsto S_x$ is uniquely specified; replacement over the set A collects its values, and union gives $\Sigma B = \bigcup_{x \in A} S_x$ as a set. When A is empty the union is empty and the conclusion still holds. No member of a fibre has been selected in this argument, so no choice axiom is used. □

The newer catalog records the dependent-sum membership interface as `DefSigmaModel` (3.38). Projection and computation additionally use the selected pairing model, as the proof shows. This carrier model has set-valued coordinates even when A or a fibre is proper: membership in a class implies sethood. A predicate model allowing proper-class coordinates needs its own pairing interpretation; it is not obtained by treating those coordinates as members of this carrier.

Theorem 31.18 (Dependent elimination for the set sum)**.**
Let A and every $B(a)$ be sets, put $S = \Sigma B$, and let $(E(p))_{p \in S}$ be a specified family of sets. Suppose a specified operation supplies $d(a, b) \in E((a, b))$ for every $a \in A$ and $b \in B(a)$. Then there is a unique function r on S with

$$r(p) \in E(p), \qquad r((a, b)) = d(a, b).$$

Taking $d(a, b) = q((a, b))$ for a given dependent section q returns $r = q$. A predicate on S holds everywhere if it holds at every such pair (a, b).

Proof.
For $p \in S$, define $r(p) = d(\mathrm{pr}_1(p), \mathrm{pr}_2(p))$. The projection theorem makes this application legitimate, and the reconstruction equation identifies its type $E((\mathrm{pr}_1(p), \mathrm{pr}_2(p)))$ with $E(p)$. In particular the value is a set, so replacement on the set S forms the graph of r. The two projection equations give the displayed computation rule. Every p is its coordinate pair, so any other solution has the same value there; function extensionality proves uniqueness. The given q satisfies the same computation equations, proving the eta assertion. Finally the coordinate decomposition substitutes directly into the assumed pair instances of a predicate, proving its assertion on all of S. The empty sum gives the empty function and a vacuous predicate assertion. The supplied operation d is data, not a choice from merely nonempty fibres. □

These are semantic equations for the fixed model. To read them as soundness of contextual typing rules, one must also supply a context interpretation and the corresponding substitution lemma. At an environment ρ satisfying a context, the value of $B[t/x]$ must be the fibre of B at the value of t; interpretations of term substitution must likewise agree with the updated environment. Under those assumptions the preceding membership checks apply at each environment. They do not establish that lemma for every proposed dependent syntax, or turn semantic beta and eta into judgmental conversion rules of an unspecified checker.

Rules for small Σ–types

For these rules, $\mathcal{U}$ denotes a chosen universe closed under the dependent sum construction. That closure is an assumption on this local universe. Formation of terms of Σ–type:

$$\Sigma : \Pi((A : \mathcal{U}) \mapsto ((A \to \mathcal{U}) \to \mathcal{U}))$$

Introduction of terms of Σ–type:

$$\mathrm{pair} : \Pi((A : \mathcal{U}) \mapsto \Pi((B : A \to \mathcal{U}) \mapsto \Pi((x : A) \mapsto (B\ x \to \Sigma\ A\ B))))$$

Elimination of terms of Σ–type:

$$\mathrm{pr}_1 : \Pi((A : \mathcal{U}) \mapsto \Pi((B : A \to \mathcal{U}) \mapsto ((\Sigma\ A\ B) \to A)))$$

$$\mathrm{pr}_2 : \Pi((A : \mathcal{U}) \mapsto \Pi((B : A \to \mathcal{U}) \mapsto \Pi((p : \Sigma\ A\ B) \mapsto B\ (\mathrm{pr}_1\ p))))$$

Computation of terms of Σ–type:

$$\mathrm{pr}_1\ (\mathrm{pair}\ A\ B\ x\ y) \equiv x$$

$$\mathrm{pr}_2\ (\mathrm{pair}\ A\ B\ x\ y) \equiv y$$

$$\mathrm{pair}\ A\ B\ (\mathrm{pr}_1\ p)\ (\mathrm{pr}_2\ p) \equiv p$$

The membership expansion of a family

For an ordinary set-valued class function B, the dependent sum has records (x, y) with $x \in \mathrm{dom}(B)$ and $y \in B(x)$. Hence

$$\Sigma B = \exists \circ B.$$

Indeed, a record in the composite supplies a set z with $(x, z) \in B$ and $y \in z$; functionality identifies $z = B(x)$, and the converse uses that same witness. In particular, for the identity class function $I = \{(x, x) : x \in \mathrm{V}\}$, this gives $\Sigma I = \exists$. These equalities concern ordinary set-valued graphs; applying the same notation to a proper-class-valued encoded family requires its separately specified expansion convention.

31.11 Implementing Π–types

Notation 31.2 (Fm)**.**
The terms for propositions are called *formulas*.

$$\mathrm{Fm} :\overset{\text{def.}}{\equiv} \mathrm{PropTerm}$$

Notation 31.3 (typecong)**.**
There are no conglomerate–level terms except conglomerate–level types and virtualisations of them. That is the reason why "typecong" is a keyword, that makes a judgment out of class–level term and conglomerate–level type. There are exactly two variants of judgements: it is either a typing $(a : A)$ or declaring a conglomerate–level type.

Notation 31.4 (conglomerate–level types)**.**
Any formula (e.g. $\langle(x \text{ set})\rangle : \mathrm{PropTerm}$), which is constructed out of a unary predicate symbol is semantically written like this:

$$(x \text{ set}),$$

while a typing judgement and variable's binding are syntactically written like this:

$$(x : \mathrm{Set}).$$

The former is the valuation of the latter. To be precise, specifically, it holds under one special assignment which maps all letters to corresponding variables of the current context.

Axioms of Π–types

Formation of terms of Π–type:
The draft describes this type-formation discipline as predicative. Its separation of class carriers from predicates on classes does not assert predicativity of the ambient comprehension schema: the local MK or admitted-NBG assumptions stated above still apply. In class–level type theory, the formation rule for a Π–type is

$$\frac{(B \text{ family})}{(\Pi B) \text{ typecong}}\ \Pi\text{–form}$$

In particular,

$$\frac{(B \text{ class–function})}{(\Pi B) \text{ typecong}}\ \Pi\text{–form}$$

$$\frac{(B \text{ family}) \qquad (\mathrm{Dom}(B) \text{ set})}{((\Pi B) \text{ class})}\ \Pi\text{–form}$$

$$\frac{(B \text{ function})}{((\Pi B) \text{ set})}\ \Pi\text{–form}$$

In set–level type theory, the formation rules for a Π–type are

$$\frac{\mathcal{W} \vdash (A : \mathrm{Class}) \qquad \mathcal{W}, (x : A) \vdash (B(x) : \mathrm{Class})}{\mathcal{W} \vdash \Pi_{(x:A)} B(x) \text{ typecong}}\ \Pi\text{–form(large)}$$

$$\frac{\mathcal{W} \vdash (A : \mathrm{Set}) \qquad \mathcal{W}, [(x : A)] \vdash (B(x) : \mathrm{Set})}{\mathcal{W} \vdash (\Pi_{(x:A)} B(x)) : \mathrm{Set}}\ \Pi\text{–form(small)}$$

For the tiny formation rule, assume additionally that the chosen universe $\mathcal{U}$ is closed under dependent products: if $A \in \mathcal{U}$ and a specified family has $B(a) \in \mathcal{U}$ for all $a \in A$, then its dependent product belongs to $\mathcal{U}$. The dependent-sum closure assumed earlier does not by itself supply this requirement.

$$\frac{\mathcal{W} \vdash (A : \mathcal{U}) \qquad \mathcal{W}, [(x : A)] \vdash (B(x) : \mathcal{U})}{\mathcal{W} \vdash \Pi_{(x:A)} B(x) : \mathcal{U}}\ \Pi\text{–form(tiny)}$$

The following implementation proposal permits the two-sort inclusion from Set to Class and omits general subtyping. This is a local design choice for these rules.

$$\frac{(A : \mathrm{Set})}{(A : \mathrm{Class})}\ (\text{Set is a subsort of Class})$$

In the contextual rules below, x is fresh for Γ, all displayed contexts and types are well formed, and substitutions avoid capture. The bound name in the eta rule is chosen fresh for f as well as for Γ. These conventions permit alpha-renaming of local binders. Introduction of terms of Π–type:

$$\frac{\Gamma, x : A \vdash b : B}{\Gamma \vdash (\lambda((x : A) \mapsto b)) : \Pi_{(x:A)} B}\ \Pi\text{-intro}$$

Elimination of terms of Π–type:

$$\frac{\Gamma \vdash (f : \Pi_{(x:A)} B) \qquad \Gamma \vdash (a : A)}{\Gamma \vdash f(a) : B[a/x]}\ \Pi\text{-elim}$$

Computation of terms of Π–type:

$$\frac{\Gamma, x : A \vdash b : B \qquad \Gamma \vdash (a : A)}{\Gamma \vdash (\lambda((x : A) \mapsto b))(a) \equiv b[a/x] : B[a/x]}\ \Pi\text{-comp}$$

Uniqueness of terms of Π–type:

$$\frac{\Gamma \vdash f : \Pi_{(x:A)} B}{\Gamma \vdash f \equiv (\lambda((x : A) \mapsto f(x))) : \Pi_{(x:A)} B}\ \Pi\text{-uniq}$$

The first model of Π–types as predicates on classes

Remark 31.1 (a section needs more than a common domain)**.**
The newer Obelisk record `DefPiPredicateModel` (3.39) displays an implication from "B and f are families with the same domain" to $\forall x \in \mathrm{Dom}(B),\ f(x) \in B(x)$. The pointwise membership condition must instead be part of the section predicate or a typing hypothesis. For example, take $B = f = \{(0, \varnothing)\}$. Both domains are $\{0\}$, but $f(0) = B(0) = \varnothing$, so the claimed membership would be $\varnothing \in \varnothing$. The predicate defined below retains the necessary condition. The generated record's verification label alone does not resolve this discrepancy in its displayed statement.

Let B be a family term. Its interpretation $B \overset{\text{def.}}{\coloneqq} [\![\boldsymbol{B}]\!]_v^{\mathrm{Family}}$ is a family of classes.
In this model, the judgement "$\boldsymbol{(f : \Pi B)}$" is interpreted as the proposition "$(f : \Pi B)$".

Definition 31.21 (judgement as a proposition)**.**

$$[\![(\boldsymbol{f} : \boldsymbol{\Pi B})]\!]_v^{\mathrm{Prop}} \overset{\mathrm{def.}}{\longleftrightarrow} ([\![\boldsymbol{f}]\!]_v^{\mathrm{Class}} : \Pi [\![\boldsymbol{B}]\!]_v^{\mathrm{Class}}),$$

Here the predicate is defined as follows:

Definition 31.22 (class–function f respects a family B)**.**

$$\begin{gathered}(f : \Pi B) \overset{\mathrm{def.}}{\longleftrightarrow} \\ \longleftrightarrow ((f \text{ class–function}) \wedge \\ \wedge(\mathrm{dom}(f) = \mathrm{Dom}(B)) \wedge \\ \wedge(\forall(x \in \mathrm{dom}(f)), (f(x) \in B(x))))\end{gathered}$$

This predicate presentation allows the domain of B to be a proper class. Any function with that domain must itself be proper and therefore cannot be a member of a class. Keeping $(f : \Pi B)$ as a predicate avoids imposing an unintended sethood condition on f.

The second model of Π–types as sets

Assume a well-formed family term $\boldsymbol{B}$ is interpreted as a specified family B, and that interpretation respects the domain constructor. This is an assumption on the interpretation of family syntax; the carrier construction below does not itself prove it. Thus

$$[\![\mathbf{Dom}(\boldsymbol{B})]\!]_v^{\mathrm{Class}} = \mathrm{Dom}([\![\boldsymbol{B}]\!]_v^{\mathrm{Family}}).$$

Let $B \overset{\mathrm{def.}}{\coloneqq} [\![\boldsymbol{B}]\!]_v^{\mathrm{Family}}$. Class comprehension suffices to collect the set-valued sections into a class. For the displayed interpretation as a set, assume additionally that B is a set-indexed family of sets. Its dependent sum and the relevant function space are then sets, and we put

$$[\![\boldsymbol{\Pi B}]\!]_v^{\mathrm{Set}} \overset{\mathrm{def.}}{\coloneqq} \{\, f \mid (f : \Pi B) \,\}$$

Further class-valued models of Π–types

The following variants collect set-sized sections. On a proper-class domain, they need not represent all instances of the predicate model. The size terminology in this proposal is local: it distinguishes syntax-level types, predicates admitting proper classes, classes of set instances, and sets of instances. Here Type is interpreted as TClass.
Large types serve to prevent primitive typing errors.
Medium types express properties of proper classes.
Small types are represented by proper classes containing the interpretations of their terms as set members.
Tiny types are interpreted as sets containing the interpretations of all their terms as members.
One class-valued candidate is:

Definition 31.23 (dependent product, variant A)**.**

$$\Pi B = \{\, f \in \mathcal{P}\Sigma B \mid (f \text{ functional}) \wedge (\mathrm{dom}(f) = \mathrm{Dom}(B)) \,\}$$

Using $\mathrm{Dom}(B)$ includes the case where B is a proper family, rather than a function.

Definition 31.24 (dependent product, variant B)**.**
For sets, the following is another definition:

$$\Pi B = \left\{ f \in \left(\bigcup \mathrm{ran}(B)\right)^{\mathrm{Dom}(B)} \;\middle|\; \forall (x \in \mathrm{dom}(f)),\ (f(x) \in B(x)) \right\},$$

This variant is intended for set-indexed, set-valued families. Extending it to class families requires a separate argument.

Definition 31.25 (dependent product, variant C)**.**
Let w be a specified family. Its collected set-sized sections are

$$\prod w \stackrel{\text{def.}}{\coloneqq} \{\, f \mid (f \text{ class–function}) \wedge (\mathrm{dom}(f) = \mathrm{Dom}(w)) \wedge (\forall x \in \mathrm{Dom}(w),\ f(x) \in w(x)) \,\}.$$

Guarded comprehension retains only set graphs. The represented domain $\mathrm{Dom}(w)$ includes indices with proper-class fibres; the smaller $\mathrm{dom}(w)$ includes only indices whose values are sets.

For example, in the canonical family convention take $w = \{(0, \{q\}) : q \in \mathrm{V}\}$. Then $\mathrm{Dom}(w) = \{0\}$ and $w(0) = \mathrm{V}$, whereas $\mathrm{dom}(w) = \varnothing$. Using $\mathrm{dom}(w)$ in the section condition would incorrectly admit the empty function, which has no value at the represented index 0. The definition above retains that index. On a proper represented domain it collects no sections, since a set graph has a set domain; the earlier predicate model still describes proper-class function sections on such domains.

Definition 31.26 (sections of a family)**.**

$$\Gamma B \stackrel{\text{def.}}{\coloneqq} \{\, s : \mathrm{Dom}(B) \to \Sigma B \mid (\mathrm{pr}_1 \circ s) \subseteq \mathrm{I} \,\}$$

Definition 31.27 (dependent product of a family)**.**

$$\Pi B \stackrel{\text{def.}}{\coloneqq} \{\, \mathrm{pr}_2 \circ s \mid s \in \Gamma B \,\}$$

Theorem 31.19 (Sections and dependent function graphs)**.**
Let A be a set and let $B(a)$ be a set for every $a \in A$. Use injective set-pair coding, and put

$$S = \{(a, b) : a \in A,\ b \in B(a)\}.$$

Then S is a set, and there is a bijection between the set of functions f on A with $f(a) \in B(a)$ and the set of sections $s : A \to S$ of the projection $\mathrm{pr}_1 : S \to A$. The mutually inverse maps are

$$f \longmapsto (a \mapsto (a, f(a))), \qquad s \longmapsto \mathrm{pr}_2 \circ s.$$

In particular the product is the set

$$\{f \in \mathcal{P}S : f \text{ is functional and } \mathrm{dom}(f) = A\}.$$

Proof.
Replacement collects the sets $\{a\} \times B(a)$, and union gives S. A function graph contained in S has exactly the required pointwise values when its domain is A. Separation inside $\mathcal{P}S$ therefore forms the product set. For such an f, replacement forms the graph of $s(a) = (a, f(a))$, and $\mathrm{pr}_1 \circ s = \mathrm{id}_A$. Conversely, for a section s, write $s(a) = (c, b) \in S$. The section equation gives $c = a$, hence $b \in B(a)$. Thus $\mathrm{pr}_2 \circ s$ is a member of the product. Applying the two constructions

in either order preserves every value, by the two projection equations and injectivity of pairing. Function extensionality proves they are inverse. □

This correspondence uses no choice axiom: it transforms supplied sections and functions. It does not assert that the product is inhabited merely because every $B(a)$ is nonempty. For arbitrary set-indexed families that assertion is precisely the set axiom of choice. When $A = \varnothing$, both the product and the set of sections contain exactly the empty function, so the correspondence includes the empty-index case.

Soundness of Π–types

Theorem 31.20 (Soundness of ΠB Formation, Abstraction, Application, and β/η Rules)**.** Use guarded class comprehension, injective set-pair coding, and replacement for the specified operations below. Let B be a family of classes over a set domain $A = \mathrm{Dom}(B)$. Let

$$\Pi B = \{\, f \in \mathcal{P}\Sigma B \mid (f \text{ functional}) \wedge (\mathrm{dom}(f) = \mathrm{Dom}(B)) \wedge (\forall (x \in \mathrm{Dom}(B)),\ (f(x) \in B(x))) \,\}.$$

This is a class of set graphs. If every $B(x)$ is a set, the power-set axiom and separation additionally make ΠB a set. Then:

1. **Introduction (λ-Abstraction):** If for every $(x \in \mathrm{Dom}(B))$ the term $t(x)$ is a set satisfying $(t(x) \in B(x))$, then the functional graph

   $$(\lambda(x \in \mathrm{Dom}(B)),\ t(x)) = \{\, (x, t(x)) \mid x \in \mathrm{Dom}(B) \,\}$$

   is a functional subclass of ΣB with domain $\mathrm{Dom}(B)$, hence $((\lambda(x \in \mathrm{Dom}(B)),\ t(x)) \in \Pi B)$.

2. **Elimination (Dependent Application):** If $(f \in \Pi B)$ and $(a \in \mathrm{Dom}(B))$, then the singleton fiber $\{\, y \mid (a, y) \in f \,\} = \{f(a)\}$ has unique element $f(a) = \mathrm{Value}(\{\, y \mid (a, y) \in f \,\}) = \bigcap \{\, y \mid (a, y) \in f \,\}$ satisfying $(f(a) \in B(a))$.

3. **β- and η-Computation:** For all $(a \in \mathrm{Dom}(B))$, $(((\lambda(x \in \mathrm{Dom}(B)),\ t(x)))(a) = t(a))$, and for every $(f \in \Pi B)$, $((\lambda(x \in \mathrm{Dom}(B)),\ f(x)) = f)$ by functional extensionality.

These rules use the explicitly restricted product defined in this theorem. They do not follow from the unrestricted same-domain implication in the generated `DefPiPredicateModel` record discussed above.

Proof.
For a specified operation t, replacement on A forms the set $g = \{(x, t(x)) : x \in A\}$. Each pair lies in ΣB by Theorem 31.17, and injectivity of pairing shows that g is functional with domain exactly A. Thus g is a set subclass of ΣB, and it satisfies every clause of the displayed product predicate. This proves introduction even when some fibres are proper classes; it does not require collecting all their members into a set.

Conversely, $f \in \Pi B$ is a functional graph with domain A. For $a \in A$, the domain clause supplies a pair $(a, y) \in f$, and functionality makes y unique. The pointwise membership clause gives $y \in B(a)$. Therefore the displayed fibre is the singleton $\{y\}$. The unique-value operation on this singleton returns y, as does its intersection: for every set z, membership in that intersection is exactly membership in y. No default value on empty or nonsingleton classes is involved.

For g constructed from t, the pair $(a, t(a))$ belongs to g; the preceding uniqueness proves $g(a) = t(a)$, which is beta. For a given $f \in \Pi B$, replacement also forms $h = \{(a, f(a)) : a \in A\}$. Every pair of h lies in f by the definition of application. Every pair of f has first coordinate in A and second coordinate its unique value, so lies in h. Extensionality gives $h = f$, proving eta.

Finally, when all fibres are sets, the sum theorem makes $S = \Sigma B$ a set. The displayed product is then a subclass of the set $\mathcal{P}S$ and is a set by separation. Its construction uses no choice axiom. In particular introduction requires a supplied operation t, rather than asserting a section exists whenever all fibres are nonempty. For $A = \varnothing$ the sole graph is the empty function, and both computation assertions remain valid. □

For contextual abstraction, the interpretation of the body at $\rho[x \mapsto a]$ supplies the operation $t(a)$ in this proof. The environment and substitution assumptions stated after Theorem 31.18 identify application and syntactic substitution with the fibre and value just computed. Under these assumptions the rules are sound in the model. The equations alone do not establish normalization or decidability of judgmental conversion for a dependent checker. Likewise, sethood of ΠB does not imply membership in an arbitrary chosen universe $\mathcal{U}$; interpreting the small formation rule requires that universe to be closed under these dependent products. For a proper-class domain the predicate model remains available, but its proper-class function graphs cannot be members of this class of set graphs.

31.12 Implementing W–types

A W-type describes well-founded trees: a label a determines the type $B(a)$ of branches. The rules first state the intended interface. The finite-stage model developed afterward is restricted to finite branching, that is, finite $B(a)$ for each label a. This restriction is what allows all branches of a node to fit in one finite stage. Infinite branching requires a further construction: after constructing the M-type, we obtain the general set-arity W-type as its well-founded part. It is not supplied by the finite-stage formulas alone.

Axioms of W–types

Formation of W–type:
Let B be a family of types.
Then $\mathrm{W}B$ is also a type.
Introduction of terms of W–type:

$$\sup(B) : \prod_{(a \in \mathrm{Dom}(B))} ((B(a) \to \mathrm{W}B) \to \mathrm{W}B)$$

Elimination of terms of W–type:

$$\begin{aligned} \mathrm{W}_{\mathrm{rect}} : & \prod_{(A:\mathcal{U})} \prod_{(B:A\to\mathcal{U})} \prod_{(P:\mathrm{W}B\to\mathcal{U})} \\ & \left(\prod_{(a:A)} \prod_{(w:B(a)\to\mathrm{W}B)} \left(\left(\prod_{(f:B(a))} P(w(f)) \right) \to P(\sup(B)(a)(w)) \right) \right) \\ & \to \left(\prod_{(w:\mathrm{W}B)} P(w) \right) \end{aligned}$$

Computation of terms of W–type:

$$\mathrm{W}_{\mathrm{rect}}(A, B, E, e, \sup(B, a, f)) \overset{\mathrm{def.}}{\coloneqq} e(a, f, \lambda(b : B(a)) \mapsto \mathrm{W}_{\mathrm{rect}}(A, B, E, e, f(b)))$$

Model of W–types

Formation of W–type:
Assume B is a family of classes with domain A. Equivalently, regard B as a dependent class indexed by $\mathrm{Dom}(B)$, where $\mathrm{Dom}(B) = A$. The following stage formulas retain that general proposal. Constructor closure below assumes finite branch sets $B(a)$; the set-model recursion theorem additionally assumes that A is a set. Neither argument asserts the unrestricted class-valued elimination rule.

$$(\mathcal{W}_{\mathrm{fun}}B)(0) \stackrel{\mathrm{def.}}{\coloneqq} \varnothing$$

$$(\mathcal{W}_{\mathrm{fun}}B)(n+1) \stackrel{\mathrm{def.}}{\coloneqq} \{\, (r,s) \mid (r \in \mathrm{Dom}(B)) \wedge (s : B(r) \to (\mathcal{W}_{\mathrm{fun}}B)(n)) \,\}$$

$$\mathcal{W}B \stackrel{\mathrm{def.}}{\coloneqq} \bigcup_{n\in\omega} ((\mathcal{W}_{\mathrm{fun}}B)(n)),$$

Equivalently, when all stages are sets, this is the union of the range:

$$\mathcal{W}B \stackrel{\mathrm{def.}}{\coloneqq} \bigcup \mathrm{ran}(\mathcal{W}_{\mathrm{fun}}B).$$

The range alone collects the stages, whereas its union collects their trees.
A variant for definable families uses sigma types and the empty type:

$$\mathrm{W}'_{\mathrm{fun}} : \mathrm{Family.Expression} \to (\mathbb{N} \to \mathrm{Type})$$

$$(\mathrm{W}'_{\mathrm{fun}}B)(0) \stackrel{\mathrm{def.}}{:\equiv} \mathrm{Void}$$

$$(\mathrm{W}'_{\mathrm{fun}}B)(n+1) \stackrel{\mathrm{def.}}{:\equiv} \sum_{(r:\mathrm{Dom}(B))} (B(r) \to (\mathrm{W}'_{\mathrm{fun}}B)(n))$$

$$\mathrm{W}' : \mathrm{Family.Expression} \to \mathrm{Type}$$

$$\mathrm{W}'B \stackrel{\mathrm{def.}}{:\equiv} \bigsqcup_{n:\mathbb{N}} ((\mathrm{W}'_{\mathrm{fun}}B)(n))$$

For the finite-branching model, the intended fixed-point equation is

$$F(X) \cong X,$$

where the functor F is the polynomial functor P_B:

$$P_B(X) :\equiv \sum_{a:A} (B(a) \to X)$$

$$F(X) :\equiv P_B(X)$$

For a polynomial functor P with finite arities, define

$$P_{\mathrm{fun}}(0) \stackrel{\mathrm{def.}}{:\equiv} \mathrm{Void}$$

$$P_{\mathrm{fun}}(n+1) \stackrel{\mathrm{def.}}{:\equiv} P(P_{\mathrm{fun}}(n))$$

and the corresponding fixed-point type is

$$\mathrm{Fix}(P) \stackrel{\mathrm{def.}}{:\equiv} \bigsqcup_{n:\mathbf{Nat}} P_{\mathrm{fun}}(n).$$

Lemma 31.3 (levels of W–types are cumulative)**.**
For all $n \in \mathbb{N}$,

$$(\mathrm{W}_{\mathrm{fun}}B)(n) \subseteq (\mathrm{W}_{\mathrm{fun}}B)(n+1).$$

Proof.
By induction on n,
Base: n=0.
Trivially, $\varnothing \subseteq (\mathrm{W}_{\mathrm{fun}}B)(1)$.
Step of induction:
Assume

$$(\mathrm{W}_{\mathrm{fun}}B)(n) \subseteq (\mathrm{W}_{\mathrm{fun}}B)(n+1).$$

To show:

$$(\mathrm{W}_{\mathrm{fun}}B)(n+1) \subseteq (\mathrm{W}_{\mathrm{fun}}B)(n+2).$$

The induction hypothesis gives

$$(B(r) \to (\mathrm{W}_{\mathrm{fun}}B)(n)) \subseteq (B(r) \to (\mathrm{W}_{\mathrm{fun}}B)(n+1)).$$

Therefore

$$\sum_{(r \in \mathrm{Dom}(B))} (B(r) \to (\mathrm{W}_{\mathrm{fun}}B)(n)) \subseteq \sum_{(r \in \mathrm{Dom}(B))} (B(r) \to (\mathrm{W}_{\mathrm{fun}}B)(n+1)).$$

Hence

$$(\mathrm{W}_{\mathrm{fun}}B)(n+1) \subseteq (\mathrm{W}_{\mathrm{fun}}B)(n+2).$$

□

Introduction of terms of W–type:

Definition 31.28 (constructor of W–type)**.**
For fixed function B,

$$\sup(B) : \prod_{(a \in \mathrm{Dom}(B))} ((B(a) \to \mathrm{W}B) \to \mathrm{W}B)$$

Assume $(a \in \mathrm{Dom}(B))$ and $(s : B(a) \to \mathrm{W}B)$. Since the branch set $B(a)$ is finite, its image $s[B(a)]$ is finite. If this image is nonempty, let m be the maximum of the least stage indices of its elements; if it is empty, take $m = 0$. Then $s[B(a)] \subseteq (\mathrm{W}_{\mathrm{fun}}B)(m)$, so $s : B(a) \to (\mathrm{W}_{\mathrm{fun}}B)(m)$. It follows that $(a, s) \in (\mathrm{W}_{\mathrm{fun}}B)(m+1)$, which is a subset of $\mathrm{W}B$, and hence $(a, s) \in \mathrm{W}B$. We can therefore define sup by

$$\sup(B) \stackrel{\mathrm{def.}}{\coloneqq} ((a \in \mathrm{Dom}(B)) \mapsto ((s : B(a) \to \mathrm{W}B) \mapsto (a, s))).$$

Elimination of terms of W–type:

Definition 31.29 (destructor of W–type)**.**

$$\mathrm{W}_{\mathrm{rect}} : \prod_{(A:\mathcal{U})} \prod_{(B:A\to\mathcal{U})} \prod_{(P:\mathrm{W}B\to\mathcal{U})} \left(\prod_{(a:A)} \prod_{(w:B(a)\to\mathrm{W}B)} \left(\left(\prod_{(f:B(a))} P(w(f)) \right) \to P(\mathrm{sup}(B)(a)(w)) \right) \right) \to \left(\prod_{(w:\mathrm{W}B)} P(w) \right)$$

```
W_rect
    : forall (A : Type) (B : A -> Type) (P : W B -> Type),
      (forall (a : A) (w : B a -> W B), (forall f : B a, P (w f)) -> P (sup B a w)) ->
      forall w : W B, P w
```

Computation of terms of W–type:
By finiteness of each branch set $B(a)$ and $(\mathrm{W}_{\mathrm{fun}}B)(0) = \varnothing$, it holds that

$$\mathrm{sup}(B, a, f) \in (\mathrm{W}_{\mathrm{fun}}B)(m+1)$$

for some $m \in \mathbb{N}$.

Theorem 31.21 (Dependent recursion for the finite-branching set model)**.**
Work in the usual set theory with powerset, replacement, and recursion on ω. Let A be a set and each $B(a)$ a finite set. Write W_n for the stage above and $W = \bigcup_{n<\omega} W_n$, using ordinary injective set-pair coding. Let $E(w)$ be a set for each $w \in W$. Suppose a specified step operation assigns

$$e(a, s, h) \in E((a, s))$$

whenever $s : B(a) \to W$ and h is a function with $h(b) \in E(s(b))$ for every $b \in B(a)$. Then there is exactly one function r on W with $r(w) \in E(w)$ and

$$r((a, s)) = e\big(a, s, b \mapsto r(s(b))\big).$$

This interprets the displayed $\mathrm{W}_{\mathrm{rect}}$ elimination and computation equations in this set model. Interpreting a formation rule that additionally asserts $\mathrm{W}B \in \mathcal{U}$ requires closure of the chosen universe under this construction. Sethood alone does not give that membership; Proposition 31.8 below provides one sufficient universe condition.

Proof.
Powerset and replacement make every function set $W_n^{B(a)}$ and hence each stage

$$W_{n+1} = \sum_{a\in A} W_n^{B(a)}$$

a set. Recursion on ω, replacement, and union give the set W. The earlier cumulative-stage and finite-branching arguments show that W is closed under (a, s) and that each node has a unique such decomposition. In particular, a node in W_{n+1} has every child in W_n.

Put $r_0 = \varnothing$. Given r_n on W_n, define on W_{n+1}

$$r_{n+1}((a,s)) = e\big(a, s, b \mapsto r_n(s(b))\big).$$

The inductive typing assertion makes the right side defined and in $E((a,s))$; replacement collects its graph. The crucial compatibility assertion is $r_{n+1}|_{W_n} = r_n$. For $n = 0$ it is vacuous. For the step, a node $(a,s) \in W_{n+1}$ has children in W_n, where r_{n+1} and r_n already agree. Thus its values under r_{n+2} and r_{n+1} are applications of the same e to identical arguments. This proves compatibility at the next stage.
The union $r = \bigcup_n r_n$ is therefore a function on W, with the required typing. Every child family fits into some W_n, so the stage equation gives the claimed computation equation. If q is another such function, induction on n proves $q|_{W_n} = r_n$: the base is empty and the step uses the computation equation on children in W_n. Their unions agree, proving uniqueness. □

Corollary 31.4 (Structural induction and initiality)**.**
If a predicate Q on W holds at (a,s) whenever it holds at every $s(b)$, then it holds everywhere on W. Furthermore, for every set X and map $\alpha : P_B(X) \to X$ there is exactly one $r : W \to X$ such that

$$r((a,s)) = \alpha(a, r \circ s).$$

Thus the constructor $P_B(W) \to W$ is an initial P_B-algebra.

Proof.
For induction, prove Q on W_n by induction on n; each node at the next stage has all children at the preceding one. For initiality, use the theorem with constant family $E(w) = X$ and $e(a,s,h) = \alpha(a,h)$. Its uniqueness clause is exactly uniqueness of the algebra morphism. The argument includes $X = \varnothing$: a step map into the empty set exists only when its domain is empty, and the theorem then still applies. □

Remark 31.2 (Stages are identified along their inclusions)**.**
In this cumulative set model, the notation $\bigsqcup_n W_n$ above must mean the union, or equivalently the colimit along the inclusions $W_n \subseteq W_{n+1}$. It cannot mean the disjoint union of the stages: that would retain a separate tagged copy of a tree at each later stage, and the proof would not identify their recursor values automatically. The compatibility argument above supplies precisely the identification needed by the model. A model with a proper class of labels or genuinely class-valued dependent outputs requires additional representation and recursion arguments and is not asserted by this set-model theorem.

Example 31.2 (Why the finite-stage proof needs finite branching)**.**
Take distinct labels z, u, c, with $B(z) = \varnothing$, $B(u) = \{0\}$, and $B(c) = \omega$. The finite trees $t_0 = (z, \varnothing)$ and $t_{n+1} = (u, \{(0, t_n)\})$ have unbounded finite heights. Their sequence is a function $s : \omega \to W$, but its image is not contained in any single W_m. Thus (c,s) belongs to $P_B(W)$ but not to the union of the finite stages. The missing root is well founded; it merely has children of unbounded finite height. This shows failure of this construction for infinite branching, not failure of the general notion of a W-type.

31.13 Implementing M–types by finite paths

An M-type allows infinite depth. The author's path-based proposal can be made precise using finite addresses: a potentially infinite tree is specified by all its finite observations. We now give a set-level model and prove its universal property. No finite-branching hypothesis is needed. Fix a set A and a set family $(B(a))_{a\in A}$. Put

$$C = \sum_{a\in A} B(a), \qquad C^{<\omega} = \bigcup_{n<\omega} C^n.$$

Elements of $C^{<\omega}$ are finite words; the empty word is ϵ, and concatenation is denoted by pq. A letter (a, b) records the parent label as well as its branch index, so overlapping branch sets do not cause ambiguity.

Definition 31.30 (Trees described by finite addresses)**.**
Let M_B consist of the partial functions $t : C^{<\omega} \rightharpoonup A$ with domain D_t satisfying:

1. $\epsilon \in D_t$;
2. D_t is prefix closed;
3. if $p \in D_t$ and $c \in C$, then

$$pc \in D_t \longleftrightarrow c = (t(p), b) \text{ for some } b \in B(t(p)).$$

These are conditions on a subset of $C^{<\omega} \times A$, so powerset and separation make M_B a set. In particular $M_B = \varnothing$ if $A = \varnothing$. Partial functions here are ordinary functional graphs; their values are used only on their stated domains.

For $t \in M_B$, let $a = t(\epsilon)$. For each $b \in B(a)$ define its subtree t_b by

$$D_{t_b} = \{q : (a, b)q \in D_t\}, \qquad t_b(q) = t((a, b)q).$$

The tree axioms immediately give $t_b \in M_B$. Thus the destructor is

$$\delta : M_B \longrightarrow P_B(M_B), \qquad \delta(t) = (a, b \mapsto t_b).$$

Lemma 31.4 (Grafting and decomposition)**.**
The destructor δ is bijective. Its inverse sends (a, s), where $s : B(a) \to M_B$, to the tree with root label a and subtrees $s(b)$.

Proof.
Define t on the set

$$D_t = \{\epsilon\} \cup \{(a, b)q : b \in B(a),\ q \in D_{s(b)}\}$$

by $t(\epsilon) = a$ and $t((a, b)q) = s(b)(q)$. Each nonempty address has a unique first letter and suffix, so this is well defined. Prefix closure and the exact-child condition follow at the root from the indexing set $B(a)$, and below it from the corresponding conditions on each $s(b)$. Hence $t \in M_B$, with $\delta(t) = (a, s)$. Conversely, decomposing any tree and grafting its subtrees recovers every nonempty address and its label, as well as the root. The constructions are inverse. □

Theorem 31.22 (Final coalgebra and corecursion)**.**
For every set X and map $\gamma : X \to P_B(X)$ there is exactly one $h : X \to M_B$ satisfying

$$\delta \circ h = P_B(h) \circ \gamma, \qquad P_B(h)(a, s) = (a, h \circ s).$$

Thus (M_B, δ) is a final coalgebra for P_B on sets. Here P_B is a functor: postcomposition preserves identities and composition, so $P_B(\mathrm{id}) = \mathrm{id}$ and $P_B(k \circ h) = P_B(k) \circ P_B(h)$.

Proof.
Write $\gamma(x) = (a_x, s_x)$, where $s_x : B(a_x) \to X$. Starting from a state x, define the state reached at a finite address recursively on its length. The empty address reaches x. If p reaches y, then precisely the letters (a_y, b) with $b \in B(a_y)$ are admissible next, and $p(a_y, b)$ reaches $s_y(b)$. A word whose prefix is inadmissible remains inadmissible. This is finite recursion with unique transitions, and replacement collects the reached states on each length; union over ω collects all valid addresses.
Label each valid address by a_y for its reached state y. The result t_x satisfies the tree axioms by construction. Its root label is a_x, and its subtree at b is $t_{s_x(b)}$, because both use the identical transition rule starting from $s_x(b)$. Thus $h(x) = t_x$ satisfies the coalgebra equation. Replacement gives a set function h.
If $k : X \to M_B$ satisfies that equation, its root at x has label a_x and its b-subtree is $k(s_x(b))$. Induction on address length therefore proves that $k(x)$ and t_x have exactly the same admissible addresses of each length and the same labels there. Every address is finite, so their graphs coincide. Consequently $k = h$. The proof uses no selection of an infinite path and no choice function. □

Theorem 31.23 (Coinduction by bisimulation)**.**
Let $R \subseteq M_B \times M_B$ have the following property: whenever $t\ R\ u$, their root labels agree, say with value a, and $t_b\ R\ u_b$ for every $b \in B(a)$. Then $t\ R\ u$ implies $t = u$. In particular, equality is the largest such relation.

Proof.
For any related pair, the root labels agree. Inductively, equal labels at a common address give exactly the same admissible next letters. At each of those letters the corresponding subtrees are again related, so their labels agree by the hypothesis. Induction on length proves identical address domains and labels at every finite level. The trees are their graphs on finite addresses, hence are equal. Equality itself has the stated property, proving the final assertion. □

Example 31.3 (Streams and the difference from finite trees)**.**
Let $A = S$ be a set of labels and $B(a) = \{0\}$ for every $a \in S$. A tree then has exactly one address at each length, so reading successive labels identifies M_B with the stream set S^ω. The destructor reads the first label and the tail. If S is nonempty, a constant stream exists, while the finite-stage W-construction is empty: it has no nullary label with which to start. If $S = \varnothing$, both sets are empty. Thus algebraic and coalgebraic fixed points have different universal properties, even for the same polynomial functor.

Proposition 31.6 (Finite trees inside the M-type)**.**
Under the finite-branching assumptions of the W-model, there is a canonical injective map $j : W \to M_B$. Its image consists exactly of trees whose valid addresses have bounded finite length.

Proof.
Use the initiality of W with the grafting algebra $\delta^{-1} : P_B(M_B) \to M_B$. The resulting map satisfies $j((a, s)) = \delta^{-1}(a, j \circ s)$. Induction on the W-stage shows that $j(w)$ has bounded address length.
To prove injectivity, use structural induction on w with the stronger assertion that for every $v \in W$, equality $j(w) = j(v)$ implies $w = v$. Write $w = (a, s)$ and $v = (d, t)$. Applying δ to the equality gives $a = d$ and $j(s(b)) = j(t(b))$ for each $b \in B(a)$. The induction hypotheses for the children $s(b)$ apply to these arbitrary children $t(b)$, giving $s(b) = t(b)$. Function extensionality then gives $s = t$ and hence $w = v$.
Conversely, induct on an upper bound n for the address lengths of a tree u. If $n = 0$, the root has no children, so its label a has $B(a) = \varnothing$ and $u = j((a, \varnothing))$. If $n > 0$, every root subtree has address lengths at most $n - 1$, hence is $j(w_b)$ for some $w_b \in W$. Injectivity makes each w_b unique, so replacement gives the function $b \mapsto w_b$ without a choice axiom. Constructor closure of W gives the node $(a, b \mapsto w_b)$, and its image under j is u by grafting. This completes the induction. □

31.14 W-types with arbitrary set arities

The finite-stage construction fails for some infinite arities, but the M-type supplies a set in which to isolate the well-founded trees. Keep A and every $B(a)$ sets; no finiteness assumption is made here.

Definition 31.31 (The well-founded part of the M-type)**.**
For $t \in M_B$, put $q \prec_t p$ when q is an immediate child address of p in D_t. Say that t is well founded if every nonempty subset of D_t has a $\prec_t$-minimal member. Define

$$W_B^{\mathrm{wf}} = \{t \in M_B : \prec_t \text{ is well founded}\}.$$

Separation makes this a set. Well-foundedness here is the minimal-member condition, not merely an assertion that no infinite path has been chosen.

Lemma 31.5 (Grafting preserves and detects well-foundedness)**.**
For $t = \delta^{-1}(a, s)$, one has

$$t \in W_B^{\mathrm{wf}} \longleftrightarrow s(b) \in W_B^{\mathrm{wf}} \text{ for every } b \in B(a).$$

Consequently grafting restricts to a bijection $P_B(W_B^{\mathrm{wf}}) \to W_B^{\mathrm{wf}}$.

Proof.
A nonempty subset of addresses of $s(b)$ lifts by prefixing (a, b) to a nonempty subset of D_t. A minimal member of the latter gives a minimal member of the former, since immediate children stay in that subtree. This proves the forward direction.
Conversely, let $S \subseteq D_t$ be nonempty. If it contains only the root, that root is minimal. Otherwise choose one nonroot address in S, and let (a, b) be its first letter. The suffixes of the addresses of S beginning with this letter form a nonempty subset of $D_{s(b)}$. Choose a minimal suffix there. Its prefixed address is minimal in all of S, because every immediate child has the same first letter. Thus t is well founded. The bijection now follows from the already proved grafting/decomposition bijection on M_B. □

Lemma 31.6 (Ordinal heights and the subtree relation)**.**
Every $t \in W_B^{\mathrm{wf}}$ has a uniquely determined ordinal height $h(t)$, with

$$h(t) = \sup_{b \in B(t(\epsilon))} \big(h(t_b) + 1\big).$$

The relation $u \prec t$ on W_B^{wf} meaning that u is an immediate root subtree of t is set-like and well founded.

Proof.
Apply the ordinal-rank construction for a founded set relation to $(D_t, \prec_t)$. It assigns ordinals $r_t(p)$ satisfying

$$r_t(p) = \sup_{q \prec_t p} (r_t(q) + 1).$$

All predecessor collections are sets, so replacement and ordinal union supply the suprema. Define $h(t) = r_t(\epsilon)$. The suffix bijection between D_{t_b} and the addresses of t beginning with $(t(\epsilon), b)$ preserves the child relation. Restriction of r_t therefore satisfies exactly the rank recursion for t_b; uniqueness of founded recursion gives $h(t_b) = r_t((t(\epsilon), b))$. This proves the displayed height formula, and in particular $h(t_b) < h(t)$.
The immediate subtrees form the image of the set $B(t(\epsilon))$, so the subtree relation is set-like. For any nonempty subset S of W_B^{wf}, replacement collects the set of its heights. Choose its least ordinal and a tree in S of that height. A predecessor in S would have smaller height, which is impossible. Hence the relation is well founded. This argument constructs heights from founded address relations first; it does not assume foundedness of the subtree relation. Finally, if another ordinal assignment satisfies the displayed height equation, founded induction on this now-established subtree relation makes it agree with h at every tree: agreement on all immediate subtrees makes the two suprema identical. □

Theorem 31.24 (Dependent recursion for arbitrary set arities)**.**
Let $(E(t))_{t \in W_B^{\mathrm{wf}}}$ be a specified family of sets. Suppose a specified operation assigns

$$e(a, s, k) \in E(\delta^{-1}(a, s))$$

whenever $s : B(a) \to W_B^{\mathrm{wf}}$ and k is a function on $B(a)$ with $k(b) \in E(s(b))$ for every $b \in B(a)$. There is exactly one function r with domain W_B^{wf}, values $r(t) \in E(t)$, and

$$r(\delta^{-1}(a, s)) = e(a, s, b \mapsto r(s(b))).$$

Proof.
Use Theorem 24.1 on the founded immediate- subtree relation. Given a function g on the predecessors of $t = \delta^{-1}(a, s)$, form $k(b) = g(s(b))$. If every such value belongs to $E(s(b))$, use $e(a, s, k)$ as the recursion step; otherwise use the empty set as a temporary default. For any other set input, including a nonfunction or a function whose domain is not the predecessor set, also return the empty set. This defines a total set-valued step, so founded recursion gives a function r. Founded induction shows that its values are correctly typed: assuming $r(u) \in E(u)$ at every predecessor, the first branch of the step applies and yields $r(t) \in E(t)$. Thus the temporary default never occurs, and the required equation holds everywhere. Replacement makes the graph a set. Two typed solutions agree on each tree as soon as they agree on its predecessors; founded induction therefore proves uniqueness. Repeated identical subtrees cause no ambiguity, since k uses the same value of g at each occurrence. □

Corollary 31.5 (Structural induction and the general initial algebra)**.**
A predicate on W_B^{wf} that is preserved by grafting from all immediate subtrees holds everywhere. The grafting map makes W_B^{wf} an initial algebra for P_B on sets.

Proof.
The induction assertion is founded induction for the subtree relation. Given any algebra $\alpha : P_B(X) \to X$, apply the recursion theorem with constant family $E(t) = X$ and $e(a, s, k) = \alpha(a, k)$. The resulting unique r satisfies $r \circ \delta^{-1} = \alpha \circ P_B(r)$, exactly the algebra-morphism condition. This includes the empty cases: existence of the specified step operation is part of the hypothesis. □

Proposition 31.7 (Agreement with the finite-stage model)**.**
If every $B(a)$ is finite, the canonical map $j : W \to M_B$ constructed above identifies W with W_B^{wf}.

Proof.
A tree with address lengths bounded by a natural number is founded: in any nonempty subset of its addresses choose one of greatest length; none of its immediate children can belong to that subset. Conversely, apply founded induction to a tree in W_B^{wf} and the property of having bounded finite address lengths. If every immediate subtree has that property, let n_b be its least natural-number bound. These bounds are uniquely specified, so replacement collects them. There are finitely many children; taking the maximum of their bounds and adding one bounds the whole tree. At a leaf the bound is zero. Thus every founded tree has bounded finite height. The preceding characterization of the image of j now proves the assertion. No equivalence between pathlessness and foundedness, and no selection of an infinite branch, is used. □
The earlier infinitely branching root with children of all finite heights belongs to W_B^{wf} although it is absent from the finite-stage union. Its address relation is founded by the grafting lemma, and its ordinal height is ω. This is precisely the extra behavior supplied by arbitrary set arities.

Lemma 31.7 (Indexed unions from the tight set-universe condition)**.**
Let U be a wellorderable transitive set containing ω as an element and closed under unordered pairing. Suppose that

$$B \subseteq U \quad \Longrightarrow \quad (B \prec U \longleftrightarrow B \in U).$$

Here, for these wellorderable sets, $B \prec U$ means $|B| < |U|$; equivalently there is no injection $U \to B$. Then for every specified set family $(X_i)_{i \in I}$ with $I \in U$ and every $X_i \in U$, one has $\bigcup_{i \in I} X_i \in U$.

Proof.
Fix a wellordering of U, and write $\kappa = |U|$. Transitivity and the displayed condition give $|I| < \kappa$ and $|X_i| < \kappa$. For any function $f : I \to U$, use the ordinary Kuratowski graph. Since $i \in I \in U$, transitivity gives $i \in U$; pairing closure then puts $(i, f(i)) = \{\{i\}, \{i, f(i)\}\}$ in U. Thus the graph of f is a subset of U of cardinality $|I| < \kappa$, and the displayed condition gives $f \in U$.
Put $Y = \bigcup_{i \in I} X_i$. Transitivity gives $Y \subseteq U$. If I is empty, then $Y = \varnothing \in U$ by the displayed condition. Otherwise suppose $Y \notin U$. Since Y and U are wellorderable, the condition implies $|Y| = \kappa$; fix a bijection $e : Y \to U$. For each $i \in I$, form

$$D_i = \{f(i) : f \in e[X_i],\ f \text{ is a function } I \to U\}.$$

This image has cardinality at most $|X_i| < \kappa$: the functions being evaluated form a subset of $e[X_i]$, and its inherited wellordering selects a least preimage for each value. Therefore $U \setminus D_i$ is

nonempty. Let $t(i)$ be its least element in the fixed wellordering. Replacement gives a function $t : I \to U$, and the graph argument above gives $t \in U$. Surjectivity of e gives $t \in e[X_j]$ for some $j \in I$, so the definition of D_j gives $t(j) \in D_j$, contrary to the choice of $t(j)$. Hence $Y \in U$. □
This lemma uses only a wellordering of the particular set U; set choice suffices to provide one, and Global Choice is not needed. Obelisk's `DefTightClass` (3.47) uses this condition, with its injection comparison (3.40, 3.42). Consider a wellorderable *set* satisfying its universe definition, `DefGrothendieckUniverse` (3.68). It has the indexed-union closure just proved. Its `Pairful` and `Powerful` clauses and $\omega \in U$ supply the other hypotheses of the following proposition. This is a proved bridge for set universes, not an assertion of cardinal comparability for arbitrary proper classes.

Proposition 31.8 (A sufficient universe condition for W- and M-types)**.**
Let $\mathcal{U}$ be a transitive set containing ω as an element. Assume that $x, y \in \mathcal{U}$ implies $\{x, y\} \in \mathcal{U}$, that $x \in \mathcal{U}$ implies $\mathcal{P}x \in \mathcal{U}$, and that for every specified set family $(X_i)_{i \in I}$ with $I \in \mathcal{U}$ and every $X_i \in \mathcal{U}$, one has $\bigcup_{i \in I} X_i \in \mathcal{U}$. If $A \in \mathcal{U}$ and every $B(a) \in \mathcal{U}$, then the finite-address carriers M_B and W_B^{wf} belong to $\mathcal{U}$. No finiteness of the arities is required. When the arities are finite, the earlier finite-stage carrier W also belongs to $\mathcal{U}$.

Proof.
First, every subset $Y \subseteq X \in \mathcal{U}$ belongs to $\mathcal{U}$: it is an element of $\mathcal{P}X \in \mathcal{U}$, and $\mathcal{U}$ is transitive. The pairing and indexed-union assumptions give binary unions. With ordinary Kuratowski pair coding, $X \times Y \subseteq \mathcal{P}\mathcal{P}(X \cup Y)$, so products of two members of $\mathcal{U}$ also belong to $\mathcal{U}$ by subset closure.
For each $a \in A$, transitivity gives $a \in \mathcal{U}$, pairing gives $\{a\} \in \mathcal{U}$, and hence $\{a\} \times B(a) \in \mathcal{U}$. The union indexed by A therefore gives $C = \sum_{a \in A} B(a) \in \mathcal{U}$. Each $n \in \omega$ belongs to $\mathcal{U}$. The function set C^n is a subset of $\mathcal{P}(n \times C)$, so belongs to $\mathcal{U}$. The union indexed by ω now gives $T = C^{<\omega} \in \mathcal{U}$. A tree is a functional graph contained in $T \times A$, so

$$M_B \subseteq \mathcal{P}(T \times A) \in \mathcal{U}.$$

Subset closure yields $M_B \in \mathcal{U}$, and then $W_B^{\mathrm{wf}} \subseteq M_B$ gives $W_B^{\mathrm{wf}} \in \mathcal{U}$. The predicates defining these subsets need not be decidable for this argument; separation and the proved subset closure suffice.
For the earlier representation, $W_0 = \varnothing \in \mathcal{U}$. If $W_n \in \mathcal{U}$, each function set $W_n^{B(a)}$ belongs to $\mathcal{U}$ by the same product and powerset argument, and its dependent sum over A gives $W_{n+1} \in \mathcal{U}$. The ω-indexed union therefore belongs to $\mathcal{U}$. Under finite arities it is the earlier W-carrier. □
These hypotheses are a sufficient condition on a local universe, not an additional assumption on every model in this chapter. The earlier sum and product closure conditions alone do not ensure W-closure. For example, the set V_ω of hereditarily finite sets is closed under dependent sums and products indexed by its members: there are only finitely many finite fibres, and all their set encodings remain hereditarily finite. But two labels, one nullary and one unary, give distinct finite trees of every natural-number height. Their W-carrier is infinite and therefore is not an element of V_ω. In particular, $\omega \notin V_\omega$, so this example does not satisfy the stronger hypotheses of the proposition. The same distinction between sethood and membership in a chosen universe applies to the M-carrier.

Relation to the larger class-valued proposal. The original sketch described collections of paths with finite or countably infinite domains and a compatibility condition on common prefixes. The finite-address construction supplies that compatibility and the destructor/corecursion rules

for set labels and set arities. Section **??** extends this construction to a possibly proper class of set labels, retaining set arities. It proves class-valued initiality and finality directly; the resulting carriers need not be sets, although every individual tree is a set. Neither construction treats proper-class arities or labels that are themselves proper classes. Nor do they assert a representation theorem for the original finite-or-infinite-path encoding. Those further proposals still require an explicit ambient size convention and their own representation arguments.

32 Suprema and infima

The first construction retains the inherited order on the class of bounds. This allows it to describe preorders, where a least element need not be unique as an object, as well as partial orders, where it is unique when it exists. For the image-based construction first, $(P, \leq)$ is a set-sized ambient order and $S \subseteq P$. Its upper and lower sections are then sets, so they can be collected by the displayed image notation. For a proper-class carrier, use the predicate-based bound classes given later; ordinary class images cannot collect proper classes as members. Whenever a carrier is restricted, the order relation is restricted to that carrier as well.

32.1 Up or down sets of an element of an order

Definition 32.1 (up set of an element of an order)**.**

$$\uparrow_{(P,\leq)} x \coloneqq \{\, y \in P \mid x \leq y \,\}$$

Definition 32.2 (down set of an element of an order)**.**

$$\downarrow_{(P,\leq)} x \coloneqq \{\, y \in P \mid y \leq x \,\}$$

32.2 Ordered set of majorants or minorants of a subset of an order

Definition 32.3 (ordered set of majorants of a subset of an order)**.**

$$\mathrm{Ma}_{(P,\leq)} S \coloneqq \left(P \cap \bigcap \uparrow_{(P,\leq)} [S], \leq\right)$$

Definition 32.4 (ordered set of minorants of a subset of an order)**.**

$$\mathrm{Mi}_{(P,\leq)} S \coloneqq \left(P \cap \bigcap \downarrow_{(P,\leq)} [S], \leq\right)$$

32.3 Ordered set of greatest or least elements of an order

Definition 32.5 (ordered set of greatest elements of an order)**.**

$$\mathrm{Greatest}_{(P,\leq)} \coloneqq \mathrm{Ma}_{(P,\leq)} P$$

Definition 32.6 (ordered set of least elements of an order)**.**

$$\mathrm{Least}_{(P,\leq)} \coloneqq \mathrm{Mi}_{(P,\leq)} P$$

32.4 Ordered set of suprema or infima of a subset of an order

Definition 32.7 (ordered set of suprema of a subset of an order)**.**

$$\mathrm{Sup}_{(P,\leq)} S \coloneqq \mathrm{Least}_{\mathrm{Ma}_{(P,\leq)} S}$$

Definition 32.8 (ordered set of infima of a subset of an order)**.**

$$\mathrm{Inf}_{(P,\leq)} S \coloneqq \mathrm{Greatest}_{\mathrm{Mi}_{(P,\leq)} S}$$

32.5 Essential supremum

Here the context changes to a measure space and a measurable real-valued function f. This value depends only on the almost-everywhere equivalence class of f: changing f on a measurable null set does not change which real numbers are essential upper bounds. Take the infimum in the extended real line, so that the expression also covers an unbounded function: an empty set of real upper bounds has infimum $+\infty$. The displayed condition means that M is an upper bound except on a set of measure zero.

Definition 32.9 (essential supremum; it depends on a measure μ)**.**

$$\operatorname{ess\,sup} f \coloneqq \inf \{\, M \in \mathbb{R} \mid \mu(\{\, x \in \operatorname{dom}(f) \mid f(x) > M \,\}) = 0 \,\}$$

With this extended-real convention, a zero-measure carrier gives $\operatorname{ess\,sup} f = -\infty$: every real number is an essential upper bound. The later L^∞ norm instead takes the infimum of *nonnegative* bounds for $|f|$, so it is zero on a zero-measure space. In these conventions its value is $\max\{0, \operatorname{ess\,sup} |f|\}$; on a space of positive measure this is simply $\operatorname{ess\,sup} |f|$, since no negative number can bound $|f|$ almost everywhere. This distinction concerns the range in which the infimum is taken, not a failure of the norm axioms.

32.6 Some theorems of the order theory

The following reformulation uses classes of bounds directly. An ordinary relational image $R[S]$ describes points above *some* member of S; upper bounds must lie above *every* member. This distinction motivates the universal definitions below. They express the intended meaning of the earlier image-based sketches.

Definition 32.10 (upper and lower bounds)**.**

$$\mathrm{UB}_R(S) \coloneqq \{\, y \in P \mid \forall x \in S,\ xRy \,\}, \qquad \mathrm{LB}_R(S) \coloneqq \{\, y \in P \mid \forall x \in S,\ yRx \,\}.$$

Context and notation:
All classes A, B, S in the following bound comparisons are subclasses of the fixed carrier P. In the real-number examples take $P = \mathbb{R}$. Here by $\leq$ we denote the standard relation "less or equal to" on real numbers.
$\leq$ for inequalities of numbers;
$\leqslant$ for inequalities of sets of numbers;
$R \coloneqq \leq$
$\leq = \geq^{-1}$
$\geq = \leq^{-1}$

Remark 32.1 (why the middle class must be nonempty)**.**
Define $A \leqslant B$ to mean that every element of A is at most every element of B. If $A \leqslant B$ and $B \leqslant C$, transitivity of $\leq$ gives $A \leqslant C$ provided the middle class B is nonempty. Without that hypothesis, both premises can hold vacuously.

Proof.
Choose $b \in B$. For arbitrary $a \in A$ and $c \in C$, we have $a \leq b \leq c$, hence $a \leq c$. For the failure with an empty middle class, take $S < H$. The familiar wordplay illustrates the two vacuous premises:

Nothing is better than eternal happiness.	$(\{H\} < \varnothing)$
A sandwich is better than nothing.	$(\varnothing < \{S\})$
Therefore, a sandwich is better than eternal happiness.	$(\{H\} < \{S\})$

We obtained $H < S$ (by definition of an order on sets). The statement together with $S < H$ and transitivity on elements contradicts the irreflexivity of the order on elements.
□

Definition 32.11 (all-to-all comparison)**.**

$$(A \leqslant B) \overset{\text{def.}}{\longleftrightarrow} (\forall a \in A \forall b \in B,\ a \leq b)$$

Definition 32.12 (bound classes)**.**

$$\mathrm{UB}(S) \coloneqq \mathrm{UB}_{\leq}(S)$$

Definition 32.13 (bound classes)**.**

$$\mathrm{LB}(S) \coloneqq \mathrm{LB}_{\leq}(S)$$

Theorem 32.1 (comparison by bounds)**.**

$$(A \leqslant B) \longleftrightarrow (B \subseteq \mathrm{UB}(A))$$

Proof. Since $B \subseteq P$, the right side says precisely that every $b \in B$ satisfies $a \leq b$ for every $a \in A$. This is the defining condition for $A \leqslant B$, with the universal quantifiers interchanged. □

Theorem 32.2 (comparison by bounds)**.**

$$(A \leqslant B) \Longleftrightarrow (A \subseteq \mathrm{LB}(B))$$

Proof. Since $A \subseteq P$, membership of every $a \in A$ in $\mathrm{LB}(B)$ means exactly that $a \leq b$ for every $b \in B$. Thus it is equivalent to $A \leqslant B$. Both comparisons include empty subclasses. □ The carrier restrictions matter: if $A = \varnothing$ and B contains an object outside P, then $A \leqslant B$ is vacuous but $B \subseteq \mathrm{UB}(A) = P$ fails.

Definition 32.14 (bound classes)**.**

$$\mathrm{Sup}(S) = \mathrm{UB}(S) \cap \mathrm{LB}(\mathrm{UB}(S))$$

Definition 32.15 (bound classes)**.**

$$\mathrm{Inf}(S) = \mathrm{LB}(S) \cap \mathrm{UB}(\mathrm{LB}(S))$$

Definition 32.16 (descriptor for the unique extremum)**.**

$$\sup(S) = (\iota x,\ x \in \mathrm{Sup}(S))$$

Definition 32.17 (descriptor for the unique extremum)**.**

$$\inf(S) \overset{\text{def.}}{\coloneqq} (\iota x,\ x \in \mathrm{Inf}(S))$$

A descriptor for a supremum or infimum denotes the desired extremum only when the corresponding bound class has exactly one member. Its behavior otherwise is governed by the descriptor convention chosen locally.

Theorem 32.3 (comparison of existing extrema)**.**
Assume the ambient order is a partial order, and both $\mathrm{Sup}(A)$ and $\mathrm{Inf}(B)$ are nonempty.

$$(A \leqslant B) \Longrightarrow (A \leqslant \mathrm{Sup}(A) \leqslant \mathrm{Inf}(B) \leqslant B)$$

Proof. Let s and i be the unique elements of $\mathrm{Sup}(A)$ and $\mathrm{Inf}(B)$. Every $b \in B$ is an upper bound of A, so the least-upper-bound property gives $s \leq b$. Thus s is a lower bound of B, and the greatest-lower-bound property gives $s \leq i$. The outer comparisons $a \leq s$ for $a \in A$ and $i \leq b$ for $b \in B$ follow from the definitions of upper and lower bounds. This proves the displayed chain. □

Theorem 32.4 (monotone convergence of a real sequence)**.**

$$(a : \mathbb{N} \to \mathbb{R}) \wedge (\{a_n\}_n \text{bdd above in } \mathbb{R}) \wedge ((a_n)_n \text{ non–decreasing}) \longrightarrow (a \longrightarrow \sup\{a_n\}_n)$$

Proof. The range of the sequence is nonempty and bounded above, so completeness of the real numbers supplies its real supremum s. For every $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound; choose N with $s - \varepsilon < a_N$. Monotonicity gives

$$s - \varepsilon < a_N \leq a_n \leq s \qquad (n \geq N).$$

Hence $|a_n - s| < \varepsilon$ for $n \geq N$, which is exactly convergence to s. This argument uses a real supremum, not an off-domain descriptor value or the ordinal infimum convention from another section. □

32.7 Bounds, closure, and fixed points

We now return from the real-number examples to a set-sized partially ordered set P. Bound classes are always restricted to this carrier. The equivalence

$$B \subseteq \mathrm{UB}(A) \quad \Longleftrightarrow \quad A \subseteq \mathrm{LB}(B) \qquad (A, B \subseteq P)$$

is the order instance of the relational polarity proved in the section "The order structure behind glare." In that notation the two maps are $p_P = \mathrm{UB}$ and $q_P = \mathrm{LB}$. In particular,

$$c(A) = \mathrm{LB}(\mathrm{UB}(A))$$

is extensive, monotone, and idempotent, and $\mathrm{UB}(\mathrm{LB}(\mathrm{UB}(A))) = \mathrm{UB}(A)$. These statements include empty subsets: $\mathrm{UB}(\varnothing) = P$. When $s = \sup A$ exists, $c(A) =\!\downarrow s$. Thus the bound construction already supplies an example of a closure operator. We next establish a more general existence theorem, which requires monotonicity but not extensivity or idempotence.

Definition 32.18 (Complete lattice and fixed-point inequalities)**.**
A set-sized partially ordered set L is a complete lattice if every subset $A \subseteq L$ has a supremum $\bigvee_L A$ and an infimum $\bigwedge_L A$ in L, including the empty subset. Write $\bot = \bigvee_L \varnothing$ and $\top = \bigwedge_L \varnothing$. For a monotone function $f : L \to L$, put

$$\begin{aligned} \mathrm{Pre}(f) &= \{x \in L : f(x) \leq x\}, \\ \mathrm{Post}(f) &= \{x \in L : x \leq f(x)\}, \\ \mathrm{Fix}(f) &= \{x \in L : f(x) = x\}. \end{aligned}$$

The first two conditions are called pre-fixed and post-fixed, respectively; this specifies the direction of the terminology.

Theorem 32.5 (Complete lattice of fixed points: Knaster–Tarski)**.**
Let $f : L \to L$ be monotone on a set-sized complete lattice. Its fixed points form a complete lattice in the inherited order. More explicitly, for $A \subseteq \mathrm{Fix}(f)$ define

$$U_A = \{x \in L : f(x) \le x \text{ and } \forall a \in A,\ a \le x\},$$
$$D_A = \{x \in L : x \le f(x) \text{ and } \forall a \in A,\ x \le a\}.$$

Then the join and meet in the fixed-point lattice are

$$\bigvee_{\mathrm{Fix}(f)} A = \bigwedge_{L} U_A, \qquad \bigwedge_{\mathrm{Fix}(f)} A = \bigvee_{L} D_A.$$

In particular, the least and greatest fixed points are

$$\mu f = \bigwedge_{L} \mathrm{Pre}(f), \qquad \nu f = \bigvee_{L} \mathrm{Post}(f).$$

The element μf lies below every pre-fixed point, and νf lies above every post-fixed point.

Proof.
The set U_A contains $\top$. Put $j = \bigwedge_L U_A$. Each $a \in A$ is a lower bound of U_A, so $a \le j$. For each $x \in U_A$, monotonicity gives $f(j) \le f(x) \le x$. Thus $f(j) \le j$. It follows that $f(f(j)) \le f(j)$, so $f(j)$ is pre-fixed. Also, for $a \in A$, $a = f(a) \le f(j)$. Hence $f(j) \in U_A$, and the infimum property gives $j \le f(j)$. Antisymmetry proves $f(j) = j$. Finally, every fixed upper bound b of A belongs to U_A, so $j \le b$. Therefore j is precisely the join among fixed points.
For completeness, the dual argument is as follows. The set D_A contains $\bot$. Put $m = \bigvee_L D_A$. Every $a \in A$ is an upper bound of D_A, so $m \le a$. If $x \in D_A$, then $x \le f(x) \le f(m)$, whence $m \le f(m)$. Therefore $f(m) \le f(f(m))$, and for each $a \in A$ we have $f(m) \le f(a) = a$. Thus $f(m) \in D_A$ and $f(m) \le m$. Antisymmetry makes m fixed. Every fixed lower bound b of A belongs to D_A, so $b \le m$. This proves the meet assertion.
Taking $A = \varnothing$ gives $U_A = \mathrm{Pre}(f)$ and $D_A = \mathrm{Post}(f)$, yielding the last assertions as well as nonemptiness of the fixed-point lattice. All bound sets used here are subsets of L. The proof uses neither a transfinite iteration nor a choice of approximating elements. □

Example 32.1 (A join can change when restricted to fixed points)**.**
On the powerset lattice of $\{a, b, c\}$, with these three elements distinct, define

$$f(X) = \begin{cases} X \cup \{c\}, & \{a, b\} \subseteq X, \\ X, & \text{otherwise.} \end{cases}$$

This map is monotone: enlarging X preserves its members and cannot turn off an already satisfied condition $\{a, b\} \subseteq X$. Both $\{a\}$ and $\{b\}$ are fixed, but their ambient join $\{a, b\}$ is not fixed. Their join among fixed points is $\{a, b, c\}$, since every fixed set containing both a and b must also contain c. Thus the theorem does not assert that fixed points are closed under the ambient joins and meets. Its displayed formulas specify the bounds in their own inherited order.

Corollary 32.1 (Least closed subset containing prescribed data)**.**
Let X be a set, $S \subseteq X$, and let $\Phi : \mathcal{P}(X) \to \mathcal{P}(X)$ be monotone. There is a least subset $C \subseteq X$ containing S and satisfying $\Phi(C) \subseteq C$. It is given by

$$C = \bigcap \{Q \subseteq X : S \subseteq Q \text{ and } \Phi(Q) \subseteq Q\}, \qquad C = S \cup \Phi(C).$$

Proof.
Apply the theorem in the inclusion-ordered complete lattice $\mathcal{P}(X)$ to $f(Q) = S \cup \Phi(Q)$. Its pre-fixed points are exactly the subsets in the displayed intersection, a nonempty family because it contains X. The least fixed point is below every such subset and is itself one of them. Its fixed-point equation is exactly $C = S \cup \Phi(C)$. □
This is the set-sized counterpart of the least-class closure theorem in Section 30. That theorem uses MK comprehension over classes; it is not obtained by treating the collection of all proper classes as a set-sized complete lattice.

Example 32.2 (Reachability as a least fixed point)**.**
Let $R \subseteq X \times X$ be a relation on a set and $S \subseteq X$. Use graph image, $R[Q] = \{y \in X : \exists x \in Q,\ (x, y) \in R\}$. This operation is monotone, so the preceding corollary gives the least R-closed subset C containing S, with $C = S \cup R[C]$. It has the concrete description

$$C = \bigcup_{n \in \mathbb{N}} C_n, \qquad C_0 = S, \quad C_{n+1} = C_n \cup R[C_n].$$

Indeed, the union contains S. If x belongs to it and $(x, y) \in R$, then $x \in C_n$ for some n and $y \in C_{n+1}$; hence the union is R-closed. Conversely, induction shows that every R-closed set containing S contains every C_n, and therefore their union. By induction on n, C_n consists precisely of points reached from S by paths of at most n edges: a new image point appends one edge, and every nonzero-length path has a penultimate point. Thus C records finite reachability, with zero-length paths retaining S. This explicit countable iteration is justified for relational image; monotonicity alone in an arbitrary complete lattice does not assert that countably many iteration stages suffice.

33 Many-valued semantic experiments

These notes explore ways to distinguish absence of a value from variation among possible values. One proposal tracks presence and stability separately; another interprets a predicate by the assignments on which it holds. In the latter proposal, correlations between subformulas matter, so a single truth table on the labels "true", "false", and "uncertain" need not determine compound formulas.
An open formula varying with its assignment is different from a formula without a classical truth value under a fixed assignment. The following small models make that distinction precise. They develop parts of the proposal; they do not assert soundness of an unspecified descriptor calculus. The original working notes remain below in the source.

33.1 Positive evidence, negative evidence, and absence

Work in the metatheory with a set D of possible values and a relation $R \subseteq D \times D$. A set $J \subseteq D \times D$ records the *jointly admissible* values of two terms. Independent candidates $S, T \subseteq D$ give $J = S \times T$, but correlations may require a smaller joint set. Define

$$E_R^+(J) \longleftrightarrow J \neq \varnothing \ \wedge \ J \subseteq R,$$
$$E_R^-(J) \longleftrightarrow J \neq \varnothing \ \wedge \ J \cap R = \varnothing.$$

These are assertions of uniform positive and uniform negative evidence. For $J = S \times T$, positive evidence is precisely the guarded universal relation from the alternative-semantics section. The second uses the complement of R on the same candidate domain. It is *not* the classical negation of the first.

Theorem 33.1 (Four mutually exclusive candidate states)**.**
Exactly one of the following holds:

1. $J = \varnothing$ (no jointly admissible value);
2. $E_R^+(J)$ (uniform positive evidence);
3. $E_R^-(J)$ (uniform negative evidence);
4. $J \cap R \neq \varnothing$ and $J \setminus R \neq \varnothing$ (mixed evidence).

If $\varnothing \neq J' \subseteq J$, either assertion of uniform evidence that holds for J also holds for J'.

Proof.
The disjoint sets $J \cap R$ and $J \setminus R$ partition J. If both are empty, J is empty. If exactly one is nonempty, that part determines which assertion of uniform evidence holds. If both are nonempty, the state is mixed. This proves exhaustiveness and exclusivity. A subset of R remains a subset of R, and a subset of a set disjoint from R remains disjoint from R. Nonemptiness of J' retains the existence guard in both cases. □
For example, take $D = \{0, 1\}$ and the equality relation $R = \{(0, 0), (1, 1)\}$. Independent candidates for two terms, each with candidate set D, give $J = D \times D$ and mixed evidence. Repeated occurrences of a *single shared value* instead give $J = \{(0, 0), (1, 1)\}$ and uniform positive evidence. Thus the marginal candidate sets do not determine the truth of equality. With $J = \varnothing$, both evidence assertions are false; with mixed J, both are again false, for a different reason. Replacing negative evidence by classical negation would conflate these two failures with uniform negative evidence.

33.2 Truth sets with an explicit absence set

A related experiment keeps track of the assignments themselves. Let Ω be a nonempty set of admissible assignments. Interpret each partial term t by a partial map $t : \Omega \rightharpoonup D$. For a binary atomic formula $R(s,t)$ put

$$\begin{aligned} H_{R(s,t)} &= \operatorname{dom}(s) \cap \operatorname{dom}(t), \\ T_{R(s,t)} &= \{\omega \in H_{R(s,t)} : (s(\omega), t(\omega)) \in R\}, \\ F_{R(s,t)} &= H_{R(s,t)} \setminus T_{R(s,t)}, \\ A_{R(s,t)} &= \Omega \setminus H_{R(s,t)}. \end{aligned}$$

Here H records presence of all required values, T and F record classical truth and falsity where values are present, and A records absence. These are sets in the metatheory, not additional classes in the object language.

One possible extension to propositional formulas propagates absence through every connective. Define negation by exchanging T and F and retaining A. Define conjunction and disjunction by

$$\begin{aligned} T_{\varphi\wedge\psi} &= T_\varphi \cap T_\psi, \\ F_{\varphi\wedge\psi} &= (F_\varphi \cap H_\psi) \cup (H_\varphi \cap F_\psi), \\ T_{\varphi\vee\psi} &= (T_\varphi \cap H_\psi) \cup (H_\varphi \cap T_\psi), \\ F_{\varphi\vee\psi} &= F_\varphi \cap F_\psi, \\ A_{\varphi\wedge\psi} &= A_{\varphi\vee\psi} = A_\varphi \cup A_\psi, \end{aligned}$$

where $H_\theta = T_\theta \cup F_\theta$. This is a precise local version of the draft's strict presence convention. The metatheory here is classical; neither Ω nor the sets of present assignments need be finite.

Lemma 33.1 (Partition and classical restriction)**.**
For every formula of this propositional fragment, T_φ, F_φ, and A_φ partition Ω. On assignments where all its atomic subformulas are present, its interpretation is classical. In particular,

$$T_{\varphi\vee\neg\varphi} = H_\varphi, \qquad A_{\varphi\vee\neg\varphi} = A_\varphi.$$

Proof.
Atomic formulas have the partition property by definition. Negation preserves it. For a binary connective, outside $H_\varphi \cap H_\psi$ the assignment lies in the specified absence set. Inside that intersection each input has exactly one classical truth value, and the displayed clauses are the ordinary Boolean clauses. This proves the partition property and classical restriction by induction. For the last display, T_φ and F_φ exchange under negation, so the disjunction's truth set is their union and its absence set is unchanged. □

This separates two reasons for an informal label "uncertain". If a formula is present everywhere but true on only part of Ω, its excluded-middle formula is true everywhere. If it is absent somewhere, that formula remains absent there under the strict interpretation. Even among formulas present everywhere, correlations between truth sets matter: on $\Omega = \{\omega_0, \omega_1\}$ take $T_p = \{\omega_0\}$. Both p and $\neg p$ vary, yet $p \vee \neg p$ is always true whereas $p \vee p$ still varies. A summary recording only "present and varying" cannot determine disjunction.

Proposition 33.1 (The presence condition for classical tautologies)**.**
Let θ be a finite formula built from atoms by negation, conjunction, and disjunction, and let $\mathrm{At}(\theta)$ be the finite set of atoms occurring in it. Then

$$H_\theta = \bigcap_{p \in \mathrm{At}(\theta)} H_p.$$

If θ is a classical propositional tautology, its strict truth set is exactly this intersection:

$$T_\theta = H_\theta = \bigcap_{p \in \mathrm{At}(\theta)} H_p.$$

Consequently a classical tautology holds at every assignment in Ω under the strict interpretation exactly when every atom occurring in it is present everywhere.

Proof.
For an atom the first equation is its definition. Negation retains presence and does not change the occurring atoms. For either binary connective, the absence union and the partition lemma give $H_{\varphi \# \psi} = H_\varphi \cap H_\psi$ for $\# \in \{\wedge, \vee\}$. The atoms of the compound are the union of the two finite atom sets, so structural induction proves the first equation. At every point of this intersection all occurring atoms have classical truth values. The partition lemma identifies evaluation there with the classical truth tables, and a tautology is true for every such valuation. Thus $H_\theta \subseteq T_\theta$. The reverse inclusion is part of the definition of presence. Finally, an intersection of subsets of Ω equals Ω if and only if each of its factors equals Ω, giving the last assertion. □
For example, with implication expanded as $\neg p \vee q$, the classical weakening axiom $p \longrightarrow (q \longrightarrow p)$ has strict truth set $H_p \cap H_q$. It is absent when p is true and q is absent. The companion catalog's classical axiom A01 and its excluded-middle axiom use a different, total truth interpretation. They agree with these clauses on the common region where the relevant atoms are present. Thus their validity in the classical calculus and their failure at absent assignments here describe different local settings; the latter does not invalidate the former, and the former supplies no unguarded axiom for this experiment.

Note 33.1 (Why guarding a partial value depends on the logic)**.**
Under the strict convention just defined, an implication with a false antecedent and an absent consequent is absent, if implication abbreviates $\neg\varphi \vee \psi$. In PFOL, by contrast, primitive atomic predicates with undefined arguments are false and the logical connectives remain classical. There $f(x) \downarrow\!\rightarrow f(x) = f(x)$ is true even off the domain. Thus the draft's desire for this guarded formula to hold selects a different treatment of absence; it is not a consequence of strict presence propagation. Neither convention changes what happens at assignments where all values are present.

Proposition 33.2 (Deduction under strict absence propagation)**.**
For this propositional fragment and any set Γ of formulas, write $\Gamma \models_T \theta$ when every assignment on which all formulas in Γ are true belongs to T_θ. Put $G = \bigcap_{\gamma \in \Gamma} T_\gamma$, with $G = \Omega$ for the empty context. If implication abbreviates $\neg\varphi \vee \psi$, then

$$\Gamma \models_T (\varphi \longrightarrow \psi)$$

holds exactly when both of the following conditions hold:

$$G \subseteq H_\varphi \cap H_\psi, \qquad \Gamma \cup \{\varphi\} \models_T \psi.$$

Proof.
The strict truth clauses give

$$T_{\varphi \longrightarrow \psi} = (F_\varphi \cap H_\psi) \cup (H_\varphi \cap T_\psi).$$

In particular, this set is contained in $H_\varphi \cap H_\psi$. If G is contained in it, then an element of $G \cap T_\varphi$ cannot belong to F_φ, so it must belong to T_ψ. This is the stated consequence with the extra assumption. Conversely, suppose both conditions hold and $\omega \in G$. Because φ is present, it is either true or false at ω. In the false case, $\omega \in F_\varphi \cap H_\psi$; in the true case, the consequence condition gives $\omega \in H_\varphi \cap T_\psi$. Thus $\omega \in T_{\varphi \longrightarrow \psi}$. □

For a concrete failure of the unqualified deduction principle, take one assignment where φ is false and ψ is absent. Then $\{\varphi\} \models_T \psi$ holds vacuously, but $\models_T (\varphi \longrightarrow \psi)$ fails because the implication is absent. Modus ponens remains sound: whenever both φ and its implication to ψ are true, the displayed truth-set formula forces ψ to be true. Thus keeping an inference rule does not automatically justify its usual deduction-theorem transformation in this local logic.

Chapter

Analysis and categorical applications

34 Analysis

This section develops topological limits along bases and their relation to one-sided limits such as $f(x+0)$ and the familiar ε–δ formulations. Amann and Escher's *Analysis I–III* provides the intended background context.
A further question is when convergence along a base can be tested by sequences. Such a comparison needs explicit hypotheses on the spaces and the base. The sequential test below supplies these hypotheses and gives a counterexample to extending the result to arbitrary topological spaces.

34.1 Limits along a proper filter base

For the limit statements in this subsection, let A be a set and $\mathcal{B}$ a nonempty family of nonempty subsets of A, such that for $B_1, B_2 \in \mathcal{B}$ there is $B_3 \in \mathcal{B}$ with $B_3 \subseteq B_1 \cap B_2$. These are the local conditions for a *proper filter base*. They exclude the empty-family case admitted by the exploratory filter axioms in the topology section.
Let $f : A \to Y$ be a function into a topological space. We write $f \to y$ along $\mathcal{B}$ when

$$\forall U \in \mathcal{N}_Y(y),\ \exists B \in \mathcal{B},\ f[B] \subseteq U.$$

The graph of f supplies a value at every point of A. If a formula for f originally denotes only a partial function, its domain must first be restricted accordingly; a default value outside the domain does not automatically belong in the limit calculation.
For a sequence $f : \mathbb{N} \to Y$, the tail sets $\{n \in \mathbb{N} \mid n \geqslant N\}$ give ordinary sequential convergence. For a right-hand limit at a real number a, use the nonempty sets $A \cap (a, a+\delta)$, $\delta > 0$, assuming that a is approached from the right by points of A. This specifies both the side of approach and the domain on which evaluation is meaningful.

Lemma 34.1 (Uniqueness of a limit in a Hausdorff space)**.**
A function has at most one limit along a proper filter base when its target space is Hausdorff.

Proof.

If distinct points y, z were limits, choose disjoint neighbourhoods U of y and W of z. There would be base members B, C with $f[B] \subseteq U$ and $f[C] \subseteq W$. A nonempty base member $D \subseteq B \cap C$ would then satisfy $f[D] \subseteq U \cap W = \varnothing$, which is impossible. □

34.2 Locality, composition, and real limits

Lemma 34.2 (Changing a function away from the approach region)**.**
Let $f, g : A \to Y$, and suppose they agree on some B_0 in a proper filter base $\mathcal{B}$. Then f and g have exactly the same limits along $\mathcal{B}$.

Proof.
If $f \to y$ and U is a neighbourhood of y, choose $B \in \mathcal{B}$ with $f[B] \subseteq U$. A base refinement $C \subseteq B \cap B_0$ satisfies $g[C] = f[C] \subseteq U$, so $g \to y$. Exchange f, g for the reverse implication. □

Proposition 34.1 (Pairing and continuous composition)**.**
If $f : A \to Y$ and $g : A \to Z$ converge along the same proper filter base to y, z, then $a \mapsto (f(a), g(a))$ converges to (y, z) in the product topology. If $h : Y \to Z$ is continuous at y and $f \to y$, then $h \circ f \to h(y)$.

Proof.
A neighbourhood of (y, z) contains $U \times V$ for neighbourhoods U, V of y, z. Choose base members making f lie in U and g lie in V, then a common refinement. On it the pair lies in the product. For composition, continuity supplies a neighbourhood U of y whose image lies in any prescribed neighbourhood of $h(y)$; the limit of f supplies a base member mapped into U. □

Theorem 34.1 (Punctured metric limits)**.**
Let $A \subseteq X$, where (X, d) is a metric space, let $a \in X$, and assume every punctured ball about a meets A. The sets

$$B_\delta = \{x \in A : 0 < d(x, a) < \delta\}, \qquad \delta > 0,$$

form a proper filter base. For a map $f : A \to (Y, e)$ into another metric space and a proposed limit $y \in Y$, its limit along this base is y exactly when

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A, \quad 0 < d(x, a) < \delta \longrightarrow e(f(x), y) < \varepsilon.$$

The value at a, when defined, has no effect on this limit.

Proof.
Nonemptiness is the approach assumption, and $B_{\min(\delta,\eta)}$ is contained in the intersection of two base members. Metric balls form a neighbourhood base in Y, so the limit definition is exactly the displayed quantifier condition. Since a belongs to none of the base members, changing its value does not change any image used here. □
For a right-hand real limit, replace $0 < d(x, a) < \delta$ by $0 < x - a < \delta$ and require that every such interval meets A. The same proof gives the one-sided epsilon–delta criterion. Without the approach assumption, an empty base member makes the quantified condition vacuous for every proposed limit, which is why it cannot be used to infer uniqueness at an isolated point.

Theorem 34.2 (Squeezing along a proper base)**.**
Let $f, h, g : A \to \mathbb{R}$ satisfy $f \le h \le g$ on some base member. If both f and g tend to L along the base, then $h \to L$.

Proof.
For $\varepsilon > 0$, refine the ordering base member with base members on which $f > L-\varepsilon$ and $g < L+\varepsilon$, respectively. On the common refinement, $L-\varepsilon < f \le h \le g < L+\varepsilon$, which is the limit condition for h. □

34.3 When sequences test a limit

A sequence (B_n) of members of a filter base $\mathcal{B}$ is *cofinal* if every $B \in \mathcal{B}$ contains some B_n. If such a sequence exists, put $C_n = \bigcap_{i=1}^{n} B_i$. Finite induction using the refinement property shows that each C_n contains a member of $\mathcal{B}$, and hence is nonempty. The decreasing sequence (C_n) is a new proper filter base, equivalent to the original: every original member contains some C_n, and every C_n contains an original member. Consequently the two bases give exactly the same limits. The sets C_n need not themselves belong to the original base. This construction requires no successive choices of refinements. The following arguments use classical logic and countable choice for the explicitly indexed nonempty sets of possible sequence terms.

Theorem 34.3 (Sequential test for a countably based limit)**.**
Let (C_n) be decreasing and cofinal in a proper filter base on A. For $f : A \to Y$ and $y \in Y$, the following are equivalent:

1. f converges to y along the base;
2. for every sequence $a_n \in C_n$, the sequence $f(a_n)$ converges to y in Y.

No countability or separation assumption on Y is required.

Proof.
If U is a neighbourhood of y, convergence gives a base member B with $f[B] \subseteq U$. Cofinality gives $C_N \subseteq B$; decreasingness gives $a_n \in C_n \subseteq C_N$ for $n \ge N$. Thus $f(a_n)$ is eventually in U. Conversely, if the base limit fails, some neighbourhood U of y contains the image of no base member. Choose $a_n \in C_n$ with $f(a_n) \notin U$ for every n. This sequence contradicts the second condition. □

Corollary 34.1 (Sequential continuity at a first-countable point)**.**
If X has a countable neighbourhood base at x, then a function $f : X \to Y$ is continuous at x if and only if $x_n \to x$ always implies $f(x_n) \to f(x)$.

Proof.
Continuity gives the forward assertion directly by inverse images of neighbourhoods. For the converse, choose a decreasing neighbourhood base C_n at x. Every sequence $x_n \in C_n$ converges to x. The theorem therefore gives convergence of f along that base to $f(x)$, which is precisely continuity at x. □

Example 34.1 (Why the countability hypothesis matters)**.**
On an uncountable set X, give X the cocountable topology: the nonempty open sets are those with countable complement. A sequence converging to x is eventually equal to x. Indeed, the set of its values different from x is countable, so its complement is a neighbourhood of x that contains no other value of the sequence. Consequently the identity from this topology to the discrete topology on X preserves all convergent sequences. It is not continuous, because a singleton is discrete-open but not cocountable-open. Thus preservation of sequences alone does not characterize continuity on an arbitrary topological space.

34.4 An elementary order argument

Theorem 34.4 (Nonnegativity from positive perturbations)**.**
For a real number a,

$$(\forall \varepsilon > 0,\ a + \varepsilon \geqslant 0) \longrightarrow a \geqslant 0.$$

Proof.
Suppose $a < 0$ and choose $\varepsilon = -a/2$. Then $\varepsilon > 0$, but

$$a + \varepsilon = a/2 < 0,$$

contrary to the hypothesis. Hence $a \geqslant 0$. □

35 Topology

This chapter explores several descriptions of topology, inspired by *Topological Vector Spaces, Distributions and Kernels* (Trèves, 1967). The point–neighbourhood relation, families of basic sets, and families of open sets provide different starting points. The conventions are stated locally: in particular, the first use of "subbase" means an intersection-closed family, while the later "generator" need not be intersection-closed.

35.1 Filter

Definition 35.1 (filter)**.**
The set F is called a *filter* in the local convention below if it satisfies the following conditions. Its carrier is taken to be $\bigcup F$, rather than supplied separately.

$$\Big(\varnothing \notin F\Big), \tag{F$_1$}$$

$$\Big(\Big(A \in F\Big) \wedge \Big(B \in F\Big)\Big) \longrightarrow \Big(\Big(A \cap B\Big) \in F\Big), \tag{F$_2$}$$

$$\Big(F \ni A \subseteq B \subseteq \bigcup F\Big) \longrightarrow \Big(B \in F\Big). \tag{F$_3$}$$

These three conditions also admit $F = \varnothing$, because the last two are then vacuous. Thus this convention includes an empty family. When a nonempty filter is required, that hypothesis will be stated separately; for a nonempty F, upward closure implies $\bigcup F \in F$.

35.2 Filter Base

Definition 35.2 (filter base with its recovered carrier)**.**
A filter base is a nonempty set $\mathcal{B}$ of nonempty sets such that for $B_1, B_2 \in \mathcal{B}$ there is $B_3 \in \mathcal{B}$ with $B_3 \subseteq B_1 \cap B_2$. Its carrier is $X = \bigcup \mathcal{B}$. Define the generated family by

$$F(\mathcal{B}) = \{A \subseteq X : \exists B \in \mathcal{B},\ B \subseteq A\}.$$

Thus the base need not itself be upward closed or closed under intersections. The refinement condition supplies a member inside each intersection of two members.

Proposition 35.1 (Generation and recovery of a nonempty filter)**.**
The family $F(\mathcal{B})$ is a nonempty filter in the preceding local convention, has carrier X, and is the least such filter containing $\mathcal{B}$. Conversely, every nonempty filter F in that convention is a filter base and satisfies $F(F) = F$.

Proof.
Every base member belongs to $F(\mathcal{B})$, so the generated family is nonempty and its union is X: containment in X follows from its definition, and the reverse containment follows from $\mathcal{B} \subseteq F(\mathcal{B})$. No member is empty, since it contains a nonempty base member. If $A_1, A_2 \in F(\mathcal{B})$, choose $B_i \in \mathcal{B}$ with $B_i \subseteq A_i$ for $i = 1, 2$. A common refinement B_3 is contained in $A_1 \cap A_2$, so this intersection belongs to $F(\mathcal{B})$. Upward closure within X is immediate. These are exactly the three filter axioms.
If a filter G contains $\mathcal{B}$, then $X \subseteq \bigcup G$. For $B \subseteq A \subseteq X$ with $B \in \mathcal{B}$, upward closure of G therefore gives $A \in G$. Hence $F(\mathcal{B}) \subseteq G$, even if G has a larger carrier. Finally, a nonempty filter is a base by its intersection axiom, and its upward closure gives $F(F) = F$. The empty filter admitted earlier is deliberately excluded from this base representation. □

35.3 Topology via a point–neighbourhood relation

The relation $x\mathcal{R}S$ is intended to mean that S is a neighbourhood of x. In this set-sized development, take $\mathcal{R}$ to be a set relation so that its images and the families defined below are sets. The carrier E is recovered from the relation. Read the point axioms for $x \in E$ and the set axioms for $S \subseteq E$. The displayed axioms specify this proposed presentation; equivalence with other presentations requires a separate argument.

Definition 35.3 (relation between points and neighbourhoods)**.**
(parameter)

$$\mathcal{R} \subseteq \mathrm{V} \times \mathrm{V}$$

Definition 35.4 (relation between neighbourhoods and points)**.**

$$\mathfrak{R} \coloneqq \mathcal{R}^T$$

Definition 35.5 (neighbourhoods of a point)**.**

$$\mathrm{tni}(x) \coloneqq \mathcal{R}[\{x\}]$$

Definition 35.6 (interior of a set)**.**

$$\mathrm{int}(S) \coloneqq \mathfrak{R}[\{S\}]$$

Definition 35.7 (Candidate family of open sets)**.**

$$\mathcal{T} \coloneqq \mathrm{ran}(\mathrm{int})$$

Definition 35.8 (Family of neighbourhood filters)**.**

$$\mathcal{N} \coloneqq \mathrm{ran}(\mathrm{tni})$$

Definition 35.9 (set of all points)**.**

$$E \coloneqq \bigcup \mathcal{T}$$

Definition 35.10 (set of all neighbourhoods)**.**

$$N \coloneqq \bigcup \mathcal{N}$$

Axiom (N_0)**.**

$$(\mathrm{tni}(x) \text{ filter})$$

Axiom (N_1)**.**

$$x \in \bigcap \mathrm{tni}(x)$$

Axiom (N_2)**.**

$$\mathrm{int}(S) \in (N \cup \{\varnothing\})$$

Scope of the proposed neighbourhood axioms. Condition (N_2) says that a nonempty interior is a neighbourhood of *some* point. This is weaker than requiring it to be a neighbourhood of every point it contains. The difference affects the intended passage from the relation to an ordinary topological interior.
For example, let $E = \{a, b, c, d\}$, with four distinct points, and put

$$U_a = \{a, b, d\}, \quad U_b = \{b, c, d\}, \quad U_c = \{c, d\}, \quad U_d = \{d\}.$$

Declare $x\mathcal{R}S$ exactly when $U_x \subseteq S \subseteq E$. Each neighbourhood family is a nonempty principal filter, and every neighbourhood of x contains x, so (N_0) and (N_1) hold. Whenever $\mathrm{int}(S)$ is nonempty, $d \in S$, hence $d \in \mathrm{int}(S)$. Consequently $\mathrm{int}(S)$ is a neighbourhood of d, proving (N_2). Nevertheless,

$$\mathrm{int}(\{a, b, d\}) = \{a, d\}, \qquad \mathrm{int}(\{a, d\}) = \{d\}.$$

Thus this interior is not idempotent. The three axioms retain a coherent weaker setting, but alone do not identify int with the interior operator of an ordinary topology.
For that identification, a natural strengthening is the pointwise clause

$$x \in \mathrm{int}(S) \longrightarrow \mathrm{int}(S) \in \mathrm{tni}(x). \tag{N_2^{pt}}$$

Read this with nonempty neighbourhood filters on the common carrier E, and with every neighbourhood a subset of E. It says that each neighbourhood contains an open neighbourhood of the same point. The subbase construction below satisfies this stronger clause too: its basic witness B lies inside $\mathrm{int}(S)$ and contains the original point x. The weaker proposal is therefore preserved without conflating its scope with that of the ordinary topological setting.

Theorem 35.1 (Recovering a topology from the pointwise axioms)**.**
Suppose every $\mathrm{tni}(x)$, $x \in E$, is a nonempty filter on E, every member contains x, and (N_2^{pt}) holds. Put $I(S) = \{x \in E \mid S \in \mathrm{tni}(x)\}$ for $S \subseteq E$. Then

$$\mathcal{T} = \{U \subseteq E \mid I(U) = U\}$$

is a topology on E, and its neighbourhood relation is the given one.

Proof.
The filter axioms give $I(E) = E$, $I(\varnothing) = \varnothing$, and $I(S \cap T) = I(S) \cap I(T)$. They also give monotonicity: $S \subseteq T$ implies $I(S) \subseteq I(T)$. The point condition gives $I(S) \subseteq S$; (N_2^{pt}) gives $I(S) \subseteq I(I(S))$. Applying the first inclusion to $I(S)$ proves equality, so I is idempotent. Its range is therefore exactly its family of fixed points, agreeing with the earlier notation $\mathcal{T} = \mathrm{ran}(I)$. Its fixed points contain $\varnothing, E$ and are closed under binary intersections. If each U_j is fixed, monotonicity gives $U_j \subseteq I(\bigcup_j U_j)$, so their union is fixed too. Thus the fixed points form a topology. Finally, if $S \in \mathrm{tni}(x)$, then $x \in I(S) \subseteq S$ and $I(S)$ is open. Conversely, an open U with $x \in U \subseteq S$ gives $U \in \mathrm{tni}(x)$ and hence $S \in \mathrm{tni}(x)$ by upward closure. This proves equality of the neighbourhood relations. □

35.4 Topology via (sub)base

Definition 35.11 (subbase)**.**
In this subsection, a *subbase* is a set $\mathcal{B}$ closed under binary intersections. This is a local naming convention: the closure condition already provides the refinement property needed for a base on the carrier $\bigcup \mathcal{B}$. Empty members are allowed.

$$\Big(A \in \mathcal{B}\Big) \wedge \Big(B \in \mathcal{B}\Big) \longrightarrow \Big(A \cap B \in \mathcal{B}\Big),$$

Definition 35.12 (topology via subbase)**.**
For any p, any S, and a subbase $\mathcal{B}$, we define: [1]

$$(p\mathcal{R}_{\mathcal{B}}S) :\Longleftrightarrow \left(\exists B \in \mathcal{B},\ p \in B \subseteq S \subseteq \bigcup \mathcal{B}\right)$$

Theorem 35.2 (topology via subbase is a topology)**.**
The relation $\mathcal{R}_{\mathcal{B}}$ between points and neighbourhoods is topological.
Proof. Put $E = \bigcup \mathcal{B}$ and fix $x \in E$.

- (N_0) Every neighbourhood contains x, so the empty set is not a neighbourhood. If S and T are neighbourhoods, choose $B_1, B_2 \in \mathcal{B}$ with $x \in B_1 \subseteq S$ and $x \in B_2 \subseteq T$. Then $B_1 \cap B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2 \subseteq S \cap T$, so $S \cap T$ is a neighbourhood. Upward closure within E follows immediately from the definition. There is some $B \in \mathcal{B}$ containing x, so E itself is a neighbourhood and this neighbourhood family is nonempty.
- (N_1) Each member S of the neighbourhood family contains a basic set containing x, hence contains x. Therefore x belongs to the intersection of the family.
- (N_2) Let $I = \mathrm{int}_{\mathcal{B}}(S)$. If $I = \varnothing$, the axiom holds. Otherwise take $x \in I$ and choose $B \in \mathcal{B}$ with $x \in B \subseteq S \subseteq E$. For every $y \in B$, the same B witnesses $y\mathcal{R}_{\mathcal{B}}S$, so $B \subseteq I \subseteq E$. Thus I is a neighbourhood of x, and consequently $I \in N_{\mathcal{B}}$.

□

Definition 35.13 (Rectangles from neighbourhoods)**.**
The following family collects neighbourhood rectangles. Its local subfamilies at a point provide the product filter bases below. Calling it an open base additionally requires the rectangles to be open.

$$\mathcal{B}_{E\times F} \coloneqq \{\, A \times B \mid (A \in N_E) \wedge (B \in N_F) \,\}$$

To describe this construction at a specified point, use the notation

Notation 35.1 (neighbourhoods of a point)**.**

$$\mathrm{nghb} \coloneqq \mathrm{tni}$$

Definition 35.14 (product filter basis function)**.**

$$\mathrm{pfb}((a,b)) \coloneqq \{\, A \times B \mid (A \in \mathrm{nghb}_E(a)) \wedge (B \in \mathrm{nghb}_F(b)) \,\}$$

[1] Here we use $\bigcup \mathcal{B}$, because otherwise any generated topology would always have a class of all sets as a carrier. Later we will also see that elements of $\mathcal{B}$ are open.

35.5 Topologies and intersection-closed structures

Here a topology or sigma-algebra carries its own underlying set: the carrier is the union of its members. This explains why a carrier does not appear as a separate argument in the definitions.

Definition 35.15 (topology)**.**
The set $\mathcal{T}$ is called *topology* iff

$$\Big(S \subseteq \mathcal{T}\Big) \longrightarrow \Big(\Big(\bigcup S\Big) \in \mathcal{T}\Big), \tag{T$_1$}$$

$$\Big(\Big(A \in \mathcal{T}\Big) \wedge \Big(B \in \mathcal{T}\Big)\Big) \longrightarrow \Big(\Big(A \cap B\Big) \in \mathcal{T}\Big), \tag{T$_2$}$$

Definition 35.16 (sigma–algebra)**.**
The set σ is called *sigma–algebra* iff

$$\Big(\Big(S \subseteq \sigma\Big) \wedge \Big(|S| \leqslant \omega\Big)\Big) \longrightarrow \Big(\Big(\bigcup S\Big) \in \sigma\Big), \tag{S$_1$}$$

$$\Big(\Big(A \in \sigma\Big) \wedge \Big(B \in \sigma\Big)\Big) \longrightarrow \Big(\Big(A \setminus B\Big) \in \sigma\Big), \tag{S$_2$}$$

$$\Big(\Big(\bigcup \sigma\Big) \in \sigma\Big). \tag{S$_3$}$$

For all the intersection statements below, Q is a *nonempty* family of the indicated sets. This ensures that $\bigcap Q$ is a set. When the structures are subspaces, closed sets, convex sets, or balanced sets, they lie in one fixed ambient vector space or topological space. Empty intersections can instead be treated relative to a stated ambient set; the unrestricted class convention $\bigcap \varnothing = \mathrm{V}$ is not used for these statements.

Theorem 35.3 (intersection of topologies)**.**

$$\Big(Q \subseteq \{\, \mathcal{T} \mid \mathcal{T} \text{ topology} \,\}\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ topology}\Big)$$

Proof.
Let $\mathcal{I} = \bigcap Q$. If $S \subseteq \mathcal{I}$, then $S \subseteq \mathcal{T}$ for every $\mathcal{T} \in Q$, so $\bigcup S \in \mathcal{T}$ for every such $\mathcal{T}$. Thus $\bigcup S \in \mathcal{I}$. Binary intersections are inherited in the same way. Nonemptiness of Q bounds $\mathcal{I}$ by a set topology, so $\mathcal{I}$ is a set and satisfies both defining axioms. □

Theorem 35.4 (intersection of sigma–algebras)**.**

$$\Big(\Big(Q \subseteq \{\, \sigma \mid \sigma \text{ sigma-algebra} \,\}\Big) \wedge \Big(\Big(\bigcup \bigcap Q\Big) \in \bigcap Q\Big)\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ sigma-algebra}\Big)$$

Proof.
Put $\mathcal{I} = \bigcap Q$. Countable unions and relative differences of members of $\mathcal{I}$ belong to every sigma-algebra in Q, hence to $\mathcal{I}$. The additional displayed hypothesis is precisely the remaining carrier axiom $\bigcup \mathcal{I} \in \mathcal{I}$. □

Theorem 35.5 (intersection of linear subspaces)**.**

$$\Big(Q \subseteq \{\, L \mid L \text{ linear subspace} \,\}\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ linear subspace}\Big)$$

Proof.
Every member of Q contains the zero vector, so their intersection does as well. If $v_1, v_2 \in \bigcap Q$ and $\alpha_1, \alpha_2 \in \mathbb{K}$, then $\alpha_1 v_1 + \alpha_2 v_2$ lies in every member of Q, hence in their intersection. Thus the intersection is a linear subspace. □

Theorem 35.6 (intersection of closed sets)**.**

$$\Big(Q \subseteq \{\, C \mid C \text{ closed} \,\}\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ closed}\Big)$$

Proof.
The complement of $\bigcap Q$ in the fixed ambient space is the union of the complements of the members of Q. Each complement is open, so their union is open and the intersection is closed. □

Theorem 35.7 (intersection of convex sets)**.**

$$\Big(Q \subseteq \{\, C \mid C \text{ convex} \,\}\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ convex}\Big)$$

Proof.
If $u, v \in \bigcap Q$ and $0 \leqslant t \leqslant 1$, then $tu + (1-t)v$ belongs to every convex member of Q. It therefore belongs to the intersection, which is convex. □

Theorem 35.8 (intersection of balanced sets)**.**

$$\Big(Q \subseteq \{\, B \mid B \text{ balanced} \,\}\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ balanced}\Big)$$

Proof.
If $v \in \bigcap Q$ and a scalar a satisfies $|a| \leqslant 1$, then av belongs to every balanced member of Q. It therefore belongs to the intersection, which is balanced. □

Theorem 35.9 (intersection of Dynkin systems/λ–systems)**.**

$$\Big(\Big(Q \subseteq \{\, \lambda \mid \lambda \text{ Dynkin} \,\}\Big) \wedge \Big(\Big(\bigcup \bigcap Q\Big) \in \bigcap Q\Big)\Big) \longrightarrow \Big(\Big(\bigcap Q\Big) \text{ Dynkin}\Big)$$

Proof.
Use the formulation of a Dynkin system by containment of its carrier, closure under differences of nested members, and closure under countable disjoint unions. The two closure properties pass to the intersection. The displayed additional hypothesis supplies its carrier, completing these three conditions. □

Example 35.1 (Why a common measurable carrier matters)**.**
If every sigma-algebra in Q has the same carrier X, then $X \in \sigma$ for every $\sigma \in Q$. Hence $X \in \mathcal{I} = \bigcap Q \subseteq \mathcal{P}X$, which gives $\bigcup \mathcal{I} = X \in \mathcal{I}$. Thus the additional hypothesis in the sigma-algebra intersection theorem is automatic on a common carrier. The same argument applies to Dynkin systems.
It is not automatic for varying carriers. For this example use the classical real line and countable choice, so a countable union of countable sets is countable. Let $Y = [0, \infty)$ and put

$$\Sigma = \{A \subseteq \mathbb{R} : A \text{ is countable or } \mathbb{R} \setminus A \text{ is countable}\},$$
$$\Lambda = \mathcal{P}Y.$$

The family Σ is a sigma-algebra on $\mathbb{R}$: complements interchange the two alternatives, and a countable union is countable if all members are countable; if one member has countable complement, the union has countable complement as well. The family Λ is a sigma-algebra on Y. Since $\mathbb{R} \setminus Y$ is uncountable, no subset of Y has countable complement in $\mathbb{R}$. Consequently

$$\Sigma \cap \Lambda = \{A \subseteq Y : A \text{ is countable}\}.$$

Every singleton of Y belongs to this intersection, so its union is Y. But Y is uncountable and is not a member. The intersection therefore fails the carrier axiom, although it is closed under countable unions and relative differences. Since sigma-algebras are Dynkin systems, the same pair also shows why the carrier hypothesis in that intersection theorem cannot be dropped. This is a difference between fixed-carrier and recovered-carrier settings, not a failure of the usual intersection theorem for sigma-algebras on one fixed space.

35.6 Topological properties

This subsection proposes a local classification by "hereditary" and "weakly hereditary" properties. The intended inheritance relation is still to be defined: it must specify which objects are compared and which operation or passage between them preserves a property. The preceding intersection theorems provide one possible setting for that question.
The proposed classification treats being a topology as hereditary, being a closed set as weakly hereditary, and being an open set as failing even weak heredity. Compactness, separability, and connectedness are also listed as candidate hereditary properties. These examples record the intended direction of the investigation; their precise statements and proofs depend on the missing definition of inheritance.
For terminological context, conventional topology distinguishes invariance under homeomorphism from inheritance by subspaces, and may further restrict attention to closed subspaces. The local terms here need an explicit definition before they can be compared with those conventions. In particular, the proposed examples should be read within that eventual local definition.

A precise preservation result for continuous images. The following supplies one definite preservation statement for three of the named properties, without identifying it with the proposed hierarchy. A space is *separable* if it has a countable dense subset, where density means meeting every nonempty open set. A space is *connected* if it cannot be partitioned into two disjoint nonempty open sets. Compactness has its open-cover meaning, stated below.

Proposition 35.2 (Continuous-image preservation)**.**
Let $f : X \to Y$ be continuous, and give $f[X]$ its subspace topology. If X is compact, separable, or connected, respectively, then $f[X]$ has that same property. No separation axiom or Choice assumption is needed.

Proof.
Compactness is the continuous-image assertion proved in Theorem 35.14. If $D \subseteq X$ is countable and dense, then $f[D]$ is countable. Every nonempty relatively open $U \subseteq f[X]$ has nonempty open inverse image in X. That inverse image meets D, so U meets $f[D]$. Thus $f[D]$ is dense in $f[X]$. Countability of the image uses no selection of preimages: an enumeration of a nonempty countable D, composed with f, enumerates its image; the empty case is immediate.
If $f[X]$ were partitioned into two nonempty disjoint relatively open sets, their inverse images would be nonempty disjoint open sets partitioning X. This contradicts connectedness of X. Empty spaces cause no exception to any of these statements. □
The operation matters. Connectedness does not pass even to closed subspaces: $\{0, 1\}$ is a disconnected closed subspace of the connected interval $[0, 1]$. For completeness, a separation of an interval would give points $a < b$ in opposite open parts. The supremum c of the first part within $[a, b]$ can belong to neither part: if it belongs to the first, openness and $c < b$ give a larger point in that part; if it belongs to the second, openness and $c > a$ exclude first-part points immediately below c, contradicting its supremum property. This proves the asserted connectedness. The two endpoints themselves have disjoint open singleton neighbourhoods in their subspace. Thus continuous-image preservation does not determine a classification by subspaces or intersections. Defining the proposed hierarchy still requires specifying the permitted operation and whether its carrier is fixed or recovered.

35.7 Topology in older notation

This subsection records the same neighbourhood–interior relationship using $\mathcal{N}_{\mathcal{T}}(x)$ for the neighbourhoods of x. The symbol Segm_{σ} denotes arbitrary unions of the displayed interval family; here the subscript is a union-closure notation, not a sigma-algebra. The relation and operator interpretations must be kept distinct in the inverse notation below.

Definition 35.17 (Basic intervals of the extended real line)**.**

$$\mathrm{Segm} = \{\, [-\infty;\beta) \mid^{\beta} \beta \in \mathbb{R} \,\} \cup \{\, (\alpha;+\infty] \mid^{\alpha} \alpha \in \mathbb{R} \,\} \cup \{\, (\alpha;\beta) \mid^{\alpha,\beta} \alpha \in \mathbb{R}, \beta \in \mathbb{R} \,\}$$

Definition 35.18 (Topology of the extended real line)**.**

$$\mathcal{T}_{[-\infty;+\infty]} = \mathrm{Segm}_{\sigma}$$

Definition 35.19 (neighbourhoods)**.**

$$\mathcal{N}_{\mathcal{T}}(x) = \left\{ S \subseteq \bigcup \mathcal{T} \;\middle|\; \exists U \in \mathcal{T},\ x \in U \subseteq S \right\}$$

Definition 35.20 (interior)**.**
Interior of a set V in a topology $\mathcal{T}$.

$$\mathrm{int}_{\mathcal{T}}(V) = \bigcup \{W \in \mathcal{T} : W \subseteq V\}$$

Theorem 35.10 (Converse point–set relations)**.**
Put $E = \bigcup \mathcal{T}$ and restrict the set arguments to $\mathcal{P}E$. Define the incidence relations

$$I = \{(x, V) \in E \times \mathcal{P}E : x \in \mathrm{int}(V)\},$$
$$N = \{(V, x) \in \mathcal{P}E \times E : V \in \mathcal{N}(x)\}.$$

Then $I = N^{-1}$ and $N = I^{-1}$. This is the precise relational meaning of the shorthand that interior and neighbourhood incidence are converse; it does not assert inverse functions.

Proof.
For $x \in E$ and $V \subseteq E$, membership in the union defining $\mathrm{int}(V)$ means that some open U satisfies $x \in U \subseteq V$. This is exactly $V \in \mathcal{N}(x)$. Thus $(x, V) \in I$ if and only if $(V, x) \in N$, proving both identities. The restriction is essential: for $E = \{0\}$ with topology $\{\varnothing, E\}$, the set $V = \{0, 1\}$ has $0 \in \mathrm{int}(V)$ but is not an allowed neighbourhood, since $V \nsubseteq E$. □

Theorem 35.11 (Common neighbourhoods and interior)**.**
Here $A, V \subseteq E = \bigcup \mathcal{T}$, and $\mathcal{N}_{\mathcal{T}}\langle A \rangle$ means $\{S \subseteq E \mid \forall x \in A,\ S \in \mathcal{N}_{\mathcal{T}}(x)\}$. For $A = \varnothing$ this is $\mathcal{P}E$, so the statement also covers that case without using an unrestricted empty intersection.

$$V \in \mathcal{N}_{\mathcal{T}}\langle A \rangle \longleftrightarrow A \subseteq \mathrm{int}(V)$$

Proof.
The left side says that V is a neighbourhood of every $x \in A$. By the preceding equivalence, this says precisely that every such x lies in $\mathrm{int}(V)$. Both conditions are vacuous for $A = \varnothing$. □

Note 35.1 (Order-theoretic interpretation)**.**
The shorthand cl ⊣ id ⊣ int records two adjunctions, with different inclusion maps understood. Closure is left adjoint to the inclusion of closed sets into all subsets; interior is right adjoint to the inclusion of open sets. Concretely, for closed C and open U,

$$\mathrm{cl}(S) \subseteq C \longleftrightarrow S \subseteq C, \qquad U \subseteq S \longleftrightarrow U \subseteq \mathrm{int}(S).$$

Theorem 35.12 (Neighbourhood–interior correspondence)**.**
Let $E = \bigcup \mathcal{T}$, $x \in E$, and $V \subseteq E$. The carrier restriction on V is needed because neighbourhoods, as defined here, are subsets of E.

$$V \in \mathcal{N}_{\mathcal{T}}(x) \longleftrightarrow x \in \mathrm{int}_{\mathcal{T}}(V)$$

Proof.
($\Rightarrow$)

$$V \in \mathcal{N}_{\mathcal{T}}(x) \implies \exists U \in \mathcal{T},\ x \in U \subseteq V \qquad \text{(nghb–elim, union–elim)}$$

$$U \in \mathcal{T}\ \ \&\ \ U \subseteq V \implies U \in \{\, W \in \mathcal{T} \mid W \subseteq V \,\} \qquad \text{(specification–intro)}$$

$$x \in U\ \ \&\ \ U \in \{\, W \in \mathcal{T} \mid W \subseteq V \,\} \implies x \in \bigcup \{\, W \in \mathcal{T} \mid W \subseteq V \,\} \qquad \text{(union–intro)}$$

$$x \in \bigcup \{\, W \in \mathcal{T} \mid W \subseteq V \,\} \implies x \in \mathrm{int}_{\mathcal{T}}(V) \qquad \text{(interior–intro)}$$

($\Leftarrow$)

$$x \in \mathrm{int}_{\mathcal{T}}(V) \implies \exists W \in \{\, U \in \mathcal{T} \mid U \subseteq V \,\},\ x \in W \qquad \text{(interior–elim, union–elim)}$$

$$W \in \{\, U \in \mathcal{T} \mid U \subseteq V \,\} \implies W \in \mathcal{T}\ \ \&\ \ W \subseteq V \qquad \text{(specification–elim)}$$

$$x \in W\ \ \&\ \ W \in \mathcal{T}\ \ \&\ \ W \subseteq V \implies \exists U \in \mathcal{T},\ x \in U \subseteq V \qquad \text{(and–intro, exists–intro)}$$

$$(\exists U \in \mathcal{T},\ x \in U \subseteq V) \implies V \in \mathcal{N}_{\mathcal{T}}(x) \qquad \text{(exists–elim, nghb–intro)}$$

□

35.8 Generating families and bases

The next two definitions retain the draft's weaker covering convention: every open set must be expressible using the family, but the family is not yet required to lie in $\mathcal{T}$. To obtain a base or subbase consisting of open sets, add $\mathfrak{B} \subseteq \mathcal{T}$ or $\mathfrak{G} \subseteq \mathcal{T}$, respectively. For finite intersections in the generator definition, the empty intersection means the fixed carrier $E = \bigcup \mathcal{T}$.

Definition 35.21 (base)**.**
A set $\mathfrak{B}$ is a base of $\mathcal{T}$ iff
$\forall U \in \mathcal{T},\ \exists W \in \mathcal{P}\mathfrak{B},\ U = \bigcup W$
or
$\mathcal{T} \subseteq \{\, \bigcup W \mid W \in \mathcal{P}\mathfrak{B} \,\}$
or
$\mathcal{T}$ is a subset of unions of subsets of $\mathfrak{B}$.

Definition 35.22 (generator)**.**
A set $\mathfrak{G}$ is a generator of $\mathcal{T}$ iff

$$\left(\forall U \in \mathcal{T},\ \exists W \in \mathcal{P}\left\{ \bigcap Q \;\middle|\; (Q \subseteq \mathfrak{G}) \wedge (Q \text{ finite}) \right\},\ U = \bigcup W\right)$$

or

$$\mathcal{T} \subseteq \left\{ \bigcup W \;\middle|\; W \in \mathcal{P}\left\{ \bigcap Q \;\middle|\; (Q \subseteq \mathfrak{G}) \wedge (Q \text{ finite}) \right\} \right\}$$

or $\mathcal{T}$ is a subset of unions of subsets of intersections of finite subsets of $\mathfrak{G}$.

With the open-set condition just stated, a generator is also called a subbase. This is the second use of that word in this chapter; unlike the earlier intersection-closed family, it need not be closed under intersections.

Definition 35.23 (The refinement preorder on families)**.**
For any two families $\mathcal{C}$ and $\mathcal{F}$ declare that $\mathcal{C} \leq \mathcal{F}$ if and only if for every $C \in \mathcal{C}$ there exists some $F \in \mathcal{F}$ such that $F \subseteq C$, in which case it is said that $\mathcal{C}$ is coarser than $\mathcal{F}$ and that $\mathcal{F}$ is finer than (or subordinate to) $\mathcal{C}$.

$$(\mathcal{C} \leq \mathcal{F}) \iff (\forall C \in \mathcal{C},\ \exists F \in \mathcal{F},\ F \subseteq C)$$

Notation 35.2 (generator/base/topology at x)**.**

$$\mathfrak{B}(x) \coloneqq \{\, U \in \mathfrak{B} \mid x \in U \,\}$$

Definition 35.24 (Pointwise refinement of bases)**.**
Consistently with the preceding direction of $\leq$, base $\mathfrak{B}_1$ is coarser than base $\mathfrak{B}_2$ iff

$$\forall x,\ (\mathfrak{B}_1(x) \leq \mathfrak{B}_2(x))$$

35.9 Alexandrov's correspondence between topology and order

Topologies closed under arbitrary intersections correspond to preorders. The result connects the point–neighbourhood presentation with the order theory developed earlier; see May and Pishevar, n.d., Section 1.6. We use the convention that open sets are *upper* sets. Reversing the preorder gives the lower-set convention used in that reference.
Fix a set X; no axiom of Choice or topological separation condition is used. All intersections below are relative to X, including the empty intersection X; no unrestricted class intersection

is intended. A topology $\mathcal{T}$ on X is *Alexandrov* if every intersection of its open sets is open. In that case each point has a smallest open neighbourhood

$$U_x = \bigcap \{U \in \mathcal{T} : x \in U\}.$$

The family being intersected contains X, and $x \in U_x$.

Theorem 35.13 (Alexandrov correspondence)**.**
There is a bijection between Alexandrov topologies on X and preorders on X. Its two directions are

$$x \preceq_{\mathcal{T}} y \quad \Longleftrightarrow \quad \forall U \in \mathcal{T},\ x \in U \longrightarrow y \in U,$$

and

$$\mathcal{T}_{\preceq} = \{U \subseteq X : \forall x \in U\ \forall y \in X,\ x \preceq y \longrightarrow y \in U\}.$$

In the corresponding topology, $U_x = \{y \in X : x \preceq y\}$.

Proof.
For any topology, the displayed relation is reflexive. If $x \preceq_{\mathcal{T}} y$ and $y \preceq_{\mathcal{T}} z$, every open set containing x contains y and then z, proving transitivity. Every open set is upper for this relation. If the topology is Alexandrov, then $y \in U_x$ is exactly $x \preceq_{\mathcal{T}} y$. Consequently an upper subset A is the union $\bigcup_{x \in A} U_x$, so it is open. Thus the preorder recovers precisely the original open sets. Conversely, upper subsets for a preorder include $\varnothing$ and X and are closed under all unions and all intersections. They therefore form an Alexandrov topology. The principal upper set $\uparrow x = \{y : x \preceq y\}$ is upper by transitivity and contains x by reflexivity. Every upper set containing x contains $\uparrow x$, so this is its smallest open neighbourhood. It follows that the relation recovered from the topology is the original preorder. Both constructions are inverse, proving the bijection, also when X is empty. □

Proposition 35.3 (Continuous maps and separation)**.**
A map between Alexandrov spaces is continuous if and only if it is monotone for their corresponding preorders. An Alexandrov space is T_0 if and only if its preorder is antisymmetric. Here T_0 means that distinct points can be distinguished by an open set containing exactly one of them.

Proof.
Let $f : X \to Y$ be continuous and $x \preceq_X x'$. For each open $V \subseteq Y$ containing $f(x)$, its inverse image is an open set containing x, hence contains x'. Thus $f(x) \preceq_Y f(x')$. Conversely, if f is monotone and V is upper in Y, then $f^{-1}[V]$ is upper in X, and so is open. This is continuity.
The two relations $x \preceq y$ and $y \preceq x$ hold together exactly when x and y belong to the same open sets. The T_0 condition therefore says precisely that this can happen only when $x = y$, which is antisymmetry. □

Corollary 35.1 (Finite spaces and the Hausdorff boundary)**.**
Every finite topological space is Alexandrov. An Alexandrov Hausdorff space is discrete; every discrete space is Alexandrov and Hausdorff.

Proof.
On a finite carrier there are only finitely many distinct open sets, so any intersection reduces to a finite one. In a Hausdorff space, for distinct x, y there is an open neighbourhood of x missing

y. Its associated preorder is therefore equality. The correspondence then makes every subset open. The converse follows directly from the definitions, including for the empty space. $\square$

For example, the preorder $0 \preceq 1$ on $\{0,1\}$ gives $\mathcal{T} = \{\varnothing, \{1\}, \{0,1\}\}$, with $U_0 = \{0,1\}$ and $U_1 = \{1\}$. This space is T_0 but not Hausdorff. The indiscrete topology instead makes every pair of points comparable in both directions. These examples show why a preorder, rather than automatically a partial order, is the correct starting point. The usual topology on $\mathbb{R}$ is not Alexandrov: the intersection of the open intervals $(-1/n, 1/n)$ over positive integers n is $\{0\}$, which is not open. Thus arbitrary intersections of open sets remain a substantive additional hypothesis.

35.10 The product topology

Let $(X_i, \mathcal{T}_{X_i})_{i \in I}$ be a set-indexed family of topological spaces. The product topology on

$$X \coloneqq \prod_{i \in I} X_i$$

is the coarsest topology for which the projections

$$\pi_i : X \to X_i$$

are continuous for all $i \in I$. ("initial topology")

$$\mathcal{A} \overset{\text{def.}}{\coloneqq} \bigcup \{\, \{\, \pi_i^{-1}[U] \mid U \in \mathcal{T}_{X_i} \,\} \mid i \in I \,\}$$

In fact $\mathcal{A}$ is a subbasis.

$$\mathcal{T}_{\mathcal{A}} \overset{\text{def.}}{\coloneqq} \bigcap \{\, \mathcal{T} \mid (\mathcal{T} \text{ topology on } X) \wedge (\mathcal{A} \subseteq \mathcal{T}) \,\}$$

Proposition 35.4 (Finite-coordinate basis and the product property)**.**
Let $\mathcal{B}$ consist of the sets

$$\bigcap_{i \in J} \pi_i^{-1}[U_i], \qquad J \subseteq I \text{ finite}, \quad U_i \in \mathcal{T}_{X_i}.$$

The intersection for $J = \varnothing$ is X. Then the product topology is exactly the family of unions of members of $\mathcal{B}$. For any topological space Y and any function $f : Y \to X$, the following are equivalent:

1. f is continuous;
2. every coordinate map $\pi_i \circ f : Y \to X_i$ is continuous.

These assertions require no Choice assumption and include empty products and products with empty carrier.

Proof.
The family $\mathcal{B}$ contains X and is closed under binary intersections. Indeed, combine the two finite sets of restricted coordinates, and at a repeated coordinate replace the two open sets by their intersection. Its unions therefore contain $\varnothing, X$ and are closed under arbitrary unions. The intersection of two such unions is the union of all pairwise intersections of their members, so these unions form a topology $\mathcal{S}$.
Every subbasic inverse image belongs to $\mathcal{B}$, hence to $\mathcal{S}$. Conversely, any topology containing $\mathcal{A}$ contains its finite intersections and all their unions. It therefore contains $\mathcal{S}$. Thus $\mathcal{S}$ equals the displayed intersection defining $\mathcal{T}_{\mathcal{A}}$; that intersection is over a nonempty set of topologies, since the discrete topology $\mathcal{P}X$ is one of them. This proves the basis assertion and the coarsest-topology characterization simultaneously. In particular each π_i is continuous.
If f is continuous, inverse images under $\pi_i \circ f$ are obtained by taking two continuous inverse images, so each coordinate map is continuous. Conversely, suppose every coordinate map is continuous. The inverse image under f of a member of $\mathcal{B}$ is the finite intersection of the corresponding open coordinate inverse images. It is open, including when the intersection is empty

and its inverse image is Y. Since inverse images commute with unions, f is continuous on every product-open set.
When $I = \varnothing$, X is the singleton containing the empty function, $\mathcal{B} = \{X\}$, and every map into it is continuous. When $X = \varnothing$, the only topology on it is $\{\varnothing\}$; a function $Y \to X$ exists only for $Y = \varnothing$, and both conditions then hold. The construction never assumes that a product of nonempty carriers has a point. □
For a specified point $x \in X$, the basis assertion says more locally: $N \subseteq X$ is a neighbourhood of x exactly when it contains some finite-coordinate basic set containing x. In a binary product $E \times F$, this is equivalently the condition

$$\exists A \in \mathrm{nghb}_E(a), \quad \exists B \in \mathrm{nghb}_F(b), \quad A \times B \subseteq N, \qquad x = (a, b).$$

Indeed an open basic rectangle gives such A, B. Conversely, choose open neighbourhoods U, V with $a \in U \subseteq A$ and $b \in V \subseteq B$; then $U \times V$ is an open neighbourhood inside N. Thus the earlier function $\mathrm{pfb}((a, b))$ really is a filter base generating precisely the product neighbourhood filter. It is nonempty, all its members contain (a, b), and intersecting two members gives another neighbourhood rectangle. This connects the pointwise filter construction with the open-set definition.

35.11 Cardinality–based topological definitions

Let $(X, \mathcal{T})$ be a topological space and $S \subseteq X$. In this subsection, $\mathcal{N}(x)$ denotes the neighbourhood family of x. The formulas define the author's cardinality-based terminology directly. In particular, "limits" requires at least two points of S in each neighbourhood; no separation assumption is implicit. The names "isolates" and "border" refer to the differences displayed below.

$$\mathrm{closure}(S) \stackrel{\text{def.}}{\coloneqq} \{\, x \in X \mid \forall N \in \mathcal{N}(x), \mathrm{card}(S \cap N) \geqslant 1 \,\}$$

$$\mathrm{exterior}(S) \stackrel{\text{def.}}{\coloneqq} X \setminus \mathrm{closure}(S)$$

$$\mathrm{interior}(S) \stackrel{\text{def.}}{\coloneqq} \mathrm{exterior}(X \setminus S)$$

$$\mathrm{limits}(S) \stackrel{\text{def.}}{\coloneqq} \{\, x \in X \mid \forall N \in \mathcal{N}(x), \mathrm{card}(S \cap N) \geqslant 2 \,\}$$

$$\mathrm{isolates}(S) \stackrel{\text{def.}}{\coloneqq} \mathrm{closure}(S) \setminus \mathrm{limits}(S)$$

$$\mathrm{border}(S) \stackrel{\text{def.}}{\coloneqq} \mathrm{closure}(S) \setminus \mathrm{interior}(S)$$

Proposition 35.5 (Comparison with the ordinary derived set)**.**
Define the ordinary derived set by

$$S' = \{x \in X : \text{every neighbourhood of } x \text{ meets } S \setminus \{x\}\}.$$

In every topological space, $\mathrm{limits}(S) \subseteq S'$. If X is T_1, meaning that every singleton is closed, then $\mathrm{limits}(S) = S'$. In that case $\mathrm{isolates}(S)$ consists exactly of the isolated points of the subspace S.

Proof.
If a neighbourhood contains two distinct points of S, at least one differs from x. This proves the first inclusion without separation assumptions. Conversely, suppose X is T_1, $x \in S'$, and N is a neighbourhood of x. If $x \in S$, then N contains x and a different point of S, as required. If $x \notin S$, choose $y \in N \cap S$. The set $N \setminus \{y\}$ is still a neighbourhood of x: it contains the intersection of an open neighbourhood inside N with the open set $X \setminus \{y\}$. Since $x \in S'$, this smaller neighbourhood meets S, giving a second point different from y. Thus every N contains at least two points of S.
If $x \in \text{closure}(S) \setminus S$, every neighbourhood meets $S \setminus \{x\} = S$, so $x \in S'$. Hence $\text{closure}(S) \setminus S' = S \setminus S'$. For $x \in S$, failure to belong to S' means that some neighbourhood meets S only in x, equivalently $\{x\}$ is open in the subspace S. This proves the assertion about isolated points. $\square$
The separation assumption cannot simply be omitted. In the indiscrete space $X = \{a, b\}$ with $a \neq b$, take $S = \{a\}$. The only neighbourhood of either point is X, so $S' = \{b\}$ whereas $\text{limits}(S) = \varnothing$. Here the author's $\text{isolates}(S) = X$ even contains a point outside S. The two conventions therefore describe different notions in this setting.

35.12 Compactness and changes of topology

In this subsection all spaces are set sized. A space is *compact* if every cover by open sets has a finite subcover. A subset is compact when it is compact with its subspace topology. These definitions do not impose a separation axiom.

Theorem 35.14 (Transport of compactness)**.**
A closed subset of a compact space is compact, and the continuous image of a compact space is compact. Every compact subset of a Hausdorff space is closed. Consequently a continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proof.
Let F be closed in a compact space X. Write each member of a relative open cover of F as $F \cap U$ for an ambient open set U. One need not choose representatives simultaneously: take all open U whose intersection with F is a member of the given cover. These sets together with $X \setminus F$ cover X. A finite subcover therefore yields a finite subcover of F from the original cover.
For a continuous $f : X \to Y$, the inverse images of an open cover of $f[X]$ cover X by open sets. A finite subcover pulls forward to a finite subcover of $f[X]$.
Let $K \subseteq Y$ be compact, with Y Hausdorff, and fix $y \in Y \setminus K$. Consider all open sets U meeting K for which there is an open neighbourhood V of y disjoint from U. Hausdorff separation makes these U cover K. Finitely many $U_1, \ldots, U_m$ cover K; choose their corresponding V_i. The finite intersection of the V_i is a neighbourhood of y disjoint from K. For $K = \varnothing$ the conclusion is immediate. Thus $Y \setminus K$ is open, proving closedness. Only finitely many choices of witnesses were used.
Finally, a continuous bijection $f : X \to Y$ as stated sends every closed subset of X to a compact, hence closed, subset of Y. This is exactly continuity of its inverse, since the inverse image under f^{-1} of a closed subset $F \subseteq X$ is $f[F]$. □
The Hausdorff hypothesis is substantive. On a two-point set, the identity from the discrete topology to the indiscrete topology is a continuous bijection between compact spaces. Its inverse is not continuous. Thus compactness alone does not recover the original topology from a continuous bijection.

35.13 Compactness in metric spaces

This subsection uses ordinary set-sized metric spaces and classical countable choice. A space is *compact* when every open cover has a finite subcover; it is *totally bounded* when, for every $\varepsilon > 0$, finitely many open balls of radius ε cover it. These are distinct definitions.

Theorem 35.15 (The metric compactness criterion)**.**
For a metric space X, the following conditions are equivalent:

1. X is compact;
2. every sequence in X has a convergent subsequence with limit in X;
3. X is complete and totally bounded.

Proof.
The empty space satisfies all three conditions, so assume X nonempty. Compactness gives total boundedness by applying the cover property to all balls of a fixed radius. If (x_n) is Cauchy,

the closed sets $K_n = \overline{\{x_j : j \geq n\}}$ are nonempty and nested. They have nonempty intersection: otherwise their open complements would cover X, and a finite subcover would make one K_n empty. Choose x in the intersection. Given $\varepsilon > 0$, a sufficiently late tail lies in the closed ball of radius $\varepsilon/2$ about one of its terms; its closure contains x. The triangle inequality then places all sufficiently late x_n within ε of x. Thus compactness implies completeness and total boundedness.
Assume completeness and total boundedness. By countable choice, fix in advance, for every $k \geq 1$, a finite ordered list of radius 2^{-k} balls covering X. Start with all sequence indices. At stage k, retain those indices in the first ball of the kth list containing infinitely many of the retained terms; such a ball exists because the list is finite. Choose n_k as the least retained index greater than n_{k-1}, starting with $n_0 = 0$. Once the lists are fixed these are uniquely specified recursive operations, not dependent choices. For $j, l \geq k$, the chosen terms lie in a common ball of radius 2^{-k}, so their distance is at most 2^{1-k}. The subsequence is Cauchy and hence converges in X.
Finally assume the subsequence condition. If total boundedness failed at some radius $r > 0$, finite induction would produce, for every positive integer n, an ordered n-tuple with pairwise distances at least r: at each finite stage some point lies outside the finitely many radius r balls about the points already obtained. Countable choice selects one such tuple for each n. Concatenate these finite tuples to obtain a sequence enumerating their union S. No finite collection of radius $r/2$ balls covers S, since each such ball contains at most one point of any selected tuple, and tuples have arbitrarily large lengths. Starting with the first enumerated point, recursively choose the first point in the enumeration outside the radius $r/2$ balls about the previously chosen points. This rule is uniquely specified and always possible by the preceding observation. It yields a sequence with pairwise distances at least $r/2$, and hence with no convergent subsequence. This contradiction proves total boundedness using only countable choice. For an open cover $\mathcal{U}$, there is a number $\delta > 0$ such that every ball $B(x, \delta)$ is contained in some member of $\mathcal{U}$. If not, choose x_n whose ball of radius $1/n$ is contained in no member. A convergent subsequence has limit x in some open $U \in \mathcal{U}$. Choose $r > 0$ with $B(x, 2r) \subseteq U$. Eventually both $d(x_{n_k}, x) < r$ and $1/n_k < r$, contradicting the choice of x_{n_k}. Now finitely many radius δ balls cover X by total boundedness, and containing cover members give a finite subcover. Hence X is compact. □

Corollary 35.2 (Relative norm compactness in a Banach space)**.**
A subset of a Banach space has compact norm closure if and only if it is totally bounded in the norm metric.

Proof.
A compact closure is totally bounded, and a subset inherits that property (replace a covering centre outside the subset by a point of an intersecting smaller ball). Conversely, the closure of a totally bounded set is totally bounded: a finite cover by radius $\varepsilon/3$ balls covers its closure by radius ε balls. The closure is complete, being closed in a complete space. Apply the metric criterion. □
For example, the closed unit ball of the infinite-dimensional Hilbert space ℓ^2 is complete and bounded, but not compact: its coordinate vectors have pairwise distance $\sqrt{2}$, so no finite collection of balls of radius less than $\sqrt{2}/2$ covers them. Total boundedness, not boundedness alone, is the needed condition. The equivalence with sequential compactness proved here is a metric theorem; it must not be applied to an arbitrary weak topology without a separate argument.

35.14 Contractions on complete metric spaces

The next result uses completeness directly and does not assume compactness or total boundedness. A contraction here has one fixed constant $q < 1$ controlling all pairs of points.

Theorem 35.16 (Banach's contraction principle and error bounds)**.**
Let (X, d) be a nonempty complete metric space and let $T : X \to X$ satisfy

$$d(Tx, Ty) \le q\, d(x, y) \qquad (x, y \in X), \qquad 0 \le q < 1.$$

There is a unique $p \in X$ with $Tp = p$. For every initial point $x_0 \in X$, the iteration $x_{n+1} = Tx_n$ converges to p. For every $n \ge 1$ it satisfies the a priori and a posteriori bounds

$$d(x_n, p) \le \frac{q^n}{1-q} d(x_1, x_0), \qquad d(x_n, p) \le \frac{q}{1-q} d(x_n, x_{n-1}).$$

More generally, every $x \in X$ satisfies the residual estimate

$$d(x, p) \le \frac{d(x, Tx)}{1-q}.$$

In particular this last estimate also applies to the initial point.

Proof.
If $q = 0$, all values of T coincide. Write $p = Tx_0$ for any $x_0 \in X$. Then $Tp = p$, and $x_n = p$ for every $n \ge 1$. The displayed iteration bounds are both zero, as required. Any fixed point equals this same value of T, so uniqueness also holds.
Now suppose $0 < q < 1$ and fix x_0. Natural-number recursion uniquely defines the iteration; after fixing x_0 no arbitrary selections are made. Put $D = d(x_1, x_0)$. Induction gives

$$d(x_{k+1}, x_k) \le q^k D \qquad (k \ge 0).$$

For integers $m > n \ge 0$, the triangle inequality and the finite geometric sum give

$$d(x_m, x_n) \le \sum_{k=n}^{m-1} q^k D = \frac{q^n(1 - q^{m-n})}{1-q} D \le \frac{q^n}{1-q} D.$$

Since $q^n \to 0$, the sequence is Cauchy; this includes $D = 0$, when all iterates are already equal. Completeness gives $x_n \to p$ for some $p \in X$. The estimate

$$d(Tp, p) \le d(Tp, Tx_n) + d(x_{n+1}, p) \le q\, d(p, x_n) + d(x_{n+1}, p)$$

tends to zero, proving $Tp = p$. If z is another fixed point, then $d(p, z) = d(Tp, Tz) \le qd(p, z)$, hence $d(p, z) = 0$ and $p = z$. This uniqueness also shows that all initial points lead to the same limit. Letting $m \to \infty$ in the preceding geometric bound proves the a priori estimate.
For either $q = 0$ or $0 < q < 1$ and any $x \in X$,

$$d(x, p) \le d(x, Tx) + d(Tx, Tp) \le d(x, Tx) + q\, d(x, p).$$

Rearranging proves the residual estimate. For $n \ge 1$, contraction gives

$$d(x_n, Tx_n) \le q\, d(x_{n-1}, x_n).$$

Applying the residual estimate to x_n now gives the a posteriori bound. Every denominator is positive; the case $q = 0$ uses no expression 0^0 in the stated iteration bounds. □

Corollary 35.3 (Stability of the fixed point)**.**
Under the theorem's assumptions, suppose another self-map $S : X \to X$ has a fixed point z and $d(Tz, Sz) \le \eta$ for some $\eta \ge 0$. Then

$$d(p, z) \le \frac{\eta}{1 - q}.$$

In particular the hypothesis holds if $d(Tx, Sx) \le \eta$ for every $x \in X$ and S has a fixed point.

Proof.
Since $Sz = z$, the residual at z is $d(z, Tz) = d(Sz, Tz) \le \eta$. Apply the residual estimate. □

Example 35.2 (Completeness and strict contraction are essential)**.**
On the incomplete space $(0, 1)$ with its usual metric, $T(x) = x/2$ is a contraction with $q = 1/2$ and maps the space into itself. Its only possible fixed point is zero, which is outside the space; its iterates converge to that missing endpoint in $\mathbb{R}$. On the complete space $\mathbb{R}$, the translation $T(x) = x + 1$ satisfies $d(Tx, Ty) = d(x, y)$ but has no fixed point. Thus the estimate with $q = 1$ is insufficient. Nonemptiness matters too: the unique self-map of the empty complete metric space satisfies every displayed contraction inequality vacuously but has no fixed point.

35.15 Proper ultrafilters and compact products

This subsection concerns set-sized topological spaces and uses classical set theory with Choice. Its filters are *proper filters on a fixed carrier*; this qualification distinguishes them from any earlier convention allowing empty filters or varying carriers. These results will supply the compactness arguments for weak topologies, where the metric subsequence criterion alone is insufficient.

Definition 35.25 (Proper filters and ultrafilters)**.**
A proper filter $\mathcal{F}$ on a set X is a family of subsets of X containing X, excluding $\varnothing$, closed under finite intersections, and upward closed among subsets of X. An ultrafilter is a proper filter maximal under inclusion among proper filters on the same carrier. A family $\mathcal{A} \subseteq \mathcal{P}X$ has the finite intersection property when every finite intersection of its members is nonempty; the empty intersection is X.

Theorem 35.17 (Ultrafilter extension and the dichotomy)**.**
Every family with the finite intersection property is contained in an ultrafilter on X. A proper filter $\mathcal{U}$ is an ultrafilter exactly when, for every $A \subseteq X$, exactly one of A and $X \setminus A$ belongs to $\mathcal{U}$.

Proof.
The supersets of finite intersections from $\mathcal{A}$ form a proper filter $\mathcal{F}_0$. Order its proper-filter extensions by inclusion. The union of a nonempty chain is a proper filter: any finitely many of its members lie together in one chain member, and none of those filters contains the empty set. For the empty chain, use $\mathcal{F}_0$ as an upper bound. Zorn's lemma gives a maximal extension $\mathcal{U}$.
If $A \notin \mathcal{U}$, adjoining A cannot generate a proper filter, by maximality. Thus some $F \in \mathcal{U}$ has $F \cap A = \varnothing$; otherwise the finite intersections from $\mathcal{U} \cup \{A\}$ would generate a proper extension. Upward closure gives $X \setminus A \in \mathcal{U}$. Both alternatives cannot occur, since their intersection is empty. Conversely, if the dichotomy holds and a proper filter extending $\mathcal{U}$ contained a new set A, it would also contain $X \setminus A \in \mathcal{U}$, a contradiction. This proves the characterization. □

Definition 35.26 (Ultrafilter convergence and direct image)**.**
An ultrafilter $\mathcal{U}$ on a topological space X converges to x if every neighbourhood of x belongs to $\mathcal{U}$. For any map $f : X \to Y$, define

$$f_*\mathcal{U} = \{B \subseteq Y : f^{-1}[B] \in \mathcal{U}\}.$$

Lemma 35.1 (Transport and uniqueness of ultrafilter limits)**.**
The family $f_*\mathcal{U}$ is an ultrafilter, without any continuity assumption on f. If f is continuous and $\mathcal{U}$ converges to x, then $f_*\mathcal{U}$ converges to $f(x)$. In a Hausdorff space an ultrafilter has at most one limit.

Proof.
Inverse images preserve the carrier, the empty set, intersections, inclusions, and relative complements. The filter axioms and the dichotomy therefore pass to $f_*\mathcal{U}$. For continuity, the inverse image of a neighbourhood of $f(x)$ is a neighbourhood of x, hence belongs to $\mathcal{U}$. Finally, disjoint neighbourhoods of two distinct Hausdorff points cannot both belong to a proper filter. □

Theorem 35.18 (Compactness characterized by ultrafilters)**.**
A topological space is compact if and only if every ultrafilter on its carrier converges to some point of the space.

Proof.
Suppose X is compact and $\mathcal{U}$ is an ultrafilter. The closed sets $\overline{F}$, $F \in \mathcal{U}$, have the finite intersection property: the corresponding finite intersection of the F is nonempty and contained in their closures. Compactness implies that their total intersection is nonempty, since otherwise the open complements would have a finite subcover. Choose x in that intersection. If a neighbourhood N of x did not belong to $\mathcal{U}$, the dichotomy would put $X \setminus N$ in $\mathcal{U}$. But an open neighbourhood O with $x \in O \subseteq N$ shows that $x \notin \overline{X \setminus N}$, a contradiction. Thus $\mathcal{U}$ converges to x.
Conversely, if an open cover has no finite subcover, its complements have the finite intersection property. Extend them to an ultrafilter. A limit x would lie in some open cover member O, putting both O and $X \setminus O$ in that ultrafilter. This is impossible. Hence the cover has a finite subcover. The empty space is covered by this argument: it is compact and carries no proper ultrafilter. □

Theorem 35.19 (Compactness of set-indexed products)**.**
Let $(X_i)_{i \in I}$ be a set-indexed family of compact spaces. Their product, with the product topology, is compact. Separation axioms on the factors are not required.

Proof.
If the product X is empty, it is compact. Otherwise let $\mathcal{U}$ be an ultrafilter on X. For each coordinate projection π_i, the ultrafilter $(\pi_i)_*\mathcal{U}$ has a limit in X_i by compactness. Choice selects such a limit x_i for each i; put $x = (x_i) \in X$. Every basic neighbourhood of x restricts only finitely many coordinates. Its coordinate neighbourhoods belong to the respective image ultrafilters, so their inverse images belong to $\mathcal{U}$. Their finite intersection therefore belongs to $\mathcal{U}$, and upward closure handles every neighbourhood of x. Thus $\mathcal{U}$ converges to x. The preceding theorem proves compactness. For $I = \varnothing$, the product is the singleton consisting of the empty function, and the same finite-coordinate argument has no conditions to impose. □
In particular, a product of closed bounded real intervals is compact. Each interval is complete as a closed subset of the complete real line and totally bounded by finite grids, so the earlier

metric criterion proves its compactness. Product compactness here is genuinely topological; no sequential compactness claim for an arbitrary uncountable product has been substituted for it. The proof uses Choice explicitly, unlike the deterministic iteration in the contraction theorem.

36 Linear algebra

These preliminary definitions describe matrices and the combination of maps before imposing algebraic operations on their values.

Definition 36.1 (Matrix over a set)**.**
A matrix A over a set S is a function

$$A : \{1, \ldots, m\} \times \{1, \ldots, n\} \longrightarrow S$$

for specified nonnegative integer dimensions m, n. Its entry in row i and column j is $A(i, j)$. If zero dimensions are allowed, the corresponding index set is empty. Addition and multiplication of matrices require further structure on S.

Definition 36.2 (Combining matching graph entries)**.**
For functions f_1 and f_2, define

$$\operatorname{combine}(f_1, f_2) \coloneqq \{\, (((m, k), (m, v)), (m, (k, v))) \mid (m, k) \in f_1 \wedge (m, v) \in f_2 \,\}.$$

This relation combines two graph entries with the same input m into a single entry with paired output (k, v). When $f_1 : M \to K$ and $f_2 : M \to L$, it records the graph-level construction of the map $m \mapsto (f_1(m), f_2(m))$ into $K \times L$. This is the pairing of maps with a common domain in the category of sets and functions.

36.1 Finite-dimensional normed spaces

The following results use set-sized real or complex vector spaces in the classical analytic setting. Completeness of $\mathbb{R}$ and $\mathbb{C}$ is assumed. The metric compactness criterion from the topology chapter supplies the only compactness implication used below. Its classical countable-choice convention is retained here; in particular, the proof of coordinate-sphere compactness uses that criterion.

Lemma 36.1 (Compact coordinate spheres)**.**
For $n \geq 1$, the set

$$S_n = \{a \in \mathbb{K}^n : \max_j |a_j| = 1\}, \qquad \mathbb{K} \in \{\mathbb{R}, \mathbb{C}\},$$

is compact in the coordinate maximum metric.

Proof.
A Cauchy sequence of vectors is Cauchy in each coordinate; scalar completeness gives a coordinatewise limit, which is also a limit in the maximum metric because there are only finitely many coordinates. The maximum norm is continuous, so S_n is closed and complete. Every bounded coordinate box is totally bounded: partition a containing real interval into finitely many intervals of length less than a prescribed ε, and take the finite product grid. In the complex case do this for real and imaginary parts, with mesh small enough to bound the complex modulus. A subset inherits total boundedness, by keeping only balls that meet the subset and recentering them at such points with twice the radius. Thus S_n is totally bounded. The metric compactness criterion gives compactness. □

Theorem 36.1 (Coordinate estimates and equivalence of norms)**.**
Let $e_1, \dots, e_n$ be a basis of a nonzero finite-dimensional normed space E over $\mathbb{K}$. There are constants $c, C > 0$ such that

$$c \max_j |a_j| \le \Big\| \sum_{j=1}^{n} a_j e_j \Big\| \le C \max_j |a_j| \qquad (a \in \mathbb{K}^n).$$

Consequently any two norms on a finite-dimensional vector space are equivalent and give the same topology.

Proof.
For the upper bound take $C = \sum_j \|e_j\| > 0$ and use the triangle inequality. The map $a \mapsto \| \sum_j a_j e_j \|$ is continuous in the coordinate maximum metric, since the reverse triangle inequality bounds its change by $C \max_j |a_j - b_j|$. On S_n it is strictly positive, by linear independence. Compactness supplies a positive lower bound: each point has a neighbourhood on which the function exceeds half its value there; finitely many such neighbourhoods cover S_n, and the minimum of their finitely many positive bounds is positive. Call it c. For $a \neq 0$, apply this bound to $a / \max_j |a_j|$ and rescale. The zero vector satisfies both inequalities directly.
Apply these estimates separately to two norms, using the same basis; each norm bounds and is bounded by positive multiples of the coordinate maximum norm, hence of the other norm. The zero-dimensional space has only its zero vector, so the conclusion there is immediate. □

Corollary 36.1 (Completeness and closedness in finite dimensions)**.**
Every finite-dimensional normed space is complete. Every finite-dimensional linear subspace of an arbitrary normed space is closed.

Proof.
For a Cauchy sequence in a finite-dimensional space, the lower coordinate estimate makes each coordinate sequence Cauchy. Scalar completeness gives limits for all coordinates, and the upper estimate proves convergence to their linear combination. This also covers the zero-dimensional space. Now let M be a finite-dimensional subspace of a normed space E, and take $x \notin M$. Adjoin x to a basis of M and apply the lower coordinate estimate in the finite-dimensional span of $M \cup \{x\}$. The coefficient of x in $x - m$ is one for every $m \in M$, so $\|x - m\| \ge c > 0$ for a fixed c. Thus the ball of radius c about x misses M. Every point outside M has such a neighbourhood, which proves that M is closed without a successive-choice argument. □
The weak topology is the coarsest topology making every continuous linear functional continuous. Basic neighbourhoods impose finitely many scalar inequalities; the functional-analysis chapter develops this definition further.

Corollary 36.2 (Finite-dimensional compactness and weak topology)**.**
In a finite-dimensional normed space, every closed bounded set is compact, and the weak topology agrees with the norm topology.

Proof.
The coordinate estimates carry bounded sets into bounded coordinate boxes and carry finite coordinate nets back to finite norm nets. Hence a bounded set is totally bounded. A closed set is complete by the preceding corollary, so closed bounded sets are compact by the metric criterion.

Each coordinate functional $\ell_j(\sum_k a_k e_k) = a_j$ is continuous, since $|\ell_j(x)| \le \|x\|/c$. For every $\varepsilon > 0$ the weakly open set

$$\{x : |\ell_j(x)| < \varepsilon/C \text{ for } 1 \le j \le n\}$$

is contained in the norm ball of radius ε by the upper estimate. Translations give the same conclusion at any centre, so every norm-open set is weakly open. Conversely every continuous linear functional is norm-continuous by definition, so the weak topology is coarser than the norm topology. They are equal. In the zero-dimensional case there is only one topology. □

Corollary 36.3 (Linear maps and finite-dimensional range)**.**
Every linear map from a finite-dimensional normed space E into a normed space F is bounded. More generally, if a linear map $T : E \to F$ is bounded and its range is finite-dimensional, then the image of the closed unit ball has compact closure in F. Neither space needs to be complete in this latter assertion.

Proof.
If $E \neq \{0\}$, choose a basis $e_1, \dots, e_n$ and the lower coordinate constant $c > 0$. Linearity and the triangle inequality give

$$\left\| T\left(\sum_j a_j e_j\right)\right\| \le \left(\sum_j \|Te_j\|\right) \max_j |a_j| \le \frac{\sum_j \|Te_j\|}{c} \left\|\sum_j a_j e_j\right\|.$$

This is boundedness; a map on the zero space is zero and is bounded too. For the second assertion, the range $M = T[E]$ is a closed finite-dimensional subspace of F. Boundedness of T makes the unit-ball image bounded in M. Its closure in F stays in M and remains bounded; it is closed in M, hence compact by finite-dimensional compactness. □
The second assertion explains why bounded finite-rank operators are compact, where an operator is called compact when its unit-ball image has compact closure. Finite rank controls the size of the range; boundedness separately controls the images of bounded sets.

36.2 The unit ball detects finite dimension

Lemma 36.2 (Riesz's separation lemma)**.**
Let M be a proper closed linear subspace of a normed space E, and let $0 < \theta < 1$. There is a unit vector x with

$$\inf_{m \in M} \|x - m\| > \theta.$$

Proof.
Choose $z \notin M$. Closedness gives $d = \inf_{m \in M} \|z - m\| > 0$. By the definition of infimum, choose $m_0 \in M$ with $\|z - m_0\| < d/\theta$, and put $x = (z - m_0)/\|z - m_0\|$. For any $m \in M$,

$$\|x - m\| = \frac{\|z - (m_0 + \|z - m_0\| m)\|}{\|z - m_0\|} \ge \frac{d}{\|z - m_0\|} > \theta.$$

The same lower bound holds for the infimum, proving the claim. □

Theorem 36.2 (Compact unit balls and dimension)**.**
In the classical countable-choice setting specified above, the closed unit ball of a normed space is compact if and only if the space is finite-dimensional. No completeness assumption on the space is needed.

Proof.
The finite-dimensional implication is the preceding compactness corollary. Conversely, suppose the closed unit ball is compact. Cover it by the open balls of radius $1/2$ centered at all its points. A finite subcover has centers $x_1, \dots, x_n$. Their span M is finite-dimensional and therefore closed. If M were proper, Riesz's lemma with $\theta = 3/4$ would supply a unit vector x whose distance from M exceeds $3/4$. But one of the covering balls contains x, giving $\|x - x_j\| < 1/2$ for a center $x_j \in M$, a contradiction. Thus $E = M$ is finite-dimensional. This reverse implication uses only a finite subcover and a single application of Riesz's lemma; it requires no successive choices. The stated countable-choice convention is used only for the metric compactness criterion in the forward implication. □

This theorem concerns the norm topology. Infinite-dimensional closed unit balls may be compact in a weaker topology; the distinction is central to the functional-analysis chapter. Nor does closedness of a subspace alone imply finite dimension: the finite-dimensional hypothesis above is a sufficient condition for closedness, not a characterization.

37 Measure theory

The functional-analytic applications require a specified measure space, not merely a collection called a sigma-algebra. The following conventions connect the earlier closure axioms with spaces of integrable functions. The Lebesgue integral is constructed below from nonnegative simple functions. Its convergence theorems then support the completeness, duality, and compactness arguments here and in the following chapter.

37.1 Carrier and measure

Fix a set Ω and a sigma-algebra $\mathcal{A} \subseteq \mathcal{P}\Omega$ with $\Omega \in \mathcal{A}$. Thus the carrier is explicit; it also equals $\bigcup \mathcal{A}$, in agreement with the earlier convention. A positive measure is a function

$$\mu : \mathcal{A} \longrightarrow [0, +\infty]$$

with $\mu(\varnothing) = 0$ and

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

for every sequence of pairwise disjoint members of $\mathcal{A}$. The measure is finite when $\mu(\Omega) < \infty$. Finiteness is an additional hypothesis, not a consequence of the carrier being a set.
A statement holds *almost everywhere* if it fails only on a subset of a measurable set of measure zero. When the measure is not complete, that subset need not itself belong to $\mathcal{A}$. Measurability requirements must therefore still be stated separately.

37.2 Generating classes and uniqueness of finite measures

All families in this subsection are families of subsets of the same fixed set Ω. A *π-system* is a family closed under binary intersections; we permit the empty family. A *Dynkin system*, or *λ-system*, is a family $\mathcal{D}$ that contains Ω, contains $\Omega \setminus A$ whenever it contains A, and contains the union of every countable pairwise disjoint sequence of its members. In particular $\varnothing \in \mathcal{D}$. For nested $A \subseteq B$ in $\mathcal{D}$, it also contains

$$B \setminus A = \Omega \setminus \big(A \cup (\Omega \setminus B)\big),$$

since the two sets inside the union are disjoint. This recovers the nested-difference formulation used in the topology chapter; conversely, that formulation gives complements by taking $B = \Omega$. Write $\sigma(\mathcal{P})$ for the smallest sigma-algebra on Ω containing a family $\mathcal{P}$, and $\lambda(\mathcal{P})$ for the smallest Dynkin system on Ω containing it. These are intersections of all the corresponding families containing $\mathcal{P}$. Such collections are nonempty, since $\mathcal{P}\Omega$ belongs to them. The earlier intersection results apply, with the common carrier Ω present in every family.

Theorem 37.1 (Dynkin's π–λ theorem)**.**
If $\mathcal{P}$ is a π-system of subsets of Ω, then $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$. Consequently, every Dynkin system containing $\mathcal{P}$ contains $\sigma(\mathcal{P})$.

Proof.
Put $\mathcal{P}_0 = \mathcal{P} \cup \{\Omega\}$. This is still a π-system, even when $\mathcal{P}$ is empty, and adjoining Ω changes neither of the generated families. Let $\mathcal{D} = \lambda(\mathcal{P}_0)$. For any fixed $A \in \mathcal{D}$ consider

$$\mathcal{D}_A = \{B \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

This is a Dynkin system on Ω. It contains Ω because $A \cap \Omega = A$. If $B \in \mathcal{D}_A$, then $\Omega \setminus B \in \mathcal{D}$ and

$$A \cap (\Omega \setminus B) = A \setminus (A \cap B) \in \mathcal{D}$$

by the nested-difference property. Thus it is closed under complements. If $B_n \in \mathcal{D}_A$ are pairwise disjoint, their union is in $\mathcal{D}$, and the sets $A \cap B_n$ are pairwise disjoint members of $\mathcal{D}$. Their union is $A \cap \bigcup_n B_n$, which belongs to $\mathcal{D}$. This proves the required disjoint-union closure.
First take $A \in \mathcal{P}_0$. Closure of $\mathcal{P}_0$ under intersections gives $\mathcal{P}_0 \subseteq \mathcal{D}_A$. Minimality of $\mathcal{D}$ then gives $\mathcal{D} \subseteq \mathcal{D}_A$. We have proved that $A \cap B \in \mathcal{D}$ whenever $A \in \mathcal{P}_0$ and $B \in \mathcal{D}$.
Next fix an arbitrary $B \in \mathcal{D}$. The preceding conclusion and symmetry of intersection give $\mathcal{P}_0 \subseteq \mathcal{D}_B$. Since $\mathcal{D}_B$ is a Dynkin system, minimality again gives $\mathcal{D} \subseteq \mathcal{D}_B$. Thus $\mathcal{D}$ is closed under intersections of any two of its members.
Complements and intersections now give finite unions and differences. For any sequence $A_n \in \mathcal{D}$, the sets

$$C_n = A_n \setminus \bigcup_{j<n} A_j$$

belong to $\mathcal{D}$, are pairwise disjoint, and have union $\bigcup_n A_n$. Disjoint-union closure therefore gives arbitrary countable unions. Hence $\mathcal{D}$ is a sigma-algebra containing $\mathcal{P}$, so $\sigma(\mathcal{P}) \subseteq \mathcal{D}$. The reverse inclusion holds because every sigma-algebra is a Dynkin system. Equality and the final assertion follow. □

Theorem 37.2 (Uniqueness from a generating π-system)**.**
Let μ, ν be finite positive measures on $(\Omega, \mathcal{A})$ with $\mu(\Omega) = \nu(\Omega)$. If $\mathcal{P} \subseteq \mathcal{A}$ is a π-system with $\sigma(\mathcal{P}) = \mathcal{A}$ and $\mu(P) = \nu(P)$ for every $P \in \mathcal{P}$, then $\mu = \nu$. If $\Omega \in \mathcal{P}$, equality of the total masses is already part of the agreement assumption.

Proof.
The agreement class

$$\mathcal{D} = \{A \in \mathcal{A} : \mu(A) = \nu(A)\}$$

contains Ω by the total-mass assumption. For $A \in \mathcal{D}$, finiteness permits subtraction and gives

$$\mu(\Omega \setminus A) = \mu(\Omega) - \mu(A) = \nu(\Omega) - \nu(A) = \nu(\Omega \setminus A).$$

For pairwise disjoint $A_n \in \mathcal{D}$, countable additivity gives

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(A_n) = \sum_n \nu(A_n) = \nu\left(\bigcup_n A_n\right).$$

Thus $\mathcal{D}$ is a Dynkin system containing $\mathcal{P}$. The π–λ theorem implies $\mathcal{A} = \sigma(\mathcal{P}) \subseteq \mathcal{D}$, proving equality on every measurable set. If $\mathcal{P}$ is empty, its generated sigma-algebra is $\{\varnothing, \Omega\}$ and the same argument applies; when Ω is empty this family has just one member. □

Example 37.1 (An arbitrary generating family is insufficient)**.**
Take $\Omega = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$, $B = \{1, 3\}$. The family $\mathcal{G} = \{\Omega, A, B\}$ generates $\mathcal{P}\Omega$ as a sigma-algebra: its singleton atoms are $A \cap B$, $A \setminus B$, $B \setminus A$, and $\Omega \setminus (A \cup B)$. But $\mathcal{G}$ is not a π-system, since $A \cap B = \{1\} \notin \mathcal{G}$. Define probability measures on all subsets by

$$\mu(C) = \tfrac{1}{2}|C \cap \{1, 4\}|, \qquad \nu(C) = \tfrac{1}{2}|C \cap \{2, 3\}|.$$

The bars denote finite cardinalities. These finite sums of point masses are countably additive: a point belongs to at most one member of a pairwise disjoint sequence. Both measures give Ω mass one and give A and B mass $1/2$, but $\mu(\{1\}) = 1/2$ and $\nu(\{1\}) = 0$. Thus agreement on generators alone does not replace agreement on a generating intersection-closed family.

The uniqueness theorem lets probability laws be compared on a smaller class of testing sets, provided that class is a generating π-system. The proof used finite total masses when taking complements; no infinite-measure uniqueness assertion is being inferred from it.

37.3 Measurable scalar functions

The Borel sigma-algebra of a topological space is the sigma-algebra generated by its open sets. A function from $(\Omega, \mathcal{A})$ to a Borel space is measurable when the inverse image of every Borel set belongs to $\mathcal{A}$. It suffices to check a generating family: the sets in the target whose inverse images belong to $\mathcal{A}$ form a sigma-algebra, since inverse images preserve complements and countable unions.

For real-valued functions, and for functions with values in the extended real line or its subspace $[0, \infty]$, measurability is equivalent to

$$\{\omega : f(\omega) > q\} \in \mathcal{A} \qquad (q \in \mathbb{Q}).$$

Here the extended line has its order topology. Rational rays generate its Borel sigma-algebra: complements and countable unions give both open rays and open intervals, including neighbourhoods of the infinite endpoints; these intervals have a countable base. For complex-valued functions, measurability is equivalent to measurability of the real and imaginary parts, since rational open rectangles form a countable base for $\mathbb{R}^2$.

These descriptions justify the closure operations used below. For a sequence of extended-real measurable functions,

$$\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}, \qquad \{\inf_n f_n < a\} = \bigcup_n \{f_n < a\}.$$

Thus countable suprema and infima are measurable, as are lower and upper limits by their iterated infimum and supremum formulas. In particular, an everywhere existing real pointwise limit is measurable; the same holds for complex limits by taking the two coordinates. The pair of two finite scalar measurable functions is measurable: inverse images of basic rectangles are intersections of the two coordinate inverse images, and every open set is a countable union of basic rectangles. Composition with a continuous scalar operation therefore preserves measurability. This proves measurability of sums, products, absolute values, real maxima and minima, and truncations of finite scalar functions, without any boundedness assumption.

For nonnegative extended-real functions, sums and products are obtained as increasing limits of the corresponding operations on finite truncations; for products this uses $0 \cdot \infty = 0$. They too are measurable. No operation $\infty - \infty$ is defined by these statements. Finally, replacing a measurable function by zero on a measurable set preserves measurability, by splitting each inverse image over that set and its complement. In particular this applies to measurable null exceptional sets; arbitrary modifications on a nonmeasurable subset of a null set need not preserve measurability for an incomplete measure.

37.4 Constructing the integral

The construction in this subsection works for every positive measure space; neither finite total measure nor completeness of the measure is assumed. All functions mentioned as integrands are measurable.

Lemma 37.1 (Continuity from below)**.**
If $A_n \subseteq A_{n+1}$ are measurable, then $\mu(\bigcup_n A_n) = \lim_n \mu(A_n)$, with infinity allowed.

Proof.
Write the union as the disjoint union of A_1 and $A_{n+1} \setminus A_n$ for $n \geq 1$. Countable additivity identifies its measure with the sum of their measures; the nth partial sum is exactly $\mu(A_n)$. No subtraction of infinite measures is involved. □

Lemma 37.2 (Continuity from above)**.**
If measurable sets satisfy $A_{n+1} \subseteq A_n$ and $\mu(A_1) < \infty$, then

$$\mu\Big(\bigcap_{n\geq 1} A_n\Big) = \lim_{n\to\infty} \mu(A_n).$$

Proof.
The sets $B_n = A_1 \setminus A_n$ increase to $A_1 \setminus \bigcap_n A_n$. Finite additivity and the finite measure of A_1 give $\mu(B_n) = \mu(A_1) - \mu(A_n)$; the same identity holds for their union and the complementary intersection. Apply continuity from below and subtract the resulting finite quantities from $\mu(A_1)$. This gives the stated limit without an expression of the form $\infty - \infty$. □
The finiteness hypothesis cannot be omitted: with counting measure on $\mathbb{N}$, the decreasing tails $A_n = \{k : k \geq n\}$ all have infinite measure, whereas their intersection is empty. On a finite measure space the hypothesis is automatic, as used later when passing from almost-everywhere convergence to convergence in measure.
For later use, countable subadditivity also follows from countable additivity. Given measurable A_n, replace them by the disjoint sets $B_n = A_n \setminus \bigcup_{k<n} A_k$, which have the same union. Monotonicity of measure follows from disjoint decomposition of an inclusion; thus $\mu(B_n) \leq \mu(A_n)$ and $\mu(\bigcup_n A_n) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n)$.
For a measurable set A, its indicator $\mathbf{1}_A$ equals one on A and zero on its complement. A nonnegative simple function is a measurable function with finitely many finite nonnegative values. For a disjoint measurable partition and $s = \sum_{j=1}^{m} a_j \mathbf{1}_{A_j}$, define

$$\int s\, d\mu = \sum_{j=1}^{m} a_j \mu(A_j), \qquad 0 \cdot \infty = 0.$$

A common refinement of two finite partitions proves independence of the representation, additivity, nonnegative homogeneity, and monotonicity for simple functions: on each cell these assertions are the corresponding scalar identities or inequalities. For $f : \Omega \to [0, \infty]$, put

$$\int f\, d\mu = \sup\Big\{ \int s\, d\mu : 0 \leq s \leq f,\ s \text{ simple} \Big\}.$$

The zero function is admissible. Simple monotonicity shows that this extends the simple integral; the definition immediately gives monotonicity for arbitrary nonnegative integrands. Integrating over A means integrating $f\mathbf{1}_A$, where its value off A is defined to be zero even if f takes the value infinity there.

Theorem 37.3 (Monotone convergence)**.**
If $0 \leq f_n \uparrow f$ pointwise, then $\int f_n\, d\mu \uparrow \int f\, d\mu$.

Proof.
Let $L = \sup_n \int f_n$. Monotonicity gives $L \leq \int f$. Fix a nonnegative simple $s \leq f$ and $0 < c < 1$. The measurable sets $E_n = \{f_n \geq cs\}$ increase to Ω: at a point where $s > 0$, the limiting value

is at least $s > cs$, and where $s = 0$ the inequality holds for every n. Since $cs\mathbf{1}_{E_n} \le f_n$, simple monotonicity and the definition give

$$L \ge c \int s\mathbf{1}_{E_n}\, d\mu.$$

Apply continuity from below to the finitely many level sets of s intersected with E_n. It follows that $L \ge c \int s$. Let $c \uparrow 1$ (the assertion is also valid if $\int s = \infty$), and then take the supremum over s. Thus $L \ge \int f$, proving equality. □

Every nonnegative measurable f admits an increasing simple approximation: round $\min(f, n)$ down to the nearest multiple of 2^{-n}, treating $\min(\infty, n) = n$. These simple functions increase to f. For $f, g \ge 0$, choose these approximations s_n, t_n. Then $s_n + t_n \uparrow f + g$, so simple additivity and monotone convergence give

$$\int (f + g)\, d\mu = \int f\, d\mu + \int g\, d\mu.$$

Likewise $\int cf = c \int f$ for $c \ge 0$, with the same zero convention. All these expressions avoid an undefined difference $\infty - \infty$. Applying monotone convergence to partial sums also proves $\int \sum_n f_n = \sum_n \int f_n$ for nonnegative measurable f_n.

Lemma 37.3 (Null sets and finite integrals)**.**
For $f \ge 0$, $\int f = 0$ if and only if $f = 0$ almost everywhere. If $\int f < \infty$, then $f < \infty$ almost everywhere. Nonnegative measurable functions equal almost everywhere have equal integrals.

Proof.
For every $a > 0$, comparison with $a\mathbf{1}_{\{f>a\}}$ gives $a\mu(\{f > a\}) \le \int f$. If the integral is zero, the measurable set $\{f > 0\} = \bigcup_{n\ge1}\{f > 1/n\}$ is null. Conversely, a simple function below f has every positive level supported on that null set, hence has integral zero. If $\int f = C < \infty$, comparison with $m\mathbf{1}_{\{f=\infty\}}$ gives $\mu(\{f = \infty\}) \le C/m$ for every positive integer m, so this set is null. Finally, decompose each of two almost-everywhere equal functions into its restrictions to a measurable null exceptional set and its complement. The null-supported parts have integral zero and the other parts agree. Additivity proves equality of their integrals. All exceptional sets used here are measurable; no completion is needed. □

Theorem 37.4 (Fatou's lemma)**.**
For nonnegative measurable f_n,

$$\int \liminf_n f_n\, d\mu \le \liminf_n \int f_n\, d\mu.$$

Proof.
The measurable functions $g_n = \inf_{k\ge n} f_k$ increase to the lower limit. For every $k \ge n$, monotonicity gives $\int g_n \le \int f_k$, hence $\int g_n \le \inf_{k\ge n} \int f_k$. Take increasing limits and apply monotone convergence to g_n. □

For a real measurable f with $\int |f| < \infty$, define $\int f = \int f^+ - \int f^-$, where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Both integrals are finite. This gives a linear integral on such functions. First, $|f + g| \le |f| + |g|$ and the nonnegative additivity already proved show that $f + g$ is integrable. Additivity then follows by integrating

$$f^+ + g^+ + (f + g)^- = f^- + g^- + (f + g)^+$$

and rearranging finite quantities; homogeneity follows by positive scaling and interchange of positive and negative parts. Define the complex integral by real and imaginary parts. In either case, $|\int f| \le \int |f|$: in the complex case choose a unit scalar making $\int f$ real nonnegative and use $\mathrm{Re}(\lambda f) \le |f|$. The real case is the same estimate with a sign.

Theorem 37.5 (Dominated convergence in norm)**.**
Let f_n, f be measurable real or complex functions, with $f_n \to f$ almost everywhere and $|f_n| \le g$ almost everywhere for some nonnegative g with $\int g < \infty$. Then f and every f_n are integrable and

$$\int |f_n - f|\, d\mu \longrightarrow 0, \qquad \int f_n\, d\mu \longrightarrow \int f\, d\mu.$$

Proof.
Remove one measurable null set containing the convergence exception, all the countably many domination exceptions, and $\{g = \infty\}$. Set all functions to zero there; the resulting functions remain measurable. The preceding null-invariance lemma applies first to their nonnegative absolute values. For the modified functions $|f| \le g$ everywhere, so they are integrable; equality of the absolute-value integrals then proves integrability of the original functions as well. Positive and negative parts (or real and imaginary parts) show that their integrals are unchanged. Put $h_n = |f_n - f|$. Then $0 \le h_n \le 2g$ and $h_n \to 0$. Fatou applied to $2g - h_n$ gives

$$2\int g \le \liminf_n \left(2\int g - \int h_n\right) = 2\int g - \limsup_n \int h_n.$$

All quantities subtracted are finite, so $\limsup_n \int h_n \le 0$. Nonnegativity proves convergence to zero. The integral inequality $|\int f_n - \int f| \le \int h_n$ proves the final assertion. □

37.5 Functions and their almost-everywhere classes

With this integral, the real space $L^1(\mu)$ consists of equivalence classes of measurable functions $f : \Omega \to \mathbb{R}$ satisfying

$$\int_\Omega |f|\, d\mu < \infty,$$

where two functions represent the same element when they agree almost everywhere. Its norm is

$$\|[f]\|_1 = \int_\Omega |f|\, d\mu.$$

The integral is unchanged by changing the representative on a null set. Point evaluation $[f] \mapsto f(\omega)$, however, generally is not well-defined on the quotient. This is a different issue from the book's choice of a default value for an unevaluated class expression.
The complex version of L^1 uses complex measurable representatives and the same absolute-value norm; all the integral estimates remain valid. The scalar field is fixed when discussing either version as a vector space.
The space $L^\infty(\mu)$ consists of essentially bounded measurable functions over that field, modulo the same equality almost everywhere. Its norm is

$$\|[g]\|_\infty = \inf\{C \ge 0 : |g| \le C \text{ almost everywhere}\}.$$

The essential bound at this infimum is attained. If its value is M, then $|g| \le M + 1/n$ almost everywhere for each positive integer n: by the definition of infimum, some admissible bound is

less than $M + 1/n$. Remove the union of these countably many measurable null exceptional sets and let $n \to \infty$ to get $|g| \le M$ there. This also proves that norm zero means the zero almost-everywhere class. Homogeneity and the triangle inequality follow by combining attained bounds off the union of the exceptional sets. Thus the expression is indeed a norm, independent of the representative; no completeness of the measure is required. For $g \in L^\infty(\mu)$ and $f \in L^1(\mu)$, the pairing

$$\langle g, f \rangle = \int_\Omega g f \, d\mu$$

is independent of representatives and satisfies $|\langle g, f\rangle| \leqslant \|g\|_\infty \|f\|_1$. It is this pairing, rather than pointwise evaluation, that enters the weak-convergence statements below. The following arguments derive completeness from the convergence theorems just proved.

37.6 Convergence and completeness

In this subsection we use Theorems 37.3 and 37.4. These integral properties do not alter the earlier choices of class evaluation. A Banach space is a normed vector space complete for its norm metric. A Hilbert space is an inner-product space complete for the norm induced by its inner product. The completeness arguments below use countable choice to select measurable representatives of a sequence of almost-everywhere equivalence classes. Subsequence indices can be chosen as the least suitable natural numbers, so their recursive selection needs no additional dependent-choice principle. These assumptions concern the quotient-space proofs; the preceding monotone and dominated convergence arguments concern already given measurable functions.

Lemma 37.4 (Tail integrals and small sets)**.**
For $f \in L^1(\mu)$ on any positive measure space,

$$\int_{\{|f|>M\}} |f| \, d\mu \longrightarrow 0 \quad (M \to \infty).$$

Consequently, for every $\varepsilon > 0$ there is $\delta > 0$ such that $\mu(A) < \delta$ implies $\int_A |f| \, d\mu < \varepsilon$.

Proof.
The functions $|f|\mathbf{1}_{\{|f|\leqslant m\}}$ increase pointwise to $|f|$ as the positive integers m tend to infinity. Monotone convergence and finiteness of $\int |f|$ imply that their complementary tail integrals tend to zero. Monotonicity in the real parameter M gives the stated limit. Choose $M > 0$ with tail integral less than $\varepsilon/2$, and put $\delta = \varepsilon/(2M)$. Splitting A into the regions $|f| \leqslant M$ and $|f| > M$ gives

$$\int_A |f| \, d\mu \leqslant M\mu(A) + \int_{\{|f|>M\}} |f| \, d\mu < \varepsilon.$$

No finiteness assumption on $\mu(\Omega)$ enters this argument. □
For $a > 0$, the same integral comparison gives $a\mu(\{|f| > a\}) \leqslant \|f\|_1$. This inequality will turn norm convergence into convergence in measure. Here convergence in measure means $\mu(\{|f_n - f| > a\}) \to 0$ for every $a > 0$; on an infinite-measure space this is a global condition, not merely convergence on each finite-measure subset.

Lemma 37.5 (A subsequence converging almost everywhere)**.**
If measurable real functions f_n converge in measure to a measurable real function f, then some subsequence converges to f almost everywhere. The measure space need not have finite measure.

Proof.
Choose increasing indices n_k with $\mu(\{|f_{n_k} - f| > 2^{-k}\}) < 2^{-k}$. If E_k denotes this exceptional set, countable subadditivity gives

$$\mu\left(\bigcap_{m\geqslant 1}\bigcup_{k\geqslant m} E_k\right) \leqslant \sum_{k\geqslant m} 2^{-k} \quad \text{for every } m.$$

Thus the limsup of the E_k is a measurable null set. Outside it, $|f_{n_k} - f| \leqslant 2^{-k}$ eventually, which proves the assertion. □

Theorem 37.6 (Completeness of L^1)**.**
For any positive measure space, $L^1(\mu)$ is a Banach space.

Proof.
Let (f_n) be Cauchy in norm. Choose a subsequence (f_{n_k}) with $\|f_{n_{k+1}} - f_{n_k}\|_1 \leqslant 2^{-k}$, and choose measurable representatives. Monotone convergence applied to the partial sums gives

$$\int_\Omega \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|\, d\mu \leqslant 1.$$

Consequently this series is finite outside a measurable null set. There the telescoping series defines a finite limit f; put $f = 0$ on the exceptional set. This gives a measurable function, even when the measure is not complete, since the exceptional set is measurable. The pointwise bound

$$|f| \leqslant |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}| \quad \text{almost everywhere}$$

shows that $f \in L^1$. Moreover,

$$\|f - f_{n_j}\|_1 \leqslant \sum_{k=j}^{\infty} 2^{-k} \longrightarrow 0.$$

The Cauchy property and the triangle inequality now imply that the whole sequence converges to f in norm. □

Theorem 37.7 (Completeness of L^∞)**.**
For any positive measure space, $L^\infty(\mu)$ is a Banach space.

Proof.
Given a norm-Cauchy sequence, choose a subsequence with successive norm distances at most 2^{-k}. For measurable representatives, the bounds $|f_{n_{k+1}} - f_{n_k}| \leqslant 2^{-k}$ and the essential bound for f_{n_1} hold simultaneously off a measurable null set: only countably many exceptional sets have been removed. The telescoping series converges uniformly on that complement to a bounded function. Extending the limit by zero on the exceptional set gives a measurable, essentially bounded function f with $\|f - f_{n_j}\|_\infty \leqslant \sum_{k\geqslant j} 2^{-k}$. As before, the original Cauchy sequence has the same norm limit. □

37.7 The Hilbert space used by truncation

For the real or complex scalar field, define $L^2(\mu)$ to be the almost-everywhere classes of measurable functions with $\int |f|^2 < \infty$. Put

$$\langle f, g\rangle = \int f\overline{g}\, d\mu, \qquad \|f\|_2 = \left(\int |f|^2\, d\mu\right)^{1/2}.$$

Conjugation is omitted over the reals. The same measurable-null-set convention applies as for L^1. Here the angle brackets denote the L^2 inner product; over complex scalars it conjugates the second factor. The earlier L^∞–L^1 pairing $\int gf\, d\mu$ is bilinear and uses no conjugation.

Lemma 37.6 (Cauchy–Schwarz and the L^2 norm)**.**
The pairing is a well-defined inner product, linear in its first argument, and

$$|\langle f, g\rangle| \le \|f\|_2\|g\|_2, \qquad \|f + g\|_2 \le \|f\|_2 + \|g\|_2.$$

Proof.
The bound $2|f\overline{g}| \le |f|^2 + |g|^2$ proves integrability of the product. Linearity, conjugate symmetry, and independence of representatives follow from the integral. A nonnegative function has zero integral precisely when it is zero almost everywhere, so the pairing is positive definite on the quotient. If $\|g\|_2 > 0$, expand $0 \le \|f - cg\|_2^2$ with $c = \langle f, g\rangle/\|g\|_2^2$ to obtain Cauchy–Schwarz. If $\|g\|_2 = 0$, its equivalence class is zero and the result is immediate. Expanding $\|f + g\|_2^2$ and applying Cauchy–Schwarz gives the triangle inequality. Closure under sums follows already from $|f + g|^2 \le 2|f|^2 + 2|g|^2$. □

Theorem 37.8 (Completeness of L^2)**.**
For any positive measure space, $L^2(\mu)$ is a Hilbert space.

Proof.
For a Cauchy sequence choose a subsequence with differences $d_k = f_{n_{k+1}} - f_{n_k}$ satisfying $\|d_k\|_2 \le 2^{-k}$, and choose measurable representatives. The triangle inequality gives

$$\left\| \sum_{k=j}^{N} |d_k| \right\|_2 \le \sum_{k=j}^{N} 2^{-k}.$$

By monotone convergence applied to the squares, the measurable nonnegative function $H_j = \sum_{k=j}^{\infty} |d_k|$ satisfies

$$\int H_j^2\, d\mu \le \left(\sum_{k=j}^{\infty} 2^{-k}\right)^2.$$

In particular H_1 is finite outside a measurable null set. The corresponding telescoping series converges there to a finite measurable function f; define $f = 0$ on the exceptional set. The bound $|f| \le |f_{n_1}| + H_1$ almost everywhere proves $f \in L^2$. Also $|f - f_{n_j}| \le H_j$ almost everywhere, hence $\|f - f_{n_j}\|_2 \le \sum_{k \ge j} 2^{-k} \to 0$. The Cauchy property supplies convergence of the whole sequence. No finiteness or completeness assumption on the measure is needed. □

38 Functional analysis

This section specifies a setting for the intended applications to Dunford–Pettis compactness. The spaces here have set-sized carriers and use classical real and complex analysis. The earlier class-theoretic framework can express their constructions, but proper-class-valued families and partial descriptors do not alter the analytic definitions by default. For Hahn–Banach and the subsequent analytic development, we work in classical set theory with Choice. Zorn's lemma is used in the extension argument; the nested constructions in Baire's theorem and the open mapping theorem use dependent choice. Some individual results need less, as their proofs indicate. This convention does not strengthen the assumptions of the separate countable-choice metric theorem.
The companion manuscript on the Dunford–Pettis theorem is not included in this checkout. The development below proves the finite-measure criterion in both sequential and topological forms, together with its analytic prerequisites. It does not reconstruct that missing manuscript or assert a formalization in Obelisk.

38.1 Linear structure and topology

Fix the scalar field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. A *topological vector space* is a vector space E equipped with a topology for which addition $E \times E \to E$ and scalar multiplication $\mathbb{K} \times E \to E$ are continuous, using the product topologies. When Hausdorffness is needed, it is an explicit assumption. A locally convex topology has a base of convex neighbourhoods of zero.
A norm $\|\cdot\|$ gives the metric $d(x, y) = \|x - y\|$ and hence a Hausdorff vector-space topology. A *Banach space* is a normed vector space in which every Cauchy sequence converges in this metric. Completeness is a condition on the specified norm topology; it is not implied by the existence of an algebraic vector-space structure.

Lemma 38.1 (Bounded linear maps and continuity)**.**
For a linear map $T : E \to F$ between normed spaces, continuity is equivalent to the existence of $C \geqslant 0$ such that

$$\|Tx\|_F \leqslant C\|x\|_E \qquad (x \in E).$$

Proof.
The inequality gives $\|Tx - Ty\|_F \leqslant C\|x - y\|_E$, hence continuity. Conversely, continuity at zero supplies $\delta > 0$ such that $\|u\|_E < \delta$ implies $\|Tu\|_F < 1$. For $x \neq 0$, apply this to $u = \delta x/(2\|x\|_E)$ and use linearity to obtain $\|Tx\|_F \leqslant (2/\delta)\|x\|_E$. The inequality at zero follows from $T0 = 0$. □An operator defined only on a subspace $D(T) \subseteq E$ has that subspace as its domain. Calling such an operator unbounded does not supply values at points outside $D(T)$.

38.2 Weak topology and the continuous dual

Let E' be the space of continuous linear maps $\ell : E \to \mathbb{K}$. The weak topology $\sigma(E, E')$ is the coarsest topology on E making every $\ell \in E'$ continuous. A basic neighbourhood of x is determined by finitely many functionals and positive tolerances:

$$\{y \in E \mid |\ell_j(y - x)| < \varepsilon_j,\ j = 1, \dots, m\}.$$

Thus a net x_i converges weakly to x exactly when $\ell(x_i) \to \ell(x)$ for every $\ell \in E'$. The same criterion applies to sequences. For a normed space E, norm convergence implies weak convergence, because $|\ell(x_i - x)| \leqslant \|\ell\|\,\|x_i - x\|$.

Weak convergence uses all continuous linear functionals on E. On a dual space E', the weak-star topology $\sigma(E', E)$ instead uses evaluations $\ell \mapsto \ell(x)$ at vectors $x \in E$. These are distinct definitions; a compactness statement must identify which topology it uses.

38.3 Dual completeness and what weak convergence detects

The operator norm on E' is

$$\|\ell\| = \sup_{\|x\| \leqslant 1} |\ell(x)|.$$

The bounded-map lemma makes this finite and gives $|\ell(x)| \leqslant \|\ell\| \, \|x\|$. It is a norm: in particular, vanishing on the unit ball implies vanishing everywhere by scaling.

Theorem 38.1 (The continuous dual is complete)**.**
For every normed space E, its continuous dual E' is a Banach space, even if E itself is not complete.

Proof.
Let (ℓ_n) be Cauchy in operator norm. For each $x \in E$,

$$|\ell_n(x) - \ell_m(x)| \leqslant \|\ell_n - \ell_m\| \, \|x\|,$$

so completeness of the scalar field gives a limit $\ell(x)$. Passing to limits in addition and scalar multiplication shows that ℓ is linear. A Cauchy sequence is norm bounded, say $\|\ell_n\| \leqslant C$, and passage to the limit gives $|\ell(x)| \leqslant C\|x\|$. Thus $\ell \in E'$. For any $\varepsilon > 0$, choose N so that $\|\ell_n - \ell_m\| < \varepsilon$ for $n, m \geqslant N$. Letting m tend to infinity gives $|\ell_n(x) - \ell(x)| \leqslant \varepsilon\|x\|$ for every x and $n \geqslant N$. Taking the supremum on the unit ball proves $\|\ell_n - \ell\| \leqslant \varepsilon$, hence norm convergence. □

Theorem 38.2 (Norm-preserving Hahn–Banach extension)**.**
Assume Zorn's lemma in the surrounding set theory. A bounded linear functional f on a linear subspace D of a real or complex normed space E extends to a bounded linear functional on E with the same norm. The subspace need not be closed.

Proof.
First work over the reals. We prove the more general extension statement for a sublinear function $p : E \to \mathbb{R}$: assume $p(x + y) \leq p(x) + p(y)$, $p(tx) = tp(x)$ for $t \geq 0$, and $f(x) \leq p(x)$ on D. If $a \notin D$, extend to $D + \mathbb{R}a$ by $g(x + ta) = f(x) + tc$. Domination by p holds exactly when

$$\sup_{y \in D} \{f(y) - p(y - a)\} \leq c \leq \inf_{x \in D} \{p(x + a) - f(x)\}.$$

The left quantities do not exceed the right quantities: for $x, y \in D$,

$$f(x) + f(y) = f(x + y) \leq p(x + y) \leq p(x + a) + p(y - a).$$

Both indexing sets are nonempty, and fixing one element on either side bounds the other. Completeness of $\mathbb{R}$ therefore supplies such a c. For $t > 0$, apply the upper inequality to x/t and multiply by t; for $t < 0$, apply the lower inequality to $x/(-t)$ and multiply by $-t$. These give $g(x + ta) \leq p(x + ta)$; the case $t = 0$ is the original bound.
Order all dominated linear extensions of f by extension of their graphs. This is a set, since each graph is a subset of $E \times \mathbb{R}$. A nonempty chain has an upper bound obtained by union:

its domains form an increasing family of subspaces, and compatibility makes the union a linear dominated extension. For the empty chain, the original f is an upper bound. Zorn's lemma gives a maximal extension. The one-dimensional construction shows its domain is all of E. Taking $p(x) = \|f\|\,\|x\|$ and applying domination to x and $-x$ gives $|g(x)| \le \|f\|\,\|x\|$. Restriction to D gives the reverse norm inequality, so the norm is preserved (also when $D = \{0\}$).
For a complex space, extend the real-linear functional $u = \operatorname{Re} f$ from the underlying real subspace of D to a real-linear $g : E \to \mathbb{R}$ with bound $\|f\|$ by the real result. Set $F(x) = g(x) - ig(ix)$. Real linearity and $F(ix) = iF(x)$ imply complex linearity. On D, the identity $\operatorname{Re} f(ix) = -\operatorname{Im} f(x)$ gives $F = f$. For $F(x) \neq 0$, choose a scalar λ with $|\lambda| = 1$ and $\lambda F(x) = |F(x)|$. Then

$$|F(x)| = \operatorname{Re} F(\lambda x) = g(\lambda x) \le \|f\|\,\|\lambda x\| = \|f\|\,\|x\|.$$

The same bound is immediate if $F(x) = 0$. Restriction again proves $\|F\| = \|f\|$. □
The choice principle enters at Zorn's lemma, not at the elementary one-dimensional extension. The following consequences use this theorem in the classical analytic setting; they are not asserted in every choice-free foundational setting explored earlier.

Proposition 38.1 (Norming functionals and weak separation)**.**
For every $x \in E$,

$$\|x\| = \sup_{\ell \in E',\ \|\ell\| \leqslant 1} |\ell(x)|.$$

Consequently $\sigma(E, E')$ is Hausdorff, and a weak limit is unique.

Proof.
The inequality from right to left follows from the operator-norm bound. For $x \neq 0$, define on the one-dimensional subspace spanned by x the functional $ax \mapsto a\|x\|$. Its norm is one over either scalar field. Hahn–Banach extends it to $\ell \in E'$ of norm one, with $\ell(x) = \|x\|$. At $x = 0$ both sides vanish. If $x \neq y$, a norming functional for $x - y$ has $\ell(x) \neq \ell(y)$. Disjoint scalar neighbourhoods of those values have disjoint weakly open inverse images containing x and y. This is Hausdorffness; uniqueness of limits follows by intersecting the corresponding eventual neighbourhood conditions. □

Corollary 38.1 (Weak lower semicontinuity of the norm)**.**
Every norm-closed ball is weakly closed. If a sequence x_n converges weakly to x, then

$$\|x\| \leqslant \liminf_n \|x_n\|.$$

Proof.
For $r \geqslant 0$, the norming formula gives

$$\{y : \|y - x\| \leqslant r\} = \bigcap_{\ell \in E',\ \|\ell\| \leqslant 1} \{y : |\ell(y - x)| \leqslant r\}.$$

Every set on the right is weakly closed, by continuity of its defining functional. For the sequential assertion, if $x \neq 0$, choose a norming functional ℓ for x. Then $|\ell(x_n)| \to \|x\|$ and $|\ell(x_n)| \leqslant \|x_n\|$. Taking lower limits proves the inequality. The case $x = 0$ is immediate. □

Proposition 38.2 (Weak neighbourhoods in infinite dimensions)**.**
If E is infinite-dimensional, every weak neighbourhood of zero contains an unbounded linear subspace. Hence the weak and norm topologies on E are different.

Proof.
Such a neighbourhood contains one defined by finitely many functionals $\ell_1, \dots, \ell_m$ and positive tolerances. It contains their common kernel. Choose $m+1$ linearly independent vectors in E. Their images under $x \mapsto (\ell_1(x), \dots, \ell_m(x)) \in \mathbb{K}^m$ are linearly dependent, so a nonzero linear combination of the chosen vectors lies in the common kernel. All scalar multiples of this nonzero vector also lie there, and their norms are unbounded. The norm unit ball therefore cannot be a weak neighbourhood of zero, although it is a norm neighbourhood. Norm convergence still implies weak convergence; the topologies need not coincide. □
This final argument uses only finite-dimensional linear dependence, not Hahn–Banach. It also explains why finitely many tests cannot control norm size in an infinite-dimensional space, even though the whole dual determines the norm by the preceding proposition.

38.4 Completeness and uniform boundedness

Lemma 38.2 (Baire's theorem for complete metric spaces)**.**
In a nonempty complete metric space, the intersection of countably many dense open sets is dense. Equivalently, such a space cannot be the union of countably many closed sets all having empty interior.

Proof.
Let G_n be dense and open, and let U be any nonempty open set. Choose a closed ball B_1 of positive radius at most 2^{-1} contained in $U \cap G_1$. Inductively choose a closed ball B_{n+1} of positive radius at most $2^{-(n+1)}$ contained in the intersection of G_{n+1} with the open ball having the same centre and radius as B_n. Density and openness make each choice possible; balls with slightly smaller radii ensure closed containment. Their centres form a Cauchy sequence because the balls are nested and their diameters tend to zero. Completeness gives a limit x. Each B_n is closed and contains all later centres, so $x \in B_n$ for every n. Thus $x \in U \cap \bigcap_n G_n$. Taking complements proves the equivalent closed-set statement. The successive choices are made in the classical set-theoretic setting of this chapter. □

Theorem 38.3 (Uniform boundedness principle)**.**
Let E be Banach, F a normed space, and $(T_i)_{i \in I}$ a set-indexed family of bounded linear maps $E \to F$. If $\sup_{i \in I} \|T_i x\| < \infty$ for every $x \in E$, then $\sup_{i \in I} \|T_i\| < \infty$.

Proof.
The empty family is immediate. Put

$$A_n = \{x \in E : \|T_i x\| \le n \text{ for every } i \in I\}, \qquad n = 1, 2, \dots.$$

Continuity makes each A_n closed, and pointwise boundedness gives $E = \bigcup_n A_n$. Baire's theorem gives an A_n containing an open ball of radius $r > 0$ about some x_0. For $\|h\| < r$, both $x_0 + h$ and x_0 belong to A_n, hence $\|T_i h\| \le \|T_i(x_0 + h)\| + \|T_i x_0\| \le 2n$ for every i. For any u with $\|u\| \le 1$, take $h = (r/2)u$ to obtain $\|T_i u\| \le 4n/r$, uniformly in i. Taking suprema proves the claim. Completeness of F is not used. □

Corollary 38.2 (Weakly convergent sequences are norm bounded)**.**
If a sequence (x_n) in a normed space E converges weakly, then $\sup_n \|x_n\| < \infty$. The space E need not be complete.

Proof.
On the Banach space E', consider the bounded linear maps $J_n : E' \to \mathbb{K}$, $J_n(\ell) = \ell(x_n)$. For each ℓ, weak convergence makes $(J_n(\ell))$ a convergent, hence bounded, scalar sequence. Uniform boundedness gives $\sup_n \|J_n\| < \infty$. The norming-functional formula gives $\|J_n\| = \|x_n\|$, proving the assertion. □
The sequential hypothesis is significant. A convergent net need not have bounded range: convergence controls only a tail, and the part outside that tail need not be finite. For example, index a net by $\mathbb{R}$ in its usual order, put $x_t = -t$ for $t < 0$ and $x_t = 0$ for $t \geq 0$. It converges in norm to zero but its range is unbounded.

Proposition 38.3 (A weakly null net with no bounded tail)**.**
In an infinite-dimensional normed space E, there is a net converging weakly to zero such that every tail is unbounded in norm. No simultaneous choice of vectors and no Hahn–Banach theorem is needed.

Proof.
Let I consist of triples (F, n, u) where F is a finite subset of E', n is a positive integer, and u is a unit vector in the common kernel of F. Such a vector exists for each finite F by the finite linear-dependence argument used for weak neighbourhoods. Include all these vectors in the index set, rather than selecting one for each F. For the empty F, the common kernel is E.
Order triples by equality or by

$$(F, n, u) < (G, m, v) \quad \Longleftrightarrow \quad F \subseteq G \text{ and } n < m.$$

This is a partial order. Two triples have an upper bound: use their union of functionals, an integer larger than both integers, and any one unit vector in the resulting common kernel. This is a single existence argument for each pair of indices, not a choice function.
Define $x_{(F,n,u)} = nu$. Given $\ell \in E'$, an index with $F = \{\ell\}$ exists, and every later index annihilates ℓ. Thus the net converges weakly to zero. Above a fixed (F, n, u) lie all (F, m, u) with $m > n$, whose vectors have norm m. Every tail is therefore unbounded. The construction is a function on the explicit set of all triples and makes no simultaneous selections. □
For each individual functional, the scalar values in this example vanish on a suitable tail. Those tails depend on the functional; they do not supply the common pointwise-bounded family needed to apply uniform boundedness. In particular, weak convergence of a general net does not imply even eventual norm boundedness.

38.5 Open mappings and closed graphs

Theorem 38.4 (Open mapping theorem)**.**
Let E, F be Banach spaces and let $T : E \to F$ be bounded, linear, and surjective. Then T maps open sets to open sets.

Proof.
Write $\overline{B}_E = \{x : \|x\| \leq 1\}$. Surjectivity gives

$$F = \bigcup_{n \geq 1} T(n\overline{B}_E) \subseteq \bigcup_{n \geq 1} \overline{T(n\overline{B}_E)} = F.$$

By Baire's theorem, one of the closed sets on the right contains an open ball $B_F(y_0, r)$, with $r > 0$. Subtracting its centre gives

$$B_F(0, r) \subseteq \overline{T(n\overline{B}_E)} - \overline{T(n\overline{B}_E)} \subseteq \overline{T(2n\overline{B}_E)}.$$

For the last inclusion, approximate the two terms by image points, subtract their preimages, and use the triangle inequality. Scaling therefore gives a number $\delta > 0$ such that

$$B_F(0, \delta) \subseteq \overline{T(\overline{B}_E)}.$$

This is only a closure statement; the next step removes the closure.
Fix y with $\|y\| < \delta/2$. By the scaled closure inclusion, choose x_1 with $\|x_1\| \leq 1/2$ and $\|y - Tx_1\| < \delta/4$. Inductively, if the residual after $k-1$ terms has norm less than $\delta 2^{-k}$, choose x_k with

$$\|x_k\| \leq 2^{-k}, \qquad \left\| y - T\sum_{j=1}^{k} x_j \right\| < \delta 2^{-(k+1)}.$$

Completeness of E makes the series converge to x with $\|x\| \leq 1$. Continuity of T and the residual estimate give $Tx = y$. Hence $B_F(0, \delta/2) \subseteq T(\overline{B}_E)$. Scaling once more yields

$$B_F(0, \delta/4) \subseteq T(\tfrac{1}{2}\overline{B}_E) \subseteq T(B_E(0, 1)).$$

Translation and scaling give, for every $a > 0$,

$$B_F(Tx_0, a\delta/4) \subseteq T(B_E(x_0, a)).$$

Every open neighbourhood of x_0 contains one of these domain balls. Taking such balls at every point proves openness. Completeness of F was used in Baire's theorem, and completeness of E in summing the lifts. □

Corollary 38.3 (Bounded inverse theorem)**.**
A bounded linear bijection between Banach spaces has a bounded linear inverse.

Proof.
The inverse is linear by bijectivity and linearity of the original map. Openness of the original map says precisely that its inverse is continuous. The bounded-map lemma then gives boundedness. □

Theorem 38.5 (Closed graph theorem)**.**
Let E, F be Banach spaces and let $T : E \to F$ be an everywhere-defined linear map. Then T is bounded if and only if its graph $G = \{(x, Tx) : x \in E\}$ is closed in $E \times F$.

Proof.
If T is bounded and $(x_n, Tx_n) \to (x, y)$, continuity gives $Tx_n \to Tx$, so $y = Tx$. This proves closedness, since the product is metrizable. Conversely, equip $E \times F$ with the norm $\|(x, y)\| = \|x\| + \|y\|$. Coordinatewise convergence of Cauchy sequences proves its completeness. A closed subspace is complete, so G is Banach. The projection $p : G \to E$, $(x, Tx) \mapsto x$, is a bounded linear bijection; surjectivity uses that T is defined on all of E. The bounded inverse theorem makes p^{-1} bounded. The second-coordinate projection $q : G \to F$ is bounded, and $T = q \circ p^{-1}$ is therefore bounded. □

Example 38.1 (Why completeness and the domain matter)**.**
Give the vector space ℓ^1 first its usual norm and then the supremum norm. The identity from the first normed space to the second is bounded and bijective, but its inverse is unbounded: a vector with n coordinates equal to one has supremum norm one and ℓ^1 norm n. The target is incomplete. The finite truncations of $(1/k)_{k\geq 1}$ are Cauchy in supremum norm but have no supremum-norm limit in ℓ^1. This does not contradict the bounded inverse theorem.
For the domain restriction, on ℓ^2 put

$$D(T) = \{x \in \ell^2 : (nx_n)_{n\geq 1} \in \ell^2\}, \qquad (Tx)_n = nx_n.$$

Its graph is closed: if $x^{(k)} \to x$ and $Tx^{(k)} \to y$ in ℓ^2, coordinate convergence gives $y_n = nx_n$; since $y \in \ell^2$, this means $x \in D(T)$ and $Tx = y$. Nevertheless $\|Te_n\| = n$ while $\|e_n\| = 1$, so T is unbounded for the inherited ℓ^2 norm on its domain. The domain is proper: $(1/n)_{n\geq 1}$ belongs to ℓ^2 but its image would be the constant sequence one. Its finite truncations belong to $D(T)$ and converge to it in ℓ^2, so $D(T)$ is not complete for its inherited norm. Thus closedness of a graph cannot replace the everywhere-defined hypothesis of the theorem. The graph norm $\|x\| + \|Tx\|$ does make $D(T)$ Banach, by its isometric identification with the closed graph.

38.6 Compact operators and norm approximation

Let E, F be normed spaces over the same scalar field. Write $B_E = \{x \in E : \|x\| \leq 1\}$ for the closed unit ball. A bounded linear map $T : E \to F$ is *compact* if $\overline{T(B_E)}$, with closure taken in F, is norm compact. For bounded linear maps use the operator norm

$$\|T\| = \sup_{x\in B_E} \|Tx\|.$$

The bounded-map lemma and scaling give $\|Tx\| \leq \|T\|\,\|x\|$. The metric compactness criterion and its classical countable-choice convention continue to apply here.

Proposition 38.4 (Sequential characterization of compact operators)**.**
A bounded linear map $T : E \to F$ is compact if and only if every bounded sequence (x_n) in E has a subsequence whose images Tx_{n_k} converge in norm to an element of F. Neither space needs to be complete.

Proof.
If T is compact and $\|x_n\| \leq R$ for some $R > 0$, the images belong to $R\overline{T(B_E)}$, a compact set. The metric compactness criterion gives a convergent subsequence. Conversely, let (y_n) be any sequence in $\overline{T(B_E)}$. Choose $x_n \in B_E$ with $\|Tx_n - y_n\| < 1/n$. The assumed property gives $Tx_{n_k} \to y$ in F, hence $y_{n_k} \to y$ as well. Closedness places y in $\overline{T(B_E)}$. This closure is sequentially compact and therefore compact by the metric criterion. The approximations x_n use countable choice. □

Corollary 38.4 (Finite rank and the identity)**.**
Every bounded linear map with finite-dimensional range is compact. The identity on a normed space is compact if and only if that space is finite-dimensional.

Proof.
The linear-algebra chapter's result on finite-dimensional ranges says exactly that the image of the unit ball has compact closure. For the identity, the image is the closed unit ball itself, so its compactness is equivalent to finite dimension by the compact-unit-ball theorem in that chapter. Neither assertion requires ambient completeness. □

Proposition 38.5 (Sums and compositions)**.**
The compact maps $E \to F$ form a vector subspace of the bounded linear maps. If $T : E \to F$ is compact and $A : D \to E$ and $B : F \to G$ are bounded linear maps between normed spaces, then $BTA : D \to G$ is compact.

Proof.
For compact S, T and a bounded sequence in E, first extract a subsequence on which the S images converge, and from it one on which the T images converge. Every fixed linear combination of the two images then converges. The sequential characterization proves closure under linear combinations; the zero map is compact too. For the composition, A sends bounded sequences to bounded sequences, T supplies a subsequence with convergent images, and continuity of B preserves that convergence. Apply the same characterization. □

Theorem 38.6 (Norm limits of compact operators)**.**
Let E be normed and F Banach. If compact operators $T_n : E \to F$ converge in operator norm to a bounded linear map $T : E \to F$, then T is compact. In particular, the compact operators form a closed subspace of the bounded linear operators in the operator norm.

Proof.
Given $\varepsilon > 0$, choose n with $\|T - T_n\| < \varepsilon/4$. Compactness of $\overline{T_n(B_E)}$ makes $T_n(B_E)$ totally bounded. Choose $x_1, \dots, x_m \in B_E$ whose images under T_n form a finite $\varepsilon/4$-net for that image. For every $x \in B_E$, some j satisfies

$$\begin{aligned}\|Tx - Tx_j\| &\le \|(T - T_n)x\| + \|T_n x - T_n x_j\| + \|(T_n - T)x_j\| \\ &< 3\varepsilon/4 < \varepsilon.\end{aligned}$$

Thus $T(B_E)$ is totally bounded. Its closure remains totally bounded and is complete because F is Banach. The metric compactness criterion therefore makes this closure compact. Sequential closedness is closedness in the operator-norm metric, giving the final assertion. □

Example 38.2 (Why completeness of the target enters)**.**
Let c_{00} denote the real or complex sequences with finite support, equipped with the ℓ^2 norm. Take $E = F = c_{00}$ and define

$$(Tx)_j = x_j/j, \qquad (T_m x)_j = \begin{cases} x_j/j, & j \le m, \\ 0, & j > m. \end{cases}$$

These maps preserve finite support. Each T_m is bounded and has finite rank, hence is compact. Directly from the norm formula,

$$\|(T - T_m)x\|_2 \le \frac{\|x\|_2}{m+1}, \qquad \|T - T_m\| = \frac{1}{m+1},$$

where equality is obtained at the coordinate vector e_{m+1}. Nevertheless T is not compact. Put $x_j^{(N)} = 1/(2j)$ for $j \le N$ and zero otherwise. These vectors belong to B_E, since $\sum_{j \ge 1} j^{-2} \le 2$ (for $j \ge 2$, compare with $1/(j(j-1))$ and telescope). Every norm-convergent subsequence of $Tx^{(N)}$ would converge coordinatewise to the sequence $(1/(2j^2))_{j \ge 1}$, which has infinite support and is not in F. There is therefore no such subsequence in F, contradicting the sequential characterization if T were compact. The target is indeed incomplete: $(x^{(N)})$ is Cauchy by convergence of $\sum j^{-2}$, but a limit in c_{00} would have coordinates $1/(2j)$ for every j, which is impossible.

Proposition 38.6 (Compact operators on weakly convergent sequences)**.**
If $T : E \to F$ is compact and $x_n \to x$ weakly in E, then $Tx_n \to Tx$ in norm. The spaces need not be Banach; this assertion uses the earlier Hahn–Banach consequences for normed spaces.

Proof.
The weakly convergent sequence is norm bounded. Suppose its images do not converge in norm to Tx. Then some subsequence satisfies $\|Tx_{n_k} - Tx\| \geq \varepsilon$ for a fixed $\varepsilon > 0$. Compactness gives a further subsequence converging in norm to some $y \in F$. For every $\ell \in F'$, the functional $\ell \circ T$ belongs to E', so weak convergence gives $\ell(Tx_{n_k}) \to \ell(Tx)$ along this further subsequence. Norm convergence gives the limit $\ell(y)$ as well. The continuous dual separates points by Hahn–Banach, hence $y = Tx$, contradicting the distance bound. Thus the whole image sequence converges in norm. □

38.7 The Schur property of ℓ^1

For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, put

$$\ell^1(\mathbb{K}) = \left\{ x = (x_j)_{j\geq 1} : \sum_{j=1}^{\infty} |x_j| < \infty \right\}, \qquad \|x\|_1 = \sum_{j=1}^{\infty} |x_j|.$$

The scalar triangle inequality makes this a normed vector space. Convergence of the nonnegative series implies $\sum_{j>m} |x_j| \to 0$ for each $x \in \ell^1$. This space is also L^1 for counting measure on the positive integers: integration is summation, by monotone convergence applied to finite truncations. Counting measure has infinite total mass. The theorem below therefore concerns a different measure setting from the finite-measure Dunford–Pettis criterion that follows.
For a bounded scalar sequence $b = (b_j)$ the formula

$$L_b(x) = \sum_{j=1}^{\infty} b_j x_j$$

defines a continuous linear functional on ℓ^1. Indeed, the series converges absolutely and $|L_b(x)| \leq (\sup_j |b_j|)\|x\|_1$; absolute convergence justifies termwise addition and scalar multiplication. Coordinate evaluation is a special case, so weak convergence implies convergence of each coordinate. We only need these explicit functionals, not a theorem identifying every member of the dual.

Theorem 38.7 (Schur's theorem)**.**
A sequence in real or complex ℓ^1 converges weakly to x if and only if it converges to x in norm.

Proof.
Norm convergence implies weak convergence in every normed space. For the converse, subtract the weak limit and write the resulting weakly null sequence as (y_n). Suppose its norms do not tend to zero. There is $\varepsilon > 0$ such that infinitely many indices satisfy $\|y_n\|_1 \geq \varepsilon$.
We construct increasing indices n_k and increasing nonnegative integers m_k, starting with $n_0 = m_0 = 0$. Having chosen m_{k-1}, coordinatewise convergence to zero ensures that, for all sufficiently large n,

$$\sum_{j \leq m_{k-1}} |(y_n)_j| < \varepsilon/8.$$

Choose $n_k > n_{k-1}$ among the infinitely many indices with norm at least ε, also satisfying that prefix bound, and set $z_k = y_{n_k}$. Since $z_k \in \ell^1$, choose $m_k > m_{k-1}$ so that its tail satisfies

$$\sum_{j \le m_{k-1}} |(z_k)_j| < \varepsilon/8, \qquad \sum_{j > m_k} |(z_k)_j| < \varepsilon/8.$$

Each integer choice may be made as the least admissible integer; the recursion requires no simultaneous selection of vectors. The mass on the intervening block therefore satisfies

$$\sum_{m_{k-1} < j \le m_k} |(z_k)_j| > 3\varepsilon/4.$$

The disjoint blocks $(m_{k-1}, m_k]$ cover the positive integers, since $m_k \ge k$. Define a single coefficient sequence by

$$b_j = \begin{cases} \overline{(z_k)_j}/|(z_k)_j|, & m_{k-1} < j \le m_k \text{ and } (z_k)_j \ne 0, \\ 0, & m_{k-1} < j \le m_k \text{ and } (z_k)_j = 0. \end{cases}$$

In the real case the first expression is the usual sign. We have $|b_j| \le 1$, so L_b is a continuous linear functional. On the kth block, $b_j(z_k)_j = |(z_k)_j|$; outside that block, the total absolute contribution is less than $\varepsilon/4$. Consequently

$$\operatorname{Re} L_b(z_k) \ge \sum_{m_{k-1} < j \le m_k} |(z_k)_j| - \sum_{j \le m_{k-1}} |(z_k)_j| - \sum_{j > m_k} |(z_k)_j| > \varepsilon/2.$$

Here Re is redundant over the reals. This contradicts weak nullity, which requires $L_b(z_k) \to 0$. Hence $\|y_n\|_1 \to 0$, proving the theorem. □

Corollary 38.5 (Weak-to-norm preservation does not imply compactness)**.**
The identity on ℓ^1 sends every weakly convergent sequence to a norm-convergent sequence, but is not compact.

Proof.
The first assertion is Schur's theorem. For the second, the coordinate vectors e_j form a bounded sequence with $\|e_j - e_k\|_1 = 2$ whenever $j \ne k$. Their images under the identity have no norm-convergent subsequence, so the sequential compact-operator criterion excludes compactness. □
The vectors e_j do converge coordinatewise to zero, but they do not converge weakly to zero: the bounded coefficient sequence $b_j = 1$ gives $L_b(e_j) = 1$. Thus coordinate tests alone would miss exactly the obstruction detected by the block functional in the proof.
Schur's theorem is a statement about sequences, not equality of topologies. The space ℓ^1 is infinite-dimensional, so its weak neighbourhoods of zero contain unbounded linear subspaces and its weak topology differs from its norm topology, as proved earlier. Proposition 38.3 even gives a weakly null net in ℓ^1 with every tail norm-unbounded. The sequence result thus does not extend to arbitrary nets. Nor does it hold in every L^1 space: the Rademacher sequence on $[0, 1)$ in the probability chapter is weakly null in L^1 and has norm one throughout. Infinite counting measure and a finite nonatomic measure produce different examples; the finite-measure compactness criterion below does not identify these settings.
A set is relatively weakly compact when its closure in the weak topology is compact. We first prove a sequential criterion: every sequence from the family has a subsequence converging weakly to an element of the ambient space. The limit need not belong to the original family. A later argument proves the topological compactness statement separately, rather than assuming that sequences characterize an arbitrary weak topology.

38.8 A finite-measure Dunford–Pettis criterion

Fix a finite positive measure space $(\Omega, \mathcal{A}, \mu)$ and use the real space $L^1(\mu)$. A family $K \subseteq L^1(\mu)$ is *uniformly integrable* in the convention used here when

$$\lim_{M\to\infty} \sup_{f\in K} \int_{\{|f|>M\}} |f|\,d\mu = 0.$$

For the empty family the condition is taken to hold. The formula is independent of the measurable representatives of its members.

Lemma 38.3 (Uniform small-set control)**.**
For a family $K \subseteq L^1(\mu)$ on this finite measure space, uniform integrability is equivalent to the conjunction of

$$\sup_{f\in K} \|f\|_1 < \infty$$

and the following condition: for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\mu(A) < \delta \longrightarrow \sup_{f\in K} \int_A |f|\,d\mu < \varepsilon \qquad (A \in \mathcal{A}).$$

Proof.
Assume uniform integrability. Choosing a positive M whose tail integrals are at most one gives $\|f\|_1 \leqslant M\mu(\Omega) + 1$ for all $f \in K$. Given $\varepsilon > 0$, instead choose $M > 0$ with all tail integrals less than $\varepsilon/2$. Then

$$\int_A |f|\,d\mu \leqslant M\mu(A) + \int_{\{|f|>M\}} |f|\,d\mu < \varepsilon$$

whenever $\mu(A) < \varepsilon/(2M)$.
Conversely, let C bound the norms and choose δ for the small-set condition. For $M > 0$, $\mu(\{|f| > M\}) \leqslant C/M$ for every $f \in K$. Choosing $M > C/\delta$ makes each such set small enough that its integral is less than ε. This is uniform integrability. The empty family satisfies both conditions by convention. □

Theorem 38.8 (Vitali convergence theorem)**.**
Suppose $f_n \in L^1(\mu)$ and f is a measurable real function, the family $\{f_n : n \geqslant 1\}$ is uniformly integrable, and $f_n \to f$ in measure, meaning

$$\mu(\{|f_n - f| > a\}) \longrightarrow 0 \quad \text{for every } a > 0.$$

Then $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.

Proof.
The small-set lemma gives $C = \sup_n \|f_n\|_1 < \infty$. By Lemma 37.5, choose a subsequence converging to f almost everywhere. Fatou's lemma gives

$$\int_\Omega |f|\,d\mu \leqslant \liminf_k \int_\Omega |f_{n_k}|\,d\mu \leqslant C.$$

Thus integrability of the limit is a conclusion, not an assumption. An individual integrable function has vanishing tail integrals, so adjoining f preserves uniform integrability. By the lemma, for a given $\varepsilon > 0$ choose $\delta > 0$ such that $\mu(A) < \delta$ implies both $\int_A |f_n|\,d\mu < \varepsilon/3$ for

every n and $\int_A |f|\,d\mu < \varepsilon/3$. Choose $a > 0$ with $a\mu(\Omega) < \varepsilon/3$. For all sufficiently large n, the set $A_n = \{|f_n - f| > a\}$ has measure less than δ. Hence

$$\|f_n - f\|_1 \leqslant a\mu(\Omega) + \int_{A_n} |f_n|\,d\mu + \int_{A_n} |f|\,d\mu < \varepsilon.$$

This proves norm convergence. □

The proof uses Fatou's lemma and the basic Lebesgue-integral fact that an integrable function has vanishing tails. A related presentation is B. Driver, *Probability Theory*, Theorem 11.31. The theorem just proved yields weak convergence as well, by the norm estimate for the L^∞ pairing. It strengthens the next criterion when convergence in measure is already known: the whole sequence then converges in norm, rather than merely possessing weakly convergent subsequences.

Proposition 38.7 (The converse and an L^1 convergence criterion)**.**
On the finite measure space above, if $f_n \to f$ in L^1, then $f_n \to f$ in measure and $\{f_n : n \geqslant 1\}$ is uniformly integrable. Consequently, for measurable real f and $f_n \in L^1$, convergence in measure together with uniform integrability is equivalent to $f \in L^1$ and convergence to f in L^1.

Proof.
For $a > 0$, the inequality $a\mu(\{|f_n - f| > a\}) \leqslant \|f_n - f\|_1$ proves convergence in measure. The sequence of norms is bounded by the triangle inequality. To prove uniform small-set control, fix $\varepsilon > 0$. Choose N so that $\|f_n - f\|_1 < \varepsilon/2$ for $n \geqslant N$. The finite family $f, f_1, \ldots, f_{N-1}$ has uniformly small integrals on sets of sufficiently small measure: for each member this follows by splitting at a large height, and one takes the minimum of finitely many positive bounds for the measure. Choose such a bound so that each integral is less than $\varepsilon/2$. Then, for $n \geqslant N$,

$$\int_A |f_n|\,d\mu \leqslant \|f_n - f\|_1 + \int_A |f|\,d\mu < \varepsilon;$$

the finitely many earlier terms satisfy the same conclusion. The small-set lemma proves uniform integrability. The reverse implication is Theorem 38.8. □

Corollary 38.6 (Domination with convergence in measure)**.**
Let f_n and f be measurable real functions. If $|f_n| \leqslant h$ almost everywhere for one $h \in L^1$, and $f_n \to f$ in measure on this finite measure space, then $f \in L^1$ and $\|f_n - f\|_1 \to 0$.

Proof.
Replace h by $|h|$ if necessary. The bound

$$\int_{\{|f_n| > M\}} |f_n|\,d\mu \leqslant \int_{\{h > M\}} h\,d\mu$$

gives uniform integrability; apply the theorem. In particular, convergence almost everywhere implies the conclusion here, since on a finite measure space the decreasing sets $\bigcup_{n \geqslant N} \{|f_n - f| > a\}$ have measures tending to zero for every $a > 0$. This last argument uses continuity of measure from above and is precisely where finiteness enters. □

38.9 Stability of uniform integrability

The finite-measure setting continues throughout this subsection. These facts distinguish norm compactness from the weak subsequence property in Theorem 38.13.

Proposition 38.8 (Closure and convex combinations)**.**
If $K \subseteq L^1(\mu)$ is uniformly integrable, then its norm-closed convex hull is uniformly integrable. More generally, the same is true for the norm closure of all finite sums $\sum_{j=1}^{m} a_j f_j$, where $f_j \in K$, $a_j \in \mathbb{R}$, and $\sum_j |a_j| \leqslant 1$.

Proof.
Let C be a bound for the norms of members of K. Every such sum has norm at most C, and for any measurable A,

$$\int_A \left| \sum_j a_j f_j \right| d\mu \leqslant \sum_j |a_j| \int_A |f_j| \, d\mu \leqslant \sup_{f \in K} \int_A |f| \, d\mu.$$

The uniform small-set condition therefore passes to these sums. It also passes to their norm closure: for a norm limit $g_n \to g$,

$$\left| \int_A |g_n| \, d\mu - \int_A |g| \, d\mu \right| \leqslant \|g_n - g\|_1.$$

For a prescribed $\varepsilon > 0$, choose the original small-set bound with $\varepsilon/2$; limits then have integrals at most $\varepsilon/2 < \varepsilon$. Norm bounds likewise pass to limits. Apply the small-set characterization. The empty-family case is immediate, taking the allowed empty sum to be zero. □

Proposition 38.9 (Norm compactness implies uniform integrability)**.**
Every relatively norm-compact subset $K \subseteq L^1(\mu)$ is uniformly integrable.

Proof.
Its norm closure is compact and hence bounded: the open balls centered at zero with positive integer radii cover it, and a finite subcover supplies a bound. For $\varepsilon > 0$, compactness also gives a finite $\varepsilon/2$-net $g_1, \ldots, g_m$ in L^1. The finite family has uniformly small integrals on small measurable sets, by the same finite-minimum argument used in the converse Vitali theorem. Choose $\delta > 0$ so that $\mu(A) < \delta$ implies $\int_A |g_j| \, d\mu < \varepsilon/2$ for every j. For any $f \in K$, choose j with $\|f - g_j\|_1 < \varepsilon/2$. Then

$$\int_A |f| \, d\mu \leqslant \|f - g_j\|_1 + \int_A |g_j| \, d\mu < \varepsilon.$$

The characterization proves uniform integrability. □
The converse is false. The Rademacher functions in the probability section are bounded by one and hence uniformly integrable, but any two distinct ones have L^1 distance one: they disagree on a set of measure $1/2$, where their difference has absolute value two. Thus no subsequence is norm Cauchy. Their weak convergence is compatible with this failure of norm compactness.

38.10 Hilbert representation and the dual of real L^1

Theorem 38.9 (Representation of a real Hilbert-space functional)**.**
Let H be a real Hilbert space. For each continuous linear functional $\ell : H \to \mathbb{R}$ there is a unique $g \in H$ such that

$$\ell(x) = \langle x, g \rangle \quad (x \in H), \qquad \|\ell\| = \|g\|.$$

Proof.
For $\ell = 0$, take $g = 0$. Otherwise the closed affine set $S = \{z : \ell(z) = 1\}$ is nonempty. Its distance d from zero satisfies $d \geq 1/\|\ell\| > 0$. Choose $z_n \in S$ with $\|z_n\| \to d$. Since $(z_n + z_m)/2 \in S$, the parallelogram identity gives

$$\|z_n - z_m\|^2 = 2\|z_n\|^2 + 2\|z_m\|^2 - 4\|(z_n + z_m)/2\|^2 \leq 2\|z_n\|^2 + 2\|z_m\|^2 - 4d^2.$$

Thus (z_n) is Cauchy. Completeness and closedness give a limit $z \in S$ of norm d. For $v \in \ker \ell$ and every real t, $z + tv \in S$, so $0 \leq 2t\langle z, v\rangle + t^2\|v\|^2$. Considering positive and negative sufficiently small t shows $\langle z, v\rangle = 0$.
For each x, the vector $x - \ell(x)z$ lies in the kernel. Hence $\langle x, z\rangle = \ell(x)\|z\|^2$, and $g = z/\|z\|^2$ gives the desired representation. Cauchy–Schwarz gives $\|\ell\| \leq \|g\|$; testing on $g/\|g\|$ gives the reverse inequality when $g \neq 0$. If g' is another representing vector, testing their difference on $x = g - g'$ gives $\|g - g'\|^2 = 0$, proving uniqueness. □

Theorem 38.10 (The dual of real L^1 on a finite measure space)**.**
Let $(\Omega, \mathcal{A}, \mu)$ be a finite positive measure space. The map

$$L^\infty(\mu) \longrightarrow (L^1(\mu))', \qquad g \longmapsto \left(f \longmapsto \int f g \, d\mu\right)$$

is a linear isometric bijection, with functions identified almost everywhere. No completeness assumption on the measure is needed.

Proof.
For $g \in L^\infty$, the integral defines a bounded functional ℓ_g with $\|\ell_g\| \leq \|g\|_\infty$. If $0 \leq r < \|g\|_\infty$, the measurable set $A = \{|g| > r\}$ has positive, finite measure. Testing on $f = \operatorname{sgn}(g)\mathbf{1}_A/\mu(A)$ gives $\|f\|_1 = 1$ and $\ell_g(f) > r$. Letting r increase to $\|g\|_\infty$ proves norm equality; for $g = 0$ it is immediate. This also proves injectivity.
For surjectivity let $\ell \in (L^1)'$ and put $C = \|\ell\|$. Cauchy–Schwarz gives $\|f\|_1 \leq \mu(\Omega)^{1/2}\|f\|_2$, so restriction of ℓ to L^2 is continuous. Theorem 37.8 and real Hilbert representation supply $g \in L^2$ with $\ell(h) = \int hg \, d\mu$ for every $h \in L^2$. For $\varepsilon > 0$, put $A_\varepsilon = \{|g| > C + \varepsilon\}$. The function $h = \operatorname{sgn}(g)\mathbf{1}_{A_\varepsilon}$ is in L^2 and L^1 because the measure is finite. Therefore

$$(C + \varepsilon)\mu(A_\varepsilon) \leq \int_{A_\varepsilon} |g| \, d\mu = \ell(h) \leq C\|h\|_1 = C\mu(A_\varepsilon).$$

It follows that $\mu(A_\varepsilon) = 0$. Taking the union over $\varepsilon = 1/n$ shows $|g| \leq C$ almost everywhere, so $g \in L^\infty$.
For any $f \in L^1$, its bounded truncations $f_m = \max(-m, \min(f, m))$ belong to L^2 and converge in L^1 to f, by the vanishing integrable-tail lemma. Continuity of ℓ and $|\int (f - f_m)g \, d\mu| \leq C\|f - f_m\|_1$ now give $\ell(f) = \int fg \, d\mu$. This proves surjectivity and the theorem. If $\mu(\Omega) = 0$, both spaces consist of the zero equivalence class, and the same conclusion is immediate. □
In particular, convergence of all the bounded-test integrals below is exactly weak convergence in real L^1, not merely convergence against an unspecified subset of its continuous dual. The finite-measure hypothesis enters both the inclusion $L^2 \subseteq L^1$ and the use of bounded test functions in L^2.

38.11 A proved direction of the compactness criterion

The following argument uses the Hilbert structure of $L^2(\mu)$ proved in Theorem 37.8. We supply the weak subsequence argument rather than assuming reflexive weak compactness. The finite-measure dual identification in Theorem 38.10 identifies convergence of all bounded integrals with weak convergence.

Lemma 38.4 (Bounded sequences in a Hilbert space)**.**
Every bounded sequence (u_n) in a Hilbert space has a weakly convergent subsequence.

Proof.
The closed linear span H_0 of the sequence has a finite or countable orthonormal basis (e_j): apply Gram–Schmidt to the ordered list $u_1, u_2, \ldots$, discarding zero remainders, and take the closed span. If no vector remains, the sequence is zero. If $\|u_n\| \le C$, repeated scalar subsequence extraction followed by diagonalization makes every coordinate converge, say $\langle u_{n_k}, e_j\rangle \to a_j$ (inner products linear in the first argument). Finite-dimensional Pythagoras gives $\sum_{j=1}^{m} |a_j|^2 \le C^2$ for every m. Thus the orthogonal series $u = \sum_j a_j e_j$ converges by completeness and has norm at most C. Coordinate convergence gives convergence against finite linear combinations of the e_j. Approximating any vector of H_0 in norm by such combinations, and using $\|u_{n_k} - u\| \le 2C$, gives convergence against all of H_0. For an arbitrary y in the Hilbert space, Bessel's inequality constructs $y_0 = \sum_j \langle y, e_j\rangle e_j$; its difference from y is orthogonal to every e_j and hence to H_0. Pairing against y is therefore the same as pairing against y_0 for these vectors.
To obtain weak convergence in the sense of *all* continuous linear functionals, let ℓ be such a functional and set $b_j = \ell(e_j)$. For $v_m = \sum_{j=1}^{m} \overline{b_j} e_j$, boundedness gives

$$\sum_{j=1}^{m} |b_j|^2 = |\ell(v_m)| \le \|\ell\| \Big(\sum_{j=1}^{m} |b_j|^2\Big)^{1/2}.$$

Thus $\sum_j |b_j|^2 \le \|\ell\|^2$. In the real case conjugation is omitted. On H_0, norm convergence of orthonormal expansions and continuity imply $\ell(v) = \sum_j \langle v, e_j\rangle b_j$. The tail after m is bounded by $\|v\|(\sum_{j>m} |b_j|^2)^{1/2}$ by Cauchy–Schwarz. Apply this bound to $v = u_{n_k} - u$, whose norm is at most $2C$; the finite part tends to zero by coordinate convergence, and the tail bound tends to zero with m. Therefore $\ell(u_{n_k}) \to \ell(u)$. This proves weak convergence without invoking Riesz representation. □

Theorem 38.11 (Uniform integrability gives weak subsequences)**.**
On the finite measure space under consideration, every uniformly integrable sequence (f_n) in real L^1 has a subsequence and an $f \in L^1$ such that

$$\int h f_{n_k}\, d\mu \longrightarrow \int h f\, d\mu \qquad (h \in L^\infty).$$

Proof.
For positive integers m let $T_m(t) = \max(-m, \min(t, m))$ and put

$$\varepsilon_m = \sup_n \|f_n - T_m(f_n)\|_1.$$

Uniform integrability implies $\varepsilon_m \to 0$, since $|t - T_m(t)| \le |t| \mathbf{1}_{\{|t|>m\}}$. For each fixed m, the functions $T_m(f_n)$ have L^2 norm at most $m\mu(\Omega)^{1/2}$. Apply the Hilbert lemma successively for

$m = 1, 2, \dots$ and take a diagonal subsequence. It gives functions $g_m \in L^2$ such that $T_m(f_{n_k})$ converges weakly in L^2 to g_m for every fixed m.
For m, l, the triangle inequality gives

$$\|T_m(f_{n_k}) - T_l(f_{n_k})\|_1 \le \varepsilon_m + \varepsilon_l.$$

The bounded measurable function $h = \operatorname{sgn}(g_m - g_l)$ belongs to L^2 because the measure is finite. Test both weak limits against this h to obtain

$$\|g_m - g_l\|_1 = \lim_k \int h\big(T_m(f_{n_k}) - T_l(f_{n_k})\big)\, d\mu \le \varepsilon_m + \varepsilon_l.$$

Here $g_m \in L^1$ follows from Cauchy–Schwarz and finite measure. Completeness of L^1 now gives $g_m \to f$ in L^1. Finally fix $h \in L^\infty$, which also belongs to L^2. For fixed m,

$$\left|\int h(f_{n_k} - f)\, d\mu\right| \le \|h\|_\infty \varepsilon_m + \left|\int h(T_m(f_{n_k}) - g_m)\, d\mu\right| + \|h\|_\infty \|g_m - f\|_1.$$

First let $k \to \infty$ and then $m \to \infty$. The bound tends to zero, proving the result. If the measure is zero all spaces here consist of the zero equivalence class and the conclusion is immediate. □

38.12 The converse: small sets and the measure algebra

The reverse implication can be proved directly from Baire's theorem, without importing the full Vitali–Hahn–Saks theorem. The auxiliary complete metric space consists of measurable sets modulo null differences.

Lemma 38.5 (Completeness of the finite measure algebra)**.**
On $\mathcal{A}$ identify A and B when $\mu(A \triangle B) = 0$. The quotient has a complete metric

$$d([A], [B]) = \mu(A \triangle B).$$

For every $h \in L^1(\mu)$, the function $[A] \mapsto \int_A h\, d\mu$ is well defined and continuous.

Proof.
The triangle inequality follows from $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$; null differences make the metric independent of representatives. Given a Cauchy sequence, select a subsequence with representatives A_k such that $\mu(A_k \triangle A_{k+1}) < 2^{-k}$. Let $A = \liminf_k A_k$, a measurable set. For each k,

$$A_k \triangle A \subseteq \bigcup_{j \ge k} (A_j \triangle A_{j+1}).$$

Indeed, if membership never changes after stage k, membership in the liminf agrees with membership at that stage. Thus $d([A_k], [A]) \le \sum_{j \ge k} 2^{-j} \to 0$. The original Cauchy sequence converges to the same limit by the triangle inequality. Finally,

$$\left|\int_A h - \int_B h\right| \le \int_{A \triangle B} |h|,$$

and absolute continuity of the integral proves continuity. These arguments use measurable representatives throughout, not completeness of the measure. □

Theorem 38.12 (Weak convergence gives uniform integrability)**.**
If f_n converges weakly to f in real $L^1(\mu)$, with μ finite, then $\{f_n : n \geq 1\}$ is uniformly integrable.

Proof.
Put $h_n = f_n - f$ and $\nu_n(A) = \int_A h_n \, d\mu$. Weak convergence against indicators gives $\nu_n(A) \to 0$ for every measurable A. Fix $\varepsilon > 0$ and, in the complete measure algebra, put

$$E_N = \{[A] : |\nu_n(A)| \leq \varepsilon \text{ for every } n \geq N\}.$$

Each E_N is closed by Lemma 38.5, and their union is the whole measure algebra by pointwise convergence. Baire's theorem supplies N, a measurable representative A_0, and $\delta > 0$ such that the open δ-ball about $[A_0]$ is contained in E_N.
If $\mu(B) < \delta$, both $[A_0 \setminus B]$ and $[A_0 \cup B]$ are in this ball, as is $[A_0]$. Additivity gives, for $n \geq N$,

$$|\nu_n(B \cap A_0)| \leq 2\varepsilon, \qquad |\nu_n(B \setminus A_0)| \leq 2\varepsilon,$$

and hence $|\nu_n(B)| \leq 4\varepsilon$. Apply this bound separately to $B \cap \{h_n \geq 0\}$ and $B \cap \{h_n < 0\}$ to obtain

$$\int_B |h_n| \, d\mu \leq 8\varepsilon \qquad (n \geq N).$$

Both subsets have measure less than δ, so the same uniform bound applies even though they depend on n.
For any desired tolerance $\eta > 0$, begin with $\varepsilon = \eta/16$. The finitely many functions $h_1, \ldots, h_{N-1}$ have a common small-set bound by absolute continuity of their integrals; shrinking δ therefore gives $\sup_n \int_B |h_n| < \eta$ whenever $\mu(B) < \delta$. Adding the single integrable function f proves the same uniform small-set property for the f_n. Their norms are bounded by the earlier weak-sequence boundedness corollary. The uniform-small-set lemma now implies uniform integrability. □

38.13 The full sequential compactness criterion

Theorem 38.13 (Dunford–Pettis, sequential finite-measure form)**.**
Let $(\Omega, \mathcal{A}, \mu)$ be a finite positive measure space and $K \subseteq L^1(\mu; \mathbb{R})$. Then K is uniformly integrable if and only if every sequence in K has a subsequence converging weakly in $L^1(\mu)$. Here weak convergence means

$$\int_\Omega g f_n \, d\mu \longrightarrow \int_\Omega g f \, d\mu \qquad \text{for every } g \in L^\infty(\mu).$$

Proof.
The forward implication is Theorem 38.11. Conversely, suppose the stated subsequence property holds but uniform integrability fails. Since tail integrals decrease as the threshold increases, there is $\varepsilon > 0$ such that, for every positive integer n, one can choose $f_n \in K$ with

$$\int_{\{|f_n| > n\}} |f_n| \, d\mu > \varepsilon.$$

(The tolerance may be halved if a supremum is not attained.) A weakly convergent subsequence f_{n_k} is uniformly integrable by Theorem 38.12. Choose M so that all its tail integrals above M are less than ε. For $n_k \geq M$, its tail above n_k is no larger, a contradiction. The empty family and zero-measure case satisfy both conditions directly. □

Both directions are now proved in the real finite-measure setting, including the needed dual identification. For comparison, the *Notes on the Dunford–Pettis compactness criterion*, Theorem 1 give a probability-space treatment whose converse invokes Vitali–Hahn–Saks. The measure-algebra argument here supplies the uniform small-set conclusion needed for this sequential result. No equivalence of sequential and topological weak compactness has been used in this argument. We now supply a separate route to topological compactness in this setting.

38.14 From uniform integrability to weak compactness

The following arguments use the proper-ultrafilter and product-compactness results of the topology chapter, in the full Choice setting already specified here. A proper ultrafilter on a set S never contains the empty set; convergence to a point means that every neighbourhood, intersected with S, belongs to the ultrafilter. This is distinct from the earlier local convention that also allowed an empty filter family.

Theorem 38.14 (Banach–Alaoglu for a normed space)**.**
Let E be a normed space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let $R \geq 0$. The closed dual ball

$$B'_R = \{\ell \in E' : \|\ell\| \leq R\}$$

is compact in the weak-star topology $\sigma(E', E)$. The space E need not be complete.

Proof.
For every $x \in E$ put $D_x = \{z \in \mathbb{K} : |z| \leq R\|x\|\}$. These are compact: over the reals they are closed bounded intervals; over the complexes they are closed bounded discs in the two-dimensional real normed space $\mathbb{R}^2$. The finite-dimensional compactness result in the linear-algebra chapter applies in both cases. Tychonoff's theorem makes

$$P = \prod_{x \in E} D_x$$

compact. Let $C \subseteq P$ be the families satisfying

$$v_{x+y} = v_x + v_y, \qquad v_{ax} = a v_x \quad (x, y \in E,\ a \in \mathbb{K}).$$

Each equation is a closed condition on finitely many scalar coordinates, so their intersection C is closed in P, and hence compact.
Every $v \in C$ defines a $\mathbb{K}$-linear functional $\ell_v(x) = v_x$ with $|\ell_v(x)| \leq R\|x\|$. It is continuous and has norm at most R. Conversely, any $\ell \in B'_R$ supplies just such a family. Thus the evaluation map

$$J : B'_R \longrightarrow C, \qquad J(\ell) = (\ell(x))_{x \in E}$$

is bijective. The relative weak-star topology on B'_R and the relative product topology on C are both generated by these same evaluations, so J is a homeomorphism. This proves compactness. The argument includes $R = 0$ and $E = \{0\}$. □
This is compactness for evaluations at vectors of E. The weak topology $\sigma(E', E'')$ tests all continuous linear functionals on E' and is not identified with the weak-star topology by this theorem. For a real Hilbert space, the Riesz representation proved earlier connects the dual evaluations with inner products; the next argument writes that identification directly in the product coordinates.

Theorem 38.15 (Weak compactness of a real Hilbert ball)**.**
If H is a real Hilbert space and $R \geq 0$, the closed ball $B_R = \{x \in H : \|x\| \leq R\}$ is compact in the weak topology.

Proof.
For each $h \in H$, let $I_h = [-R\|h\|, R\|h\|]$ and define

$$J : B_R \longrightarrow \prod_{h \in H} I_h, \qquad J(x)_h = \langle h, x \rangle.$$

The product is compact by Theorem 35.19, since its factors are compact real intervals. Its subset consisting of families v with

$$v_{h+k} = v_h + v_k, \qquad v_{th} = t v_h \quad (h, k \in H,\ t \in \mathbb{R})$$

is closed: each equation is a closed condition on finitely many real coordinates. Such a family defines the continuous linear functional $\ell(h) = v_h$, with $|\ell(h)| \leq R\|h\|$. By Theorem 38.9, it has the form $\ell(h) = \langle h, x \rangle$ for a unique $x \in H$ with $\|x\| = \|\ell\| \leq R$. Thus this closed subset is exactly $J[B_R]$.
The map J is injective: equality of all coordinates, tested on the difference of two points, gives equality of the points. The weak topology on H is precisely the topology induced by these coordinates, since real Hilbert representation supplies every continuous linear functional. Hence J is a homeomorphism onto its image. That image is a closed subset of a compact product, so B_R is weakly compact. The argument includes $R = 0$ and the zero space. □

Lemma 38.6 (Uniform integrability passes to weak closure)**.**
On a finite positive measure space, the weak closure in real L^1 of a uniformly integrable family is uniformly integrable.

Proof.
For a measurable set A, the function $q_A(f) = \int_A |f|\, d\mu$ satisfies

$$q_A(f) = \sup \left\{ \int f g \, d\mu : g \in L^\infty,\ |g| \leq 1 \text{ a.e.},\ g\mathbf{1}_A = g \text{ a.e.} \right\}.$$

The bound follows from $|g| \leq 1$ on A; equality is attained by $g = \operatorname{sgn}(f)\mathbf{1}_A$. Each displayed test is weakly continuous, so $\{f : q_A(f) \leq r\}$ is weakly closed, being the intersection of its closed linear half-spaces. This also applies to $q_\Omega(f) = \|f\|_1$.
Let W be the weak closure of the given family K. Its uniform norm bound passes to W by the closedness just proved. For $\varepsilon > 0$, choose $\delta > 0$ such that $\mu(A) < \delta$ implies $q_A(f) < \varepsilon/2$ for every $f \in K$. For each such A, the weakly closed set $\{f : q_A(f) \leq \varepsilon/2\}$ contains K and therefore W. Thus the members of W satisfy the uniform bound $q_A(f) \leq \varepsilon/2 < \varepsilon$. The small-set characterization now makes W uniformly integrable. Empty families cause no exception. □

Lemma 38.7 (Sequences in a weakly compact set)**.**
Every sequence in a weakly compact subset K of a real normed space E has a weakly convergent subsequence with limit in K. Completeness of E is not needed for this implication.

Proof.

First, every norm-closed linear subspace $M \subseteq E$ is weakly closed. If $x \notin M$, its distance d from M is positive. On $M + \mathbb{R}x$ the functional $\ell_0(m + tx) = t$ is well defined and satisfies

$$|\ell_0(m + tx)| \leq d^{-1}\|m + tx\|.$$

For $t \neq 0$, this follows from $\|m + tx\| \geq |t|d$; for $t = 0$ it is immediate. Hahn–Banach extends ℓ_0 to a continuous linear functional on E that vanishes on M but not on x. Thus M is the intersection of the kernels of the continuous functionals vanishing on it, proving weak closedness. Given a sequence (x_n) in K, let M be its norm-closed linear span. Finite rational linear combinations of its terms form a countable dense subset of M. If $M = \{0\}$, the conclusion is immediate. Otherwise, normalizing the nonzero elements of that dense subset gives a countable dense sequence (u_j) in the unit sphere of M. Choose norm-one functionals $\ell_j \in E'$ with $\ell_j(u_j) = 1$ by Hahn–Banach. They separate points of M: for $z \neq 0$, choose u_j within $1/2$ of $z/\|z\|$; then $|\ell_j(z/\|z\|)| > 1/2$.
The set $C = K \cap M$ is weakly compact, as a closed subset of K. The map

$$C \longrightarrow \mathbb{R}^{\mathbb{N}}, \qquad x \longmapsto (\ell_j(x))_j$$

is continuous and injective. By the compact-to-Hausdorff theorem, it is a homeomorphism onto its image. The countable product is metrizable, for example by

$$d(a, b) = \sum_{j \geq 1} 2^{-j} \min(1, |a_j - b_j|).$$

Finite-coordinate conditions control the initial part of this sum, and its uniformly small tails give the converse neighbourhood comparison, so this metric induces the product topology. Thus C, with its weak topology, is compact and metrizable. The metric compactness criterion supplies a convergent subsequence of (x_n) with limit in $C \subseteq K$, proving the claim. This argument concerns sequences inside a weakly compact set; it does not metrize the weak topology on all of E or identify general nets with sequences. □

Theorem 38.16 (Dunford–Pettis, finite-measure weak compactness)**.**
Let $(\Omega, \mathcal{A}, \mu)$ be a finite positive measure space and $K \subseteq L^1(\mu; \mathbb{R})$. The following are equivalent:

1. K is uniformly integrable;
2. its weak closure is weakly compact;
3. every sequence in K has a subsequence converging weakly to an element of $L^1(\mu)$.

Proof.
The first and third conditions are equivalent by Theorem 38.13. The second implies the third by Lemma 38.7, applied to the weak closure. It remains to prove that the first implies the second. Let W be the weak closure of K. By Lemma 38.6, it is uniformly integrable. If W is empty, it is compact. Otherwise, fix an arbitrary proper ultrafilter $\mathcal{U}$ on W. Set $\varepsilon_j = 2^{-j}$ and choose positive integers m_j such that, with the pointwise truncation $T_j f = \max(-m_j, \min(f, m_j))$,

$$\sup_{f \in W} \|f - T_j f\|_1 \leq \varepsilon_j.$$

Uniform integrability gives these bounds because truncation error is bounded by the corresponding tail integral. Almost-everywhere equal representatives have almost-everywhere equal truncations, so T_j is a well-defined map on L^1 equivalence classes into L^2. Each $T_j f$ lies in the L^2

ball of radius $m_j\mu(\Omega)^{1/2}$. This ball is weakly compact by Theorem 38.15 and completeness of L^2. The image ultrafilter under T_j therefore converges weakly in L^2 to some u_j in that ball. The limit is unique because the weak Hilbert topology is Hausdorff. No continuity of the truncation map from weak L^1 to weak L^2 is asserted or needed: it is simply a map used to push forward an ultrafilter.
Every $g \in L^\infty$ is in L^2 on the finite measure space, so the map $h \mapsto \int gh\,d\mu$ is a continuous functional on L^2. For every $f \in W$,

$$\left|\int g(T_j f - T_k f)\,d\mu\right| \le \|g\|_\infty(\varepsilon_j + \varepsilon_k).$$

Passing to the scalar limits along $\mathcal{U}$ preserves this closed inequality and gives the same bound with $u_j - u_k$ in place of $T_j f - T_k f$. Here sums and differences of scalar ultrafilter limits are justified by intersecting the finitely many required filter sets. Both u_j, u_k belong to L^1 by the finite-measure inclusion $L^2 \subseteq L^1$. Taking $g = \operatorname{sgn}(u_j - u_k)$ gives

$$\|u_j - u_k\|_1 \le \varepsilon_j + \varepsilon_k.$$

Completeness of L^1 yields $u_j \to f_*$ in norm for some $f_* \in L^1$; letting k tend to infinity in this estimate gives $\|u_j - f_*\|_1 \le \varepsilon_j$.
For each $g \in L^\infty$ and $f \in W$,

$$\left|\int g(f - f_*)\,d\mu\right| \le 2\|g\|_\infty\varepsilon_j + \left|\int g(T_j f - u_j)\,d\mu\right|.$$

First choose j making the first term arbitrarily small; the second tends to zero along $\mathcal{U}$ by the weak L^2 limit. Therefore $\int gf$ tends along $\mathcal{U}$ to $\int gf_*$ for every bounded test g. The dual identification in Theorem 38.10 and finite intersections of filter sets show that $\mathcal{U}$ converges to f_* in the weak topology of L^1.
Since W is weakly closed, this limit lies in W: otherwise its open complement would give an empty trace on W in a proper ultrafilter. Every proper ultrafilter on W thus converges in W. Theorem 35.18 makes W weakly compact. The zero-measure case is included, since both function spaces then consist of their zero equivalence class. □
This establishes topological weak compactness in the finite-measure L^1 setting, not an unrestricted theorem that every sequentially compact subset of every topological vector space is compact. The ultrafilter proof uses weak compactness of Hilbert balls and uniform truncation; the reverse implication uses the separable span of a given sequence. These extra arguments are the bridge that a bare reference to convergence of sequences would omit.

38.15 Passing from real to complex scalars

The preceding real proofs also give the complex finite-measure criterion. The passage uses the underlying real space, rather than silently changing the scalar field in a representation theorem.

Lemma 38.8 (Realification preserves the weak topology)**.**
Let E be a complex normed space and $E_{\mathbb{R}}$ its underlying real normed space. Their weak topologies on the common carrier agree.

Proof.

The real and imaginary parts of every continuous complex-linear functional are continuous real-linear functionals. Thus that complex functional is continuous for the real weak topology. Conversely, for a continuous real-linear functional ℓ put

$$L(x) = \ell(x) - i\ell(ix).$$

Real linearity and $L(ix) = iL(x)$ show complex linearity, and $|L(x)| \le 2\|\ell\|\,\|x\|$ gives continuity. Since $\ell = \operatorname{Re} L$, every real test is continuous for the complex weak topology. The two defining families therefore induce the same topology. No completeness assumption is needed. □

For a finite positive measure μ, write $R = L^1(\mu;\mathbb{R})$ and $E = L^1(\mu;\mathbb{C})$. The maps

$$J : E_{\mathbb{R}} \longrightarrow R \times R, \qquad J(f) = (\operatorname{Re} f, \operatorname{Im} f), \qquad J^{-1}(u,v) = u + iv$$

are well defined on almost-everywhere classes and are inverse real-linear bounded maps. Indeed,

$$\|u + iv\|_1 \le \|u\|_1 + \|v\|_1 \le 2\|u + iv\|_1.$$

Every continuous real-linear functional on the product with its sum norm has the form $(u,v) \mapsto \ell_1(u) + \ell_2(v)$, by restriction to the two axes; conversely every such sum is continuous. Hence its weak topology is the product of the two real weak topologies. Composing functionals with J and J^{-1} and using the lemma shows that J is a homeomorphism from complex weak L^1 onto this weak product.

In particular, a net converges weakly in complex L^1 precisely when its real and imaginary parts converge weakly in real L^1. Equivalently, all integrals $\int gf\,d\mu$ converge for $g \in L^\infty(\mu;\mathbb{C})$. For the converse in this last assertion it suffices to take real bounded g: the real and imaginary parts of these integrals test both real coordinates, whose duals were identified above. For the forward direction, $f \mapsto \int gf\,d\mu$ is complex-linear and bounded by $\|g\|_\infty\|f\|_1$. This pairing is bilinear; no conjugation of g is implicit. A sesquilinear notation would replace g by its conjugate and give the same family of tests.

Theorem 38.17 (Dunford–Pettis for complex scalars)**.**
For $K \subseteq L^1(\mu;\mathbb{C})$ on a finite positive measure space, the following are equivalent:

1. K is uniformly integrable, with tails defined using $|f|$;
2. its closure in the complex weak topology is compact;
3. every sequence in K has a subsequence converging weakly to an element of complex L^1.

Proof.
Put $K_1 = \{\operatorname{Re} f : f \in K\}$ and $K_2 = \{\operatorname{Im} f : f \in K\}$. First, K is uniformly integrable if and only if both coordinate families are. One implication follows from $|\operatorname{Re} f|, |\operatorname{Im} f| \le |f|$: each coordinate tail integral above M is bounded by the corresponding $|f|$ tail integral. Conversely, uniform integrability of K_1, K_2 gives uniform norm bounds and uniform small-set integral bounds. The inequality $|u + iv| \le |u| + |v|$ gives both bounds for K. If C bounds its norms, then $\mu(\{|f| > M\}) \le C/M$; applying the small-set bound to these sets proves the required uniform tail estimate. This is the same small-set argument as in the real case and uses no ordering on complex values.

If K is uniformly integrable, the real theorem makes the weak closures C_1, C_2 of K_1, K_2 compact. Their product is compact and Hausdorff. The weak closure of $J[K]$ is contained in $C_1 \times C_2$, since this product is closed, and is closed there. It is therefore compact. The weak homeomorphism J

proves the second condition. Conversely, if the weak closure of K is compact, its two coordinate images are compact, hence closed in the Hausdorff real weak spaces. They contain K_1, K_2, so each coordinate weak closure is a closed subset of a compact set. The real theorem gives coordinate uniform integrability, and the preceding equivalence gives uniform integrability of K. For the sequential assertion, coordinate uniform integrability first gives a subsequence with weakly convergent real parts. From it extract a further subsequence with weakly convergent imaginary parts. The real parts still converge, and J^{-1} gives a complex weak limit. Conversely, suppose every sequence in K has such a subsequence. Any sequence in K_1 can be lifted to a sequence in K by choosing one preimage of each term, using the Choice convention of this chapter. A complex weakly convergent subsequence gives a real weakly convergent subsequence of the original coordinate sequence. The same holds for K_2. The real sequential theorem makes both coordinate families uniformly integrable, and hence makes K uniformly integrable. If K is empty all conditions hold directly; if $\mu(\Omega) = 0$, both real and complex spaces consist only of their zero class. □

Why the hypotheses matter. The small-set lemma shows why both norm boundedness and control of concentration matter. An L^1 bound alone does not control concentration on small sets. For example, on $[0, 1]$ with Lebesgue measure, let $f_n = n\mathbf{1}_{(0,1/n)}$. Every f_n has norm one, but for $n > M$ its tail integral above M is still one. This family therefore fails the displayed uniform-integrability condition. It also has no weakly convergent subsequence: testing against $\mathbf{1}_{[\delta,1]}$, for each $\delta > 0$, forces a putative limit to vanish on $(0, 1]$, while the constant test function one would give integral one. More explicitly, the same test works for every measurable subset of $[\delta, 1]$, so the conclusion is almost-everywhere vanishing, not merely zero integral on an interval.
Nor should the finite-measure hypothesis be dropped silently. On $\mathbb{R}$ with Lebesgue measure, the functions $f_n = \mathbf{1}_{[n,n+1)}$ satisfy the tail condition, since they are bounded by one. They have no weakly convergent subsequence in L^1: a putative weak limit has integral zero over every bounded measurable set and hence is zero almost everywhere, whereas testing against the constant function one would give integral one. This example exhibits mass escaping the finite regions of the carrier. A criterion for an infinite measure space needs additional hypotheses or a stronger uniform-integrability convention.
Here "Dunford–Pettis" denotes the equivalent weak-compactness and sequential criteria proved for subsets of real or complex L^1 on a finite measure space. The compact-operator results earlier in the chapter concern norm-compact images and their consequences for weakly convergent sequences. An operator property or an integrability notion for vector-valued functions requires its own definition and hypotheses, even when it shares the Dunford–Pettis name.

39 Probability theory

For the measure-theoretic setting, a probability space is a measure space $(\Omega, \mathcal{A}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A real random variable is a measurable function $X : \Omega \to \mathbb{R}$. Its expectation is the integral $\mathbb{E}X = \int_\Omega X \, d\mathbb{P}$ when X is integrable; no default value for an undefined signed integral is implicit. In $L^1(\mathbb{P})$, random variables are identified when they agree almost surely. The finite-measure conventions of the functional-analysis section therefore apply directly. In particular, uniform integrability controls more than boundedness of the expectations of absolute values.
Two uses of "weak convergence" should be distinguished. Weak convergence in the Banach space $L^1(\mathbb{P})$ tests $\mathbb{E}(ZX_n)$ against each fixed $Z \in L^\infty(\mathbb{P})$. Convergence in distribution of real random variables instead tests $\mathbb{E}(h(X_n))$ against bounded continuous functions $h : \mathbb{R} \to \mathbb{R}$. The latter concerns their laws; it does not even require the random variables to be defined on a common probability space. A later theorem comparing these notions must state its assumptions rather than identifying them by terminology.

39.1 Two separations between modes of convergence

The following examples make the distinction concrete; all variables are defined on a single probability space.

Example 39.1 (The same law need not give weak L^1 convergence)**.**
Let X take values 1 and -1, each with probability $1/2$, and put $X_n = (-1)^n X$. Every X_n has the same law as X, so their laws converge to the law of X. Nevertheless, testing weak L^1 convergence against the bounded random variable $Z = X$ gives

$$\mathbb{E}(ZX_n) = (-1)^n,$$

which does not converge. Thus the sequence has no weak L^1 limit. Even uniform boundedness of the variables does not repair this implication: information about joint position on the probability space is absent from a law.

Example 39.2 (Weak L^1 convergence need not give convergence in probability)**.**
On $[0, 1)$ with Lebesgue probability measure, let

$$r_n(t) = (-1)^{\lfloor 2^n t \rfloor}, \qquad n \geqslant 1.$$

The functions r_n have norm one in L^2 and are pairwise orthogonal. Indeed, for $m < n$, r_m is constant on each dyadic interval of length 2^{-m}, whereas r_n takes each sign on half of that interval. For any real $g \in L^\infty$, put $c_n = \int_0^1 g r_n$. Expanding a square gives, for each N,

$$0 \leqslant \left\| g - \sum_{n=1}^{N} c_n r_n \right\|_2^2 = \|g\|_2^2 - \sum_{n=1}^{N} c_n^2.$$

Thus $c_n \to 0$. The finite-measure identification of the L^1 dual with L^∞ shows $r_n \rightharpoonup 0$ in L^1. But $\mathbb{P}(|r_n| > 1/2) = 1$, so there is no convergence in probability to zero. Their laws are all the symmetric two-point law, not a sequence converging to the law of zero: the bounded continuous test function $h(x) = \min(1, |x|)$ has expectation one for every r_n and value zero at zero.

In contrast, convergence in probability together with uniform integrability gives L^1 convergence by Theorem 38.8. It then gives convergence of expectations because $|\mathbb{E}X_n - \mathbb{E}X| \leqslant \mathbb{E}|X_n - X|$. The concentration example $X_n = n\mathbf{1}_{(0,1/n)}$ shows why uniform integrability is needed: $X_n \to 0$ in probability while $\mathbb{E}X_n = 1$ for all n.

Corollary 39.1 (Weak convergence together with convergence in probability)**.**
On a probability space, suppose $X_n \rightharpoonup X$ in L^1 and $X_n \to Y$ in probability, where Y is a measurable real random variable. Then $Y = X$ almost surely and $\|X_n - X\|_1 \to 0$.

Proof.
Theorem 38.12 gives uniform integrability of $\{X_n : n \geq 1\}$. Vitali's theorem gives $Y \in L^1$ and $\|X_n - Y\|_1 \to 0$. For every bounded Z, both convergences then give $\mathbb{E}(ZX) = \mathbb{E}(ZY)$. Taking $Z = \operatorname{sgn}(X - Y)$ yields $\mathbb{E}|X - Y| = 0$, so $X = Y$ almost surely. Replacing Y by its equal L^1 class proves the norm-convergence statement. □

39.2 Conditional expectation from Hilbert representation

Fix a finite positive measure space $(\Omega, \mathcal{A}, \mu)$ and a sub-sigma-algebra $\mathcal{G} \subseteq \mathcal{A}$ containing Ω. All functions in this subsection are real-valued. Write $L^p(\mathcal{G})$ for the space formed using the restricted measure $\mu|_{\mathcal{G}}$. Its natural map into $L^p(\mu)$ is isometric and injective: for two $\mathcal{G}$-measurable representatives their difference and its nonzero set are $\mathcal{G}$-measurable, and the restricted measure has the same value on that set. We identify these spaces with their images. An element of $L^p(\mathcal{G})$ is required to have a $\mathcal{G}$-measurable representative; arbitrary modifications on null subsets are not asserted to preserve that measurability.

Theorem 39.1 (Existence and uniqueness of conditional expectation)**.**
For each $f \in L^1(\mu)$ there is a unique $g \in L^1(\mathcal{G})$ such that

$$\int_A g \, d\mu = \int_A f \, d\mu \qquad (A \in \mathcal{G}).$$

Write $P_{\mathcal{G}} f = g$, also denoted $\mathbb{E}_\mu[f \mid \mathcal{G}]$. This is a linear positive projection onto $L^1(\mathcal{G})$, preserves the total integral, and is a contraction in L^1:

$$\|P_{\mathcal{G}} f - P_{\mathcal{G}} k\|_1 \leq \|f - k\|_1.$$

The measure need not be complete, and its total mass need not be one.

Proof.
First establish uniqueness using only the displayed characterization. If g_1, g_2 satisfy it, take $\mathcal{G}$-measurable representatives and put $v = g_1 - g_2$. For $A_n = \{v \geq 1/n\} \in \mathcal{G}$,

$$0 = \int_{A_n} v \, d\mu \geq \mu(A_n)/n,$$

so each A_n is null. Applying the same argument to $-v$ gives $v = 0$ almost everywhere. This proves uniqueness in the quotient.
For existence first suppose $f \in L^2(\mu)$. On the real Hilbert space $L^2(\mathcal{G})$ the functional

$$u \longmapsto \int f u \, d\mu$$

is continuous by Cauchy–Schwarz. Completeness of L^2 for the restricted measure and Theorem 38.9 give $g \in L^2(\mathcal{G})$ with $\int ug \, d\mu = \int uf \, d\mu$ for every $u \in L^2(\mathcal{G})$. Finite measure implies

$g, f \in L^1$ and puts every $\mathcal{G}$-indicator in $L^2(\mathcal{G})$, giving the required characterization. The sign of a $\mathcal{G}$-measurable representative of g is also in this Hilbert space, so

$$\|g\|_1 = \int \operatorname{sgn}(g) g \, d\mu = \int \operatorname{sgn}(g) f \, d\mu \le \|f\|_1.$$

Uniqueness makes this construction linear on L^2: linear combinations of characterized outputs characterize the corresponding combinations of inputs. Consequently it contracts L^1 distances between L^2 inputs.
For arbitrary $f \in L^1$, let $f_n = \max(-n, \min(f, n))$. These are well-defined equivalence classes in $L^2(\mu)$ and converge to f in L^1, by integrable-tail control and finite measure. The preceding contraction makes their outputs g_n Cauchy in $L^1(\mathcal{G})$. Completeness for the restricted measure supplies $g \in L^1(\mathcal{G})$ with $g_n \to g$ in that norm, hence also in ambient L^1. For every $A \in \mathcal{G}$,

$$\int_A g \, d\mu = \lim_n \int_A g_n \, d\mu = \lim_n \int_A f_n \, d\mu = \int_A f \, d\mu.$$

The limits follow from the L^1 bound for integrals over a set. This proves existence without invoking Radon–Nikodym.
Linearity now follows from uniqueness for all L^1 inputs. Taking $A = \Omega$ proves preservation of the integral. If $f \ge 0$ almost everywhere, then $\int_A g \, d\mu \ge 0$ for every $A \in \mathcal{G}$. The negative level sets $\{g \le -1/n\}$ show, just as in the uniqueness argument, that $g \ge 0$ almost everywhere. Thus the map is positive. If $f \in L^1(\mathcal{G})$, it itself satisfies the characterization, so $P_{\mathcal{G}} f = f$; this proves the projection assertion. Finally the construction and L^1 convergence give $\|P_{\mathcal{G}} f\|_1 = \lim_n \|g_n\|_1 \le \lim_n \|f_n\|_1 = \|f\|_1$. Apply this to $f - k$ and use linearity for the distance estimate. The zero-measure case is included: each of these quotient spaces then consists of its zero class. □

Example 39.3 (Conditioning on a finite partition)**.**
Let $A_1, \dots, A_m$ be a finite measurable partition of Ω, and let $\mathcal{G}$ consist of its unions of atoms. For $f \in L^1(\mu)$,

$$P_{\mathcal{G}} f = \sum_{\substack{1 \le j \le m \\ \mu(A_j) > 0}} \frac{\int_{A_j} f \, d\mu}{\mu(A_j)} \mathbf{1}_{A_j} \quad \text{as an almost-everywhere class.}$$

Indeed the displayed function is $\mathcal{G}$-measurable and integrable. On a positive-measure atom its integral agrees with that of f by construction; on a null atom both integrals are zero. Finite additivity therefore gives the defining identity on every union of atoms, and uniqueness proves the formula. The displayed representative is zero on null atoms. Any other finite constants on those atoms give the same class and remain $\mathcal{G}$-measurable; there is no division by their zero measures. Thus conditioning on this finite partition averages the input over each observable atom without assigning a determined numerical average on a null event.

For $f \in L^2$, the construction also gives the familiar orthogonality identity $\int u(f - P_{\mathcal{G}} f) \, d\mu = 0$ for $u \in L^2(\mathcal{G})$. Taking $u = P_{\mathcal{G}} f$ and expanding the square gives

$$\|f\|_2^2 = \|P_{\mathcal{G}} f\|_2^2 + \|f - P_{\mathcal{G}} f\|_2^2.$$

The extension to L^1 retains the integral characterization; it does not assert that an arbitrary L^1 input has finite L^2 norm.

Proposition 39.1 (Bounded tests, pullout, and the tower identities)**.**
For $f \in L^1(\mu)$ and $u \in L^\infty(\mathcal{G})$,

$$\int uP_{\mathcal{G}} f \, d\mu = \int uf \, d\mu, \qquad P_{\mathcal{G}}(uf) = uP_{\mathcal{G}} f.$$

If $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{A}$ are sub-sigma-algebras on the same carrier, then

$$P_{\mathcal{G}} P_{\mathcal{H}} f = P_{\mathcal{G}} f, \qquad P_{\mathcal{H}} P_{\mathcal{G}} f = P_{\mathcal{G}} f.$$

Proof.
The first identity holds for $\mathcal{G}$-indicators by definition and therefore for real $\mathcal{G}$-measurable simple functions by linearity. A bounded representative of u is uniformly approximated by such simple functions. The approximation error in the two integrals is at most $\|u - s\|_\infty(\|P_{\mathcal{G}} f\|_1 + \|f\|_1)$, which tends to zero. This proves the bounded-test identity.
The product $uP_{\mathcal{G}} f$ belongs to $L^1(\mathcal{G})$. Apply the first identity to $u\mathbf{1}_A$ for $A \in \mathcal{G}$: its integral on A equals $\int_A uf \, d\mu$. Uniqueness of conditional expectation proves pullout.
For the first tower identity, take $A \in \mathcal{G}$. It also belongs to $\mathcal{H}$, so successive uses of the characterization give

$$\int_A P_{\mathcal{G}} P_{\mathcal{H}} f \, d\mu = \int_A P_{\mathcal{H}} f \, d\mu = \int_A f \, d\mu.$$

The result is $\mathcal{G}$-measurable, hence equals $P_{\mathcal{G}} f$ by uniqueness. For the second identity, $P_{\mathcal{G}} f$ already has an $\mathcal{H}$-measurable representative, so the projection property applies. □

Corollary 39.2 (Uniform integrability under varying conditional expectations)**.**
On the fixed finite positive measure space, let $K \subseteq L^1(\mu; \mathbb{R})$ be uniformly integrable and let $(\mathcal{G}_i)_{i \in I}$ be any set-indexed family of sub-sigma-algebras of $\mathcal{A}$, each containing Ω. Then

$$\{P_{\mathcal{G}_i} f : f \in K, \ i \in I\}$$

is uniformly integrable. In particular this holds for a fixed sub-sigma-algebra.

Proof.
The conclusion is immediate if K or I is empty. Otherwise finite measure and uniform integrability give $C = \sup_{f \in K} \|f\|_1 < \infty$. Positivity and linearity, applied to $-|f| \le f \le |f|$, give

$$|P_{\mathcal{G}_i} f| \le P_{\mathcal{G}_i} |f| \quad \text{almost everywhere.}$$

Choose a $\mathcal{G}_i$-measurable representative of the output, and for $M > 0$ put $A = \{|P_{\mathcal{G}_i} f| > M\} \in \mathcal{G}_i$. The L^1 contraction and Markov's inequality give $\mu(A) \le C/M$. For any $R > 0$, the defining integral identity for conditional expectation, now applied to $|f|$, yields

$$\begin{aligned} \int_A |P_{\mathcal{G}_i} f| \, d\mu &\le \int_A P_{\mathcal{G}_i} |f| \, d\mu = \int_A |f| \, d\mu \\ &\le \int_{\{|f| > R\}} |f| \, d\mu + R\mu(A) \le \int_{\{|f| > R\}} |f| \, d\mu + \frac{RC}{M}. \end{aligned}$$

Given $\varepsilon > 0$, choose R making the first term less than $\varepsilon/2$ uniformly for $f \in K$, and then choose M making $RC/M < \varepsilon/2$. These choices are independent of both f and i, proving uniform integrability. This direct estimate uses neither weak compactness nor a choice of representatives simultaneously for the whole indexed family. □

39.3 Probability densities and compactness

Fix a finite positive measure space $(\Omega, \mathcal{A}, \mu)$. A probability density relative to μ is an $f \in L^1(\mu)$ with $f \geq 0$ almost everywhere and $\int f \, d\mu = 1$. It determines the probability measure $\nu_f(A) = \int_A f \, d\mu$. If $\mu(\Omega) = 0$, there are no such densities; the statements about a sequence of them then have no instances.

Proposition 39.2 (The density set is weakly closed)**.**
The set

$$\mathcal{D}_\mu = \{f \in L^1(\mu; \mathbb{R}) : f \geq 0 \text{ a.e.}, \ \int f \, d\mu = 1\}$$

is weakly closed in $L^1(\mu; \mathbb{R})$.

Proof.
The indicator of every measurable set A is a bounded test function, so $f \mapsto \int_A f \, d\mu$ is weakly continuous. The same holds for the total integral, using the constant test function one. Therefore

$$\left\{ f : \int f \, d\mu = 1 \right\} \cap \bigcap_{A \in \mathcal{A}} \left\{ f : \int_A f \, d\mu \geq 0 \right\}$$

is weakly closed. We claim it equals $\mathcal{D}_\mu$. Every density belongs to it. Conversely, for a member f choose a measurable representative and put $A_k = \{f \leq -1/k\}$. The asserted integral inequalities give

$$0 \leq \int_{A_k} f \, d\mu \leq -\mu(A_k)/k,$$

so $\mu(A_k) = 0$. Their countable union is $\{f < 0\}$, proving $f \geq 0$ almost everywhere. Its total integral is already one, which proves the claimed equality and weak closedness. □

Corollary 39.3 (Weak limits and subsequences of densities)**.**
Every weak L^1 limit of a net of probability densities is a probability density. In particular every uniformly integrable sequence of probability densities has a subsequence converging weakly in L^1 to a probability density.

Proof.
The weakly closed density set contains every limit of a net taking values in it. For a uniformly integrable sequence, Theorem 38.13 supplies a weakly convergent subsequence; its limit belongs to $\mathcal{D}_\mu$ by closedness. □

Corollary 39.4 (Weak compactness of families of densities)**.**
For $K \subseteq \mathcal{D}_\mu$, uniform integrability is equivalent to compactness of its weak closure in L^1. When these conditions hold, every member of that compact closure is still a probability density. In particular, a weakly closed uniformly integrable family of probability densities is weakly compact.

Proof.
The equivalence is Theorem 38.16. Since $\mathcal{D}_\mu$ is weakly closed and contains K, it contains the weak closure of K. If K itself is weakly closed, that closure is K. The empty family, including the only possible family when $\mu(\Omega) = 0$, satisfies the conclusions directly. □

Lemma 39.1 (Integrable observables on a dominated interval)**.**
Let $h \in L^1(\mu;\mathbb{R})$ be nonnegative almost everywhere and put

$$D_h = \{f \in L^1(\mu;\mathbb{R}) : 0 \le f \le h \text{ a.e.}\}.$$

If ψ is a measurable real function with $\int |\psi| h \, d\mu < \infty$, then

$$\Phi_\psi : D_h \longrightarrow \mathbb{R}, \qquad \Phi_\psi(f) = \int \psi f \, d\mu$$

is continuous for the relative weak topology of L^1.

Proof.
Choose a nonnegative measurable representative of h. The estimate $|\psi f| \le |\psi| h$ almost everywhere makes every displayed integral finite and independent of the representative of f. Put $\psi_m = \max(-m, \min(\psi, m))$. Each bounded test Φ_{ψ_m} is weakly continuous on all of L^1, and

$$\sup_{f \in D_h} |\Phi_\psi(f) - \Phi_{\psi_m}(f)| \le \int_{\{|\psi| > m\}} |\psi| h \, d\mu \longrightarrow 0.$$

Dominated convergence proves the limit, since ψ is finite-valued and $|\psi|h$ is integrable. Thus the restrictions of the bounded-test functionals converge uniformly to Φ_ψ on D_h. Explicitly, given a tolerance, choose m so that both approximation errors are less than one third of it, and then a relative weak neighbourhood controlling the difference of the two Φ_{ψ_m} values by its remaining third. This proves continuity, for nets as well as sequences. □
The conclusion is restricted to the dominated interval. An unbounded observable need not define a finite continuous functional on all of L^1; domination supplies the uniform approximation used in the proof.

Theorem 39.2 (Attainment under domination and moment constraints)**.**
Let $h \in L^1(\mu;\mathbb{R})$ be nonnegative almost everywhere. Let $(\varphi_i)_{i \in I}$ be a set-indexed family of measurable real functions with $\int |\varphi_i| h \, d\mu < \infty$ for every i, and let $(c_i)_{i \in I}$ be a family of real numbers. Suppose the family

$$C = \left\{ f \in \mathcal{D}_\mu : f \le h \text{ a.e.}, \quad \int \varphi_i f \, d\mu = c_i \text{ for every } i \in I \right\}$$

is nonempty. Then C is weakly compact. For every measurable real observable ψ with $\int |\psi| h \, d\mu < \infty$, there are $f_-, f_+ \in C$ such that

$$\int \psi f_- \, d\mu \le \int \psi f \, d\mu \le \int \psi f_+ \, d\mu \qquad (f \in C).$$

No countability or finiteness condition on I is required.

Proof.
The condition $f \le h$ almost everywhere is equivalent to

$$\int_A f \, d\mu \le \int_A h \, d\mu \qquad (A \in \mathcal{A}).$$

One implication is monotonicity of the integral. For the converse, apply the positivity argument from Proposition 39.2 to $h - f$: its integral on every measurable set is nonnegative, so $h - f \geq 0$ almost everywhere. Each displayed inequality defines a weakly closed half-space. Together with the weakly closed density set, these inequalities make $C_0 = \{f \in \mathcal{D}_\mu : f \leq h \text{ a.e.}\}$ weakly closed. All members of C_0 satisfy $0 \leq f \leq h$ almost everywhere. Consequently

$$\sup_{f \in C_0} \int_{\{f > M\}} f \, d\mu \leq \int_{\{h > M\}} h \, d\mu \longrightarrow 0.$$

The vanishing-tail lemma makes C_0 uniformly integrable. Theorem 38.16 makes C_0 weakly compact. By Lemma 39.1, each moment map is continuous on $C_0 \subseteq D_h$. Therefore C is an intersection of relatively closed subsets of C_0, and is compact, even for arbitrarily many set-indexed constraints. The same lemma makes the objective continuous on C, so its image is a nonempty compact subset of $\mathbb{R}$. Such a subset has a minimum and maximum: it is bounded, and its infimum and supremum belong to it by closedness. Their preimages in C give f_- and f_+. □

Bounded observables and bounded moment functions satisfy these weighted integrability hypotheses automatically, since $h \in L^1$. Thus the theorem includes the bounded-test case while also permitting unbounded observables whose absolute values are integrable against the common dominating density bound.

The nonemptiness hypothesis is a genuine feasibility condition on the moments. Without moment constraints, the dominated density family is nonempty exactly when $\int h \, d\mu \geq 1$. Necessity follows by integrating $f \leq h$. For sufficiency put $H = \int h \, d\mu \geq 1$ and take $f = h/H$: this is a density and $0 \leq f \leq h$. Extra prescribed moments may make the family empty, so the theorem does not assert feasibility of arbitrary numerical data.

Example 39.4 (Attaining expectations of an unbounded observable)**.**
On $(0, 1)$ with Lebesgue measure, take $h = 2$ and $\psi(x) = x^{-1/2}$. Although ψ is unbounded, $\int_0^1 |\psi| h \, dx = 4 < \infty$. Among all probability densities with $0 \leq f \leq 2$ almost everywhere, the maximum and minimum expectations are attained by

$$f_+ = 2\mathbf{1}_{(0,1/2)}, \qquad f_- = 2\mathbf{1}_{(1/2,1)},$$

respectively, and their values are

$$\int_0^1 \psi f_+ \, dx = 2\sqrt{2}, \qquad \int_0^1 \psi f_- \, dx = 4 - 2\sqrt{2}.$$

Both functions are feasible because their integrals are one. To verify optimality without assuming it from the shape of ψ, subtract the threshold value $\psi(1/2) = \sqrt{2}$. Since all feasible densities have the same mass,

$$\int \psi(f - f_+) \, dx = \int (\psi - \sqrt{2})(f - f_+) \, dx \leq 0.$$

On $(0, 1/2)$ the first factor is positive and the second is nonpositive; on $(1/2, 1)$ their signs reverse. Both products are integrable by the common bound h. Replacing f_+ by f_- makes the product nonnegative on both intervals, proving the minimum assertion. Finally, $\int_0^b x^{-1/2} \, dx = 2\sqrt{b}$ gives the two displayed values. Thus domination permits optimization of this unbounded test even though it is not an element of the ambient L^∞ dual-test space.

Proposition 39.3 (Setwise testing for densities)**.**
For probability densities f_n and f, the following are equivalent:

$$f_n \rightharpoonup f \text{ in } L^1(\mu), \qquad \nu_{f_n}(A) \longrightarrow \nu_f(A) \text{ for every } A \in \mathcal{A}.$$

Proof.
Weak convergence implies the displayed convergence by testing indicators. Conversely, the assumption gives convergence of integrals against every real measurable simple function. For $g \in L^\infty$, choose a bounded representative and a simple s with $\|g - s\|_\infty < \varepsilon$. Then

$$\left|\int g(f_n - f)\,d\mu\right| \le \left|\int s(f_n - f)\,d\mu\right| + \varepsilon(\|f_n\|_1 + \|f\|_1) = \left|\int s(f_n - f)\,d\mu\right| + 2\varepsilon.$$

Take the upper limit in n, then let ε decrease to zero. The dual identification in Theorem 38.10 gives weak L^1 convergence. The uniform norm bound, here exactly one, is what allows indicators to be extended to all bounded tests. □

39.4 Convergence tested on a generating algebra

Continue with the finite positive measure space $(\Omega, \mathcal{A}, \mu)$. An *algebra of sets* on Ω contains Ω and is closed under complements relative to Ω and finite unions. It is therefore a π-system as well. Let $\mathcal{B} \subseteq \mathcal{A}$ be such an algebra with $\sigma(\mathcal{B}) = \mathcal{A}$. No countability assumption on $\mathcal{B}$ is imposed.

Lemma 39.2 (Approximation by a generating algebra)**.**
For every $A \in \mathcal{A}$ and every $\varepsilon > 0$ there is $B \in \mathcal{B}$ with $\mu(A \triangle B) < \varepsilon$.

Proof.
Let $\mathcal{D}$ consist of the measurable sets with this approximation property for every positive tolerance. It contains $\mathcal{B}$ and hence Ω. It is closed under complements because

$$(\Omega \setminus A) \triangle (\Omega \setminus B) = A \triangle B,$$

and complements of members of $\mathcal{B}$ remain in $\mathcal{B}$.
Suppose $A_n \in \mathcal{D}$ are pairwise disjoint and put $A = \bigcup_n A_n$. Since μ is finite, countable additivity gives $\sum_n \mu(A_n) = \mu(A) < \infty$. Given $\varepsilon > 0$, choose $N \ge 1$ with $\mu(\bigcup_{n>N} A_n) < \varepsilon/2$. For $1 \le n \le N$, choose $B_n \in \mathcal{B}$ with $\mu(A_n \triangle B_n) < \varepsilon/(2N)$. The set $B = \bigcup_{n=1}^{N} B_n$ is in $\mathcal{B}$, and

$$A \triangle B \subseteq \bigcup_{n>N} A_n \;\cup \bigcup_{n=1}^{N} (A_n \triangle B_n).$$

Subadditivity gives $\mu(A \triangle B) < \varepsilon$. Thus $\mathcal{D}$ is a Dynkin system containing the π-system $\mathcal{B}$. Theorem 37.1 gives $\mathcal{A} = \sigma(\mathcal{B}) \subseteq \mathcal{D}$, proving the claim. Only finitely many approximating sets were chosen for each fixed tolerance; no enumeration of $\mathcal{B}$ was needed. □

Theorem 39.3 (A generating-algebra test for weak L^1 convergence)**.**
Let $f_n, f \in L^1(\mu; \mathbb{R})$. Then f_n converges weakly to f if and only if $\{f_n : n \ge 1\}$ is uniformly integrable and

$$\int_B f_n\,d\mu \longrightarrow \int_B f\,d\mu \qquad \text{for every } B \in \mathcal{B}.$$

The functions need not be positive or normalized densities.

Proof.
Weak convergence gives the displayed limits by testing indicators, and gives uniform integrability by Theorem 38.12.
For the converse, adjoin f to the uniformly integrable family. Its vanishing tail integrals show that the enlarged family is still uniformly integrable. The uniform-small-set lemma therefore gives, for every $\eta > 0$, a $\delta > 0$ such that

$$\mu(C) < \delta \implies \int_C |f_n|\,d\mu < \eta/2 \text{ for all } n, \quad \int_C |f|\,d\mu < \eta/2.$$

For an arbitrary $A \in \mathcal{A}$, choose $B \in \mathcal{B}$ with $\mu(A \triangle B) < \delta$ using the lemma. Then

$$\left|\int_A (f_n - f)\,d\mu\right| \le \left|\int_B (f_n - f)\,d\mu\right| + \int_{A\triangle B} (|f_n| + |f|)\,d\mu.$$

The first term tends to zero by hypothesis and the second is less than η, uniformly in n. Since η is arbitrary, $\int_A (f_n - f) \to 0$ for every measurable A. Taking finite linear combinations gives convergence against every real measurable simple function.
Uniform integrability on the finite measure space also gives $C = \sup_n \|f_n\|_1 + \|f\|_1 < \infty$. For $g \in L^\infty$, choose a bounded measurable representative. For each $a > 0$, rounding its values to a sufficiently fine finite grid gives a measurable simple s with $\|g - s\|_\infty < a$. Consequently

$$\left|\int g(f_n - f)\,d\mu\right| \le \left|\int s(f_n - f)\,d\mu\right| + aC.$$

Take the upper limit in n, then let a decrease to zero. Thus all bounded-test integrals converge, and the dual identification in Theorem 38.10 proves weak L^1 convergence. □
For densities, this criterion permits setwise convergence to be checked on a generating algebra once uniform integrability is established. The four-point example in the measure chapter only had agreement on an arbitrary generating family. Its generated algebra also contains $\{1\}$, which distinguishes the two measures. Here closure under the algebra operations supplies the needed testing sets, while uniform integrability makes approximation in μ-measure control the integrals uniformly throughout the sequence.

Example 39.5 (Algebra tests and bounded norms do not replace uniform integrability)**.**
Let $\Omega = [0,1]$ with its Borel sigma-algebra and Lebesgue measure restricted to that sigma-algebra. Let $\mathcal{B}$ be the algebra formed by finite Boolean combinations, relative to Ω, of the sets $\Omega \cap (p,q)$ with rational $p < q$. It generates the Borel sigma-algebra: these sets form a countable base for the relative usual topology, so every relatively open set is a countable union of them; conversely all the generators are Borel. We use the Borel space here, without asserting that this algebra generates the completed Lebesgue sigma-algebra.
Fix an irrational $a \in (0,1)$ and put

$$\delta_n = \frac{\min(a, 1-a)}{n+1}, \qquad f_n = \delta_n^{-1}\big(\mathbf{1}_{(a,a+\delta_n)} - \mathbf{1}_{(a-\delta_n,a)}\big).$$

Both intervals lie in $[0,1]$, are disjoint, and have length δ_n. Hence $\int f_n = 0$ and $\|f_n\|_1 = 2$.
Each fixed $B \in \mathcal{B}$ is constant in membership on some open interval about a. Indeed, its finite Boolean expression involves only finitely many rational endpoints, none equal to a; choose a small interval avoiding all these endpoints. Every generator in that expression, and therefore B

itself, either contains this entire interval or misses it. For all sufficiently large n, the support of f_n lies in that interval. Thus

$$\int_B f_n \, d\mu = 0 \quad \text{eventually for every fixed } B \in \mathcal{B}.$$

These tests converge to the corresponding integrals of $f = 0$. Nevertheless the bounded Borel test $g = \mathbf{1}_{(a,1]}$ satisfies $\int g f_n \, d\mu = 1$ for every n, so f_n does not converge weakly to zero. The family also fails uniform integrability directly: for any $M > 0$, choose n with $\delta_n^{-1} > M$. Then

$$\int_{\{|f_n| > M\}} |f_n| \, d\mu = 2.$$

The bounded norms and convergence on the entire generating algebra therefore do not suffice without the uniform-integrability condition.

39.5 Total variation of probability densities

Theorem 39.4 (Total variation and Scheffe's lemma)**.**
For probability densities f, g, define the total variation distance by

$$d_{\mathrm{TV}}(\nu_f, \nu_g) = \sup_{A \in \mathcal{A}} |\nu_f(A) - \nu_g(A)|.$$

Then $d_{\mathrm{TV}}(\nu_f, \nu_g) = \frac{1}{2}\|f - g\|_1$. If probability densities f_n converge almost everywhere to a probability density f, then $\|f_n - f\|_1 \to 0$, hence their measures converge in this distance. The assumption that the limit has integral one is essential.

Proof.
Choose nonnegative measurable representatives of the densities; changing them on their measurable null sets does not change any conclusion. Put $h = f - g$ and $h^+ = \max(h, 0)$, $h^- = \max(-h, 0)$. Since $\int h = 0$, their integrals agree and each is $\frac{1}{2}\int |h|$. For every measurable A,

$$-\int h^- \le \int_A h \le \int h^+.$$

The upper bound is attained at $A = \{h \ge 0\}$, proving the distance identity. Next, $\min(f_n, f) \to f$ almost everywhere and $0 \le \min(f_n, f) \le f$. Dominated convergence gives $\int \min(f_n, f) \to 1$. The elementary identity $|f_n - f| = f_n + f - 2\min(f_n, f)$ therefore yields

$$\|f_n - f\|_1 = 2 - 2\int \min(f_n, f) \, d\mu \longrightarrow 0.$$

The concentration densities $n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$ converge almost everywhere to zero, whose integral is zero, and their norms stay one. Thus the conclusion fails if the normalization of the limit is dropped. □

Corollary 39.5 (Convergence in measure of normalized densities)**.**
For probability densities f_n, f relative to the fixed finite measure, the following are equivalent:

$$f_n \to f \text{ in measure}, \qquad \|f_n - f\|_1 \to 0, \qquad d_{\mathrm{TV}}(\nu_{f_n}, \nu_f) \to 0.$$

In particular, convergence in measure to a normalized density supplies uniform integrability of the sequence; it need not be assumed separately.

Proof.
Norm convergence implies convergence in measure by

$$\mu(\{|f_n - f| > a\}) \le a^{-1}\|f_n - f\|_1 \qquad (a > 0),$$

and the total variation identity proves the last equivalence. Suppose now that convergence in measure holds but norm convergence does not. Choose $\varepsilon > 0$ and a subsequence with $\|f_{n_k} - f\|_1 \ge \varepsilon$ for all k. The almost-everywhere subsequence lemma supplies a further subsequence converging to f almost everywhere. Scheffe's lemma, Theorem 39.4, makes its L^1 distances tend to zero, a contradiction. Finally, the converse Vitali implication proved in the functional-analysis section says that an L^1-convergent sequence is uniformly integrable. Both positivity and preservation of total mass enter through Scheffe's lemma; convergence in measure of arbitrary integrable functions alone does not give this conclusion. □

Example 39.6 (Setwise convergence need not be total variation convergence)**.**
For the Rademacher functions above, $f_n = 1 + r_n$ is a probability density on $[0,1)$: it takes values zero and two and has integral one. The proved weak convergence $r_n \rightharpoonup 0$ gives $f_n \rightharpoonup 1$, hence setwise convergence of the associated measures. But $\|f_n - 1\|_1 = \|r_n\|_1 = 1$, so their total variation distance from Lebesgue probability measure is always $1/2$. Even uniform boundedness of the densities does not turn setwise convergence into total variation convergence. It does supply uniform integrability and hence compactness of the weak closure, as well as the weak subsequence conclusion above. Neither conclusion is norm compactness.

40 The free heat equation and uniqueness

Work on $\mathbb{R}^d$, $d \geq 1$, with Lebesgue measure. In this chapter write $L = \sum_{j=1}^{d} \partial_j^2$ for the differential Laplacian. Thus the nonnegative operator used in the next chapters is $H_0 = -L$, and its heat equation is $u_t = Lu$. We use elementary multivariable differentiation and Lebesgue integration, including Tonelli's theorem and dominated convergence. The Gaussian integral is computed below; the L^2 Fourier multiplier identification uses the imported Fourier–Plancherel theorem. The point of the uniqueness argument is its explicit growth class: a whole-space initial-value problem does not acquire unrestricted uniqueness from a bounded-domain maximum principle. Jost's Theorem 5.1.2 is precisely a maximum principle with such a growth restriction, while his Section 5.2 constructs the fundamental solution (Jost, 2013).

40.1 The Gaussian and its convolution law

For $t > 0$ put

$$p_t(x) = (4\pi t)^{-d/2} \exp\bigl(-|x|^2/(4t)\bigr).$$

Lemma 40.1 (Elementary Gaussian identities)**.**
The function p_t is positive, has integral one, satisfies $\partial_t p_t = L p_t$, and obeys

$$\int_{\mathbb{R}^d} p_s(x-z) p_t(z-y)\, dz = p_{s+t}(x-y) \qquad (s,t>0).$$

Proof.
For completeness, the one-dimensional Gaussian integral follows from Tonelli and polar coordinates. If $I = \int_{\mathbb{R}} e^{-q^2}\, dq$, then

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dx\, dy = 2\pi \int_0^\infty r e^{-r^2}\, dr = \pi.$$

Positivity gives $I = \sqrt{\pi}$. The substitution $x = 2\sqrt{t}\, q$ and the product formula in d coordinates now give $\int p_t = 1$. Direct coordinate differentiation gives

$$\partial_t p_t(x) = \left(-\frac{d}{2t} + \frac{|x|^2}{4t^2}\right) p_t(x), \qquad \partial_j^2 p_t(x) = \left(-\frac{1}{2t} + \frac{x_j^2}{4t^2}\right) p_t(x),$$

so summing the second identity proves the differential equation. For convolution complete the square:

$$\frac{|x-z|^2}{s} + \frac{|z-y|^2}{t} = \frac{s+t}{st}\left|z - \frac{tx+sy}{s+t}\right|^2 + \frac{|x-y|^2}{s+t}.$$

The remaining Gaussian integral is $(4\pi st/(s+t))^{d/2}$. Multiplying by the two normalization factors yields the asserted formula. All these integrals are absolutely convergent; their positive integrands also permit Tonelli's theorem. □

Lemma 40.2 (The Gaussian Fourier multiplier)**.**
For the unitary Fourier normalization $\widehat{f}(\xi) = (2\pi)^{-d/2} \int e^{-ix\cdot\xi} f(x)\, dx$,

$$\widehat{p_t}(\xi) = (2\pi)^{-d/2} e^{-t|\xi|^2}.$$

Consequently convolution by p_t has Fourier multiplier $e^{-t|\xi|^2}$.

Proof.
In one dimension let $F(a) = \int_{\mathbb{R}} e^{-x^2/(4t)} e^{-iax}\,dx$. The integrable function $|x|e^{-x^2/(4t)}$ permits differentiation under the integral. Since $xe^{-x^2/(4t)} = -2t\,\frac{d}{dx}e^{-x^2/(4t)}$, integration by parts, with vanishing Gaussian boundary terms, gives

$$F'(a) = 2it \int_{\mathbb{R}} \frac{d}{dx}\bigl(e^{-x^2/(4t)}\bigr) e^{-iax}\,dx = -2taF(a).$$

Together with $F(0) = \sqrt{4\pi t}$, this scalar differential equation gives $F(a) = \sqrt{4\pi t}\,e^{-ta^2}$. Taking the product in d coordinates proves the displayed transform. For an integrable function f, Fubini applied to the convolution integral gives $\widehat{p_t * f} = (2\pi)^{d/2}\widehat{p_t}\,\widehat{f}$, which proves the multiplier assertion on integrable functions. For $f \in L^2$, Jensen's inequality for the probability density p_t and Tonelli give

$$\|p_t * f\|_2^2 \le \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} p_t(z)|f(x-z)|^2\,dz\,dx = \|f\|_2^2.$$

Approximation by $L^1 \cap L^2$ functions and Fourier–Plancherel therefore extend the multiplier identity to L^2. □

Theorem 40.1 (The solution for bounded continuous initial data)**.**
If $f : \mathbb{R}^d \to \mathbb{R}$ is bounded and continuous, then

$$u(x,t) = \int_{\mathbb{R}^d} p_t(x-y) f(y)\,dy \quad (t>0), \qquad u(x,0) = f(x)$$

is continuous on $\mathbb{R}^d \times [0,\infty)$, smooth for $t > 0$, and solves $u_t = Lu$. Moreover $|u(x,t)| \le \|f\|_\infty$ and $u(\cdot,t) \to f$ locally uniformly as $t \downarrow 0$. The operators $P_t f = u(\cdot,t)$ satisfy $P_s P_t = P_{s+t}$.

Proof.
Absolute convergence and the bound follow from $\int p_t = 1$. Every spatial or time derivative of the Gaussian is a polynomial in $x - y$ times a Gaussian, with coefficients bounded when t stays in a compact subinterval of $(0,\infty)$. For x in a fixed compact set these derivatives admit an integrable majorant in y of the form $C(1+|y|^N)e^{-c|y|^2}$. Differentiation under the integral is therefore legitimate to every order; Lemma 40.1 gives the equation.
For a compact set K and a number $\varepsilon > 0$, continuity of f on a compact neighbourhood of K supplies $\delta > 0$ such that $|f(x-z) - f(x)| < \varepsilon$ whenever $x \in K$ and $|z| < \delta$. Consequently

$$\sup_{x\in K} |P_t f(x) - f(x)| \le \varepsilon + 2\|f\|_\infty \int_{|z|\ge\delta} p_t(z)\,dz.$$

After $z = \sqrt{t}\,q$, the last integral is a Gaussian tail outside $|q| \ge \delta/\sqrt{t}$, and tends to zero. This proves the initial trace and continuity at $t = 0$. Fubini, boundedness of f, and the Gaussian convolution identity prove the semigroup law. For complex initial data apply the construction to real and imaginary parts. □

Corollary 40.1 (Uniform initial trace for uniformly continuous data)**.**
Let $\mathrm{BUC}(\mathbb{R}^d)$ be the bounded uniformly continuous functions with supremum norm. The operators P_t preserve this space and form a strongly continuous contraction semigroup on it:

$$\lim_{t\downarrow 0} \|P_t f - f\|_\infty = 0 \qquad (f \in \mathrm{BUC}(\mathbb{R}^d)).$$

Proof.
Uniform continuity lets the preceding Gaussian-tail argument use one δ for all $x \in \mathbb{R}^d$, so it proves the asserted norm limit. Moreover

$$|P_t f(x+h) - P_t f(x)| \le \sup_y |f(y+h) - f(y)|,$$

which tends to zero with h and proves preservation of uniform continuity. The contraction bound and semigroup law were proved above. For $t > s \ge 0$, they give $\|P_t f - P_s f\|_\infty \le \|P_{t-s} f - f\|_\infty$, so continuity at zero also gives continuity at every time. □

Example 40.1 (Bounded continuous data need not have a uniform initial trace)**.**
The locally uniform conclusion of Theorem 40.1 cannot be replaced by convergence in supremum norm on $C_b(\mathbb{R}^d)$, the space of bounded continuous functions. In one dimension take $f(x) = \sin(x^2)$. For a fixed $t > 0$, change variables in the convolution of the complex function e^{iy^2}:

$$P_t(e^{i(\cdot)^2})(x) = e^{ix^2} \int_{\mathbb{R}} a_t(z) e^{-2ixz}\, dz, \qquad a_t(z) = p_t(z) e^{iz^2}.$$

Here $a_t(z) \to 0$ at both infinities and

$$a_t'(z) = \Big(-\frac{z}{2t} + 2iz\Big)\, a_t(z), \qquad a_t' \in L^1(\mathbb{R}).$$

Integration by parts on finite intervals, followed by the limit of their endpoints, therefore gives, for $x \neq 0$,

$$\left| \int a_t(z) e^{-2ixz}\, dz \right| \le \frac{\|a_t'\|_1}{2|x|}.$$

Thus $P_t f(x) \to 0$ as $|x| \to \infty$ for each fixed $t > 0$. At $x_n = \sqrt{\pi/2 + 2\pi n}$ one has $f(x_n) = 1$, whence $\|P_t f - f\|_\infty \ge 1$ for every $t > 0$. This does not contradict the locally uniform initial trace: the test points escape every compact set. It also shows why strong continuity on C_b in its supremum norm must not be inferred from the kernel formula.

40.2 Why the growth condition gives uniqueness

A classical subsolution on a time slab $[0, T]$ means a real function continuous up to $t = 0$, with continuous second spatial derivatives and first time derivative for $t > 0$, satisfying $u_t - Lu \le 0$. At the final time derivatives can be understood from the left.

Lemma 40.3 (Maximum principle on a bounded cylinder)**.**
Let $B_R = \{x : |x| < R\}$ and let u be a classical subsolution on $\overline{B}_R \times [0, T]$. Its supremum is at most its supremum on the parabolic boundary $(\overline{B}_R \times \{0\}) \cup (\partial B_R \times [0, T])$.

Proof.
For $\eta > 0$ consider $w = u - \eta t$, which satisfies $w_t - Lw \le -\eta$. If w exceeded the parabolic-boundary supremum of u, compactness would give a maximum at (x_0, t_0) with $x_0 \in B_R$ and $t_0 > 0$. At this point $Lw \le 0$ and $w_t \ge 0$: at $t_0 = T$ the latter follows from the left derivative, and at an interior time it is zero. This contradicts the strict differential inequality. Thus $u \le \sup_{\partial_p} u + \eta T$; letting $\eta \downarrow 0$ proves the claim. □

Theorem 40.2 (Whole-space maximum principle with Gaussian growth)**.**
Let $T > 0$ be finite and let u be a classical subsolution on $\mathbb{R}^d \times [0, T]$. Suppose there are $M, \lambda > 0$ with

$$u(x,t) \le M e^{\lambda |x|^2} \quad (0 \le t \le T), \qquad m = \sup_{x \in \mathbb{R}^d} u(x,0) < \infty.$$

Then $u(x,t) \le m$ throughout the slab.

Proof.
First suppose the slab length S satisfies $S < 1/(4\lambda)$. Choose τ with $S < \tau < 1/(4\lambda)$ and define

$$b(x,t) = (\tau - t)^{-d/2} \exp\left(\frac{|x|^2}{4(\tau - t)}\right), \qquad 0 \le t \le S.$$

Direct differentiation gives $b_t = Lb$. For a fixed $\varepsilon > 0$, $w = u - m - \varepsilon b$ is a subsolution and $w(x,0) \le 0$. On $|x| = R$ the uniform bound

$$w(x,t) \le M e^{\lambda R^2} + |m| - \varepsilon \tau^{-d/2} e^{R^2/(4\tau)}$$

is negative for all sufficiently large R, since $1/(4\tau) > \lambda$. The bounded-cylinder lemma implies $w \le 0$ inside every such cylinder. For each fixed (x,t) choose a cylinder containing it and then let $\varepsilon \downarrow 0$. Hence $u \le m$ on this short slab.
For an arbitrary finite T, partition $[0, T]$ into finitely many slabs of length less than $1/(4\lambda)$. The same growth constants work on each translated slab. The first step bounds the initial values of the next slab by m, so induction gives the conclusion on all of them. No limit as a spatial boundary tends to infinity has been taken without the barrier estimate. □

Corollary 40.2 (Uniqueness in the stated growth class)**.**
Two real classical solutions with the same continuous initial data coincide if, on every finite time slab, each satisfies an absolute bound $|u(x,t)| \le M e^{\lambda |x|^2}$ for some slab-dependent constants. In particular the bounded solution in Theorem 40.1 is unique among bounded classical solutions. The same conclusion holds for complex solutions.

Proof.
On a fixed slab the difference w has zero initial data and an absolute Gaussian-growth bound, after increasing the two constants if necessary. Apply the maximum principle to w and to $-w$. Both are at most zero, hence $w = 0$. For complex solutions apply this to the real and imaginary parts of the difference. □
The Gaussian here is a kernel of the heat semigroup, not a statement that the differential operator L has only one distributional kernel or that all unrestricted classical Cauchy problems are unique. The L^2 interpretation uses Fourier–Plancherel and the spectral calculus in Section 42; its initial condition is an L^2 limit, whereas the present theorem assumes continuous initial data and proves a locally uniform limit.

41 A criterion for separately continuous kernels

The criterion below proves a locally finite Radon-measure version of Simon's Theorem A.1.3 (Simon, 1982), with boundedness of the kernel explicit. The source uses Baire measures, asks for a norm-continuous and weakly continuous extension, and imposes full support for uniqueness. We retain the existence criterion without full support, and state precisely where support is needed. Its conclusion is separate continuity, not joint continuity. We give the proof, including the dual-pair argument needed on a noncompact space.

41.1 Measures, representatives, and the dual pair

Let (X, d) be a nonempty separable locally compact metric space, and let ν be a positive locally finite Radon measure on X. Write $C_b(X)$ for the complex bounded continuous functions, with supremum norm, and $\mathcal{M}(X)$ for the finite complex Radon measures, with total variation norm. A function $f \in L^1(X, \nu)$ determines the measure $f\nu \in \mathcal{M}(X)$, with $\|f\nu\|_{\mathrm{TV}} = \|f\|_1$. Local compactness and separability make X a countable union of compact sets; in particular ν is σ-finite. The next lemma justifies the Radon property of the finite density measures, including for the wider measure convention discussed below. All measures here are on the Borel sets. In this metric setting the Baire and Borel sigma algebras coincide: each closed set is the zero set of its continuous distance function.

Lemma 41.1 (Regularity of the finite measures used here)**.**
Every finite positive Borel measure μ on X is outer regular by open sets and inner regular by compact sets. Consequently every finite complex Borel measure on X is a finite Radon measure.

Proof. Separability and metrizability give a countable base. Local compactness then gives a countable cover by relatively compact open sets: choose a relatively compact neighborhood at each point and extract a countable subcover using the base. Finite unions of their closures form increasing compact sets K_n with union X.
We first establish approximation by closed and open sets. Let $\mathcal{R}$ be the class of Borel sets E for which, for every $\epsilon > 0$, there are a closed F and an open G with $F \subseteq E \subseteq G$ and $\mu(G \setminus F) < \epsilon$. Every open set $G \neq X$ belongs to $\mathcal{R}$: the closed sets $\{x : d(x, X \setminus G) \geq 1/n\}$ increase to G, and continuity from below gives the required approximation. The set X itself is immediate. The class $\mathcal{R}$ is closed under complements by interchanging the complements of F, G.
For $E = \bigcup_{j\geq 1} E_j$ with $E_j \in \mathcal{R}$, choose $F_j \subseteq E_j \subseteq G_j$ with $\mu(G_j \setminus F_j) < \epsilon 2^{-j}/3$. Since μ is finite, choose N with $\mu(E \setminus \bigcup_{j\leq N} E_j) < \epsilon/3$. Then $F = \bigcup_{j\leq N} F_j$ is closed, $G = \bigcup_{j\geq 1} G_j$ is open, and

$$\mu(G \setminus F) \leq \sum_{j\geq 1} \mu(G_j \setminus E_j) + \mu\left(E \setminus \bigcup_{j\leq N} E_j\right) + \sum_{j\leq N} \mu(E_j \setminus F_j) < \epsilon.$$

Thus $\mathcal{R}$ is a sigma algebra containing the open sets, so it contains every Borel set. For a closed inner approximant F, $F \cap K_n$ is compact and increases to F; continuity from below therefore improves closed inner approximation to compact inner approximation. The same argument applied to the finite positive total variation of a complex measure proves the final assertion. □
Use the bilinear pairing

$$\langle \mu, g \rangle = \int_X g \, d\mu, \qquad \mu \in \mathcal{M}(X), \quad g \in C_b(X).$$

For $x \in X$, the Dirac measure δ_x is the unit point mass at x, so $\langle \delta_x, g \rangle = g(x)$. Here $\sigma(\mathcal{M}, C_b)$ means convergence against every fixed $g \in C_b$, and $\sigma(C_b, \mathcal{M})$ means convergence against every fixed $\mu \in \mathcal{M}$. These definitions apply to nets, not only to sequences. They specify a dual pair; they do not identify $\mathcal{M}(X)$ with the full Banach dual of $C_b(X)$, an identification that is generally false on a noncompact space. For completeness, the pairing separates both spaces. Dirac measures separate continuous functions. If a finite complex measure integrates every bounded continuous function to zero, apply this to $\max(0, 1 - nd(x, F))$ for each nonempty closed F. Dominated convergence gives $\mu(F) = 0$. Closed inner approximation with respect to $|\mu|$ then gives $\mu(E) = 0$ for every Borel E, so $\mu = 0$.
Let $J : C_b(X) \to L^\infty(X, \nu)$ take the almost-everywhere class. A bounded operator $T : L^1(\nu) \to L^\infty(\nu)$ has an extension $S : \mathcal{M}(X) \to C_b(X)$ in the sense used here if

$$J(S(f\nu)) = Tf \qquad (f \in L^1(\nu)).$$

This is an equality of equivalence classes. Full support means that every nonempty open subset of X has positive ν-measure. If ν has full support, J is injective and isometric: a continuous function exceeding a bound at one point exceeds it on an open set of positive measure. Without full support, continuous representatives need not be unique.

Lemma 41.2 (Continuous linear functionals for the specified weak topology)**.**
Every continuous complex-linear functional F on $(\mathcal{M}(X), \sigma(\mathcal{M}, C_b))$ has the form $F(\mu) = \int g \, d\mu$ for a unique $g \in C_b(X)$.

Proof. Continuity at zero gives $g_1, \dots, g_n \in C_b(X)$ and $\epsilon > 0$ such that $|\int g_j \, d\mu| < \epsilon$ for all j implies $|F(\mu)| < 1$. If all those integrals vanish, every scalar multiple of μ belongs to this neighborhood; linearity forces $F(\mu) = 0$. Thus F factors through the map $\mu \mapsto (\int g_1 \, d\mu, \dots, \int g_n \, d\mu)$ into $\mathbb{C}^n$. A linear functional on its image extends to $\mathbb{C}^n$ by extending a basis, so it is a linear combination of its coordinates. The same combination of the g_j gives g. Testing Dirac measures δ_x proves uniqueness. □

41.2 The criterion and its proof

Theorem 41.1 (Bounded separately continuous kernel criterion)**.**
Let $T : L^1(X, \nu) \to L^\infty(X, \nu)$ be bounded and linear. The following are equivalent:

1. There is a bounded function $K : X \times X \to \mathbb{C}$, continuous in each variable separately, such that
$$(Tf)(x) = \int_X K(x, y) f(y) \, d\nu(y)$$
for ν-almost every x, for every $f \in L^1(\nu)$.

2. There is a bounded linear extension $S : \mathcal{M}(X) \to C_b(X)$ of T which is continuous from $\sigma(\mathcal{M}, C_b)$ to $\sigma(C_b, \mathcal{M})$.

In particular these conditions imply $\operatorname{Ran} T \subseteq J(C_b(X))$. Kernels and extensions correspond by

$$K(x, y) = (S\delta_y)(x), \qquad (S\mu)(x) = \int_X K(x, y) \, d\mu(y),$$

and $\|S\| = \sup_{x,y} |K(x, y)|$. If ν has full support, both the kernel and the extension are unique, and their norm is also $\|T\|_{L^1 \to L^\infty}$.

Proof. First suppose K satisfies condition 1, and put $B = \sup_{x,y} |K(x,y)|$. The function K is jointly Borel measurable, although it need not be jointly continuous. To verify measurability, choose a countable dense sequence (a_j) in X. For each positive integer n, let $q_n(x)$ be the first a_j with $d(x,a_j) < 1/n$. Its level sets are Borel. Consequently $K(q_n(x), y)$ is Borel on $X \times X$, since it is given on a countable Borel partition in x by continuous functions of y. Continuity in the first variable gives its pointwise limit $K(x,y)$.

Define S by the asserted integral formula. It is linear and $|(S\mu)(x)| \le B\|\mu\|_{\mathrm{TV}}$. If $x_n \to x$, separate continuity and dominated convergence with respect to $|\mu|$ give $(S\mu)(x_n) \to (S\mu)(x)$. Since X is metric, this proves $S\mu \in C_b(X)$. The kernel representation of T proves the extension property.

For $\lambda \in \mathcal{M}(X)$, define

$$g_\lambda(y) = \int_X K(x,y)\, d\lambda(x).$$

The same dominated-convergence argument shows $g_\lambda \in C_b(X)$. Fubini applies to finite total variation measures, because K is bounded and jointly measurable, and gives

$$\langle \lambda, S\mu \rangle = \int_X g_\lambda \, d\mu.$$

Each defining scalar test of the target weak topology is therefore a continuous scalar test of the source weak topology. This proves the required continuity for nets, not merely sequential continuity.

Conversely, suppose condition 2 holds and define $K(x,y) = (S\delta_y)(x)$. For fixed y, this is continuous in x because $S\delta_y \in C_b(X)$. For fixed x, the map $F_x : \mu \mapsto (S\mu)(x)$ is continuous for $\sigma(\mathcal{M}, C_b)$: evaluation at x is the target test δ_x, and S is weakly continuous. By Lemma 41.2, there is $g_x \in C_b(X)$ such that $F_x(\mu) = \int g_x \, d\mu$. Evaluating at δ_y identifies $g_x(y) = K(x,y)$. This proves continuity in y and, for every measure, the asserted integral formula for S. Moreover $|K(x,y)| \le \|S\|$, since $\|\delta_y\|_{\mathrm{TV}} = 1$. Putting $\mu = f\nu$ and using the extension property proves condition 1. The formula bounds $\|S\|$ by $\sup|K|$, while Dirac measures give the reverse inequality. The two constructions are inverse, since the formula for $S\mu$ was derived for all measures.

Now assume full support. For fixed $y \in X$, choose relatively compact open neighborhoods U_n of y, contained in the radius $1/n$ ball. Then $0 < \nu(U_n) < \infty$. Put $f_n = \mathbf{1}_{U_n}/\nu(U_n)$, so $\|f_n\|_1 = 1$. The continuous representative $S(f_n\nu)$ has supremum norm $\|Tf_n\|_\infty \le \|T\|$. For fixed x, continuity of $K(x,\cdot)$ gives

$$(S(f_n\nu))(x) = \frac{1}{\nu(U_n)} \int_{U_n} K(x,u)\, d\nu(u) \longrightarrow K(x,y).$$

Hence $|K(x,y)| \le \|T\|$, and integration gives the reverse operator-norm inequality, proving norm equality.

If K_1, K_2 represent the same T, the functions $\int (K_1(x,y) - K_2(x,y)) f(y)\, d\nu(y)$ are continuous in x and vanish almost everywhere, so full support makes them zero at every x. Substitute the same f_n and take the limit to get $K_1(x,y) = K_2(x,y)$ for every x, y. The already proved integral formula then gives equality of the corresponding extensions. □

The measure convention and a wider formulation. The source allows a Baire measure which need not have finite total mass. Our locally finite Radon convention already includes Lebesgue measure on $\mathbb{R}^d$. In fact the bounded-kernel equivalence above holds for every positive Borel measure ν on the same space X, without local finiteness: each $f \in L^1(\nu)$ defines a finite complex Borel measure $f\nu$, and finite Borel measures on a separable locally compact metric space

are Radon. The proof integrates against these finite measures, so its dominated-convergence and Fubini arguments remain valid; it does not need a product of two possibly non-sigma-finite copies of ν.

For the uniqueness and norm-equality conclusions it suffices that every nonempty open set U contain a Borel set E with $0 < \nu(E) < \infty$. Replace U_n in the proof by such a subset $E_n \subset U_n$ and use $\mathbf{1}_{E_n}/\nu(E_n)$; continuity still makes its averages tend to the value at the center. This condition holds for locally finite measures of full support, and more generally for full-support semifinite measures. Here semifiniteness means that each measurable set of infinite measure contains a measurable subset of finite positive measure.

A measure convention is necessary for an unrestricted full-support statement. The Borel measure assigning $\nu(\varnothing) = 0$ and $\nu(E) = \infty$ to every nonempty E has full support but $L^1(\nu) = \{0\}$. Both $K = 0$ and $K = 1$ then represent its zero operator. They yield different bounded weakly continuous extensions, namely $S\mu = 0$ and $S\mu = \mu(X)\mathbf{1}$. Thus full support alone does not ensure uniqueness for arbitrary nonsemifinite Borel measures. This explains the measure hypotheses rather than assigning a broader meaning to the source's Baire-measure convention.

41.3 Why weak continuity is part of the hypothesis

Norm boundedness of an extension alone is insufficient, even when the reference measure has full support. On $X = \mathbb{R}$ with Lebesgue measure, let $T : L^1 \to L^\infty$ be zero and define

$$(S\mu)(x) = \mu(\{0\}) \qquad (x \in \mathbb{R}).$$

This is a bounded linear map $\mathcal{M}(\mathbb{R}) \to C_b(\mathbb{R})$ of norm one. It extends T, since every density $f\,dx$ assigns zero mass to the singleton. Nevertheless it differs from the zero extension.

To see exactly which hypothesis fails, put

$$d\mu_n(x) = \frac{n}{2}\mathbf{1}_{(-1/n,1/n)}(x)\,dx.$$

For every $g \in C_b(\mathbb{R})$,

$$\left|\int g\,d\mu_n - g(0)\right| \le \sup_{|x|<1/n} |g(x) - g(0)| \longrightarrow 0.$$

Thus $\mu_n \to \delta_0$ in $\sigma(\mathcal{M}, C_b)$, whereas $S\mu_n = 0$ and $S\delta_0 = \mathbf{1}$. Evaluation at zero is a test in the target weak topology, so S fails the required weak continuity.

This approximation is not convergence in total variation: $\|\mu_n - \delta_0\|_{\mathrm{TV}} = 2$. The partition into $\{0\}$ and its complement gives the lower bound two, and the triangle inequality gives the upper bound. Consequently the uniqueness of the extension in the theorem comes from the specified weak topology, rather than from a nonexistent norm density of L^1 densities among all finite measures.

41.4 What the criterion does and does not assert

No full-support assumption was used for the equivalence. It was used only for uniqueness and comparison with the norm of T. For example, let $X = \{0, 1\}$ with the discrete topology and $\nu = \delta_0$. Both $K = 0$ and the kernel which is one at $(1, 1)$ and zero elsewhere represent the zero operator on $L^1(\nu)$, but their extensions to $\mathcal{M}(X)$ differ. In particular the extension norm cannot be recovered from $\|T\|$ in this setting. Boundedness of the kernel in the theorem is explicit: with $X = \mathbb{R}$, $\nu = \delta_0$, the unbounded separately continuous function $K(x, y) = y$ also represents the

zero operator, but it does not define a bounded extension on all finite measures. Statements about arbitrary representatives outside the support require care.
Separate continuity cannot be strengthened to joint continuity by this argument. On $X = \mathbb{R}$ with Lebesgue measure, set

$$K(x,y) = \begin{cases} xy/(x^2+y^2), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

This kernel is separately continuous and bounded by $1/2$, but $K(t,t) = 1/2$ for $t \neq 0$, so it is not jointly continuous at the origin. It defines a bounded $L^1 \to L^\infty$ operator and a weakly continuous extension exactly as in the theorem. Full support makes this its unique separately continuous kernel; there is no alternative jointly continuous kernel for the same operator. Obtaining joint continuity for a heat semigroup therefore requires an additional argument using that semigroup's analytic structure.

42 Schrödinger semigroups with bounded potentials

This chapter works on $\mathbb{R}^d$, for a positive integer d, with Lebesgue measure and complex L^2. Its analytic inputs are the Fourier–Plancherel theorem, the spectral calculus for self-adjoint operators, and the usual product-integration theorems. We take these standard analytic results as prerequisites and prove the bounded-potential perturbation and kernel statements below directly. In particular, no pointwise limit of product formulas is used to assert existence or continuity of a kernel. We use the inner product $\langle f, g\rangle = \int f\overline{g}$, linear in its first variable. An operator norm written without subscripts means the norm from $L^2(\mathbb{R}^d)$ to itself. The imported spectral calculus says that a self-adjoint operator A has a projection-valued spectral measure E_A with $A = \int \lambda\, dE_A(\lambda)$ on the domain determined by $\int \lambda^2\, d\langle E_A(\lambda)f, f\rangle < \infty$; bounded Borel functions of A act by the corresponding spectral integrals. Fourier–Plancherel says that the normalized transform below extends to a unitary operator on L^2. These standard background theorems are used, not reproved here; see (Weidmann, 1980). The concrete joint-continuity conclusion is also treated in (Simon, 1982, Theorem B.7.1); the bounded-potential proof below supplies its own argument, distinct from the abstract separate-continuity criterion in Simon's Section A.1.

42.1 The operator and its sign convention

Use the unitary Fourier transform

$$\widehat{f}(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\, dx$$

initially on integrable square-integrable functions and then by Plancherel extension. Define the nonnegative Laplacian by

$$H_0 = -\sum_{j=1}^{d} \partial_j^2, \qquad D(H_0) = \{f \in L^2 : |\xi|^2 \widehat{f}(\xi) \in L^2\}, \qquad \widehat{H_0 f} = |\xi|^2 \widehat{f}.$$

The differential expression agrees with this definition on smooth functions of compact support. Throughout, V is a real essentially bounded measurable function, $M = \|V\|_\infty$, and multiplication by V is denoted by the same symbol. Choose a representative with $|V(x)| \le M$ everywhere. None of the resulting operators or kernels depends on this choice on null sets.

Proposition 42.1 (Bounded real perturbations)**.**
The operator $H = H_0 + V$ on $D(H_0)$ is self-adjoint and bounded below by $-M$. Its spectral semigroup $T_t = e^{-tH}$, $t \ge 0$, is strongly continuous and satisfies $\|T_t\|_{2\to 2} \le e^{Mt}$.

Proof.
The real multiplication operator $u \mapsto |\xi|^2 u$ on its maximal domain is self-adjoint. To check the domain assertion, if v belongs to its adjoint domain with adjoint value w, test against all L^2 functions supported in $\{|\xi| \le n\}$. On that set the multiplier is bounded, so these tests give $w = |\xi|^2 v$ there. Letting n increase proves $|\xi|^2 v \in L^2$, exactly the original domain. The reverse inclusion follows by the integral pairing. Plancherel conjugation therefore makes H_0 self-adjoint and nonnegative; its domain is dense by frequency truncation.
Multiplication by real V is bounded self-adjoint. In general, for a densely defined operator A and bounded operator B, a vector belongs to $D((A+B)^*)$ exactly when it belongs to $D(A^*)$: the bounded functional contributed by B can be subtracted from the defining adjoint identity.

Thus $(A+B)^* = A^* + B^*$ on that domain. Applying this with $A = H_0$ proves self-adjointness of H. Also

$$\langle Hf, f\rangle = \int |\xi|^2 |\widehat{f}(\xi)|^2 \, d\xi + \int V(x)|f(x)|^2 \, dx \geq -M\|f\|_2^2.$$

The spectral theorem now gives T_t, its semigroup law and norm bound. Strong continuity follows by dominated convergence in each vector's spectral measure, since $e^{-t\lambda} \to 1$ and the spectrum lies in $[-M, \infty)$. □

This is the bounded-perturbation case of the self-adjoint perturbation principle: $\|Vf\|_2 \leq M\|f\|_2$ gives relative bound zero with respect to H_0. The direct adjoint-domain proof avoids needing the more general Kato–Rellich theorem for unbounded relatively bounded perturbations; see (Kato, 1995) for that broader setting.

Put $P_t = e^{-tH_0}$. The Gaussian Fourier calculation gives

$$(P_t f)(x) = \int_{\mathbb{R}^d} p_t(x-y) f(y) \, dy, \qquad p_t(r) = (4\pi t)^{-d/2} \exp\bigl(-|r|^2/(4t)\bigr), \quad t > 0.$$

Indeed the Fourier multiplier is $e^{-t|\xi|^2}$; Lemma 40.2 computes the Gaussian transform with precisely this normalization. In particular

$$p_t \geq 0, \qquad \int p_t = 1, \qquad \int p_{t-s}(x-z) p_s(z-y) \, dz = p_t(x-y) \quad (0 < s < t).$$

The last identity follows either by the Fourier multipliers or by the square-completion calculation in Lemma 40.1. Notice the sign: the heat generator is $-H_0 = \sum_j \partial_j^2$.

42.2 Duhamel's equation and its kernel series

All time integrals below are taken after applying the operator to a fixed vector. For a continuous map $u : [a, b] \to L^2$, its integral is the norm limit of Riemann sums: uniform continuity makes these sums Cauchy, and completeness of L^2 supplies their limit. The triangle inequality for sums passes to the limit and gives

$$\left\| \int_a^b u(s) \, ds \right\|_2 \leq \int_a^b \|u(s)\|_2 \, ds.$$

Bounded linear maps commute with this integral by passage through the sums. The same inequality and scalar dominated convergence justify the limits of continuous vector-valued integrands used below; the fundamental theorem for a continuously differentiable L^2-valued map follows by the usual Riemann-sum argument. Thus no operator-norm continuity of the free semigroup at zero is required to form these strong integrals.

Lemma 42.1 (The bounded-perturbation expansion)**.**
For $t \geq 0$,

$$T_t = P_t - \int_0^t P_{t-s} V T_s \, ds = \sum_{n=0}^{\infty} A_n(t),$$

where $A_0(t) = P_t$ and

$$A_n(t) = -\int_0^t P_{t-s} V A_{n-1}(s) \, ds, \qquad \|A_n(t)\|_{2\to 2} \leq \frac{M^n t^n}{n!}.$$

The integrals are strong integrals after application to an L^2 vector, and the series converges in operator norm, uniformly on bounded time intervals.

Proof.
For $f \in D(H_0) = D(H)$, differentiate $P_{t-s}T_s f$ for $0 < s < t$. The spectral calculus preserves $D(H)$ under T_s, and $H_0 T_s f = HT_s f - VT_s f$ is continuous in s in L^2. The product rule therefore gives

$$\frac{d}{ds}(P_{t-s}T_s f) = -P_{t-s}VT_s f.$$

Integrating and using strong continuity at the endpoints proves the first equation on the dense domain. The uniform operator bound on the integrand extends it to every L^2 vector.
The recursive integrals defining A_n exist inductively by strong continuity. More explicitly, substitute $s = t\theta$ on the fixed interval $0 \le \theta \le 1$. The maps $P_{t(1-\theta)}VA_{n-1}(t\theta)f$ are continuous in t for each θ, by the induction hypothesis and the uniform bound on P. On bounded time intervals the induction bound supplies an integrable constant majorant. Dominated convergence therefore gives strong continuity of $A_n(t)$; its value at zero is zero for $n \ge 1$. Induction and $\|P_s\| \le 1$ give

$$\|A_n(t)\| \le M \int_0^t \frac{M^{n-1}s^{n-1}}{(n-1)!}\, ds = \frac{M^n t^n}{n!}.$$

Consequently the series converges with the stated uniformity and satisfies the same integral equation, by termwise integration. For uniqueness, the difference D_t of two locally bounded operator solutions satisfies $D_t = -\int_0^t P_{t-s}VD_s\, ds$. On $0 \le t \le b$, iteration bounds its norm by $C_b M^n t^n/n!$ for every n, where $C_b = \sup_{s\le b}\|D_s\|$. This tends to zero. Thus the sum equals T_t.
□
Define measurable kernels, for $t > 0$, by

$$k_0(t,x,y) = p_t(x-y), \qquad k_n(t,x,y) = -\int_0^t \int_{\mathbb{R}^d} p_{t-s}(x-z)V(z)k_{n-1}(s,z,y)\, dz\, ds.$$

Values at the two time endpoints need not be assigned in these Lebesgue integrals. Induction using Gaussian convolution proves absolute convergence and the pointwise estimate

$$|k_n(t,x,y)| \le \frac{M^n t^n}{n!} p_t(x-y). \tag{K_n}$$

For the induction step, the spatial integral after taking absolute values is bounded by

$$\frac{M^n s^{n-1}}{(n-1)!} p_t(x-y).$$

Integration over s gives the result. Changing V on a null set changes none of these integrals.

Theorem 42.1 (A continuous Gaussian-bounded kernel)**.**
For real $V \in L^\infty(\mathbb{R}^d)$ the function

$$k_V(t,x,y) = \sum_{n=0}^{\infty} k_n(t,x,y)$$

is jointly continuous on $(0,\infty)\times\mathbb{R}^d\times\mathbb{R}^d$ and is an integral kernel for e^{-tH}, with

$$|k_V(t,x,y)| \le e^{Mt} p_t(x-y).$$

For every $f \in L^2$, the integral $\int k_V(t,x,y)f(y)\, dy$ is absolutely convergent at every x and gives a continuous representative of $T_t f$.

Proof.
We first prove continuity of each term; measurability of V alone is sufficient. The assertion is clear for k_0. In the recursion for k_n, put $s = t\theta$, and restrict first to $\varepsilon \le \theta \le 1-\varepsilon$. On a compact set of parameters with $0 < a \le t \le b$ and bounded x, y, both Gaussian time parameters are bounded away from zero. The induction hypothesis makes the integrand continuous in t, x, y for each θ, z, even though its fixed factor $V(z)$ is only measurable. Estimate (K_n) bounds it by an integrable Gaussian envelope $C\exp(-c|z|^2)$, with constants depending on this compact set and ε. Dominated convergence proves continuity of the truncated integral.
For the two omitted time intervals, the convolution estimate gives the uniform bound

$$\frac{M^n t^n}{(n-1)!} p_t(x-y) \int_{(0,\varepsilon)\cup(1-\varepsilon,1)} \theta^{n-1}\, d\theta.$$

On the stated compact parameter set this tends to zero with ε. Thus k_n is a locally uniform limit of continuous functions and is continuous. Finally, on $a \le t \le b$,

$$|k_n(t,x,y)| \le \frac{(Mb)^n}{n!}(4\pi a)^{-d/2}.$$

The series is therefore locally uniformly convergent and has a jointly continuous sum. Summing the sharper Gaussian estimates gives the asserted bound.
We will use the following elementary density fact. For $1 \le p < \infty$ and $f \in L^p(\mathbb{R}^d)$, put

$$f_m = f\mathbf{1}_{\{|x|\le m,\ |f(x)|\le m\}}.$$

Each f_m is bounded and compactly supported. The functions $|f - f_m|^p$ tend to zero almost everywhere and are bounded by $|f|^p$, so dominated convergence gives $\|f - f_m\|_p \to 0$. In particular these functions are dense in both L^1 and L^2, and $L^1 \cap L^2$ is dense in L^1. This argument does not claim density in L^∞.
To identify the operator, first take a bounded compactly supported function f. The same estimates and Tonelli's theorem for absolute values justify interchanging all integrals in the recursion; thus $A_n(t)f(x) = \int k_n(t,x,y)f(y)\,dy$ almost everywhere, inductively. They also justify summation under this integral. The integral operator defined by the sum is bounded on L^2: its absolute value is at most $e^{Mt}P_t|f|$, whose L^2 norm is at most $e^{Mt}\|f\|_2$. Density and the operator-norm series in Lemma 42.1 identify it with T_t on all of L^2.
For an arbitrary $f \in L^2$, Cauchy–Schwarz and the Gaussian bound give absolute convergence at each x. On any bounded set of x, the rows $k_V(t,x,\cdot)$ have a common Gaussian envelope in L^2. Their pointwise continuity and dominated convergence therefore imply continuity in L^2 as x varies. Pairing these rows with f proves continuity of the displayed representative. Its pointwise bound also gives $\|T_t f\|_\infty \le e^{Mt}\|p_t\|_2\|f\|_2$. □

The integral inequality used for smoothing. For clarity, $L^p = L^p(\mathbb{R}^d)$ consists of measurable functions modulo equality almost everywhere, with

$$\|f\|_p = \left(\int |f|^p\right)^{1/p} \quad (1 \le p < \infty), \qquad \|f\|_\infty = \operatorname{ess\,sup}|f|.$$

All operator bounds below refer to these norms. The kernel supplies specified continuous representatives when its spatial regularity is used.

For conjugate exponents $1 < \alpha, \beta < \infty$, meaning $1/\alpha + 1/\beta = 1$, elementary one-variable minimization gives

$$ab \le a^\alpha/\alpha + b^\beta/\beta \qquad (a, b \ge 0).$$

Indeed the minimum of $a^\alpha/\alpha - ab$ as a function of a is $-b^\beta/\beta$. Apply this inequality to $|u|/\|u\|_\alpha$ and $|v|/\|v\|_\beta$ and integrate to obtain Hölder's inequality $\int |uv| \le \|u\|_\alpha \|v\|_\beta$. Zero norms are handled directly; at the conjugate endpoints $1, \infty$ the inequality follows from the almost-everywhere essential bound.
We also use its three-factor form: if $r_j \in [1, \infty]$ and $\sum_{j=1}^3 1/r_j = 1$, then $\int |u_1 u_2 u_3| \le \prod_{j=1}^3 \|u_j\|_{r_j}$. For completeness, put $1/r = 1/r_1 + 1/r_2$. When $r < \infty$, apply the two-factor inequality to $|u_1|^r, |u_2|^r$ with conjugate exponents $r_1/r, r_2/r$, giving $\|u_1 u_2\|_r \le \|u_1\|_{r_1} \|u_2\|_{r_2}$. For $r = \infty$ this follows directly from essential bounds. A second application with conjugate exponents r, r_3 proves the claim, including all endpoint cases.

Corollary 42.1 (Quantitative smoothing between Lebesgue spaces)**.**
Let $1 \le p \le q \le \infty$, and define $r \in [1, \infty]$ by $1 + 1/q = 1/p + 1/r$, with $1/\infty = 0$. The kernel defines a consistent bounded operator $T_t : L^p \to L^q$ extending the spectral operator on $L^p \cap L^2$, and

$$\|T_t f\|_q \le c_r e^{Mt} (4\pi t)^{-\frac{d}{2}(\frac{1}{p} - \frac{1}{q})} \|f\|_p, \qquad c_r = \begin{cases} r^{-d/(2r)}, & r < \infty, \\ 1, & r = \infty. \end{cases}$$

In particular the estimate includes $p = 1, q = \infty$ and $p = q = \infty$. No strong-continuity assertion on L^∞ at time zero is needed here. This is the bounded-potential instance of the smoothing property in (Cycon et al., 1987, Theorem 2.1).

Proof.
The Gaussian bound gives $|T_t f| \le e^{Mt} p_t * |f|$. We record the convolution estimate rather than requiring an interpolation theorem. For finite q, the exponent relation gives $p, r \le q$. Hölder's inequality, applied to the three factors

$$\big(|f(y)|^{p/q} |g(x-y)|^{r/q}\big), \quad |f(y)|^{1-p/q}, \quad |g(x-y)|^{1-r/q},$$

with reciprocal exponents $1/q$, $1/p - 1/q$, and $1/r - 1/q$, gives

$$|(g * f)(x)|^q \le \left(\int |f(y)|^p |g(x-y)|^r \, dy \right) \|f\|_p^{q-p} \|g\|_r^{q-r}.$$

Factors whose reciprocal exponent is zero are omitted. Integrating in x and using Tonelli gives $\|g * f\|_q \le \|g\|_r \|f\|_p$. Zero functions are immediate; truncation justifies the calculation if an intermediate pointwise integral is initially infinite. For $q = \infty$, the relation says $1/p + 1/r = 1$, and the same estimate is Hölder directly, including its L^1–L^∞ endpoints.
Gaussian integration gives

$$\|p_t\|_r = r^{-d/(2r)} (4\pi t)^{-\frac{d}{2}(1 - 1/r)} \quad (r < \infty), \qquad \|p_t\|_\infty = (4\pi t)^{-d/2}.$$

This proves the displayed bound and absolute convergence almost everywhere of the defining kernel integral for every $f \in L^p$. The kernel is the same for all these spaces, so the operators agree wherever their domains overlap. On $L^p \cap L^2$, the preceding theorem identifies that integral with the spectral semigroup. □

42.3 Pointwise identities and positivity

An integral operator has a unique continuous kernel within the Gaussian-bounded class just constructed. Indeed, equality of two such operators gives zero integral of the kernel difference against every product of bounded compactly supported test functions in x and y. If a continuous difference were nonzero at a point, multiplication by a scalar of modulus one and sufficiently small product neighbourhoods would make its real part have one strict sign there. Their indicator tests give a contradiction. This observation upgrades operator identities to pointwise kernel identities; it is stronger than merely choosing unrelated almost-everywhere representatives at each time.

Proposition 42.2 (Symmetry, composition, and two-sided bounds)**.**
For $s, t > 0$ and $x, y \in \mathbb{R}^d$,

$$k_V(t, x, y) = k_V(t, y, x) \in \mathbb{R},$$
$$k_V(t + s, x, y) = \int_{\mathbb{R}^d} k_V(t, x, z) k_V(s, z, y)\, dz,$$
$$e^{-Mt} p_t(x - y) \le k_V(t, x, y) \le e^{Mt} p_t(x - y).$$

In particular the semigroup preserves positivity.

Proof.
Every term in the kernel series is real. Self-adjointness of T_t identifies its kernel with the transposed real kernel, and continuous kernel uniqueness gives symmetry everywhere. The spatial composition integral converges absolutely by Gaussian convolution. It is continuous in x, y: on compact parameter sets the product has an integrable Gaussian envelope in z, so dominated convergence applies. Fubini identifies its operator with $T_t T_s = T_{t+s}$. Kernel uniqueness gives the second equation pointwise.
For positivity put $W = M - V$, so $0 \le W \le 2M$ almost everywhere. Apply the preceding kernel construction to the potential $-W$. All of its Dyson terms are nonnegative, since each recursion has factor $-(-W) = W \ge 0$. Its kernel is therefore at least p_t. The operators satisfy

$$H_0 + V = H_0 - W + M, \qquad e^{-t(H_0+V)} = e^{-Mt} e^{-t(H_0-W)}.$$

Continuous kernel uniqueness gives $k_V = e^{-Mt} k_{-W}$, and hence the lower bound. The upper bound is the already proved absolute Gaussian bound. Integrating the nonnegative kernel against any nonnegative L^2 function proves positivity preservation. □

A normalization check: constant potentials. If $V(x) = c$ for a real constant c, spectral calculus gives $e^{-t(H_0+c)} = e^{-ct} P_t$. Continuous kernel uniqueness therefore gives

$$k_c(t, x, y) = e^{-ct} (4\pi t)^{-d/2} \exp\bigl(-|x - y|^2/(4t)\bigr).$$

The same formula follows directly from the series: Gaussian convolution gives $k_n = (-c)^n t^n p_t(x - y)/n!$, and summation gives $e^{-ct} p_t$. For a prescribed $M \ge 0$, the choices $c = M$ and $c = -M$ attain, respectively, the lower and upper Gaussian bounds. Thus the time factor and the sign of the potential in those bounds cannot be improved uniformly over all real potentials with $\|V\|_\infty \le M$.

42.4 The heat equation in its precise solution class

For a merely measurable potential, a continuous kernel need not give classical second spatial derivatives. The natural equation here is an L^2 evolution equation, whose spatial differential expression is understood in distributions. As in the free heat chapter, write $L = \sum_{j=1}^{d} \partial_j^2$, so that $H_0 = -L$. If $f \in D(H_0)$ and $\varphi \in C_c^\infty(\mathbb{R}^d)$, integration by parts in the Fourier transform gives $\widehat{-L\varphi} = |\xi|^2 \widehat{\varphi}$. Plancherel then gives

$$\langle H_0 f, \varphi \rangle = \langle |\xi|^2 \widehat{f}, \widehat{\varphi} \rangle = \langle \widehat{f}, |\xi|^2 \widehat{\varphi} \rangle = \langle f, -L\varphi \rangle.$$

All pairings are finite by Cauchy–Schwarz: both $H_0 f$ and $-L\varphi$ belong to L^2. This identity is exactly the distributional interpretation of $H_0 f = -Lf$, without requiring pointwise derivatives of f.

Proposition 42.3 (The strong solution and its distributional equation)**.**
For every $f \in L^2$ and $b > 0$, $u(t) = T_t f$ is the unique function in

$$C([0, b]; L^2) \cap C^1((0, b); L^2)$$

such that $u(0) = f$, $u(t) \in D(H) = D(H_0)$ for $0 < t < b$, and $u'(t) = -Hu(t)$ in L^2. At every positive time,

$$\|HT_t\|_{2\to 2} \le Me^{Mt} + \frac{1}{et}.$$

Its kernel representative is jointly continuous in (t, x) for $t > 0$. For every $\varphi \in C_c^\infty(\mathbb{R}^d)$ it satisfies

$$\frac{d}{dt} \int u(t, x)\overline{\varphi(x)}\, dx = \int u(t, x)\overline{L\varphi(x)}\, dx - \int V(x)u(t, x)\overline{\varphi(x)}\, dx.$$

The initial condition is convergence to f in L^2.

Proof.
The spectrum of H is contained in $[-M, \infty)$. For $\lambda \ge 0$, $\lambda e^{-t\lambda} \le 1/(et)$, whereas for $-M \le \lambda < 0$ one has $|\lambda| e^{-t\lambda} \le Me^{Mt}$. Thus spectral calculus gives the displayed bound. It also gives

$$\int \lambda^2 e^{-2t\lambda}\, d\langle E_H(\lambda) f, f \rangle < \infty,$$

so $T_t f \in D(H)$ for every $t > 0$. On any compact positive-time interval the scalar multipliers $\lambda e^{-t\lambda}$ have a common bound. The difference quotients of $e^{-t\lambda}$ are dominated by the same type of bound after slightly enlarging that interval. Dominated convergence in the spectral measure therefore proves $u'(t) = -HT_t f$ in L^2 and continuity of this derivative. Strong continuity at zero was already established.
The preceding distributional identity, applied to $u(t)$, gives the tested differential equation. The kernel formula gives a jointly continuous representative: on a compact set of (t, x) with $t > 0$, its rows have a common Gaussian envelope in L^2 as functions of y. Joint kernel continuity and dominated convergence give continuity of these rows in L^2, and their pairing with f preserves continuity.
For uniqueness let w be the difference of two solutions in the stated class. The lower bound $H \ge -M$ gives, for $0 < t < b$,

$$\frac{d}{dt} \|w(t)\|_2^2 = -2 \operatorname{Re} \langle Hw(t), w(t) \rangle \le 2M \|w(t)\|_2^2.$$

Consequently $e^{-2Mt} \|w(t)\|_2^2$ is nonincreasing on $(0, b)$. For $0 < s < t < b$ this implies $\|w(t)\|_2^2 \le e^{2M(t-s)} \|w(s)\|_2^2$. Let $s \downarrow 0$ and use $w(0) = 0$ and continuity; then $w(t) = 0$. Continuity also gives equality at b. This proves uniqueness without requiring differentiability at time zero. □

42.5 First spatial derivatives of the kernel

Bounded measurable potentials permit more spatial regularity than mere continuity of the kernel. The following estimate requires no derivative of the potential. It does not assert the existence of classical second spatial derivatives.

Proposition 42.4 (Continuous first derivatives and Gaussian bounds)**.**
Put $b_d = 2^{d/2}/\sqrt{e}$. For every real $V \in L^\infty$, the first derivatives of $k_V(t,x,y)$ in x and in y exist and are jointly continuous for $t > 0$. With $M = \|V\|_\infty$,

$$|\nabla_x k_V(t,x,y)|,\ |\nabla_y k_V(t,x,y)| \le B_{d,M}(t)\, p_{2t}(x-y),$$
$$B_{d,M}(t) = b_d e^{Mt} t^{-1/2}\big(1 + 2^{d/2+1} Mt\big).$$

The norm of each gradient is its Euclidean norm; no mixed derivative is asserted.

Proof.
Direct differentiation and comparison of two Gaussian scales give

$$|\nabla p_r(w)| = 2^{d/2}\frac{|w|}{2r} e^{-|w|^2/(8r)} p_{2r}(w) \le b_d r^{-1/2} p_{2r}(w).$$

Indeed, the maximum of $\rho e^{-\rho^2/(8r)}$ for $\rho \ge 0$ is $2\sqrt{r/e}$. Absolute summation of the kernel recursion gives the pointwise integral equation

$$k_V(t,x,y) = p_t(x-y) - \int_0^t \int p_{t-s}(x-u) V(u) k_V(s,u,y)\, du\, ds.$$

Summation and integration can be interchanged because the sum of the absolute integrands is bounded after spatial integration by $Me^{Ms}p_t(x-y)$, which is integrable in s.
The candidate derivative integral is absolutely convergent. Gaussian convolution and $t \le 2t-s \le 2t$ give

$$\int |\nabla p_{t-s}(x-u)|\, |V(u)|\, |k_V(s,u,y)|\, du \le b_d M e^{Ms}(t-s)^{-1/2} p_{2t-s}(x-y)$$
$$\le 2^{d/2} b_d M e^{Mt}(t-s)^{-1/2} p_{2t}(x-y).$$

For the last step, the normalization ratio is at most $2^{d/2}$, and the Gaussian exponential at time $2t-s$ is at most the one at time $2t$. Integration in s gives the bound $2^{d/2+1} b_d M e^{Mt}\sqrt{t}\, p_{2t}(x-y)$. Adding the free derivative bound and using $e^{Mt} \ge 1$ yields the asserted coefficient $B_{d,M}(t)$.
We justify both differentiation and continuity, rather than only estimating a formal derivative. Substitute $s = t\theta$ and initially restrict to $\varepsilon \le \theta \le 1-\varepsilon$. On compact sets of (t,x,y) with $t > 0$, the truncated integral and its derivative in x have integrable Gaussian majorants in u. Their integrands are continuous in t, x, y for each θ, u; the factor $V(u)$ is fixed. Differentiation under the integral and dominated convergence therefore give truncated kernels that are continuously differentiable in x, with their x-gradients jointly continuous in (t,x,y).
The omitted parts of the undifferentiated integral tend uniformly to zero on these compact sets by its bound $Me^{Ms}p_t(x-y)$. For the derivative, the preceding estimate reduces the endpoint errors, up to a compact-set constant, to

$$\int_0^\varepsilon (1-\theta)^{-1/2}\, d\theta + \int_{1-\varepsilon}^1 (1-\theta)^{-1/2}\, d\theta,$$

which tends to zero. The truncated gradients thus converge locally uniformly to a continuous function G. To identify G as the gradient, apply the fundamental theorem of calculus to each truncated kernel along a coordinate segment. Uniform convergence of both the functions and their derivatives passes this identity to the limit. Dividing by the segment length and using continuity of G proves that $\nabla_x k_V = G$. Finally symmetry $k_V(t,x,y) = k_V(t,y,x)$ proves the same assertions in y. □

Consequently the integral representative of $T_t f$ belongs to $C^1(\mathbb{R}^d)$ for every $f \in L^p$, $1 \le p \le \infty$, at each fixed $t > 0$. On a compact set of x, the kernel and its first derivatives have common Gaussian envelopes in y. Their products with $|f(y)|$ are integrable by Hölder's inequality, including its endpoints. Differentiation and continuity therefore pass under the integral. For $1 \le p \le q \le \infty$ and $1 + 1/q = 1/p + 1/r$, the convolution estimate already proved gives

$$\|\nabla T_t f\|_q \le B_{d,M}(t)\, \|p_{2t}\|_r\, \|f\|_p.$$

Here the L^q norm of the gradient uses its pointwise Euclidean norm, and $\|p_{2t}\|_r$ is the explicit Gaussian norm from Corollary 42.1 with t replaced by $2t$.

42.6 Uniform initial traces on spaces of continuous functions

The kernels diverge on the diagonal as time tends to zero, but their integrals against suitable initial data have a uniform initial trace. This remains true when the bounded potential is merely measurable.

Proposition 42.5 (Strong continuity on C_0 and BUC)**.**
Let V be real and essentially bounded, with $M = \|V\|_\infty$. For $t > 0$, define $T_t f$ by the continuous kernel integral, and put $T_0 f = f$. On each of $C_0(\mathbb{R}^d)$ and $\mathrm{BUC}(\mathbb{R}^d)$ these operators form a strongly continuous semigroup in the supremum norm, with

$$\|T_t\| \le e^{Mt}, \qquad \|T_t f - f\|_\infty \le \|P_t f - f\|_\infty + (e^{Mt} - 1)\|f\|_\infty.$$

Here C_0 consists of continuous functions vanishing at infinity, and BUC consists of bounded uniformly continuous functions. Both carry the supremum norm. The operator norm in this display is taken on the indicated function space.

Proof.
Summing the Dyson estimates for orders $n \ge 1$ gives

$$|k_V(t,x,y) - p_t(x-y)| \le (e^{Mt} - 1) p_t(x-y).$$

Consequently, for every bounded measurable f,

$$\|T_t f - P_t f\|_\infty \le (e^{Mt} - 1)\|f\|_\infty, \qquad \|T_t f\|_\infty \le e^{Mt}\|f\|_\infty.$$

The gradient bound in Proposition 42.4 and differentiation under the kernel integral give

$$\sup_x |\nabla T_t f(x)| \le B_{d,M}(t) \|f\|_\infty.$$

Integration along line segments therefore makes $T_t f$ globally Lipschitz for every fixed $t > 0$. In particular it belongs to BUC.

If $f \in C_0$, fix $\epsilon > 0$ and choose R so that $|f(y)| \le \epsilon$ for $|y| > R$. Gaussian domination gives

$$|T_t f(x)| \le e^{Mt} \left[\epsilon + \|f\|_\infty \int_{|y| \le R} p_t(x-y)\, dy \right].$$

For fixed t, R, the last integral tends to zero as $|x| \to \infty$: it is at most the volume of the ball times $(4\pi t)^{-d/2} \exp(-(|x| - R)^2/(4t))$ when $|x| > R$. Letting $\epsilon \downarrow 0$ proves $T_t f \in C_0$.
For bounded f, the double integral defining $T_t T_s f(x)$ is absolutely convergent: the integral of its absolute integrand is at most $e^{M(t+s)}\|f\|_\infty$, using Gaussian convolution and mass one. Fubini and the pointwise kernel composition law thus give $T_t T_s f = T_{t+s} f$ for $s, t > 0$; the law at zero follows by definition. The earlier free uniform-trace result, Corollary 40.1, and the displayed perturbation estimate imply $\|T_t f - f\|_\infty \to 0$ for $f \in \mathrm{BUC}$. Every function in C_0 is uniformly continuous: use uniform continuity on a large closed ball, and smallness of the function outside a slightly smaller ball for pairs outside it. Thus the same limit holds on C_0. Finally, for $t > s \geq 0$, the semigroup law gives

$$\|T_t f - T_s f\|_\infty \leq e^{Ms} \|T_{t-s} f - f\|_\infty.$$

This proves continuity at every time from continuity at zero. □

Corollary 42.2 (The maximal space with a uniform initial trace)**.**
For a real bounded measurable potential and $f \in L^\infty$,

$$\lim_{t \downarrow 0} \|T_t f - f\|_\infty = 0 \quad \Longleftrightarrow \quad f \text{ has a bounded uniformly continuous representative.}$$

In particular, for $f \in C_b$ the uniform initial trace holds exactly when $f \in \mathrm{BUC}$.

Proof.
Sufficiency follows from Proposition 42.5. Conversely, suppose the norm limit holds, and let f_n be the continuous kernel representative of $T_{1/n} f$, for $n \geq 1$. The gradient estimate makes every f_n bounded and uniformly continuous. The assumed limit makes this sequence Cauchy in L^∞. A bounded continuous function on $\mathbb{R}^d$ has supremum norm equal to its essential supremum norm, since every nonempty open set has positive Lebesgue measure. Applied to $f_n - f_m$, this shows that the sequence is uniformly Cauchy. Its uniform limit g is bounded and uniformly continuous, and convergence in L^∞ gives $g = f$ almost everywhere. If f itself is continuous, equality almost everywhere between f and g implies equality everywhere by the same full-support argument. □
Strong continuity on all of C_b or L^∞ would be false already for $V = 0$. The example $f(x) = \sin(x^2)$ in Section 40 gives $\|P_t f - f\|_\infty \geq 1$ for every $t > 0$ in dimension one. The difference is continuous, so its supremum equals its essential supremum; the example applies to both spaces. In higher dimensions use $f(x) = \sin(x_1^2)$ and the product Gaussian formula. The proposition concerns the integral semigroup; it requires no claim that multiplication by a discontinuous V preserves either C_0 or BUC.

42.7 Time derivatives and the limit of classical spatial regularity

The time variable has a different regularity mechanism: at positive times the spectral semigroup can be differentiated between two fixed smoothing operators. This avoids differentiating the potential.

Proposition 42.6 (Continuous time derivatives of every order)**.**
For each integer $n \geq 1$, the derivative $\partial_t^n k_V(t, x, y)$ exists and is jointly continuous for $t > 0$. It satisfies

$$|\partial_t^n k_V(t, x, y)| \leq (2\pi t)^{-d/2} \left[e^{Mt} M^n + e^{Mt/2} \left(\frac{2n}{et} \right)^n \right].$$

This estimate is uniform in x, y; it is not an off-diagonal Gaussian derivative estimate.

Proof.
For $s > 0$, spectral calculus gives the bounded operator $H^n T_s$ and

$$\|H^n T_s\|_{2\to 2} \le M^n e^{Ms} + \Big(\frac{n}{es}\Big)^n.$$

On the nonnegative part of the spectrum, maximize $\lambda^n e^{-s\lambda}$ at $\lambda = n/s$; on $[-M, 0)$ use $|\lambda|^n e^{-s\lambda} \le M^n e^{Ms}$. On any compact positive s interval, the same bounds with $n+1$ and $n+2$ control the scalar Taylor remainders uniformly in $\lambda \ge -M$. The spectral operator norm estimate therefore proves that $s \mapsto T_s$ is infinitely differentiable in operator norm, with derivative $(-H)^n T_s$ of order n.
Fix $a > 0$. The real functions $r_x(u) = k_V(a, u, x) = k_V(a, x, u)$ belong to L^2, depend continuously on x in that norm, and obey

$$\|r_x\|_2 \le e^{Ma}\|p_a\|_2, \qquad \|p_a\|_2^2 = (8\pi a)^{-d/2}.$$

The norm continuity follows from joint kernel continuity and a common L^2 Gaussian envelope when x ranges over a compact set. The pointwise composition law, applied twice, gives for $t > 2a$

$$k_V(t, x, y) = \langle T_{t-2a} r_y, r_x\rangle.$$

Here the inner product is linear in its first variable; the reality and symmetry of the rows explain the displayed order. Since a is fixed, only the middle operator depends on t. Differentiating in operator norm gives

$$\partial_t^n k_V(t, x, y) = \langle (-H)^n T_{t-2a} r_y, r_x\rangle.$$

Operator-norm continuity and the L^2 continuity of the rows prove joint continuity on $t > 2a$. Every positive time has a neighborhood of this form, proving the assertion throughout $t > 0$.
To estimate at a particular time t, we may now choose $a = t/4$ in the already proved identity. Cauchy–Schwarz and the operator bound with $s = t/2$ give

$$e^{Mt/2}(2\pi t)^{-d/2}\left[M^n e^{Mt/2} + \left(\frac{2n}{et}\right)^n\right],$$

as required. This choice is made after differentiation; the smoothing rows were held fixed when taking the derivative. □

Example 42.1 (A bounded potential need not give continuous second derivatives)**.**
In dimension one, take $V(x) = \mathbf{1}_{(0,\infty)}(x)$. For every $t > 0$ and $y \in \mathbb{R}$, the function $x \mapsto k_V(t, x, y)$ is not C^2 on any neighborhood of zero, although its first spatial derivative and all its time derivatives are continuous.

Proof.
We first justify the spatial distributional equation for a kernel column. Fix $0 < a < t$ and write

$$k_V(t, \cdot, y) = T_{t-a} k_V(a, \cdot, y).$$

The initial column is in L^2, so Proposition 42.3 applies. Pairing with a compactly supported smooth test function identifies its strong L^2 derivative with the pointwise time derivative from the preceding proposition: on the test function's compact support, that derivative is continuous

and uniformly bounded on a compact time interval, so differentiation passes under the integral. Consequently

$$\partial_x^2 k_V(t,\cdot,y) = \partial_t k_V(t,\cdot,y) + V\, k_V(t,\cdot,y)$$

as distributions.
If the column were C^2 near zero, its continuous second derivative would equal $\partial_t k_V(t,x,y)$ for $x < 0$ and $\partial_t k_V(t,x,y) + k_V(t,x,y)$ for $x > 0$. Distributional equality gives equality almost everywhere, and continuity on each open half interval makes it pointwise there. The limits of these two expressions at zero differ by $k_V(t,0,y)$, which is strictly positive by the Gaussian lower bound. This contradicts continuity of the second derivative at zero. □

42.8 The extension to finite measures

For a fixed $t > 0$, the kernel just constructed connects the concrete semigroup to Theorem 41.1. To make the domains precise, its integral formula defines a bounded operator $T_t^{(1)} : L^1(\mathbb{R}^d) \to L^\infty(\mathbb{R}^d)$, since

$$\left| \int k_V(t,x,y) f(y)\, dy \right| \le e^{Mt} (4\pi t)^{-d/2} \|f\|_1.$$

On $L^1 \cap L^2$ this agrees with the spectral semigroup T_t, by its proved kernel representation. This intersection is dense in L^1, so $T_t^{(1)}$ is the unique bounded extension of that restriction with the indicated domain and codomain.
Lebesgue measure has full support, and the kernel is bounded and jointly, hence separately, continuous. The criterion therefore gives the unique extension

$$S_t : \mathcal{M}(\mathbb{R}^d) \longrightarrow C_b(\mathbb{R}^d), \qquad (S_t\mu)(x) = \int k_V(t,x,y)\, d\mu(y),$$

which is both bounded in norm and continuous from $\sigma(\mathcal{M}, C_b)$ to $\sigma(C_b, \mathcal{M})$. In particular

$$(S_t\delta_y)(x) = k_V(t,x,y), \qquad \|S_t\| = \|T_t^{(1)}\|_{1\to\infty} = \sup_{x,y} |k_V(t,x,y)| \le e^{Mt}(4\pi t)^{-d/2}.$$

Thus evaluation on a point mass has a precise meaning even though the point mass is not an L^1 density. Joint continuity still comes from the Dyson-series proof above; the abstract criterion alone only yields separate continuity.

42.9 The Trotter product formula for bounded potentials

The product formula provides a second way to approximate the semigroup; the original product-formula paper is Trotter (1959). We prove the precise bounded-potential case needed here, rather than importing a general product theorem with additional domain hypotheses. In particular the multiplication operator e^{-hV} need not preserve $D(H_0)$.

Theorem 42.2 (Strong product approximation, uniformly on bounded time intervals)**.**
For the operators of this chapter and every $f \in L^2$ and $b > 0$,

$$\lim_{n\to\infty} \sup_{0\le t\le b} \left\| \left(P_{t/n} e^{-(t/n)V} \right)^n f - T_t f \right\|_2 = 0.$$

The exponential e^{-hV} is multiplication by the bounded measurable function $x \mapsto e^{-hV(x)}$.

Proof.
Put $F(h) = P_h e^{-hV}$ for $h \geq 0$. Then $F(0) = I$ and

$$\|F(h)\| \leq e^{Mh}, \qquad \|T_h\| \leq e^{Mh}.$$

Let $D = D(H_0) = D(H)$, with graph norm $\|u\|_D = \|u\|_2 + \|H_0 u\|_2$. For $u \in D$ split the difference quotient in the order

$$\frac{F(h)u - u}{h} = P_h \frac{e^{-hV} - I}{h} u + \frac{P_h u - u}{h}.$$

The first term tends to $-Vu$: boundedness of V gives $(e^{-hV} - I)/h \to -V$ in operator norm, and P_h tends strongly to I. The second tends to $-H_0 u$ by spectral calculus on $D(H_0)$. Likewise $(T_h u - u)/h \to -Hu$. Consequently the linear maps

$$E_h : D \longrightarrow L^2, \qquad E_h u = \frac{F(h)u - T_h u}{h}, \quad h > 0,$$

converge to zero on each fixed $u \in D$. This calculation never applies H_0 to $e^{-hV}u$.
We need this convergence uniformly on one compact set, so record the necessary bound. For $0 < h \leq b$, scalar spectral estimates and the semigroup derivative on $D(H)$ give

$$\begin{aligned}
\left\| \frac{P_h u - u}{h} \right\|_2 &\leq \|H_0 u\|_2, \\
\left\| \frac{e^{-hV} - I}{h} u \right\|_2 &\leq M e^{Mb} \|u\|_2, \\
\left\| \frac{T_h u - u}{h} \right\|_2 &\leq e^{Mb} \|Hu\|_2 \leq e^{Mb} \big(\|H_0 u\|_2 + M \|u\|_2 \big).
\end{aligned}$$

For the third displayed estimate, one can integrate $\frac{d}{ds} T_s u = -T_s H u$ from 0 to h; the integrand is strongly continuous. It follows that

$$\|E_h u\|_2 \leq (1 + e^{Mb}) \|H_0 u\|_2 + 2M e^{Mb} \|u\|_2 \leq C_b \|u\|_D$$

with a constant independent of h.
Here is the compactness argument explicitly. If linear maps E_h are uniformly bounded by C_b from D to L^2 and converge to zero at every fixed vector, then they converge uniformly on each graph-norm compact subset Q of D. Indeed, choose a finite ε-net $u_1, \ldots, u_N$ in Q. For $u \in Q$ and a net point within ε,

$$\|E_h u\|_2 \leq C_b \varepsilon + \max_{1 \leq j \leq N} \|E_h u_j\|_2.$$

The finite maximum tends to zero; then let ε tend to zero.
Fix $f \in D$ and use $Q_f = \{T_s f : 0 \leq s \leq b\}$. This set is compact in the graph norm. In fact $s \mapsto T_s f$ is continuous in L^2, and

$$H_0 T_s f = T_s H f - V T_s f$$

is continuous there as well, by the spectral domain invariance and commutation identity $H T_s f = T_s H f$. Thus Q_f is the continuous image of a compact interval. In particular

$$\omega_f(\delta) = \sup_{\substack{u \in Q_f \\ 0 < h \leq \delta}} \|E_h u\|_2 \longrightarrow 0 \quad (\delta \downarrow 0).$$

For $0 < t \le b$ put $h = t/n$. The algebraic telescoping identity, which requires no commutation between $F(h)$ and T_h, reads

$$F(h)^n f - T_h^n f = \sum_{j=0}^{n-1} F(h)^{n-1-j}(F(h) - T_h)T_h^j f.$$

Since $T_h^j f = T_{jh} f \in Q_f$ and $T_h^n = T_t$, its norm is at most

$$nh\, e^{Mb}\omega_f(b/n) \le b e^{Mb}\omega_f(b/n),$$

which tends to zero independently of t. At $t = 0$ the difference vanishes exactly. This proves the theorem on D.
Finally D is dense in L^2. For $f \in L^2$ and $g \in D$, the uniform bounds on both approximants give

$$\sup_{0\le t\le b} \|F(t/n)^n f - T_t f\|_2 \le 2e^{Mb}\|f - g\|_2 + \sup_{0\le t\le b} \|F(t/n)^n g - T_t g\|_2.$$

Take the limit superior in n and then approximate f by such g. □
This theorem proves strong operator convergence, uniformly in the stated time range. It does not assert convergence in operator norm, nor pointwise convergence of the product kernels. Kernel existence, joint continuity and Gaussian domination were proved independently above, so none of them rests on an unjustified interchange of the product limit with point evaluation.
For comparison with contraction-semigroup formulations, replace V by $V + M$. Both P_t and multiplication by $e^{-t(V+M)}$ are contractions, and $H + M \ge 0$. The exact scalar identities are

$$\big(P_{t/n}e^{-(t/n)(V+M)}\big)^n = e^{-Mt}\big(P_{t/n}e^{-(t/n)V}\big)^n,$$
$$e^{-t(H+M)} = e^{-Mt}T_t.$$

Multiplication by e^{Mt} transfers the shifted approximation to the original one, preserving uniform strong convergence on each $[0, b]$.
The Gaussian normalization, the factor $e^{t\|V\|_\infty}$, and the restriction $t > 0$ are part of these statements. At time zero the semigroup is the identity; the heat kernels approach a point mass, not a continuous function kernel of the same form. Unbounded potentials require additional hypotheses and are not included in this bounded perturbation argument.

43 Dynamically defined Schrödinger kernels

We address the parameter-regularity question for a compact dynamical system. The argument gives a quantitative Lipschitz estimate under the stated linear growth bound for the action. It is a proof under these hypotheses, not a claim that the result is new in the literature.
The baseline quantitative results use Lipschitz sampling and the linear action bound (D); isometric actions give sharper constants. The Hölder extension retains that action bound and weakens only the sampling assumption, whereas the exponential-action extension retains Lipschitz sampling and replaces the action bound. These extensions are stated separately. The localized operator-norm Lipschitz estimate uses the baseline assumptions. Finally, Proposition 43.4 requires only continuous sampling and a continuous action on a compact metric space, with no quantitative action bound. The final quantitative subsystem spectral estimates return to the baseline Lipschitz and linear-growth assumptions; their qualitative counterpart requires only continuity.

43.1 The hypotheses and translation convention

Let (Z, d_Z) be a nonempty compact metric space, and let $T : \mathbb{R}^d \times Z \to Z$ be a continuous action:

$$T(0, z) = z, \qquad T(g, T(h, z)) = T(g + h, z).$$

Assume that a constant $C \geq 0$ satisfies

$$d_Z(T(g, z), T(g, w)) \leq C(1 + |g|)d_Z(z, w). \tag{D}$$

Let $v : Z \to \mathbb{R}$ have Lipschitz constant L, and put $M = \|v\|_\infty = \sup_{z\in Z} |v(z)| < \infty$. Define

$$V_z(g) = v(T(g, z)), \qquad H_z = H_0 + V_z, \qquad H_0 = -\sum_{j=1}^{d} \partial_j^2.$$

Each V_z is continuous and bounded by M. Write $K_z(t, g, h) = k_{V_z}(t, g, h)$ for its continuous kernel and set

$$k_t(z, r) = K_z(t, 0, r), \qquad t > 0.$$

The actual potential difference is bounded by $2M$ and also satisfies

$$|V_z(u) - V_w(u)| \leq LC(1 + |u|)d_Z(z, w). \tag{P}$$

The latter bound grows in u; it will be integrated against a Gaussian, not used as a uniform bound for a multiplication operator.

Proposition 43.1 (Pointwise covariance)**.**
For $t > 0$ and $g, h \in \mathbb{R}^d$,

$$K_z(t, g, h) = k_t(T(g, z), h - g). \tag{C}$$

Consequently, for every $\psi \in L^2(\mathbb{R}^d)$,

$$(e^{-tH_z}\psi)(g) = \int_{\mathbb{R}^d} k_t(T(g, z), h - g)\psi(h)\, dh,$$

where the left side denotes the continuous representative constructed in Theorem 42.1.

Proof.
The action law gives $V_{T(g,z)}(u) = V_z(u+g)$. Translation invariance of the Gaussian and a change of variable in each Dyson integral give, by induction on its order,

$$k_n^{V_{T(g,z)}}(t,a,b) = k_n^{V_z}(t,a+g,b+g).$$

Summing the absolutely convergent series and taking $a = 0$, $b = h - g$ proves the pointwise identity. The integral against ψ converges absolutely by the Gaussian bound and Cauchy–Schwarz; the kernel theorem identifies it with the stated continuous representative. □
The sign is fixed by the potential convention. If instead $\widetilde{V}_z(g) = v(T(-g,z))$, the same proof gives

$$(e^{-t\widetilde{H}_z}\psi)(g) = \int_{\mathbb{R}^d} \widetilde{k}_t(T(-g,z), h-g)\psi(h)\,dh.$$

Thus a covariant kernel formula containing $T(-g,z)$ corresponds to the negative-action potential convention. Keeping $V_z(g) = v(T(g,z))$ requires the plus sign. All estimates below hold in either convention, since the action bound also applies to $-g$.

43.2 A pointwise Duhamel comparison

We temporarily return to arbitrary bounded real potentials. The comparison identity below is independent of the dynamical assumptions; applying it to V_z and V_w will provide the parameter estimates.

Lemma 43.1 (Comparison of continuous kernel representatives)**.**
Let V, W be real bounded measurable potentials with norms at most M. At every $t > 0$ and every $x, y \in \mathbb{R}^d$,

$$k_V(t,x,y) - k_W(t,x,y) = -\int_0^t \int_{\mathbb{R}^d} k_V(t-s,x,u)[V(u) - W(u)] \cdot k_W(s,u,y)\,du\,ds. \tag{DK}$$

The integral is absolutely convergent and jointly continuous in t, x, y.

Proof.
The two operators have common domain $D(H_0)$. Differentiating $e^{-(t-s)(H_0+V)}e^{-s(H_0+W)}f$ there, as in Lemma 42.1, and then extending by density gives

$$e^{-t(H_0+V)} - e^{-t(H_0+W)} = -\int_0^t e^{-(t-s)(H_0+V)}(V-W)e^{-s(H_0+W)}\,ds.$$

This is initially a strong operator identity. Gaussian convolution bounds the absolute spatial integral of its proposed kernel by $e^{Mt}\|V - W\|_\infty p_t(x-y)$, independently of s. The double integral is therefore absolutely convergent and bounded by $te^{Mt}\|V - W\|_\infty p_t(x-y)$.
We check continuity before evaluating at a particular point such as $x = 0$. Put $s = t\theta$ and first retain $\varepsilon \le \theta \le 1 - \varepsilon$. On compact parameter sets with t bounded away from zero, both kernels are continuous and the integrand has an integrable Gaussian envelope in u. Dominated convergence gives continuity of the truncated integral. The omitted intervals contribute at most

$$2\varepsilon t e^{Mt}\|V - W\|_\infty p_t(x-y),$$

uniformly on these compact sets. The full integral is continuous. Fubini against bounded compactly supported test functions identifies its operator with the strong Duhamel integral. Continuous-kernel uniqueness now proves the formula everywhere, not just almost everywhere. □

Corollary 43.1 (Potential order and scalar shifts)**.**
Let V, W be real bounded measurable potentials. If $V \le W$ almost everywhere, then, for every $t > 0$ and $x, y \in \mathbb{R}^d$,

$$k_V(t, x, y) \ge k_W(t, x, y).$$

Consequently, if real constants m, b satisfy $m \le V \le b$ almost everywhere, then

$$e^{-bt} p_t(x - y) \le k_V(t, x, y) \le e^{-mt} p_t(x - y).$$

For every real constant c, one also has

$$k_{V+c}(t, x, y) = e^{-ct} k_V(t, x, y).$$

Proof.
Choose a common bound for the norms of V, W and apply Lemma 43.1. Both kernel factors in its integrand are nonnegative by Proposition 42.2. Since $V - W \le 0$ almost everywhere, the leading minus sign makes their integral nonnegative. This proves the comparison at every x, y. Comparison with the constant potentials b and m, whose kernels are $e^{-bt} p_t$ and $e^{-mt} p_t$, gives the two-sided estimate. Finally, scalar spectral translation gives

$$e^{-t(H_0+V+c)} = e^{-ct} e^{-t(H_0+V)}.$$

Both sides have continuous kernels, so continuous-kernel uniqueness proves the shift identity pointwise. □
The comparison also says that $e^{-t(H_0+V)} f \ge e^{-t(H_0+W)} f$ for every nonnegative $f \in L^2$, using the continuous representatives. This is order on nonnegative functions; no inequality between quadratic forms of these operators is asserted.
If a family of potentials has a common lower bound m, its kernels are bounded above by $e^{-mt} p_t$. In a Duhamel comparison the two kernel factors are therefore bounded by

$$e^{-m(t-s)} p_{t-s}(x - u)\, e^{-ms} p_s(u - y) = e^{-mt} p_{t-s}(x - u) p_s(u - y).$$

Thus in the Duhamel-derived pointwise, weighted-norm, and localized operator-norm parameter estimates below, the factor e^{Mt} can be replaced by e^{-mt}, including when it is combined with other exponential factors. For the original dynamical family one may take $m = \min_{z \in Z} v(z)$, which exists by continuity and compactness. This improvement concerns these parameter comparison estimates; it does not assert the same substitution in the separate spatial or time derivative estimates.

43.3 The Gaussian moment controlling parameter regularity

Lemma 43.2 (Weighted Gaussian convolution)**.**
For $0 < s < t$ and $r \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} (1 + |u|) p_{t-s}(u) p_s(r - u)\, du \le p_t(r) \left(1 + \frac{t - s}{t} |r| + \sqrt{\frac{2ds(t - s)}{t}} \right).$$

Consequently, with $c_d = \pi\sqrt{2d}/8$,

$$\int_0^t \int_{\mathbb{R}^d} (1+|u|) p_{t-s}(u) p_s(r-u)\, du\, ds \le p_t(r) \left[t\left(1 + \frac{|r|}{2}\right) + c_d t^{3/2} \right].$$

Proof.
Completing the square gives

$$p_{t-s}(u) p_s(r-u) = p_t(r) p_a(u-m), \qquad a = \frac{s(t-s)}{t}, \quad m = \frac{t-s}{t} r.$$

The density p_a has mass one and second moment $\int |q|^2 p_a(q)\, dq = 2da$, by one-dimensional Gaussian integration in each coordinate. The triangle inequality and Cauchy–Schwarz give

$$\int |u| p_a(u-m)\, du \le |m| + \sqrt{2da}.$$

This proves the first estimate. The mean term integrates to $t|r|/2$, and the square-root term integrates to

$$\sqrt{2d}\, t^{3/2} \int_0^1 \sqrt{\theta(1-\theta)}\, d\theta = \frac{\pi\sqrt{2d}}{8} t^{3/2}.$$

Indeed, $\theta = (1+q)/2$ makes the integral one quarter of the area under the upper unit semicircle, namely $\pi/8$. Adding the integral t of the constant term proves the result. □

Theorem 43.1 (Lipschitz dependence on the dynamical parameter)**.**
Under (D), the canonical kernel satisfies

$$\begin{aligned} |k_t(z,r) - k_t(w,r)| &\le LCe^{Mt} d_Z(z,w) p_t(r) \\ &\quad \cdot \left[t\left(1 + \frac{|r|}{2}\right) + c_d t^{3/2} \right], \qquad c_d = \frac{\pi\sqrt{2d}}{8}. \end{aligned} \tag{L}$$

It is jointly continuous on $(0,\infty) \times Z \times \mathbb{R}^d$.

Proof.
Apply Lemma 43.1 to V_z, V_w at $x = 0, y = r$. Their actual difference is bounded, so that lemma applies. In its absolute integrand use the sharper bound (P) and both Gaussian kernel bounds. The result is at most

$$LCe^{Mt} d_Z(z,w) \int_0^t \int (1+|u|) p_{t-s}(u) p_s(r-u)\, du\, ds.$$

Lemma 43.2 proves the estimate. Its coefficient is bounded on compact positive time intervals and bounded sets of r. For fixed z, continuity in t, r is already proved. At a converging triple (t,z,r), insert and subtract $k_t(z_0,r)$; this locally uniform Lipschitz bound and continuity for fixed z_0 prove joint continuity. □

43.4 Exponential decay and weighted norms

For $a \ge 0$, let $C_a(\mathbb{R}^d)$ consist of continuous functions f such that $e^{a|r|} f(r)$ tends to zero as $|r| \to \infty$, with norm

$$\|f\|_a = \sup_{r \in \mathbb{R}^d} e^{a|r|} |f(r)|.$$

Multiplication by $e^{a|r|}$ identifies this with the Banach space $C_0(\mathbb{R}^d)$. Completeness follows because uniform limits preserve continuity and vanishing at infinity.

Corollary 43.2 (Weighted decay and Lipschitz bounds)**.**
For every $t > 0$ and $a \geq 0$, the map $z \mapsto k_t(z, \cdot)$ takes values in C_a and is Lipschitz. More precisely,

$$\|k_t(z,\cdot)\|_a \leq (4\pi t)^{-d/2} e^{(M+a^2)t},$$

and

$$\|k_t(z,\cdot) - k_t(w,\cdot)\|_a \leq LC(4\pi t)^{-d/2} e^{(M+2a^2)t} \cdot \left[t + (c_d + e^{-1/2})t^{3/2}\right] d_Z(z,w).$$

The map $(t, z) \mapsto k_t(z, \cdot)$ is continuous into C_a for $t > 0$. The Lipschitz constants are uniform on every $[\tau, b]$ with $0 < \tau \leq b < \infty$.

Proof.
The Gaussian bound gives weighted decay. Completing the square gives $a\rho - \rho^2/(4t) \leq a^2 t$, proving the first norm bound. For the second use

$$a\rho - \frac{\rho^2}{4t} \leq 2a^2 t - \frac{\rho^2}{8t}, \qquad \sup_{\rho \geq 0} \rho e^{-\rho^2/(8t)} = 2\sqrt{t/e},$$

in (L), and take the supremum.
On $[\tau, b]$ there is the uniform weighted tail bound

$$e^{a|r|}|k_t(z,r)| \leq e^{Mb}(4\pi\tau)^{-d/2} \exp\left(a|r| - \frac{|r|^2}{4b}\right).$$

It tends to zero uniformly in t, z outside large balls. On a fixed compact ball, joint continuity is uniform on $[\tau, b] \times Z$ by compactness. Splitting the norm into this ball and its complement proves norm continuity. The Lipschitz coefficient is bounded on $[\tau, b]$. □
Exponential decay therefore holds for every fixed exponential rate, with Lipschitz parameter dependence in that weighted norm. There is no continuity at $t = 0$ in these norms: $k_t(z, 0) \geq e^{-Mt}(4\pi t)^{-d/2}$ diverges as $t \downarrow 0$. Nor does this establish operator-norm Lipschitz dependence on z on all of L^2: the covariant formula uses $T(g, z)$, whose distortion need not be bounded uniformly in g.

43.5 Isometric actions and a torus example

Proposition 43.2 (Sharper estimate for an isometric action)**.**
Suppose, in addition, that every $T(g, \cdot)$ is an isometry of Z. For $t > 0$, $r \in \mathbb{R}^d$, and $z, w \in Z$,

$$|k_t(z,r) - k_t(w,r)| \leq Lte^{Mt} p_t(r) d_Z(z,w).$$

Consequently, for every $a \geq 0$,

$$\|k_t(z,\cdot) - k_t(w,\cdot)\|_a \leq Lt(4\pi t)^{-d/2} e^{(M+a^2)t} d_Z(z,w).$$

Proof.
The isometry assumption gives the uniform spatial estimate

$$|V_z(u) - V_w(u)| \leq L d_Z(T(u,z), T(u,w)) = L d_Z(z,w).$$

Apply Lemma 43.1 and the Gaussian bounds to obtain

$$\begin{aligned}|k_t(z,r) - k_t(w,r)| &\le Le^{Mt} d_Z(z,w) \int_0^t \int_{\mathbb{R}^d} p_{t-s}(u) p_s(r-u)\, du\, ds \\ &= Lte^{Mt} p_t(r) d_Z(z,w).\end{aligned}$$

The equality is Gaussian convolution. Multiplying by $e^{a|r|}$ and using $a|r| - |r|^2/(4t) \le a^2 t$ proves the norm estimate. Membership in C_a was already established above. □

For a concrete nonconstant family, let $m \ge 1$, set $Z = \mathbb{R}^m/\mathbb{Z}^m$, and equip this torus with

$$d_Z([x],[y]) = \min_{n \in \mathbb{Z}^m} |x - y - n|.$$

The minimum exists since only finitely many lattice points can lie within any fixed bounded distance of $x - y$. Changing representatives just reindexes the lattice. This defines a metric: positivity and symmetry follow directly, and adding minimizing lattice vectors gives the triangle inequality. The quotient map is distance-decreasing, and its restriction to $[0,1]^m$ is surjective; hence Z is compact. For any real $m \times d$ matrix A, define

$$T(g,[x]) = [x + Ag], \qquad v([x]) = \cos(2\pi x_1).$$

Translation preserves the metric, and the group-action identities are immediate. Joint continuity follows from

$$d_Z(T(g,z), T(h,w)) \le d_Z(z,w) + \|A\|\, |g - h|.$$

The sampling function is well defined and nonconstant. For a minimizing lattice vector n, periodicity and the real mean-value bound give

$$|v([x]) - v([y])| \le 2\pi |x_1 - y_1 - n_1| \le 2\pi d_Z([x],[y]).$$

Thus $M = 1$ and $L = 2\pi$ are valid constants. The potentials are

$$V_{[x]}(g) = \cos\bigl(2\pi(x_1 + (Ag)_1)\bigr).$$

If the first row of A is nonzero, each is spatially nonconstant. The proposition gives the explicit estimate

$$\|k_t(z,\cdot) - k_t(w,\cdot)\|_a \le 2\pi t (4\pi t)^{-d/2} e^{(1+a^2)t} d_Z(z,w).$$

Here the parameter distortion has no spatial growth, so the Gaussian first-moment term needed for a general action disappears. This is a worked application of the proved comparison formula, not a claim of novelty for torus potentials.

43.6 An optional extension to Hölder sampling

In this subsection only, replace the Lipschitz assumption on v by

$$|v(z) - v(w)| \le L_\alpha d_Z(z,w)^\alpha, \qquad 0 < \alpha \le 1.$$

Keep compactness of Z, continuity of the action, and (D). The function v is still continuous and bounded; write $M = \|v\|_\infty$. Thus all bounded-potential kernel constructions and the covariance formula remain applicable.

Theorem 43.2 (Hölder dependence on the parameter)**.**
Under these local assumptions, for $t > 0$,

$$|k_t(z,r) - k_t(w,r)| \le L_\alpha C^\alpha d_Z(z,w)^\alpha e^{Mt} t p_t(r) \cdot \left(1 + \frac{|r|}{2} + c_d\sqrt{t}\right)^\alpha, \qquad c_d = \frac{\pi\sqrt{2d}}{8}.$$

For every $a \ge 0$, the kernels belong to C_a, and

$$\|k_t(z,\cdot) - k_t(w,\cdot)\|_a \le L_\alpha C^\alpha d_Z(z,w)^\alpha (4\pi t)^{-d/2} e^{(M+2a^2)t} \cdot \left[t + (c_d + e^{-1/2})t^{3/2}\right].$$

These Hölder constants are uniform on every compact positive-time interval. The map $(t,z) \mapsto k_t(z,\cdot)$ is continuous into C_a.

Proof.
Put $\delta = d_Z(z,w)$. The potential difference obeys

$$|V_z(u) - V_w(u)| \le L_\alpha C^\alpha \delta^\alpha (1 + |u|)^\alpha.$$

Its actual supremum is at most $2M$, so the pointwise Duhamel comparison applies without any unbounded-perturbation assumption. Gaussian bounds reduce the desired difference to at most

$$L_\alpha C^\alpha \delta^\alpha e^{Mt} \int_0^t \int_{\mathbb{R}^d} (1 + |u|)^\alpha p_{t-s}(u) p_s(r-u)\, du\, ds.$$

For fixed t, r, define the probability measure on $(0,t) \times \mathbb{R}^d$ by

$$d\nu_{t,r}(s,u) = \frac{p_{t-s}(u) p_s(r-u)}{t p_t(r)}\, du\, ds.$$

Its mass is one by Gaussian convolution, and its first spatial moment is finite by Lemma 43.2. In particular,

$$\int (1 + |u|)\, d\nu_{t,r} \le 1 + \frac{|r|}{2} + c_d\sqrt{t}.$$

The function $x \mapsto x^\alpha$ is increasing and concave on $[0,\infty)$. Jensen's inequality, applied to this probability measure, gives

$$\int (1 + |u|)^\alpha\, d\nu_{t,r} \le \left(\int (1 + |u|)\, d\nu_{t,r}\right)^\alpha \le \left(1 + \frac{|r|}{2} + c_d\sqrt{t}\right)^\alpha.$$

Multiplication by $t p_t(r)$ proves the pointwise estimate.
The base in its last factor is at least one, so its α-power is bounded by the base itself. The two elementary weighted Gaussian inequalities used in Corollary 43.2 therefore give exactly the displayed, deliberately nonoptimal, weighted bound. Gaussian domination still proves membership in C_a and a uniform weighted tail on $[\tau, b]$, $0 < \tau \le b < \infty$. For fixed z, kernel continuity is already known. The pointwise Hölder bound is uniform on compact sets of (t,r), so insertion of a fixed-parameter kernel proves joint continuity in (t,z,r). On a compact spatial ball this continuity is uniform; outside that ball the uniform weighted tail tends to zero. This proves continuity into C_a. The displayed coefficient is bounded on $[\tau, b]$, proving the asserted uniformity. □
For $\alpha = 1$ the pointwise estimate is precisely (L). For $0 < \alpha < 1$ it supplies a proved extension to less regular sampling functions, without changing the assumptions of the preceding Lipschitz results.

43.7 An extension to locally uniformly Lipschitz actions

The linear growth assumption gives particularly useful constants, but the method also handles exponential growth of the action's Lipschitz constants. This subsection replaces the bound (D); it retains the compact metric space, the continuous action, and the Lipschitz sampling function from the original setting.

Theorem 43.3 (Exponential action growth)**.**
Suppose instead that $C \geq 0$ and $\kappa \geq 0$ satisfy

$$d_Z(T(u,z),T(u,w)) \leq Ce^{\kappa|u|}d_Z(z,w).$$

For every $t > 0$ and $r \in \mathbb{R}^d$,

$$|k_t(z,r) - k_t(w,r)| \leq LC\,2^{d/2}t\,e^{(M+\kappa^2/2)t}e^{\kappa|r|}p_t(r)d_Z(z,w).$$

In particular, for every $a \geq 0$ the map $z \mapsto k_t(z,\cdot)$ is Lipschitz into C_a, with

$$\|k_t(z,\cdot) - k_t(w,\cdot)\|_a \leq LC\,2^{d/2}t(4\pi t)^{-d/2}e^{[M+\kappa^2/2+(a+\kappa)^2]t}d_Z(z,w).$$

The map $(t,z) \mapsto k_t(z,\cdot)$ is continuous into C_a for $t > 0$, and the Lipschitz constants are uniform on compact positive time intervals.

Proof.
For $b > 0$ and $q \in \mathbb{R}^d$, completing a scalar square gives

$$\kappa|q| \leq \frac{|q|^2}{8b} + 2\kappa^2 b, \qquad e^{\kappa|q|}p_b(q) \leq 2^{d/2}e^{2\kappa^2 b}p_{2b}(q).$$

Hence, for the bridge parameters $b = s(t-s)/t$ and $m = (t-s)r/t$, translation and the triangle inequality give

$$\begin{aligned}\int e^{\kappa|u|}p_b(u-m)\,du &\leq e^{\kappa|m|}\int e^{\kappa|q|}p_b(q)\,dq \\ &\leq 2^{d/2}e^{\kappa|r|+\kappa^2 t/2},\end{aligned}$$

because $|m| \leq |r|$ and $b \leq t/4$. The potential difference is bounded both by $2M$ and by $LCe^{\kappa|u|}d_Z(z,w)$. The pointwise Duhamel comparison is therefore applicable. Its Gaussian bounds and the bridge identity reduce its absolute value to at most

$$LCe^{Mt}d_Z(z,w)p_t(r)\int_0^t\int e^{\kappa|u|}p_{s(t-s)/t}(u-(t-s)r/t)\,du\,ds.$$

The preceding moment bound and integration over an interval of length t prove the pointwise assertion. The time endpoints are null sets and require no bridge density of variance zero.
Multiplication by $e^{a|r|}$ and $(a+\kappa)|r| - |r|^2/(4t) \leq (a+\kappa)^2 t$ give the weighted bound. Membership in C_a and uniform weighted tails still follow from the original Gaussian kernel estimate, independently of the action's distortion. The pointwise parameter bound is uniform on compact sets of positive time and bounded r. Combining it with fixed-parameter kernel continuity proves joint continuity there; splitting into a compact spatial ball and the uniform tail then gives continuity into C_a. The displayed coefficient is bounded on every compact positive time interval. □

It is enough to assume a uniform Lipschitz bound $B \geq 1$ for the maps $T(u,\cdot)$ with $|u| \leq 1$, while retaining joint continuity of the action. If $u \neq 0$, set $n = \lceil |u| \rceil$. The action identity writes $T(u,\cdot)$ as a composition of n copies of $T(u/n,\cdot)$, each with Lipschitz constant at most B. Thus

$$\operatorname{Lip}(T(u,\cdot)) \leq B^n \leq B\, e^{(\log B)|u|}.$$

At $u = 0$, the identity map obeys the same bound. The theorem applies with $C = B$ and $\kappa = \log B$. This yields the weighted Lipschitz conclusion under a local uniform bound. The earlier estimates remain available when their stronger hypotheses hold; their different constants can be compared in the particular regime of interest.

43.8 Symmetry and the covariant semigroup law

For $s, t > 0$, the canonical kernels satisfy

$$k_t(z,r) = k_t(T(r,z),-r), \qquad k_{t+s}(z,r) = \int_{\mathbb{R}^d} k_t(z,u) k_s(T(u,z), r-u)\, du.$$

Symmetry of $K_z(t,0,r)$ and covariance at $(r,0)$ prove the first identity. The pointwise composition law at $(0,r)$ and covariance of its second factor prove the second. Gaussian convolution gives absolute convergence. In the negative-action convention replace $T(r,z)$ by $T(-r,z)$ in the first identity and $T(u,z)$ by $T(-u,z)$ in the second, and use the corresponding kernels $\widetilde{k}$.

43.9 A counterexample to mixing translation signs

For bounded real V continuous at g, put $M = \|V\|_\infty$. Its Dyson expansion gives, as $t \downarrow 0$,

$$\frac{k_V(t,g,g)}{p_t(0)} = 1 - tV(g) + o(t). \tag{S}$$

Here are the details needed for this use of a short-time expansion. After division by $p_t(0)$, terms of order at least two are bounded by $e^{Mt} - 1 - Mt = O(t^2)$. The first-order term divided by $tp_t(0)$ is the negative average of V against Gaussians centered at g with covariance $2t\theta(1-\theta)I$, integrated over $0 < \theta < 1$. For each such θ, scaling to a fixed Gaussian and bounded convergence give the limit $V(g)$. Bounded convergence in θ then proves the displayed expansion.
Take $d = 1$, $Z = \mathbb{R}/(2\pi\mathbb{Z})$ with its circle metric, $T(g,z) = z + g$, and $v(z) = \cos z$. The action is isometric, so it satisfies (D) with $C = 1$, and v is Lipschitz. At $z = g = \pi/4$, one has $V_z(g) = 0$. The incorrect negative-action formula for this same potential would equate $K_z(t,g,g)$ with $k_t(T(-g,z),0)$, whose local potential at zero is $v(0) = 1$. Their first-order coefficients in (S) disagree. The obstruction is mixing two conventions; either consistently defined family is valid.

43.10 Lipschitz operator bounds after spatial localization

Return to the original linear action bound (D) and Lipschitz sampling function. The kernel estimate also controls the full operator after a spatial cutoff. This provides a direct quantitative connection with local operator-norm continuity, while keeping global operator-norm continuity as a separate question. Write $S_z(t) = e^{-tH_z}$.

Proposition 43.3 (Local operator-norm Lipschitz estimate)**.**
Let $R \geq 0$ and let B be a Lebesgue-measurable subset of $\{g \in \mathbb{R}^d : |g| \leq R\}$. Denote multiplication by its indicator by χ_B. For $t \geq 0$, both

$$\|\chi_B(S_z(t) - S_w(t))\|_{2\to 2}, \qquad \|(S_z(t) - S_w(t))\chi_B\|_{2\to 2}$$

are at most

$$LCe^{Mt}\left[t(1+R)+\left(c_d+\sqrt{d/2}\right)t^{3/2}\right]d_Z(z,w).$$

The estimate is uniform on bounded time intervals, including time zero.

Proof.
For $t>0$, use the pointwise Duhamel formula at (g,h) and the potential bound $LC(1+|u|)d_Z(z,w)$. Put $r=h-g$ and $u=g+v$. Since $1+|g+v|\le 1+|g|+|v|$, Gaussian convolution and Lemma 43.2 give

$$\begin{aligned}|K_z(t,g,h)-K_w(t,g,h)| &\le LCe^{Mt}d_Z(z,w)p_t(h-g)\\ &\quad\cdot\left[t\left(1+|g|+\frac{|h-g|}{2}\right)+c_dt^{3/2}\right].\end{aligned}$$

For $g\in B$, replace $|g|$ by R. Define the nonnegative, even, integrable function

$$q_t(r)=LCe^{Mt}d_Z(z,w)p_t(r)\left[t\left(1+R+\frac{|r|}{2}\right)+c_dt^{3/2}\right].$$

The preceding estimate, and the fact that the localized output is zero outside B, imply

$$|\chi_B(S_z(t)-S_w(t))f|\le q_t*|f|.$$

Young's inequality in the case $L^1*L^2\to L^2$ gives an operator norm bound $\|q_t\|_1$. The Gaussian has mass one and second moment $2dt$, so Cauchy–Schwarz gives $\int|r|p_t(r)\,dr\le\sqrt{2dt}$. Substitution proves the displayed coefficient. Both $S_z(t)-S_w(t)$ and χ_B are self-adjoint; taking adjoints therefore gives the same norm for input localization. At $t=0$ the semigroup difference is zero. The coefficient is continuous on every $[0,b]$, which proves uniformity there. □

43.11 Strong parameter continuity and a norm-discontinuous family

The continuity of a canonical kernel in C_a controls a specified spatial row. It also implies joint continuity of the full kernel in (t,z,g,h) for $t>0$: use the continuous action and covariance $K_z(t,g,h)=k_t(T(g,z),h-g)$. The following statements distinguish this from continuity of the full operator in operator norm. The localized estimate above holds even for the family whose global norm continuity fails below.

Proposition 43.4 (Parameter continuity without a quantitative action bound)**.**
For this proposition it suffices that Z be compact metric, the action be continuous, and $v:Z\to\mathbb{R}$ be continuous. If $z_n\to z$, then, for every $f\in L^2$ and $b>0$,

$$\sup_{0\le t\le b}\|(S_{z_n}(t)-S_z(t))f\|_2\longrightarrow 0.$$

Neither a Lipschitz bound nor the action bound (D) is needed here. Under the same hypotheses, $K_z(t,g,h)$ is jointly continuous in (t,z,g,h) for $t>0$, and $(t,z)\mapsto k_t(z,\cdot)$ is continuous into every C_a, $a\ge 0$, on positive times. For any bounded Lebesgue-measurable set B, both localized operator norms

$$\sup_{0\le t\le b}\|\chi_B(S_{z_n}(t)-S_z(t))\|_{2\to 2},\qquad \sup_{0\le t\le b}\|(S_{z_n}(t)-S_z(t))\chi_B\|_{2\to 2}$$

also tend to zero. These qualitative continuity conclusions require no rate of continuity for the sampling function or the action.

Proof.
Strong continuity. Compactness gives $M = \|v\|_\infty < \infty$, and continuity gives $V_{z_n}(x) \to V_z(x)$ at each x. The operator Duhamel identity gives

$$(S_{z_n}(t) - S_z(t))f = -\int_0^t S_{z_n}(t-s)(V_{z_n} - V_z)S_z(s)f\,ds.$$

Consequently the supremum in the statement is at most

$$e^{Mb}\int_0^b \|(V_{z_n} - V_z)S_z(s)f\|_2\,ds.$$

For each fixed s, dominated convergence in the spatial variable makes the integrand tend to zero. It is bounded by $2Me^{Mb}\|f\|_2$. Dominated convergence in s proves the claim.
Kernel and weighted-norm continuity. The function $(z,u) \mapsto V_z(u) = v(T(u,z))$ is continuous and bounded in absolute value by M. Track z through the proof of the Dyson kernel series. Its zeroth coefficient is the Gaussian. In the induction step, first restrict the relative time variable to $\varepsilon \le s/t \le 1-\varepsilon$. The integrand is continuous in t, z, g, h for each integration point u, and on compact positive time and spatial ranges it has the same integrable Gaussian majorant as before, uniformly in z. Dominated convergence proves continuity of the truncated integral. The omitted endpoint bounds depend only on M, t, g, h, so they tend to zero uniformly on these compact ranges. Induction proves joint continuity of every coefficient. The factorial estimate makes the series locally uniformly convergent, proving the joint continuity of K_z.
On $t \in [\tau, b]$, $0 < \tau \le b$, the original Gaussian bound gives

$$e^{a|r|}|k_t(z,r)| \le e^{Mb}(4\pi\tau)^{-d/2}\exp\left(a|r| - \frac{|r|^2}{4b}\right).$$

The right side tends to zero as $|r| \to \infty$, independently of t, z. On a compact spatial ball, the joint continuity just proved is uniform on $[\tau, b] \times Z$. Splitting into that ball and its uniform tail proves continuity into C_a.
Localized operator norms. Choose R with $B \subseteq \{|g| \le R\}$ and fix $R' > R$. Uniform continuity of $v \circ T$ on $\{|u| \le R'\} \times Z$ gives

$$\eta_{n,R'} = \sup_{|u| \le R'} |V_{z_n}(u) - V_z(u)| \longrightarrow 0.$$

Split the potential difference in the Duhamel integral into its part on this ball and its complement. The first part contributes at most $be^{Mb}\eta_{n,R'}$ to the supremum of the localized operator norm over $0 \le t \le b$.
For the other part put $D = R' - R > 0$. The kernel of $\chi_B S_{z_n}(\sigma)\chi_{\{|u|>R'\}}$, $\sigma > 0$, is in absolute value bounded by

$$e^{M\sigma}p_\sigma(g-u)\mathbf{1}_{\{|g-u|>D\}}.$$

Convolution domination and Young's inequality bound its L^2 operator norm by

$$e^{M\sigma}\int_{|r|>D} p_\sigma(r)\,dr.$$

Gaussian scaling shows that this tail increases with σ, so for $0 < \sigma \le b$ it is at most the tail at b. At $\sigma = 0$ the two spatial cutoffs are disjoint, and their product is zero. The potential difference

on the complement has norm at most $2M$, and $\|S_z(s)\| \le e^{Ms}$. Thus

$$\sup_{0\le t\le b} \|\chi_B(S_{z_n}(t) - S_z(t))\|_{2\to 2} \le b e^{Mb} \left[\eta_{n,R'} + 2M \int_{|r|>R'-R} p_b(r)\, dr \right].$$

First choose R' large enough to make the Gaussian tail small, and then choose n large enough to make $\eta_{n,R'}$ small. This proves convergence of the output-localized operator norms. Taking adjoints gives the input-localized statement. □

The earlier regularity hypotheses supply the displayed quantitative Lipschitz and Hölder rates. The present proposition separates those rates from qualitative continuity, which follows from weaker assumptions. For comparison, in dimensions $d \ge 2$ the localized continuity is also a bounded-potential consequence of Broderix et al. (2000, Theorem 5.1): uniformly bounded potentials converging locally uniformly satisfy their local Kato convergence and uniform negative-part hypotheses. Their kinetic normalization is $-\frac{1}{2}\sum_j \partial_j^2$, so the source potential and time must be $V/2$ and $2t$ to represent $H_0 + V$ here. The proof above applies directly in every dimension $d \ge 1$ and keeps the localization and Gaussian-tail estimates explicit.

Here is a family satisfying even the original Lipschitz hypotheses for which operator-norm continuity fails. Let $Z = \mathbb{R}^2/\mathbb{Z}^2$ have the quotient Euclidean metric and set

$$T(g,[x,y]) = [x, y + g\sin(2\pi x)], \qquad v([x,y]) = \cos(2\pi y), \qquad g \in \mathbb{R}.$$

Periodicity makes both definitions independent of representatives. The first coordinate stays fixed, so adding the second-coordinate increments proves the action law; the formulas also give continuity. For representatives whose difference realizes the quotient distance, the sine Lipschitz bound and the triangle inequality give

$$d_Z(T(g,z),T(g,w)) \le (1 + 2\pi|g|) d_Z(z,w) \le 2\pi(1+|g|) d_Z(z,w).$$

Thus $C = 2\pi$, $M = 1$, and $L = 2\pi$ are valid constants.

Put $z_0 = [0,0]$, $z_\varepsilon = [\varepsilon, 0]$, and $a_\varepsilon = \sin(2\pi\varepsilon)$, with $0 < \varepsilon < 1/4$. Then $z_\varepsilon \to z_0$ as $\varepsilon \downarrow 0$, whereas

$$V_{z_0}(g) = 1, \qquad V_{z_\varepsilon}(g) = \cos(2\pi a_\varepsilon g), \qquad \|V_{z_\varepsilon} - V_{z_0}\|_\infty = 2.$$

In fact, for every fixed $t > 0$,

$$\lim_{\varepsilon\downarrow 0} \|S_{z_\varepsilon}(t) - S_{z_0}(t)\|_{2\to 2} = e^t - e^{-t} > 0. \tag{N}$$

To prove the lower bound, let $b_\varepsilon = 1/(2a_\varepsilon)$ and conjugate by the unitary translation $(U_b f)(x) = f(x+b)$. It commutes with H_0 and sends multiplication by V_{z_ε} to multiplication by

$$W_\varepsilon(u) = V_{z_\varepsilon}(u + b_\varepsilon) = -\cos(2\pi a_\varepsilon u) \longrightarrow -1.$$

Figure .1 illustrates the two spatial windows used in this argument.

More explicitly, the conjugation identities are

$$U_{b_\varepsilon}(H_0 + V_{z_\varepsilon})U_{b_\varepsilon}^{-1} = H_0 + W_\varepsilon,$$
$$U_{b_\varepsilon} S_{z_\varepsilon}(t) U_{b_\varepsilon}^{-1} = e^{-t(H_0+W_\varepsilon)}.$$

The potentials W_ε are bounded by one. The same Duhamel and dominated-convergence argument just proved, now comparing these potentials directly with the constant -1, gives

$$e^{-t(H_0+W_\varepsilon)} f \longrightarrow e^t P_t f \qquad (f \in L^2).$$

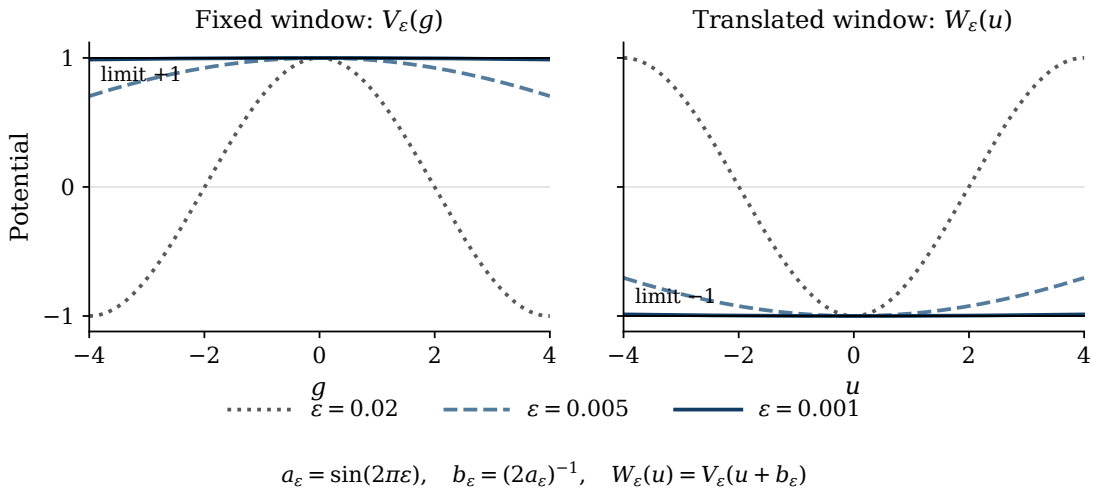


Figure .1: Fixed and translated windows for the shear-torus potentials, with $V_\varepsilon = V_{z_\varepsilon}$ in the plot. As $\varepsilon \downarrow 0$, V_{z_ε} tends to 1 on each fixed bounded window, while $W_\varepsilon(u) = V_{z_\varepsilon}(u + b_\varepsilon)$ tends to -1, with $b_\varepsilon \to \infty$. The curves show the explicit potentials, not numerical approximations to their semigroups.

The member $S_{z_0}(t) = e^{-t}P_t$ is unchanged by translation. Unitary invariance of the norm therefore implies, for any unit vector f,

$$\liminf_{\varepsilon \downarrow 0} \|S_{z_\varepsilon}(t) - S_{z_0}(t)\| \geq (e^t - e^{-t})\|P_t f\|_2.$$

Here $\|P_t\|_{2\to 2} = 1$: the upper bound is known, and a unit vector whose Fourier transform is supported in $\{|\xi| < \delta\}$ gives $\|P_t f\|_2 \geq e^{-t\delta^2}$. Letting $\delta \downarrow 0$ proves the desired lower bound.
For the upper bound, the two-sided Gaussian estimates with $M = 1$ give

$$0 \leq K_{z_\varepsilon}(t, x, y) - e^{-t}p_t(x - y) \leq (e^t - e^{-t})p_t(x - y).$$

Integration against $|f|$ and the L^2 contraction property of P_t show that the operator norm of the difference is at most $e^t - e^{-t}$. This proves (N). Nevertheless the strong-continuity proposition applies, and the earlier weighted Lipschitz estimate gives convergence of $k_t(z_\varepsilon, \cdot)$ to $k_t(z_0, \cdot)$ in every C_a. The loss of operator-norm continuity comes from spatial translates moving out to infinity, which are not controlled uniformly by convergence of this single row.

43.12 Application to spectra over invariant subsystems

The preceding counterexample concerns operator norm at individual phases. There is a different useful consequence of the kernel estimates: spectral continuity when entire invariant subsystems are compared. We give a direct Euclidean heat-kernel proof of the localization argument in Beckus and Takase (2025, Lemmas 4.2–4.3 and Theorem 4.6). Smooth cutoffs, the Gaussian first moment, and the localized operator estimate supply the needed bounds. Only the self-adjoint spectral calculus already used in the thesis is imported in this proof.

Let $\mathcal{S}$ be the collection of nonempty closed subsets of Z invariant under every $T(g,\cdot)$. For $X, Y \in \mathcal{S}$, write

$$d_H(X,Y) = \max\left\{ \sup_{x\in X}\inf_{y\in Y} d_Z(x,y), \sup_{y\in Y}\inf_{x\in X} d_Z(x,y) \right\}.$$

For each fixed $t > 0$ define the compact spectral set

$$\Sigma_t(X) = \overline{\bigcup_{z\in X} \sigma(e^{-tH_z})} \subseteq [0, e^{Mt}].$$

The overline denotes closure in $\mathbb{R}$. Each spectrum is nonempty, and operator positivity from spectral calculus and the norm bound for the heat operator give the displayed inclusion, hence compactness.

Lemma 43.3 (Averaged localization of a heat operator)**.**
Fix a real function $\chi \in C_c^1(B(0,1))$ with $\|\chi\|_2 = 1$, and put $G_\chi = \|\nabla\chi\|_2$. For $R > 0$ and $a \in \mathbb{R}^d$, let $\chi_{a,R}(x) = R^{-d/2}\chi((x-a)/R)$, also denoting the corresponding multiplication operator. Write $S_z = e^{-tH_z}$ for a fixed $t > 0$, and define

$$B_t = e^{Mt}\sqrt{2dt}\, G_\chi.$$

For every $f \in L^2$,

$$\int \|\chi_{a,R} f\|_2^2 \, da = \|f\|_2^2, \qquad \left(\int \|[S_z, \chi_{a,R}]f\|_2^2 \, da\right)^{1/2} \le \frac{B_t}{R}\|f\|_2.$$

Proof.
Tonelli and the change of variable $u = (x-a)/R$ give the first identity. Write $\chi_R = \chi_{0,R}$. The fundamental theorem of calculus along a segment, followed by the triangle inequality in L^2, gives

$$\|\chi_R(\cdot + r) - \chi_R\|_2 \le |r|\|\nabla\chi_R\|_2 = \frac{|r|G_\chi}{R}.$$

The commutator has the absolutely convergent kernel expression

$$([S_z, \chi_{a,R}]f)(x) = \int K_z(t, x, x+r)[\chi_R(x+r-a) - \chi_R(x-a)]f(x+r)\, dr.$$

Absolute convergence at each a, x follows from the Gaussian bound, boundedness of χ_R, and Cauchy–Schwarz against f. All these integrands are jointly measurable. Put $m_t = \int |r| p_t(r)\, dr \le \sqrt{2dt}$. Scalar Cauchy–Schwarz, with weight $|r|p_t(r)$, gives

$$|([S_z, \chi_{a,R}]f)(x)|^2 \le e^{2Mt} m_t \int \frac{p_t(r)}{|r|} |\chi_R(x+r-a) - \chi_R(x-a)|^2 |f(x+r)|^2\, dr.$$

The value at $r = 0$ is irrelevant. This inequality is valid with an infinite right side; the next calculation proves its integrability. For each fixed r, integration in a of the squared difference equals $\|\chi_R(\cdot + r) - \chi_R\|_2^2$, independently of x. Tonelli and translation invariance in x therefore give

$$\begin{aligned}\int \|[S_z, \chi_{a,R}]f\|_2^2 \, da &\le e^{2Mt} m_t \|f\|_2^2 \int \frac{p_t(r)}{|r|} \|\chi_R(\cdot + r) - \chi_R\|_2^2 \, dr \\ &\le \frac{e^{2Mt} m_t^2 G_\chi^2}{R^2}\|f\|_2^2 \le \frac{B_t^2}{R^2}\|f\|_2^2.\end{aligned}$$

This proves the commutator estimate using only scalar Cauchy–Schwarz and Tonelli. □

Corollary 43.3 (Spectral continuity of invariant subsystems)**.**
Under the original Lipschitz sampling and linear action-growth hypotheses, fix the normalized cutoff from Lemma 43.3 and put

$$Q_t = LCe^{Mt}, \qquad b_d = c_d + \sqrt{d/2}, \qquad A_t = \max\{e^{Mt},\ B_t + Q_t(2t + b_d t^{3/2})\}.$$

For all nonempty closed invariant subsets X, Y of Z,

$$\operatorname{dist}_H(\Sigma_t(X), \Sigma_t(Y)) \le A_t d_H(X,Y)^{1/2}.$$

Proof.
Let $\delta = d_H(X,Y)$. If $\delta = 0$, compactness and closedness give $X = Y$, and the claim is immediate. First suppose $0 < \delta \le 1$. Fix $z \in X$ and $\lambda \in \sigma(S_z)$. Since S_z is bounded self-adjoint, for every $\epsilon > 0$ there is a unit vector f with $\|(S_z - \lambda)f\|_2 < \epsilon$. To see this from the spectral theorem, its spectral projection on $(\lambda - \epsilon, \lambda + \epsilon)$ is nonzero; otherwise the spectral integral of $(s - \lambda)^{-1}$ would supply a bounded inverse to $S_z - \lambda$. Normalize a nonzero vector in that projection's range.
For any $R > 0$, use

$$(S_z - \lambda)\chi_{a,R} f = \chi_{a,R}(S_z - \lambda)f + [S_z, \chi_{a,R}]f.$$

The localization lemma and the triangle inequality in $L^2(da; L^2)$ give

$$\left(\int \|(S_z - \lambda)\chi_{a,R} f\|_2^2 \, da\right)^{1/2} \le \epsilon + B_t/R, \qquad \int \|\chi_{a,R} f\|_2^2 \, da = 1.$$

These are measurable functions of a; for example, $a \mapsto \chi_{a,R} f$ is norm continuous, since translates of the compactly supported continuous function χ_R vary continuously in supremum norm. For every $\eta > 0$, there is therefore an a with $\chi_{a,R} f \neq 0$ and

$$\frac{\|(S_z - \lambda)\chi_{a,R} f\|_2}{\|\chi_{a,R} f\|_2} \le \epsilon + B_t/R + \eta.$$

Otherwise the squared numerator would be at least $(\epsilon + B_t/R + \eta)^2$ times the squared denominator almost everywhere, contradicting the two integral bounds.
Normalize this localized vector and translate it with $(U_a u)(x) = u(x + a)$. The resulting unit vector v is supported in $B(0, R)$. Translation of the potential gives

$$U_a S_z U_a^{-1} = S_{T(a,z)}.$$

Thus $z' = T(a,z) \in X$ and $\|(S_{z'} - \lambda)v\|_2 \le \epsilon + B_t/R + \eta$. Compactness of Y supplies $w \in Y$ with $d_Z(z', w) \le \delta$. Proposition 43.3, applied to the ball of radius R, yields

$$\|(S_{z'} - S_w)v\|_2 \le Q_t\,[t(1 + R) + b_d t^{3/2}]\delta.$$

Spectral calculus for the self-adjoint operator S_w gives $\operatorname{dist}(\lambda, \sigma(S_w)) \le \|(S_w - \lambda)v\|_2$. Consequently

$$\operatorname{dist}(\lambda, \Sigma_t(Y)) \le \epsilon + \eta + B_t/R + Q_t[t(1 + R) + b_d t^{3/2}]\delta.$$

Let ϵ, η decrease to zero; the resulting bound is independent of the chosen localizations and of w. Choose $R = \delta^{-1/2}$ and use $\delta \le \sqrt{\delta}$ to bound the right side by

$$[B_t + Q_t(2t + b_d t^{3/2})]\sqrt{\delta}.$$

The bound is uniform in $z \in X$ and $\lambda \in \sigma(S_z)$. Distance to the closed set $\Sigma_t(Y)$ is continuous, so it extends to every $\lambda \in \Sigma_t(X)$. Interchanging X, Y gives the other directed distance, proving the desired Hausdorff bound for $0 < \delta \le 1$.
Finally, if $\delta > 1$, both nonempty spectral sets lie in $[0, e^{Mt}]$. Their Hausdorff distance is at most $e^{Mt} \le e^{Mt}\sqrt{\delta}$. The stated maximum therefore works for every δ. □

Corollary 43.4 (Qualitative spectral continuity under continuous sampling)**.**
Suppose only that Z is a nonempty compact metric space, T is a continuous action of $\mathbb{R}^d$ on Z, and $v : Z \to \mathbb{R}$ is continuous. For each fixed $t > 0$, the map $X \mapsto \Sigma_t(X)$ on nonempty closed invariant subsets of Z is uniformly continuous for Hausdorff distance.

Proof.
Put $M = \|v\|_\infty$, and retain the cutoff and constant B_t from Lemma 43.3. That lemma uses only the uniform Gaussian bound, so it remains valid under the present hypotheses. For $R > 0$, let χ_{B_R} be multiplication by the indicator of $B(0, R)$, and define

$$\omega_R(\delta) = \sup_{\substack{z,w \in Z \\ d_Z(z,w) \le \delta}} \|(S_z(t) - S_w(t))\chi_{B_R}\|_{2\to 2}, \qquad \delta \ge 0.$$

Proposition 43.4 gives operator-norm continuity of $z \mapsto S_z(t)\chi_{B_R}$. Compactness of Z makes this map uniformly continuous, and hence $\omega_R(\delta) \to 0$ as $\delta \downarrow 0$ for each fixed R.
The localization and recentering argument in the proof of Corollary 43.3 uses the quantitative hypotheses only when comparing the two operators on a vector supported in $B(0, R)$. Replacing that comparison by the definition of ω_R gives, for every $R > 0$,

$$\operatorname{dist}_H(\Sigma_t(X), \Sigma_t(Y)) \le \frac{B_t}{R} + \omega_R(d_H(X, Y)).$$

Indeed, a localized approximate spectral vector for a phase in X can be translated into $B(0, R)$; invariance keeps its translated phase in X, and compactness of Y supplies a phase within $d_H(X, Y)$. The displayed modulus bounds the operator difference on that vector. Letting the approximate-vector errors vanish, passing to spectral closures, and interchanging X, Y proves the inequality. Given a positive error tolerance, first choose R large and then choose $d_H(X, Y)$ small. Both choices are independent of X, Y, which proves uniform continuity. □

Comparison with the general group theorem. The quantitative proof uses the original metric and linear action bound. The following normalization additionally explains how the setting fits Beckus and Takase (2025, Theorem 1.3(a)), whose action bound has coefficient one at the group identity.

Lemma 43.4 (An adapted metric for the action)**.**
Under the original linear-growth hypothesis, put $C_0 = \max\{1, C\}$ and

$$d_*(z, w) = \sup_{g \in \mathbb{R}^d} \frac{d_Z(T(g, z), T(g, w))}{1 + |g|}.$$

Then d_* is a metric with the same compact topology as d_Z, and

$$d_Z \le d_* \le C_0 d_Z, \qquad d_*(T(h, z), T(h, w)) \le (1 + |h|)d_*(z, w).$$

The sampling function v remains Lipschitz with constant at most L.

Proof.
The assumed growth bound makes the supremum finite. Symmetry and the triangle inequality follow from those of d_Z, taking suprema; the term $g = 0$ gives $d_*(z, w) \geq d_Z(z, w)$ and hence definiteness. The two metric inequalities imply equivalence of their topologies and preserve compactness. With $u = g + h$, the triangle inequality in $\mathbb{R}^d$ gives

$$1 + |u| \leq (1 + |u - h|)(1 + |h|).$$

Reindexing the supremum proves the action estimate. The sampling bound is

$$|v(z) - v(w)| \leq L d_Z(z, w) \leq L d_*(z, w).$$

□

Let $d_{H,*}$ denote Hausdorff distance induced by d_*. The metric comparison gives $d_H \leq d_{H,*} \leq C_0 d_H$ on nonempty closed subsystems. With $\tau(g, z) = T(-g, z)$ and $\kappa_t(r, z) = k_t(z, r)$, the operator convention in that theorem is

$$(A_z f)(g) = \int \kappa_t(h - g, \tau(-g, z)) f(h)\, dh = e^{-tH_z} f(g).$$

The adapted action has bound $1+|g|$. Applying Theorem 43.1 in d_* gives an integrable parameter-Lipschitz envelope, since it is a Gaussian times a linear polynomial in $|r|$. The Gaussian bound also gives uniform integrability and a uniform first moment, including suprema along translated orbits. The heat operators are self-adjoint and hence normal. Finally, $\mathbb{R}^d$ with its proper translation-invariant Euclidean metric is unimodular and has ball volume $\omega_d r^d$, so the group's required two-sided volume-growth condition holds. Thus the general theorem gives another route to the result; its kernel hypotheses are verified rather than assumed for the Schrödinger family.

Corollary 43.5 (The bottom of the subsystem spectrum)**.**
Under the baseline quantitative hypotheses, for $X \in \mathcal{S}$ let

$$E(X) = \inf \bigcup_{z \in X} \sigma(H_z).$$

Then $-M \leq E(X) \leq M$, and for every $t > 0$,

$$|E(X) - E(Y)| \leq \frac{e^{tM}}{t} A_t d_H(X, Y)^{1/2}.$$

This statement concerns a spectral infimum; it does not assert that there is an eigenvector at that energy.

Proof.
The operator inequality $H_z \geq -M$ proves the lower bound. For any $\delta > 0$, choose a unit vector whose Fourier transform is supported in $\{|\xi|^2 < \delta\}$. It belongs to $D(H_0) = D(H_z)$ and satisfies

$$\langle H_z f, f \rangle = \int |\xi|^2 |\widehat{f}(\xi)|^2\, d\xi + \int V_z(x) |f(x)|^2\, dx \leq \delta + M.$$

Spectral calculus gives $\inf \sigma(H_z) \leq \langle H_z f, f \rangle$. Letting $\delta \downarrow 0$ proves the upper bound for each phase, and hence for the nonempty subsystem. Spectral mapping and the definition of infimum give

$$\max \Sigma_t(X) = e^{-tE(X)} \in [e^{-tM}, e^{tM}].$$

Maxima of nonempty compact real sets differ by at most their Hausdorff distance. On this interval the derivative of $-(1/t)\log s$ has absolute value at most e^{tM}/t. Applying this Lipschitz bound to the two maxima and then Corollary 43.3 proves the estimate. □
The objects compared here are invariant subsystems, not arbitrary pairs of individual phases. Moreover the spectral sets belong to the bounded heat operators. By spectral mapping,

$$\sigma(e^{-tH_z}) = \overline{\{e^{-tE} : E \in \sigma(H_z)\}}.$$

Passing back to unbounded energies uses a logarithm whose derivative is unbounded near zero. No unweighted Hausdorff estimate for the unbounded Schrödinger spectra, or for their intersections with a sharp energy window, is asserted. The direct Euclidean argument specializes the cited localization method; no new general spectral-continuity principle is claimed.

Example 43.1 (The shear family and its limiting subsystem)**.**
For the shear-torus family above, fix $t > 0$ and define

$$X_\varepsilon = \{[\varepsilon, y] : y \in \mathbb{R}/\mathbb{Z}\}, \qquad X_0 = \{[0, y] : y \in \mathbb{R}/\mathbb{Z}\}.$$

For $0 < \varepsilon < 1/4$, the nonzero speed $a_\varepsilon = \sin(2\pi\varepsilon)$ makes the orbit of z_ε equal to X_ε: as g ranges over $\mathbb{R}$, ga_ε ranges over $\mathbb{R}$. The quotient Euclidean metric gives $d_H(X_\varepsilon, X_0) = \varepsilon$, by matching second coordinates and minimizing the first-coordinate distance. But the orbit of z_0 is only the singleton $\{z_0\}$, not X_0.
All phases in X_ε give unitarily equivalent operators: translation by b conjugates H_z to $H_{T(b,z)}$, and the action is transitive on this circle. Thus $\Sigma_t(X_\varepsilon) = \sigma(S_{z_\varepsilon}(t))$. On X_0 each potential is the constant $\cos(2\pi y)$. The Fourier multiplier $|\xi|^2$ has essential range $[0,\infty)$, so $\sigma(H_0) = [0,\infty)$. Spectral mapping therefore gives

$$\sigma(e^{-t(H_0+c)}) = [0, e^{-tc}] \quad (c \in \mathbb{R}), \qquad \Sigma_t(X_0) = [0, e^t].$$

Corollary 43.3 now proves

$$\operatorname{dist}_H\big(\sigma(S_{z_\varepsilon}(t)), [0, e^t]\big) \le A_t\sqrt{\varepsilon}.$$

In contrast, $\sigma(S_{z_0}(t)) = [0, e^{-t}]$. The triangle inequality for Hausdorff distance, applied with this fixed interval, yields

$$\lim_{\varepsilon\downarrow 0} \operatorname{dist}_H\big(\sigma(S_{z_\varepsilon}(t)), \sigma(S_{z_0}(t))\big) = e^t - e^{-t}.$$

Thus even the phase spectra are discontinuous here. The subsystem spectra converge to the spectrum of the limiting circle, which is larger than the orbit of the limiting phase. This conclusion follows from the spectral localization argument; the earlier operator-norm limit has its separate direct proof.

44 Category theory

This section develops the author's Hom-functor construction and its Yoneda correspondence. The essential distinction is between *precomposition*, which changes the source of a morphism, and *postcomposition*, which changes its target.
We write C for a category and, when the context is clear, also for its collection of objects Ob_C. Composition is read from right to left: if $g : X \to Y$ and $f : Y \to Z$, then $f \circ g : X \to Z$. Thus its typed form is

$$\circ : \mathrm{Hom}_C(Y, Z) \times \mathrm{Hom}_C(X, Y) \longrightarrow \mathrm{Hom}_C(X, Z).$$

The opposite category has the same objects and $\mathrm{Hom}_{C^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_C(Y, X)$, with composition reversed.
We assume that C is locally small: each $\mathrm{Hom}_C(X, Y)$ is a set. The notation Set denotes the category of sets in the chosen size convention. Functors and natural transformations are considered in an ambient framework large enough to contain them; local smallness alone does not make the collection of all functors a set. One may, for example, work with a small C relative to a chosen universe and use a larger ambient universe for the functor category. The calculations below are componentwise and do not depend on this particular choice of framework.

Note 44.1 (Comparison with the Obelisk Yoneda summary)**.**
The newer portal's `yoneda_ai.html` gives the same correspondence $\eta \mapsto \eta_A(1_A)$ and inverse $u \mapsto (f \mapsto G(f)(u))$ proved below. Its discussion of "functor classes" needs a size interpretation: an individual functor may have a class-sized graph, while a collection of all such graphs can lie at a higher level. Impredicative class comprehension still has sets as members, so it does not by itself turn proper-class graphs into members of a class. The ambient convention stated above makes this distinction explicit without changing the componentwise Yoneda argument.

44.1 Functors and the contravariant Hom-functor

Definition 44.1 (Functor)**.**
A covariant functor from C to D consists of an object map F and maps on morphisms

$$\widetilde{F}^{A,B} : \mathrm{Hom}_C(A, B) \longrightarrow \mathrm{Hom}_D(F(A), F(B))$$

that preserve identities and composition:

$$\widetilde{F}^{A,A}(1_A) = 1_{F(A)}, \qquad \widetilde{F}(f \circ g) = \widetilde{F}(f) \circ \widetilde{F}(g).$$

When no ambiguity arises, we write $F(f)$ for $\widetilde{F}(f)$.

Definition 44.2 (Functor category)**.**
The notation $[C, D]$ denotes the category whose objects are functors $C \to D$ and whose morphisms are natural transformations. A natural transformation $\eta : F \to G$ consists of components $\eta_A : F(A) \to G(A)$ satisfying

$$G(f) \circ \eta_A = \eta_B \circ F(f) \quad \text{for every } f : A \to B.$$

Definition 44.3 (Morphism action of the Hom-functor)**.**
Fix $x \in C$. On objects, put $h_x(A) = \mathrm{Hom}_C(A, x)$. For a morphism $f : A \to B$ of C, define precomposition by

$$\widetilde{\mathrm{Hom}}_C^{A,B}(f, x) : \mathrm{Hom}_C(B, x) \longrightarrow \mathrm{Hom}_C(A, x), \qquad g \longmapsto g \circ f.$$

This is the morphism action of h_x on the corresponding arrow $B \to A$ of C^{op}. The notations $\mathrm{Hom}_C(-,x)$ and h_x denote the same functor; $\widetilde{\mathrm{Hom}}_C(f,x)$ makes its morphism action explicit.

Lemma 44.1 (Contravariant Hom-functor)**.**
For each $x \in C$, the construction above gives $h_x = \mathrm{Hom}_C(-,x) \in [C^{\mathrm{op}}, \mathrm{Set}]$.

Proof.
The object values are sets by local smallness. Precomposition has the displayed source and target, because composition in C is defined for these morphisms. It preserves identities since $g \circ 1_A = g$. For $f : A \to B$, $k : B \to D$, and $g : D \to x$, associativity gives

$$\widetilde{\mathrm{Hom}}_C(k \circ f, x)(g) = g \circ (k \circ f) = (g \circ k) \circ f = \big(\widetilde{\mathrm{Hom}}_C(f,x) \circ \widetilde{\mathrm{Hom}}_C(k,x)\big)(g).$$

This reversed order is exactly the composition law for a functor on C^{op}. Equality on each argument gives equality of the corresponding maps. □

44.2 The Yoneda functor

Definition 44.4 (Yoneda functor)**.**
On objects, the Yoneda functor sends x to $h_x = \mathrm{Hom}_C(-,x)$. For $f : x \to y$, define $h_f : h_x \to h_y$ by postcomposition:

$$(h_f)_w : \mathrm{Hom}_C(w,x) \longrightarrow \mathrm{Hom}_C(w,y), \qquad g \longmapsto f \circ g.$$

This component family is the transformation denoted by η in the construction below.

Lemma 44.2 (Naturality of postcomposition)**.**
For every $f : x \to y$, the family $\eta = h_f$ is a natural transformation.

Proof.
An arrow $m : a \to b$ in C^{op} is an arrow $m : b \to a$ in C. For $g : a \to x$, the two sides of the naturality square give

$$\big(\eta_b \circ \widetilde{\mathrm{Hom}}_C(m,x)\big)(g) = f \circ (g \circ m),$$

$$\big(\widetilde{\mathrm{Hom}}_C(m,y) \circ \eta_a\big)(g) = (f \circ g) \circ m.$$

Associativity makes them equal. This also identifies the source and target of both maps as $\mathrm{Hom}_C(a,x) \to \mathrm{Hom}_C(b,y)$. □

Lemma 44.3 (Functoriality of the Yoneda construction)**.**
The assignments $x \mapsto h_x$ and $f \mapsto h_f$ define a functor $h_- : C \to [C^{\mathrm{op}}, \mathrm{Set}]$.

Proof.
For every $g : w \to x$, $(h_{1_x})_w(g) = 1_x \circ g = g$, so identities are preserved. If $f : x \to y$ and $k : y \to z$, then

$$(h_{k \circ f})_w(g) = (k \circ f) \circ g = k \circ (f \circ g) = \big((h_k)_w \circ (h_f)_w\big)(g).$$

Natural transformations compose componentwise, so composition is preserved as well. □

44.3 The Yoneda correspondence

Lemma 44.4 (Yoneda lemma, contravariant version)**.**
Let $G : C^{\mathrm{op}} \to \mathrm{Set}$ and $A \in C$. There is a bijection

$$\mathrm{Nat}(h_A, G) \longrightarrow G(A), \qquad \Psi \longmapsto \Psi_A(1_A).$$

Its inverse sends $u \in G(A)$ to the transformation Φ^u with

$$\Phi^u_x(f) = G(f)(u) \qquad (f : x \to A \text{ in } C).$$

Here $G(f) : G(A) \to G(x)$ denotes the action on the corresponding arrow of C^{op}.

Proof.
First verify that the inverse formula defines a natural transformation. For $m : y \to x$ and $f : x \to A$ in C, functoriality gives

$$G(m)\big(\Phi^u_x(f)\big) = G(m)\big(G(f)(u)\big) = G(f \circ m)(u) = \Phi^u_y(f \circ m).$$

This is the naturality condition for $\Phi^u : h_A \to G$. Its component at the identity recovers the chosen element:

$$\Phi^u_A(1_A) = G(1_A)(u) = u.$$

Thus different choices of u give different transformations.
Conversely, let $\Psi : h_A \to G$ be natural and put $u = \Psi_A(1_A)$. For any $f : x \to A$, naturality on the corresponding arrow $A \to x$ in C^{op} gives

$$\Psi_x(f) = \Psi_x(1_A \circ f) = G(f)\big(\Psi_A(1_A)\big) = G(f)(u) = \Phi^u_x(f).$$

The equality holds for every x and every f, so $\Psi = \Phi^u$. The two displayed constructions are therefore inverse bijections. □
The correspondence is natural in both parameters. A transformation $\alpha : G \to H$ sends $\Psi_A(1_A)$ to $\alpha_A(\Psi_A(1_A))$, which is the element corresponding to $\alpha \circ \Psi$. For an arrow $k : A \to B$, precomposing $\Psi : h_B \to G$ with h_k gives the element

$$(\Psi \circ h_k)_A(1_A) = \Psi_A(k) = G(k)\big(\Psi_B(1_B)\big).$$

These identities state precisely how the bijection respects a change of functor or representing object.

Lemma 44.5 (Yoneda embedding)**.**
The functor $h_- : C \to [C^{\mathrm{op}}, \mathrm{Set}]$ is fully faithful.

Proof.
Apply the Yoneda correspondence with $A = x$ and $G = h_y$. It gives

$$\mathrm{Nat}(h_x, h_y) \longrightarrow \mathrm{Hom}_C(x, y), \qquad \eta \longmapsto \eta_x(1_x).$$

The inverse sends $f : x \to y$ to the transformation with components $g \mapsto f \circ g$, exactly h_f. Hence the map induced by h_- on each Hom-set is a bijection: it is surjective (full) and injective (faithful). □

44.4 Universal elements and uniqueness of representations

A representation is more than an object whose elements have the desired shape: it includes a natural identification. The following formulation makes that extra structure explicit.

Definition 44.5 (Universal element)**.**
For $G : C^{\mathrm{op}} \to \mathrm{Set}$, an element $u \in G(A)$ is universal if, for every object X and $v \in G(X)$, there is exactly one morphism $f : X \to A$ with $G(f)(u) = v$.

Proposition 44.1 (Universal-element criterion)**.**
The functor G is naturally isomorphic to h_A if and only if it has a universal element in $G(A)$. If (A, u) and (B, v) are two such choices, there is a unique isomorphism $k : A \to B$ satisfying $G(k)(v) = u$.

Proof.
The Yoneda transformation $\Phi^u_X : f \mapsto G(f)(u)$ is natural. Universality says exactly that every component is bijective. The inverses are natural: from $G(m)\Phi^u_X = \Phi^u_Y h_A(m)$ for $m : Y \to X$, compose with the inverse components to obtain $(\Phi^u_Y)^{-1}G(m) = h_A(m)(\Phi^u_X)^{-1}$. Conversely, for a natural isomorphism $\Phi : h_A \to G$, put $u = \Phi_A(1_A)$. Yoneda gives $\Phi = \Phi^u$, and componentwise bijectivity gives universality.
For two universal elements there are unique arrows $k : A \to B$ and $l : B \to A$ with $G(k)(v) = u$ and $G(l)(u) = v$. Contravariance gives

$$G(l \circ k)(u) = G(k)G(l)(u) = u, \qquad G(k \circ l)(v) = G(l)G(k)(v) = v.$$

Uniqueness for (A, u) and (B, v) respectively implies $l \circ k = 1_A$ and $k \circ l = 1_B$. The condition on k already determines it uniquely. □

Example 44.1 (Products as representing objects)**.**
Fix objects B, D in C and define $G(X) = \mathrm{Hom}_C(X, B) \times \mathrm{Hom}_C(X, D)$, with $G(m)(b, d) = (b \circ m, d \circ m)$. Identity and composition laws follow componentwise from those in C. An element of $G(P)$ is a pair of arrows $p : P \to B$, $q : P \to D$. Its universality says exactly that every pair $b : X \to B$, $d : X \to D$ factors uniquely as

$$p \circ f = b, \qquad q \circ f = d \qquad (f : X \to P).$$

Thus a categorical product represents this functor. The preceding proposition proves uniqueness up to the unique isomorphism preserving both projections. It does not assert that products exist in every category.

Proposition 44.2 (Yoneda detects isomorphisms)**.**
An arrow $f : A \to B$ is an isomorphism if and only if every map

$$\mathrm{Hom}_C(X, A) \longrightarrow \mathrm{Hom}_C(X, B), \qquad g \longmapsto f \circ g$$

is bijective.

Proof.
An inverse to f supplies the inverse maps by postcomposition. Conversely, surjectivity at $X = B$ gives $g : B \to A$ with $f \circ g = 1_B$. Then $f \circ (g \circ f) = f = f \circ 1_A$. Injectivity at $X = A$ gives $g \circ f = 1_A$, so g is an inverse. □

The quantifier over all test objects is essential. In the category of abelian groups, the zero group is terminal, and there is just one arrow from it to any group. Testing only that object therefore makes every arrow appear bijective on these Hom-sets, including the noninvertible arrow from the zero group to $\mathbb{Z}$. In Set, by contrast, maps from a singleton identify with elements, and this particular test object does detect bijections.

44.5 Cancellation and pointwise tests

Proposition 44.3 (What Hom-functors detect about cancellation)**.**
An arrow $f : A \to B$ is a monomorphism if and only if postcomposition $\mathrm{Hom}_C(X, A) \to \mathrm{Hom}_C(X, B)$ is injective for every X. It is an epimorphism if and only if precomposition $\mathrm{Hom}_C(B, X) \to \mathrm{Hom}_C(A, X)$ is injective for every X. By contrast, postcomposition is surjective for every X if and only if f has a right inverse.

Proof.
The first two equivalences are precisely left and right cancellation, respectively: $f \circ g = f \circ h$ implies $g = h$, or $g \circ f = h \circ f$ implies $g = h$, for each permissible source or target. If postcomposition is surjective at $X = B$, the identity of B has a preimage $s : B \to A$, so $f \circ s = 1_B$. Conversely, such an s lifts every $g : X \to B$ to $s \circ g : X \to A$, proving surjectivity at all X. □

Example 44.2 (An epimorphism need not give pointwise surjectivity)**.**
In the category of abelian groups, the quotient $q : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is an epimorphism: two homomorphisms agreeing after q agree on every element, because q is surjective as a set map. But it has no right inverse. Any homomorphism $s : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ sends the generator to an integer n satisfying $2n = 0$, so $n = 0$ and $s = 0$. Thus postcomposition with q is not surjective on the Hom-set with source $\mathbb{Z}/2\mathbb{Z}$. This distinguishes cancellation in the category from pointwise surjectivity of the corresponding natural transformation h_q; the latter is the stronger split-epimorphism condition proved above.

44.6 Hom-functors and universal constructions

There are two variances to keep apart. For a fixed source X, the *covariant* functor $\mathrm{Hom}_C(X, -)$ sends a target arrow to postcomposition. For a fixed target A, the earlier $h_A = \mathrm{Hom}_C(-, A)$ is contravariant. The next statement concerns the former functor.

Proposition 44.4 (Products and equalizers tested by Hom)**.**
If P with projections $p : P \to B$, $q : P \to D$ is a product, then for every X the map

$$\mathrm{Hom}_C(X, P) \longrightarrow \mathrm{Hom}_C(X, B) \times \mathrm{Hom}_C(X, D), \qquad u \longmapsto (p \circ u, q \circ u)$$

is bijective. If $e : E \to B$ is an equalizer of $f, g : B \to D$, then postcomposition with e gives a bijection

$$\mathrm{Hom}_C(X, E) \longrightarrow \{u \in \mathrm{Hom}_C(X, B) : f \circ u = g \circ u\}.$$

Both bijections are natural in the test object X.

Proof.
The product property gives existence and uniqueness of u with any prescribed two components, which proves the first bijection. The equalizer property says exactly that every arrow u satisfying

the displayed equality has a unique factorization $u = e \circ v$, proving the second. For $k : Y \to X$, precomposing u or v by k precomposes all displayed components by k, by associativity. This proves naturality. □

These statements prove preservation of the indicated limits by $\mathrm{Hom}_C(X, -)$; they do not assume those limits exist in every category. Dually, the contravariant h_A takes an existing coproduct to a product of Hom-sets and an existing coequalizer to an equalizer of Hom-sets. The proof is the same factorization argument with arrows reversed. All Hom-sets here are sets by local smallness; no collection of all proper-class-valued diagrams is being formed.

44.7 Small limits from products and equalizers

A *small diagram* here is a functor $D : J \to C$, where the objects and arrows of the indexing category J form sets in the chosen size convention. No smallness of all of C is assumed. A cone from X to D is a family $u_j : X \to D(j)$ satisfying

$$D(\alpha) \circ u_i = u_j \qquad (\alpha : i \to j \text{ in } J).$$

A limit is a cone $(L, (\ell_j))$ such that each cone $(X, (u_j))$ has a unique factorization $u_j = \ell_j \circ h$ for all j, with $h : X \to L$. For a set-indexed family, a product has the same factorization property without compatibility equations between its components; an empty product is therefore a terminal object.

Theorem 44.1 (Construction of a small limit)**.**
For the specified small diagram D, suppose the two products

$$A = \prod_{j \in \mathrm{Ob}_J} D(j), \qquad B = \prod_{\alpha : i \to j \text{ in } J} D(j)$$

exist in C. Let $p_j : A \to D(j)$ be the first product's projections, and let $q_\alpha : B \to D(j)$ be the second product's projections. There are unique arrows $s, t : A \to B$ determined by

$$q_\alpha \circ s = D(\alpha) \circ p_i, \qquad q_\alpha \circ t = p_j \qquad (\alpha : i \to j).$$

If an equalizer $e : L \to A$ of s, t exists, then $\ell_j = p_j \circ e$ makes L a limit of D. Consequently a category with all small products and equalizers has limits of every small diagram.

Proof.
The product property of B gives s, t from the displayed families of arrows with common source A. Since $s \circ e = t \circ e$, for every $\alpha : i \to j$ we have

$$D(\alpha) \circ \ell_i = q_\alpha \circ s \circ e = q_\alpha \circ t \circ e = \ell_j.$$

Thus the proposed family is a cone.
Given any cone $(X, (u_j))$, the product property of A gives a unique $u : X \to A$ with $p_j \circ u = u_j$ for all j. For each $\alpha : i \to j$,

$$q_\alpha \circ s \circ u = D(\alpha) \circ u_i = u_j = q_\alpha \circ t \circ u.$$

Uniqueness in the product property of B gives $s \circ u = t \circ u$. The equalizer property therefore gives a unique $h : X \to L$ with $e \circ h = u$, and hence $\ell_j \circ h = u_j$. If $h' : X \to L$ has the same cone components, then $p_j \circ e \circ h' = u_j$ for all j. Product uniqueness gives $e \circ h' = u$, and equalizer uniqueness gives $h' = h$. This proves the limit property.

If J has no objects, it has no arrows. Both products are terminal objects, and $s = t$ is the unique arrow between them. One may take $e = 1_A$ as their equalizer, giving the terminal object A as the limit of the empty diagram. More generally the preceding argument shows that any chosen equalizer in this case is terminal. The proof constructs the limit for one specified diagram from specified products and an equalizer; it requires no simultaneous choice of limits for a class of diagrams. □

Proposition 44.5 (Hom preserves and detects the limit property)**.**
A cone $(L, (\ell_j))$ to a small diagram $D : J \to C$ is a limit if and only if, for every $X \in C$, the map

$$\begin{aligned} \mathrm{Hom}_C(X, L) &\longrightarrow \mathrm{Cone}(X, D), \\ h &\longmapsto (\ell_j \circ h)_{j \in \mathrm{Ob}_J} \end{aligned}$$

is bijective. The set on the right consists of the compatible families in the product

$$\prod_{j \in \mathrm{Ob}_J} \mathrm{Hom}_C(X, D(j)).$$

These bijections are natural in X. In particular, the covariant functor $\mathrm{Hom}_C(X, -)$ preserves every existing small limit.

Proof.
Local smallness and the set-sized index make the displayed product of Hom-sets, and its subset of compatible families, sets in the stated ambient convention. A cone determines the displayed map because precomposing its compatibility equations by h gives those for the resulting family. Surjectivity is existence of a factorization for each cone, and injectivity is its uniqueness. This proves the first assertion in both directions.
The compatible-family set is the limit in Set of the diagram $j \mapsto \mathrm{Hom}_C(X, D(j))$, whose arrow action is postcomposition by $D(\alpha)$. Indeed, the coordinate projections form a cone; any compatible family of functions from a set S determines a unique function from S into the compatible-family set by evaluation at each $s \in S$. Thus the preceding bijection says precisely that $\mathrm{Hom}_C(X, -)$ preserves the given limit. For an arrow $k : Y \to X$, both routes send $h : X \to L$ to the family $(\ell_j \circ h \circ k)_j$, by associativity. This proves naturality in the test object. In the empty-diagram case the compatible-family set is the singleton containing the empty family, so the criterion reduces to having exactly one arrow $X \to L$ from each X, as required for a terminal object. □
The indexing restriction is essential to the size claim: local smallness controls individual Hom-sets, not arbitrary proper-class products of them. These theorems assert the stated universal properties inside the chosen framework and do not construct limits of every class-indexed diagram.

44.8 Initial algebras, final coalgebras, and fixed points

Let $P : C \to C$ be an endofunctor. A P-algebra is an arrow $a : P(A) \to A$. An algebra morphism from (A, a) to (B, b) is an arrow $u : A \to B$ with

$$u \circ a = b \circ P(u).$$

Identities satisfy the equation because $P(1_A) = 1_{P(A)}$. For algebra morphisms $u : (A, a) \to (B, b)$ and $v : (B, b) \to (D, c)$,

$$(v \circ u) \circ a = v \circ b \circ P(u) = c \circ P(v) \circ P(u) = c \circ P(v \circ u).$$

Thus composition is inherited from C, and its associativity and identity laws are inherited as well. These objects and arrows therefore form an algebra category in the chosen ambient size convention. An algebra is *initial* when it has exactly one algebra morphism to every algebra. Dually, a coalgebra is an arrow $d : M \to P(M)$, a coalgebra morphism $u : (M, d) \to (N, e)$ satisfies $e \circ u = P(u) \circ d$, and a coalgebra is *final* when every coalgebra has exactly one morphism to it. The analogous composition calculation, with the structure arrows reversed, verifies the coalgebra category laws.

Lemma 44.6 (Lambek's lemma)**.**
The structure map of an initial algebra is an isomorphism. The structure map of a final coalgebra is also an isomorphism.

Proof.
Let (A, a) be initial. The arrow $P(a) : P(P(A)) \to P(A)$ is another algebra, so there is a unique algebra morphism $h : (A, a) \to (P(A), P(a))$. The arrow a itself is an algebra morphism in the reverse direction: its required equation is the identity $a \circ P(a) = a \circ P(a)$. Thus $a \circ h$ is an endomorphism of the initial algebra, and uniqueness gives $a \circ h = 1_A$. The equation for h now gives

$$h \circ a = P(a) \circ P(h) = P(a \circ h) = 1_{P(A)}.$$

Hence h is an inverse to a.
For a final coalgebra (M, d), let $k : (P(M), P(d)) \to (M, d)$ be its unique coalgebra morphism. The arrow d is a coalgebra morphism in the other direction, so finality gives $k \circ d = 1_M$. Its morphism equation yields

$$d \circ k = P(k) \circ P(d) = P(k \circ d) = 1_{P(M)}.$$

Thus k is an inverse to d. □

Proposition 44.6 (Uniqueness with structure)**.**
Two initial P-algebras are uniquely isomorphic as algebras, and two final P-coalgebras are uniquely isomorphic as coalgebras.

Proof.
For initial algebras, take their unique morphisms in the two directions. Each composite is an endomorphism of an initial object, hence equals its identity. The two morphisms are therefore inverse, and uniqueness is part of initiality. Reverse the arrows for final coalgebras. □
For the polynomial functor $P_B(X) = \sum_{a \in A} X^{B(a)}$ on sets, the W-type construction in Theorem 31.24 supplies an initial algebra, and Theorem 31.22 supplies the finite-address M-type as a final coalgebra. Lambek's lemma explains their fixed-point equations without treating a bare equation $P_B(X) \cong X$ as a substitute for the universal property. The recursion and corecursion theorems supply the unique structure-preserving maps; the preceding proposition then makes the constructions independent of their particular encodings, up to those unique isomorphisms.

44.9 Fusion, induction, and coinduction

Keep the same endofunctor $P : C \to C$. For an initial algebra (A, a), write $\mathrm{fold}_b : A \to B$ for the unique morphism to an algebra (B, b). For a final coalgebra (M, d), write $\mathrm{unfold}_c : X \to M$ for the unique morphism from a coalgebra (X, c). These names refer to the universal properties, not to an additional algorithm or a choice of encodings.

Proposition 44.7 (Recursion and corecursion fusion)**.**
If $h : (B, b) \to (D, e)$ is an algebra morphism, then

$$h \circ \mathrm{fold}_b = \mathrm{fold}_e \,.$$

If $h : (X, c) \to (Y, e)$ is a coalgebra morphism, then

$$\mathrm{unfold}_e \circ h = \mathrm{unfold}_c \,.$$

Proof.
For the algebra assertion, the defining equations and functoriality give

$$\begin{aligned}(h \circ \mathrm{fold}_b) \circ a &= h \circ b \circ P(\mathrm{fold}_b) \\ &= e \circ P(h) \circ P(\mathrm{fold}_b) \\ &= e \circ P(h \circ \mathrm{fold}_b).\end{aligned}$$

Thus the composite is an algebra morphism from the initial algebra to (D, e), so uniqueness identifies it with fold_e. For the coalgebra assertion, similarly,

$$\begin{aligned}d \circ (\mathrm{unfold}_e \circ h) &= P(\mathrm{unfold}_e) \circ e \circ h \\ &= P(\mathrm{unfold}_e) \circ P(h) \circ c \\ &= P(\mathrm{unfold}_e \circ h) \circ c.\end{aligned}$$

The composite is therefore a coalgebra morphism to the final coalgebra, and uniqueness identifies it with unfold_c. □
For the polynomial W-types, the first identity says that transforming a recursively computed result by a map preserving the constructor operations gives the same result as the corresponding recursion into the new target. For the M-types, the second says that a map of state systems preserving their labelled transitions does not change their unfolded trees. The preservation equations for h are essential; arbitrary maps need not satisfy these identities.

Proposition 44.8 (Subalgebra and quotient rigidity)**.**
Suppose (A, a) is initial and an algebra morphism $m : (S, s) \to (A, a)$ has an underlying arrow $m : S \to A$ that is a monomorphism in C. Then m is an isomorphism of algebras. Dually, suppose (M, d) is final and a coalgebra morphism $q : (M, d) \to (Q, e)$ has an underlying arrow $q : M \to Q$ that is an epimorphism in C. Then q is an isomorphism of coalgebras.

Proof.
Initiality supplies an algebra morphism $h : (A, a) \to (S, s)$. The composite $m \circ h$ is an endomorphism of the initial algebra, so it equals 1_A. Consequently

$$m \circ (h \circ m) = (m \circ h) \circ m = m = m \circ 1_S.$$

Left cancellation by m gives $h \circ m = 1_S$. Thus m and h are inverse algebra morphisms.
For the dual assertion, finality supplies a coalgebra morphism $k : (Q, e) \to (M, d)$, and $k \circ q = 1_M$ by uniqueness of the endomorphism of the final coalgebra. Hence

$$(q \circ k) \circ q = q \circ (k \circ q) = q = 1_Q \circ q.$$

Right cancellation by q gives $q \circ k = 1_Q$, so q and k are inverse coalgebra morphisms. □

Both assertions assume that the displayed structure on the subobject or quotient already exists and makes the indicated arrow a morphism. They do not claim that every subobject is closed under the operations, or that every quotient admits a compatible coalgebra structure. No preservation of monomorphisms or epimorphisms by P is assumed.

In sets, this gives a direct form of structural induction for the W-type in Theorem 31.24. Let $S \subseteq W$ be a subset such that grafting any labelled family of children in S again gives an element of S. Restricting grafting defines $s : P_B(S) \to S$, and the inclusion $m : S \hookrightarrow W$ satisfies $m \circ s = a \circ P_B(m)$. It is a monomorphism, so the proposition makes it an isomorphism. Since it is the inclusion, this says $S = W$. This derives induction from initiality for predicates whose extension is a specified subset, independently of a particular tree encoding.

There is also a useful direct coinduction consequence of finality. If (T, c) is a coalgebra and $u, v : (T, c) \to (M, d)$ are coalgebra morphisms, then $u = v$, since the target is final. In particular, let $R \subseteq M_B \times M_B$ satisfy the polynomial bisimulation condition: for each $(t, u) \in R$, the root labels agree, say at a, and $(t_b, u_b) \in R$ for every $b \in B(a)$. Then

$$(t, u) \longmapsto \big(a,\ b \longmapsto (t_b, u_b)\big)$$

defines a coalgebra structure on the set R. Both projections from R to M_B are coalgebra morphisms, by the displayed matching of labels and children. Finality in Theorem 31.22 makes the projections equal, so every related pair has $t = u$. The structure on R is explicitly constructed; no arbitrary lifting of relations through an endofunctor or choice of matching successors is being assumed. This recovers the tree coinduction principle from its universal property.

Example 44.3 (A fixed-point equation alone does not give initiality)**.**
Take P to be the identity functor on sets. The map $1_X : X \to X$ is an isomorphism for every set X, so every such X satisfies the bare fixed-point equation. But the singleton algebra $(\{0\}, 1_{\{0\}})$ is not initial: there is no map from it to the empty algebra. Dually, the empty coalgebra is not final because the singleton coalgebra has no map into it. For this functor the empty identity algebra is initial, and the singleton identity coalgebra is final. Thus even an invertible structure map is strictly less information than the relevant universal property.

Chapter

Various logic–related ideas

45 De Bruijn indices in the untyped lambda calculus

The following Standard ML sketch represents a bound variable by its distance from the binder: index zero refers to the nearest enclosing abstraction. The constructors represent variables, abstractions, and applications. The natural-number constructors and support functions belong to the surrounding prototype and are not defined in this excerpt.
This is a record of an implementation experiment. A complete substitution routine must preserve abstraction nodes and shift free indices when crossing a binder; the displayed helper does not yet implement those requirements. Likewise, the normalizer needs a stopping case for applications whose head cannot reduce. The final self-application illustrates the separate fact that untyped lambda terms need not have a normal form. These distinctions matter when turning the sketch into an executable evaluator.

```
datatype Exp =
  Var of nat (*todo: Nat.t*)
| Lam of Exp
| CApp of Exp * Exp
;

fun App x y = CApp (x,y);

val zero = ze;
val one = su zero;
val two = su one;

val id = Lam (Var zero);
val fst = Lam (Lam (Var one));
val snd = Lam (Lam (Var zero));


val tst = App id fst;


fun pred (n:nat) : nat =
  case n of
    ze => raise (excErr "Nat: zero minus one?")
```

```
  | su y => y
;

fun apply_hlp (M:Exp) (N:Exp) (mem:nat) : Exp =
  case M of
    Var n => if (n=mem) then N else (if (Nat.toInt n)>(Nat.toInt mem) then (Var (pred n)
        ) else (Var n))
  | Lam M2 => apply_hlp M2 N (su mem)
  | CApp (e1,e2) => (App (apply_hlp e1 N mem) (apply_hlp e2 N mem))
;

(* The first time, without the optimisation. *)
fun apply (LM:Exp) (N:Exp) : Exp =
  case LM of
    Var n => (App LM N)
  | Lam M => apply_hlp M N zero
  | CApp (e1,e2) => (App LM N)
;

val q = apply id fst;

(* The first time, without the optimisation. *)
fun normalize (e:Exp) : Exp =
  case e of
    Var n => Var n
  | Lam M => Lam (normalize M)
  | CApp (e1,e2) => (
    let
      val e1' = (normalize e1)
      val e2' = (normalize e2)
    in
      (normalize (apply e1' e2'))
    end)
;

val w = normalize tst;

val e = Lam (App (Var zero) (Var zero));

val ee = App e e;


val w = normalize ee; (* it will freeze the computations*)
```

46 Programming language with dependent types

A related implementation of a dependent type library in Standard ML is available at https://github.com/georgydunaev/set_theory_prover/blob/main/program/lambda.ML.

47 Primitive programming metalanguage [OUTDATED]

47.1 Definitions

All inductive definitions in this section are simultaneous. This historical design interprets classes as types and expressions as sets. The convention is local to this section; it records a different design from the predicates-as-types interpretation used elsewhere in the book.

Notation 47.1 (Strings and judgments)**.**
The relations :::, ::, and : below relate syntactic strings at successive levels: kinds, kinded expressions, and typed expressions. Quotation marks around literal names such as Kind, Type, VarType, ConstType, Term, and Fm are suppressed. Single Latin or Greek letters, possibly with subscripts, are metavariables for strings; the symbols ι and ε retain their operator meanings.
A string is derivable when it is generated by the stated inductive rules. The arrows $\Rightarrow$ and $\Longrightarrow$ express inference from the judgments on the left to the judgment on the right. Commas and semicolons group premises and bind more tightly than these arrows. The relation $\rightsquigarrow$ specifies a proposed rewrite of an expression inside a derivable text. Its use under binders requires capture-avoiding substitution.

Definition 47.1 (Kind)**.**
Inductive definition of kinds:

1. (Type ::: Kind).
2. (Expr ::: Kind).
3. (VarType ::: Kind).
4. (ConstType ::: Kind).
5. (OptionType ::: Kind).
6. If $(K_1$ ::: Kind) and $(K_2$ ::: Kind), then $((K_1$ -$\!\!\longrightarrow K_2)$ ::: Kind).

Axiom 20 (Kinding values of functional types)**.**

1. If $(A :: K_1)$ and $(F :: (K_1$ -$\!\!\longrightarrow K_2))$ then $(F(A) :: K_2)$.

Definition 47.2 (ConstType)**.**
Inductive definition of some constant atomic types which are main entities for which the framework is developed:

1. (Term :: ConstType).
2. (Fm :: ConstType).
3. (Bool :: ConstType).

Definition 47.3 (OptionType)**.**
Inductive definition of optional type results:

1. (None :: OptionType).
2. $(\alpha :: \text{Type}) \Longrightarrow (\text{Some}(\alpha) :: \text{OptionType})$.

Definition 47.4 (Type)**.**
Inductive definition of types: for any (strings) α and β,

1. $(\alpha :: \text{ConstType}) \Longrightarrow (\alpha :: \text{Type})$.
2. $(\alpha :: \text{Type})\ ;\ (\beta :: \text{Type}) \Longrightarrow ((\alpha \to \beta) :: \text{Type})$.
3. $(\alpha :: \text{Type})\ ;\ (\beta :: \text{Type}) \Longrightarrow ((\alpha \times \beta) :: \text{Type})$.

Definition 47.5 (VarType)**.**
Inductive definition of a kind of typed variables:

1. $(\alpha :: \text{Type}) \Longrightarrow (\text{Var}(\alpha) :: \text{VarType})$.

Definition 47.6 (Expr)**.**
Inductive definition of expressions:

1. $(\alpha :: \text{Type}) \Longrightarrow (\text{var}^{\alpha} :: \text{Expr})$.
2. $(\alpha :: \text{Type})\ ;\ (\beta :: \text{Type}) \Longrightarrow ((x^{\alpha}.s) :: \text{Expr})$.
3. $(\alpha :: \text{Type})\ ;\ (\beta :: \text{Type}) \Longrightarrow (\text{pr}_1^{\alpha\times\beta} :: \text{Expr})$.
4. $(\alpha :: \text{Type})\ ;\ (\beta :: \text{Type}) \Longrightarrow (\text{pr}_2^{\alpha\times\beta} :: \text{Expr})$.
5. $(a :: \text{Expr})\ ;\ (b :: \text{Expr}) \Longrightarrow ((a, b) :: \text{Expr})$.

47.2 Definitions related to lambda–calculus

Axiom 21 (typing of Expr)**.**
Typing axioms for the expressions. Let α and β be arbitrary types, then

1. $(\text{var}^{\alpha} : \text{Var}(\alpha) \to \alpha)$.
2. Lambda–abstraction: $(x : \text{Var}(\alpha))\ ,\ (s : \beta)\ \Rightarrow\ ((x.s) : \alpha \to \beta)$.
3. Application: $(g : \alpha \to \beta)\ ,\ (t : \alpha)\ \Rightarrow\ (g(t) : \beta)$.
4. OPairIntro: $(a : \alpha)\ ,\ (b : \beta)\ \Rightarrow\ ((a, b) : \alpha \times \beta)$.
5. FirstProj: $(\text{pr}_1^{\alpha\times\beta} : \alpha \times \beta \to \alpha)$.
6. SecondProj: $(\text{pr}_2^{\alpha\times\beta} : \alpha \times \beta \to \beta)$.

Definition 47.7 (conseq :: (Type -−→ OptionType))**.**
This operation extracts the range type β of a function type $\alpha \to \beta$. Define its optional result by cases on γ:

1. If γ is an atomic constant type, return None.

2. If the type γ is a function type $\alpha \to \beta$ for some α and β, then return Some(β).

3. If the type γ is a pair type $\alpha \times \beta$ for some α and β, then return None.

Every case returns a value of kind OptionType, so the following inference rule is admissible:

$$(\alpha :: \text{Type}) \Longrightarrow (\text{conseq}(\alpha) :: \text{OptionType}),$$

and conseq(α) is computable for every explicitly given type α.

Definition 47.8 (TypeOf :: (Expr -−→ OptionType))**.**
The intended type-recovery procedure returns Some(α) when it recovers a type and None otherwise. The clauses below are a partial specification of that procedure. A complete checker also needs clauses for pairs and projections, and must check the argument type in an application. Type uniqueness and correctness of the completed procedure are separate metatheoretic obligations.

1. In case it is $\text{var}^{\alpha}(x)$, then
$$\text{TypeOf}(\text{var}^{\alpha}(x)) \coloneqq \text{Some}(\alpha).$$

2. In case it is $(x^{\alpha}.s)$, then determine TypeOf($(x^{\alpha}.s)$) by cases on the recursively computed result TypeOf(s):

   (a) In case it is None, then return None.

   (b) In case it is Some(β), then return Some($\alpha \to \beta$).

3. In case it is $(g(t))$, then determine TypeOf($g(t)$) by cases on the recursively computed result TypeOf(g):

   (a) In case it is None, then return None.

   (b) In case it is Some(γ), then return conseq(γ).

The intended result kind is OptionType. Once the missing cases and checks are supplied, the specification should justify the rule

$$(e :: \text{Expr}) \Longrightarrow (\text{TypeOf}(e) :: \text{OptionType}),$$

and TypeOf(e) is intended to be computable for every explicitly given expression e.

Definition 47.9 (well–typed expression)**.**
The proposed computational criterion calls e well–typed when a completed, verified TypeOf(e) returns Some(α) for some type α. The partial clauses above alone do not establish this criterion.

Definition 47.10 (Capture-avoiding substitution in the simple fragment)**.**
A finite simultaneous substitution θ assigns replacement expressions to distinct variable names. On a variable x, put $x\theta = \theta(x)$ if listed, and $x\theta = x$ otherwise. Apply substitution componentwise to applications, pairs, and projections. Replacement expressions are inserted unchanged, not substituted into a second time.
For an annotated abstraction, first rename its bound variable x to a fresh y, outside the free variables of the replacement expressions and the substitution domain, and fresh for the original body. Then put
$$(\lambda x : \alpha.e)\theta = \lambda y : \alpha.\big((e[y/x])\theta\big).$$

Here the preliminary renaming changes precisely the occurrences bound by that abstraction. Simple type annotations contain no term variables and are unchanged. An infinite supply of names and finite syntax guarantee a fresh name; different choices give alpha-equivalent expressions. Single substitution $e[t/x]$ is the case with sole assignment $x \mapsto t$. These clauses define substitution for the completed simple grammar below; extra binders in the broader language require their own clauses.

Definition 47.11 (transformation)**.**
The following are the intended expression transformations. In alpha-conversion, y is fresh for the body; in projection reduction, $a : \alpha$ and $b : \beta$. All substitutions are capture-avoiding.

1. Alpha–conversion: $(x.M) \rightsquigarrow (y.M[y/x])$, where y is fresh for M.
2. Beta–reduction: $((x.M)N) \rightsquigarrow (M[N/x])$.
3. Proj1–reduction: $\mathrm{pr}_1^{\alpha\times\beta}((a, b)) \rightsquigarrow a$.
4. Proj2–reduction: $\mathrm{pr}_2^{\alpha\times\beta}((a, b)) \rightsquigarrow b$.

The freshness condition is substantive. A tempting scheme $(x.M[x/v]) \rightsquigarrow (y.M[y/v])$ is invalid unless the initial binder x is also fresh for M apart from the designated occurrences of v. With $M = x$ and $v \neq x$, it changes the identity abstraction $x.x$ into $y.x$, leaving x free. The displayed rule instead renames exactly the occurrences bound by the original binder.

47.3 A completed checker for the simple fragment

The application clause above exposes a genuine gap: if g has type $\alpha \to \beta$ and t has a different type γ, it still returns Some(β). Thus it recognizes a codomain, not a well-typed application. We now complete this part of the proposal under an explicit syntax convention. The result concerns simple types, not the dependent type library mentioned at the start of the section.
Take atomic types, arrow types, and product types as freely generated finite syntax trees, with syntactic equality of types and decidable equality of atomic type symbols (for example, finite identifier strings). A context Γ is a finite assignment of types to distinct variable names. Expressions are variables, annotated abstractions $\lambda x : \alpha.e$, applications $g(t)$, pairs (a, b), and projection operations $\pi_1(e), \pi_2(e)$. Here variables denote the earlier variable-access expressions; the projection operation abbreviates application of the relevant typed projection once the operand type has been recovered. Bound names are renamed fresh when necessary.
Typing uses the variable rule $\Gamma \vdash x : \Gamma(x)$ for $x \in \mathrm{dom}(\Gamma)$, the usual abstraction and application rules, and the product introduction and projection rules displayed above, now carrying their contexts. In particular, application requires *both* $\Gamma \vdash g : \alpha \to \beta$ and $\Gamma \vdash t : \alpha$. There is no implicit subtyping or conversion between different type trees in this fragment.

Definition 47.12 (Context-indexed type recovery)**.**
Define $T_\Gamma(e)$ recursively, returning either a type or failure.

1. For x, return $\Gamma(x)$ if present, and failure otherwise.
2. For $\lambda x : \alpha.e$, compute $T_{\Gamma,x:\alpha}(e)$ with x fresh for Γ. Return $\alpha \to \beta$ if that result is β, and failure otherwise.
3. For $g(t)$, compute $T_\Gamma(g)$ and $T_\Gamma(t)$. Return β exactly when the results are respectively $\alpha \to \beta$ and α for the same type α. Otherwise return failure.

4. For (a, b), return $\alpha \times \beta$ exactly when the two recursive results are α and β; otherwise fail.
5. For $\pi_i(e)$, return the ith component type exactly when $T_\Gamma(e)$ is a product type; otherwise fail.

The option-valued version wraps a returned type in Some and represents failure by None.

Theorem 47.1 (Termination, correctness, and uniqueness)**.**
The procedure terminates on every expression in this grammar, and

$$T_\Gamma(e) = \alpha \longleftrightarrow \Gamma \vdash e : \alpha.$$

Consequently the fragment has unique types in a fixed context and decidable typability.

Proof.
Each recursive call uses a proper expression subtree, so induction on tree size proves termination. Type equality is decidable by structural comparison of finite trees. For correctness, induct on e. The variable and abstraction clauses exactly match their typing rules, using the induction hypothesis for the body in the extended context. An application has a typing derivation precisely when its last rule is application with some domain α and codomain β. The two induction hypotheses identify its premises with the two successful recursive results and the required domain match. The pair and projection cases similarly invert their only possible final typing rules. This proves both directions in every case. Since the deterministic procedure returns at most one type, two derivable types must coincide. Running it decides typability. □

Lemma 47.1 (Typed substitution in the simple fragment)**.**
If $\Gamma, x : \alpha \vdash e : \beta$ and $\Gamma \vdash t : \alpha$, where x is not declared in Γ, then $\Gamma \vdash e[t/x] : \beta$ for capture-avoiding substitution.

Proof.
First, adding fresh declarations preserves typing, by induction on a typing derivation: existing variable lookups are unchanged and all other rules retain their premises. Now induct on the derivation for e. If its last rule is a variable rule, the variable is either x, in which case use the derivation for t, or a variable from Γ, whose rule is unchanged. For abstraction, rename the binder y fresh for t, x and Γ. Weakening types t under $\Gamma, y : \gamma$; the induction hypothesis substitutes in the body under that context. Abstraction then restores the same arrow type. Application, pairing, and projections follow by applying the induction hypotheses to their immediate premises and then the same typing rule. □

Theorem 47.2 (Preservation under the proposed reductions)**.**
Beta reduction and the two projection reductions preserve types in this fragment, including reduction inside an expression.

Proof.
A typed redex $(\lambda x : \alpha.e)(t)$ has premises $\Gamma, x : \alpha \vdash e : \beta$ and $\Gamma \vdash t : \alpha$ by inverting its application and abstraction rules. The substitution lemma types its reduct $e[t/x]$ at β. Inverting the pair and projection rules similarly gives $\pi_1(a, b) \rightsquigarrow a$ and $\pi_2(a, b) \rightsquigarrow b$ with unchanged types. For a reduction in a proper subexpression, induct on its surrounding expression: the unchanged outer typing rule can be reapplied because the replaced premise has exactly the same type. Under an abstraction, apply this argument in the extended context. □

Theorem 47.3 (Progress for the closed simple fragment)**.**
Call an abstraction a value, and call a pair a value when both its components are values. Use left-to-right call-by-value evaluation: first evaluate the operator and then the argument of an application, then contract $(\lambda x : \alpha.e)(v)$ when v is a value; evaluate pair components from left to right; and evaluate the operand of a projection before projecting from a pair of values. A closed well-typed expression is either a value or admits an evaluation step.

Proof.
First note the canonical forms: by inspecting the grammar of values and the last typing rule, a value of arrow type is an abstraction and a value of product type is a pair. These use the free type constructors: an arrow type cannot equal a product type.
Induct on a typing derivation in the empty context. There is no variable case, because the context has no declarations. An abstraction is a value. For an application, the induction hypothesis either steps its operator or makes the operator a value. In the latter case, apply the induction hypothesis to the argument: it either steps, or is a value. If both are values, the operator has arrow type, so its canonical form is an abstraction and the beta step applies. A pair either steps in its first component, steps in its second after the first is a value, or is itself a value. For a projection, its operand either steps or is a value of product type; in the latter case the projection step applies. These exhaust the grammar. □
Together with preservation, this proves that a closed well-typed expression cannot reduce in finitely many evaluation steps to a stuck nonvalue. Closedness is preserved: beta substitution introduces no free variables outside those of the redex, and the other reductions plainly preserve closedness. Induction on the number of steps therefore retains the empty-context typing judgment, after which progress applies again. This statement does not claim that every evaluation terminates; progress and termination are separate properties.
These results repair the checker and justify the three reductions for this specified fragment. They do not assert termination of reduction, normal-form uniqueness, or correctness for dependent types; those are different metatheorems.

47.4 A set interpretation of the simple fragment

Choose a set D_a for each atomic type a. Interpret type constructors by

$$D_{\alpha\times\beta} = D_\alpha \times D_\beta, \qquad D_{\alpha\to\beta} = \{f : f \text{ is a function } D_\alpha \to D_\beta\}.$$

These are sets by the usual powerset and separation axioms. An environment ρ for Γ assigns $\rho(x) \in D_{\Gamma(x)}$ to each declared variable. Atomic sets may be empty; a context with no such environment then has no semantic instance, rather than an arbitrarily chosen value.
Interpret variables by lookup, pairs by pairing, projections by their coordinate maps, and application by function application. Interpret an annotated abstraction by the function

$$[\![\lambda x : \alpha.e]\!]_\rho : a \longmapsto [\![e]\!]_{\rho[x\mapsto a]} \qquad (a \in D_\alpha).$$

Bound variables are renamed fresh for the context. This is a semantic valuation in sets; it is distinct from a syntactic translation into class expressions such as the earlier μ_A.

Theorem 47.4 (Typing soundness and semantic substitution)**.**
If $\Gamma \vdash e : \alpha$ and ρ is an environment for Γ, then $[\![e]\!]_\rho$ is defined and belongs to D_α. For the substitution assertion, suppose $\Gamma, x : \alpha \vdash e : \beta$ and $\Gamma \vdash t : \alpha$, then

$$[\![e[t/x]]\!]_\rho = [\![e]\!]_{\rho[x\mapsto[\![t]\!]_\rho]}.$$

Here x is fresh for Γ and substitution is capture-avoiding.

Proof.
For typing soundness, induct on the typing derivation. Variables have the assigned type by the definition of environment. Pairing and projection use the Cartesian-product clauses. In application, the induction hypotheses give a function with domain D_α and an argument in that domain, so application is defined with the required result type. In abstraction, the body induction hypothesis applies to every updated environment. Replacement collects its graph as a set function on D_α, which therefore belongs to the arrow interpretation.
For substitution, induct on the expression e. At x, the equation is the definition of the updated environment; other variables are unchanged. Application, pairing, and projection follow by applying the same semantic operation to the inductively equal values. For an abstraction, rename its binder y fresh for t, x, Γ. At every $a \in D_{\mathrm{type}(y)}$, the updates at x and y commute, and evaluating t is unaffected by the update at y. The latter coincidence property follows by the same structural induction since y is not free in t. The body induction hypothesis now gives equality of the two function values at each a; function extensionality proves equality of the abstraction interpretations. □

Corollary 47.1 (Reduction preserves denotation)**.**
For well-typed expressions of this fragment, alpha-conversion, beta reduction, and either projection reduction preserve the set interpretation. So do reductions performed inside a well-typed expression context.

Proof.
Fresh binder renaming only changes the name used to update an environment. For beta reduction, evaluating the abstraction at the argument value gives the updated-environment expression in the substitution theorem, hence the interpretation of the substituted body. Projection reduction is the Cartesian-pair projection identity. Finally, each surrounding constructor is interpreted by an operation on the meanings of its subexpressions; equal meanings therefore remain equal in that context. For abstraction contexts apply this argument at every argument value and use extensionality. □

Proposition 47.1 (Semantic eta laws)**.**
If $\Gamma \vdash f : \alpha \to \beta$ and x is fresh for Γ, f, then $\lambda x : \alpha.f(x)$ and f have equal denotations in every environment for Γ. If $\Gamma \vdash p : \alpha \times \beta$, then $(\pi_1(p), \pi_2(p))$ and p likewise have equal denotations.

Proof.
For the function equation, at every $a \in D_\alpha$ the abstraction has value $[\![f]\!]_\rho(a)$, because updating the fresh variable does not affect the value of f. Extensionality gives equality of the two functions, including when D_α is empty. The product equation is reconstruction of a Cartesian pair from its two coordinates. □

Proposition 47.2 (Simultaneous substitution and environments)**.**
Let θ assign to each variable $x : \alpha$ of Γ a term $\theta(x)$ with $\Delta \vdash \theta(x) : \alpha$. For an environment ρ for Δ, define the environment $\theta^*\rho$ for Γ by $(\theta^*\rho)(x) = [\![\theta(x)]\!]_\rho$. Then

$$[\![e\theta]\!]_\rho = [\![e]\!]_{\theta^*\rho} \qquad (\Gamma \vdash e : \beta).$$

If σ similarly assigns Ξ-terms to the variables of Δ, and $(\theta\sigma)(x) = \theta(x)\sigma$, then

$$(\theta\sigma)^*\rho = \theta^*(\sigma^*\rho)$$

for every environment ρ for Ξ. Identity substitution induces the identity on environments.

Proof.
The same induction first gives $\Delta \vdash e\theta : \beta$, so its interpretation is defined. The first formula then follows by structural induction exactly as for single substitution: variables use the definition of $\theta^*\rho$, the nonbinding constructors act componentwise, and fresh abstraction binders allow assignment updates to commute with all replacements. Typing soundness ensures that every assigned value is in its specified set. For composition, evaluate at each x and apply the first formula to $\theta(x)$ and σ:

$$[\![\theta(x)\sigma]\!]_\rho = [\![\theta(x)]\!]_{\sigma^*\rho}.$$

These are exactly the two environment values to compare. For identity substitution the value at x is $[\![x]\!]_\rho = \rho(x)$. □
This proves soundness of the stated equations in the set interpretation. It does not say that all semantically equal terms reduce to the same syntax, or that all reduction sequences terminate. The next subsection supplies the separate normalization argument; semantic completeness is not claimed.

47.5 Strong normalization of the simple fragment

Here reduction means full compatible beta and product-projection reduction:

$$(\lambda x : \alpha.s)u \longrightarrow s[u/x], \qquad \pi_i((u_1, u_2)) \longrightarrow u_i,$$

with reduction allowed in every subexpression, including abstraction bodies. Terms are considered modulo alpha-conversion; renaming itself is not a reduction step. There are no recursive constants or additional rewrite rules. This relation includes the earlier call-by-value steps, but permits more reductions. The proof below is a reducibility argument, not an appeal to termination of the type checker.
Work in the ordinary classical metatheory of finite syntax. Supply a disjoint infinite supply of variable names for each simple type: a free name has one fixed type throughout this argument. Consider open well-typed terms in finite contexts drawn from these supplies. Their free-variable contexts therefore have compatible unions. Bound variables are renamed fresh before combination to avoid capture. This convention loses no generality: a finite typing derivation can be renamed into these supplies. A term is *strongly normalizing*, written $t \in \mathrm{SN}$, if it admits no infinite reduction sequence. Since each term has only finitely many one-step reducts up to alpha-conversion, a strongly normalizing term has a finite maximum reduction length $h(t)$. Indeed, unbounded finite lengths in a finitely branching reduction tree has a child with unbounded remaining lengths at every such node. Order the finitely many redex positions lexicographically and always take the first child with that property. Recursion on the natural numbers would then produce an infinite branch, a contradiction. Alpha-renaming choices do not create extra branches, since terms are considered modulo alpha. Thus $t \to t'$ implies $h(t') < h(t)$ whenever t is strongly normalizing.

Definition 47.13 (Reducible terms)**.**
Define classes of well-typed open terms by induction on the type:

$$\begin{aligned}
\mathcal{R}_a &= \{t : a \mid t \in \mathrm{SN}\},\\
\mathcal{R}_{\alpha\times\beta} &= \{t : \alpha \times \beta \mid \pi_1(t) \in \mathcal{R}_\alpha, \quad \pi_2(t) \in \mathcal{R}_\beta\},\\
\mathcal{R}_{\alpha\to\beta} &= \{t : \alpha \to \beta \mid tu \in \mathcal{R}_\beta \text{ for every } u \in \mathcal{R}_\alpha\}.
\end{aligned}$$

A term is *neutral* here if its outer constructor is neither an abstraction nor a pair. In particular, an application is neutral even when it is a beta redex. This convention matters in the expansion argument below.

Lemma 47.2 (Closure properties of reducibility)**.**
For every simple type α:

1. $\mathcal{R}_\alpha \subseteq \mathrm{SN}$;
2. if $t \in \mathcal{R}_\alpha$ and $t \to t'$, then $t' \in \mathcal{R}_\alpha$;
3. if $t : \alpha$ is neutral and every one-step reduct of t lies in $\mathcal{R}_\alpha$, then $t \in \mathcal{R}_\alpha$.

Consequently every variable is reducible at its declared type.

Proof.
Induct on α, establishing the three assertions together. For an atomic type they are the elementary closure properties of strong normalization: an infinite reduction must start with a one-step reduct.
For a product type, an infinite reduction of t would lift to one of $\pi_1(t)$, contradicting the first property for the component type. A reduction $t \to t'$ induces $\pi_i(t) \to \pi_i(t')$, so the second component properties give reduction closure. For neutral t, each $\pi_i(t)$ is neutral and its reducts are precisely $\pi_i(t')$ for $t \to t'$: there is no root projection contraction because t is not a pair. By hypothesis t' is reducible at the product type; its projections are therefore reducible. Neutral expansion at the component type proves the desired two projection conditions.
For an arrow type, a fresh variable $z : \alpha$ is reducible by neutral expansion at the smaller type, since it has no reducts. If $t \in \mathcal{R}_{\alpha\to\beta}$, then $tz \in \mathcal{R}_\beta$ is strongly normalizing; any infinite reduction of t would lift to tz. Reduction closure follows from $tu \to t'u$ and closure at β. Finally let t be neutral with reducible one-step reducts. For each $u \in \mathcal{R}_\alpha$, prove $tu \in \mathcal{R}_\beta$ by induction on $h(u)$. A reduct $t'u$ is reducible by the hypothesis on t'. A reduct tu' is reducible by the induction on $h(u)$ and reduction closure for u'. These are all reducts: t is not an abstraction, so there is no root beta step. Neutral expansion at β now applies to tu, proving the arrow condition. Applying the third property to a variable proves the last assertion. □

Lemma 47.3 (Introduction preserves reducibility)**.**
If $u \in \mathcal{R}_\alpha$ and $v \in \mathcal{R}_\beta$, then $(u, v) \in \mathcal{R}_{\alpha\times\beta}$. If $s[w/x] \in \mathcal{R}_\beta$ for every $w \in \mathcal{R}_\alpha$, then $\lambda x : \alpha.s \in \mathcal{R}_{\alpha\to\beta}$.

Proof.
For the pair, use induction on $h(u) + h(v)$ to prove that each $\pi_i((u, v))$ is reducible. Its root reduct is the corresponding reducible component. Its other reducts reduce one component, so the induction hypothesis and reduction closure apply. Neutral expansion then gives reducibility of both projections.
For abstraction, first choose a fresh variable $z : \alpha$. The hypothesis makes $s[z/x]$ strongly normalizing; renaming free x to fresh z faithfully transports reductions, so s is strongly normalizing. We use the following beta-expansion fact: if s, u are strongly normalizing and $s[u/x] \in \mathcal{R}_\beta$, then $(\lambda x : \alpha.s)u \in \mathcal{R}_\beta$. Prove it by induction on $h(s) + h(u)$. The root reduct is the assumed reducible contractum. If $s \to s'$, then $s[u/x] \to s'[u/x]$, so the new contractum is reducible and the smaller-height induction applies. If $u \to u'$, reducing each of the finitely many substituted occurrences gives $s[u/x] \to^* s[u'/x]$ (possibly zero steps if x is absent). Again closure and the

smaller-height induction apply. These exhaust the immediate reducts of the application, so neutral expansion proves the fact. Stability of reduction under capture-avoiding substitution used here follows by checking the two contractions and then the surrounding constructors. For a beta contraction, first rename its binder y fresh for u, x. The required syntactic identity is

$$(r[v/y])[u/x] = (r[u/x])[v[u/x]/y],$$

which follows by structural induction on r with fresh binders. Thus the substituted redex contracts to the substituted contractum. The projection case follows directly by substituting in both pair components. The earlier typed substitution lemma ensures all these terms retain their indicated types.
For each reducible u, the hypothesis supplies $s[u/x]$ reducible, and the first closure property supplies u strongly normalizing. Beta expansion therefore gives the required arrow condition. □

Theorem 47.5 (Strong normalization)**.**
Every well-typed term of the simple arrow/product fragment is strongly normalizing for full beta and projection reduction.

Proof.
Prove the stronger substitution assertion by induction on a typing derivation: replacing every free variable $x : \alpha$ by a reducible term of type α yields a reducible term of the original result type. The variable case is the substitution hypothesis. Application and projections use the defining elimination clauses for reducibility. Pairing uses the introduction lemma. For abstraction, rename the binder fresh for the replacements; extending the substitution by any reducible argument makes the body reducible by the induction hypothesis, so the abstraction clause of that lemma applies. This covers every typing rule.
The identity substitution replaces variables by themselves, and all variables are reducible. Thus every well-typed term is reducible and, by the first closure property, strongly normalizing. In particular, repeatedly selecting any available beta or projection redex terminates at a normal form. The earlier call-by-value evaluator also terminates, since each of its steps is among these reductions. □
This theorem concerns the explicitly specified simple fragment. Confluence and uniqueness of normal forms require the separate argument below. The reduction relation does not include eta expansion, and the theorem does not imply termination after adding general recursion or arbitrary proof-assistant conversion rules. The reducibility method is also discussed in the primary lecture notes at https://www.cs.cmu.edu/~fp/courses/97-comp-ded/lectures/lecture21.html; the proof here includes the product and full-reduction cases explicitly.

47.6 Confluence and decidable conversion of the simple fragment

Continue with the full reduction relation just defined, on well-typed simple terms modulo alpha-conversion. Write $s \to^* t$ for zero or more steps. Two terms are *joinable* if they reduce in zero or more steps to a common term. Local confluence means that the two results of any one-step fork are joinable; confluence makes the same assertion for arbitrary finite reduction sequences.

Lemma 47.4 (Local confluence)**.**
If $t \to u$ and $t \to v$, then there is a term w with $u \to^* w$ and $v \to^* w$.

Proof.

Represent an occurrence of a subterm by its position in the finite syntax tree. Choose fresh bound names consistently when performing substitutions. If both contractions have the same position, their results agree modulo alpha: at a given position at most one of beta contraction and the two projection contractions applies. If the positions are disjoint, neither replacement changes the other subtree. Performing the two replacements in either order gives the same term modulo alpha.
It remains to consider strictly nested positions. Remove their common surrounding context, so that the outer contraction is at the root. There are two possible shapes of that outer redex.

1. Suppose it is $(\lambda x : \alpha.s)a$. The inner contraction is in either s or a; neither the binder nor the fixed type annotation has reductions. If $s \to s'$, the inner-first branch contracts next to $s'[a/x]$. The outer-first branch is $s[a/x]$, and substitution stability, proved above, gives $s[a/x] \to s'[a/x]$. If instead $a \to a'$, the inner-first branch contracts next to $s[a'/x]$. The outer-first branch is $s[a/x]$. Each free occurrence of x in s has been replaced by a copy of a. These occurrences were leaves at distinct positions, so the substituted copies are disjoint. Reduce the specified redex once in each of these finitely many copies, without contracting any surrounding redex. This gives $s[a/x] \to^* s[a'/x]$. If there are no free occurrences, the terms already agree. Freshness of the bound names ensures that these are precisely the capture-avoiding substitutions.

2. Suppose the outer redex is $\pi_i((a_1, a_2))$. An inner step reduces some a_j to a'_j. If $j = i$, both branches reach a'_i: the outer-first branch uses that inner step and the inner-first branch uses the projection step. If $j \neq i$, both branches reach a_i, because the projection discards the component that changed.

These are all strictly nested cases: the immediate pair inside a projection is not itself a redex, and the immediate abstraction inside a beta redex is not itself a redex. Reinsert the surrounding context; compatibility allows all the displayed joining steps there. Subject reduction keeps every joining term well typed in the original context. □

Theorem 47.6 (Confluence)**.**
For well-typed simple terms, if $t \to^* u$ and $t \to^* v$, then u and v are joinable.

Proof.
We give the terminating local-confluence argument explicitly. Induct on the maximum reduction length $h(t)$, which exists by strong normalization and finite branching. If one of the given sequences has length zero, its endpoint is t, so the other endpoint is a common reduct. Otherwise write their first steps as

$$t \to t_1 \to^* u, \qquad t \to t_2 \to^* v.$$

Local confluence gives a term w such that $t_1 \to^* w$ and $t_2 \to^* w$. Since $h(t_1), h(t_2) < h(t)$, the induction hypothesis at t_1 joins u and w at some p, and that at t_2 joins v and w at some q. Now both p and q are reducts of w. Moreover $h(w) \leq h(t_1) < h(t)$, even when $w = t_1$. The induction hypothesis at w therefore joins p and q at some r. Concatenating the resulting sequences gives $u \to^* r$ and $v \to^* r$, as required. This proves the form of Newman's lemma needed here without imposing a particular evaluation strategy. □

Corollary 47.2 (Unique normal forms)**.**
Every well-typed simple term has exactly one full beta/projection normal form modulo alpha-conversion.

Proof.
Existence follows by repeatedly contracting, for example, the first available redex in a fixed traversal of the syntax tree. Strong normalization makes this process terminate. If n and m are normal forms of the same term, confluence joins them. Neither admits a step, so their common reduct is both n and m, modulo alpha-conversion. □

Theorem 47.7 (Decidable beta/projection conversion)**.**
For two well-typed simple terms in a fixed context and of the same type, it is decidable whether they are connected by a finite sequence of forward or backward beta/projection reductions through terms well typed in that context and of that type.

Proof.
Compute the full normal form of each term using the terminating procedure above. All its operations are effective on finite syntax: redex recognition, fresh renaming, capture-avoiding substitution, and contraction. Compare the resulting normal forms modulo alpha-conversion. This comparison is decidable: encode each bound occurrence by the number of intervening binders to its binder, preserve free-variable names, and compare the resulting finite labelled trees, including their type annotations. Variable-symbol equality and equality of the finite simple-type trees are decidable as assumed for the type checker.
To verify the decision, a single forward reduction preserves the unique normal form: a normal form of its target is also a normal form of its source. The same statement holds for a backward reduction. Consequently every finite conversion sequence preserves the normal form. Conversely, if the two normal forms agree modulo alpha, reducing both terms to that form and reversing one sequence exhibits their conversion. Thus the comparison returns exactly the required answer. If terms are supplied without typing derivations, the earlier checker first decides whether they meet the context and same-type hypotheses. □
Here conversion includes only the stated beta and projection equations. For instance, with a free variable $f : \alpha \to \beta$, the terms f and $\lambda x : \alpha.f(x)$, with x fresh, are distinct normal forms and hence are not convertible in this sense. Their set-theoretic denotations nevertheless agree by the semantic eta law. Thus decidability of this syntactic conversion does not assert that denotational equality is captured by it. No result in this subsection concerns eta expansion as a reduction, recursive constants, or a dependent type theory with additional conversion rules.

47.7 Completing the Boolean operational extension

We now extend the proved simple calculus, rather than merely declare a Boolean type. Add a fresh primitive type Bool, distinct from all other atomic types and from arrow and product types, and add terms

$$\mathsf{false}, \qquad \mathsf{true}, \qquad \mathsf{if}(b; s; t).$$

The last expression has a guard and two branches; it introduces no binder. Type annotations and all pre-existing constructors retain their previous meanings. The added typing rules, in a common context, are

$$\Gamma \vdash \mathsf{false} : \mathsf{Bool}, \qquad \Gamma \vdash \mathsf{true} : \mathsf{Bool},$$
$$\frac{\Gamma \vdash b : \mathsf{Bool} \qquad \Gamma \vdash s : \alpha \qquad \Gamma \vdash t : \alpha}{\Gamma \vdash \mathsf{if}(b; s; t) : \alpha}.$$

Both branches must have the same type, including when the guard is a literal. Thus $\mathsf{if}(\mathsf{true}; s; t)$ is not admitted merely because s has the desired type while t fails to type check. This is a simply typed conditional, not the dependent Boolean eliminator of the set-model chapter.

Proposition 47.3 (The extended checker and substitution)**.**
Extend T_Γ by returning Bool for either Boolean constant. For a conditional, compute the three recursive results; return α precisely when they are $\mathsf{Bool}, \alpha, \alpha$, and return failure otherwise. The extended checker terminates and returns α exactly when $\Gamma \vdash e : \alpha$. Types remain unique and typability decidable. Weakening and capture-avoiding typed substitution remain valid.

Proof.
The new recursive calls inspect proper subexpressions, so the previous size argument still proves termination. Constants have just their stated typing rule. Inverting a conditional typing derivation gives exactly the three premises checked by the algorithm. Conversely, successful recursive checks give those premises by induction, so the conditional rule produces the returned type. These are all new cases of the correctness proof; the old syntax cases have the same final rules and recursive checks. Determinism of the checker gives uniqueness.
For weakening, Boolean constants type in every context, and weakening the three conditional premises permits the same conditional rule. For substitution, constants are unchanged. The new structural equation is

$$\mathsf{if}(b; s; t)[u/x] = \mathsf{if}(b[u/x]; s[u/x]; t[u/x]).$$

The three induction hypotheses preserve the guard type and the common branch type, so the substituted expression has its original type. No new binder case is introduced; abstraction still uses fresh names as in the earlier substitution proof. The same reasoning proves the simultaneous-substitution version needed below. □

Reduction and evaluation

Extend full compatible reduction by exactly the two new contractions

$$\mathsf{if}(\mathsf{true}; s; t) \longrightarrow s, \qquad \mathsf{if}(\mathsf{false}; s; t) \longrightarrow t.$$

For this full relation, reduction is allowed in the guard and in either branch, as well as in all earlier expression positions. Boolean constants themselves have no reducts. In particular we do not add Boolean eta reduction or a rule discarding a conditional merely because its two branches are syntactically identical.
For call-by-value evaluation, both Boolean constants are values in addition to abstractions and pairs of values. Evaluate the guard of a conditional until it is a Boolean constant, then select the indicated branch without evaluating the discarded branch. Evaluation subsequently continues in the selected expression. The other evaluation clauses remain left-to-right as before.

Theorem 47.8 (Preservation and progress with Booleans)**.**
Full compatible reduction preserves typing in the extended calculus. Every closed well-typed term is a value or has a call-by-value step. The specified call-by-value step is deterministic.

Proof.
For a root Boolean contraction, typing inversion supplies both branches at the conditional's result type, hence supplies the selected reduct at that type. A step within any of the three subexpressions preserves its type by the induction hypothesis; the conditional rule then reconstructs the result. The preceding substitution proposition supplies the beta case, and the earlier product argument supplies projection. Induction on the surrounding expression proves preservation for every compatible position, including beneath binders.

The new canonical-form fact is that a value of Boolean type is either false or true. Indeed, an abstraction has arrow type and a pair has product type, and the freely generated type constructors are disjoint. For a closed conditional, the guard induction hypothesis either supplies a step or makes the guard a value. Its Boolean type then makes one of the two contractions applicable. Constants are already values; all earlier progress cases use the same canonical forms for arrows and products, which the new constants do not alter. Closedness is preserved by the new contractions because a selected branch has no free variables outside the whole conditional. For determinism, induction on the expression shows that the evaluation clauses choose at most one subexpression to evaluate. In a conditional it is the guard until the guard is a value; after that exactly one of the two distinct constants selects a branch. Applications, pairs, and projections likewise have the unique evaluation position specified by their left-to-right clauses. At that position the beta, projection, and Boolean contraction shapes are disjoint and have unique results modulo alpha-conversion. Thus two evaluation steps have the same result. □

The two-point interpretation

Keep the earlier interpretations of the other atomic types, arrows, and products, and put $D_{\mathsf{Bool}} = \{0, 1\}$. Interpret false by 0, true by 1, and define

$$[\![\mathsf{if}(b; s; t)]\!]_\rho = \begin{cases} [\![s]\!]_\rho, & [\![b]\!]_\rho = 1, \\ [\![t]\!]_\rho, & [\![b]\!]_\rho = 0. \end{cases}$$

This use of $0, 1$ is a semantic choice of two distinct set values; it does not add arithmetic operations to the term grammar.

Proposition 47.4 (Soundness of the Boolean interpretation)**.**
Typing soundness, semantic substitution, and preservation of denotation under full reduction hold for the extended calculus.

Proof.
For typing soundness, constants lie in the designated two-point set. The guard induction hypothesis gives one of its two values, and both branch induction hypotheses give values in the same D_α. The displayed selection therefore has the required value and type. This proves the new case, while all old constructors keep their previous semantic clauses.
For semantic substitution, the three structural induction hypotheses identify the guard meanings and both branch meanings before and after the environment update. Equal guard meanings select the same branch, whose meanings are equal by the corresponding hypothesis. Constants are independent of the environment. Fresh binder renaming handles the old abstraction case exactly as before.
A root Boolean contraction preserves denotation by the selection formula. A reduction in the guard preserves its Boolean value; a reduction in a branch preserves that branch's value and hence the conditional's value, whether or not that branch is selected. The previous beta proof uses the now-established substitution theorem, and projection uses the unchanged product interpretation. Induction on the expression context completes full reduction soundness. □

Normalization, including open conditional guards

Write $\mathcal{R}^B_\alpha$ for reducibility computed using the *extended* full reduction relation. For every old atomic type and for Bool, take all strongly normalizing well-typed terms of that type. Retain

the arrow and product clauses:

$$\mathcal{R}^B_{\alpha\times\beta} = \{t : \alpha \times \beta : \pi_1(t) \in \mathcal{R}^B_\alpha,\ \pi_2(t) \in \mathcal{R}^B_\beta\},$$
$$\mathcal{R}^B_{\alpha\to\beta} = \{t : \alpha \to \beta : tu \in \mathcal{R}^B_\beta \text{ for every } u \in \mathcal{R}^B_\alpha\}.$$

Use disjoint infinite variable supplies for these extended types, as in the earlier proof, so finite free-variable contexts can be combined. Call a term neutral when its outer constructor is neither abstraction, pair, nor Boolean constant. In particular every conditional is neutral, even when its guard is a constant and it has a root contraction. Reduction is still finitely branching modulo alpha-conversion. Thus a strongly normalizing term has a finite maximum reduction height $h(t)$ by the earlier ordered-redex-position argument, now for this relation.

Lemma 47.5 (Reducibility closure for the extended relation)**.**
For every type, reducible terms are strongly normalizing, reducibility is closed under one-step reduction, and a neutral well-typed term whose immediate reducts are all reducible is itself reducible. Variables and both Boolean constants are reducible.

Proof.
Induct on the type, proving the three closure properties together. For an old atomic type or Bool, reducibility is strong normalization. Reducts of a strongly normalizing term are strongly normalizing; if every immediate reduct is strongly normalizing, an infinite reduction from the term is impossible because it would begin with one of those reducts.
For a product, an infinite reduction of t lifts through π_1, contradicting strong normalization of its reducible projection. If $t \to t'$, each $\pi_i(t) \to \pi_i(t')$, so reduction closure at the component types proves closure at the product type. If t is neutral, its projections have no root projection contraction: their operand is not a pair. Every immediate reduct is $\pi_i(t')$ for a reduct t' of t, and is reducible by the product hypothesis on t'. Neutral expansion at the component types proves both projection conditions. This includes the case in which t is a conditional.
For an arrow, a fresh variable z at the domain type is reducible by neutral expansion at that smaller type. Thus tz is strongly normalizing whenever t is reducible, which rules out an infinite reduction of t. Closure follows by applying codomain closure to $tu \to t'u$ for every reducible u. For neutral expansion, fix such a u and induct on $h(u)$. The reducts of tu are $t'u$, reducible by the assumption on t', and tu', reducible by the induction hypothesis and closure for u'. There is no root beta contraction because t is not an abstraction. Neutral expansion at the codomain therefore makes tu reducible. This proves the arrow clause, also when t is a conditional with its own root reduction. Variables have no reducts and are neutral, while the two Boolean constants are strongly normalizing at the Boolean base type. □

Lemma 47.6 (Conditional elimination preserves reducibility)**.**
If $b \in \mathcal{R}^B_{\mathsf{Bool}}$ and $s, t \in \mathcal{R}^B_\alpha$, then $\mathsf{if}(b; s; t) \in \mathcal{R}^B_\alpha$.

Proof.
All three terms are strongly normalizing by closure. Induct on $h(b) + h(s) + h(t)$. A root reduct, when available, is s or t and is reducible by hypothesis. Any other immediate reduct reduces one of b, s, t; reduction closure preserves its reducibility and strictly decreases the indicated height sum. The induction hypothesis then makes the resulting conditional reducible. These exhaust the full compatible one-step reducts. Since the conditional is neutral, the neutral-expansion property proves the assertion. This reasoning does not assume that an open guard reduces to a

Boolean constant; it also covers guards whose normal form is a variable or another neutral term. □

The pair and abstraction introduction properties also hold for $\mathcal{R}^B$. Here are the additional checks needed to use their earlier proofs with the extended relation. Capture-avoiding substitution commutes with the conditional constructor. Substituting into either new contraction retains its literal Boolean guard and contracts to the substituted selected branch. Hence reduction stability under substitution includes the new root cases. The structural substitution identity used for beta reduction also includes a conditional by applying the identity to each of its three immediate subterms. Reducing a term inserted at finitely many variable occurrences still reduces disjoint copies, now possibly inside guards or branches.

Consequently the pair proof by induction on $h(u) + h(v)$ has exactly the same outer reducts: the selected component and reductions in a component. Closure and neutral expansion therefore prove both projection conditions. For abstraction, the hypothesis that every reducible substitution instance of the body is reducible first makes the body strongly normalizing by substituting a fresh variable. The beta-expansion induction on the sum of body and argument heights then applies: its root contractum is reducible; a body reduction preserves that property by substitution stability; an argument reduction preserves it by reducing its finitely many substituted copies. Each non-root case has smaller height sum. Neutral expansion gives the application reducible, proving the arrow introduction property. No new contraction can occur at the root of a pair, abstraction, or application merely because their subterms contain Booleans.

Theorem 47.9 (Strong normalization with Booleans)**.**
Every well-typed term of the extended calculus is strongly normalizing for full compatible beta, projection, and Boolean reduction.

Proof.
Induct on a typing derivation to prove the reducible-substitution assertion: replacing each free variable by a reducible term of its type yields a reducible result. The variable case is the substitution hypothesis; Boolean constants are reducible by the closure lemma. Application and projections use the arrow and product definitions. Pairing and abstraction use the introduction properties just proved, with fresh abstraction binders and arbitrary reducible arguments. For a conditional, the induction hypotheses make the substituted guard and both branches reducible, and the conditional-elimination lemma gives the result. This checks every typing constructor of the extended grammar. Identity substitution is reducible, so every typed term is reducible and therefore strongly normalizing. In particular the specified call-by-value evaluation terminates, since all its steps belong to full reduction; preservation and progress show that a closed well-typed term terminates at a value. □

Confluence and the extended decision procedure

Theorem 47.10 (Confluence and unique normal forms with Booleans)**.**
Full reduction in the extended calculus is confluent on well-typed terms modulo alpha-conversion. Every such term has a unique normal form. Equality of these normal forms decides conversion by the stated beta, projection, and Boolean equations in a fixed context and type.

Proof.
First prove local confluence by the positions of two contracted redexes. At the same position the root constructor and, for a conditional, its distinct literal guard determine a unique contractum. At disjoint positions the replacements commute. For nested positions, remove their common

context. If the outer contraction is beta, an inner step in the body commutes by substitution stability, including its new Boolean cases. An inner step in the argument is reproduced in every substituted copy, or discarded when there are none. If the outer contraction is a projection, an inner step in the selected component is reproduced after projection, while a step in the other component is discarded. These are the earlier joins, and allow inner Boolean contractions as well.
The new outer-redex cases are $\mathsf{if}(\mathsf{true}; s; t)$ and $\mathsf{if}(\mathsf{false}; s; t)$. Their literal guard has no reduct, so an inner step is in one of the branches. If it is in the selected branch, one path selects that branch and takes the inner step, while the other takes the inner step and selects the reduced branch. Both reach the same term. If it is in the discarded branch, both paths reach the unchanged selected branch. These exhaust nested positions; compatibility permits the joins to be put back in the common context. Preservation keeps all joining terms in the fixed context and type.
Strong normalization and finite branching now supply the finite heights needed by the preceding proof of Newman's lemma. That proof uses only termination and local confluence, both now established for the extended relation, so it gives confluence here. Repeatedly selecting the first available redex terminates at a normal form; confluence makes that normal form unique modulo alpha-conversion. The extended syntax still has effective redex recognition, substitution, fresh renaming, and alpha comparison. Normalizing two checked terms and comparing their normal forms therefore decides conversion: each forward or backward step preserves the unique normal form, and agreement exhibits a conversion by reversing one normalization path. □

Proposition 47.5 (Conservativity over the original simple calculus)**.**
Let Γ, α, and the terms in question use only the original simple signature: no Bool in types or annotations, and no Boolean constants or conditional constructors. Such a judgement $\Gamma \vdash e : \alpha$ is derivable in the extended calculus if and only if it is derivable in the original calculus. For two such terms typed at α in Γ, conversion in the extended calculus holds if and only if original beta–projection conversion holds, even when an extended conversion path passes through new syntax.

Proof.
The extended checker has unchanged clauses on every original constructor. Structural induction therefore gives the same result on an original term in an original context. Correctness of the two checkers proves the typing claim; the converse also follows by retaining the original typing derivation.
Every reduct of an original term is again original: beta substitution and projection introduce no new constructors or type annotations. No Boolean contraction is available in such a term. Thus its old and extended reduction sequences, and hence its normal form, agree. If two original terms are convertible in the extension, extended confluence makes these normal forms equal modulo alpha-conversion. Their original normalization paths then witness original conversion. Conversely every original conversion step is an allowed extended step. This proves the claim without assuming that an arbitrary extended conversion path stays inside the original syntax. □
A closed Boolean term has normal form either false or true. Indeed, a full normal form permits no call-by-value step, so progress makes it a value, and Boolean canonical forms give these two possibilities. Open Boolean normal forms need not be constants. For example, in the context $b : \mathsf{Bool}$, both b and $\mathsf{if}(b; \mathsf{true}; \mathsf{false})$ are distinct normal forms. They have the same two-point denotation but are not convertible by the rules just proved. Thus the extension still distinguishes semantic equations from its specified syntactic conversion relation. It adds neither recursive constants nor a dependent conditional.

48 Uniqueness and its properties

The quantifier ! means *at most one*; it does not assert existence. The notation ∃! combines existence with this uniqueness condition. This distinction is essential when a defining condition has no solution.

48.1 Uniqueness quantifier

Definition 48.1 (uniqueness quantifier)**.**
Uniqueness quantifier for classes:
Choose distinct fresh variables Y_1, Y_2 that are free for Y in φ. Then:

$$(!Y,\ \varphi) \longleftrightarrow (\forall Y_1 \forall Y_2,\ \varphi[Y_1/Y] \wedge \varphi[Y_2/Y] \longrightarrow Y_1 = Y_2) \qquad \text{(def:uniqueness quantifier)}$$

Note 48.1 (Scope of descriptions)**.**
The description constructions considered in these notes are intended to select sets. This is a local design choice: class comprehension collects set witnesses, and a default value must be specified when no witness exists. Other sections explore descriptions of classes and partial semantics under their own assumptions. Large objects can alternatively be handled relative to a chosen universe, when the required universe assumption is available.

Definition 48.2 (restricted uniqueness quantifier)**.**
Restricted uniqueness quantifier for sets only

$$\begin{aligned}(!(y \text{ set}),\ \varphi) &\longleftrightarrow (!y,\ (y \text{ set}) \wedge \varphi) \\ &\longleftrightarrow (\forall(y_1 \text{ set})\forall(y_2 \text{ set}),\ \varphi[y_1/y] \wedge \varphi[y_2/y] \longrightarrow y_1 = y_2)\end{aligned}$$

Note 48.2 (Predicate bodies in the Obelisk export)**.**
The newer catalog calls the property $(!x,\ x \in s)$ "semi-singleton" (`DefSemiSingleton`, 3.35), and combines it with $s \subseteq A$ in `DefSemiSingletonSubclass` (3.36). The exported formulas display universal closures of these predicate bodies without the predicate heads. Here they are read as the bodies of local definitions, rather than assertions that every class has at most one member or that every class is a subclass of every other class. For example, $\{\varnothing, \{\varnothing\}\}$ has two distinct members. The explicit at-most-one definition above avoids this ambiguity.

48.2 Singleton set

Definition 48.3 (Singleton (functional symbol))**.**
The bound variable x must be fresh for the expression w; substitution is capture-avoiding. Distinct variable names here are syntactic names, not an assertion that their values are distinct. In particular, replacing w by x requires renaming the bound variable first.

$$\{w\} = \{\, x \mid x = w \,\}$$

For a set w, this class has the sole member w. If w is a proper class, the same comprehension is empty, because only sets can be members.

Definition 48.4 (singleton (a predicate symbol))**.**

$$(A \text{ singleton}) \longleftrightarrow (\exists! a,\ a \in A)$$

48.3 Properties

Theorem 48.1 (Comprehension with a uniqueness condition)**.**
Conjoining a formula with the assertion that it has at most one witness produces a class with at most one member. Choose w distinct from y and not free in φ, so the outer uniqueness binder does not change the parameters of the comprehension.

$$(!w,\ w \in \{\, y \mid \varphi \wedge (!y,\ \varphi) \,\}) \qquad \text{(thm:class with at most one object)}$$

Proof.
Choose distinct witnesses w_1, w_2 fresh for φ, y, w. Assume $w_1 \in \{\, y \mid \varphi \wedge (!y,\ \varphi) \,\}$ and $w_2 \in \{\, y \mid \varphi \wedge (!y,\ \varphi) \,\}$.
To show: $w_1 = w_2$.

$$w_1 \in \{\, y \mid \varphi \wedge (!y,\ \varphi) \,\} \implies (w_1 \text{ set}) \wedge \varphi[w_1/y] \wedge (!y,\ \varphi)$$

$$w_2 \in \{\, y \mid \varphi \wedge (!y,\ \varphi) \,\} \implies (w_2 \text{ set}) \wedge \varphi[w_2/y] \wedge (!y,\ \varphi)$$

$$(!y,\ \varphi)\ \ \&\ \ (w_1 \text{ set}) \wedge \varphi[w_1/y]\ \ \&\ \ (w_2 \text{ set}) \wedge \varphi[w_2/y] \implies (w_1 = w_2)$$

□

The freshness condition is necessary. If φ were $y = w$, the comprehension would be $\{w\}$, and the displayed assertion would become $!w,\ w \in \{w\}$: at most one set exists. With two distinct sets this is false, since the bound w would vary the class being tested.

Theorem 48.2 (Uniqueness passes to subclasses)**.**

$$(A \subseteq B) \wedge (!x,\ x \in B) \longrightarrow (!x,\ x \in A)$$

Proof.
Assume $(A \subseteq B)$ and $(!x,\ x \in B)$.

$$(!x,\ x \in B) \implies (\forall x_1 \forall x_2,\ x_1 \in B \wedge x_2 \in B \longrightarrow x_1 = x_2) \qquad \text{(by def:uniqueness quantifier)}$$

The claim: $(\forall x_1 \forall x_2,\ x_1 \in A \wedge x_2 \in A \longrightarrow x_1 = x_2)$.
Fix x_1 and x_2. Assume $x_1 \in A$ and $x_2 \in A$.

$$A \subseteq B\ \ \&\ \ x_1 \in A \implies x_1 \in B$$

$$A \subseteq B\ \ \&\ \ x_2 \in A \implies x_2 \in B$$

$$(!x,\ x \in B)\ \ \&\ \ x_1 \in B\ \ \&\ \ x_2 \in B \implies x_1 = x_2$$

The claim is proven.

$$(\forall x_1 \forall x_2,\ x_1 \in A \wedge x_2 \in A \longrightarrow x_1 = x_2) \implies (!x,\ x \in A) \qquad \text{(by def:uniqueness quantifier)}$$

□

48.4 Unique witnesses and replacement

Theorem 48.3 (Unique choice requires no choice axiom)**.**
Work in classical class theory with the comprehension instance for $\varphi(x, y)$, replacement, and ordinary set-pair coding. Suppose A is a set and

$$\forall x \in A,\ \exists!(y \text{ set}),\ \varphi(x, y).$$

Then there is a unique function f with domain A, set values, and $\varphi(x, f(x))$ for every $x \in A$. Its graph and range are sets.

Proof.
Comprehension forms the class relation

$$R = \{\, (x, y) \mid x \in A \wedge (y \text{ set}) \wedge \varphi(x, y) \,\}.$$

The hypothesis makes it functional with domain exactly A. Replacement applied to its set-valued outputs gives the set $B = \{y : \exists x \in A,\ \varphi(x, y)\}$. Apply replacement once more, now to the uniquely determined set pair $x \mapsto (x, y)$ with $\varphi(x, y)$. Its image is exactly the graph R, so the graph is a set without a further appeal to separation or a product-set construction. It is the required f. Any other function satisfying these conditions has the same output at every input by uniqueness of witnesses, and therefore has the same graph by extensionality. If A is empty, the graph and range are both empty and the conclusion still holds. □
The theorem separates two operations often both called "choosing". When every input has a unique output, comprehension and replacement collect already determined values. A choice axiom addresses a family of nonempty sets whose possible outputs need not be unique. Also, replacing "unique set witness" above by "unique class witness" does not justify the set conclusion: a proper class cannot be a value in an ordinary set-coded graph. The singleton input with the prescribed value V already demonstrates this size obstruction.

Lemma 48.1 (Inhabited semi-singletons)**.**
Assume the empty set and singleton sets exist. A class S with at most one member is either empty or a singleton set. Consequently S itself is a set.

Proof.
If S has no member, extensionality gives $S = \varnothing$. Otherwise choose a witness $s \in S$; membership makes s a set. At-most-one uniqueness implies that every member of S equals s, while s belongs to S. Thus $S = \{s\}$ by extensionality. The case split uses classical logic and a single existential witness, not a global choice function. □

49 Partial function semantics following W. Farmer

This section uses partial first-order logic (PFOL). The notation $t\downarrow$ asserts that t denotes, and $s \simeq t$ denotes Kleene equality: the terms either denote the same value or are both undefined. Variables denote; compound terms may fail to denote. Function evaluation is defined by a unique description, independently of the total union-based evaluation used in other sections.

Note 49.1 (Three conventions for an absent value)**.**
The newer Obelisk catalog's `DefValueAtArg` (3.25) uses a total conditional evaluation: it returns V when there is no represented output. Plain union evaluation, used in another local construction of this book, returns $\bigcup \varnothing = \varnothing$ in that case. PFOL instead leaves the unique description undefined. These are three distinct conventions. For the empty relation, for example, the value at any argument is respectively V, $\varnothing$, or undefined. The definedness assertions in this section belong to the last convention.

49.1 A proof in PFOL

The argument follows the notation of William M. Farmer and Joshua D. Guttman, "A Set Theory with Support for Partial Functions". The PFOL axiom numbers below refer to that presentation, especially Section 2.3; its guarded application constructor appears in Section 3.2. [1] For the set-theoretic statements below, indices and output variables are sets, ordered pairs use the usual injective encoding, and relations may be classes. This convention avoids treating a proper class as a component of a set-coded pair.

Definition 49.1 (value of a function)**.**

$$f(x) \simeq (\iota y, \langle x, y\rangle \in f)$$

Here the descriptor tests uniqueness at the particular input x. Farmer–Guttman's application constructor instead includes the additional conjunct that f is a function. The two agree for functions; they can differ for other relations. For example,

$$r = \{\langle 0,0\rangle, \langle 1,0\rangle, \langle 1,1\rangle\}$$

has local value $r(0) = 0$ and no local value at 1, while the globally guarded constructor is undefined at every input because r is not univocal. The local convention is intentional in the following results.

Theorem 49.1 (Evaluation of a univocal relation)**.**
If f is univocal (at most one output for each input), and x, y are variables, then

$$(f(x) \simeq y) \longleftrightarrow (\langle x, y\rangle \in f)$$

Proof.
($\Leftarrow$) We use the following form PFOL10′ of PFOL10:

$$(\iota x,\ \varphi)\downarrow \quad \overset{\text{PFOL10}'}{\Longleftrightarrow} \quad (\exists! x,\ \varphi)$$

[1] William M. Farmer and Joshua D. Guttman, "A Set Theory with Support for Partial Functions", *Studia Logica* 66 (2000), 59–78. https://web.cs.wpi.edu/~guttman/pubs/set_theory_partial_fns.pdf.

So the proof is the following:

$$(\langle x, y\rangle \in f) \overset{\exists\text{–intro}}{\Longrightarrow} (\exists w,\ \langle x, w\rangle \in f)$$

$$f \text{ univocal} \overset{\text{def}}{\Longrightarrow} (!w,\ \langle x, w\rangle \in f)$$

$$(\exists w,\ \langle x, w\rangle \in f)\ \ \&\ \ (!w,\ \langle x, w\rangle \in f) \overset{\exists!\text{–intro}}{\Longrightarrow} (\exists! w,\ \langle x, w\rangle \in f)$$

$$(\exists! w,\ \langle x, w\rangle \in f) \overset{\text{PFOL10}'}{\Longrightarrow} (\iota w,\ \langle x, w\rangle \in f)\downarrow$$

$$(\iota w,\ \langle x, w\rangle \in f)\downarrow \overset{\text{PFOL11}}{\Longrightarrow} \langle x, (\iota w,\ \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w,\ \langle x, w\rangle \in f)\rangle \in f\ \ \&\ \ (\langle x, y\rangle \in f) \overset{f \text{ univocal}}{\Longrightarrow} (\iota w,\ \langle x, w\rangle \in f) \simeq y$$

$$f(x) \simeq y$$

($\Rightarrow$)

$$(y\downarrow)\quad (\text{by PFOL8})$$

$$(y\downarrow)\ \ \&\ \ ((\iota w,\ \langle x, w\rangle \in f) \simeq y) \overset{\text{PFOL7}}{\Longrightarrow} (\iota w,\ \langle x, w\rangle \in f)\downarrow$$

$$(\iota w,\ \langle x, w\rangle \in f)\downarrow \overset{\text{PFOL11}}{\Longrightarrow} \langle x, (\iota w,\ \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w,\ \langle x, w\rangle \in f)\rangle \in f\ \ \&\ \ ((\iota w,\ \langle x, w\rangle \in f) \simeq y) \overset{\text{PFOL7}}{\Longrightarrow} (\langle x, y\rangle \in f)$$

□

Theorem 49.2 (Domain of unique-description evaluation)**.**

$$f(x)\downarrow \longleftrightarrow x \in \{\, x \mid \exists! y,\ \langle x, y\rangle \in f \,\}$$

Proof.
By the definition of $f(x)$ and PFOL10′, it denotes exactly when $\exists! y,\ \langle x, y\rangle \in f$. The displayed class collects precisely these inputs, giving the claimed equivalence. □

Theorem 49.3 (The graph retained by local evaluation)**.**
For any class relation r of ordered pairs of sets, define

$$U(r) = \{\langle x, y\rangle \in r : \forall z \in V\ (\langle x, z\rangle \in r \to z = y)\}.$$

Then $U(r)$ is univocal, $U(U(r)) = U(r)$, and $U(r)(x) \simeq r(x)$ for every set x. Two relations have the same local evaluations at every set input if and only if their U-graphs are equal.

Proof.
At an input with exactly one output, U retains that output. At an input with no output or with at least two outputs, it retains none. This proves univocality and equality of the partial evaluations. Applying U again changes no singleton or empty fibre, proving idempotence. If the retained graphs are equal, equality of local evaluations follows. Conversely, if all local evaluations agree, each retained pair on one side supplies a denoting value on both sides and hence is retained on the other. Extensionality gives equality of the retained graphs. □
This operation is not monotone for inclusion of relations. With $0 \neq 1$, let $r = \{\langle 0, 0\rangle\}$ and $s = r \cup \{\langle 0, 1\rangle\}$. Then $r \subseteq s$, but $U(r) = r$ and $U(s) = \varnothing$. Adding a possible output can destroy local definedness. Consequently the extensionality theorem below requires univocal graphs: evaluation intentionally forgets every ambiguous fibre of a general relation.

49.2 Equality and composition of partial evaluations

Strict equality $s = t$ is true only when both terms denote the same value. Kleene equality therefore satisfies

$$s \simeq t \longleftrightarrow (s = t) \vee (s\!\uparrow \wedge t\!\uparrow), \qquad t = t \longleftrightarrow t\!\downarrow,$$

where $t\!\uparrow$ abbreviates $\neg(t\!\downarrow)$. These identities follow by separating the cases in which both, one, or neither term denotes. Consequently the denoting-variable hypothesis in the evaluation theorem matters: if $f(x)$ and a term t are both undefined, $f(x) \simeq t$ is true, while strict pair membership $\langle x, t\rangle \in f$ is false.

Theorem 49.4 (Extensionality of partial function graphs)**.**
For univocal class relations f, g of ordered pairs of sets,

$$f = g \longleftrightarrow \forall x \in V, \quad f(x) \simeq g(x).$$

Proof.
Equal relations have the same fibres, so their evaluations either both fail to denote or denote the same unique member. Conversely, suppose all their evaluations are Kleene equal. If $\langle x, y\rangle \in f$, then $f(x)$ denotes y. Kleene equality forces $g(x)$ to denote the same y, so the evaluation theorem gives $\langle x, y\rangle \in g$. This proves $f \subseteq g$; exchanging the relations proves the reverse inclusion. Class extensionality gives equality. □

Theorem 49.5 (Composition and its domain)**.**
Let f, g be univocal class relations of ordered pairs of sets, and put

$$h = g \circ f = \{\langle x, z\rangle : \exists y \in V\ (\langle x, y\rangle \in f \wedge \langle y, z\rangle \in g)\}.$$

Then h is univocal, and for every set x,

$$h(x) \simeq g(f(x)), \qquad h(x)\!\downarrow \longleftrightarrow \exists y \in V\ (\langle x, y\rangle \in f \wedge y \in \mathrm{dom}(g)).$$

Proof.
Two outputs of h at x have intermediate witnesses y, y'. Univocality of f gives $y = y'$, and then univocality of g identifies the outputs. If f has no output at x, the composite graph has none; also $f(x)$ is undefined. Strict pair membership makes every witness condition in $g(f(x))$ false, so this term is undefined too. If $f(x)$ denotes y, the fibre of h at x is exactly the fibre of g at y. It is either empty on both sides or has the same unique output on both sides. This proves the Kleene equality and the stated domain criterion. □
Univocality cannot be omitted from the composition theorem. Take

$$f = \{\langle 0, 0\rangle, \langle 0, 1\rangle\}, \qquad g = \{\langle 0, 0\rangle, \langle 1, 0\rangle\}.$$

Then $g \circ f = \{\langle 0, 0\rangle\}$, whose value at 0 is 0. But $f(0)$ is undefined under unique-description evaluation, so $g(f(0))$ is undefined. A multivalued intermediate relation can lose its ambiguity after composition; local uniqueness does not commute with that operation.

49.3 Extension by definitions

The following formulas describe how a descriptor can be introduced under an existence hypothesis. They are derived schemata, not further PFOL axioms. In particular, unique existence must be asserted under the domain hypothesis: placing that hypothesis inside an implication governed by $\exists!y$ would impose uniqueness even when the hypothesis is false. The intended

conditional unique-existence premise is

$$\forall f \forall x,\ ((x \in \mathrm{dom}(f)) \wedge (f \text{ univocal at } x)) \to (\exists! y,\ (x, y) \in f)$$

from which the intended evaluation rule is

$$\forall f \forall x,\ ((x \in \mathrm{dom}(f)) \wedge (f \text{ univocal at } x)) \to (x, f(x)) \in f$$

A total extension by a function symbol requires a unique value for every input. A partial extension instead retains the domain hypothesis above. The distinction is the reason for using partial descriptions in this section.
In general, a closed formula

$$\forall \vec{x},\ \exists! y,\ \varphi$$

implies

$$\forall \vec{x},\ \varphi[(\iota y,\ \varphi)/y]$$

.
The first rule below is admissible by PFOL10 and PFOL11: unique existence yields definedness of the descriptor, and definedness yields its witness formula. The second additionally requires a choice descriptor with its own witness axiom. In both rules, substitution must be free for y, with bound variables renamed where necessary. No generalization is performed, so $y \notin \mathrm{FV}(\Gamma)$ is not needed for these witness rules. Freshness conditions for a newly introduced symbol or an eigenvariable belong to different rules.

$$\frac{\Gamma \vdash \exists! y,\ \varphi}{\Gamma \vdash \varphi[(\iota y,\ \varphi)/y]}\ \text{Unique witness}$$

and

$$\frac{\Gamma \vdash \exists y,\ \varphi}{\Gamma \vdash \varphi[(\epsilon y,\ \varphi)/y]}\ \text{Choice witness}$$

Theorem 49.6 (A guarded partial definition)**.**
Let $A(\vec{x})$ and $\varphi(\vec{x}, y)$ be old-language formulas whose free variables are among those displayed, with y not free in A. Suppose a theory T entails

$$\forall \vec{x}\ (A(\vec{x}) \to \exists! y\ \varphi(\vec{x}, y)).$$

Assume $\vec{x}$ is a nonempty tuple of distinct variables. Add a new partial operator k of that arity with the axioms

$$\forall \vec{x}\ (k(\vec{x}) \downarrow \longleftrightarrow A(\vec{x})), \qquad \forall \vec{x}\ (A(\vec{x}) \to \varphi(\vec{x}, k(\vec{x}))).$$

Every model of T has exactly one expansion interpreting k subject to these axioms. Its value is described by

$$k(\vec{x}) \simeq \big(\iota y,\ A(\vec{x}) \wedge \varphi(\vec{x}, y)\big).$$

The extension is conservative for old-language semantic consequences.

Proof.
Fix a model of T and a tuple of denoting arguments. If A is true, there is exactly one witness for φ; assign that witness as the value of k. If A is false, leave k undefined. This defines a partial operation and satisfies both axioms. Any operation satisfying them must be undefined

in the second case and equal the unique witness in the first, proving uniqueness. The displayed descriptor has exactly this behavior: its predicate has one witness when A holds and no witness when A fails.

An old-language sentence has the same truth value in a model and its expansion. Thus every old-language countermodel to a proposed consequence of T expands to a countermodel of the extension. No new old-language semantic consequence is obtained. This is a model argument; a claim about derivability additionally uses completeness of the chosen calculus. □

The domain axiom is essential. A guarded witness axiom alone allows arbitrary values away from A and does not determine a unique partial operation. Likewise, inserting $A \to \varphi$ instead of $A \wedge \varphi$ inside the descriptor gives the wrong off-domain predicate: when A is false, every object is a witness. In a one-object model the resulting descriptor even denotes there. Finally, if one substitutes possibly undefined terms for $\vec{x}$, strict operator application remains undefined when an argument is undefined; the theorem and its displayed defining equations quantify over denoting variables.

50 Evaluation under alternative semantic conventions

This section compares partial evaluation with two encodings of families. The opening discussion belongs to a semantics that distinguishes defined, underdefined, and uncertain expressions. The later semisingleton calculation explicitly switches to total union evaluation. Each convention should be read locally; the same notation $F(x)$ does not identify their treatment of arguments outside the domain.

Definition 50.1 (value)**.**

$$F(x) :\simeq \begin{pmatrix} \text{if}((F \text{ family}) \wedge (x \in \text{dom}(F))) \\ \quad \text{then } \bigcup \{\, s \mid (x,s) \in F \,\} \text{ else undef} \\ \text{exception_else undef} \end{pmatrix}$$

50.1 Value

The following notes explore how equality should interact with partial values. The first two entries are unfinished proposals, not axioms used in the later calculation:

Definition 50.2 (semi–equality)**.**
The interpretation of $I_{\mathrm{Fm}}(s \approx t)$ remains to be specified. Its rule must distinguish the intended behavior when either argument lacks a unique value.

Definition 50.3 (equality: specification pending)**.**
For candidate-valued expressions, the symbol $s = t$ still needs a rule specifying whether it compares unique values, candidate collections, or values under each admitted selection. The comparisons below make these alternatives explicit; this unfinished entry fixes none of them.

Definition 50.4 (quasi–equality in the partial-function convention)**.**

$$(s \simeq t) : \Longleftrightarrow \ (((s \text{ defined}) \vee (t \text{ defined})) \longrightarrow (s = t))$$

Proposition 50.1 (Three precise comparisons of partial values)**.**
In a local model, let each term have a candidate set $S_t \subseteq D$, where D is a specified domain, and call it defined exactly when S_t is a singleton. Put v_t equal to its sole member when defined. The following comparisons have different purposes:

1. Strict equality holds when both terms are defined and their values are equal. It is symmetric and transitive, but is reflexive only on defined terms.

2. Kleene equality holds when both are undefined, or both are defined with equal values. It is an equivalence relation. It identifies an empty candidate set and a nonsingleton candidate set as undefined; it does not retain their distinct information.

3. Candidate equality means $S_s = S_t$. It is an equivalence relation and retains that information, but answers a different question from whether two expressions have the same unique value.

Proof.
For strict equality, symmetry and transitivity reduce to equality in D, and reflexivity is exactly the definedness condition. For Kleene equality, partition terms into one undefined class and one

class for each defined value; equality of these classes is an equivalence relation. For candidate equality, use equality of subsets of D. □
These are explicit local completions of possible equality conventions, not an identification of the unfinished semi-equality symbol with all three. Mere overlap of candidates is another possible comparison, but it is not transitive: $\{0\}$ overlaps $\{0,1\}$, and $\{0,1\}$ overlaps $\{1\}$, while $\{0\}$ and $\{1\}$ do not overlap when $0 \neq 1$. Hence overlap cannot support unrestricted substitution of equals. The displayed implication defining quasi-equality has the Kleene reading only when the surrounding propositional operations and strict equality have the stated classical partial-function interpretation; a three-valued implication needs its own table.

Theorem 50.1 (When candidate images preserve Kleene equality)**.**
Let $g : D \to E$ be a total function between sets. Extend it to candidate sets by direct image $g_*(S) = g[S]$. Interpret definedness and Kleene equality as in the preceding proposition. Then g_* preserves Kleene equality of all candidate sets if and only if g is injective. Candidate equality is preserved by every g_*.

Proof.
If g is injective, an empty set has empty image, a singleton has its corresponding singleton image, and a set with at least two elements has at least two image elements. Thus undefined inputs remain undefined, and two equal defined values have equal defined images. These are the two cases of Kleene equality. Conversely, suppose $x \neq y$ but $g(x) = g(y)$. The candidate sets $\{x, y\}$ and $\varnothing$ are both undefined and therefore Kleene equal. Their images are respectively $\{g(x)\}$ and $\varnothing$, which are not Kleene equal. This contradicts preservation. Finally, $S = T$ always implies $g[S] = g[T]$ by extensionality. □
An equivalence relation is therefore not automatically a congruence for all proposed term constructors. Strict partial application, which first requires a unique input and propagates undefinedness, behaves differently from taking the image of all candidates. The theorem identifies precisely when these two ways of forgetting information are compatible with the Kleene comparison; it does not prohibit either locally chosen semantics.

Definition 50.5 (underdefined expression)**.**
An expression with no possible value is distinguished here from one with several possible values. Both may lack a unique value, but they have different interpretations.

$$\text{undef} :\simeq (\varepsilon x,\ \bot)$$

For the proposed conditional, an uncertain condition is intended to yield an underdefined result. Thus if $I(\varphi) = U$, the proposed rule is

$$(I((\text{if } \varphi \text{ then } t_1 \text{ else } t_2)) = I(\text{undef}))$$

In the set-valued interpretation of possibilities, $I(\text{undef}) = \varnothing$: this means there are no possible values. It does not mean that the expression denotes the empty class as its unique value.

50.2 A comparison on ordinary graph fibres

Let $R \subseteq \mathrm{V} \times \mathrm{V}$ and put $R_x = \{y \in \mathrm{V} : (x, y) \in R\}$ for a set input x. Three operations examined in the book can be compared without identifying their meanings:

Fibre	Unconditional union	Guarded union	Unique-fibre value
$\varnothing$	$\varnothing$	V	undefined
$\{v\}$	v	v	v
At least two members	$\bigcup R_x$	$\bigcup R_x$	undefined

Here the guarded default is specifically V, and the last column is the local unique-fibre convention; Farmer's globally guarded application has an additional functionhood condition. Every entry follows directly from its definition. In particular, a singleton fibre whose value is the empty set is represented and defined, despite having empty union.

Proposition 50.2 (What a fixed default can distinguish)**.**
Suppose a partial map has values in a set D, and a totalization replaces undefined values by a fixed $d \in D$. The totalized value alone cannot determine definedness uniformly over all such partial maps. If instead one adjoins a tagged absent value disjoint from all tagged present values, the totalized value determines both definedness and the original value.

Proof.
At a fixed input, compare an undefined map and a map defined with value d. Both totalizations return d, while definedness differs. For the second statement, put $0 = \varnothing$, $1 = \{0\}$ and use ordinary injective set pairs. Encode absence by $(0, 0)$ and a defined value v by $(1, v)$. Since $0 \neq 1$, no present code equals the absent code, and equality of present codes forces equality of their values. Testing the first coordinate recovers definedness; the second recovers a defined value. □
For the class-theoretic guarded union convention, a proper-class default similarly cannot equal a set value. But when a family is allowed to have proper-class values, the default itself may also be a represented value. Then the represented domain must be retained separately, as in the family extensionality theorem. The tagged set construction above explains the information issue; it does not encode arbitrary proper classes as members of an ordinary set-valued graph.

50.3 Singleton–based implementation of families (preferred local encoding)

In both graph encodings below, stored coordinates are sets and the pair coding is injective on set coordinates. Accordingly the displayed binder $(x : \mathrm{Class})$ collects records only at set inputs: the record comprehensions explicitly bind their coordinates over V. Any proposed equation quantifying over all class inputs needs an additional encoding: an ordinary class of set-pair records cannot store a proper-class argument. The beta equations below assert evaluation only at set inputs.

Definition 50.6 (singleton)**.**

$$(s \text{ singleton}) : \longleftrightarrow (\exists!(y : \mathrm{Class}),\ y \in s)$$

Side condition:
Variable y is not among free variables of a class term s.

Definition 50.7 (family constructor)**.**
The auxiliary bound variable s is fresh for t and distinct from x. The variable x is the abstraction variable; all substitutions avoid capture.

$$(\Lambda(x : \text{Class}),\ t) \coloneqq \left\{ (x, s) \;\middle|^{x,s:\mathrm{V}}\; \begin{array}{c} ((t \text{ proper}) \wedge (s \text{ singleton}) \wedge (s \subseteq t)) \\ \vee((t \text{ set}) \wedge (s = t)) \end{array} \right\}.$$

(or, later, maybe

$$(\Lambda(x : \text{Class}),\ t) \coloneqq (\epsilon(F : \text{Class}),\ (F \text{ family}) \wedge \forall(x : \text{Class}),\ (F(x) = t))$$

)

Proposition 50.3 (Reconstructing the preferred family encoding)**.**
Work in the classical class interpretation of the displayed constructor, with its comprehension instance, empty and singleton sets, and injective ordered pairs of sets. Let $F = (\Lambda(x : \text{Class}), t)$ use the preferred encoding above. For every set input z, with capture-avoiding substitution, put $A = t[z/x]$. Its stored fibre is

$$\{s \in \mathrm{V} : (z, s) \in F\} = \begin{cases} \{A\}, & A \text{ set}, \\ \{\{q\} : q \in A\}, & A \text{ proper}. \end{cases}$$

In either case the fibre is nonempty and its union is A.

Proof.
Rename the abstraction variable and the auxiliary variable freshly before substituting z. Injectivity of the set-pair coding identifies the fibre with the set records satisfying the constructor's condition at A. If A is a set, that condition is exactly $s = A$; the union of its singleton fibre is A, including when A is empty. If A is proper, the condition says that s is a singleton subclass of A. Such records are precisely the singleton sets of members of A. Every member therefore belongs to the union of the records, and every member of a record belongs to A. Extensionality gives the claimed union. A proper class is nonempty because the empty class is a set, so this fibre is also nonempty. □

This proposition establishes reconstruction from records; it does not fix the domain test in the opening partial-evaluation proposal. If that test uses the small domain of the axioms chapter, it admits only arguments whose reconstructed value is a set. For example, the constant body $t = \mathrm{V}$ gives a nonempty fibre at every set input, with union V, but its small domain is empty. The opening conditional would then return undef at every input. Using the represented domain instead would admit those proper-class values, but that is a separate choice of partial-evaluation convention. Neither domain lets an ordinary record graph store a proper-class input; the alternative descriptor formula above, which quantifies over all class inputs, remains a proposal requiring an additional interpretation.

50.4 Semisingleton–based implementation of families (earlier encoding)

This earlier encoding represents each value by all of its semisingleton subclasses, including the empty class. Its calculations remain meaningful under that convention, even though the preceding construction uses a different canonical fiber. The label "outdated" records the development history, not a change to the assumptions of the following proofs.

Definition 50.8 (some outdated syntactic sugar)**.**

$$(x \text{ any}) \overset{\text{def.}}{\longleftrightarrow} \top$$

Informal context: (x:Var)

Definition 50.9 (semi–singleton)**.**

$$(s \text{ ss}) : \longleftrightarrow (!y,\ y \in s)$$

Side condition:
Variable y is not in free variables of a class term s.

Definition 50.10 (semi–singleton subset relation)**.**

$$(s \text{ sss } t) : \longleftrightarrow ((s \text{ ss}) \wedge (s \subseteq t))$$

Definition 50.11 (function constructor)**.**
This graph represents a value only at set inputs where t is a set; proper-class values require one of the family encodings.

$$(\lambda(x : \text{Class}),\ t) \coloneqq \{\, (x,t)^{x:\mathrm{V}} \mid t \text{ set} \,\}$$

Context: (x:Var) (t:Class)
Comment: Class term t can contain x as a free variable.
Side conditions:
(variable x is bound by class collection in r.h.s)
(variable x is bound by lambda abstraction in l.h.s)

Definition 50.12 ((outdated) family constructor)**.**

$$(\Lambda(x : \text{Class}),\ t) \coloneqq \{\, (x,s)^{x:\mathrm{V},\, s:\mathrm{V}} \mid s \text{ sss } t \,\}$$

Term t can contain x as a free variable.
Side conditions:
Variables x and s are different.
Variable s is not in free variables of t.
As term s can contain x as a free variable.

Definition ((LOCAL,OUTDATED) definition of a value of a function, value of a family, for the next theorem)**.**

$$f(x) \coloneqq \bigcup \{\, s \mid (x,s) \in f \,\}$$

Theorem 50.2 (evaluation of a function)**.**
Context:
(x:Var)(t:Class)(z:Class)

$$(z \text{ set}) \wedge (t[z/x] \text{ set}) \longrightarrow$$
$$(\lambda x,\ t)(z) = t[z/x]$$

Proof.
Rename the bound variable to a fresh u before evaluating at the term z, and put $v = t[z/x]$. Capture-avoiding substitution and injectivity of the chosen set-pair coding give, for every set b,

$$(z,b) \in (\lambda x, t) \longleftrightarrow b = v.$$

For the forward implication, a graph witness has first coordinate z; substitution in its second coordinate therefore yields v. Conversely, z and v are sets by hypothesis, so their pair is a legitimate graph member. Thus the fibre is exactly $\{v\}$ and its union is v. This argument uses semantic substitution, not the generally false claim that $t[z/x]$ has no free occurrences of x: the replacement term z itself may contain x. □

Theorem 50.3 (evaluation of family)**.**
Let z be a set, and assume substitution of z for x is capture-avoiding. Under the semisingleton encoding and the local total union evaluation,

$$(\Lambda x,\ t)(z) = t[z/x],$$

and in particular, when x denotes a set:

$$(\Lambda x,\ t)(x) = t.$$

Proof.
Write $A = t[z/x]$. The fiber of the semisingleton encoding at z consists of the set subclasses $s \subseteq A$ with at most one member. Union evaluation reconstructs A from those subclasses: for any set q,

$$q \in (\Lambda x,\ t)(z) \longleftrightarrow q \in \bigcup \{\, s \mid (z, s) \in (\Lambda x,\ t) \,\} \longleftrightarrow \exists s,\ (q \in s \wedge (z, s) \in (\Lambda x,\ t)).$$

By pair injectivity and the comprehension $(\Lambda x,\ t) = \{\, (x, s) \mid s \in \mathrm{V} \wedge s \text{ sss } t \,\}$ (under capture-avoidance $\mathrm{FFI}(z, v_x, (s \text{ sss } t))$), $(z, s) \in (\Lambda x,\ t)$ holds if and only if s is a set satisfying $(s \text{ sss } t[z/x])$. If $q \in s$ and $s \subseteq t[z/x]$, then $q \in t[z/x]$; conversely, if $q \in t[z/x]$, then q is a set and $s \coloneqq \{q\}$ is a semi-singleton subclass of $t[z/x]$ with $q \in \{q\}$. Thus:

$$q \in (\Lambda x,\ t)(z) \longleftrightarrow q \in t[z/x].$$

By class extensionality, $((\Lambda x,\ t)(z) = t[z/x])$. Taking $z \coloneqq x$ where $t[x/x] = t$ immediately yields $((\Lambda x,\ t)(x) = t)$. □

50.5 Earlier graph-based lambda operator

Definition 50.13 (lambda operator)**.**
This graph encoding applies to set-valued terms on set arguments. It does not by itself represent proper-class values as members of a graph.

$$(\lambda x,\ t) = \{\, p \mid \exists (x \in \mathrm{V}),\ (t \text{ set}) \wedge (p = \langle x, t \rangle) \,\} \qquad \text{(def:lambda operator)}$$

The bound on x is essential for the Metamath angle-pair convention, which returns the empty class if either coordinate is proper. Without that bound, even the constant body $t = \varnothing$ would admit the witness $x = \mathrm{V}$ and insert $\langle \mathrm{V}, \varnothing \rangle = \varnothing$ into the graph. The empty class is not a Kuratowski pair of sets. The sethood condition on t likewise excludes records for proper-class values.

51 Descriptors

This section explores descriptors whose values need not be determined uniquely by their defining predicates. Its local semantics distinguishes absence of a candidate, multiplicity of candidates, and a selected value. It also allows an uncertain truth value U. The purpose is to make those distinctions visible in notation, especially when a descriptor occurs inside a formula. The final subsection records a separate proposal for axioms of set-valued descriptors.

51.1 Existence, uniqueness, and selection

A predicate can have no satisfying objects (the *nullary* case), exactly one (the *singular* case), or several (the *plural* case). These are properties of its possible witnesses. A choice interpretation adds a further ingredient: it selects one witness when candidates exist. Several possible witnesses therefore need not make a term ambiguous once a fixed choice interpretation has been supplied. Here "nullary" refers to the absence of witnesses, not to the number of arguments of the predicate. The draft sometimes uses "defined" to mean "well-defined under the chosen interpretation". We keep the distinction explicit: existence concerns whether a value is available; uniqueness concerns whether the predicate determines it; selection concerns which available value an interpretation chooses. Invoking choice requires specifying the collection on which it operates. In particular, a choice principle for sets does not by itself specify a selector for every predicate on arbitrary classes.

51.2 Three conventions for a default value

The superscript in τ^D describes the admissible default values when the predicate has no witness. It does not select among several actual witnesses. There are three conventions:

1. $\tau^{\varnothing}$ supplies no default value. In the nullary case, the value is absent; "at most one" must not be confused with existence.
2. $\tau^{\{K\}}$ supplies a specified default K. The notation $\tau_K^{\exists!}$ emphasizes that the default is uniquely specified.
3. $\tau^{\mathcal{U}}$ allows any class as a default, where $\mathcal{U}$ is the conglomerate of all classes. This convention leaves the default undetermined until an interpretation selects it.

Here the superscript is semantic notation for a collection of admissible values. If K is a proper class, $\{K\}$ in this notation is not an object-theoretic singleton class containing K.
Our conventions are:

- $\tau := \tau^{\mathcal{U}}$;
- $\epsilon := \tau_{\varnothing}^{\exists!} = \tau^{\{\varnothing\}}$;
- $\varepsilon := \tau^{\varnothing}$.

The total ϵ-operator uses the empty class when no witness exists; the partial ε-operator supplies no value in that case. The operators are intended here for definable conglomerates, namely collections of classes described by predicates in the chosen language. The precise language and the interpretation of choice are part of the local setting.

51.3 Quantifiers and total choice

The following notation separates existence from the condition of having at most one witness:

$$((\varepsilon x : \varphi)\ \text{unique}) \iff (!x : \varphi), \qquad ((\varepsilon x : \varphi)\ \text{exists}) \iff (\exists x : \varphi).$$

Here $!x : \varphi$ means that there is at most one witness, whereas $\exists! x : \varphi$ asserts both existence and uniqueness. The displayed conditions describe the witness collection; they are not claims that a partial term can be substituted without a definedness condition.
With a fixed total choice interpretation, Hilbert-style quantifier clauses use the total operator ϵ:

$$(\exists x : \varphi) : \iff \varphi[(\epsilon x : \varphi)/x], \qquad (\forall x : \varphi) : \iff \varphi[(\epsilon x : \neg\varphi)/x].$$

Substitution is capture-avoiding. The existential clause selects a witness when one exists. The universal clause tests a selected counterexample when one exists; if there is none, every value satisfies φ, including the default. This explanation assumes a nonempty domain and ordinary two-valued predicates under the fixed selection. Extending these clauses to predicates that themselves take the value U requires further semantic clauses.
The corresponding notation for universal satisfaction is

$$((\varepsilon x : \varphi)\ \text{forall}) : \iff (\neg((\varepsilon x : \neg\varphi)\ \text{exists})).$$

If a partial descriptor has no determined value, substituting it into a predicate may instead produce the uncertain value U. In the strict connective convention considered here, any connective with an uncertain argument has uncertain output, including equivalence. Consequently, an object-language equivalence involving such terms cannot automatically serve as a two-valued metalinguistic definition. This is the reason to distinguish the total choice clauses from the partial-descriptor experiment.

51.4 A two-candidate example

Let a and b be distinct sets, and consider

$$c \coloneqq (\varepsilon x :\ x \in \{a, b\}).$$

The predicate does not uniquely determine c. With a fixed choice interpretation, however, c denotes one of the two witnesses, so

$$(a = c) \vee (b = c)$$

holds under that interpretation. Its individual disjuncts may vary when the choice interpretation varies. This distinguishes a statement about all permitted selections from a statement about one selected value.
An alternative under investigation leaves c unresolved, treating it as a descriptor with several candidates rather than as an already selected witness. Then the meanings of $a = c$, $a \simeq c$, and their disjunctions must be specified before those formulas can be evaluated. A strict uncertainty convention need not validate the same formulas as fixed choice. The original question can therefore be stated precisely: which laws of substitution and equality express membership among the candidates when no selection has been made?
Introducing a new constant can keep the descriptor visible in the syntax, but does not by itself answer this semantic question. Likewise, replacing ordinary equality by $\simeq$ only helps after its interpretation has been given. These are open design questions for this local calculus, rather than counterexamples to the ordinary laws of fixed choice.

51.5 A finite selection model for comparison

The two-candidate example admits a precise model that keeps track of correlations between occurrences of a descriptor. This model is a comparison with the strict U convention, not a replacement for it.

Let $\Omega = \{\omega_a, \omega_b\}$ be two permitted selections. Interpret c as a at ω_a and as b at ω_b, keeping the distinct parameters a, b fixed. Evaluate formulas classically at each selection, and use the *same* selection at every occurrence of c in a formula. For a formula ψ, let

$$[\psi] = \{\omega \in \Omega : \psi \text{ is true at } \omega\}.$$

These brackets denote a set of selections, not an object-theoretic truth predicate for the whole language. Define a summary value by

$$v(\psi) = \begin{cases} T, & [\psi] = \Omega, \\ F, & [\psi] = \varnothing, \\ U, & \varnothing \subsetneq [\psi] \subsetneq \Omega. \end{cases}$$

Thus U means variation among permitted classical selections; it does not mean failure of classical evaluation at any one selection.

Theorem 51.1 (Uncertainty summaries lose correlations)**.**
In this model, $v(a = c) = v(b = c) = U$, but

$$v\big((a = c) \vee (b = c)\big) = T.$$

There is no binary truth function on $\{T, F, U\}$ that computes the summary of every disjunction from the two summaries alone.

Proof.
Put $p \equiv (a = c)$. Its truth set is $\{\omega_a\}$, while the truth set of $b = c$, equivalently $\neg p$ in this model, is $\{\omega_b\}$. Both summaries are U, and the union of their truth sets is Ω. On the other hand, $p \vee p$ has the same truth set as p, so its summary is U. The same input pair (U, U) would therefore require both outputs T and U, which no truth function can provide. □

The full truth sets do determine the connectives:

$$[\neg\psi] = \Omega \setminus [\psi], \qquad [\psi \vee \chi] = [\psi] \cup [\chi], \qquad [\psi \wedge \chi] = [\psi] \cap [\chi].$$

These equations follow by evaluating at an arbitrary selection. They make precise the information that is discarded when a truth set is replaced by one of the three labels. In particular, $c = c$ is true at every selection, even though c has two candidates. Treating its two occurrences as independent choices would describe a different experiment.

The strict convention introduced earlier instead propagates U through every connective. It can therefore assign U to the displayed disjunction. The two answers express different meanings of uncertainty: strict propagation from an unresolved term, versus invariance across complete selections. A calculus should say which meaning it implements before adopting either set of laws.

51.6 Substitution must transport the selections

The shared-selection interpretation distinguishes truth throughout one specified experiment from logical validity under all assignments. This distinction matters when a rule permits substitution.

The following result does not require finitely many selections: Ω is a supplied set of complete assignments. It does not assert that every family of nonempty candidate sets admits a complete selection, and thus does not introduce a hidden choice principle.

Theorem 51.2 (Transport of truth sets under substitution)**.**
Work with a classical first-order structure whose terms have total values, and let Ω be a nonempty set of assignments. For a finite, sort-correct, capture-avoiding simultaneous substitution σ, define

$$F_\sigma(v)(x) = [\![x\sigma]\!]_v \quad (v \in \Omega), \qquad \Omega_\sigma = F_\sigma[\Omega].$$

Here F_σ has domain Ω and codomain Ω_σ; the inverse image below is taken for this restricted map. Writing $[\varphi]_S$ for the assignments in S satisfying φ,

$$[\varphi\sigma]_\Omega = F_\sigma^{-1}\big([\varphi]_{\Omega_\sigma}\big).$$

In particular, $\varphi\sigma$ holds throughout Ω exactly when φ holds throughout Ω_σ. If $\Omega_\sigma \subseteq \Omega$, truth throughout Ω is preserved by this substitution.

Proof.
The ordinary substitution lemma gives $v \models \varphi\sigma$ if and only if $F_\sigma(v) \models \varphi$. This proves the inverse-image equality. Every member of Ω_σ has a preimage by its definition, so universal truth on the two sides is equivalent. The inclusion in the last claim lets truth on Ω be restricted to Ω_σ. □

Example 51.1 (Universal truth in an experiment is not substitution validity)**.**
Let $D = \{0, 1\}$, and let Ω consist of assignments in which $x = y$, allowing both common values. Then $x = y$ is true throughout Ω. Substituting the constants 0 and 1 for x and y produces $0 = 1$, which is false throughout Ω. The transported assignments have $x = 0, y = 1$ and lie outside Ω. There is no conflict with classical substitution of valid formulas: $x = y$ was valid only on this restricted experiment, not on all assignments.

Proposition 51.1 (Marginal equality does not justify correlated replacement)**.**
Equality of the candidate sets of two terms need not permit substitution of one for the other in shared-selection formulas.

Proof.
On $\Omega = \{\omega_0, \omega_1\}$, take $c(\omega_i) = i$ and $d(\omega_i) = 1 - i$. Both terms have candidate set $\{0, 1\}$. Nevertheless, the context $z = c$ is true everywhere after substituting c for z, and false everywhere after substituting d for z. Their marginal sets agree, but their joint values with c do not. □
This does not invalidate equality of candidate sets in the alternative powerset-domain model: there a candidate set is itself the semantic object, and no further selection correlation is retained. In a shared-selection model, a sufficient condition for replacement is pointwise agreement $s(v) = t(v)$ at every admitted assignment (together with the ordinary binder freshness conditions). Equality of their ranges is weaker.

51.7 Proposed axioms for set-valued descriptors

This subsection uses descriptors restricted to set witnesses. Read the following formulas as proposed local axiom schemes. The notation defined concerns the descriptor's value; the earlier strict propagation rule for U does not supply a semantics for these schemes on its own.

Axiom 22 (Choice of an existing set witness)**.**

$$\exists(x \text{ set}) : \varphi \longrightarrow (\varepsilon(x \text{ set}) : \varphi) \in \{\, x \mid \varphi \,\}$$

Axiom 23 (Existence required for partial choice)**.**

$$(\varepsilon(x \text{ set}) : \varphi) \text{ defined} \longrightarrow \exists(x \text{ set}) : \varphi$$

Axiom 24 (Description of a unique set witness)**.**

$$(\exists!(x \text{ set}) : \varphi) \longrightarrow (\iota(x \text{ set}) : \varphi) \in \{\, x \mid \varphi \,\}$$

Axiom 25 (Uniqueness required for description)**.**

$$(\iota(x \text{ set}) : \varphi) \text{ defined} \longrightarrow \exists!(x \text{ set}) : \varphi$$

A related encoding represents an unsuccessful description by a proper-class default, so that sethood can detect success. Under that additional convention one proposes

$$(\iota(x \text{ set}) : \varphi) \text{ defined} \longleftrightarrow (\iota(x \text{ set}) : \varphi) \text{ set}.$$

This is an encoding of definedness for set-valued descriptions, not a claim that proper classes are intrinsically undefined.
The intended combined specification is

$$(\iota(x \text{ set}) : \varphi) \text{ defined} \longleftrightarrow \exists!(x \text{ set}) : \varphi,$$

with the witness property

$$(\iota(x \text{ set}) : \varphi) \text{ defined} \longrightarrow (\iota(x \text{ set}) : \varphi) \in \{\, x \mid \varphi \,\}.$$

For example, if x is a set satisfying φ and there is at most one set witness, the description is defined:

$$(x \text{ set}) \wedge \varphi \wedge (!(x \text{ set}) : \varphi) \longrightarrow (\iota(x \text{ set}) : \varphi) \text{ defined}.$$

The free occurrence of x in the premise supplies existence. The sethood condition makes explicit the witness domain used by this subsection.

52 [OUTDATED]Alternative semantics

This historical proposal interprets a term by a collection of possible classes rather than by a single class. It investigates how membership, choice, and definedness might be expressed under that interpretation. The label records its place in the development; the definitions below are to be read in this local setting.

52.1 Assignments and truth

Let K denote the conglomerate of all classes in the intended model M. Both K and $\mathcal{P}(K)$ are metatheoretic collections here; the notation does not assert the existence of a class of all classes inside NBG or MK. An assignment has the form

$$A : \mathrm{Var} \to \mathcal{P}(K).$$

Thus a variable, and more generally a term, may have no candidate value, one candidate, or several candidates. The subscripts in I_A^{Term} and I_A^{Fm} distinguish term interpretation from formula interpretation under the same assignment.
The proposed definedness clause is

$$I_A^{\mathrm{Fm}}(S\downarrow) \longleftrightarrow (I_A^{\mathrm{Term}}(S) \neq \varnothing).$$

Here definedness means nonemptiness of the candidate collection, not uniqueness of its member. Formula interpretation is intended to remain two-valued:

$$\mathrm{Bool} \coloneqq \{\mathrm{False}, \mathrm{True}\}.$$

Satisfaction under an assignment and validity in the model are distinguished by

$$M, A \vDash \varphi \overset{meta}{\Longleftrightarrow} I_A^{\mathrm{Fm}}(\varphi) = \mathrm{True},$$

$$(M \vDash \varphi) \overset{meta}{\Longleftrightarrow} (\forall(A : \mathrm{Assignment}),\ M, A \vDash \varphi).$$

The quantifier in the second display ranges over the assignments admitted by this semantics.

52.2 Candidate membership and choice

One proposed clause for membership, called "weak" in the draft, is

$$M, A \vDash X \in Y \quad \Longleftrightarrow \quad \forall C \in I_A^{\mathrm{Term}}(X)\, \forall D \in I_A^{\mathrm{Term}}(Y),\ C \in D.$$

Here X and Y are arbitrary terms. Validity of $X \in Y$ additionally quantifies over assignments. The clause compares every candidate on the left with every candidate on the right. With ordinary universal quantification it is vacuously true if either candidate collection is empty; whether to add a nonemptiness condition is a further design choice.

A precise relational fragment. To isolate what this clause says, work in the metatheory with a set D of candidate objects and a binary relation $R \subseteq D \times D$. No assumption that D contains all classes is needed for this fragment. For $S, T \subseteq D$, put

$$R^{\forall}(S,T) \longleftrightarrow \forall s \in S\, \forall t \in T,\ (s,t) \in R.$$

The proposed membership clause is the same construction with the actual membership relation on its candidate domain.

Lemma 52.1 (Singletons, refinement, and absence)**.**
The lifted relation has the following properties:

1. $R^{\forall}(\{s\},\{t\})$ holds exactly when $(s,t) \in R$.
2. If $R^{\forall}(S,T)$, $S' \subseteq S$, and $T' \subseteq T$, then $R^{\forall}(S',T')$.
3. $R^{\forall}(\varnothing,T)$ and $R^{\forall}(S,\varnothing)$ hold for every S,T.

Proof.
In the first claim the two quantifiers range over one pair. For the second, each pair in $S' \times T'$ is already in $S \times T$, where the relation holds by hypothesis. In the third, there is no pair on which the relation could fail. □
This gives a conservative reading of atomic formulas on single-valued assignments and explains why narrowing the candidate collections preserves a true atomic membership claim. It also proves that this membership clause alone cannot imply definedness. For example, assigning no candidates to X makes $X \in Y$ true while $X\downarrow$ is false.
If an application needs membership to imply existence of both values, it can instead use the separately named guarded lift

$$R^{+}(S,T) \longleftrightarrow (S \neq \varnothing) \wedge (T \neq \varnothing) \wedge R^{\forall}(S,T).$$

The singleton property remains valid, and refinement preserves truth when the refined candidate collections remain nonempty. These conclusions follow from the lemma and the two guards. The guarded lift is an optional variant; the original unguarded clause is retained above.

Repeated occurrences and equality. A candidate collection records possible values but not correlations among occurrences. If equality is lifted by the same independent-pair rule, then

$$\mathrm{Eq}^{\forall}(S,S) \longleftrightarrow \forall s,t \in S,\ s = t.$$

For nonempty S, this holds exactly when S is a singleton: fix one $s \in S$ and use the displayed condition to identify every other member with it. Thus a term with two distinct candidates would fail reflexivity under independent-pair equality. This is not an objection to the proposed membership clause; it shows why the equality clause requires a separate choice. Equality of candidate collections is reflexive, while evaluation under a shared selection preserves the correlation between repeated occurrences. Those alternatives answer different questions and must not be silently identified.
The proposed choice principle is

$$P(\varepsilon x,\ P) \longleftrightarrow \exists x,\ P(x).$$

The letter P denotes a predicate, leaving A for assignments. This formula is a hypothesis about a future descriptor interpretation. The membership and definedness clauses alone do not establish it; the meaning of ε and of substitution into P must also be specified.

52.3 A finite model of candidate choice

The preceding clauses admit a complete small interpretation once the quantifier range and the descriptor are fixed. This is a local model of a fragment, not a model of the intended class theory or its comprehension scheme.

Fix a finite nonempty set D, a relation $R \subseteq D \times D$, and a linear ordering of $\mathcal{P}(D)$. Use the set $\mathcal{P}(D)$ as the domain of semantic objects. Variables range over candidate sets $S \subseteq D$; a constant c_S names each such set. Interpret the binary relation symbol by $R^{\forall}$, equality by equality of candidate sets, and $t \downarrow$ by nonemptiness of the candidate set denoted by t. Give Boolean connectives and quantifiers their ordinary classical meanings on this domain. Thus $\exists x$ ranges over *all candidate sets*, including the empty candidate set, rather than just over individual candidates in D.
For a formula φ and assignment a, let

$$\mathcal{W}_{\varphi,a} = \{S \subseteq D : M, a[x \mapsto S] \vDash \varphi\}.$$

Interpret the descriptor $\varepsilon x, \varphi$ as the first member of $\mathcal{W}_{\varphi,a}$ in the fixed ordering if this set is nonempty, and as $\varnothing$ otherwise. The formula inside a descriptor is a shorter expression, so these clauses define term and formula interpretation simultaneously by recursion. No choice axiom is needed for this finite model.

Lemma 52.2 (Substitution and extensionality of finite choice)**.**
For a term t, write $[\![t]\!]_a$ for its candidate-set value. If substitution is capture-avoiding, then

$$[\![r[t/z]]\!]_a = [\![r]\!]_{a[z \mapsto [\![t]\!]_a]},$$

and the analogous equivalence holds for satisfaction of formulas. Furthermore, formulas with the same witness collection under a have equal descriptor values under a.

Proof.
Use simultaneous structural induction on terms and formulas. Variables, constants, predicates, Boolean connectives, and quantifiers obey the usual assignment-update clauses. At a descriptor $\varepsilon x, \psi$, rename its binder so that $x \neq z$ and x is not free in t. For each candidate set S, the induction hypothesis identifies the truth of $\psi[t/z]$ under $a[x \mapsto S]$ with the truth of ψ under $a[z \mapsto [\![t]\!]_a][x \mapsto S]$. Here the value of t is unchanged by updating x: a simultaneous induction on expressions proves that values and satisfaction depend only on free variables. Thus the two witness collections are equal. Their first members in the fixed ordering are equal, or both collections are empty and both defaults are empty. This proves the descriptor case. The same reasoning proves the last assertion directly, without any substitution: the choice rule depends on the witness collection, not on the syntax used to describe it. □
This extensionality is a property of the specified selector. A bare axiom saying only that a descriptor chooses some witness would not force two syntactically different descriptions with the same multiple-witness collection to make the same choice.

Choice and a change of labels. Extensional choice need not be invariant under every relabelling of its objects. The order used above is additional structure, and it must be transported when the model is transported.

Proposition 52.1 (No fully permutation-invariant selector)**.**
Let B be a set with at least two elements. There is no function

$$c : \mathcal{P}(B) \setminus \{\varnothing\} \longrightarrow B, \qquad c(W) \in W,$$

such that $c(\pi[W]) = \pi(c(W))$ for every permutation π of B and every nonempty $W \subseteq B$.

Proof.
Put $b = c(B)$ and choose $b' \in B$ different from b. The permutation that exchanges b and b' and fixes all other elements satisfies $\pi[B] = B$. The required identity would give $b = c(B) = c(\pi[B]) = \pi(c(B)) = b'$, a contradiction. □
This obstruction also applies to the candidate model even if only relabellings of the underlying D are allowed. When D has at least two elements, the witness collection $W = \{\{d\} : d \in D\}$ is invariant under every such relabelling. A selector must choose one singleton; exchanging its member with another member of D fixes W but moves that singleton. Thus even equality of the transported witness collections cannot by itself ensure equivariance of the selected value. This is a statement about selectors on all nonempty collections; it does not assert that every such collection is definable by the formulas of a given smaller language.

Proposition 52.2 (Transport of ordered finite choice)**.**
Let $(B, \leq)$ and $(B', \leq')$ be finite nonempty linearly ordered sets, and let $h : B \to B'$ be an order-isomorphism. For each nonempty $W \subseteq B$, put $c_{\leq}(W) = \min_{\leq} W$. Then

$$c_{\leq'}(h[W]) = h(c_{\leq}(W)).$$

If the selectors are extended to empty collections by defaults $b_0 \in B$ and $b_0' \in B'$, the same identity holds for every $W \subseteq B$ exactly when $b_0' = h(b_0)$.

Proof.
The element $m = \min_{\leq} W$ belongs to W and satisfies $m \leq w$ for all $w \in W$. Hence $h(m)$ belongs to $h[W]$ and is below each of its elements in the order $\leq'$. It is therefore its minimum. For the empty collection the identity is precisely $b_0' = h(b_0)$. □
For example, a bijection $f : D \to D'$ induces $h : \mathcal{P}(D) \to \mathcal{P}(D')$, $h(S) = f[S]$. Transport the relation by $R' = \{(f(d), f(e)) : (d, e) \in R\}$, the assignment by $a'(x) = h(a(x))$, and each constant c_S to $c_{h(S)}$. Transport the order by declaring $h(S) \leq' h(T)$ exactly when $S \leq T$. The default is respected because $h(\varnothing) = \varnothing$. Simultaneous induction on terms and formulas now shows that term values are carried by h and formula truth is preserved. The relation case follows by transporting every pair of candidates; quantifiers use the bijectivity of h. At a descriptor, the induction hypothesis transports its witness collection to its direct image under h, and the preceding proposition transports the selected value.
Keeping the old order while permuting the objects is a different operation: it need not preserve descriptor values. None of this conflicts with the substitution lemma, which compares assignments inside one fixed ordered model and identifies the witness collections there literally. Nor does renaming a bound variable change the order on semantic objects. Syntactic renaming, substitution, and transport of a semantic structure are three distinct operations.

Theorem 52.1 (Choice and candidate presence are distinct)**.**
For this interpretation, with capture-avoiding substitution,

$$M, a \vDash \varphi[(\varepsilon x, \varphi)/x] \quad \Longleftrightarrow \quad M, a \vDash \exists x\, \varphi.$$

Nevertheless, the truth of either side need not imply $(\varepsilon x, \varphi)\downarrow$.

Proof.
If there is a witness candidate set, the descriptor denotes one of the witness sets, so substitution yields a true formula. If there is none, φ is false at every candidate set, including the default $\varnothing$,

so the substituted formula is false. The substitution step follows by induction on expressions, as in ordinary first-order semantics; the descriptor case uses the same set of witnesses under the two corresponding assignments.
For the last claim, take $\varphi = \neg(x\downarrow)$. Its unique witness in $\mathcal{P}(D)$ is the empty candidate set. The existential formula and the descriptor witness formula are therefore true, but the descriptor's candidate-presence assertion is false. □

Corollary 52.1 (Choice with an explicit presence guard)**.**
Define the term

$$e_\varphi = (\varepsilon x,\ x\downarrow \wedge \varphi).$$

Then

$$e_\varphi\downarrow \longleftrightarrow \exists x\,(x\downarrow \wedge \varphi), \qquad e_\varphi\downarrow \Longrightarrow\ \varphi[e_\varphi/x].$$

Proof.
When a nonempty witness candidate set exists, the descriptor selects such a set and it satisfies φ. Otherwise the witness collection for the guarded predicate is empty and the descriptor has the empty default, so its presence assertion is false. □
In this model every term denotes a semantic object, which may itself be an empty collection of candidates. Thus $t\downarrow$ is an internal presence predicate, not the metatheoretic assertion that term interpretation exists. PFOL makes a different distinction: an undefined term has no denotation. The finite model establishes consistency of the candidate clauses with their stated choice principle, while identifying exactly which meaning of quantification makes that claim true. It does not settle a semantics in which quantifiers range only over singleton candidate sets.

52.4 A proposed comprehension scheme

The class-existence principle motivates a syntactic operation setify on formulas. Its intended role is to restrict the relevant variables and quantifiers to sets. A recursive definition must state how it handles binders and free class parameters; those clauses are not yet given in this draft.
The proposed characterization was

$$(\varphi \in \mathrm{setify}[\mathrm{Fm}]) \overset{meta}{\Longleftrightarrow} (\vdash (\{\, x \mid \varphi \,\}\downarrow)).$$

The right-hand side is a claim of provability, so the equivalence requires a specified calculus as well as semantics. In particular, the reverse direction says that the syntactic transformation captures *every* provably defined comprehension, which is stronger than a sufficient formation condition. The display records that intended characterization; its two directions remain separate proof obligations.

52.5 A precise quantifier restriction in the candidate model

The following construction isolates one part of the preceding proposal. It specifies a restriction of bound variables in the finite candidate model; it does not identify that restriction with the intended setify operation or interpret comprehension.
Fix the earlier finite model with semantic domain $B = \mathcal{P}(D)$. Work first in its descriptor-free relational language: terms are variables and constants c_S, and atomic formulas use candidate equality, the lifted relation $R^{\forall}$, and candidate presence. Choose any subset $E \subseteq B$ of allowed bound-variable values. Expand the language by a fresh unary predicate G_E, interpreted by

$$M, a \vDash G_E(t) \quad\Longleftrightarrow\quad [\![t]\!]_a \in E.$$

This is an explicitly supplied predicate interpretation, not an assertion that E is definable in the original language.
Define $M, a \vDash_E \varphi$ by keeping the existing atomic and Boolean clauses, but letting every quantifier range over E. For example,

$$M, a \vDash_E \forall x\, \psi \quad \Longleftrightarrow \quad \forall S \in E,\ M, a[x \mapsto S] \vDash_E \psi.$$

Assignments and constants still have values in B; in particular, free parameters are not silently restricted to E. This is a semantics with restricted bound-variable ranges, not necessarily an induced substructure with domain E.
Define a translation $\varphi \mapsto \varphi^E$ recursively. It fixes atomic formulas, commutes with Boolean connectives, and has the binder clauses

$$(\forall x\, \psi)^E = \forall x\, (G_E(x) \longrightarrow \psi^E),$$
$$(\exists x\, \psi)^E = \exists x\, (G_E(x) \wedge \psi^E).$$

The superscript denotes a syntactic translation using the new predicate symbol. Since that symbol has no free parameters, its inserted occurrence introduces no new free variable and captures none from ψ.

Proposition 52.3 (Agreement for restricted quantifiers)**.**
For every descriptor-free formula φ and every assignment $a : \mathrm{Var} \to B$,

$$M, a \vDash_E \varphi \quad \Longleftrightarrow \quad M, a \vDash \varphi^E.$$

This includes $E = \varnothing$, with universal quantifiers vacuous and existential quantifiers false in the restricted interpretation.

Proof.
Induct on the formula. Atoms have identical term values and identical interpretations on the two sides, so the assertion holds for atoms. Each Boolean step follows by applying its ordinary two-valued truth operation to the equivalences supplied by induction.
For a universal quantifier, the left side means that ψ holds in the restricted interpretation under every update $a[x \mapsto S]$ with $S \in E$. By induction this is equivalent to truth of ψ^E under exactly those updates. In the original domain B, requiring $G_E(x) \longrightarrow \psi^E$ at every update imposes the same condition: updates outside E satisfy the implication vacuously. For an existential quantifier, a witness update in E satisfies ψ in the restricted interpretation exactly when it satisfies ψ^E by induction. This is precisely a witness in B for $G_E(x) \wedge \psi^E$. These arguments also handle $E = \varnothing$, completing the induction. □

Different restrictions give different theories. Take $D = \{0, 1\}$ with $0 \neq 1$. Three possible ranges are

$$E_{\mathrm{all}} = \mathcal{P}(D), \qquad E_{\mathrm{present}} = \mathcal{P}(D) \setminus \{\varnothing\}, \qquad E_{\mathrm{single}} = \{\{0\}, \{1\}\}.$$

The sentence $\forall x\, (x \downarrow)$ is false for E_{all} (the empty candidate set is a witness to failure) and true for E_{present} and E_{single}. The sentence

$$\forall x\, (x = c_{\{0\}} \vee x = c_{\{1\}})$$

is true for E_{single} and false for E_{present}, since $\{0, 1\}$ is not equal to either singleton candidate set. Thus ranging over arbitrary candidate collections, nonempty candidate collections, and single-valued candidates are genuinely different interpretations even when every underlying object in D is a set. None is selected by the word "set" alone in the original proposal.

Why descriptors and comprehension remain separate. The lemma deliberately contains no descriptor term. To extend it, one must decide whether an ε binder also restricts its witnesses to E, and what happens when there is no such witness. The previous finite choice model returns the empty candidate set for $\varepsilon x, \bot$. This lies outside both E_{present} and E_{single}. Therefore a semantics whose term values must stay inside either restricted domain cannot retain that default unchanged. One can instead keep the ambient value domain B while restricting only selected quantifiers, or specify a different descriptor/default policy; those are additional choices, not consequences of relativization.

Likewise, the chapter supplies no semantic clause for a comprehension term $\{\, x \mid \varphi \,\}$ and no calculus for its presence assertions. The proved translation lemma therefore does not establish either direction of the proposed provability characterization of setify[Fm]. It identifies a binder mechanism that can be adopted once the intended range, parameter policy, and term constructors are fixed, while leaving that original design question open.

53 [OUTDATED] Variants of Iota Definitions

These notes investigate two related questions: how a relation can represent a family with class-valued members, and how a definite descriptor can signal that no unique set witness exists. The subsections record successive local conventions, rather than cumulative definitions of one evaluation operation. In particular, union evaluation, protected-family evaluation, and the intersection-based iota descriptor have different default behavior.
The symbol ! means at-most-one uniqueness. Class comprehension collects set witnesses, even when the surrounding variables range over classes. In the final construction, $\bigcap \varnothing = V$ supplies a proper-class default, so that sethood detects successful description. That convention belongs to this construction and is independent of the partial-logic treatment elsewhere in the book.

53.1 First Attempt: Evaluating Relations and Families

The initial proposal compares several encodings. Its central difficulty is preserving an index whose intended value is empty: a representation made only of members of that value loses the index when there are no members.

Definition 53.1 (value of relation/function/protected family at x)**.**

$$F(x) \coloneqq \bigcup \{\, b \mid (x, b) \in F \,\}$$

Definition 53.2 (value of a simple family at x)**.**

$$F(x) \coloneqq \{\, b \mid (x, b) \in F \,\}$$

This is the fibre at x, whereas $\operatorname{ran}(F)$ collects second components at all indices. With the ordinary relational domain, every represented fibre is nonempty. Declaring every set to be an index permits empty fibres, but then the domain no longer records which indices were intended. The following encoding introduces a separate record for an empty value. This distinction also matters in the newer Obelisk catalog: `DefValueAtArg` returns V on an absent fibre, whereas the unconditional union convention in this subsection returns $\varnothing$. The two conventions agree when a record exists. A theorem transferring between them must therefore state its domain hypothesis.

Definition 53.3 (value of a protected family at x)**.**

$$F(x)_p \coloneqq \{\, b \mid (x, \{b\}) \in F \,\} = \bigcup \{\, s \mid (x, s) \in F \,\}$$

For a protected family, the second components are empty sets or singletons, so the two displayed expressions agree. The union expression motivates a common evaluation notation for the encodings under consideration.

Definition 53.4 (Domain of relation/function/family/protected family)**.**

$$\operatorname{Dom}(F) \coloneqq \{\, x \mid \exists s \in V,\ (x, s) \in F \,\} =: \operatorname{dom}(F)$$

Definition 53.5 (Range of relation)**.**
The relational range collects the stored second components. It can differ from the collection of values obtained by evaluating the family.

$$\operatorname{ran}(R) \coloneqq \{\, y \mid \exists x \in V,\ (x, y) \in R \,\}$$

Definition 53.6 (in range of relation/function/family)**.**
This predicate describes evaluated values, including proper classes. It does not collect proper classes as members of an object-theoretic class.

$$(K \text{ in classrange of } F) \iff \exists x,\ (x \in \mathrm{Dom}(F)) \wedge (F(x) = K)$$

Definition 53.7 (in range of protected family)**.**
This predicate describes evaluated values, including proper classes. It does not collect proper classes as members of an object-theoretic class.

$$(K \text{ in classrange of protected } F) \iff \exists x,\ (x \in \mathrm{Dom}(F)) \wedge (F(x)_p = K)$$

Definition 53.8 ((Set–)Range of function/family. "F–Ran","Ran", "big range")**.**

$$\mathrm{Ran}(F) \coloneqq \{\, y \mid y \text{ in classrange of } F \,\}$$

Definition 53.9 (function)**.**
A function is a relation which is univocal.

Definition 53.10 (protected family)**.**
A protected family is a relation whose second components are subsets of singletons, hence either empty sets or singletons.

Definition 53.11 (r/f/F is defined at x)**.**

$$(F \text{ defined at } x) \iff (x \in \mathrm{dom}(F))$$

A witness property for $(\varepsilon x, \varphi)$ requires an existence hypothesis; a unique-witness property for $(\iota x, \varphi)$ requires existence and uniqueness. A total interpretation can still assign values when these hypotheses fail, using defaults D_1 and D_2 respectively.
A default that is also a legitimate output cannot, by its value alone, distinguish a failed description from a successful one. For example, if D_1 is the empty class, an empty output need not indicate failure. A success predicate therefore needs information about the construction or its context, unless the default lies outside the intended output range. This motivates recording a validity condition alongside the value.
The relational inverse always exists as a relation. The protected-family encoding is asymmetric, however: reversing its pairs need not produce another protected family. Thus inversion of a representation and inversion within a chosen representation format are different questions.

53.2 Nonempty Values and Relational Domains

For the next comparison, fix an injective ordered-pair encoding of sets, and let $F \subseteq V \times V$ be a class relation. Write

$$U_F(x) = \bigcup \{\, y \mid (x, y) \in F \,\}$$

for its unconditional union value, to distinguish this evaluation from the fibre and descriptor conventions elsewhere. The domain still means the relational domain: $x \in \mathrm{Dom}(F)$ if some record (x, y) occurs in F.

Definition 53.12 (Relation with nonempty evaluated values)**.**
In this encoding, F is a family of nonempty values when

$$\forall x \in \mathrm{Dom}(F), \quad U_F(x) \neq \varnothing.$$

A represented fibre may be nonempty while its evaluated value is empty. For instance, the one-record relation $\{(x, \varnothing)\}$ has domain $\{x\}$ but value $\varnothing$ at x. This is exactly the case that an encoding of nonempty values alone cannot retain.

53.3 Canonical Representations of Functions and Families

Definition 53.13 (conglomerate of all classes)**.**
Here $\mathcal{U}$ denotes the conglomerate of all classes in the metatheory. It is not a class-denoting term of the local theory (MK with the proposed universe axiom).

The following candidate canonicalizations encode membership using singleton records. Their treatment of empty values distinguishes the variants:

Definition 53.14 (Canonicalization of a family of nonempty classes)**.**

$$\text{canonize0}(F) = \{\, (x, \{q\}) \mid \exists y \in V,\ ((x,y) \in F) \wedge (q \in y) \,\}$$

Definition 53.15 (Canonicalization allowing empty values)**.**

$$\text{canonize1}(F) = \{\, (x, \{q\}) \mid \exists y \in V,\ ((x,y) \in F) \wedge (q \in y) \,\} \cup \{\, (x, \varnothing) \mid x \in \mathrm{Dom}(F) \,\}$$

$$\begin{aligned} \text{canonize2}(F) = {} & \{\, (x, \{q\}) \mid \exists y \in V,\ ((x,y) \in F) \wedge (q \in y) \,\} \\ & \cup \{\, (x, \varnothing) \mid (x \in \mathrm{Dom}(F)) \wedge (F(x) = \varnothing) \,\}. \end{aligned}$$

$$\text{canonize3}(F) = \{\, (x, s) \mid \exists y \in V,\ ((x,y) \in F) \wedge (s \subseteq y) \wedge (!q,\ q \in s) \,\}$$

In these formulas, $F(x)$ means $U_F(x)$, and all indices and stored components are sets. Singleton formation and replacement ensure that the new records are sets, so class comprehension supplies the displayed relations. Write $C_i(F) = \text{canonize}i(F)$ for $i = 0, 1, 2, 3$.

Theorem 53.1 (Values and domains of the canonicalizations)**.**
For every set x and each $i = 0, 1, 2, 3$,

$$U_{C_i(F)}(x) = U_F(x).$$

Moreover,

$$\mathrm{Dom}(C_0(F)) = \{\, x \in \mathrm{Dom}(F) \mid U_F(x) \neq \varnothing \,\},$$

whereas $\mathrm{Dom}(C_i(F)) = \mathrm{Dom}(F)$ for $i = 1, 2, 3$. In fact $C_1(F) = C_3(F)$.

Proof.
Put $B = U_F(x)$. The nonempty records at x in $C_0(F)$ are precisely $(x, \{q\})$ for $q \in B$: membership in the union means that q lies in some original component y. The union of those singleton components is B, and a record exists exactly when B is nonempty.
The first variant adds the record $(x, \varnothing)$ at every original domain point. This does not change the union and restores precisely the indices lost by C_0. The second adds it only when $B = \varnothing$; when $B \neq \varnothing$, a singleton record already retains the index. Thus both variants preserve the original domain and all values.
For the third variant, an at-most-one set $s \subseteq y$ is either empty or a singleton $\{q\}$ with $q \in y$. A singleton record therefore occurs exactly when $q \in B$, and an empty record occurs exactly when some original component y exists. These are precisely the records of $C_1(F)$. □

Corollary 53.1 (Idempotence and the remaining representational choice)**.**
Each C_i is idempotent: $C_i(C_i(F)) = C_i(F)$. The canonicalizations C_1 and C_2 have the same values and domain, but need not be equal.

Proof.
The records of $C_0(F)$ are determined solely by the values $U_F(x)$, which C_0 preserves. The records of $C_1(F)$ and $C_2(F)$ are determined by those values together with the domain, both of which they preserve. This proves idempotence for the first three maps, and $C_3 = C_1$ gives the fourth case.
If $U_F(x) \neq \varnothing$, $C_1(F)$ contains $(x, \varnothing)$ while $C_2(F)$ does not. For example, take $F = \{(x, \{q\})\}$ for sets x, q. Both canonicalizations evaluate to $\{q\}$ at x, but only the first stores the empty marker there. □
The two conventions therefore solve the same value-and-domain problem with different normal forms. Choosing one is a representation decision, not a change to the represented class-valued family.

Theorem 53.2 (Complete normal forms for values with a domain)**.**
For class relations $F, G \subseteq V \times V$, the following are equivalent:

1. $\mathrm{Dom}(F) = \mathrm{Dom}(G)$ and $U_F(x) = U_G(x)$ for every set x;
2. $C_1(F) = C_1(G)$;
3. $C_2(F) = C_2(G)$.

By contrast, $C_0(F) = C_0(G)$ is equivalent just to equality of all union values, without equality of the original domains.

Proof.
Equal values determine the same singleton records. Equal domains then determine the same empty records for C_1, and, together with the values, for C_2. This proves the first condition implies both others. Conversely, equality of either canonicalization implies equality of its domain and union values, which are those of the original relation by the preceding theorem. For C_0, its only records are the singletons of members of the union values, and it preserves those values; hence each direction of the last equivalence follows too. □
Thus C_1 and C_2 each select exactly one representative of a family specified by its domain and values. This is an assertion about the representations, not a claim that a quotient collection of all such class families is an object of the theory.
Here "select" does not require a choice function on arbitrary collections of representatives. Each $C_i(F)$ is defined explicitly from the records of F, and the theorem proves that it is independent of which relation with the same domain and values was supplied. Neither a well-order of the universe nor a choice of one member from each nonempty value is used. Indeed, all members of a value contribute singleton records. This explains why canonicalizing a family here is a different operation from choosing a witness in the finite candidate model of the preceding section.

Theorem 53.3 (Encoding prescribed membership data)**.**
Let A be a class of set indices and $E \subseteq A \times V$ a class relation describing the intended members of each value. Put

$$E_x = \{q : (x, q) \in E\}, \qquad P = \{(x, \{q\}) : (x, q) \in E\} \cup \{(x, \varnothing) : x \in A\}.$$

Then P is a protected family with domain A, with $U_P(x) = E_x$ for every set x, and $C_1(P) = P$. It is the unique C_1-fixed relation with this domain and these values.

Proof.
Every record has a set index in A and an empty or singleton component. The empty record ensures that each $x \in A$ is in the domain, and no record has an index outside A. The members of the union value at x are exactly the q for which the singleton record $(x, \{q\})$ is present, namely the members of E_x. Its records are already precisely those prescribed by C_1, so P is fixed. If Q is another fixed relation with the same data, the normal-form theorem gives $Q = C_1(Q) = C_1(P) = P$. □
The data E are important: an arbitrary metatheoretic assignment of a class to every index need not be definable by a class relation in the chosen theory. The theorem assumes such a relation is supplied; it does not assert unrestricted class-valued replacement. It does allow proper class values. For example, for a set a, take $A = \{a\}$ and $E = \{(a, q) : q \in V\}$. Then $U_P(a) = V$, although every individual record of P is a set.
The next definition describes an indexed relation before a canonical representation has been chosen.

Definition 53.16 (family)**.**

$$(F : A \to \mathcal{U}) \iff (F \subseteq V \times V) \wedge (\mathrm{Dom}(F) = A)$$

Definition 53.17 (canonical family)**.**
One can interpret a family through C_1 or C_2, or require its representative to be fixed by the chosen canonicalization. The theorem above justifies preservation of values and domains, and idempotence ensures that applying the same convention twice makes no further change.

Definition 53.18 (family)**.**
In this alternative convention, a family is already a protected relation. Its permitted second components form the class

$$\mathsf{SS} = \{\varnothing\} \cup \{\, \{q\} \mid q \in V \,\},$$

consisting of the empty set and all singletons of sets. Equivalently, these are exactly the set subsets of singletons. With this explicit interpretation of the earlier schematic range notation $\{-\}[\,]$, put

$$(F \text{ family}) \iff (F \subseteq V \times \mathsf{SS}).$$

These representations have different scopes. A class relation can encode set-indexed families with proper-class values through their members, whereas a function symbol may also take proper classes as arguments. An ordinary class function has set values because its graph stores ordered pairs of sets.
The singleton notation raises a related question: how should $\{V\}$ be interpreted when proper classes cannot be elements? An empty-class default can signal failure of singleton formation, since a successfully formed singleton is nonempty. That reasoning does not transfer directly to function evaluation, where the empty class may be an ordinary value. The relevant criterion is whether the default is distinguishable from successful outputs.

53.4 Value of a function via set–descriptors

This subsection uses the intersection-based set descriptor defined below. Throughout this subsection assume that V is proper and that $\bigcap \varnothing = V$, as in that construction. Its unsuccessful value is therefore not a set, so a set-valued result certifies successful description. This is different from the earlier union default.

Definition 53.19 (Descriptor value of a relation)**.**
Let F be an arbitrary class.

$$F(x) = (\iota y,\ \langle x, y\rangle \in F) \tag{def:value}$$

Definition 53.20 (univocal at)**.**

$$(F \text{ univocal at } x) \longleftrightarrow (!y, \langle x, y\rangle \in F)$$

Definition 53.21 (univocal)**.**

$$(F \text{ univocal}) \longleftrightarrow (\forall x, F \text{ univocal at } x)$$

Theorem 53.4 (pair in function)**.**
If $(F \text{ univocal at } x) \wedge (y \text{ set})$, then

$$(F(x) = y) \longleftrightarrow (\langle x, y\rangle \in F)$$

$$\begin{aligned}(F \text{ univocal at } x) \wedge (y \text{ set}) &\longrightarrow ((\langle x, y\rangle \in F) \longrightarrow (F(x) = y))\\ (y \text{ set}) &\longrightarrow ((F(x) = y) \longrightarrow (\langle x, y\rangle \in F))\end{aligned}$$

Proof.
Write $P(z)$ for $\langle x, z\rangle \in F$, choosing z, w fresh for the fixed parameters. Use the final set-descriptor definition:

$$Q = \{z : P(z) \wedge (!(w \text{ set}), P(w))\}, \qquad F(x) = \bigcap Q.$$

The uniqueness guard ranges over sets; comprehension likewise retains only set witnesses. The variable w is independent of the outer comprehension variable z.
First assume $P(y)$, y set, and F univocal at x. Local univocality implies that at most one set satisfies P, and y supplies such a set witness. Thus $Q = \{y\}$. The singleton-intersection identity gives $F(x) = \bigcap\{y\} = y$, proving the forward membership-to-value implication.
For the converse assume only $F(x) = y$ and y set; no univocality hypothesis is used. If Q were empty, then $F(x) = V$, contrary to sethood of y and properness of V. Choose $a \in Q$. Comprehension gives a set, $P(a)$, and at most one set satisfying P. Every member of Q is such a set, so $Q = \{a\}$. Hence $y = F(x) = \bigcap Q = a$, and equality substitution gives $P(y)$, which is $\langle x, y\rangle \in F$. This proves the separately stated reverse implication with only its sethood hypothesis. □

Theorem 53.5 (Witness property of a defined set descriptor)**.**

$$((\iota y,\ \varphi) \text{ defined}) \longrightarrow \varphi[(\iota y,\ \varphi)/y]$$

Proof.
Use the local convention that a set descriptor is defined exactly when its value is a set. Put

$$Q = \{\, y \mid \varphi \wedge (!(y \text{ set}), \varphi)\,\}.$$

Its intersection is the value of the descriptor. If Q were empty, this value would be V, which is a proper class. Definedness therefore implies that Q is nonempty. Its uniqueness condition makes Q a singleton, say $Q = \{a\}$. Then $\bigcap Q = a$, and membership of a in Q gives the required instance of φ. Substitution is understood to avoid capture of free variables. □

Remark 53.1 (Only uniqueness among sets is needed)**.**
In the membership-to-value implication, the stronger assumption F univocal at x can be replaced by $!(w \text{ set}),\ \langle x, w\rangle \in F$. The proof uses exactly this set-restricted uniqueness together with the given set witness y. Even if the predicate also holds of a proper class, that additional object is not collected by Q and does not affect the descriptor. The reverse implication instead uses sethood of the result to exclude V; it needs neither uniqueness assumption in advance and would not be valid with an arbitrary set default.

53.5 Earlier Set-Descriptor Axioms

Axiom 26 (Choice witness)**.**

$$\exists(x \text{ set}),\ \varphi \longrightarrow (\varepsilon x,\ \varphi) \in \{\, x \mid \varphi \,\}$$

Axiom 27 (Sethood signals existence)**.**

$$(\varepsilon x,\ \varphi) \text{ set} \longrightarrow \exists(x \text{ set}),\ \varphi$$

Axiom 28 (Unique description witness)**.**

$$\exists!(x \text{ set}),\ \varphi \longrightarrow (\iota x,\ \varphi) \in \{\, x \mid \varphi \,\}$$

Axiom 29 (Sethood signals unique existence)**.**

$$(\iota x,\ \varphi) \text{ set} \longrightarrow \exists!(x \text{ set}),\ \varphi$$

The next construction asks whether an intersection can describe a unique class, including a proper class:

$$(Iy, \varphi) = \bigcap \{\, y \mid \varphi \wedge (!y, \varphi) \,\}.$$

The obstacle is local to this encoding: class comprehension collects only set witnesses. If the unique satisfying object is a proper class, it is not collected. Moreover, uniqueness among all classes and uniqueness among sets are different conditions when φ has witnesses of both kinds. The construction below therefore restricts uniqueness to set witnesses. A descriptor for arbitrary classes would require a different construction or an explicitly enlarged semantic setting.

Definition 53.22 (iota operator for sets)**.**

$$(\iota y,\ \varphi) = \bigcap \{\, y \mid \varphi \wedge (!(y \text{ set}),\ \varphi) \,\}$$

Theorem 53.6 (Sethood characterizes unique set description)**.**

$$((\iota y,\ \varphi) \text{ set}) \longleftrightarrow (\exists!(y \text{ set}),\ \varphi)$$

Proof.
Put $Q = \{\, y \mid \varphi \wedge (!(y \text{ set}), \varphi) \,\}$. By definition, $\iota y, \varphi = \bigcap Q$.
If there is exactly one set witness a, the uniqueness condition holds and $Q = \{a\}$. The singleton-intersection identity gives $\bigcap Q = a$, which is a set.

Conversely, suppose $\bigcap Q$ is a set. Since $\bigcap \varnothing = V$ and V is proper, Q cannot be empty. Choose $a \in Q$. By comprehension, a is a set satisfying φ, and the second conjunct in the definition of Q states that there is at most one such set. Thus there is exactly one set witness. □

Corollary 53.2 (The successful value and the default)**.**
If a is the unique set satisfying φ, then $(\iota y, \varphi) = a$. If there is no unique set satisfying φ, then $(\iota y, \varphi) = V$.

Proof.
The successful case was proved above. In the other case, a member of Q would supply both existence and uniqueness, a contradiction. Hence $Q = \varnothing$ and its intersection is V. □
For example, a predicate with the empty set as its unique set witness returns $\varnothing$, whereas a predicate with no set witness returns V. If a predicate holds of exactly one set and also of a proper class, this set-restricted descriptor still returns the set. This example explains why its uniqueness quantifier must explicitly range over sets: it solves the set-description problem, not the different problem of selecting a unique class witness.

54 [OUTDATED] Proof assistant based on the developed theory

This section preserves an earlier presentation of the proof assistant and illustrates its intended use of contexts. A sequent $\Gamma \vdash \varphi$ means that φ is derivable from the assumptions listed in Γ. The proof trees below explain ordinary structural and logical steps; they do not assert that the displayed source listing verifies every theorem in the book.

54.1 The encoded theory

The accompanying Standard ML listing records the local implementation. Its definitions and proof constructors must be read with the assumptions of that implementation.

```
(*
 * Documentation update: openrouter/nvidia/nemotron-3.5-lightning:free (Capability Tier 1) PRESENTATION_ML_GUIDE.
     md
 * Ticket: ticket_7192_zone_delegated.md
 *
 * This file presents the underlying theory of Class-Theoretic Foundations based on Morse-Kelley class theory and
      Tarski's axiom.
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 * SML pseudo-code documentation:
 * - MK + Tarski's axiom: Foundations with Morse-Kelley class theory and Tarski-style universe axioms
 * - Sets : Term: Universal class of sets (proper classes are not members)
 * - c : Term -> Term: "Super global choice" descriptor/choice operator
 * - Compr : Bound Fm -> Term: Impredicative class comprehension {x | phi}
 * - Dunaev Lam x.t: Big lambda forming class-families/graphs, not SML functions
 * - Complexity weights (compl. = n): Provisional syntactic weights, not runtime costs
 *
 * Ticket 7436 audit: arbitrary-class pairing uses Dunaev tags and semi-singleton
 * subsets; no Kuratowski encoding of proper classes is introduced.
 * Verified by opencode-go/kimi-k3.1-pro (Capability Tier 2).
 *
*)
(*
 * File: presentation.ML
 * Author: mistral/codestral-2508 (Capability Tier 2)
 * Description: This file presents the underlying theory of Class-Theoretic Foundations based on Morse-Kelley
     class theory and Tarski's axiom.
 * It includes constructors of terms and formulas, complexity annotations, and definitions.
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 * Complete Presentation of underlying theory of Class-Theoretic Foundations:
Morse-Kelley class theory + Tarski's axiom

Syntax:
a,b,c,...,x,y,z denote variables and terms; phi,psi,... denote formulas
phi[a/x] denote the substitution.
(only for cases when a is free for x in phi("ffi a x phi"))
Technical axioms for substitutions of binders are ommited:
e.g. {x|phi}={w|phi[w/x]} (ffi w x phi).

Constructors of terms and formulas:
Sets : Term
c : Term -> Term (* "super global choice" *)
 _ : Var -> Term
-[-/-] : Var * Term * Fm -> Fm (*order: x,t,phi for phi[t/x]; raises exception if subst isn't possible)*)
Bot : Fm (* falsehood *)
(- <-> -), (- & -), (- \/ -), (- -> -) : Fm * Fm -> Fm
forall, exists : Var * Fm -> Fm
{-|-} : Var * Fm -> Term
{-,-}, (-,-): Term * Term -> Term
{-}, pow(-), dom(-), ran(-), union(-), inter(-) : Term -> Term
(- in -),(- = -), (- C -),(- from -),(- onto -),(- smaller than -) : Term * Term -> Fm
(- func),(- transitive),(- hereditary),(- powerful),(- limited),(- universe) : Term -> Fm
(!-:-): Var * Fm -> Fm

Todo: measure the complexity of the definition in depth of amount of constructors/connectives
or in... ?
Definitions: (constructors of type Prf; rules of the 1st order logic are ommited)
{t|phi, x bound} := {z| exists x : t = z /\ phi } (* z is not free neither in t not in phi *)
                 = Class (z. (exists x : t = z /\ phi))
Compr : Bound Fm -> Term

   { } = {x|Bot} (* compl. = 2 *)
    V = {x|Top}
(A x B) := {(a,b)| (a in A) /\ (b in B)} (* compl. = 4 *)
(F relation) :<-> (F C (V x V)) (* compl. = 2 *)
(A C B) <-> (forall x: x in A -> x in B) (* :Var*Term*Term->Prf *)
(**s"A C B" means "A is a subclass of B"**)

(if (X empty) then V else union X)
{t|phi, }
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{z|phi} = Class (z.phi)

(unite x.t) := {(union t)| (x any) /\ (t inhabited)} (*x is a bound variable for both sides*)
            = {z| exists x: (z=(union t))/\(t inhabited) }
            = {z| exists x': (z=(union t[x'/x]))/\(t inhabited) }
            = {z| exists x': (z=(union (x.t)(x') ))/\(t inhabited) }

W = {w|(w inhabited)}
"x.t" is surely a bounded term

(* s.phi := x. x=s /\ phi[s/x] ? *)

(Lam x t) = {(x,s)|(s sss t)}
(Lam x.t) = {(x,s)|(s sss t)}
          = {z| exists x: ((z=(x,s)) /\ (s sss t))}
          = Class z.(exists x.((z=(x,s)) /\ (s sss t)) )
F(x) := (union {s|(x,s) in F})

(pow B) := {A|(A C B)} (* :Var*Term->Prf *) (* compl. = 2 *)
(dom R) := {x|exists s: (x,s) in R} (* compl. = 3 *)
(ran R) := {y|exists x: (x,y) in R} (* compl. = 3 *)
(Dom F) := dom(F)
(Ran F) := { F(x) | (x in Dom(F)) }
         = Repl (x |-> F(x)) Dom(F)
         = F[Dom F]
         = (unite {s|(x,s) in F} for x)
 ?
F|A = F n fst^{-1}[A]


  F|A := {(a,s) in F | a in A}

 F[A] := Ran F|A
dom F := fst[F]

union : Term -> Term

 [-]: Term -> ((Term -> Term) -> Term)
-[Q]: (Term -> Term) -> Term

U:Term -> Term
"U[Q]":= {union m| m in Q}

up : Term -> (Term -> Term)
  (up F) := (fn x => F(x))

down : (Term -> Term) -> Term
(down S) := { (x, S(x)) | x any }

up_opt : Term -> (Term -> Prf -> (Term option))
  (up F) := (fn x => if (x in (dom F)) then SOME F(x) else NONE)

down_opt : (Term -> (Term option)) -> Term
(down S) := { (x,z) | (x any) /\ (case S(x) of SOME y => z=y ; NONE => Bot) }

"-[-]" : (Term -> Term) -> Term -> Term

"(F |--> (x |--> F(x))" : Term -> (Term -> Term)


F|A = F n (A x V)
F[A] = (Ran (F|A))
Ran F|(A n dom(F))

   F/A/ := {F(a) | (a in A)}
(Ran F) := F/dom(F)/
   F[A] := Ran F|A

   F[A] := {F(a) | (a in A)/\(a in dom(F))}
   F[A] := {F(a) | (a in (A n dom(F)))}
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  F[A] := {F(a) | (a in A)/\(a in dom(F))}

(Ran F) := { F(x) | (x in dom(F)) }
(Ran F) := F[V]
(Ran F) := F[dom(F)]

(Ran F) := {y|exists (x in dom(F)): F(x) = y} (* compl. = 4 *)
(Ran F) := {y|exists x: (exists s: (x,s) in F)/\ F(x) = y} (* compl. = 4 *)

(Ran F) := {y|exists x: (exists s: (x,s) in F) /\ (union {s|(x,s) in F} = y)} (* compl. = 4 *)
(Ran F) := {(union m)|exists x: (exists s: (x,s) in F) /\ ({s|(x,s) in F} = m)}
(Ran F) := {(union m)|exists x: (m inhabited) /\ ({s|(x,s) in F} = m)}
(Ran F) := {(union {s|(x,s) in F})| (x any) /\ ({s|(x,s) in F} inhabited)}
(qqq t) := {(union t)| (x any) /\ (t inhabited)}
(Ran F) := unite_inhabited ( x.{s|(x,s) in F} )


(m inhabited) <-> (exists q: q in m) (* compl. = 2 *)
F[A] = {s| exists (a in A): (a,s) in F}
F[{x}] = {s|(x,s) in F}

(F from A) <-> (dom(F) = A) (* :Term*Term->Prf *)
(F onto B) <-> (ran(F) = B) (* :Term*Term->Prf *)
(not phi) <-> (phi -> Bot) (* negation *)
(z smaller than U) <-> (not (exists f: (f func) & (f from z) & (f onto U))) (* :Var*Term*Term->Prf *)
(U transitive) <-> (forall (B in U): (B C U)) (* :Var*Term->Prf *) (*B not free in U*)
(U hereditary) <-> (forall (B in U): forall (X C B): (X in U)) (* :Var*Var*Term->Prf *)
(U powerful) <-> (forall (B in U): ((pow B) in U)) (* :Var*Term->Prf *)
(U limited) <-> (forall (z C U): ((z smaller than U) <-> (z in U))) (* :Var*Term->Prf *)
(U universe) <-> (U transitive, hereditary, powerful, limited) (* :Term->Prf *)
SetUniverses = {U|U universe} (* :Var->Prf *)
(union X) = {z|exists (y in X): z in y} (*:Var*Var*Term -> Prf*) (*order: z,y,X*)

Axioms: (constructors of type Prf) (* first 4 are given axiomatically, but they can be defined *)
 ({W|phi} exists)
 (A C B) & (B C A) -> (A = B)
 forall A, B: (A = B) -> (phi[A/v] -> phi[B/v])
 forall W: ((phi) & (W in Sets)) <-> ((W in {W|phi})) (*:Var*Fm->Prf*)
 ((phi) & (W in Sets)) <- ((W in {W|phi}) & (W exists)) (*:Var*Fm->Prf*)
 ((phi) & (W in Sets)) -> ((W in {W|phi}) ) (*:Var*Fm->Prf*)
 ((pow K) C K) -> (Sets C K) (*:Term->Prf*)
 (Sets universe) (*:Prf*)
 Sets C (union SetUniverses) (*:Prf*)

Logical notes:
1) (A = B) -> (A unique).
2) (exists B:(A = B)) <-> (A unique).
 ((phi) & (W in Sets) & (W exists)) <-> ((W in {W|phi}) & (W exists)) (*:Var*Fm->Prf*)

Def. ("s is a semi-singleton subset of x")
 (s sss x) <-> (s C x) /\ (s ss)

Axioms for constructions:
Sometimes objects can be defined by explicit term, but often they are working only in certain
conditions, and we don't want the details, and keep focus on important properties.
The main idea is that you cannot say anything about term, if it is unspecified.
# Singleton (3 equivalent variants.)
(
intro: (A set) -> (A in {A})
elim: (A set) -> ((B in {A}) -> (B = A))
,
(A set) -> ((B in {A}) -> (B = A))
(A set) -> ((B = A) -> (B in {A}))
,
(A set) -> ((B in {A}) <-> (B = A))
)

(A set) -> ({A} set)
Projection class-functions:
  fst := {z|exists x,y: z = ((x,y),x)}
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  snd := {z|exists x,y: z = ((x,y),y)}
# Composition ?
# union of two classes
  A u B := {x|(x in A) \/ (x in B) }

# Ordered pair of sets.
  forall (a1, b1, a2, b2 sets): (((a1, b1) = (a2, b2)) <-> ((a1 = a2) /\ (b1 = b2)))
  forall (a, b sets): ((a, b) set)

  ([a, b] sets) -> ((a, b) set)
# Family constructor (big lambda)
  (x |-> t) := {(x,s) | (s C t) /\ (s ss)} := {(x,s)|(s sss t)}
  (both x and s are bound by class comprehension on r.h.s., variable s is not free in t)
# Identity:
  Function: id = {(x,x) | x any}
  Family: (note: here Id = {(x,s) | (s C x) /\ (s ss)}) ("sic! preimage of the inverse of sss relation class")
          Id := (x |-> x). (*GOOD for family! "(Lambda x x)" *)
# (Binary) Product of families of classes:
  Classes: A x B := {(a,b)|(a in A)/\(b in B)}
  Families: F x G := ...
  it implies that there is an ordered pair of proper classes. E.g. [F(a), G(a)]. ... sth is shady

# Binary product of functions
  For usual functions:
  (f1 times f2) := {(x,(y1,y2)) | ((x,y1) in f1) /\ ((x,y2) in f2)}
  For families(WIP):
  (F1 times F2) := {(x,[y1,y2]) | (F1(x)=y1) /\ ((F2(x)=y2))}
  (F1 times F2) := {(x,s) | (x in dom(F1)) /\ (dom(F1)=dom(F2)) /\ (s ss) /\ (s C [y1,y2]) /\ (F1(x)=y1) /\ (F2(x)
      =y2)}
  (F times G) := (x |-> [F(x),G(x)])
  For any classes A and B,
  [A, B] := {(1,s)|(s sss A)}u{(0,s)|(s sss B)} (*Note the reversed order! It is correct.*)
         = (x |-> A)|`{1} u (x |-> B)|`{0}
# Lists of classes(!).
  Intended use:
   "([a,b,c] are sets) -> ..."
  Constructors: (nil:Term), (cons:Term->Term)
  Notation:
       :: := cons (infix op)
       [] := nil
      [c] := c::nil
    [b,c] := b::[c]
  [a,b,c] := a::[b,c]
  "(h :: t) increases first elements of pairs in t by one(1+alpha), and adds {(0,{q})|q in t}."

  Semantics:
      nil := emptyclass
  (h :: t) := {(0,{q})|q in h} u ((1+_) times Id)[t]
  (h :: t) := {(0,{q})|q in h} u {(1+n,x)|(n,x) in t}
(*" 'n+1':= n u {n} "
    '1+n':=?
  By Transfinite recursion:
    1+0 := 1
    1+(q+1) = (1+q)+1
    (1+alpha) := sup {1+m| m < alpha}. (if n is a limit ordinal)
          = if (alpha non-zero limit ordinal) then alpha else (alpha+1).
          note: Ord is a limit ordinal proper class.
# Transfinite recursion (yet for functions only, which is enough for the previous application)

    G:V^On -> V ?
    K={f|in V^On| \forall x in (dom f): f(x)=G(f|`x)}
    F:=(union K)
    1) (F mapping), (dom(F) = On)
    2) forall (alpha ordinal): F(alpha) = G(F|`alpha)
    3) Mapping with properties (1) and (2) is unique.

An ordinal alpha is called a limit ordinal, if there is not ordinal beta such, that alpha = beta + 1.
The semantics of the supremum on ordinals is union.)
*)
  ([x1,x2,...] are sets) :<-> forall (j in (dom [x1,x2,...])): ([x1,x2,...](j) is set)
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# Descriptors
  (exists X: phi) -> phi[(eps X: phi)/X]
  (exists_unique X: phi) -> phi[(iota X: phi)/X]
  (basically the extension by definitions embedded in the system)
  (if (not (exists X: phi)) then (eps X: phi) is not specified.)

# Dependent classes
  E = {(x,y)| x in y} (*A relation, it can not be a family, since it is not functional. *)
  Sigma P = {(x,y)| y in P(x)} (* good for both cases: function P and family P *)

# image of a function

(* Every class can be enumerated by an ordinal class.
   Therefore, it is possible to define family of classes as an image of some family.
   Basically, it is a restriction of original family to any class.
*)
# Value of a function
  (x in (dom f)) -> f(x) := (U(f[{x}]))

End of foundations. **)
```

54.2 Proof trees

The first example unfolds the definition of inclusion and then uses the membership assumption to instantiate the restricted universal statement. For readability, write $\Gamma = (x \in A,\ A \subseteq B)$ for the common context.

$$\dfrac{\dfrac{\dfrac{x \in A \vdash x \in A}{\Gamma \vdash x \in A}\ \text{weakening} \qquad \dfrac{\dfrac{\dfrac{A \subseteq B \vdash A \subseteq B}{A \subseteq B \vdash \forall u \in A,\ u \in B}\ \text{definition of inclusion}}{\Gamma \vdash \forall u \in A,\ u \in B}\ \text{weakening}}{\Gamma \vdash x \in B}\ \text{restricted universal elimination}}{\vdash ((x \in A) \wedge (A \subseteq B)) \to (x \in B)}\ \text{discharge and conjunction of assumptions}$$

The next tree starts from an already established derivation of R under three assumptions. Its top sequent is a premise of the transformation, not an identity axiom asserting an arbitrary conclusion.

$$\dfrac{\dfrac{\dfrac{P_3, P_2, P_1 \vdash R}{P_3, P_2 \vdash P_1 \to R}\ \text{implication introduction}}{P_3 \vdash (P_2 \wedge P_1) \to R}\ \text{discharge and propositional equivalence}}{\vdash (P_3 \wedge P_2 \wedge P_1) \to R}\ \text{discharge and propositional equivalence}$$

Finally, sethood follows directly from membership in a class. The variable Y in the existential statement is chosen fresh.

$$\dfrac{\dfrac{\dfrac{B \in A \vdash B \in A}{B \in A \vdash \exists Y,\ B \in Y}\ \text{existential introduction, witness } A}{B \in A \vdash B \text{ set}}\ \text{definition of sethood}}{\vdash (B \in A) \to (B \text{ set})}\ \text{implication introduction}$$

54.3 Why the context matters for generalisation

Lemma 54.1 (Sound generalisation under assumptions)**.**
In ordinary classical first-order semantics, suppose every assignment satisfying all formulas in Γ satisfies φ. If x is not free in any formula of Γ, then every such assignment also satisfies $\forall x\, \varphi$.

Proof.
Let v satisfy Γ. For every value a of the sort of x, the assignment $v[x \mapsto a]$ still satisfies Γ, by coincidence of assignments and the freshness hypothesis. The premise therefore gives φ under each $v[x \mapsto a]$. This is exactly the truth condition for $\forall x\, \varphi$ under v. □
The freshness hypothesis cannot be dropped. In a two-element domain, interpret P as a singleton and assign its member to x. Then $P(x)$ is true but $\forall x\, P(x)$ is false. Hence the otherwise tempting step from $P(x) \vdash P(x)$ to $P(x) \vdash \forall x\, P(x)$ is unsound. Discharging that assumption would give the false formula $P(x) \to \forall x\, P(x)$. This counterexample explains why the discharge steps in the preceding trees are purely propositional, and why the logical chapter's deduction principle places a condition on any intervening generalisation. Quoted contexts and reflection rules need separate checks; this elementary semantic argument covers neither by mere reuse of the sequent notation.

54.4 Composing finite derivations

The following results concern ordinary total first-order natural deduction with finite proof trees, an unlimited supply of variables of each sort, and the usual introduction and elimination rules. They are statements about mathematical derivations, not a correctness certificate for the Standard ML listing. Formula substitution is capture-avoiding. Assumptions are occurrences at leaves, with discharge labels recording exactly which occurrences an introduction or elimination rule discharges. Two leaves bearing the same formula need not have the same discharge label. An occurrence is *open* if no rule above it discharges it.
We use the parameter form of the quantifier rules. Universal introduction passes from $\varphi(a)$ to $\forall x\, \varphi(x)$, with the parameter a absent from the conclusion and all open assumptions of its premise. Existential elimination has a subsidiary derivation from a temporary assumption $\varphi(a)$; its parameter a must be absent from the conclusion, the existential premise, and the other open assumptions of the subsidiary derivation. The indicated occurrences of $\varphi(a)$ are discharged. These conditions allow consistent renaming of the local parameter in its premise subtree, including its temporary assumptions, while leaving the rule's conclusion and the external assumptions unchanged.

Lemma 54.2 (Finite freshening)**.**
Given a finite derivation and a finite collection of other finite expressions, its local eigenparameters and discharge labels can be renamed to avoid the latter collection and to be pairwise distinct. The conclusion and formulas at open assumption occurrences are unchanged.

Proof.
There are only finitely many local parameters, labels, and symbols to avoid. At each quantifier rule choose a variable of the required sort outside the symbols occurring in the trees already under consideration. Rename the eigenparameter throughout its local premise subtree. Its side condition says that this parameter is absent from the external formulas that must be preserved. Renaming bound variables first if necessary makes the operation capture-avoiding; every rule instance remains an instance of the same rule. Repeating this for the finitely many quantifier rules provides distinct fresh parameters. Discharge labels can independently be replaced throughout each labelled group of leaves and its discharging rule. Labels do not occur in formulas, so this changes no formula. □

Theorem 54.1 (Grafting derivations; admissibility of cut)**.**
Let $\mathcal{D}$ be a finite derivation of B. Select any finite collection of its open assumption occurrences. For each selected occurrence with formula A_i, supply a finite derivation $\mathcal{E}_i$ of that same formula. There is a derivation of B whose open leaves are precisely the unselected open leaves of $\mathcal{D}$, together with the open leaves of the supplied derivations, with multiplicity. In particular, for finite contexts,

$$\Gamma, A \vdash B, \qquad \Delta \vdash A \quad \Longrightarrow \quad \Gamma, \Delta \vdash B.$$

Here the context notation permits repeated use of assumptions; it does not require the resulting proof to use each permitted assumption.

Proof.
Make a separate copy of a supplied proof for every selected occurrence. Freshen the local parameters of $\mathcal{D}$ away from every symbol in these copies, and then freshen the local parameters of the copies away from the resulting tree and from each other. Rename all discharge labels so that

distinct original discharge groups and distinct inserted copies have distinct labels. Replace each selected leaf by its corresponding proof, identifying its conclusion A_i with the old leaf formula. We verify the resulting tree from the replaced leaves upwards. Inside an inserted proof all rule instances are unchanged. At an original propositional rule, the premise conclusions have not changed; its rule instance therefore still applies. Its original discharge labels still identify exactly the original temporary assumptions: no inserted open leaf shares those labels. The same observation handles the discharged assumptions of existential elimination. At an original universal introduction or existential elimination, any newly open assumptions come from inserted proofs. The original eigenparameter is absent from all their formulas by freshening. Its other freshness requirements, including absence from the rule conclusion, are unchanged. Thus these quantifier rules also remain valid. Quantifier eliminations and existential introductions retain their original terms and formulas, so their substitution side conditions are unchanged. This checks every rule of the chosen calculus, and the stated description of open leaves follows from the preserved discharge labels.
For the displayed contextual instance, select all open leaves licensed by the extra assumption A and insert a copy of the proof from Δ at each. If there are no such leaves, nothing is inserted. If several are present, the proof from Δ is copied finitely many times; ordinary natural deduction permits this reuse. Any other open leaves are licensed by Γ. Hence all resulting open leaves are licensed by Γ, Δ. □

Corollary 54.1 (Weakening and finite support)**.**
A finite derivation can be used in any larger context of permitted assumptions. If $\Gamma \vdash B$ has a finite derivation, then $\Gamma_0 \vdash B$ for some finite $\Gamma_0 \subseteq \Gamma$.

Proof.
For weakening, keep the same tree: every open leaf licensed by the old context is licensed by the larger one. Quantifier side conditions here concern the actual open assumptions of a premise subtree, so merely permitting additional assumptions changes no rule instance. In a notation that writes a finite permitted context at every node, freshen the local parameters away from the additional context first; this gives the usual explicit weakening rule with its freshness conditions. For finite support, a finite tree has finitely many open leaves. Take Γ_0 to be the set of their formulas. All leaves are still licensed and all discharge and eigenparameter conditions remain the same. □
For an infinite Γ, the notation $\Gamma \vdash B$ in this paragraph means that a finite tree has its open leaves in Γ; it does not require infinitely many unused assumptions to be written at every node. Finite support is a syntactic fact about finitary proofs. It does not, without a completeness theorem, establish semantic compactness.

Example 54.1 (Reusing the inclusion proof)**.**
The first displayed tree gives a derivation of $x \in B$ with open assumptions $x \in A$ and $A \subseteq B$. Suppose separate derivations establish $\Delta \vdash x \in A$ and $\Theta \vdash A \subseteq B$. Grafting them at the corresponding leaves gives

$$\Delta, \Theta \vdash x \in B.$$

No new membership axiom is involved: the established derivations supply the inputs of the existing proof. Likewise, grafting this result into an inclusion proof with $B \subseteq C$ gives $\Delta, \Theta, B \subseteq C \vdash x \in C$. This is composition of proofs through their open assumptions, rather than substitution of terms for free variables.

The occurrence bookkeeping is essential. An assumption introduced inside a proof and later discharged cannot be replaced as though it were an open dependency, and an inserted open

assumption must not accidentally acquire an old discharge label. For a calculus with quoted sequents or reflection, each additional rule needs its own preservation argument: replacing a derivation can change the context represented in a quotation or the data tested by a reflection rule. The grafting theorem above therefore does not assert cut admissibility for arbitrary encoded reflection principles or for the earlier implementation merely because they display the same sequent symbol.

55 Restricted quantifiers: notation and a syntactic proposal

For an explicitly specified bound variable x that is not free in A, restricted quantifiers abbreviate ordinary quantification with a condition:

$$(\forall x \in A,\ Q) \longleftrightarrow (\forall x,\ x \in A \longrightarrow Q), \qquad (\exists x \in A,\ Q) \longleftrightarrow (\exists x,\ x \in A \wedge Q).$$

Likewise, $(!x \in A,\ Q)$ means $(!x,\ x \in A \wedge Q)$, with ! expressing at most one witness. All substitutions are capture-avoiding.
The commented draft below explores a more compressed syntax in which a bound variable would be recovered from the first term of the restriction. That proposal needs an explicit rule for selecting the variable and handling terms with several free variables. Until such a rule is fixed, the explicit-binder notation above determines the meaning unambiguously.

55.1 A precise version of inferred binders

One way to make the proposed compressed notation unambiguous is to supply a syntactic parameter context P: a finite set of variable names that are parameters, not candidates for the new binder. For a restriction formula R, compute $\mathrm{FV}(R) \setminus P$. Permit the abbreviation only when this set is a singleton $\{x\}$. Then expand

$$\forall[R]\,Q \quad \text{as} \quad \forall x\,(R \longrightarrow Q), \qquad \exists[R]\,Q \quad \text{as} \quad \exists x\,(R \wedge Q).$$

The analogous at-most-one abbreviation expands to $!x\,(R \wedge Q)$. When there are no candidates or several, require an explicit binder; do not choose a variable from its alphabetical order or an accidental presentation order. This is one formal realization of the proposal, not a claim that the current checker implements it.

Proposition 55.1 (Hygiene of the elaborated restriction)**.**
Once the binder has been determined, capture-avoiding substitution for parameters commutes with the displayed expansion, provided the binder is first renamed fresh for all replacement terms. Bijective renaming of variable names, applied also to P, preserves the inference rule.

Proof.
For the universal abbreviation, expansion yields $\forall x(R \longrightarrow Q)$. Freshness makes its substitution exactly $\forall x(R\sigma \longrightarrow Q\sigma)$, which is the expansion with the same explicitly retained binder and substituted components. Existential and at-most-one cases use the identical argument with conjunction. A bijective renaming ρ sends the candidate set to $\rho(\mathrm{FV}(R) \setminus P)$, because free occurrences and parameter names are renamed together. A singleton remains the singleton $\{\rho(x)\}$, so the inferred binder is renamed consistently. □

Example 55.1 (Inference precedes substitution)**.**
With $P = \{A\}$, the restriction $x \in A$ has the unique candidate x. If A is replaced by the variable y while the parameter context is left unchanged, the unelaborated restriction $x \in y$ has two candidate names. Repeating inference would reject it or select a different binder under an arbitrary heuristic. Elaborating first retains the intended binder x, and ordinary capture-avoiding substitution gives $\forall x(x \in y \longrightarrow Q\sigma)$. A parameter-context update could also record y explicitly; it must not be implicit in the mathematical meaning of an already parsed expression.

Chapter

Topics related to Dunford–Pettis theorem

56 Preamble

This work is about Dunford–Pettis theorem, the idea of which firstly appeared in **pettis1938integration**, as

$$(x \text{ measurable}) \iff ((x \text{ weakly measurable}) \wedge (x \text{ separably valued})),$$

where $x : S \to X$, and the following definition is used

Definition (x separably valued)**.**

$$\exists (D \subseteq B \text{ separable subspace}), \exists (S_1 \subseteq S), \mu(S_1) = \mu(S) \wedge \forall s \in S_1, x(s) \in D.$$

In this text, a variations of this statement and its corollaries are written, following texts "Schrödinger semigroups"(**Simon1982**) and "Topological vector spaces"(Trèves, 1967). Without defining details, it is proven later in the text in the following form:

Theorem ("A.1.1" from **Simon1982**)**.**
Let (M, μ) be a separable measure space and E a separable Banach space. Let A be a bounded operator from E to $L^\infty(M, d\mu)$. Then there exists a unique (up to sets of measure zero) weakly measurable function, a, from M to E' so that for each $f \in E$ and a.e. $x \in M$,

$$(Af)(x) = \langle a(x), f \rangle,$$

where $\langle a(x), f \rangle$ is a notation for $(a(x))(f)$. Moreover, $\|a\|_\infty = \|A\|$.

Notation 56.1 (styles)**.**
The following LaTeX math font styles are used in this text.

1. Unfactorised spaces of functions and sigma–algebras are denoted with style "mathfrak".
2. Factorised spaces of function are denoted with style "mathrm". I.e. the normal style: e.g. "B" and "L".
3. Linear operator spaces, topologies, set of functions which is zero almost everywhere are denoted with style "mathcal".
4. Nonspecified spaces with topological, linear or other structure, are denoted with style "mathbf".

57 Preliminaries

57.1 Linear transformations notation

Definition 57.1 (homogeneous function)**.**
Let $f : V_1 \to V_2$. Then

$$(f : V_1 \to V_2 \text{ homogeneous}) :\Longleftrightarrow (\forall(\alpha \in \mathbb{K})\forall(x \in V_1) : f(\alpha x) = \alpha f(x))$$

Definition 57.2 (additive function)**.**
Let $f : V_1 \to V_2$. Then

$$(f : V_1 \to V_2 \text{ additive}) :\Longleftrightarrow (\forall(\alpha \in \mathbb{K})\forall(x, y \in V_1) : f(x + y) = f(x) + f(y))$$

Definition 57.3 (space of linear operators between vector spaces over)**.**
Let V_1, and V_2 be modules over some commutative ring $\mathbb{K}$, Let V_1, and V_2 be vector spaces over some field $\mathbb{K}$,

$$(f \text{ linear}) :\Longleftrightarrow ((f \text{ homogeneous}) \wedge (f \text{ additive}))$$

Definition 57.4 (space of linear functions between VS)**.**

$$\text{Linear}(V_1, V_2) \coloneqq \{\ f : V_1 \to V_2 \mid f \text{ linear function}\}.$$

Definition 57.5 (space of continuous linear functions between TVS)**.**

$$\mathcal{L}(V_1, V_2) \coloneqq \{\, f : V_1 \to V_2 \mid f \text{ continuous linear operator}\,\}.$$

Definition 57.6 (form/functional on module)**.**
Let be V be module over ring $\mathbb{K}$, then the elements of the module $\text{Linear}(V, \mathbb{K})$ are called functionals or linear forms.

$$V^* \overset{\text{def.}}{\coloneqq} \text{Linear}(V, \mathbb{K})$$

$$V' \overset{\text{def.}}{\coloneqq} \mathcal{L}(V, \mathbb{K})$$

Theorem 57.1 (linear functions are continous iff bounded)**.**
Let $T : X \to Y$ be a linear operator between normed spaces. TFAE:

1. T continuous,
2. T continuous at 0,
3. T bounded,
4. T uniformly continuous.

Notation 57.1 (?)**.**
Treves denotes the space of continuous linear functionals as E', while Simon uses E^*. Due to the last theorem, in this text, it is possible to use the following agreement:

Definition 57.7 (algebraic dual space)**.**

$$E^* \overset{\text{def.}}{\coloneqq} \{\, x : E \to \mathbb{C} \mid x \text{ linear}\,\},$$

Definition 57.8 (topological dual space)**.**

$$E' \stackrel{\text{def.}}{\coloneqq} \{\, x : E \to \mathbb{C} \mid x \text{ linear and bounded} \,\}.$$

Any vector space V has a corresponding dual vector space, consisting of all linear forms on V, together with the vector space structure of pointwise addition and scalar multiplication by constants.
The dual space as defined above is defined for all vector spaces, and to avoid ambiguity may also be called the algebraic dual space.
When defined for a topological vector space, there is a subspace of the dual space, corresponding to continuous linear functionals, called the continuous dual space.

57.2 Closed graph theorem

Definition 57.9 (closed map, continuous map)**.**
(FA1, 4.4.1)
Context:

$$(x_n) : \mathbb{N} \to D$$

$$T : D \to Y$$

Definitions:

$$A(x) \overset{\text{def.}}{\longleftrightarrow} \Big((x \in X) \wedge \Big(\lim_{n\to\infty} \|x_n - x\| = 0 \Big) \Big)$$

$$B(y) \overset{\text{def.}}{\longleftrightarrow} \Big((y \in Y) \wedge \Big(\lim_{n\to\infty} \|Tx_n - y\| = 0 \Big) \Big)$$

$$\text{closed}(T) \longleftrightarrow (\forall x,\ \forall y,\ A(x) \wedge B(y) \longrightarrow (Tx = y) \wedge (x \in D))$$

$$\text{continuous}(T) \longleftrightarrow (\forall x,\ \exists y,\ A(x) \longrightarrow B(y) \wedge (Tx = y) \wedge (x \in D))$$

Theorem 57.2 (theorem of Hellinger–Toeplitz, 7.4.13)**.**
Let H be a Hilbert space. If $T : H \to H$ be a linear map satisfying

$$\langle Tx, y\rangle = \langle x, Ty\rangle$$

for all $x, y \in H$, then T is continuous and self–adjoint.

57.3 Bounded operators

57.4 Bounded and unbounded operators

Definition 57.10 (set of unbounded linear operators between Hilbert spaces)**.**

$$\text{UnbLin}(H_1, H_2) \overset{\text{def.}}{\coloneqq} \left\{ T \;\middle|\; \exists D,\ (\text{cl}_{\mathcal{T}_{H_1}}(D) = H_1) \wedge (D \subseteq H_1 \text{ vector subspace}) \wedge (T : D \to H_2 \text{ linear}) \right\}$$

Definition 57.11 (adjoint of unbounded operator)**.**

$$T^* \overset{\text{def.}}{\coloneqq} (\iota(S \in \text{UnbLin}(H_2, H_1)),\ (\forall x \in H_1,\ \forall y \in H_2,\ \langle y, Tx\rangle = \langle Sy, x\rangle) \wedge (\|S\| \leqslant \|T\|))$$

Theorem 57.3 (?)**.**

$$(D(T^*) = H_2) \longleftrightarrow (T \text{ bounded})$$

Proof. (TODO) □

57.5 Topological lemmas

To Show: "The image of a dense set under a map, which is continuous and surjective (i.e., onto), is dense."
Trivially, the map u is a surjective onto $u[E]$.

Definition 57.12 (topological property/homeomorphism invariant)**.**
The property P is called topological, if it respects homeomorphisms between any pair of defined topological spaces $\mathbf{X}$ and $\mathbf{Y}$:

$$(\mathbf{X} \cong \mathbf{Y}) \implies (P(\mathbf{X}) \longleftrightarrow P(\mathbf{Y}))$$

Definition 57.13 (hereditary topological property)**.**
The topological property P is called hereditary, if whenever a topological space has that property, then so does every subspace of it.

$$\forall S, ((X, \mathcal{T}) \text{ topological space}) \wedge (S \subseteq X)) \implies (P((X, \mathcal{T})) \implies P((S, \mathcal{T}|_S)))$$

Definition 57.14 (weakly hereditary topological property)**.**
The topological property P is called weakly hereditary, if whenever a topological space has that property, then so does every closed subspace of it.

$$\forall S,\ ((X, \mathcal{T}) \text{ topological space}) \wedge (S \subseteq X) \wedge (S \text{ closed})) \implies (P((X, \mathcal{T})) \implies P((S, \mathcal{T}|_S)))$$

Remark 57.1 (?)**.**
For example, to be a topology is a hereditary topological property. To be a closed set is a [trivially] weakly hereditary topological property. To be an open set is a topological property, but not even weakly hereditary. Several hereditary topological properties: being compact, being separable, being connected etc. Also the last implication becomes equivalence in case of compactness.

The statements of several useful topological lemmas about topological subspaces being understood, while proving statements which follows.

Lemma 57.1 (?)**.**

$$(A \subseteq X \text{ topological subspace}) \implies ((f : X \to Y \text{ continuous}) \implies (f : A \to Y \text{ continuous}))$$

(checked; no written proof)

Lemma 57.2 (?)**.**

$$(f[X] \subseteq Y \text{ topological subspace}) \implies ((f : X \to f[X] \text{ continuous}) \iff (f : X \to Y \text{ continuous}))$$

(checked; no written proof)

Lemma 57.3 (?)**.**

$$(K \text{ cpt w.r.t. } (X, \mathcal{T})) \overset{\text{lemma}}{\iff} (K \text{ cpt w.r.t. } (K, \mathcal{T} \restriction K)) \overset{\text{def.}}{\iff} ((K, \mathcal{T} \restriction K) \text{ cpt})$$

(checked; no written proof)

Lemma 57.4 (?)**.**

$$(V \text{ closed w.r.t. } (Y, \mathcal{T} \restriction Y)) \iff (\exists W : (W \text{ closed w.r.t. } (X, \mathcal{T})) \wedge (V = W \cap Y))$$

(checked; no written proof)

Lemma 57.5 (?)**.**

$$\mathrm{cl}_{(W,T\restriction W)}(D) = W \cap \mathrm{cl}_{(X,T)}(D)$$

(checked; no written proof)

Lemma 57.6 (?)**.**

$$(R \subseteq \mathbf{Y_1} \overset{\text{top.sub.}}{\subseteq} \mathbf{Y_2}) \implies ((Q \text{ dense in } R \text{ w.r.t. } \mathbf{Y_1}) \iff (Q \text{ dense in } R \text{ w.r.t. } \mathbf{Y_2}))$$

(checked; no written proof)

Lemma 57.7 (?)**.**

$$(E, F \text{ topological spaces}) \wedge (f : E \to F \text{ continuous}) \wedge (S \text{ dense w.r.t. } E) \implies (f[S] \text{ dense in } f[E] \text{ w.r.t. } F)$$
(lem:contdenden)

Proof. Our goal is to show that from $(f : E \to F$ continuous) it follows, that

$$\mathrm{cl}_E(S) \supseteq E \implies \mathrm{cl}_F(f[S]) \supseteq f[E].$$

Firstly,

$$y \in f[E] \implies \exists x \in E : f(x) = y.$$

$$x \in E \subseteq \mathrm{cl}(S) \implies x \in \bigcap_{C \text{ closed in } E, S \subseteq C} C$$

To Show:

$$y = f(x) \in \bigcap_{K \text{ closed in } F, S \subseteq K} K$$

Fix K closed in $(F, \mathcal{T}_F)$, s.t. $f[S] \subseteq K$.
The set $f^{-1}[K]$ is closed in $(E, \mathcal{T}_E)$. (

$$(F \setminus K) \text{ open}, f \text{ continuous} \implies f^{-1}[(F \setminus K)] \text{ open}$$
$$\implies F \setminus f^{-1}[(F \setminus K)] \text{ closed}$$

)

$$(S \subseteq \mathrm{dom}(f)) \& (f[S] \subseteq K) \implies (S \subseteq f^{-1}[]K)$$

It follows from assumption, that $x \in C \coloneqq f^{-1}[K]$.

$$x \in f^{-1}[K] \implies f(x) \in K.$$

And we are done. □

Lemma 57.8 (the image of a dense subset is dense in range)**.**
The image of a dense set under a map, which is continuous and surjective (i.e., onto), is dense. (exercise p.13 TVS Trèves, 1967[p.13])(it is needed for the next theorem)
Proof. Prove that the image of a dense set under a map, which is continuous and surjective (i.e., onto), is dense.
1) Lemma lem:contdenden states:

$$(E, F \text{ topological spaces}) \wedge (f : E \to F \text{ continuous}) \wedge (S \text{ dense w.r.t. } E) \implies (f[S] \text{ dense in } f[E] \text{ w.r.t. } F).$$

2) The new statement:

$$((f : X \to Y \text{ continuous}) \ \& \ (A \text{ dense in } S \text{ w.r.t. } X)) \implies (f[A] \text{ dense in } f[S] \text{ w.r.t. } Y)$$

new:

$$((f : X \to Y \text{ continuous}) \ \& \ (S \text{ dense in } E \text{ w.r.t. } X)) \implies (f[S] \text{ dense in } f[E] \text{ w.r.t. } Y)$$

$$(S \text{ dense w.r.t. } E) \implies (S \text{ dense in } E \text{ w.r.t. } E)$$

Let $E \coloneqq S$
Let $f' \coloneqq (f \restriction \dots)$

$$(f : X \to Y \text{ continuous}) \ \& \ (A \subseteq \mathrm{cl}_X(S)) \implies f[A] \subseteq \mathrm{cl}_Y(f[S])$$

$$\mathrm{cl}(S) = \mathrm{dom}(f) \implies \mathrm{cl}(f[S]) = \mathrm{ran}(f)$$

statement:

$$(f : X \to Y \text{continuous onto}) \ \& \ (S \text{ is dense in } A \text{ w.r.t. } A) \implies (f[S] \text{ is dense in } Y \text{ w.r.t. } Y)$$

Using the previous lemma: $f[S] \dots$

□

57.6 Space of bounded functions

Definition 57.15 (bounded set)**.**

$$(\mathbf{P} \coloneqq (P, <) \text{ preorder}) \wedge (B \subseteq P)) \implies (B \text{ bounded set w.r.t. } \mathbf{P}) : \iff (\sup_{\mathbf{P}}(B) < +\infty_{\mathbf{P}})$$

Definition 57.16 (bounded function)**.**

$$(h \text{ bounded function w.r.t. } \mathbf{P}) :\iff (\text{range}(h) \text{ bounded set w.r.t. } \mathbf{P})$$

Let X be a set in this subsection.
Let $(Y, |\cdot|_Y)$ be a normed space in this subsection.

Definition 57.17 ($\mathfrak{B}^\infty(X, Y)$, the space of bounded normed–space–element–valued functions on X)**.**

$$\mathfrak{B}^\infty(X, Y) \coloneqq \{\, f : X \to Y \mid (|\cdot|_Y \circ f) \text{ is a } (\mathbb{R}, <)\text{–bounded function} \,\}.$$

Definition 57.18 ($\mathfrak{B}^\infty(X)$, the space of bounded complex–valued functions on X)**.**
Since $(\mathbb{C}, |\cdot|)$ is a complete normed space, we can define:

$$\mathfrak{B}^\infty(X) \coloneqq \mathfrak{B}^\infty(X, (\mathbb{C}, |\cdot|)).$$

Definition 57.19 ($\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$, a norm in $\mathfrak{B}^\infty(X, Y)$)**.**

$$\|f\|_{\mathfrak{B}^\infty(X,Y)} \coloneqq \sup_{x \in X} |f(x)|_Y$$

Theorem 57.4 ($\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$ is a (semi)norm)**.**
If $(Y, |\cdot|_Y)$ is a (semi)normed space, then the function $\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$ is indeed a (semi)norm on $\mathfrak{B}^\infty(X, Y)$.
Proof. The proof is divided into steps:

1. Functionality of $\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$ and its totality on $\mathfrak{B}^\infty(X, Y)$.
   From definition,
   $$\|\cdot\|_{\mathfrak{B}^\infty(X,Y)} = \sup \circ (|\cdot|[X]).$$
   Therefore it follows from $\left(\sup : \mathcal{P}\mathbb{R} \to \overline{\mathbb{R}}_{\geqslant 0}\right)$ and $(f[X] \subseteq \mathbb{R})$ that
   $$\|\cdot\|_{\mathfrak{B}^\infty(X,Y)} : \mathfrak{B}^\infty(X, Y) \to \overline{\mathbb{R}}.$$

2. The range of $\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$ is a subset of $\mathbb{R}$.
   Let $f \in \mathfrak{B}^\infty(X, Y)$.
   Locally define $|f|$:
   $$|f| \coloneqq (|\cdot|_Y \circ f),$$
   $$|f| : X \to \mathbb{R}.$$
   (function f is bounded from above in $\mathbb{C}$) $\overset{\text{def.}}{\iff}$ (function $|f|$ is bounded from above in $\mathbb{R}$)
   (function $|f|$ is bounded from above in $\mathbb{R}$) $\implies$ $((\sup_{x \in X} |f(x)|) \in \mathbb{R})$ (by the least upper bound property of $\mathbb{R}$)

3. Triangle inequality.
   To show: $\|f+g\|_{\mathfrak{B}^\infty(X,Y)} \leqslant \|f\|_{\mathfrak{B}^\infty(X,Y)} + \|g\|_{\mathfrak{B}^\infty(X,Y)}$

$$\begin{aligned}
&\|f+g\|_{\mathfrak{B}^\infty(X,Y)} \\
&= \sup_{x\in X} |(f+g)(x)|_Y && \text{(by definition)} \\
&= \sup_{x\in X} |f(x)+g(x)|_Y && \text{(by def of pointwise sum)} \\
&\leqslant \sup_{x\in X} (|f(x)|_Y + |g(x)|_Y) && \text{(supremum respects pointwise inequality(here triangle inequality))} \\
&\leqslant \sup_{x\in X} (|f(x)|_Y) + \sup_{x\in X} (|g(x)|_Y) && (X \neq \emptyset, \text{ so sup of a pointwise sum of functions} \leqslant \text{sum of sups}[1])
\end{aligned}$$

4. Definiteness(if $(Y, |\cdot|_Y)$ is a normed space).
   To show: $\|f\|_{\mathfrak{B}^\infty(X,Y)} = 0_{\mathbb{R}} \implies f = 0_{\mathfrak{B}^\infty(X,Y)}$.

$$\sup_{x\in X} |f(x)|_Y = 0 \implies |f|[X] \leqslant 0$$
$$0 \leqslant |f|[X] \ \&\ |f|[X] \leqslant 0 \implies |f|[X] = 0$$

   So
$$(\forall x \in X)|f(x)|_Y = 0.$$
   Therefore by definiteness property of $|\cdot|_Y$ norm,
$$(\forall x \in X) f(x) = 0_Y.$$
   Therefore
$$f = (x \in X \mapsto 0_Y) = 0_{\mathfrak{B}^\infty(X,Y)}$$

5. Absolute homogeneity: $\|\alpha f\|_{\mathfrak{B}^\infty(X,Y)} = |\alpha| \|f\|_{\mathfrak{B}^\infty(X,Y)}$.

$$\sup_{x\in X} |\alpha f(x)| = \sup_{x\in X} |\alpha| |f(x)|_Y = |(|\alpha|)| \sup_{x\in X} (\operatorname{sgn}(|\alpha|) |f(x)|_Y) = |\alpha| \sup_{x\in X} |f(x)|_Y$$

□

Theorem 57.5 (???)**.**
If $(Y, |\cdot|_Y)$ is Banach space, then the space $(\mathfrak{B}^\infty(X,Y), \|\cdot\|_{\mathfrak{B}^\infty(X,Y)})$ is also a Banach space.
(The proof here is an elaboration of a text from https://math.stackexchange.com/a/168167/251394)
Proof. 1) $\|\cdot\|_{\mathfrak{B}^\infty(X,Y)}$ is a norm, by previous theorem.
2) $\mathfrak{B}^\infty(X,Y)$ is Cauchy–complete wrt that norm.

If $(f_n)_{n\in\mathbb{N}_0}$ is a Cauchy sequence, Let (f_n) be a Cauchy sequence in $\mathfrak{B}^\infty(X,Y)$. Then it has limit that is also in $\mathfrak{B}^\infty(X,Y)$.

1. Define:
$$f(x) \coloneqq \lim_{n\to\infty} f_n(x).$$
   Now we need to show that for each x, the value $f(x)$ is defined.

[1] https://en.wikipedia.org/wiki/Infimum_and_supremum

Since $(f_n)_{n\in\mathbb{N}_0}$ is a Cauchy sequence in $\mathfrak{B}^\infty(X,Y)$, then for all $x \in X$, $(f_n(x))_{n\in\mathbb{N}_0}$ is a Cauchy sequence in $\mathbb{R}$.

Due to that and since $\mathbb{R}$ is Cauchy complete, then the limit

$$\left(\lim_{n\to\infty} f_n(x)\right)$$

exists and is in $\mathbb{R}$.

2. Claim: f is bounded.

$$\Big((f_n) \text{ is Cauchy}\Big) \implies \left(\exists N : \forall n, m \geqslant N : \|f_n - f_m\|_\infty < \frac{1}{2}\right)$$

Fix $x \in X$,

$$|f(x)| \leqslant |f(x) - f_N(x)| + |f_N(x)| \leqslant \|f - f_N\| + \|f_N\|.$$

$$f_N \in \mathfrak{B}^\infty(X,Y) \implies \|f_N\|_\infty < \infty$$

Claim: $\|f - f_N\|_\infty \leqslant \frac{1}{2}$.
Fix $x \in X$,

$$|f(x) - f_N(x)|_Y = \left|\left(\lim_{n\to\infty} f_n(x)\right) - f_N(x)\right|_Y = \lim_{n\to\infty} |f_n(x) - f_N(x)|_Y$$

$$((\forall n \geqslant N)\, |f_n(x) - f_N(x)|_Y < 1/2) \implies \lim_{n\to\infty} |f_n(x) - f_N(x)|_Y \leqslant 1/2$$

We obtained, that

$$(\forall x \in X)|f(x) - f_N(x)| \leqslant \frac{1}{2}$$

Therefore the claim is proven:

$$\|f - f_N\|_\infty = \sup_{x\in X} |f(x) - f_N(x)| \leqslant \frac{1}{2}.$$

Using the claim, we finally obtain that function f is bounded

$$|f(x)| \leqslant \|f - f_N\| + \|f_N\| < \infty.$$

3. Convergence in norm: to show

$$\|f_n - f\|_\infty \overset{n\to\infty}{\longrightarrow} 0.$$

Since, (f_n) is a Cauchy sequence,

$$(\forall \varepsilon > 0)(\exists N)(\forall m, n \geqslant N)\|f_n - f_m\|_\infty \leqslant \varepsilon.$$

By the property of supremum,

$$(\forall x \in X)|f_m(x) - f_n(x)| \leqslant \sup_{w\in X} |f_m(w) - f_n(w)|.$$

From previous two formulas, the following follows by transitivity:

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N)(\forall x \in X)(\forall m \geqslant N)|f_m(x) - f_n(x)| \leqslant \varepsilon.$$

A limit of a sequence with a bounded from above tail is less than or equal to the bound:

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N)(\forall x \in X) \lim_{m \to N} |f_m(x) - f_n(x)| \leqslant \varepsilon.$$

Subtraction and absolute value function are continuous function, therefore:

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N)(\forall x \in X) \left| \left(\lim_{m \to N} f_m(x) \right) - f_n(x) \right| \leqslant \varepsilon.$$

By definition of a limit function f,

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N)(\forall x \in X) |f(x) - f_n(x)| \leqslant \varepsilon.$$

Since supremum is the least upper bound,

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N) \sup_{x \in X} |f(x) - f_n(x)| \leqslant \varepsilon.$$

By definition of norm in $\mathfrak{B}^\infty(X, Y)$,

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geqslant N) \|f - f_n\|_\infty \leqslant \varepsilon.$$

That meant that,

$$\|f - f_n\|_\infty \overset{n \to \infty}{\longrightarrow} 0 \text{ in } \mathbb{R},$$

which is a definition of convergence in norm in $\mathfrak{B}^\infty(X, Y)$:

$$f_n \overset{n \to \infty}{\longrightarrow} f \text{ in } \mathfrak{B}^\infty(X, Y).$$

□

57.7 Measure–theoretic notions

Definition 57.20 (measure space)**.**
A measure space is a triple $(X, \mathcal{A}, \mu)$, where
X is a set
$\mathcal{A}$ is a σ–algebra on the set X
μ is a measure on $(X, \mathcal{A})$

Definition 57.21 (integrable)**.**
A set is called *integrable* if it is measureable and its measure is finite.

$$(S \text{ integrable}) \stackrel{\text{def.}}{:=} ((S \in \mathcal{A}) \wedge (\mu(S) \in [0; +\infty)))$$

Definition 57.22 (separable topological space)**.**
A topological space is called *separable* iff it contains a countable, dense subset.

Note 57.1 (?)**.**
For measure spaces that are also topological spaces various compatibility conditions can be placed for the measure and the topology. Most measures met in practice in analysis (and in many cases also in probability theory) are Radon measures. Radon measures have an alternative definition in terms of linear functionals on the locally convex topological vector space of continuous functions with compact support. This approach is taken by Bourbaki (2004) and a number of other sources. For more details, see the article on Radon measures.
(https://en.wikipedia.org/wiki/Measure_(mathematics))

Definition 57.23 (Radon operator)**.**
Let X be a compact space.
Let $\mathbb{K}$ be a (not necessarily commutative) field $\mathbb{K}$.
Let $C(X, \mathbb{K})$ be a space of continuous functions.
Let $\mathbb{K}_0$ be one–dimensional vector space over $\mathbb{K}$.
If $A \in \text{LinCont}(C(X,), \mathbb{K}_0)$, then A is called a Radon operator (...).

Definition 57.24 (?)**.**
Radon measure, is a measure on the σ–algebra of Borel sets of a Hausdorff topological space X that is finite on all compact sets, outer regular on all Borel sets, and inner regular on open sets.[1] These conditions guarantee that the measure is "compatible" with the topology of the space, and most measures used in mathematical analysis and in number theory are indeed Radon measures.
(https://en.wikipedia.org/wiki/Radon_measure)

Definition 57.25 (inner regular set)**.**
Let (X, T) be a topological space and let Σ be a σ–algebra on X. Let μ be a measure on (X, Σ). A measurable subset A of X is said to be *inner regular* if

$$\mu(A) = \sup \{ \mu(F) \mid (F \subseteq A) \wedge (F \text{ compact and measurable}) \}$$

(https://en.wikipedia.org/wiki/Regular_measure)

Definition 57.26 (outer regular set)**.**
It is said to be outer regular if

$$\mu(A) = \inf \{ \mu(G) \mid G \supseteq A, G \text{ open and measurable} \}$$

(https://en.wikipedia.org/wiki/Regular_measure)

Definition 57.27 (regularity of a measure)**.**
A measure is called *inner regular* if every measurable set is inner regular. Some authors use a different definition: a measure is called inner regular if every open measurable set is inner regular.
A measure is called *outer regular* if every measurable set is outer regular.
A measure is called [just] *regular* if it is outer regular and inner regular.
(https://en.wikipedia.org/wiki/Regular_measure)

Note 57.2 (?)**.**
"The following are all examples of Radon measures: Lebesgue measure on Euclidean space (restricted to the Borel subsets); ... "

An L^p space may be defined as a space of measurable functions for which the p–th power of the absolute value is Lebesgue integrable, where functions which agree almost everywhere are identified.

Definition 57.28 ($\mathfrak{L}^p(M, d\mu)$)**.**
Let (S, Σ, μ) be a measure space and $1 \leqslant p \leqslant \infty$. When $p \neq \infty$, consider the set $\mathcal{L}^p(S,\, \mu)$ of all measurable functions f from S to $\mathbb{C}$ or $\mathbb{R}$ whose absolute value raised to the p–th power has a finite integral, or in symbols:

$$\|f\|_p \stackrel{\text{def}}{=} \left(\int_S |f|^p \, \mathrm{d}\mu\right)^{1/p} < \infty.$$

(https://en.wikipedia.org/wiki/Lp_space#Lp_spaces_and_Lebesgue_integrals)

Definition 57.29 (separable measure space)**.**

The following two definitions are slightly different. They seem to coincide for the reflexive spaces, but in this text only the first one will be used.

Definition 57.30 (weakly measurable function)**.**
Let E be a Banach space, and E' its (topological) dual with its Banach space. A function $f : X \to E'$ is said to be scalarly measurable if, for every $e \in E$, the complex–valued function

$$(x \mapsto \langle f(x), e\rangle)$$

is measurable,
where $\langle a(x), f\rangle$ is a notation for $(a(x))(f)$. (Definition 46.2 from Trèves, 1967, p.471)

Definition (weakly measurable function(but stronger definition))**.**
A weakly measurable function taking values in a Banach space is a function whose composition with any element of the dual space is a measurable function in the usual (strong) sense. For separable spaces, the notions of weak and strong measurability agree. https://en.wikipedia.org/wiki/Weakly_measurable_function

57.8 Locally convex Hausdorff topological vector spaces

Definition 57.31 (topological vector spaces / TVS)**.**
Let $\mathbf{E} = (E, \mathcal{T}, \mathbb{K}, +, \cdot)$ be both complex vector space and topological space. If the addition and the left multiplication with a scalar are continuous wrt product topologies on $E \times E$ and $\mathbb{K} \times E$, then the topology is called *compatible with its linear structure*, a $\mathbf{E}$ becomes a *topological vector space*. (reformulation of Trèves, 1967, p.57)

Example 57.1 (strong operator topology)**.**
Let H be Hilbert space. Let the set ρ be

$$\rho \overset{\text{def.}}{\coloneqq} \{ (T \in \mathcal{L}(H) \mapsto \|Tx\|_H) \mid x \in H \}.$$

The coarses topology on $\mathcal{L}(H)$, such that all functions form ρ are continuous, is called *strong operator topology*.

Definition 57.32 (?)**.**

$$(T_n \to T \text{ strongly}) \overset{\text{def.}}{\longleftrightarrow} (\forall x \in H,\ \|T_n(x) - T(x)\| \to 0)$$

Definition 57.33 (non–degenerate pairing)**.**

Definition 57.34 (locally convex Hausdorff space / LCH–space)**.**
A topological vector space $\mathbf{E}$ is said to be *locally convex Hausdorff space* if it is Hausdorff and there is a basis of neighbourhoods of $\mathbf{E}$ consisting of convex sets.
(Trèves, 1967, p.57)

Definition 57.35 (family of seminorms)**.**
Let N be a neighbourhood from the basis of neighbourhoods of $\mathbf{E}$.

$$p_N(x) \coloneqq \inf \{ \lambda > 0 \mid x \in \lambda N \}$$

Definition 57.36 (space of essentially bounded by seminorms LCH–valued functions)**.**
Let E be locally convex Hausdorf space.

$$\mathfrak{F}^\infty(X, E) \coloneqq$$

$$\{ f : X \to E \mid (\forall p \text{ continuous seminorm on } E), \exists M_p \geqslant 0, \exists N_p \in \mathcal{N}(X, E), \forall x \in (X \setminus N_p), p(f(x)) \leqslant M_p \}$$

(Trèves, 1967, p.468)

In particular,

$$\mathfrak{F}^\infty(X, \mathbb{C}) = \{ f : X \to \mathbb{C} \mid \exists m \geqslant 0, \exists N \in \mathcal{N}(X, \mathbb{C}), \forall x \in (X \setminus N), |f(x)| \leqslant m \}.$$

Definition 57.37 (space of measurable essentially bounded by seminorms LCH–valued functions)**.**

$$\mathfrak{L}^\infty(X, E) \coloneqq \{ f \in \mathfrak{F}^\infty(X, E) \mid f \text{ measurable} \}$$

Definition 57.38 (canonical surjection to $\mathrm{L}^\infty(X)$)**.**

$$c_L : \mathfrak{L}^\infty(X) \to \mathrm{L}^\infty(X)$$
$$c_L(q) \coloneqq [q] \coloneqq (q + \mathcal{N})$$

It is also a homomorphism of vector spaces, since it is a linear mapping.

57.9 Lebesgue space L^∞ of essentially bounded integrable functions

Definition and properties of $\mathfrak{L}^\infty$

Let (X, Σ, μ) be a positive measure space.
We require μ is σ–finite, i.e., that X is the union of a sequence of μ–integrable sets.

"We denote by $\mathfrak{L}^\infty$ the vector space of all complex–valued, measurable functions f in X such that there is a finite constant $M \geqslant 0$ with the following property:

(11.5) There is a subset N of X, with measure zero, such that $|f(x)| \leqslant m$ for all $x \in X \setminus N$."(Trèves, 1967, p.105)

Definition 57.39 (almost everywhere bounded by m)**.**
(parametrised by measure μ)

$$\varphi(\mu, f, m) :\Longleftrightarrow (\exists N \subseteq X : \mu(N) = 0 \wedge \forall x \in (X \setminus N) : |f(x)| \leqslant m)$$

Definition 57.40 ($\mathfrak{L}^\infty$–space)**.**

$$\mathfrak{L}^\infty(X, \mu) \coloneqq \{\, f \mid \exists m \geqslant 0 : \varphi(\mu, f, m) \,\}$$

FA1–inspired, p.26:

Definition 57.41 ($\mathfrak{B}^\infty$–norm)**.**

$$\|f\|_{\mathfrak{B}^\infty(W,\mu)} \overset{\text{def.}}{\coloneqq} \sup_{x \in W} |f(x)|_{\mathbb{K}}$$

(note, that the subscript here is a part of a name, and is not an application, because the corresponding space is not yet defined)

Definition 57.42 ($\mathfrak{B}^\infty$–space)**.**

$$\mathfrak{B}^\infty(W, \mu) \coloneqq \left\{\, f \;\middle|\; (f \text{ measurable}) \wedge (\|f\|_{\mathfrak{B}^\infty(W,\mu)} < \infty) \,\right\}$$

Definition 57.43 ($\mathfrak{L}^\infty$–norm)**.**

$$\|f\|_{\mathfrak{L}^\infty(X,\mu)} \overset{\text{def.}}{\coloneqq} \inf_{N \in \mathcal{A}, \mu(N)=0} \sup_{x \in X \setminus N} |f(x)|_{\mathbb{K}}$$

or, equivalently,

$$\|f\|_{\mathfrak{L}^\infty(X,\mu)} \overset{\text{def.}}{\coloneqq} \inf_{N \in \mathcal{A}, \mu(N)=0} \|f \restriction (X \setminus N)\|_{\mathfrak{B}^\infty(X \setminus N, \mu \restriction (\mathcal{P}(X \setminus N)))}$$

$$\|f\|_{\mathfrak{L}^\infty(X,\mu)} \overset{\text{def.}}{\coloneqq} \inf_{N \in \mathcal{A}, \mu(N)=0} \|f \restriction (X \setminus N)\|_{\mathfrak{B}^\infty(X,\mu)|_{X \setminus N}}$$

where for any measurable set W,

$$(X, \mu)\big|_W \overset{\text{def.}}{\coloneqq} (W, \mu \restriction (\mathcal{P}W))$$

(note, that the subscript here is a part of a name, and is not an application, because the corresponding space is not yet defined)

Lemma 57.9 (?)**.**
If (X, μ) is a measure space and $(W \subseteq X)$ is a measurable set of it, then $(X, \mu)|_W$ is a measure space.

Proof. 1) To show: $\mathrm{dom}(\mu \restriction (\mathcal{P}W))$ is a σ–algebra.

$$\mathrm{dom}(\mu \restriction (\mathcal{P}W)) = \mathrm{dom}(\mu) \cap (\mathcal{P}W)$$

Due to the third of the next lemmas,

$$(W \in \mathrm{dom}(\mu)) \;\&\; (\mathrm{dom}(\mu)\ \sigma\text{–algebra}) \implies ((\mathrm{dom}(\mu) \cap (\mathcal{P}W))\ \sigma\text{–algebra})$$

2) To show: $\mu \restriction (\mathcal{P}W)$ is a measure.
All defining properties of a measure are preserved on $(\mathrm{dom}(\mu) \cap (\mathcal{P}W))$, simple exercise. □

Lemma 57.10 (?)**.**
If $(W \in \mathcal{A})$, then $\bigcup(\mathcal{A} \cap \mathcal{P}W) = W$.

Proof. $(\subseteq)$: Fix x. Assume there exists $A \in \mathcal{A}$, $A \subseteq W$, $x \in A$. Then $x \in W$.
$(\supseteq)$: elementary exercise. □

Lemma 57.11 (?)**.**
If $(W \in \mathcal{A})$, then $\bigcup(\mathcal{A} \cap (\mathcal{P}W)) \in \mathcal{A} \cap (\mathcal{P}W)$.

Proof.

$$\begin{aligned}&(W \in \mathcal{A}) \implies (W \in \mathcal{A} \cap (\mathcal{P}W))\\ &\implies \bigcup(\mathcal{A} \cap (\mathcal{P}W)) \in \mathcal{A} \cap (\mathcal{P}W)\end{aligned}$$

□

Lemma 57.12 (?)**.**

$$(\mathcal{A}\ \sigma\text{–algebra}) \wedge (W \in \mathcal{A}) \longrightarrow ((\mathcal{A} \cap \mathcal{P}W)\ \sigma\text{–algebra})$$

Proof. via previous lemmas and "Theorem 27.3 (intersection of sigma–algebras)" □

Definition 57.44 ($\mathfrak{L}^\infty$–space)**.**

$$\mathfrak{L}^\infty(X, \mu) \coloneqq \left\{ f \;\middle|\; (f \text{ measurable}) \wedge (\|f\|_{\mathfrak{L}^\infty(X,\mu)} < \infty) \right\}$$

Notation 57.2 (?)**.**

$$\mathfrak{L}^\infty \coloneqq \mathfrak{L}^\infty(X, d\mu)$$

Definition 57.45 (essential supremum)**.**

$$\operatorname{ess\,sup}(f) \coloneqq \inf \{\, m \geqslant 0 \mid \varphi(\mu, f, m) \,\}$$

Definition 57.46 ($\|\cdot\|_{\mathfrak{L}^\infty}$)**.**
"We denote by $\|f\|_{\mathrm{L}^\infty}$ the infimum of all numbers $m \geqslant 0$ with property (11.5)".

$$\|f\|_{\mathfrak{L}^\infty} \coloneqq \begin{cases} \operatorname{ess\,sup}(f) & \text{if } \mu(S) \neq 0 \\ 0 & \text{if } \mu(S) = 0 \end{cases}$$

Definition 57.47 (functions are equal a.e.)**.**

$$(f \sim g) :\Longleftrightarrow (\operatorname{ess\,sup}(f-g) = 0) :\Longleftrightarrow (\varphi(\mu, f-g, 0))$$
$$:\Longleftrightarrow (\exists N \subseteq X,\ \mu(N) = 0 \wedge \forall x \in (X \setminus N),\ |f(x) - g(x)| = 0)$$

Theorem 57.6 (equal a.e. is an equivelaence relation)**.**

$$(f \sim f)$$

$(N \coloneqq \emptyset, g \coloneqq f, \mu(N) = 0)$

$$(f \sim g) \implies (g \sim f)$$

$(N \coloneqq \emptyset, |f(x) - g(x)| = |g(x) - f(x)|)$

$$(f \sim g) \wedge (g \sim h) \implies (f \sim h)$$

$(N \coloneqq \emptyset, |f(x)-g(x)| = 0 \wedge |g(x)-h(x)| = 0 \implies 0 \leqslant |f(x)-h(x)| \leqslant |f(x)-g(x)+g(x)-h(x)| \leqslant |f(x)-g(x)| + |g(x)-h(x)| \leqslant 0+0 \leqslant 0)$

Definition 57.48 (equivalence class)**.**

$$[f] \coloneqq \{\, g \mid f \sim g \,\} =$$
$$= \{\, g \mid \|f-g\|_{\mathfrak{L}^\infty} = 0 \,\}$$

Definition 57.49 (canonical projection)**.**

$$\pi \colon \mathfrak{L}^\infty \to L^\infty$$
$$\pi(f) \coloneqq [f]$$

(surjective map, returning an equivalennce class)

Just factorisation of function in classes, in which the norm of a substraction of any elemtn with any other is zero. (show that it follows that all elements in the class have the same norm)

$$\|f\|_{\mathfrak{L}^\infty} \coloneqq \operatorname{ess\,sup}(f)$$

(from 11.5 we can derive, that essential supremum is well–defined on f from $\mathfrak{L}^\infty$).
Note: "$\|\cdot\|_{\mathfrak{L}^\infty}$" is only a seminorm.

$$L^\infty \coloneqq \{\, [f] \mid f \in \mathfrak{L}^\infty \,\}$$

Definition of the L^∞–norm

The normed space associated to the seminormed space $\mathfrak{L}^\infty$ will be denoted by L^∞, and its norm by $f \mapsto \|f\|_{L^\infty}$;

Definition 57.50 ($\|f\|_{L^\infty}$)**.**

$$\|[f]\|_{L^\infty} := \|f\|_{\mathfrak{L}^\infty}$$

Theorem 57.7 ($\|\cdot\|_{L^\infty}$ is well–defined)**.**
Every element of class has the same $\mathfrak{L}^\infty$–norm, so L^∞–norm on classes is well–defined.
Proof. Assume $g \in [f]$. To show: $\|g\|_{\mathfrak{L}^\infty} = \|f\|_{\mathfrak{L}^\infty}$.

Since $g \in [f]$, then $\|f - g\|_{\mathfrak{L}^\infty} = 0$.
It follows, that

$$(\exists N \subseteq X,\ \mu(N) = 0 \wedge \forall x \in (X \setminus N),\ |(f-g)(x)| \leqslant 0)\,,$$

i.e. that f and g are equal almost everywhere.

$$\|f\| = \|f - g + g\| \leq \|f - g\| + \|g\| = 0 + \|g\| = \|g\|$$

$$\|g\| = \|g - f + f\| \leq \|g - f\| + \|f\| = 0 + \|f\| = \|f\|$$

Therefore $\|f\| = \|g\|$. □

"
one often writes L_∞, instead of L^∞. The elements of L^∞ are not functions in X, but equivalence classes of functions modulo the relation "to be equal almost everywhere". However, we shall often deal with them as if they were functions. An element f of $\mathfrak{L}^\infty$ (or of L^∞) is often said to be *essentially bounded* in X (with respect to the measure dx) and $\|f\|_{\mathfrak{L}^\infty}$ is called its *essential supremum.* "(p.105) "

Notation 57.3 (?)**.**
The following variant of notation is also used:

$$\mathfrak{L}^\infty(S, \Sigma, \mu) = \{\, f : (S, \Sigma) \to (\mathbb{C}, \mathfrak{L}_{\mathbb{C}}) \text{ measurable} \mid \|f\|_\infty < \infty \,\}\,,$$

where $\sigma(\mathcal{S})$ denotes the smallest sigma algebra generated by sets from $\mathcal{S}$, and

$$\mathrm{L}^\infty(S, \Sigma, \mu) := \{\, [f] \mid f \in \mathfrak{L}^\infty(S, \Sigma, \mu) \,\}\,.$$

57.10 Bounded functions on measurable space

Let (X, Σ) be a measurable space in this subsection.
Let $(Y, |\cdot|)$ be a Banach space in this subsection.

Definition 57.51 (support)**.**

$$\operatorname{supp}(f) \overset{\text{def.}}{\coloneqq} \{\, x \in X \mid f(x) \neq 0_Y \,\}$$

(https://en.wikipedia.org/wiki/Support_(mathematics))

Definition 57.52 (functions vanishing outside of a set of measure zero)**.**

$$\mathcal{N}(X, Y) \coloneqq \{\, f \in \mathfrak{B}^\infty(X) \mid dx(\{\, x \in X \mid f(x) \neq 0 \,\}) = 0_Y \,\}$$

Theorem 57.8 (?)**.**
The set $\mathcal{N}$ is a linear subspace of $\mathfrak{B}^\infty(X)$.

Proof.

1. Additivity:
   $\{\, x \in X \mid (f_1 + f_2)(x) \neq 0 \,\} \subseteq \{\, x \in X \mid f_1(x) \neq 0 \,\} \cup \{\, x \in X \mid f_2(x) \neq 0 \,\}$ By monotonicity of positive measure,

   $$0 \leqslant dx(\{\, x \in X \mid (f_1 + f_2)(x) \neq 0 \,\}) \leqslant dx(\{\, x \in X \mid f_1(x) \neq 0 \,\} \cup \{\, x \in X \mid f_2(x) \neq 0 \,\}).$$

   By countable subadditivity of measure,

   $$\leqslant dx(\{\, x \in X \mid f_1(x) \neq 0 \,\}) + dx(\{\, x \in X \mid f_2(x) \neq 0 \,\}) = 0 + 0 = 0.$$

2. Homogeneity:

   $$\{\, x \in X \mid (\alpha f)(x) \neq 0 \,\} \subseteq \{\, x \in X \mid f(x) \neq 0 \,\}$$

   By monotonicity of positive measure,

   $$0 \leqslant dx(\{\, x \in X \mid (\alpha f)(x) \neq 0 \,\}) \leqslant dx(\{\, x \in X \mid f(x) \neq 0 \,\}) = 0.$$

□

Lemma 57.13 (support of f is subset of union of supports)**.**

$$\operatorname{supp}(f) \subseteq \bigcup_{k \in \omega} \operatorname{supp} f_k$$

Proof.

$$x \in \operatorname{supp}(f) \implies f(x) \neq 0$$

To Show: $\exists k \in \omega : x \in \operatorname{supp}(f_k)$ i.e. $f_k(x) \neq 0$ for some k.
Assume oppposite: $(\neg\exists \dots)$. Therefore $\forall k \in \omega,\ f_k(x) = 0$.
Set $\varepsilon \coloneqq (|f(x)|/2)$.
Therefore $|f_k(x) - f(x)| = |0 - f(x)| = |f(x)| < \varepsilon = (|f(x)|/2)$.
That's a contradiction.
Therefore $\exists k \in \omega : f_k(x) \neq 0$. Moreover, there exist an infinite amount of such indexes, via the same argument, but for a subsequence of functions starting with f_{k+1}. □

Theorem 57.9 (set of functions vanishing outside of set of measure zero is closed)**.**
$\mathcal{N}(X, Y)$ is closed.

Proof. Indeed, if a sequence $\{f_k\}$ in $\mathcal{N}(X, Y)$ converges to f in $\mathfrak{B}^\infty(X, Y)$, each f_k must vanish outside some set of measure zero, N_k, therefore $f = 0_Y$ outside the union $\bigcup_k N_k$ which has also measure zero.

$$\mu \coloneqq dx$$

For any k:

$$N_k \coloneqq \operatorname{supp}(f_k) \coloneqq \{\, x \in X \mid f_k(x) \neq 0_Y \,\}$$
$$f_k \in \mathfrak{B}^\infty(X) \implies dx(N_k) = 0_{\mathbb{R}}$$

By countable subadditivity and since the sum is converging,

$$dx\left(\bigcup_k N_k\right) \leqslant \left(\sum_k dx(N_k)\right) = \left(\sum_k 0\right) = 0_{\mathbb{R}}.$$

By previous lemma,

$$\operatorname{supp}(f) \subseteq \bigcup_k N_k$$

By monotonicity of dx,

$$dx(\operatorname{supp}(f)) \leqslant dx\left(\bigcup_k N_k\right) = 0_{\mathbb{R}}$$

□

Definition 57.53 ($\mathrm{B}^\infty(X, Y)$)**.**
Let $\mathbf{X} = (X, \Sigma)$ be a measure space,
let $\mathbf{Y} = (Y, |\cdot|_Y)$ be a Banach space,
the set of classes of bounded functions unequal almost nowhere is denoted as

$$\mathrm{B}^\infty(X, Y) \coloneqq \mathfrak{B}^\infty(X, Y)/\mathcal{N}(X, Y).$$

Definition 57.54 (canonical surjection to $\mathrm{B}^\infty(X, Y)$)**.**

$$c_B : \mathfrak{B}^\infty(X) \to \mathrm{B}^\infty(X)$$
$$c_B(q) \coloneqq [q] \coloneqq (q + \mathcal{N})$$

It is also a homomorphism of vector spaces, since it is a linear mapping.

57.11 Linear isometric lift from separable subspaces of $\mathrm{B}^\infty(X)$

Lemma 57.14 (lift possession)**.**
Every separable linear subspace V of $\mathrm{B}^\infty(X)$ possesses a lift into $\mathfrak{B}^\infty(X)$ which is a linear isometry:

$$\forall\,(V \subseteq \mathrm{B}^\infty(X))\,(V \text{ linear}) \wedge (V \text{ separable}) \implies \exists\,(\sigma : V \to \mathfrak{B}^\infty)\,,\ (c_B \circ \sigma = \mathrm{id}_V) \wedge (\sigma \text{ linear isometry}).$$

(lem:possess_lift)

(46.1 Trèves, 1967, p.470)
Proof. Let A be a countable and dense subspace of V.
Let W be the set of all finite linear combinations of elements of A with coeficients in the field $\mathbb{Q}$ of rational numbers.

$$W \coloneqq \mathrm{span}_{\mathbb{Q}}(A) \coloneqq \left\{ \sum_{j=1}^{n} \alpha_j a_j \;\middle|\; n \in \mathbb{N}_0, \alpha_j \in \mathbb{Q}, a_j \in A \right\}$$

Plan of the proof: let τ be an arbitrary lift of W into $\mathfrak{B}^\infty$ which is linear: such a lift is obtained by lifting in an arbitrary manner an algebraic basis of W and extending the lift by $\mathbb{Q}$–linearity.

1. Given
$$A \overset{\text{dense}}{\subseteq} V \overset{\text{linear}}{\subseteq} \mathrm{B}^\infty(X)$$
$$(A \subseteq V \text{ is dense and countable}, V \text{ is a linear subspace of } \mathrm{B}^\infty(X)).$$

2. Define
$$W \coloneqq \mathrm{span}_{\mathbb{Q}}(A).$$

3. Therefore W is a vector space(over the field $\mathbb{Q}$).

4. Every vector space has an algebraic basis(Hamel basis), choose one.

5. Denote M as a chosen algebraic basis of W. Since it is basis of W, then $W = \mathrm{span}_{\mathbb{Q}}(M)$.

6. The function τ is defined by arbitrary choice on basis M, in the following way: The set $\mathrm{B}^\infty(X)$ consists of equivalence classes. Due to reflexivity of an equivivalence relation, the equivalence classes are nonempty. Therefore $\mathrm{B}^\infty(X)$ is a set of nonempty sets, so by the choice axiom, there is a choice function $c : \mathrm{B}^\infty(X) \to \mathfrak{B}^\infty(X)$, which picks element from each equivalence class. Define $\tau \overset{\text{def.}}{\coloneqq} (c \restriction M)$.
$$\tau : M \to \mathrm{B}^\infty(X)$$

7. Then the function τ is extended by $\mathbb{Q}$–linearity to W. For any $n \in \mathbb{N}$, $\vec{q} \in \mathbb{Q}^n$, $\vec{m} \in \mathbb{M}^n$,
$$\tau_{\text{new}}(\vec{q} \cdot \vec{m}) \overset{\text{def.}}{\coloneqq} \sum_{i=1}^{n} q_i \tau_{\text{old}}(m_i)$$
It is indeed an extension, because $\tau_{\text{new}}(m_j) = \sum_{i=1}^{n} \delta_j^i \tau_{\text{old}}(m_i) = \tau_{\text{old}}(m_j)$.

8.
$$N \overset{\text{def.}}{\coloneqq} \{\, n : X \to \mathbb{C} \mid \exists M \in \Sigma, \mu(M) = 0 \wedge \forall x \in (X \setminus M), n(x) = 0 \,\}$$
$$[f] \overset{\text{def.}}{\coloneqq} (f + N)$$

$$\|[f]\|_{\mathrm{B}^\infty(X)} \overset{\text{def.}}{\coloneqq} \inf_{(n\in N)} \|f+n\|_{\mathfrak{B}^\infty(X)}$$

The via the fact, that $\forall S, \mathrm{LB}(S) \leqslant \mathrm{Inf}(S) \leqslant S$, the following holds:

$$\|[f]\|_{\mathrm{B}^\infty(X)} \leqslant \left\{ \|f+n\|_{\mathfrak{B}^\infty(X)} \;\middle|\; n \in N \right\}$$

So for any $w \in \mathrm{B}^\infty(X)$,

$$(\forall \varepsilon > 0)(\exists N_\varepsilon \in \Sigma), \mu(N_\varepsilon) = 0 \wedge ((\forall x \in X \setminus N_\varepsilon), |\tau(w)(x)| \leqslant \|w\|_{\mathrm{B}^\infty(X)} + \varepsilon)$$

9. Set $\varepsilon_n \coloneqq \frac{1}{n}$.
10. Set $N \coloneqq \bigcup_{n=1}^{\infty} N_{\varepsilon_n}$.
11. By countable subadditivity $dx(N) \leqslant 0$.
12. Set

$$\begin{aligned} &\sigma(w)(x) = 0 && \text{if } x \in N \\ &\sigma(w)(x) = \tau(w)(x) && \text{if } x \in X \setminus N \end{aligned}$$

13. Note: the condition $x \in N$ doesn't depend on w. Therefore, it is either true or not, so in both cases

$$\begin{aligned} &\sigma(\alpha_1 w_1 + \alpha_2 w_2)(x) = 0 = \alpha_1 0 + \alpha_2 0 = \alpha_1 \sigma(w_1)(x) + \alpha_2 \sigma(w_2)(x) && \text{if } x \in N \\ &\sigma(\alpha_1 w_1 + \alpha_2 w_2)(x) = \tau(\alpha_1 w_1 + \alpha_2 w_2)(x) = \alpha_1 \tau(w_1)(x) + \alpha_2 \tau(w_2)(x) = \alpha_1 \sigma(w_1)(x) + \alpha_2 \sigma(w_2)(x) && \text{if } x \notin N \end{aligned}$$

14. **Therefore σ is linear**.
15. The map σ is a lift.
16. Therefore

$$\|w\|_{\mathrm{B}^\infty(X)} \leq \sup_{x \in X} |\sigma(w)(x)|$$

17. For all $x \in X$, $|\sigma(w)(x)| \leqslant \|w\|_{\mathrm{B}^\infty(X)}$.
18. **Therefore $\|\sigma(w)\|_{\mathfrak{B}^\infty(X)} = \|w\|_{\mathrm{B}^\infty(X)}$, so σ is an isometry**.

□

57.12 Boundedness of TVS–valued functions

Let X be a set.
Let μ be a measure on X.
Let E be a locally convex Hausdorf space.
Let f be a function such that $f : X \to E$.
For any p that is a continuous seminorm on E,

$$((x.\psi) \text{ is } \mu\text{–a.e. on } X) \overset{\text{def.}}{\iff} (\exists N_p \subseteq X : \mu(N_p) = 0 \wedge \forall\, (x \in X \setminus N_p)\,, \psi(x))$$

$$\psi_{p,f,m}(x) \overset{\text{def.}}{\iff} (p(f(x)) \leqslant m)$$

A real number m bounds f almost everywhere with respect to seminorm p

$$P(X, \mu, f, m, p) \overset{\text{def.}}{\iff} ((x.(\psi_{p,f,m}(x)) \ \mu \text{ - a.e. on } X))$$

$$\|f\|_{\mathfrak{F}^\infty((X,\mu),E),p} \overset{\text{def.}}{\coloneqq} \inf \{\, m \geqslant 0 \mid P(X, \mu, f, m, p) \,\}$$

Set of almost everywhere bounds of f. ...

$$\|f\|_{\mathfrak{F}^\infty((X,\mu),E),p} \overset{\text{def.}}{\coloneqq} \inf \{\, m \geqslant 0 \mid \,\}$$

Now p is not fixed.

Definition 57.55 (boundedness almost everywhere)**.**
Let X be a set.
Let E be a locally convex Hausdorf space.

$$(f \text{ is } \mathfrak{F}^\infty(X, E)\text{–bdd}) \overset{\text{def.}}{\iff} (\forall (p \text{ continuous seminorm on } E),\ \|f\|_{\mathfrak{F}^\infty(X,E),p} < +\infty)$$

$$\mathfrak{F}^\infty(X, E) = \left\{\, f : X \to E \;\middle|\; \forall (p \text{ continuous seminorm on } E),\ \|f\|_{\mathfrak{F}^\infty(X,E),p} < +\infty \,\right\}$$

$$\mathfrak{F}^\infty(X, E) = \{\, f : X \to E \mid f \text{ a.e. bdd} \,\}$$

new2:

$$\|f\|_{\mathrm{L}^\infty} \overset{\text{def.}}{\coloneqq} \|f\|_{\mathfrak{F}^\infty(X,\mathbf{C}),|\cdot|}$$

58 Dunford–Pettis theorem

In this section,
Let E be a Banach space,
Let X be a measurable space,
Let $\mathrm{L}^\infty(X,\mu)$ be a set of equivalence classes of Lebesgue measurable essentially bounded complex–valued functions on X,
Let A be a continuous linear map of E into L^∞.

58.1 Separability of subspaces

Lemma 58.1 (?)**.**
The image of a dense set under a map, which is continuous and surjective (i.e., onto), is dense.
(Trèves, 1967, p.13)

Lemma 58.2 (?)**.**
Let E be a Banach space.
Let A be a continuous linear map of E into L^∞.
If E is separable, then $A[E] \subseteq \mathrm{L}^\infty$ is a separable subspace.
Proof.

1. Goal: $A[E]$ is a subspace.
   - Additivity.
     Assume that $v_1 \in A[E]$, $v_2 \in A[E]$
     Then there exist $e_1 \in E$ and $e_1 \in E$, such that $v_1 = A(e_1)$, $v_2 = A(e_2)$.
     Claim: $v_1 + v_2 = A(e_1 + e_2)$.
     By assumption, A is a continuous linear map.
     Therefore $u[E] \ni A(e_1 + e_2) = A(e_1) + A(e_2)$.
     The image of a linear transformation of a linear subspace(whole E) is a linear subspace.
   - Homogeneity.
     Given: $v \in A[E]$, and $\alpha \in \mathcal{K}$.
     There exist $e \in E$, such that $v = A(e)$.
     By assumption, u is a continuous linear map.
     Therefore $A[E] \ni A(\alpha e) = \alpha A(e)$.
2. Goal: $A[E]$ is separable.
   E is separable, therefore there exists a countable subset D, such that $E \subseteq \overline{D}$.
   here we will use the previous lemma 58.1,and we are done.

□

Since $A[E]$ is separable, then $(h \circ A)[E] \subseteq \mathrm{B}^\infty$ is a separable subspace.
(todo: write the proof explicitly).

58.2 Definition of functions g_e and g

Definition 58.1 (σ)**.**
"by virtue of Lemma 46.1, there is a lift σ of $(h \circ A)(E)$ into $\mathfrak{B}^\infty$ which is a linear isometry"
(Trèves, 1967, lemma 46.1)

$(\forall V \subseteq \mathrm{B}^\infty(X))\,(V \text{ linear}) \wedge (V \text{ separable}) \implies (\exists \sigma : V \to \mathfrak{B}^\infty)\,(c_B \circ \sigma = \mathrm{id}_V) \wedge (\sigma \text{ linear isometry})$.

Definition 58.2 (definition of h)**.**

Using lemma 10.1, we obtained that

Definition 58.3 ($g_e(x)$)**.**

$$h : \mathrm{L}^\infty \to \mathrm{B}^\infty$$
$$A : E \to \mathrm{L}^\infty$$
$$e \in E$$
$$g_e(x) \coloneqq \sigma((h \circ A)(e))(x) \in \mathbb{C}$$
$$g_e = \sigma((h \circ A)(e)) \in \mathfrak{B}^\infty$$

This map differs from any other representative of $A(e) \in \mathrm{L}^\infty$ only on a set of measure zero.

Definition 58.4 (**g**)**.**

$$\mathbf{g} \coloneqq (x \in X \mapsto (e \in E \mapsto g_e(x)))$$
$$\mathbf{g} : X \to E'$$

58.3 Linearity of $(e \in E \mapsto g_e)$

Definition 58.5 (linear form)**.**
If V is a vector space over a field k, the set of all linear functionals from V to k is itself a vector space over k with addition and scalar multiplication defined pointwise. This space is called the dual space of V, or sometimes the algebraic dual space, when a topological dual space is also considered. It is often denoted $\mathrm{Hom}(V, k)$, or, when the field k is understood, V^* https://en.wikipedia.org/wiki/Linear_form

Note 58.1 (?)**.**
"The scalars are always complex numbers; no other field is ever considered (we shall switch to the field of real numbers for a very short moment, when talking about the theorem of Hahn–Banach, in Part II, and switch quickly back to complex numbers)."
(Trèves, 1967, p.7)

Theorem 58.1 (?)**.**
The function

$$g\colon E \times X \to \mathbb{K}$$
$$g(e, x) \coloneqq g_e(x)$$

is linear in the first component.

Proof. It is linear as a composition of two linear functions: A(given) and σ(proven in the step 14 of the lemma lem:possess_lift). □

$\forall\,(V \subseteq \mathrm{B}^\infty(X))\,,\ (V \text{ linear}) \wedge (V \text{ separable}) \implies \exists\,(\sigma : V \to \mathfrak{B}^\infty)\,,\ (c_B \circ \sigma = \mathrm{id}_V) \wedge (\sigma \text{ linear isometry})$.

58.4 Function g_e is weakly measurable

Definition 58.6 (evaluation map)**.**
Let X, Y be any classes, define

$$\text{ev.} : X \to ((X \to Y) \to Y),$$

$$\text{ev}_x(f) \coloneqq f(x).$$

Definition 58.7 (scalarly measurable("weak* measurable"))**.**
A function $(e \in E \mapsto g_e)$ is called *scalarly measurable* if and only if for any $e \in E$ the map $g_e \coloneqq (x \mapsto g_e(x))$ is measurable.

$$(g. \text{ is scalarly measurable}) \iff (\forall e \in E, ((x \in M \mapsto g_e(x)) \text{ is measurable}).$$

Lemma 58.3 ($g.$ is scalarly measurable)**.**
Let .. be .. .
...
Then function $(e \in E \mapsto g_e)$ is scalarly measurable.
Proof.
It is enough to show that for any $e \in E$ the map $g_e \coloneqq (x \mapsto g_e(x))$ is measurable.

Fix $e \in E$.

$$\begin{aligned}
g_e &= (x \mapsto g_e(x)) && (\eta\text{–transformation}) \\
&= (x \mapsto ((b \mapsto g_b(x))(e))) && (\beta\text{–transformation}) \\
&= (x \mapsto ((b \mapsto g_b(x))(e))) && (\beta\text{–transformation}) \\
&= (x \mapsto (\text{ev}_e(b \mapsto g_b(x)))) && (\delta\text{–transformation}) \\
&= (x \mapsto (\text{ev}_e(b \mapsto g_b(x)))) && (\delta\text{–transformation}) \\
&= (x \mapsto (\text{ev}_e(\mathbf{g}(x)))) && (\delta\text{–transformation}) \\
&= (x \mapsto \langle \mathbf{g}(x), e \rangle) && (\text{notation}) \\
&= \text{ev}_e \circ \mathbf{g} && (\delta\text{–transformation})
\end{aligned}$$

Therefore, as a composition of evaluation map of an element e of E and a measurable function, it is weakly measurable by definition. (Note: still by the weak definition of weakly measurable functions given by Trèves, 1967, p.471.)

$$\begin{aligned}
\sigma((h \circ A)(e)) &= (\sigma \circ h \circ A)(e) && () \\
&= (\sigma \circ c_{\text{B}} \circ r \circ \tau \circ A)(e) && () \\
&= (\text{id}_{(h \circ A)[E]} \circ r \circ \tau \circ A)(e) && () \\
&= (r \circ \tau \circ A)(e) && ()
\end{aligned}$$

Since $(\tau \circ A)(e) \in \mathfrak{L}^\infty(M, d\mu)$, it follows that $(\tau \circ A)(e)$ is measurable by definition of $\mathfrak{L}^\infty(M, d\mu)$.
The function r preserves measurability.(add ref to why it is so)
So $r((\tau \circ A)(e))$ is measurable.
So $(r \circ \tau \circ A)(e)$ is measurable.
Therefore $\sigma((h \circ A)(e))$ is measurable.
Therefore g_e is measurable, and the proof is completed. □

58.5 Essential boundedness of g_e

Lemma 58.4 (essential boundedness of g_e)**.**
For any fixed $e \in E$, it holds that $g_e \in \mathfrak{L}^\infty$.
Proof.

$$g_e = \sigma((h \circ A)(e))$$

$$\sigma : (h \circ A)[E] \to \mathfrak{B}^\infty$$

So g_e in $\mathfrak{B}^\infty$.

$$A : E \to Linf$$

Subgoal: the function r preserves measurable functions.
Characterization of r:

$$r(f)(x) = \inf \{\, g_n(x) \mid n \in N \,\}$$

(moreover, it's a minimum, but it's unrelated).
Fact 1: The pointwise infimum of any countable sequence of measurable functions is also measurable.
Fact 2: If both real and imaginary parts of complex–valued function are (real)measurable, then the function is also (complex)measurable.
Fact 3: (characterization of measurable functions) https://proofwiki.org/wiki/Characterization_of_Measurable_Functions

□

58.6 Finding the bound of g_e

Definition 58.8 (?)**.**
" The second term is also commonly used: if we have a map f of a set E *onto* a set F, a *lift* of a subset V of F is a mapping $g\colon V \to E$ such that $f \circ g$ is the identity of V (if one prefers, g is a right inverse of f on V, $f \circ g = \mathrm{id} \restriction V$). "

In the setting of Dunford–Pettis theorem, we use this definition with the following assignment of variables:

$$\text{“}E\text{”} \coloneqq \mathfrak{B}^\infty(X)$$
$$\text{“}V\text{”} \coloneqq (h \circ A)[E]$$
$$\text{“}f\colon E \to F\text{”} \coloneqq c_B\colon \mathfrak{B}^\infty(X) \to \mathrm{B}^\infty(X)$$
$$\text{“}F\text{”} \coloneqq \mathrm{B}^\infty(X)$$
$$\text{“}g\colon V \to E\text{”} \coloneqq \sigma\colon (h \circ A)[E] \to \mathfrak{B}^\infty(X)$$

Definition 58.9 (lift from the image of $(h \circ A)$)**.**
Lift from $(h \circ A)[E]$ into $\mathfrak{B}^\infty(X)$ w.r.t. canonical homomorphism]

$$\sigma\colon (h \circ A)[E] \to \mathfrak{B}^\infty(X)$$

(this σ is obtained via the lifting lemma)

Definition 58.10 ($g_e(x)$)**.**
For each $e \in E$ and $x \in X$, define

$$g_e(x) \coloneqq \sigma((h \circ A)(e))(x). \tag{def:gex}$$

Theorem 58.2 ($g_e(x)$ is bounded as a function of the argument $e \in E$)**.**
For each $x \in X$, $(e \mapsto g_e(x))$ is a bounded operator.
Proof.

$$\begin{aligned}
|g_e(x)| &= |\sigma((h \circ A)(e))(x)| && \text{(by definition)}\\
&\leqslant \sup_{y \in X} |\sigma((h \circ A)(e))(y)| && \text{(since sup is the least upper bound)}\\
&= \|\sigma((h \circ A)(e))\|_{\mathfrak{B}^\infty(M,\mathbb{C})} && \text{(by def of norm in } \mathfrak{B}^\infty(M,\mathbb{C}))\\
&= \|(h \circ A)(e)\|_{\mathrm{B}^\infty(M,\mathbb{C})} && \text{(since } \sigma \text{ is an isometry)}\\
&= \|h(A(e))\|_{\mathrm{B}^\infty(M,\mathbb{C})} && \text{(by unfolding composition)}\\
&= \|A(e)\|_{\mathrm{L}^\infty(M,\mathbb{C})} && \text{(since } h \text{ is an isometry)}\\
&\leqslant \|A\|_{\mathcal{L}(E,\mathrm{L}^\infty)} \|e\|_E && \text{(since } A \text{ is a bounded operator)}
\end{aligned}$$

□

Lemma 58.5 (equality of two supremums)**.**
Let E be a Banach space,

$$\sup_{x \in X} \left(\sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \right) = \sup_{x \in X} \left(\sup_{e \in E, \|e\|=1} \|(h \circ A)(e)\|_{\mathfrak{F}^\infty} \right) \tag{lem:eq2sup}$$

Proof.

1. ($\leq$)

$$\forall(x \in X), \forall(e \in E, \|e\| = 1), |\sigma((h \circ A)(e))(x)| \leq \|(h \circ A)(e)\|_{\mathfrak{F}^\infty} \qquad \text{(by "}A \leq \sup A\text{")}$$

$$\forall(x \in X), \sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \leq \sup_{e \in E, \|e\|=1} \|(h \circ A)(e)\|_{\mathfrak{F}^\infty} \qquad \text{(by "}(\forall e : k_1(e) \leqslant k_2(e)) \Rightarrow \sup k_1[E] \leq \sup k_2[E]\text{")}$$

$$\sup_{x \in X} \sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \leq \sup_{x \in X} \sup_{e \in E, \|e\|=1} \|(h \circ A)(e)\|_{\mathfrak{F}^\infty} \qquad \text{(by "}(\forall x : k_1(x) \leqslant k_2(x)) \Rightarrow \sup k_1[X] \leq \sup k_2[X]\text{")}$$

2. ($\geq$)

$$\forall(b \in E, \|b\| = 1), \left(\sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \right) \geq |\sigma((h \circ A)(b))(x)| \qquad \text{(by "}A \leqslant \sup(A)\text{")}$$

$$\forall(b \in E, \|b\| = 1), \sup_{x \in X} \left(\sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \right) \geq \sup_{x \in X} |\sigma((h \circ A)(b))(x)| \qquad \text{(sup respects pointwise ineq.)}$$

$$= \|\sigma((h \circ A)(b))\|_{\mathfrak{B}^\infty} \qquad \text{(by definition of } \|\cdot\|_{\mathfrak{B}^\infty}\text{)}$$

$$= \|(h \circ A)(b)\|_{\mathrm{B}^\infty} \qquad \text{(by } \|\cdot\|_{B^\infty} \text{ being well–defined)}$$

$$= \|A(b)\|_{\mathrm{L}^\infty} \qquad \text{(since } h \text{ is an isometry)}$$

$$\sup_{x \in X} \left(\sup_{e \in E, \|e\|=1} |\sigma((h \circ A)(e))(x)| \right) \geq \sup_{b \in E, \|b\|=1} \sup_{x \in X} |\sigma((h \circ A)(b))(x)| \qquad \text{(by "}A \geqslant B \Longrightarrow A \geqslant \sup B\text{")}$$

□

Theorem 58.3 (equal norms)**.**
Let E be a Banach space,
Let $\boldsymbol{X} = (X, \Sigma, \mu)$ be a measurable space,
Let $\mathrm{L}^\infty \coloneqq \mathrm{L}^\infty(\boldsymbol{X}, d\mu)$ denote a set of equivalence classes of Lebesgue measurable essentially bounded complex–valued functions on X,
Let A be a continuous linear map of E into L^∞,
Let $\mathbf{g}$ be $(x \in X \mapsto (e \in E \mapsto g_e(x)))$,
Let $\mathcal{L}$ Then the following holds:

$$\|\mathbf{g}\|_{\mathfrak{B}^\infty(X, E')} = \|A\|_{\mathcal{L}(E, \mathrm{L}^\infty)} \qquad \text{(thm:eq_norms)}$$

Proof.

$$
\begin{aligned}
&\|\mathbf{g}\|_{\mathfrak{B}^\infty(X,E')} && \text{(given)}\\
=&\sup_{x\in X}\|\mathbf{g}(x)\|_{E'} && \text{(by def of } \mathbf{g})\\
=&\sup_{x\in X}\|\mathbf{g}(x)\|_{\mathcal{L}(E,\mathbb{C})} && \text{(by def of } E')\\
=&\sup_{x\in X}\left(\sup_{e\in E,\|e\|=1}|g_e(x)|\right) && \text{(by def of operator norm)}\\
=&\sup_{x\in X}\left(\sup_{e\in E,\|e\|=1}|\sigma((h\circ A)(e))(x)|\right) && \text{(by def def:gex)}\\
=&\sup_{x\in X}\left(\sup_{e\in E,\|e\|=1}\|A(e)\|_{\mathrm{L}^\infty}\right) && \text{(by lemma of equality of two supremums lem:eq2sup)}\\
=&\left(\sup_{e\in E,\|e\|=1}\|A(e)\|_{\mathrm{L}^\infty}\right) && \text{(since the argument doesn't depend on } x)\\
=&\|A\|_{\mathcal{L}(E,\mathrm{L}^\infty)} && \text{(by one of equvivalent defs of operator's norm)}
\end{aligned}
$$

□

59 Statements from "Schrödinger Semigroups" by Barry Simon

Theorem (A.1.1)**.**
Let (M, μ) be a separable measure space and E a separable Banach space. Let A be a bounded operator from E to $\mathrm{L}^\infty(M, d\mu)$. Then there exists a unique (up to sets of measure zero) weakly measurable function, a, from M to E' so that for each $f \in E$ and a.e. $x \in M$,

$$(Af)(x) = \langle a(x), f \rangle.$$

where $\langle a(x), f \rangle$ is a notation for $(a(x))(f)$. Moreover, $\|a\|_\infty = \|A\|$.
(**Simon1982**, in sligthly different notation)
Proof.

1. Since (M, μ) is a measure space, then by definition the set $\Sigma \stackrel{\text{def.}}{\coloneqq} \mathrm{dom}(\mu)$ is a sigma–algebra on M.

   Definition 59.1 (generated σ–algebra)**.**

   $$\sigma(P) \stackrel{\text{def.}}{\coloneqq} \bigcap \{\, \Sigma \mid (P \subseteq \Sigma) \wedge (\Sigma \text{ is a sigma–algebra}) \,\}$$

   Here it's denoted as

   Definition 59.2 (Borel σ–algebra)**.**
   The smallest σ–algebra that contains its open set, is called a Borel σ–algebra.

   $$\mathcal{B}_M \stackrel{\text{def.}}{\coloneqq} \mathcal{B}(\tau_M).$$

   Since (M, μ) is a separable measure space, then there exists a topology τ_M, such that (M, τ_M) is a separable topological space, such that $\Sigma \supseteq \mathcal{B}(\tau_M)$.

2. Supports

   $$\mathrm{supp}(f) \stackrel{\text{def.}}{\coloneqq} \{\, x \in M \mid f(x) \neq 0 \,\}$$
   $$\overline{\mathrm{supp}}(f) \stackrel{\text{def.}}{\coloneqq} \overline{\mathrm{supp}(f)}$$

3. Define $\mathcal{N}(M, \mathbb{C})$.

   Definition 59.3 (functions vanishing outside of a set of measure zero)**.**

   $$\mathcal{N}(M, \mathbb{C}) \coloneqq \{\, f \in \mathfrak{B}^\infty(X) \mid \mu(\mathrm{supp}(f)) = 0_{\mathbb{C}} \,\}$$

4. Define c_{B}, a canonical surjection from $\mathfrak{B}^\infty(M, \mathbb{C})$ to $\mathrm{B}^\infty(M, \mathbb{C})$.

   Definition 59.4.

   $$c_B : \mathfrak{B}^\infty(M, \mathbb{C}) \to \mathrm{B}^\infty(M, \mathbb{C})$$
   $$c_B(q) \coloneqq [q] \coloneqq (q + \mathcal{N}(M, \mathbb{C}))$$

5. Prove that

$$\forall \varepsilon > 0 \Rightarrow \forall f \in \mathfrak{F}^{\infty}(M, \mathbb{C}) \Rightarrow \exists g \in \mathfrak{B}^{\infty}(M, \mathbb{C}), \|f\|_{\mathrm{L}^{\infty}(M,\mathbb{C})} \leqslant \sup_{x \in M} |g(x)| \leqslant \|f\|_{\mathrm{L}^{\infty}(M,\mathbb{C})} + \varepsilon$$

Proof.

$$\mathfrak{F}^{\infty}(X, \mathbb{C}) = \{\, f : X \to \mathbb{C} \mid \exists m \geqslant 0, \exists N \in \Sigma, (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \,\}$$

$$\begin{aligned}
(f \in \mathfrak{F}^{\infty}(M, \mathbb{C})) &\implies \exists m \geqslant 0, \exists N \in \Sigma, (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \\
&\implies \{\, m \geqslant 0 \mid \exists N \in \Sigma, (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \,\} \neq \emptyset \\
&\implies \inf \{\, m \geqslant 0 \mid \exists N \in \Sigma, (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \,\} < +\infty \\
&\implies \inf \operatorname{essUB}(f) < +\infty \\
&\implies \|f\|_{\mathfrak{F}^{\infty}} < +\infty \\
&\implies \forall \varepsilon > 0 \Rightarrow \exists m \in \operatorname{essUB}(f), m \leqslant \|f\|_{\mathfrak{F}^{\infty}} + \varepsilon
\end{aligned}$$

Fix $\varepsilon > 0$.
Obtain $m \in \operatorname{essUB}(f)$.
Define

$$N \coloneqq \operatorname{choice}(\{\, N \in \Sigma \mid (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \,\}).$$

(The set is non–empty, so the choice is possible here.)
Define

$$g_{\varepsilon}(x) \overset{\text{def.}}{\coloneqq} \begin{cases} f(x), & \text{if } (x \in M \setminus N) \\ 0, & \text{if } (x \in N) \end{cases} \qquad (g_{\varepsilon}\text{–def})$$

Obviously, $\|g_{\varepsilon}\|_{\mathfrak{B}^{\infty}(M,\mathbb{C})} \leqslant m$, because it is a supremum, and g_{ε} is zero on N, and equal to $f(x)$ outside of N, while $f(x)$ being bounded by m outside of N.

$$\|g_{\varepsilon}\|_{\mathfrak{B}^{\infty}(M,\mathbb{C})} \leqslant m \leqslant \|f\|_{\mathfrak{F}^{\infty}} + \varepsilon$$

Finally, by transitivity,

$$\|g_{\varepsilon}\|_{\mathfrak{B}^{\infty}(M,\mathbb{C})} \leqslant \|f\|_{\mathfrak{F}^{\infty}} + \varepsilon$$

It is left to prove the left inequality, i.e. to show that

$$\|f\|_{\mathfrak{F}^{\infty}} \leqslant \|g_{\varepsilon}\|_{\mathfrak{B}^{\infty}(M,\mathbb{C})},$$

It is true, since otherwise $\|f\|_{\mathfrak{F}^{\infty}}$ is not a precise lower bound(in this case $\|g_{\varepsilon}\|_{\mathfrak{B}^{\infty}(M,\mathbb{C})}$ would be less than the infimum, so contradiction). The proof is competed. □

6. Goal: Define r.

-
- a)

$$\left(g_n \overset{\text{a.e.}}{\underset{\mu}{=}} f\right) \overset{\text{def.}}{\longleftrightarrow} \left(\|f - g_n\|_{\mathfrak{F}^{\infty}(M,\mathbb{C}),|\cdot|} = 0\right)$$

- b)

$$g_n \in \mathfrak{B}^\infty$$

Therefore there exist a function $N : \{g_n\}_{n\in\mathbb{N}} \to \Sigma$, such that if $N_{g_n} \overset{\text{def.}}{:=} N(g_n)$ then $\mu(N_{g_n}) = 0$ and ....

$$g_n \overset{\text{a.e.}}{\underset{\mu}{=}} f$$

$$\|f\|_{\mathfrak{F}^\infty(M,\mathbb{C})} \leqslant \sup_{x\in M} |g_n(x)| \leqslant \|f\|_{\mathfrak{F}^\infty(M,\mathbb{C})} + \frac{1}{n} \tag{$*$}$$

- c) Lets define a local constant "$N_{r(f)}$".

$$\text{“}N_{r(f)}\text{”} \overset{\text{def.}}{:=} \bigcup_{n\in\mathbb{N}} N_{g_n}$$

- d)

Definition 59.5 (truncation function)**.**

$$r(f)(x) \overset{\text{def.}}{:=} \begin{cases} f(x), & \text{if } (x \in M \setminus N_{r(f)}) \\ 0, & \text{if } (x \in N_{r(f)}) \end{cases} \tag{$r(f)$–def}$$

- e)

$$r : \mathfrak{F}^\infty(M, d\mu) \to \mathfrak{B}^\infty(M, \mathbb{C})$$

7. Prove that r is an isometry.
To show: $\forall f \in \mathfrak{F}^\infty(M, d\mu), \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} = \|f\|_{\mathfrak{F}^\infty(M,d\mu)}$.
Fix $f \in \mathfrak{F}^\infty(M, d\mu)$.
($\leqslant$) To show: $\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \leqslant \|f\|_{\mathfrak{F}^\infty(M,d\mu)}$.
It is easy to see that

$$\forall n \in \mathbb{N}, \forall x \in M, |r(f)(x)| \leqslant |g_n(x)|$$

Therefore

$$\forall n \in \mathbb{N}, \sup_{x\in M} |r(f)(x)| \leqslant \sup_{x\in M} |g_n(x)|.$$

I.e.

$$\forall n \in \mathbb{N}, \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \leqslant |g_n(x)|_{\mathfrak{B}^\infty(M,\mathbb{C})}.$$

Now using the inequality $(*)$, we obtain

$$\forall n \in \mathbb{N}, \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \leqslant |g_n(x)|_{\mathfrak{B}^\infty(M,\mathbb{C})}. \leqslant \|f\|_{\mathfrak{F}^\infty(M,\mathbb{C})} + \frac{1}{n}$$

By transitivity,

$$\forall n \in \mathbb{N}, \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \leqslant \|f\|_{\mathfrak{F}^\infty(M,\mathbb{C})} + \frac{1}{n}.$$

Finally, since the left hand side of inequality doesn't depend on n, it is bounded by the infimum of the right hand side,

$$\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \leqslant \|f\|_{\mathfrak{F}^\infty(M,\mathbb{C})}.$$

($\geqslant$) To show: $\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \geqslant \|f\|_{\mathfrak{F}^\infty(M,d\mu)}$.

$$\|f\|_{\mathfrak{F}^\infty(M,d\mu)} = \inf\{\, m \geqslant 0 \mid \exists N \in \Sigma, (\mu(N) = 0) \wedge \forall x \in (M \setminus N), |f(x)| \leqslant m \,\} < +\infty$$

Fact 1:

$$\mu(N_{r(f)}) = 0.$$

Fact 2:

$$\forall x \in M, |f(x)| \leqslant \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})}.$$

Weakening of the fact 2:

$$\forall x \in (M \setminus N_{r(f)}), |f(x)| \leqslant \|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})}$$

Therefore

$$\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \in (\mu\,\mathrm{essUB}(f)).$$

Therefore

$$\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \geqslant \inf(\mu\,\mathrm{essUB}(f)).$$

I.e.

$$\|r(f)\|_{\mathfrak{B}^\infty(M,\mathbb{C})} \geqslant \|f\|_{\mathfrak{F}^\infty(M,d\mu)}$$

The proof that r is an isometry is completed.

8. Define h, such that

$$h : \mathrm{L}^\infty(M, d\mu) \to \mathrm{B}^\infty(M, \mathbb{C})$$

Firstly, we already have that:

$$r : \mathfrak{F}^\infty(M, d\mu) \to \mathfrak{B}^\infty(M, \mathbb{C}).$$

So since $\mathfrak{L}^\infty(M, d\mu) \subseteq \mathfrak{F}^\infty(M, d\mu)$, define $r_1 \stackrel{\text{def.}}{\coloneqq} (r \restriction \mathfrak{L}^\infty(M, d\mu))$, so now

$$r_1 : \mathfrak{L}^\infty(M, d\mu) \to \mathfrak{B}^\infty(M, \mathbb{C}).$$

Secondy, equivalence classes are non–empty, therefore choice can be made, so also define

$$\tau \stackrel{\text{def.}}{\coloneqq} (\mathrm{choice} \restriction \mathrm{L}^\infty(M, d\mu)),$$

where $\forall S, \mathrm{choice}(S) \in S$.
Therefore

$$\tau : \mathrm{L}^\infty(M, d\mu) \to \mathfrak{L}^\infty(M, d\mu).$$

The third map is already defined:

$$c_{\mathrm{B}} : \mathfrak{B}^\infty(M, \mathbb{C}) \to \mathrm{B}^\infty(M, \mathbb{C}).$$

Finally,

Definition 59.6 (function h from L^∞ to B^∞)**.**

$$h \stackrel{\text{def.}}{\coloneqq} (c_{\mathrm{B}} \circ r_1 \circ \tau).$$
$$h : \mathrm{L}^\infty(M, d\mu) \to \mathrm{B}^\infty(M, \mathbb{C})$$

9.

Lemma 59.1 (h is an isometry)**.**
Prove that h is an isometry, as a composition of isometries.
Proof.

- τ is an isometry, because a norm of an element of $\mathrm{L}^\infty(M, d\mu)$ is "defined this way".
- r_1 is an isometry, because it is a restriction of the isometry r.
- c_B is an due to the definition of $\mathrm{B}^\infty(M, \mathbb{C})$

□

10.

Lemma 59.2 (h is continuous)**.**

Proof.

$$(h \text{ isometry}) \implies (h \text{ continuous})$$

(add details here) □

11. Since E is a separable topological space, and h continuous, and A continuous, therefore $(h \circ A)[E]$ is separable.

12. Define σ.
Statement:

$$(\forall V \subseteq \mathrm{B}^\infty(M, d\mu))\,(V \text{ linear subspace}) \wedge (V \text{ separable in } \mathfrak{B}^\infty(M, \mathbb{C})) \implies$$
$$\implies (\exists \sigma_V : V \to \mathfrak{B}^\infty)\,(c_B \circ \sigma_V = \mathrm{id}_V) \wedge (\sigma \text{ linear isometry}).$$

Note: the topology of $\mathfrak{B}^\infty(M, \mathbb{C})$, in which V is a separable subspace, is the one that is generated by the norm $\|\cdot\|_\infty \overset{\text{def.}}{\coloneqq} (f \mapsto \sup_{x \in M} |f(x)|)$.

13. So using this statement, we define the σ, for particular V, which is

$$V \coloneqq (h \circ A)[E],$$

In order to use the stattement, it is necessary to prove the following lemma firstly:

Lemma 59.3.
$(h \circ A)[E]$ is a separable linear subspace.
Proof. TODO □

Definition 59.7 (A lift σ from a subset of B^∞ to $\mathfrak{B}^\infty$)**.**

$$\sigma \overset{\text{def.}}{\coloneqq} \sigma_{(h \circ A)[E]}.$$

Due to the used statement the following is also obtained:

$$(c_B \circ \sigma = \mathrm{id}_{(h \circ A)[E]})$$

$$(\sigma \text{ linear isometry})$$

14. Define $g_e(x)$.
$$(e \in E) \implies ((h \circ A)(e) \in (h \circ A)[E])$$
$$\implies ((h \circ A)(e) \in \operatorname{dom}(\sigma))$$
$$\implies (\sigma((h \circ A)(e)) \in \mathfrak{B}^\infty(M, \mathbb{C}))$$

$$g_e(x) \coloneqq \sigma((h \circ A)(e))(x) \in \mathbb{C}$$
$$g_e = \sigma((h \circ A)(e)) \in \mathfrak{B}^\infty(M, \mathbb{C})$$

15. Define a.
$$a(x)(e) \overset{\text{def.}}{\coloneqq} \sigma((h \circ A)(e))(x)$$

16. Modify a on a set of measure 0 and the properties will be the same. (todo?)

17. The function a is weakly measurable by lemma 58.3.
...

18.

19. OLD
Using Dunford–Pettis theorem.
Let $u \coloneqq A$.
Fix $f \in E$, and let $e \coloneqq f$.
Let $a \coloneqq (e \in E \mapsto g_e)$.

The function a is weakly measurable by lemma 58.3.
The equality $\|a\|_\infty = \|A\|$ in the notation of the current document is $\|\mathbf{g}\|_{\mathfrak{B}^\infty(M,E')} = \|A\|_{\mathcal{L}(E,\mathrm{L}^\infty)}$.
The equality $\|a\|_\infty = \|A\|$ holds due to isometricity of σ, via the theorem thm:eq_norms.

□

Corollary (A.1.2)**.**
If A is a bounded operator on $\mathrm{L}^p(M, d\nu)$ and A is bounded also from L^p to L^∞ then there is a measurable function a on $M \times M$ obeying

$$\sup_x \left[\int |a(x,y)|^q \, dy \right]^{1/q} (= \|A\|_{p;\infty}) < \infty \tag{A10}$$

(here $1/p + 1/q = 1$) so that, for any $f \in \mathrm{L}^p$,

$$(Af)(x) = \int a(x,y) f(y) \, d\mu(y). \tag{A11}$$

Conversely, if $A : \mathrm{L}^p \to \mathrm{L}^\infty$ has an internal kernel, a, in the sense of $(A11)$ obeying $(A10)$, then A is a bounded map from L^p to L^∞.
(**Simon1982**)
Proof. Here we use Dunford–Pettis theorem, but for special case, where

$$E \coloneqq \mathrm{L}^p(X, d\nu).$$

Using theorem (A.1.1) we obtain an element A of E'.
Then via Hölder's inequality: Let (S, Σ, μ) be a measure space and let $p, q \in [1, \infty]$ with $1/p + 1/q = 1$. Then for all measurable complex–valued functions f and g on S,

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

$$\int |a(x,y)f(y)|d\mu(y) \leq \left(\int |a(x,y)|^q \, d\mu(y)\right)^{\frac{1}{q}} \left(\int |f(y)|^p| \, d\mu(y)\right)^{\frac{1}{p}}$$

which is by (A10)

$$\leqslant \|A\|_{p;\infty} \|f\|_{\mathrm{L}^p} < \infty$$

Therefore $\int a(x,y)f(y)d\mu(y)$ converges absolutely, exists and finite.
In "Conversely" part, we also use the same inequality. It follows that A is bounded, via triangle inequality, and finiteness of the supremum.

$$\left|\int a(x,y)f(y)\, d\mu(y)\right| \leq \int |a(x,y)f(y)|\, d\mu(y)$$

and

$$\left(\int |a(x,y)|^q \, dy\right)^{1/q}$$

$$\leq \sup_{x \in M} \left[\int |a(x,y)|^q \, dy\right]^{1/q} (= \|A\|_{p;\infty}) < \infty.$$

□

Definition 59.8 ($\sigma(C, \mathfrak{M})$)**.**
By $\sigma(C, \mathfrak{M})$ we mean the weak topology on C determined by the functions

$$f \mapsto \int f \, d\mu.$$

Definition 59.9 (Weak topology)**.**
Suppose (X, Y, b) is a pairing of vector spaces over a topological field $\mathbb{K}$ (i.e. X and Y are vector spaces over $\mathbb{K}$ and $b\colon X \times Y \to \mathbb{K}$ is a bilinear map).
(https://en.wikipedia.org/wiki/Weak_topology#Weak_topology_with_respect_to_a_pairing)

Definition 59.10 (weak *–topology)**.**
*Weak *–topology* is the coarsest topology such that the maps T_x, defined by $T_x(\varphi) = \varphi(x)$ from X^* to the base field $\mathbb{R}$ or $\mathbb{C}$ remain continuous. (https://en.wikipedia.org/wiki/Weak_topology#Weak-*_topology)

Definition 59.11 ($\sigma(\mathfrak{M}, C)$)**.**
$\sigma(\mathfrak{M}, C)$ the weak *–topology on $\mathfrak{M}$

Theorem (A.1.3)**.**
Let X be a separable locally compact space and ν a Baire measure. Let T be bounded map from $\mathrm{L}^1(X, d\nu)$ to $\mathrm{L}^\infty(X, d\nu)$. Then T has an integral kernel which is separately continuous in x, y if and only if $\mathrm{Ran}(T) \subseteq C(X)$ and T has an extension f from $\mathfrak{M}(X)$ to $C(X)$ which is continuous in norm and also when $\mathfrak{M}$ is given the $\sigma(\mathfrak{M}, C)$ topology and C the $\sigma(C, \mathfrak{M})$ topology. If $\mathrm{supp}(\nu) = X$, then such a $\widetilde{T}$ is unique.
(**Simon1982**)
(without proof)

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